

GENERAL CONTENTS

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ORIGINAL TECHNICAL™
PUBLICATIONS



GENERAL FITTING INSTRUCTIONS

SAFETY PRECAUTIONS

Check that any lifting equipment used has adequate capability and is fully serviceable. Never rely on a jack alone to support a vehicle. Use supports carefully placed at the jacking points to provide a rigid location. Ensure that a suitable form of fire extinguisher is conveniently located.

Inspect power leads of any mains electrical equipment for damage and check it is properly earthed. Disconnect the earth (grounded) terminal of the vehicle battery.

Do not disconnect any pipes in the air conditioning refrigerant system unless trained and instructed to do so. A refrigerant is used which can cause blindness if allowed to contact eyes. Ensure adequate ventilation is provided when volatile de-greasing agents are being used.

WARNING

**FUME EXTRACTION EQUIPMENT MUST BE IN OPERATION WHEN SOLVENTS ARE USED
E.G. TRICHLOROETHANE, WHITE SPIRIT, SPB3,
METHYLENE CHLORIDE.
DO NOT SMOKE IN THE VICINITY OF VOLATILE
DE-GREASING AGENTS**

Strictly adhere to handling and safety instructions found on containers and labels. Containers whose capacity is over 25 litres (5 gallons) require a bund wall in order to contain spillages.

Keep oils and solvents away from naked flames and other sources of ignition e.g. welders.

Do not apply heat in an attempt to free stiff nuts or fittings; as well as causing damage to protective coatings, there is a risk of damage to electronic equipment and brake lines from stray heat.

Do not leave tools, equipment, spilt oil, etc. on or around the work area.

Used Engine Oils

Prolonged and repeated contact with mineral oil will result in the removal of natural fats from the skin, leading to dryness, irritation and dermatitis. In addition, used engine oil contains potentially harmful contaminants which may cause skin cancer. Adequate means of skin protection and washing facilities should be provided.

Health Precautions

1. Avoid prolonged and repeated contact with oils, particularly used engine oils.
2. Wear protective clothing, including impervious gloves where applicable
3. Do not put oily rags in pockets.
4. Avoid contaminating clothes with oil.
5. Overalls must be cleaned regularly. Discard unwashable clothing and oil impregnated footwear.
6. First aid treatment should be obtained immediately for open cuts or wounds.
7. Use barrier creams, applying before each work period, to help the removal of oil from the skin.
8. Wash with soap and water to ensure all oil is removed (skin cleansers and nail brushes will help). Preparations containing lanolin replace the natural skin oils which have been removed.
9. Do not use petrol, kerosene, diesel fuel, gas oil, thinners or solvents for washing skin.
10. If skin disorders develop, obtain medical advice.
11. Where practicable, de-grease components prior to handling.
12. Where there is risk of eye contact, eye protection should be worn, for example, chemical goggles or face shields; in addition an eye wash facility should be provided.

Environmental Protection Precautions

It is illegal to pour used oil on to the ground, down sewers or drains, or into water courses. The burning of used engine oils in small space heaters or boilers is not recommended unless emission control equipment is fitted; in cases of doubt, contact the Local Authority for advice on disposal facilities.

GENERAL FITTING INSTRUCTIONS

Always fit covers to protect the body and paintwork before commencing work on the vehicle. Cover the seats and carpets, wear clean overalls and wash hands before working inside the vehicle. Avoid spilling hydraulic fluid or battery acid on paintwork. In the event of spillage, wash off with water immediately. Use polythene sheets

in the boot to protect the carpet. Always use a recommended service tool where specified.

Preparation

Before disassembly, clean the surrounding area as thoroughly as possible. Blank off any openings exposed by component removal using greaseproof paper and masking tape. Seal fuel, oil and hydraulic lines immediately when separated, using plastic caps or plugs, to prevent loss of fluid and the ingress of foreign matter. Close the open ends of oil ways exposed by component removal with tapered hardwood plugs or readily visible plastic plugs.

Immediately a component is removed, place it in a suitable container: use a separate container for each component and its associated parts.

Before dismantling a component clean it thoroughly with a recommended cleaning agent: check that the agent is suitable for all materials of the component.

Clean the bench and provide marking materials, labels, containers and locking wire before dismantling a component.

Dismantling

Observe scrupulous cleanliness when dismantling components, particularly when brake, fuel or hydraulic system parts are being worked on. A particle of dirt or a cloth fragment could cause a dangerous malfunction if trapped in these systems.

Blow out all tapped holes, crevices, oilways and fluid passages with dry air. Ensure that any 'O' rings used in sealing are correctly replaced or renewed if disturbed. Mark mating parts, using marking ink, to ensure that they are replaced as dismantled.

Wire together mating parts where necessary to prevent accidental interchange (e.g. roller bearing components). Wire labels onto all parts to be renewed, and to parts requiring further inspection before being passed for reassembly. Place labelled parts and other parts for rebuild in separate containers.

Inspection - General

Never inspect a component for wear or dimensional check unless it is absolutely clean; a slight smear of grease can conceal and incipient failure. When a component is to be checked dimensionally against figures quoted for it, use the correct equipment (surface plates, micrometers, dial gauges, etc.) in serviceable condition.

Makeshift checking equipment can be dangerous. Reject a component if its dimensions are outside the limits quoted, or if damaged. A part may however, be refitted if its critical dimension is exactly limit size, and is otherwise satisfactory.

Ball and Roller Bearings

NEVER REPLACE A BALL OR ROLLER BEARING WITHOUT FIRST ENSURING THAT IT IS AS-NEW CONDITION.

Remove all traces of lubricant from the bearing under inspection by washing it in petrol or a suitable de-greaser, maintain absolute cleanliness throughout the operations.

Inspect visually for markings of any form on roller elements, raceways, outer surfaces of outer rings or inner surface of inner rings. Reject any bearings found to be marked, since any markings in these areas indicates onset of wear.

Holding the inner race between fingers and thumb of one hand, spin the outer race and check that it revolves absolutely smoothly. Repeat, holding the outer race and spinning the inner race.

Rotate the outer ring with a reciprocating motion, while holding the inner ring; feel for any check or obstruction to rotation, and reject the bearing if action is not perfectly smooth.

Lubricate the bearing with the correct quantity and specification of lubricant on installation. Inspect the shaft and bearing housing for discolouration or other markings suggesting that movement has taken place between bearing and seatings. If markings are found use Loctite in installation of replacement bearing.

Ensure that the shaft and housing are clean and free from burrs before fitting the bearing. If one bearing of a pair shows imperfection it is generally advisable to renew both bearings. An exception could be made only if the faulty bearing had covered a low mileage, and it could be established that damage was confined to it.

When fitting the bearing to shaft, apply force only to the inner ring of the bearing, and only to the outer ring when fitting into a housing.

Always mark components of separable bearings, e.g. roller bearings, in dismantling to ensure correct reassembly. Never fit new rollers in a used cup.

Oil Seals

Always use new oil seals when rebuilding an assembly. It is not physically possible to replace a new oil seal exactly as it has been bedded down.

Examine the seal before fitment to ensure that it is clean and undamaged.

Where applicable, smear the sealing lips with clean grease, pack dust excluder seals with grease, and heavily grease duplex seals in cavity between sealing lips.

Ensure that the seal spring, if provided, is correctly fitted.

Always use the correct service tool when fitting seals. If a service tool is not available, use a suitable drift approximately 0.4mm (0.015in) smaller than the outside diameter of the seal. Use a hammer VERY GENTLY on the drift if a press is not suitable.

Press or drift the seal into the depth of housing if the housing is shouldered, or flush with the face of the housing where no shoulder is provided.

NOTE

Most cases of failure or leakage of oil seals are due to careless fitting, and resulting in damage to both seals and sealing surfaces. Care in fitting is essential if good results are to be obtained.

Joints and Joint Faces

Always the correct gaskets where they are specified.

Use jointing compound only when recommended. Otherwise fit dry joints.

When jointing compound is used, apply a thin uniform film to metal surfaces; take great care to prevent it entering oilways, pipes or blind tapped holes.

Remove all traces of old jointing materials prior to reassembly. Inspect jointing faces for scratches or burrs and remove with a fine file or oil-stone. Do not allow swarf or dirt to enter tapped holes or enclosed parts. Blow out any pipes, channels or crevices with dry compressed air, renewing any 'O' rings or seals displaced by air blast.

Flexible Hydraulic Hoses

Before removing any brake or clutch hose, clean the end fittings and surrounding area as thoroughly as possible. Obtain blanking caps before detaching hose fittings so that ports can be covered to exclude dirt.

Clean the hose externally and blow through with dry compressed air. Examine carefully for cracks, security of end fittings and external damage. Reject any faulty hose.

When refitting a hose, ensure that no unnecessary bends are introduced and that the hose is not twisted before or during tightening of unions.

Containers for hydraulic fluid must be kept absolutely clean. Do not store hydraulic fluid in an unsealed container. It will absorb water and fluid in this condition would be dangerous to use due to a lowering of its boiling point.

Do not allow hydraulic fluid to be contaminated with mineral oil, or use a container which has previously contained contaminated mineral oil.

Do not re-use fluid bled from the system. Always use clean fluid of the correct specification to clean hydraulic components.

Fit a blanking cap to a hydraulic union and a plug to its socket after removal to prevent excessive loss of fluid and the ingress of dirt.

Absolute cleanliness must be observed with hydraulic components at all times.

After any work on hydraulic systems, carefully inspect for leaks underneath the vehicle while a second person applies maximum pressure to the brakes (engine running if servo is fitted) and/or clutch.

FIXINGS

Special Fixings

Bolts used in the door lock mechanism have a coating applied to the thread. Should any of these bolts be moved or removed the **MUST** be replaced with a new bolt.

Nyloc nuts are also used on the vehicle. If removed these **MUST** be replaced with a new nut.

Where self-locking nuts have been removed, replace them with new ones.

NOTE

Where bearing pre-load is involved nuts should be tightened in accordance with special instructions.

Screw Threads

Both UNF and Metric threads to ISO standards are used.

Damaged threads must always be discarded. Cleaning up threads with a die or tap impairs the strength and closeness of fit of the threads and is not recommended.

Always ensure that replacement fixings are JaguarSport approved parts.

Do not allow oil, grease or jointing compound to enter blind threaded holes. The hydraulic action on screwing in the bolt or stud could split the housing.

Always tighten a nut or bolt to the recommended torque figure. Damaged or corroded threads can affect the torque reading.

To check or re-tighten a bolt or screw to a specified torque figure, first slacken a quarter of a turn, then re-tighten to the correct figure.

Tabwashers

Fit new tabwashers in all places where they are used. Never replace a used tabwasher. Ensure that the new tabwasher is a JaguarSport approved part.

Split pins

Fit new split pins throughout when replacing any unit.

Always fit split pins where split pins were originally used. Do not substitute spring washers.

All split pins should be fitted as shown unless otherwise stated.

Locking wire

Fit new locking wire of the correct type for all assemblies incorporating it.

Arrange wire so that its tension tends to tighten the bolt heads to which it is fitted.

Aluminium Hose fittings

All Aluminium Hose fittings should be lightly lubricated, on assembly, with copper based grease to prevent any possibility of damage to the aluminium threads.



GENERAL SPECIFICATION

ENGINE

See Engine Tuning and Data

COOLING SYSTEM

Coolant		50% water/anti freeze
Water pumps	Type	2 centrifugal vane
	Drive	Gear driven from the crankshaft
Radiator		Front mounted incorporating two electric fans
Thermostat		Opening 88°C (190°F)

FUEL INJECTION EQUIPMENT

	Type	Electronic sequential Multi-point with electronic boost control
Fuel	Grade	95 RON Unleaded/98 RON Super Unleaded
Fuel Pumps	Type	Twin, electric volumetric high pressure pumps,
		Twin, electric low pressure pumps (in tank)
Fuel pressure		3.8bar (55lb/sq.in) modulated by manifold pressure

TRANSAXLE

Gearbox	Type	Manual, 5-speed with synchromesh on forward gears	
	Ratios	First	3.00:1
		Second	1.95:1
		Third	1.42:1
		Fourth	1.09:1
		Fifth	0.085:1
	"	Reverse	2.58:1
Clutch	Type	Hydraulically operated 8.5" twin dry plate	
Final Drive	Ratio	3.36:1	

FRONT SUSPENSION

Coil Springs	Type	Independent unequal length wishbones, coil spring and dampers	
	Wire dia.	120.7mm (0.420in)	
	No. of coils	11.6	
	Spring rate	225lbs/ins	
	Toe-in	0 deg 20 min $\pm$ 05 min	
	Camber	-1 deg 00 min $\pm$ 10 min	
	Castor	5 deg 00 min $\pm$ 30 min	

REAR SUSPENSION

Coil Spring	Type	Independent unequal length wishbones, coil springs and dampers
	Wire dia	12.3mm (0.484in)
	No. of coils	9.8
	Spring rate	450lbs/ins
	Toe-in	0 deg 05 min $\pm$ 05 min
	Camber	-1 deg 00 min $\pm$ 10 min

STEERING

Type	Mechanical rack and pinion
Turning circle	15.2m (49.8ft) kerb to kerb

BRAKES

Handbrake	Type	Drilled, ventilated discs with bridge-type calipers
	Type	Mechanical, separate calipers operating on rear discs
Disc diameter	Front	330.2mm (13in)
	Rear	300mm (11.8in)
Disc thickness	Front	35.5mm (1.4in)
	Rear	33.5mm (1.3in)

WHEELS AND TYRES

Tyres	Type	Unidirectional, asymmetrical tread pattern
	Size	Front 255/45ZR17
		Rear 345/35ZR18
Wheels	Type	One piece, lightweight aluminium alloy
	Size	Front 9in x 17in
		Rear 14in x 18in

ELECTRICS

Fuse Box A

Fuse No.	Colour	Rating	Maxi Fuse No.	Protected Circuit
1	Red	10A	3	Instruments and door mirror heaters
2	Red	10A	3	Reverse lamps and Direction indicators
3	Red	10A	3	Ignition coil, Alarm
4	Red	10A	3	Modules, Relays and Air conditioning mode
5	Light blue	15A	3	Wipers
6	Red	10A	3	Radio and Door mirrors

Fuse No.	Colour	Rating	Maxi Fuse No.	Protected Circuit
7	Yellow	20A	4	LH Window lift motor
8	Yellow	20A	4	RH Window lift motor
9	Tan	7.5A	4	Interior lamps and Clock
10	Tan	5A	2	RH Side lamps and Number plate lamp
11	Tan	5A	2	LH Side lamps and Number plate lamp
12	Tan	5A	2	Rear fog lamps
13	-	-	-	Not in use
14	Yellow	20A	5	Radio amplifier
15	Brown	7.5	5	Mag clutch
16	-	-	-	Spare
17	Red	10A	5	Hazard lamps
18	Light blue	15A	5	Horn
19	Red	10A	2	LH Headlamp (dipped beam)
20	Red	10A	2	RH Headlamp (dipped beam)
21	Red	10A	2	LH Headlamp (main beam)
22	Red	10A	2	RH Headlamp (main beam)

Fuse Box B

Fuse No.	Colour	Rating	Maxi Fuse No.	Protected Circuit
23	Light blue	15A	2	LH Headlamp cover motor
24	Light blue	15A	2	RH Headlamp cover motor
25	Tan	5A	2	Stop lamps

Fuse Box C

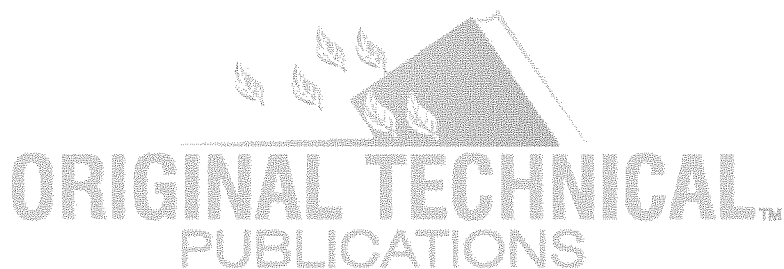
Fuse No.	Colour	Rating	Maxi Fuse No.	Protected Circuit
1	Brown	7.5A	1	Low pressure pumps O2 sensors
2	Light blue	15A	1	High pressure pump No.1
3	Tan	5A	1	Purge system
4	Red	10A	1	Engine management system
5	Light blue	15A	1	High pressure pump No. 2

Fuse Box D

Fuse No.	Colour	Rating	Maxi Fuse No.	Protected Circuit
1	Yellow	20A	-	C.D.L, Alarm
2	Tan	5A	-	Alarm arming system
3	Tan	5A	-	Clock

Bulbs

Location	Capacity	Type
Headlamp - main beam lamp	12V 55W	PK22s
Headlamp - dip beam lamp	12V 55W	P14.5s
Front side lamps	12V 5W	BA15s
Front direction indicator lamps	12V 21W	BA5s
Side repeater indicator lamps	12V 5W	Capless
Door lamps	12V 3W	Festoon
Footwell lamps	12V 5w	Festoon
Main interior lamp (roof console)	12V 5W	Festoon
Map lights (roof console)	12V 5W	Capless
Engine compartment lamps	12V 5w	BA15s
Luggage compartment lamps	12V 5w	Festoon



RECOMMENDED LUBRICANTS, FLUIDS, CAPACITIES AND DIMENSIONS

LUBRICANTS AND FLUIDS

The following products are recommended as they were used during the development and testing of the vehicle. Other products, which must meet the relevant specifications, may be used if desired.

Component	SAE Viscosity rating	Recommended products
Engine oil	10w60	Castrol Formula RS Castrol Syntrol Castrol Pure Racing
Transaxle	EP 75w90	Castrol Racing XH21
Brakes		Castrol SRF
Clutch		Castrol SRF
Grease Points		Castrol LMX
Oil can points		Light engine oil

CAPACITIES

	Litres	Imperial gallons		Litres	Imperial gallons
Fuel Tank	90	19.8	Cooling system	16	3.52
Engine oil			<i>Note: Always top up the cooling system with the 50% concentration water/anti-freeze NEVER with water only</i>		
-incl. filter	12	2.64			
-excl. filter	11.5	2.53			
Transaxle	4.25	0.935	Washer reservoir	8 approx.	1.76 approx.

DIMENSIONS

Overall Length	4930mm (194.1in)	Ground clearance	127mm (5in)
Overall Width	2220mm (87.4in)	Turning circle	15.2m (598.5in), kerb to kerb 15.5m (610.3in), wall to wall
Overall height	1150mm (45.3in)	Front ramp angle	7 degrees
Wheelbase	2640mm (103.9in)	Rear ramp angle	17 degrees
Track	Front 1710mm (67.3in) Rear 1588mm (62.5in)		

SERVICING SCHEDULE

1 INTRODUCTION

This section details the procedures to be carried out and the intervals for periodic servicing of the vehicle.

2 PRECAUTIONS

WARNINGS:

- 1 **ERECT NO SMOKING SIGNS AROUND THE VEHICLE AND KEEP A SERVICEABLE DRY POWDER FIRE EXTINGUISHER ON HAND.**
- 2 **ENSURE ALL LIFTING EQUIPMENT IS SERVICEABLE AND SATISFIES LOCAL HEALTH AND SAFETY REGULATIONS.**

CAUTIONS:

- 1 **Remove all traces of foreign matter from the vehicle body and fit JaguarSport approved body covers before any work is commenced.**
- 2 **Ensure the vehicle is correctly positioned on the lifting apparatus.**
- 3 **Apply barrier cream to the hands.**

NOTES:

- 1 *Ensure absolute cleanliness of work area and tools.*
- 2 *Ensure correct use of tools.*
- 3 *Inspect components as they are removed and label for possible repair or rejection where applicable.*

ALWAYS USE WING COVERS, INTERNAL PROTECTION MATERIALS AND THE CORRECT GRADES OF LUBRICANTS AND FLUIDS.

ORIGINAL TECHNICAL
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3	6	12	Months
1500	3000	6000	Miles
2500	5000	10000	Km
X			Renew engine oil
		X	Renew engine oil filter
		X	Clean engine sump filter gauze
X	X	X	Check engine for oil leaks
		X	Check crankcase breather system for leaks and turbo inlet for oil deposits
		X	Renew air cleaner elements
X			Renew transaxle oil and clean filter gauze
X	X		Check transaxle for oil leaks - top up if necessary
X	X	X	Check/top up cooling system - assure specific gravity
X	X	X	Check cooling/heating system for leaks

3	6	12	Months
1500	3000	6000	Miles
2500	5000	10000	Km
FLUIDS, FILTERS AND LEAK CHECK			
X	X	X	Check for fuel leaks
		X	Renew brake fluid
X	X	X	Check/top up brake fluid reservoir
		X	Renew clutch fluid
X	X	X	Check/top up clutch fluid reservoir
X	X	X	Visually check brake hydraulic hoses, pipes and unions for leaks
	X	X	Check suspension dampers for fluid leaks and wear
X	X	X	Check/top up windscreen washer reservoir
X	X	X	Check induction system vacuum hoses, pipes and unions for leaks
X	X	X	Check exhaust, catalyst and turbo system for leaks
		X	Spanner check exhaust manifold and head outtrigger nuts
X	X	X	Check air conditioning sight glass and check system for leaks
		X	Replace air conditioning drier bottle, check oil level and recharge

MECHANICAL FUNCTION CHECK

X	X	X	Lubricate all doors locks and hinges (not steering lock)
	X	X	Grease all points excluding hubs
	X	X	Check hub clearance and report
		X	Check ancillary drive belt tension, renew if necessary
X	X	X	Lubricate handbrake mechanical linkage and cables
X	X	X	Check security of handbrake (Release fully after checking)
	X	X	Check condition of handbrake pads/deglaze
	X	X	Inspect brake pads for wear and discs for condition/deglaze
X		X	Spanner check driveline, suspension, steering, brakes
	X	X	Lubricate accelerator control linkage and check adjustment
	X	X	Check windscreen wiper arms and blades, renew if necessary
		X	Clean out radiator matrix and condenser
X	X	X	Check condition of all seat belts
X	X	X	Check exhaust system and turbos for security
X	X	X	Check brake hoses, pipes and unions for chafing, cracks and corrosion
		X	Clean out intercooler ducts and matrix
X	X	X	Check security and condition of fuel hoses and unions
X	X	X	Check cooling system hoses for security and condition
X	X	X	Check air conditioning hoses for security and condition
X	X	X	Check condition and security of steering unit joints and gaiters
X	X	X	Check condition and security of suspension fixings, noddors and dampers
		X	Check tightness of driveshaft coupling bolts
X	X	X	Check rear view mirrors for security and function

3	6	12	Months
1500	3000	6000	Miles
2500	5000	10000	Km
MECHANICAL FUNCTION CHECK			
	X	X	Check front suspension lower ball joints, adjust if necessary
		X	Check the condition of the underbody anti-corrosion protection
		X	Check drain tubes, air conditioning and battery
	X	X	Check adjustment of turbo boost control valves
	X	X	Check condition and security of engine and transaxle mounts

ELECTRICAL FUNCTION CHECK

		X	Check engine management settings
X		X	Check throttle potentiometer setting
	X	X	Renew spark plugs
X	X	X	Push high tension leads firmly home
		X	Check condition of rotor arm
		X	Check/top up battery electrolyte and check charge condition
X	X	X	Clean and grease battery connections and check operation of battery isolator
X	X	X	Check and adjust headlamp alignment and covers
X	X	X	Check operation of lamps
X	X	X	Check operation of air conditioning and radiator fans
X	X	X	Check operation of warning indicators and hazard lamps
X	X	X	Check audible warnings
X	X	X	Check operation of horns
X	X	X	Check operation of windscreen wipers and mechanism
X	X	X	Check operation of windscreen washers
X	X	X	Check operation of audio system
X	X	X	Clean radio aerial mast
X	X	X	Check operation of vehicle security system and locks

GEOMETRY, WHEELS AND TYRES CHECK

X	X	X	Check that tyres are the right size, type and rotation
X	X	X	Check tyre tread depth
X	X	X	Check tyres for uneven and excessive wear and damage
X	X	X	Check/adjust tyre pressures
X	X	X	Check tightness of road wheel nuts, refit security clips
X	X	X	Check and balance all road wheels
		X	Check and adjust steering and suspension geometry

3	6	12	Months
1500	3000	6000	Miles
2500	5000	10000	Km
X	X	X	Apply Jaguar screen clean paste to the exterior of the windscreen

GENERAL

ROAD TEST CHECK

X	X	X	Carry out road test - report all faults
X	X	X	Check operation of seat belts and seat adjustments

RECOMMENDATIONS

Every 24 months 12,000 miles 20,000Km	Replace engine cam drive belts and check idler pulleys Replace ancillary drive belt and check idler pulleys Replace clutch components, release and pinion nose bearing Replace engine and transaxle mounting bushes Renew engine coolant - assure specific gravity Renew transaxle oil and clean filter gauze Clean engine compartment Renew fuel filters Check engine compartment heat and noise insulation
Every 48 months 24,000 miles 40,000 Km	Check engine valve clearances
Every 6 years	Replace fuel bag tank Replace engine coolant hoses

ENGINE MANAGEMENT FAULT FINDING

INTRODUCTION

This section is intended as an aid to be used in conjunction with the Engine Management Fault Finding diagnostic programme.

The vehicle's engine management control unit (ECU) is located in the driver/passenger compartment, on the RH sill panel, towards the rear bulkhead.

CONNECT AND SET-UP PROCEDURE

Firstly identify the equipment to be used during fault diagnosis:-

1. Toshiba T1000 Se Laptop Computer
2. Computer to interface box cable - SPD 1040. One end has a 9-pin 'D' type plug which connects to the computer, the other end has a round plug which connects to the interface box.
3. Interface box - SPD 1039. This interface box has two sockets at one end, one connects to the computer (cable SPD 1040), and the other connects to the ECU (cable SPD 1038).
4. Interface to ECU cable - SPD 1038. The interface end of this cable has a round 5-pin socket, and the ECU end has an AMP-PM4 plug with 4 pins. This connects to the vehicle ECU by a similar female AMP-PM4 plug. Connect the two plugs by lining up the tie loops on each AMP-PM4 plug then push the two together firmly.

STARTING

Open the computer, insert the programme disk and turn on the computer. The computer will 'beep' and then run its own internal memory check. On completion of the memory check, A:\>EMSAN1.EXE will be displayed on the screen and the computer will load information from the disk. The screen will display the name of the programme in use.

When ready the computer will show the ZyteK logo. The programme will ask the operator to wait a short period and then 'beep' prompting the user to press any key. On pressing any key the display will change to a page giving copyright details:

ZYTEK ENGINE MANAGEMENT SYSTEM DIAGNOSTICS

Copyright © 1992 ZyteK Systems Ltd.

Configured for use with:- JAGUARSFORD ECU No. P220-P-81-001

Press any key to continue

Press any key again and the display will change to a seven page menu.

Dynamic display of system parameters monitored by ECU
 Diagnostics
 Display of Oxygen Sensor Feedback Data/Throttle position
 Display program & identification details
 Set up engine
 Display status of speed and sync. sensors
 Quit program and return to DOS

Use the directional keys to select the page required and press "ENTER".

COMPUTER SOFTWARE

Dynamic display of system parameters monitored by the ECU.

This screen is intended as a visual display of the engine parameters. The parameters can be displayed with the engine stationary or running, although the **ignition must be on** in order for it to give any information.

Parameters are displayed giving their current values as measured by the ECU.

ENGINE SPEED: 1070 rpm

<u>LOAD</u>	<u>A BANK</u>	<u>B BANK</u>	<u>MISCELLANEOUS</u>
MANIFOLD DEPRESSION:	450	464mbar	ROAD SPEED: OMPH
INJECTOR OPEN TIME:	4.64	4.71msec	THROTTLE OPENING: 295mV
IGNITION ADVANCE:	10.25	10.0 deg	BATTERY VOLTAGE: 13.5V

<u>BOOST</u>	<u>A BANK</u>	<u>B BANK</u>	<u>OUTPUTS</u>
VALVE MARK/SPACE	1	1	AIR CON: DISABLED
			FAN: ENABLED

<u>LAMBDA SENSORS</u>	<u>A BANK</u>	<u>B BANK</u>	
SENSOR VOLTAGES:	0.12	0.78	PURGE FLOW: 2%
			IDLE SPEED VALVE: 23%

TEMPERATURES

SPEED/SYNC STATUS: speed/sync. OK	WATER: 91 deg C
	AIR-BANK A: 32 deg
	AIR-BANK B: 30 deg

DYNAMIC DISPLAY OF SYSTEM PARAMETERS

<Q>-QUIT

SCREEN 1

Engine Speed

This display shows either:-

- a) the actual speed of the engine, if it is running
- or
- b) the word 'STOPPED'

If the engine is not running the word 'STOPPED' should appear. When the engine is started the word 'STOPPED' should be replaced by the actual speed of the engine shown in rpm.

If 'STOPPED' is displayed during cranking this indicates that the ECU is not receiving any engine speed signal from the speed sensor on the end of "a" bank camshaft.

Speed/Sync Status

This is an indication of the RPM and synchronising sensor operation. The synchronising sensor identifies No. 1 cylinder to the ECU.

When the engine is running:

- a) This display should show 'speed/sync ok', therefore indicating that the sensor arrangement is operating correctly.
- b) If 'No sync pulse' is displayed this indicates an open circuit, either in the wiring or the sync sensor itself.
- c) If 'Sync pulse every speed pulse' is displayed this indicates a short circuit to ground, either in the wiring or the sync sensor itself.

Pressure (mbar)

Manifold pressure is measured for both banks of the engine as the main load input from the engine. This is achieved by means of a pipe running from each plenum of the engine to pressure sensors inside the ECU. With the ignition switch on, the display should show atmospheric pressure of the day.

When the engine is started and run at idle speed, the pressure value should drop immediately to a low pressure value around 450mbar, and as the engine speed rises and falls the pressure should vary.

When the car is driven on the road, the load on the engine changes continually as the throttle is opened and closed, which is measured by the ECU which meters the fuel and spark timing according to this pressure.

Any faults relating to the pressure measurement will be shown as a static or fluctuating reading of pressure, which does not respond to varying engine speeds and loads, resulting in the engine running very erratically. Check the connections at both ends of the pressure pipe to ensure they are attached and there is no leaks. If all appears satisfactory, suspect the pressure sensor in the ECU.

Injector Pulse (msec)

This display indicates the actual injector pulse width being used at a given engine operating condition. This can be used to ensure the pulse width increases and decreases as engine speed and pressure are varied. Under low load conditions (during acceleration) only the primary injectors are operating.

When the injectors reach 18msec pulse time the ECU activates the secondary injectors and reduces the primary injector pulse time to 9msec to coincide with the secondary injectors pulse time. Maximum injector pulse time is 23msec for each injector bank.

On reduction of engine speed both injector banks are reduced to a pulse time of 6.5msec. On reaching 6.5msec the secondary injectors are switched off and the primary injectors pulse time is increased to 13msec.

Ignition Advance

This display shows the actual ignition timing the engine is running. This can be used to check that the ignition timing seen on the engine timing marks by a strobe light, corresponds to the timing seen on the computer.

If a difference between engine and computer timing is apparent, the ignition timing of the engine should be matched to the computer by adjusting the speed/sync sensing disc.

Boost Control Valves

The boost control valves modulate the pressure in the waste gates to maintain the correct boost pressure to the engine.

With the ignition on, engine stopped, the reading on the computer should be 255.

The value will change above 2000 rpm to 10, although actual control is only possible when the engine is under load.

If 243 is displayed then there is a fault with the valves themselves, or another boost control related component

Lambda A/Lambda B (one per bank)

These display the outputs from the feedback lambda sensors located in the exhaust system.

With the ignition on, engine stopped (and cold) there could be three levels

- i) 4 to 5 volts - this is correct
- ii) 0 volts - indicating circuit fault, either in the wiring or the sensor itself.
- iii) If the engine has been started these values may be between 0.2 volts and 5 volts, and should drift back towards 5 volts as the sensors cool.

On starting the engine the voltages shown for Lambda A/B should fall as the sensors warm up.

At idle, with the engine warm, the sensor outputs should operate between 0.1 and 0.8 volts, switching continuously.

If either of the outputs remain at 5 volts then there is an open circuit between the sensor and the ECU. Check the wiring.

If either of the outputs change but remain at a constant level between 4 and greater than 5 volts then the sensor heaters are not working.

Each lambda sensor has an internal heater which raises the temperature of the sensing element, enabling it to operate at the idle condition. The lambda heater has a 12 volt supply via the fuel pump relay, so it will only operate when the engine is running. To check this supply, remove the 2-pin heater connector at the sensor and start the engine. Check the voltage across the two pins is 12 to 14 volts.

Road Speed

This input is fed from the instrument pack to the ECU and is used primarily for idle speed control. At idle the reading should be zero (0).

Throttle Opening

This display measures the throttle sensor voltage on a scale of 0 to 5000mV. With the throttle closed the voltage reading will be low. When the throttle is opened, the voltage reading will rise smoothly with the opening throttle. If the throttle is held open at full throttle, the voltage reading should be high and constant.

If there is a fault present with the throttle then the computer will display 2500mV.

Battery Volts

This display shows battery voltage measured by the ECU. An indication of a discharged/faulty battery can be seen, as well as a faulty alternator.

Air Conditioning

The ECU has final control of the air conditioning compressor. This is mainly for idle speed control. At idle the screen should show either ENABLED or DISABLED.

Fan

The ECU controls the cooling fans based on coolant temperature. The fans are also switched on if the air conditioning is requested and enabled. At idle, with the air conditioning off, the fans will read either:-

DISABLED - if the coolant temperature is below 85°C and the fans should be off.

or

ENABLED - if the coolant temperature is above 90°C and the fans should be on.

To ENABLE the fans the coolant temperature must increase to above 90°C before switching on, and then will not switch off until the coolant temperature falls below 85°C. Therefore it is possible to be between 85 and 90°C and for the fans to be on.

Purge Flow (%) (if fitted)

This refers to the valve which controls the flow of fuel vapour into the engine from the charcoal canister. At idle, when the engine is warm and feedback operating, this will be running between 0 to 10%.

Idle Speed Control Valve

This controls the air requirement of the engine at starting and normal running to maintain the desired idle speed.

At idle with the engine warm, the valve should run at 9 to 25%. (See Engine Tuning for details of the correct setting).

If DISABLED is displayed on the screen then a fault has occurred with either the control valve or another input to the ECU.

Air Temperature 'A' Bank(°C)

Indication of air temperature for 'A' bank can be shown in three ways:-

- a) Actual temperature of the air.
- b) S/C, indicating a short circuit of the sensor or the wiring.
- c) O/C, indicating an open circuit of the sensor or the wiring.

Fault Finding - Thermistors

These faults will also be recorded on the diagnostic screen.

With the ignition,

If S/C is shown on the display, disconnect the plug to the sensor and note if the display changes to O/C.

If the display does change to O/C, replace the thermistor.

If the display stays at S/C, check the wiring continuity to the ECU for a short circuit

If O/C is shown on the display, disconnect the plug to the sensor and using a short piece of wire short the terminals in the

plug together. The display should now change to S/C indicating an open circuit in the sensor itself. Replace the sensor. If the display stays at O/C whilst the terminals are shorted together this indicates an open circuit in the wiring to the ECU. Test the continuity of the wiring to the ECU.

Coolant Temperature (°C)

As with the air temperature there are three possible displays:-

- a) Actual temperature of the coolant
- b) S/C, indicating an short circuit of the sensor, either in the wiring or the sensor itself. This condition would cause difficulty in starting and running with low coolant temperatures.
- c) O/C, indicating an open circuit of the sensor, either in the wiring or the sensor itself. This condition would cause the same problems as a short circuited sensor.

Refer to the 'Fault Finding - Thermistors' section under Air Temperature 'A' Bank for fault finding explanation.

If this thermistor is faulty then the feedback, idle speed control, purge system, fan control and air conditioning all revert to default conditions and will not operate as normal. This fault must be rectified before setting the engine.

Air Temperature 'B' Bank (°C)

Indication of the air temperature for 'B' bank can be shown three ways:-

- a) Actual temperature of the ambient air.
- b) S/C, indicating a short circuit of the sensor either in the wiring or the sensor itself.
- c) O/C, indicating an open circuit of the sensor, either in the wiring or the sensor itself. Check all the wiring of the sensors and continuity of the sensor for faults. Refer to 'Fault Finding - Thermistors' section under Air Temperature 'A' Bank for fault finding explanation.

SCREEN 2

This displays faults found by the ECU which have occurred in the last 50 starts of the engine.

FAULT CODE ANALYSIS

Total number of faults found is 3

CODE	FAULT
27	WATER THERMISTOR
34	BOOST CONTROL VALVE - A BANK
42	PRESSURE FAULT - B BANK

FAULT DESCRIPTION

Description of the fault highlighted in the above list

PgDn PgUp> - Select Fault<Q>-Quit

If there are any faults on the system then these are displayed giving the number of different faults, a list, and then a brief description of the faults.

The descriptions are selected by positioning the inverted cursor on the listed fault using the 'PgDn' and 'PgUp' keys.

For information to assist in tracing the fault refer to the Fault Diagnosis Analysis.

Once a fault has been rectified the code must be cleared out of the ECU memory from this screen using the 'D' key. The computer will request the user to confirm that the fault code is to be cleared to avoid accidental clearing of codes.

SCREEN 3

Display of Oxygen Sensor Feedback Data/Throttle Position

This display takes the form of bar graphs indicating the output from the feedback sensors, the position of the throttle.

Feedback Operation

Before checking the operation of the feedback sensors ensure the correct set-up and operation of the throttle potentiometer. (See Set-up Engine).

When the engine is running at idle, and the coolant temperature has reached a set level (50°C), the outputs from the ECU should start to switch, controlling the fuelling to the engine. The ECU output is represented by a solid bar on the graph and correct operation can be seen as switching between two levels.

If no solid bar is present for either bank, then check the output voltages of the sensors as displayed on Screen 1 of the menu options (with the engine running).

If the output voltages are varying between 0.10 volts and 0.90 volts, but there is still no switching on the bar graph then, with the engine running at idle check:-

- coolant temperature greater than 50°C

- pressure less than 600 mbar

- throttle volts less than 0.4 volts

If the pressure and/or throttle outputs are not below these values then check their operation.

If the coolant temperature is below 50°C then allow the engine to warm up further.

NOTE

If the engine is started warm there is a delay of approximately 30 seconds before feedback operates.

SCREEN 4

Display Programme & Data Identification Details

This page is provided so that an operator can identify the programme within the ECU. With the ignition on and after pressing 'ENTER' the following will be displayed:-

- Date and time when the programme was compiled.
- Initials of the operator who constructed the programme.
- Comment line, giving some information about the programme.

SCREEN 5

Set-up Engine

This details the process to follow in order to set the ignition and throttle to ensure correct idle and full load running.

Each screen prompts actions necessary to ensure that the engine is set and operating at its specified point.

Once the action requested on each screen has been satisfactorily completed press the SPACE bar to proceed to the next screen. If at any point a fault occurs on the system the programme will jump out of this set procedure and prompt the user to address the fault.

SCREEN 6

Display Status of Speed and Sync Sensors

This displays the status of the speed and sync sensors and can only be viewed with the ignition on.

This is intended to be used in conjunction with the timing data if the ignition timing requires adjustment, or the speed and sync sensors are changed.

SCREEN 7

Quit Programme and Return to DOS

This page allows the operator to quit the Zytec programme.

NOTE

Please ensure the battery pack contained with the computer is kept fully charged. Failure to do so will shorten the life of the battery and may mean that the equipment cannot be used when most needed.

FAULT CODE DESCRIPTION

Code	Parameter or component checked	Code	Parameter or component checked
1	Lambda sensor/wiring fault - 'A' bank	23	Secondary purge valve - boost protection
2	Lambda sensor/wiring fault - 'B' bank	24	Throttle potentiometer
5	Sync/speed pulse missing	26	Thermistor ground return wiring fault
6	Ignition drive	27	Water thermistor
7	Specific injector fault - injector 1,1	28	Air thermistor - 'A' bank
8	Specific injector fault - injector 6,1	29	Air thermistor - 'B' bank
9	Specific injector fault - injector 3,1	32	Fuel pump relay
10	Specific injector fault - injector 5,1	33	Purge valve/associated wiring
11	Specific injector fault - injector 2,1	34	Boost control valve - 'A' bank
12	Specific injector fault - injector 4,1	36	Road speed input
13	Specific injector fault - injector 1,2	37	Idle speed valve
14	Specific injector fault - injector 6,2	38	Air conditioning relay drive
15	Specific injector fault - injector 3,2	39	Fan relay drive
16	Specific injector fault - injector 5,2	40	Boost control valve - 'B' bank
17	Specific injector fault - injector 2,2	41	Pressure fault - 'A' bank
18	Specific injector fault - injector 4,2	42	Pressure fault - 'B' bank
19	Global fuel injector fault	43	Pressure imbalance between banks
		44	ECU supply open circuit

FAULT CODE ANALYSIS

The following information provides guidelines for rectifying problems found with the XJ220 ZyteK Engine Management System (EMS) in service. It should be used in conjunction with the system schematic wiring diagram. The format assumes the ability of the engineer/technician to follow the circuits of interest on the schematic diagram through the various harness connectors. Not all possible faults are covered at this time. The aim has been to concentrate on the most likely reasons for each individual Electronic Control Unit (ECU) fault code being set in service, together with an appreciation of some of the faults which cannot generate codes. It will be seen that in most sections, a faulty ECU is listed as the final possibility, in reality the ECU is one of the most reliable components in the system and is included in the analysis for completeness.

Fault Code	Test Carried Out	Possible Cause
1	The EMS Checks the Lambda sensor status 2 minutes after the engine start if the coolant temperature is greater than 46°C. If the sensor voltage seen is less than 1.5V, then the code is set	1. Sensor not plugged in. 2. No 12v supply to the sensor heaters (wiring harness, low pressure fuel pump fuse, etc.). 3. Open circuit in sensor output line to ECU pin 33. 4. Faulty sensor, either heater coil or sensing element failure. 5. Faulty ECU
2	This code is identical in operation as Code 1.	As Code 1. Test ECU pin 53 for a sensor output line open circuit.
5	The ECU checks for correct sensor operation when the engine speed is above 500rpm. The speed and sync outputs are independent and so the speed/sync sensor set up page in the service diagnostic software should be used to confirm which sensor is actually faulty.	1. Open circuit in 12V supply from ECU pin 17 to sensors. 2. Open circuit sensor ground connection. 3. Open circuit sensor output line to ECU pin 55 & 12 Speed) and ECU pin 13 (Sync). 4. Speed or sync sensor failure. 5. ECU failure.
6	The test is carried out when engine speed is greater than 800rpm. The ECU checks for 6 output signals in every two revolution period. This function does not test for faults in the HT side of the ignition system i.e. distributor cap, plug leads etc.	1. Short circuit to ground or to positive supply between ECU pin 31 and the ignition amplifier. 2. Faulty amplifier. 3. Faulty ECU

Fault Code	Test Carried Out	Possible Cause
7	Code 7 together with Code 8-18 inclusive refer to failures of individual injectors. These tests are run by the ECU following the determination of a code 19 (Global fuel injector fault). Thus codes 7-18 will always occur in conjunction with code 19. NOTE: <i>Setting of these injector failure codes is very rare</i> These tests are made between 800 and 1400rpm and between load site 4 and 8 i.e. around idle.	1. Open circuit in the injector signal line to ECU pin 21. 2. Short circuit to ground of the line to ECU pin 21. 3. Faulty injector coil.
8	See Code 7.	1. ECU pin 41
9	See code 7	1. ECU pin 41
10	See Code 7	1. ECU pin 42
11	See Code 7	1. ECU pin 23
12	See Code 7	1. ECU pin 43
13	See Code 7	1. ECU pin 60
14	See Code 7	1. ECU pin 40
15	See Code 7	1. ECU pin 59
16	See Code 7	1. ECU pin 39
17	See Code 7	1. ECU pin 58
18	See Code 7	1. ECU pin 38

Fault Code	Test Carried Out	Possible Cause
19	This test is carried out by the ECU when engine speed is greater than 800rpm and the load site is less than 24. The system checks for 6 injections every two revolutions of the engine. If all 12 injectors are operating, then faults in both primary and secondary injector systems will be found. Once a code 19 has been set, the closed loop fuel control system is disabled and the ECU will attempt to identify the particular injector or injectors which has caused the fault to be flagged. This will be done the next time the engine enters the speed and load range defined in code 7. NOTE: <i>This code is seen very rarely.</i>	
23	The secondary purge valve closes just below atmospheric pressure so that positive manifold pressure does not leak into the evaporative emission canister (when fitted). The ECU checks that the valve drive circuit is in the state intended. The valve is supplied with an ignition feed on one side. The other side of the valve is then grounded to energise the system by ECU pin 48. The valve is normally closed when not energised and the ECU commands it open when the engine is off boost.	1. The valve is incorrectly connected. 2. Harness failed open circuit. This would indicate no 12V available on the supply side of the valve and a lack of continuity between the valve and ECU pin 48 on the signal side of the circuit. 3. Valve failed open circuit. 4. Harness failed short circuit. 5. Valve failed short circuit. 6. ECU failed.
24	The throttle potentiometer is used to identify transient fuelling requirements and, in conjunction with road speed, engine and idle speed control valve to identify idle conditions. The potentiometer must be set within the specified range using the service software. The test is carried out by the ECU whenever the ignition is switched on. The test checks that the potentiometer output voltage is between 0.1v and 4.95V. The system adopts a default setting of 2.5V when the code is active and all functions on which the throttle position has an influence are disabled (acceleration enrichment, deceleration enrichment, idle recognition and deceleration fuel cut-off).	1. Potentiometer output 0V i.e. Potentiometer has not been set up. 2. Throttle potentiometer is not connected. 3. Potentiometer failed open circuit. 4. Harness failed open circuit. This would mean no 5V supply to the potentiometer from ECU pin 16, no continuity to the potentiometer connector from ECU pin 35 (signal ground) or no continuity to the potentiometer connector from ECU pin 6 (signal). 5. Harness failed short circuit. 6. ECU failure.

Fault Code	Test Carried Out	Possible Cause
27	This test is carried out whenever the ignition is on. If a fault is detected, the system will adopt a default setting as a "limp home" function. The screen of the EMS computer will show whether the sensor input is open circuit (O/C) or short circuit (S/C).	1. Sensor not connected correctly. Care is needed when connecting the water thermistor as access is restricted around the thermostat housing. 2. Sensor failed open circuit. This is the usual sensor failure confirmed by an open circuit reading across the sensor. 3. Harness failed open circuit. The water thermistor circuit runs from the sensor to ECU pin 25 on one side and to ECU pin 36 (ground connection) on the other. 4. Harness failed short circuit. This would entail a signal wire becoming shorted out to the sensor ground return line and is an unlikely failure. 5. ECU failure.
28	This test is carried out whenever the ignition is switched on. If a fault is detected, the system will use the remaining air thermistor for both banks of the engine. The screen of the EMS computer will show either the sensor input is open circuit (O/C) or short circuit (S/C).	1. Sensor not connected correctly. Care is needed when connecting the air thermistor as access is restricted under each plenum. 2. Sensor failed open circuit. This is the usual sensor failure confirmed by an open circuit reading across the sensor. 3. Harness failed open circuit. The air thermistor circuit runs from ECU pin 27 through the thermistor and returns to ECU pin 36. 4. Harness failed short circuit. This would entail a signal wire becoming shorted out to the sensor ground return line and is an unlikely failure. 5. ECU failure.

Fault Code	Test Carried Out	Possible Cause
29	This test is carried out whenever the ignition is switched on. If a fault is detected, the system will use the remaining air thermistor for both banks of the engine. The screen of the EMS computer will show either the sensor input is open circuit (O/C) or short circuit (S/C).	1. Sensor not connected correctly. Care is needed when connecting the air thermistor as access is restricted under each plenum. 2. Sensor failed open circuit. This is the usual sensor failure confirmed by an open circuit reading across the sensor. 3. Harness failed open circuit. The air thermistor circuit runs from ECU pin 45 through the thermistor and returns to ECU pin 36. 4. Harness failed short circuit. This would entail a signal wire becoming shorted out to the sensor ground return line and is an unlikely failure. 5. ECU failure.
32	The fuel pump relays are controlled by the ECU, the ignition switch and the inertia switch. If the ignition is on and the inertia switch closed, then the ECU energises the relays by driving ECU pin 24 to ground. The pumps will run for approximately 2 seconds following ignition switch on and continuously during cranking and running. A fault will be flagged if the ECU finds that the driver circuit is not in the state intended.	1. Harness open circuit. No 12V supply to the relay pin 5 or no continuity from relay pin 2 to ECU pin 24. Check inertia switch is closed. 2. Harness short circuit to ground between relay and ECU.
33	This fault code diagnoses the BOSCH purge valve and associated wiring. The Bosch unit is a pulse width modulated valve which provides control of the flow rate of fuel vapour leaving the Canister and entering the engine. The ECU checks that its output is in the state intended. The valve is supplied by an ignition feed and its duty cycle (flowrate) is controlled from ECU pin 44.	1. Valve incorrectly connected. 2. Harness failed open circuit. This would mean a lack of the 12V supply on the feed side of the valve or lack of continuity between the valve and ECU pin 44 on the signal side. Check fuse No. C3. 3. Valve failed short circuit. 4. Harness failed short circuit. 5. ECU failure.

Fault Code	Test Carried Out	Possible Cause
34	The test carried out on the boost control valve system is to ensure that the output from the ECU is in the state intended. Each SEM valve is supplied with an ignition controlled feed. This is then subject to a pulse width modulated signal from the ECU which allows the valve position to be modified up to 12 times each second.	1. Valve not correctly plugged into the harness connector. 2. Valve failed open circuit. (Normal valve winding resistance is approx.. 27 Ohms, measured across the two pins of the valve connector). 3. Harness failed open circuit. On the ignition feed side of the circuit, this would mean no 12V supply available at the valve. Check fuse No. C3. On the ECU side of the circuit, no continuity between the valve and the ECU pin 50. 4. Valve failed short circuit. 5. ECU failure. Once a code 34 (or 40 for B bank) has been set, a default minimum boost level only is allowed. This is defined by a basic setting of the wastegate capsule at around 600mB gauge pressure.
36	The road speed input is received by the ECU from the instrument pack (IP). The information is used in the idle speed control logic. A test is carried out when the engine speed is greater than 1000 rpm and manifold pressure more than 750mB. This test looks for road speed greater than zero (0), while a second test, run at all other times constantly monitors that the speed is less than 300mph. If either of these tests fail a code is flagged.	1. Harness fails open circuit. Road speed information leaves the IP on pin B9 and reaches the ECU on pin 9. 2. Check for correct speedometer operation. If inoperative, check for fault in VSS circuit to the IP or faulty sensor.
37	The Idle Speed Control Valve (ISCV) is controlled by the ECU with data from engine speed, road speed and throttle position. The ISCV is powered by an ignition feed and receives a pulse width modulated signal from ECU pin 8.	1. ISCV not correctly connected. 2. Harness open circuit. Either no 12V ignition feed to the ISCV pin 1 or no continuity between ISCV pin 2 and ECU pin 8. 3. Harness short circuit between ISCV and ECU. 4. ISCV failed open circuit.

Fault Code	Test Carried Out	Possible Cause
38	The air conditioning (A/C) relay drive controls the status of the A/C compressor clutch. The A/C clutch is activated by the ignition switch, driver demand (from the A/C request line to ECU pin 7) and the control signal from ECU pin 26. The system checks that the ECU output is in the state intended.	1. Harness failed open circuit. Either no 12V feed to the compressor relay or no continuity between the compressor relay pin A and ECU pin 26. 2. Compressor relay coil failing open circuit. 3. Harness short circuit between relay and ECU.
39	The radiator cooling fans are driven from a single relay controlled by the ECU. The relay has an ignition feed and is energised by the ECU grounding pin 28. The fans are enabled at 91°C and disabled at 86°C. The ECU checks that its output is in the state intended.	1. Harness open circuit. Either no 12V feed to the relay coil or no continuity between the relay pin A and ECU pin 28. 2. Fan relay coil failing open circuit. 3. Harness short circuit between relay and ECU.
40	The test carried out on the boost control valve system is to ensure that the output from the ECU is in the state intended. Each SEM valve is supplied with an ignition controlled feed. This is then subject to a pulse width modulated signal from the ECU which allows the valve position to be modified up to 12 times each second.	1. Valve not correctly plugged into the harness connector. 2. Valve failed open circuit. (Normal valve winding resistance is approx.. 27 Ohms, measured across the two pins of the valve connector). 3. Harness failed open circuit. On the ignition feed side of the circuit, this would mean no 12V supply available at the valve. On the ECU side of the circuit, no continuity between the valve and the ECU pin 4. 4. Valve failed short circuit. 5. ECU failure. Once a code 40 (or 34 for A bank) has been set, a default minimum boost level only is allowed. This is defined by a basic setting of the wastegate capsule at around 600mB gauge pressure.



Fault Code	Test Carried Out	Possible Cause
41	This test is carried out when the engine has started and is running at more than 500 rpm. The ECU expects to see constant movement of the pressure signal from the inlet manifold. The signal is monitored over 5 second periods and a fault is flagged if no change in signal is seen NOTE: <i>The signal will never be completely steady even at a fixed throttle opening.</i>	1. Pressure signal pipe detached at the manifold or ECU. 2. Pressure signal pipe detached internally within the ECU. 3. ECU (pressure transducer) failure. If either code 41 or 42 is set, the ECU uses the other to calculate fuel and ignition requirements. Minimum boost only is allowed in this default condition. If both codes 41 and 42 are set, then a "limp home" default setting, based on throttle position, is enabled with minimum boost allowed and a maximum allowable engine speed of 2000 rpm.
42	See Code 41	See Code 41
43	This test looks at the balance of both bank to bank plenum pressures. If a pressure difference of 500mB occurs between the two plenums for more than 5 seconds, then code 43 is set and only minimum boost is allowed. This code will often accompany codes 41 and 42. If 41 and 42 are not present, this code is diagnosing the mechanical system	1. Incorrect throttle adjustment bank to bank. 2. Turbocharger failure. (Compressor/turbine/wastegate). 3. Boost control system failure. In this case 43 would be accompanied by codes 34 and 40. 4. Restriction in air inlet system on one side, e.g. collapsed inlet ducting or air filter, blocked intercooler or charge pipe. 5. Restriction in exhaust system on one side, e.g. failed catalyst as a result of a misfire.
44	This is a test for the ECU power supply on pin 57. This test may flag a code 5 (speed/sync fault) as well as code 44, as the system cannot distinguish between a speed sensor fault and the loss of the ECU supply. The speed sensor code appearing with code 44 implies a speed signal interference problem. This can result in an apparent speed reading in excess of 20,000 rpm on the computer screen while the engine may be at idle or even cranking. Repeated occurrences of this code should be brought to the attention of the JaguarSport Service Department at Bloxham who will be able to advise further.	

FAULTS OUTSIDE THE SCOPE OF ECU DIAGNOSTICS

This section deals with faults and operating problems which may not set fault codes.

Failure to Run

If the engine cranks normally but does not start, checks should be made in the following way;

1. Check for any fault codes which may have been set and refer to the relevant diagnosis.
2. If no fault codes are logged, check security of connections to the coil and ignition amplifier.
3. HT system. Check for spark at plug and then at king lead aiming to eliminate, distributor cap and rotor arm.

CAUTION:

Care should be taken while handling HT ignition system components.

4. If there is no spark, then the power feeds to the coil and ignition amplifier should be checked.
5. At this point, coil and amplifier can be eliminated by substitution, paying particular attention to the ground tag of the ignition amplifier signal lead which should be attached to the amplifier securing bolt.
6. If no ignition problems are apparent, the fuel system should be investigated. The main fuel pumps run for 2 seconds following ignition on. If the pumps cannot be heard in this way check the inertia switch is closed, and then fuses C2 and C5 should be checked.
7. Check low pressure fuel pump fuse C1.
8. The fuel pressure for each engine bank should then be checked. The nominal fuel pressure is 3.8bar (52psi).
9. With ignition and fuel systems checked, the engine should now start. If this is not so, re-check the fault codes using the service software, particularly with regard to speed and sync sensor codes.
10. If the engine fires but will not run such that faulty ignition timing is suspected, this can be checked using a timing light. The engine timing marks are located on the periphery of the flywheel with the reference mark on the centreline of the sump lower rear edge. Cranking ignition timing is 6 degrees BTDC. Consistent timing well away from this point denotes a slipped speed sensing disc (at the rear of A bank exhaust camshaft) while inconsistent timing indicates a faulty rotor arm drive.

Misfire

The sparking plugs fitted to the XJ220 engine (Champion C57C) are relatively cold to ensure correct operation at high engine speeds and large power outputs. Repeated cold starting without allowing the engine to warm thoroughly can lead to misfire due to spark plug fouling. The only effective answer which does not compromise the engine's performance and safety at high speed is to fit a new set of the correct grade of plug. It must be stressed that many cold starts without the engine reaching normal running temperature are required to reach a foul condition. Infrequent vehicle use should not be sufficient to promote this condition.

Poor Low Speed Driveability

Poor low speed driveability, in particular a harsh throttle response both opening and closing at low vehicle speed is often caused by an incorrect throttle set up. The potentiometer on the B bank throttle body should first be checked for correct setting using the appropriate page of the service software package. The synchronisation of the two throttles should then be checked to ensure that they are both open together. Any differences should be removed using cable adjustments.

Loss Of Power

In the event of a loss of power, the fault code log should be consulted to eliminate boost control valve failures etc. from the investigation. If no fault codes are set, the most likely reason for a power loss is a low boost pressure on one or both banks. This is best confirmed by a road test using full throttle in a high gear (to avoid rapid accelerations while maintaining moderate and legal speeds). Using the service software the boost for each bank should be monitored to ensure that between 1950 and 2050mB is achieved by 3000rpm. Testing of boost levels at speed in excess of this should be conducted off the public road. If a low boost level is confirmed, the following points should be checked;

1. Security of upper and lower charge pipes between compressors and intercoolers. Leaks at these joints often cause noises which aid tracking down the area of concern.
2. The lack of boost on one bank may not be severe enough to set a code 43 as described in the code analysis. The possibilities of collapsed air inlet, failed catalyst, blocked air filter need still to be considered.
3. A wastegate actuator capsule may have relaxed. This will not allow the boost control system to attain the commanded pressure. This will be seen on the display screen as the boost control system position number reaching 1 when the boost is not above 1950mB. The capsule preload then needs to be measured and adjusted. The capsule pushrod length should be adjusted until the wastegate valve lever just moves (0.010") when a pressure of 600mB is applied to the capsule using a hand held pump.

Overboost Cut-Out

The ECU has a fuel cut-off feature which operates if an overboost condition is detected. This is a safety feature which prevents the engine running at boost pressures which are unsafe. This ECU function is calibrated so that the cut-out boost level is not achieved in normal operation. The overboost cut-out is felt as a hard cut in the engine power delivery, usually occurring between 3000 and 4500 rpm while accelerating with large throttle openings. The causes of an overboost cut-out are;

1. Foreign body ingress into the boost control pipework and restrictor system.
2. Incorrect routing of the boost control pipework.
3. Base setting of a turbocharger wastegate actuator too high.
4. Mechanical failure of a boost control (SEM) valve.

If such a cut is experienced, the display page of the service software should be used to confirm which bank of cylinders is overboosting. The relevant boost control pipework and restrictors should be inspected and their cleanliness ensured. The routing of the pipework should also be checked, particularly if the problem occurs following service operations which might have affected that area of the engine bay. Having confirmed the system cleanliness and pie routing, the preload or base settings of the wastegate actuators should be checked together with the smoothness of operation of the wastegate. If no problems with these items are found, then the relevant SEM valve should be tested by substitution.

Further Actions

As outlined previously this is not a complete fault-finding guide to the XJ220 Engine Management System. If it is found that insufficient information is contained here to resolve a vehicle problem, then the JaguarSport Service Department at Bloxham should be contacted in the first instance. This will provide access to the specialist resources which can provide further more detailed advice.

Fig. 1

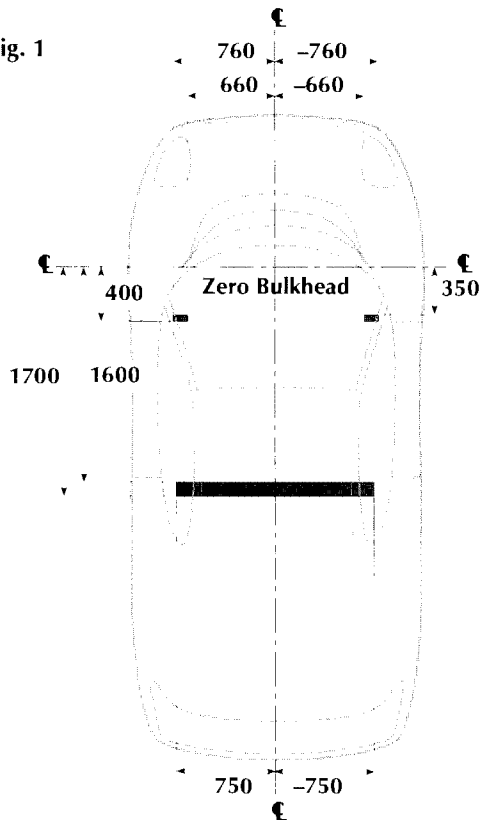
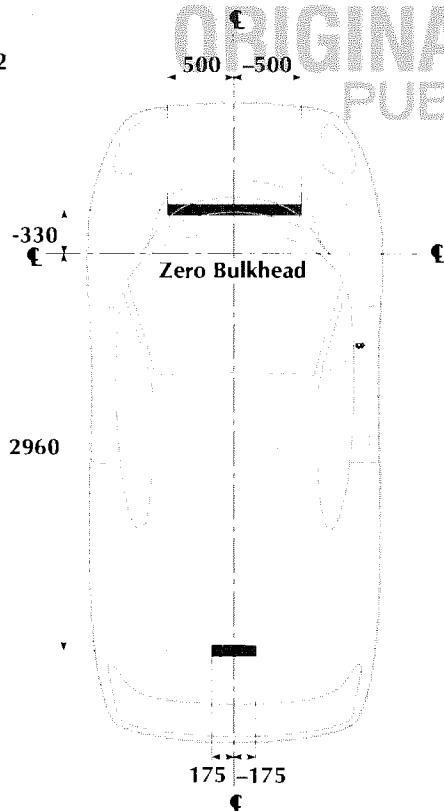


Fig. 2



LIFTING AND JACKING

When carrying out any work on the vehicle which requires all four wheels to be raised, it is essential that 'wheels free' lifting equipment is used in conjunction with additional stands/supports placed under the rear uprights or rear engine plate.

LIFTING

Position the forward and rear adjustable lifting arms within the shaded areas shown in Fig 1. Place a suitable block of hardwood between the lifting arms and the vehicle underbody. Slowly take the weight of the vehicle and ensure the vehicle is stable on the lifting equipment. Check that the vehicle underbody is parallel to the floor and the rear engine plate is perpendicular. When carrying out work beneath the vehicle place additional support beneath the rear suspension uprights or rear engine plate as applicable.

JACKING

When jacking the front or rear wheels clear of the ground the shaded areas in Fig 2 should be used. Always chock the wheels as well as applying the handbrake following jacking operations.

To lift the vehicle proceed as follows;

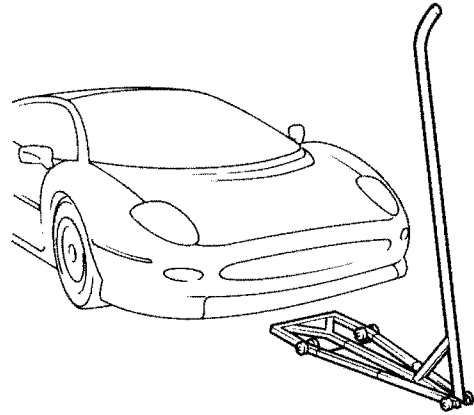
- (1) Ensure the handbrake is off and the transaxle is in neutral selection.
- (2) Ensure the jack lift plate is free from debris and place a suitable block of hardwood between the jack and the vehicle underbody.

NOTE:

During jacking operations the vehicle will move toward the jack.

- (3) For front end jacking
Position jack on the jacking point directly under zero bulkhead, ensuring the centreline of the jack aligns with the centreline of the vehicle and lift the vehicle.
- (3a) For rear end jacking
Position jack directly under the rear engine bay plate ensuring the centreline of the jack aligns with the centre line of the vehicle and lift the vehicle.

- (4) Ensure the vehicle is stable on the jack and place additional support beneath the front or rear suspension uprights to provide a secure platform.



VEHICLE TOWING AND RECOVERY

Recovery

The preferred methods of vehicle recovery are either by towing or by special flatbed transporter.

CAUTION:

Due to the low body clearance, 127mm (5in), and the long frontal overhang, 1200mm (47in), of the vehicle, the maximum recommended transporter ramp angle is 6.5 degrees.

NOTE:

The vehicle is not suited for suspended towing.

Attach tow webbing to the tow loop located below the front grille. Attach the transporter winch cable hook to the tow webbing. Do not attach the cable to any other points on the vehicle.

NOTE:

Never attach the winch cable directly to the towing loop.

CAUTION:

Take care not to damage the front splitter or the surrounding paintwork.

Slowly winch the vehicle on to the transporter continually checking front and rear ground clearance.

Once loaded onto the transporter apply the vehicle handbrake and chock the wheels. It is recommended that webbing straps are placed over the road tyres to tie down the vehicle for maximum security during transit. Do not attach tie down hooks to the towing loop.

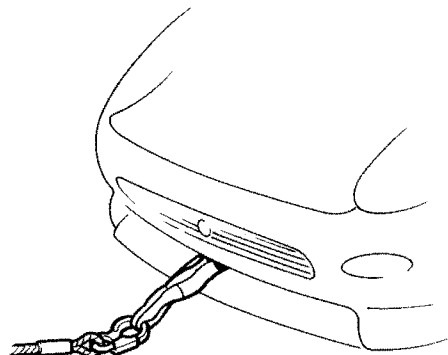
Towing

The vehicle is equipped with a towing loop at the front only. The towing loop is situated below the front grille.

CAUTION:

The towing loop is designed for a horizontal forward pull only, avoid uneven pulls, i.e. sideways or upwards.

To tow the vehicle attach tow webbing to the loop and then attach the tow rope or transporter winch cable hook



to the webbing. Do not attach the rope or cable to any other points on the vehicle.

NOTE:

Never attach the tow rope or winch cable directly to the towing loop.

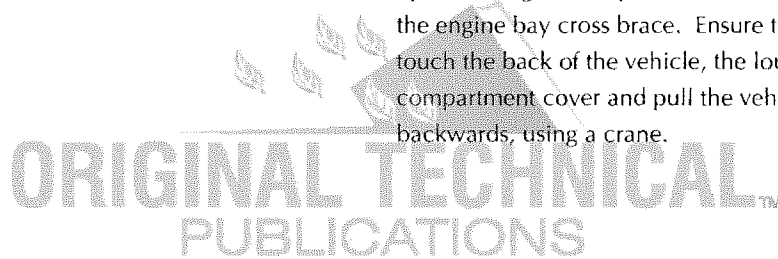
CAUTION:

Take care not to damage the front splitter or surrounding paintwork.

The vehicle may be towed for short distances provided a speed of 48km/h (30mph) is not exceeded. When being towed the vehicle's gear lever must be in neutral and the ignition key turned to position 'II' to release the steering lock and render the indicators, horn and brake lights operational.

Always adhere to the local towing regulations; in certain countries the registration number of the towing vehicle and on 'ON TOW' sign or warning triangle must be displayed in a prominent position on the rear of the vehicle being towed.

If it is necessary to recover the vehicle from the rear end, open the engine compartment cover and attach a rope to the engine bay cross brace. Ensure the rope does not touch the back of the vehicle, the louvres or the engine compartment cover and pull the vehicle carefully backwards, using a crane.



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DESCRIPTION

The XJ220 Body-in-White is constructed from aluminium external panels mounted on an aluminium honeycomb chassis assembly.

The chassis, formed from aluminium honeycomb and machined bulkheads, is bonded and riveted using ASVT (Aluminium Structural Vehicle Technology).

The primary structure, or occupant protection cell, consists of a steel roll cage directly mounted on to the honeycomb chassis with bolted fastenings.

The roll cage is then enveloped in aircraft specification pressed aluminium panels which are spot welded together and riveted to the chassis, again employing ASVT Technology.

The frontal impact absorption structure or crush tube assembly is formed from a heavy gauge aluminium construction riveted, welded and bonded with ASVT Technology.

The exterior pressed panels are joined by conventional spot, seam welding and riveting. Attachment of these panels to the chassis is through bolted or riveted joints and structural adhesive.

Body closures are manufactured from pressed aluminium panels. The doors are mounted on conventional lift-off hinges whilst bonnet and boot closures are mounted with fixed hinge systems.

1 BODY ASSEMBLY

Due to the complexity of the body assembly and the exacting procedures required to carry out repairs to the body and paintwork, JaguarSport recommend that they be contacted prior to any repairs being undertaken when they will advise on whether the repair can be carried out in the field or whether the vehicle should be returned to the factory.

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2 PAINT PROCESS

1 External Body Shaping all Aluminium Outer Skins

NOTE:

Before using any R-M product for the first time consult the R-M Health and Safety Guide. Always read the risks and precautions stated on the product label.

- (1) Bake for 30 minutes at 80°C.
- (2) Thoroughly degrease aluminium areas to be filled with Pre-Kleano 900 Heavy Duty Degreaser using Red Scotchbrite to wipe on, wipe off using a clean lint free cloth. Ensure the degreaser is removed with contaminants

NOTE:

Do not allow the degreaser to evaporate.

- (3) After thoroughly degreasing, bake for 60 minutes at 70°C.
- (4) Bare metal must be mechanically abraded thoroughly with P120-P180 grit production paper, or sand blasted, to ensure good adhesion of subsequent materials.
- (5) Reclean substrate with Pre-Kleano 900 as described in paragraph 2.
- (6) Repeat operation 6 at least once to ensure a clean surface is obtained.
- (7) Mix R-M Stop Zinc 52-22-130 with 1-3% Hardener 52-22-101 on a non-absorbent surface, e.g. Formica. Mix thoroughly.

NOTE

- 1 *Do not exceed 3% hardener as the excess peroxide may cause bleaching/discolouring of subsequent top coats.*
- 2 *R-M Stop Zinc when mixed with hardener has a pot life of 7-8 minutes at 20°C.*
- (8) Using a plastic or rubber applicator, apply the Stop Zinc in a thin skim to dents/imperfect surfaces working filler well to avoid trapping air.
- (9) Allow to dry for approximately 30 minutes at 20°C or 4 minutes 50% infra-red shortwave at 60 cm distance, then dry sand P80, followed by P240 production paper.
- (10) Blow off dust with dry, oil-free compressed air.

- (11) Repeat operations 7 to 10 until the required surface profile is obtained making sure no air is trapped between skims and all pinholes are filled.
- (12) Check the film build of R-M Stop Fillers applied, the total film build must not exceed 3mm.
- (13) Areas which have had R-M Stop Fillers applied should be recorded for each vehicle silhouette.

2 Exterior Priming Preparation

NOTE:

Plastic gloves must be worn by all personnel who are likely to come into contact with bodywork to avoid contamination likely to cause blistering between substrate and subsequent paint coats.

- (1) Degrease the entire bodywork/part with Pre-Kleano 900 heavy duty degreaser using Red Scotchbrite wipe on technique, wipe off using a clean lint-free cloth. Ensure degreaser is removed with contaminants.

NOTE:

Do not allow degreaser to evaporate.

- (2) Blow out joints with dry, oil-free compressed air.
- (3) Degrease again using wipe on wipe off technique using clean lint-free cloth soaked in degreaser in one hand and a dry lint-free cloth in the other hand. Ensure the degreaser is removed with contaminants. Change cloths regularly.

NOTE:

Do not allow the degreaser to evaporate.

3 Epoxy Adhesion Priming

- (1) Pre-heat all aluminium outer skins for 10 minutes at 60°C. Allow to cool to ambient temperature (20°C).

NOTE:

Rubber gloves must be worn from this stage onwards

- (2) Mix Euroxy Wash Primer Filler in the ratio 10:5:3 by volume with Euroxy Hardener and 2K ER54 Thinner. Filter paint prior to use.
- (3) Apply two coats of Euroxy allowing 15 minutes flash off between coats achieving a dry film build of 50 microns. Gun setup- EX fluid tip and needle, 30 air cap with spraying pressure of 45-50 psi at the gun.
- (4) Allow to flash off for 15 minutes before baking for 45 minutes at 60°C metal temperature.

- (5) Fill any pinholes or minor imperfections using R-M Stop Extra Fin mixed with 1-3% appropriate hardener on a non-absorbent surface e.g. Formica.

NOTE:

Do not exceed 3% hardener addition as excess peroxide may cause bleaching/discolouration of subsequent top coats.

- (6) Remove from the oven and after cooling, dry flat with P320 using an orbital sander and use Red Scotchbrite for difficult areas i.e. under bonnets, edges etc.
- (7) Thoroughly degrease the entire body with Panel Clean Degreaser using wipe on wipe off technique.
- (8) Total dry film build should be in the region of 50 microns.

4 Primer Filling

- (1) Mix R-M Fast Filler tinted to compliment top coat with D85 Hardener and 2K ER54 Thinner. Ratio by volume as follows:

500 Fast Filler
50 Solo Base
100 Hardener
200 Thinner

If required, the quantity of thinner may be reduced to 100 parts. Filter material before spraying.

NOTES:

- 1 For spraying temperatures over 25°C use BC030 Thinner
- 2 R-M Fast Filler has a pot life of 2 hours at 20°C.

- (2) Apply 3 coats of primer allowing 10 minutes flash off between coats to achieve a dry film build of Fast Filler of approximately 100 microns. Refer to Technical Sheet for gun settings, spraying pressure.
- (3) Allow 10 minutes flash off before baking at 60°C substrate temperature for 30 minutes.

5 Filler Surface Preparation

NOTE:

Observe all recommendations within R-M Health and Safety Guide pertinent to spraying operations.

- (1) Mix R-m 178-2026 Cellulose Black Thinner 199-18 in the ratio 1:9 by volume.

- (2) Apply one very light spraycoat as a guidecoat for subsequent sanding operations.
- (3) Wet flat the entire shell and hinge panels with P400 grit wet and dry. Re-guidecoat as per operations 1 and 2 and wet flat again with P800 grit wet and dry.
- (4) Bake primer coat at 60°C for 20 minutes to ensure removal of all traces of water.
- (5) Total dry film build of paint should be in the region of 150 microns.
- (6) Sealing process to take place at this stage

6 Preparation for Finishing

- (1) Totally mask off all apertures etc. e.g. door shuts that do not require finishing.

NOTE:

Plastic gloves must be worn during the following operations.

- (2) Thoroughly degrease the entire body with Panel Clean Degreaser using wipe on wipe off technique.
- (3) Repeat operation 2 to ensure surface is thoroughly clean.
- (4) Thoroughly dry out any traces of Panel Clean (15 minutes minimum) before applying finishing coats.
- (5) Blow off any dust with dry, oil-free compressed air.
- (6) Clean surface to remove any residual dust/static dust particles.

7 Finishing - Wet on Wet Primer

NOTE:

Observe all recommendations within the R-M Health and Safety Guide pertinent to spraying operations.

- (1) Mix Hi-Fi Primer with D85 Hardener and S30ER54 Thinner in the ratio 10:1.5:10 by volume using an R-M mixing stick. Filter material before spraying.
- (2) Apply 2 light/medium wet coats with 5-10 minutes flash off between coats to achieve a dry film thickness of Hi-Fi Primer of 30 microns. Refer to Technical Sheet for gun settings and spraying pressure.

- (3) Allow 15 minutes flash off before application of the Diamont Basecoat.

NOTE:

Do not allow flash off period to exceed 30 minutes.

- (4) The total dry film build of paint should be in the region of 180 microns.

8 Finishing - Basecoat and Clear Lacquer Finish

The following operations apply to standard clear over metallic basecoat finishes (COB) and to clear over solid colour basecoat finishes (COSB).

NOTE:

Observe all recommendations within the R-M Health and Safety Guide pertinent to spraying operations.

- (1) Add 5% D76 Hardener to R-M Diamont basecoat. Then thin the basecoat with R-M Thinner BC020 in ration 5:3 by volume. Filter the paint and check the viscosity is in the range of 18-22 seconds BSB4 flow cup at 20°C.

NOTE:

At spraying temperatures in excess of 25°C use Retard Thinner BC030 in place of BC020.

- (2) Apply, by spray, 2-3 single coats to give desired opacity, allowing 5-10 minutes flash off between coats. Refer to Technical sheet for gun settings and spraying pressures.
- (3) Allow to flash off for 10 minutes before application of Diamontop Clear Lacquer.

NOTE:

Do not allow the flash off period to exceed 2 hours before application of Diamontop Clear. If a 2 hour period is exceeded, re-coat with Diamont Base.

- (4) The total dry film build of paint should be in the region of 200 microns.
- (5) Mix R-M Diamontop with D76 Hardener in the ratio 5:1 by volume. Filter paint before application.
- (6) Apply 2 coats allowing 20 minutes flash off between coats to achieve a dry film build of 60 microns. Flash off for 10 minutes prior to baking. Refer to Technical Sheet for gun settings and spray pressures.
- (7) Bake at 80°C metal/substrate temperature for 45 minutes.
- (8) Allow to cool slightly and remove all masking, taking care not to tear paint from adjacent areas.
- (9) The total dry film build of paint should be in the region of 260 microns.

9 Finishing - Black Out Areas (Satin Black)

- (1) Abrade substrate with P1000 grit wet and dry. Degrease with Panel Clean using wipe on wipe off technique.
- (2) Mix R-M Solo Pure Black HS SPB with R-M Special Matt 05 in the following ratio:

	Weight
Solo Pure Black HS SPB	1 litre
Special Matt 05	417gms
D5000 Hardener	270gms
HS300 Thinner	270gms

Mix thoroughly and filter paint before applying.

NOTE:

Special Matt 05 should be kept on the mixing machine to ensure thorough dispersion.

- (3) Apply 2 coats allowing 10 minutes flash off between coats, achieving a dry film build of 50 microns. Refer to Technical Sheet for gun settings and spray pressure.
- (4) Bake at 60°C substrate temperature for 30 minutes.
- (5) Allow to cool slightly and remove all masking, taking care not to tear the paint from adjacent areas.

10 Repairs to Finishes

- (1) Wet flat damaged areas as required.
- (2) Bake the vehicle at 60°C for 30 minutes to remove all traces of moisture.
- (3) Repaint the damages surface, dependent on damage, using R-M products as follows:
 - (a) Damage to bare metal, re-apply Euroxy, Fast Filler, Hi-Fi and Diamont as in operations 2, 3, 4, 5, 6, 7 and 11. Spray primers only inside the sanded area, extending the sprayed area with each coat but keeping the area as small as possible.
 - (b) Damage to primer, re-apply Fast Filler, Hi-Fi and Diamont as in operations 4, 5, 6, 7 and 11. Spray primers only inside the sanded area, extending the sprayed area with each coat but keeping the area as small as possible.
 - (c) Damage in topcoat, re-apply Diamont as in operation 11.

11 Blend-in Repairs

- (1) Degrease repair area using Panel Clean.
- (2) Compound with Pate Bril 852 well beyond the intended fade-out area.
- (3) Degrease thoroughly with Panel Clean to remove all traces of compound residue.
- (4) Mix Diamont basecoat as per operation 8.1 and Diamontop Clear as per operation 8.4.
- (5) Apply first coat of Diamont colour to the repair area, flash off for 5 minutes. Apply second coat of Diamont colour extending the sprayed area so that the colour merges completely into the surrounding original finish.
- (6) Flash off for 10-15 minutes before applying Diamontop Clear.
- (7) Apply first coat of Diamontop Clear to cover the 2 coats of base colour. Flash off for 5-10 minutes. Apply second coat of Diamontop Clear extending the first coat wet edge into the prepared fade out area. Flash off for 2 minutes.
- (8) Carry out one of the following operations;
 - (a) Reduce the air pressure to approximately 30 psi at the gun and melt in the second clear coat wet edge using neat BC020 Thinner.
 - or
 - (b) Leave 1 part activated clear in the spray gun and add 5 parts Spot 50 and mix well. Reduce the air pressure to approximately 30 psi at the gun and apply the Spot 50 mix to melt in the second coat wet edge.
- (9) Bake at 60°C metal temperature for 30 minutes. Polish the fade out area with Bril 852P and Brilstar 848 once the vehicle has fully cooled.

12 Undersealing and Sound Deadening (Alochrome 40747)

NOTE:

Plastic gloves must be worn by all personnel who are likely to come into contact with bodywork to avoid contamination likely to cause blistering between substrate and subsequent paint coats.

- (1) Mask off all areas not designated for undersealing or sound deadening.
- (2) Underbody and areas designated for this process must be prepared as in operation 2.3. Repeat the process.

Do not abrade the surface, the blue Alochrome treatment must not be removed. If the Alochrome surface is not intact, the aluminium must be thoroughly scuffed with P120. Degrease again as in operation 2.3 prior to painting.

- (3) Apply 4-6 coats Glasurit coarse texture stone chip and underbody protection 1109-1240/4 by undersealing or Schutz gun to give a film build of 0.5-1mm.

NOTE:

Refer to Technical Sheet for application by airless spray, conventional gun plus pressure pot and brush/roller application.

- (4) Allow to dry at least 6 hours at 20°C or 60 minutes at 60°C substrate temperature.
- (5) Mix R-M Diamontop Clear Lacquer with R-M Hardener D76 in the ratio of 5:1 by volume. Filter the material prior to application.
- (6) Apply 2 coats allowing 10 minutes flash off between coats, to achieve a dry film build of 60 microns. No flash off is required prior to baking. Refer to Technical Sheet for gun settings and spray pressure.
- (7) Bake at 60°C metal substrate temperature for 30 minutes.
- (8) Allow to cool slightly and de-mask areas previously masked, taking care not to tear edges of paint from the Adjacent areas.

13 Polishing

- (1) All polishing operations should be carried out mechanically (i.e. mop) using Bril 852 Fine Compound.
- (2) Final polishing may be carried out using silicon free Brilstar 848.

14 Film Builds

- (1) The recommended dry film builds are detailed in each relevant section. They are as follows:
- | | |
|------------------------------|-------------|
| Euroxy (2 coats) | 50 microns |
| Fast Filler (3 coats) | 100 microns |
| Hi-Fi (2 coats) | 30 microns |
| Diamont Basecoat (2-3 coats) | 20 microns |
| Diamontop (2 coats) | 60 microns |
| | 260 microns |
| Satin Black (2 coats) | 50microns |
| Total | 310 microns |
- (2) The minimum dry film build covered by warranty is 180 microns, the maximum is 350 microns which makes allowance for some repairs.
- (3) Film builds at each stage should be checked and recorded for each vehicle.
- (4) after assembly of the vehicle, scratches in the paintwork may need to be repaired. a further allowance of 50 microns is made to cover such repairs. After repair the total film build covered by warranty is 400 microns.

4 Repair of Stone Chips

- (1) Inspect regularly for stone chips and scratches, especially the vulnerable areas on the front of the vehicle or where stones are thrown up by other vehicles.
- (2) Apply touch-up paint to protect against further deterioration or rust spots occurring.

5 Tar Spots

- (1) Remove as soon as possible after contamination with a quality brand of Tar Remover and then polish back to pristine condition.

6 Garaging

- (1) Whilst regular garaging will help to keep the paintwork in excellent condition, a wet car should always be dried off before being left in an enclosed area.

AFTERCARE OF VEHICLE PAINTWORK

1 Washing of Fresh Paintwork

- (1) It is recommended that fresh paintwork i.e. up to 3 months old is washed with a soft sponge and cold or lukewarm water only, followed by rinsing off thoroughly. Finally dry off with a clean chamois leather.

2 Normal Washing

- (1) In subsequent months, the addition of a good quality car shampoo to the washing water will add to the removal of traffic film and grime.

3 Polishing

- (1) To replace the paintwork's natural protection and to assist in maintaining the paintwork in "showroom condition", it is recommended that the regular use of a good quality "ammonia-free" wax polish will guard against the influences of weather, traffic grime and pollution e.g. acid rain, etc.

Product	Flash Point	Max pot Life	Hardener	Thinner	Mixing Ratio	Drying Time	Panel Temp	Sanding Grit	RFU Viscosity	Gun Pressure	Coats
Pre-Kleano	Below 21°C	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
Panel Clean	Below 21°C	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
Stop Zinc	Over 32°C	7-8 mins	55-22-101	N/A	100:2-3	30 mins	20°C	P80	N/A	N/A	A/R
Stop Extra Fin	Over 32°C	5 mins	Supplied with Product	N/A	100:3	20 mins	20°C	P240	N/A	N/A	A/R
Euroxy Wash	Over 21°C	Over 8 hrs	Euroxy Hardener	SM 2K ER 54	2:1:1	N/A	20°C	N/A		45-50 psi	1
Fast Filler	22-23°C	2 hrs	D85	S 30 ER 54	5:1:2	30 mins	60°C	P400 dry machine or P800 wet by hand	30-32 secs	50-60 psi	3
Spraygloss	Below 21°C	N/A	N/A	199-18	1:9	N/A	N/A		N/A	50-60 psi	1 light
Hi-Fi Wet on	22-23°C	2 hrs	D85	2K ER 54	10:1:10	N/A	20°C	N/A	20-22 secs	50 psi	2
Wet Primer											
Diamont	Above 21°C	N/A	N/A	BC020/ BC030	5:2	10 mins	20°C	N/A	18-25 secs	45-55 psi	2 - 3
Diamontop	22-23°C	4 hrs	D76	None	5:1	30 mins	60°C	N/A	approx. 19"	45-55 psi	2
Jet 5 Accelerator		Reduces pot life	Dependent on product used								

PAINT FORMULATIONS

Spa Silver-JS0001 (Silver Frost, no mica in this formulation)

	1 Litre Weight	4 Litre Weight	Volume
BC020	88	352	100
BC171	893.8	3575.2	974
BC250	912.4	3649.6	994
BC400	917.8	3671.2	1000

Le Mans Blue-JS0002 (Blue Mica)

	1 Litre Weight	4 Litre Weight	Volume
BC020	88	352	100
BC406	406.9	1627.6	447
BC400	609	243.6	666
BC118	778.9	3115.6	842
BC200	923	3692	1000

Silverstone Green-JS0003 (Green Mica)

	1 Litre Weight	4 Litre Weight	Volume
BC020	88	352	100
BC500	500.8	2003.2	542
BC400	636.6	2546.4	689
BC200	734.2	2936.8	796
BC105	842.7	3370.8	913
BC171	850.4	3401.6	921
BC111	930	3720	1000

Monza Red-JS0004 (Red mica)

	1 Litre Weight *	4 Litre Weight	Volume
BC020	88	350	100
BC820	442.5	1768	483
BC840	721	2882	789
BC115	836.5	3344	906
BC200	922	3686	1000

Daytona Black-JS0005 (Anthracite, no mica in this formulation)

	1 Litre Weight	4 Litre Weight	Volume
BC020	88	352	100
BC250	546.4	2185.6	594
BC200	763.5	3054	832
BC400	835.6	3342.4	910
BC140	885.4	3541.6	964
BC170	909.4	3637.6	990
BC300	918.6	3674.4	1000

Gloss Black-JS0009 (Black, no mica in this formulation)

	1 Litre Weight	4 Litre Weight	Volume
BC020	88	352	
BC200	760.1	3040.4	
BC406	909.6	3638.4	

SBK Satin Black-JS0008 (Windscreen areas)

Solo pure black HS SPB - 1 Litre
Special Matt 05 - 400 gms

Gunmetal-JS0006 (Bumper/Rear Louvre)

	1 Litre Weight	4 Litre Weight	Volume
BC020	88	352	100
BC111	336	1344	346
BC200	557.6	2230.4	589
BC250	775.7	3102.8	824
BC600	868	3472	922
BC118	927.7	3710.8	984
BC170	937.8	3751.2	995
BC500	942.5	3770	1000

Black-JS0007 (Blackout areas)

	1 Litre Weight	4 Litre Weight	Volume
HS300	90	360	100
HS25	356	1424	373
HS69	529.5	2118	549
HS99	678.5	2750	697
HS10	974.5	3898	1000

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1 INTRODUCTION

This section details the procedures for replacement of body assembly components. Replacement of components or disassembly instructions not covered in this section require the component to be returned to JaguarSport.

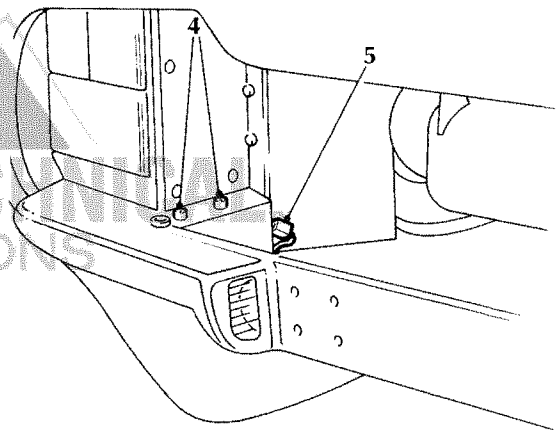
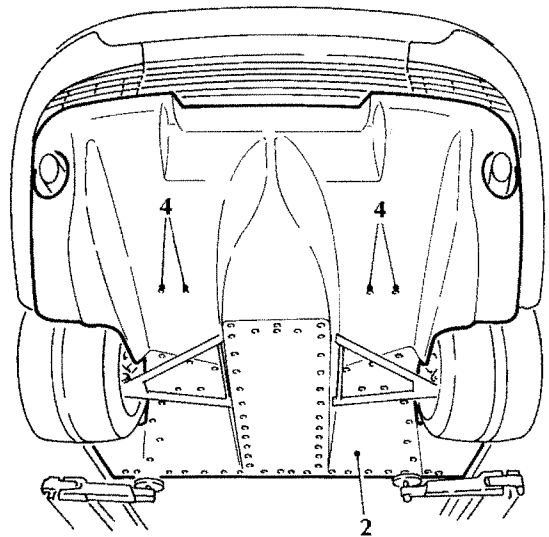
2 BUMPER AND VENTURI

A Removal Procedure

NOTES:

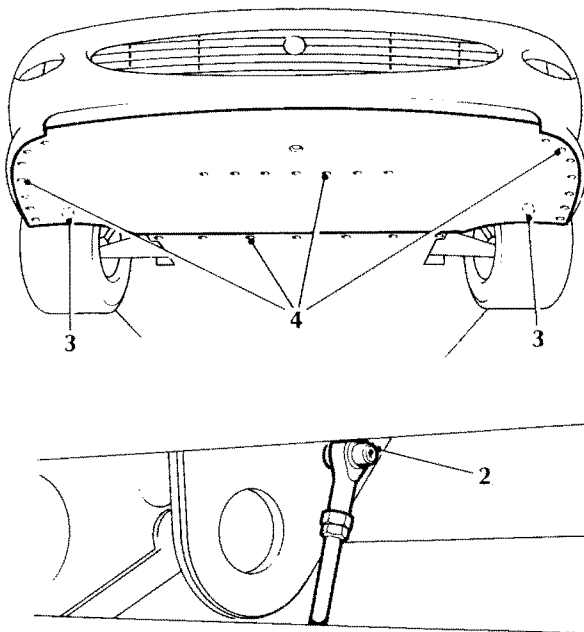
- 1 *Ensure absolute cleanliness of work area and tools.*
- 2 *Observe correct use of tools.*

- (1) Correctly position the vehicle on the lifting apparatus and raise the vehicle to a suitable working height (Refer to Section 1.1).
- (2) Remove the screws and washers securing the venturi panel and remove the panel. (Refer to Fig for screw location).
- (3) Remove the rear grille and wheelarch liner to bumper fixings. (Refer to Sections 4 & 11).
- (4) Support the bumper and remove the screws washers securing the bumper.
- (5) Support the bumper and disconnect the number plate light electrical connectors.
- (6) Remove the roll-over valve drain tube and engine coolant expansion tank drain tube and remove the bumper.



B Installation Procedure

- (1) Support bumper close to underside of vehicle and connect the number plate light electrical connector.
- (2) Position bumper on underside of vehicle and secure with screws, nuts and washers. Switch on the lights and ensure number plate light is illuminated.
- (3) Fit the rear grille and secure the wheelarch liners to the bumper. (Refer to Sections 4 & 11).
- (4) Position venturi on underside of vehicle and secure with screws and washers. Ensure the drain tubes are located correctly in the panel.
- (5) Lower lifting apparatus.
- (6) Remove vehicle from lifting apparatus.



3 FRONT SPLITTER

A Removal Procedure

NOTES:

- 1 Ensure absolute cleanliness of work area and tools.
- 2 Observe correct use of tools.

- (1) Position the vehicle on lifting apparatus and raise to a suitable working height.
- (2) Remove the bolt, nut and washer securing the splitter ball joint to the front cross member.
- (3) Remove the wheelarch liner to splitter fixings.
- (4) Support the splitter and remove the screws and washers attaching the splitter to the vehicle underbody.

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B Installation Procedure

- (1) Align the splitter with the vehicle underbody and secure the splitter with screws and washers.
- (2) Secure the ball joint to the front cross member using bolt, washer and nut.
- (3) Fit the wheelarch liner to splitter fixings.
- (4) Lower the vehicle and remove from the lifting apparatus.

4 WHEEL ARCH LINERS

A Removal Procedure

Special Tools

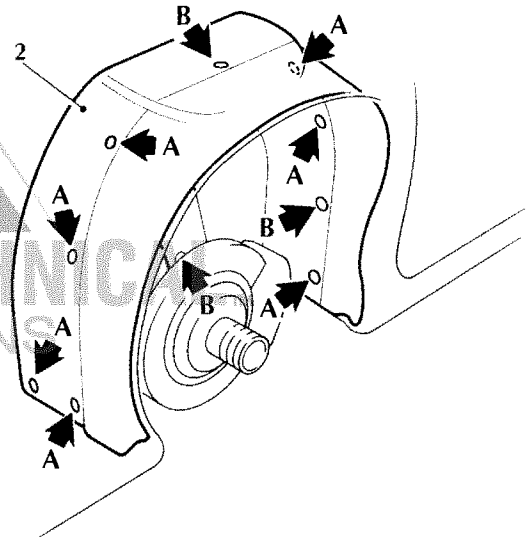
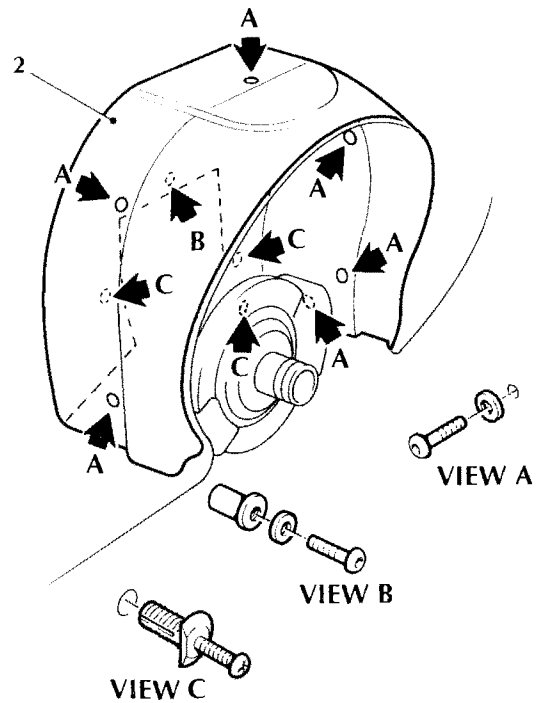
The following special tools will be required

Torque wrench 0 - 15Nm (0 - 11 ftlb)

NOTES:

- 1 Ensure absolute cleanliness of work area and tools.
- 2 Observe correct use of tools.

- (1) Remove the relevant wheel (Refer to Section 4.1).
- (2) Remove screws, washers and fixings securing the inner wheel arch liner. Note the position of the shorter screws.
- (3) Remove the wheel arch inner liner .



B Installation Procedure

- (1) Fit the inner wheel arch liner.
- (2) Secure wheel arch liner with screws, washers and fixings.
- (3) Torque load the shorter screws 4 to 5.5Nm (3 to 4ftlb) and the longer screws 2.4 to 2.9Nm (1.8 to 2.1ftlb).
- (4) Fit the wheel (Refer to Section 4.1).

5 BONNET LOUVRES

A Removal Procedure

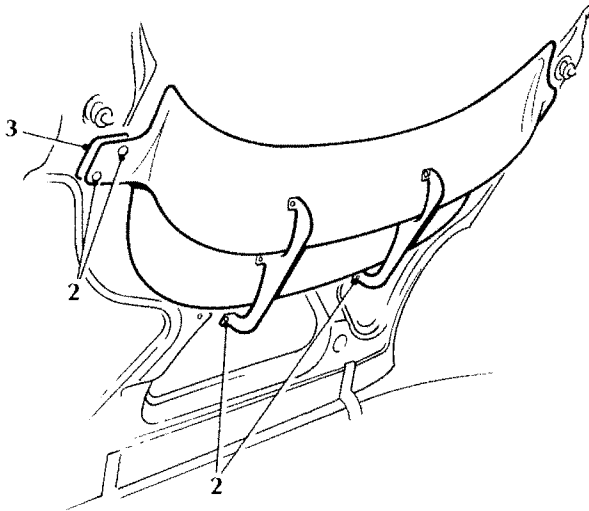
Special Tools

The following special tools will be required

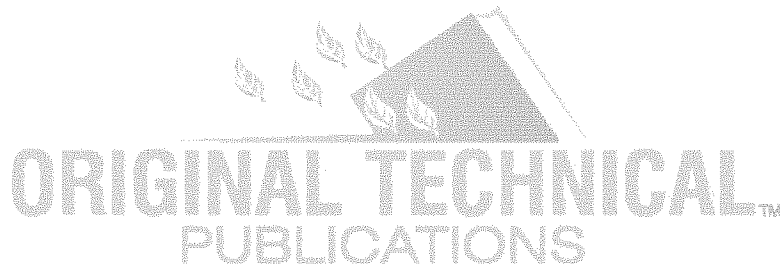
Torque wrench 0 to 15Nm (11.5ftlb)

NOTES:

- 1 Ensure absolute cleanliness of work area and tools.
- 2 Observe correct use of tools.



- (1) Open the bonnet
- (2) Support the louvre assembly and remove the button head screws securing louvres to the bonnet.
- (3) Remove the louvres and collect the spacer from between the louvres.



B Installation Procedure

- (1) Locate the spacer between the two louvres
- (2) Fit the louvre assembly to the bonnet and secure the button head screws.
- (3) Torque load the screws 7.5 to 9Nm (5.5 to 6.5ftlb).

6 BONNET

A Removal Procedure

Special Tools

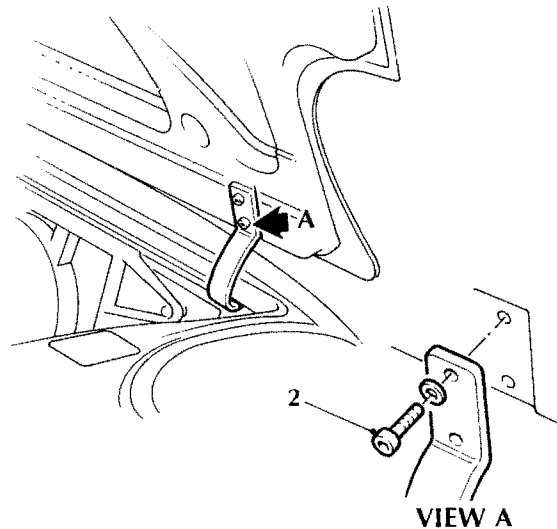
The following special tools will be required

Torque wrench 65Nm (48ftlb)

NOTES:

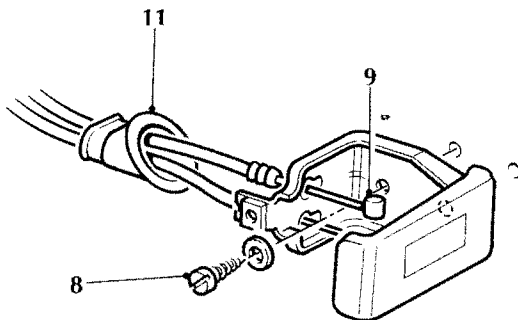
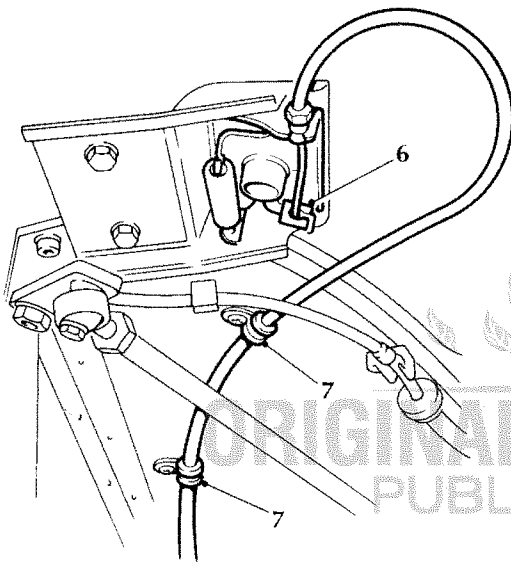
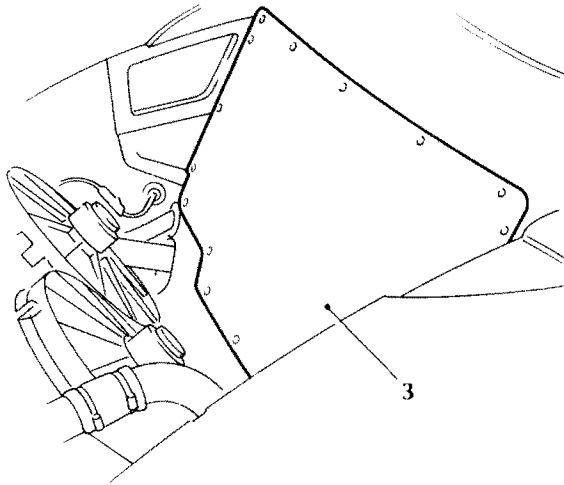
- 1 Ensure absolute cleanliness of work area and tools.
- 2 Observe correct use of tools.

- (1) Pull the bonnet release handle on the tunnel side in the passenger's footwell and open the bonnet.
- (2) Support the bonnet and remove the four bolts and shims attaching the bonnet to the hinge assembly. Note the shim thickness and position.
- (3) Remove bonnet.



B Installation Procedure

- (1) Position the bonnet on the hinges and fit the four screws with packing shims as required.
- (2) Torque load the cap head screws 20 to 23Nm (15 to 17ftlb).
- (3) Close the bonnet and ensure the bonnet is faired with the front fenders and headlamp covers.
- (4) Open and close the bonnet to ensure positive locking and unlocking. Rectify any faults as necessary.



7 BONNET RELEASE/LATCH MECHANISM

A Removal Procedure

Special Tools

The following special tools will be required

Torque wrench 12 to 14Nm (9 to 10.5ftlb)

WARNING:

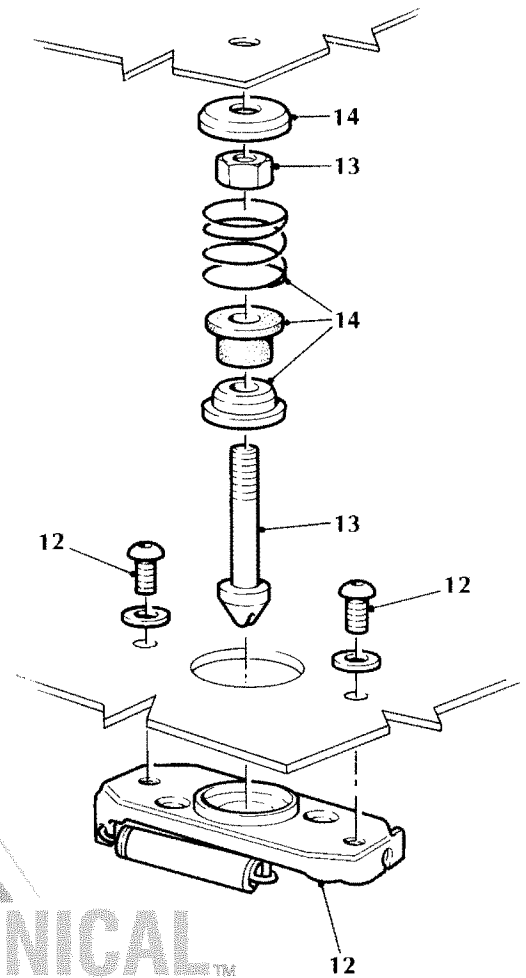
WHEN HANDLING CABLE ASSEMBLIES BEWARE OF FRAYED OR BROKEN STRANDS. WEAR PROTECTIVE GLOVES.

NOTES:

- 1 Ensure absolute cleanliness of work area and tools.
- 2 Observe correct use of tools.

- (1) Pull the bonnet release handle on the passenger side of the tunnel (in passenger footwell).
- (2) Open the bonnet.
- (3) Remove the radiator duct access panel.
- (4) Remove the structural hatch.
- (5) Remove the front wheels and wheel arch liners. (Refer to Section 4).
- (6) Press the bonnet latch lever to remove the spring pressure and disconnect the cables.
- (7) Remove the P clips securing the bonnet release cable to the vehicle body.
- (8) Remove the two screws securing the release handle to the tunnel.
- (9) Remove the release cables from the handle assembly.
- (10) Mark the LH and RH side cables separately for identification and attach a cord to each cable end to enable new cables to be refitted in the tunnel without trim removal.
- (11) Remove the grommet from the tunnel aperture and remove the cables complete with grommet into the passenger footwell.

- (12) Remove the button head screws and washers securing bonnet latch to vehicle and remove the latch.
- (13) Hold the bonnet striker with a screw driver and loosen the lock nut.
- (14) Unscrew the bonnet striker from the bonnet and collect the spring, spring bush and bonnet overslam bush.



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B Installation Procedure

- (1) Fit the bonnet striker assembly to the bonnet and screw in until spring pressure is felt.

NOTE:

Do not tighten lock nut until after adjustment.

- (2) Fit the bonnet latch and secure with button head screws.
- (3) Torque load screws 7 to 8.5Nm (5.5 to 6.5ftlb).
- (4) Fit the cables to the grommet, route the cables through the tunnel and Zero bulkhead and fit the grommet to the tunnel aperture.

NOTE

Ensure the barrel ends of the cables are in the passenger footwell

- (5a) On RHD vehicles, fit the short bonnet release cable to the LH latch and the long bonnet release cable to the RH latch.
- (5b) On LHD vehicles, fit the short bonnet release cable to the RH latch and the long bonnet release cable to the LH latch.
- (6) Connect bonnet release cables to bonnet release handle.
- (7) Fit the release handle to the tunnel with two screws and washers.
- (8) Secure bonnet release cables to vehicle body with P clips ensuring the cables have smooth, kink-free paths.
- (9) Operate bonnet release handle through several cycles to ensure unrestricted movement and adjust the outer cable, at the handle end, for unlatching travel.
- (10) Close the bonnet and check for positive locking.
- (11) Check the bonnet is faired with the front wings of the vehicle.
- (12) Open the bonnet, hold striker with screw driver and tighten lock nut.
- (13) Repeat items (10) and (11).
- (14) If unacceptable conditions result in items (10) and (11) adjust the length of the striker to obtain a satisfactory condition.
- (15) Fit the wheel arch liners and wheels. (Refer to Section 4.1).
- (16) Fit the structural hatch and secure the screws.
- (17) Fit the radiator duct access hatch and secure the fasteners.

8 FRONT GRILLE

A Removal Procedure

Special Tools

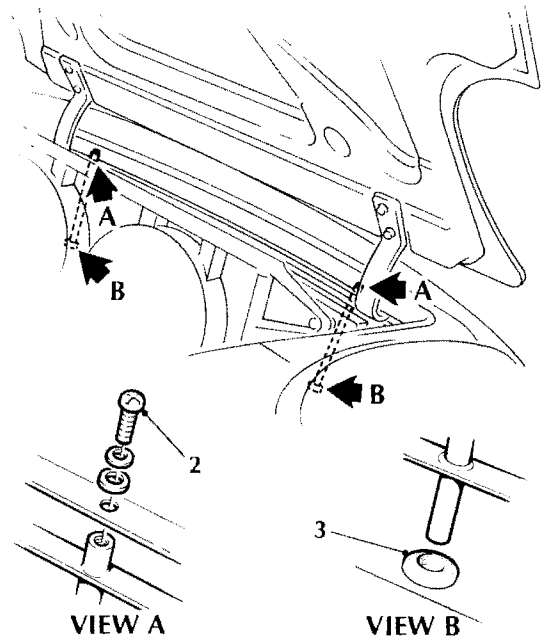
The following special tools will be required

Torque wrench 0 to 8Nm (6ftlb)

NOTES:

- 1 Ensure absolute cleanliness of work area and tools.
- 2 Observe correct use of tools.

- (1) Pull bonnet release handle and open bonnet.
- (2) Remove the screws and washers securing the grille.
- (3) Tilt the grille forwards and lift out of the rubber grommets and remove the grill.



B Installation Procedure

- (1) Locate the grille in the rubber grommets and fit the grille.
- (2) Fit the screws and washers and torque load the screws 2.5 to 4Nm (2 to 3ftlb).
- (3) Close the bonnet and ensure there are no obstructions.

9 ENGINE BAY LOUVRES

A Removal Procedure

Special Tools

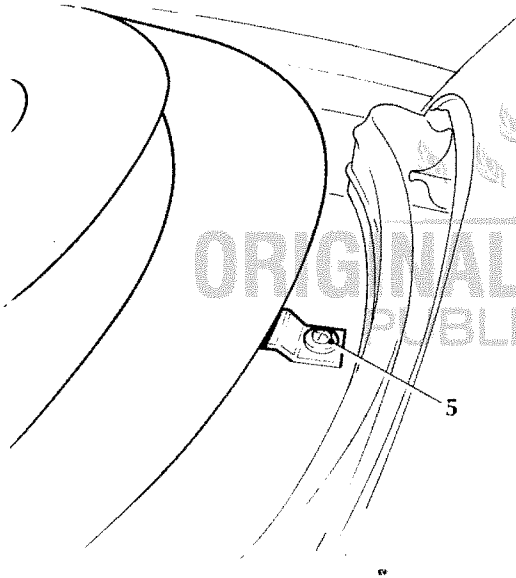
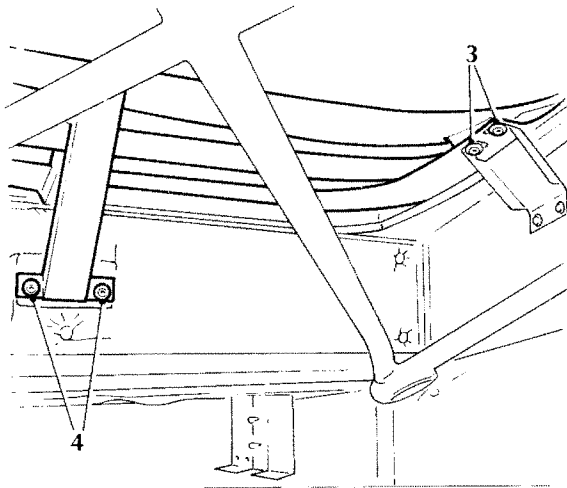
The following special tools will be required

Torque wrench 0 to 15Nm (11.5ftlb)

NOTES:

- 1 *Ensure absolute cleanliness of work area and tools.*
- 2 *Observe correct use of tools.*

- (1) Open the engine compartment hatch.
- (2) Fit JaguarSport protective covers.
- (3) Remove button head screws and washers securing the louvres to the side brackets
- (4) Support the louvres and remove the button head screws and washers securing the centre gutter support to the boot front panel.
- (5) Raise the boot lid and remove the screw attaching the rear of the support bracket to the vehicle body.
- (6) Lift the louvre assembly from the vehicle.
- (7) Drill out the rivets attaching the centre gutter to the louvres and separate the louvres.



B Installation Procedure

- (1) Assemble the louvres and secure to the centre gutter using countersunk rivets.
- (2) Fit the centre gutter to the support and secure with washer and screw.
- (3) Locate the louvre assembly in the engine compartment.
- (4) Secure the centre gutter support to the boot front panel with button head screws and washers.
- (5) Torque load screws 7 to 8.5Nm (5.5 to 6.5ftlb).
- (6) Fit the louvres to the brackets and secure using button head screws and washers.
- (7) Check the clearance between the engine louvre and the tailgate glass, adjust if necessary.
- (8) Torque the screws 7 to 8.5Nm (5.5 to 6.5ftlb)
- (9) Remove the protective covers and close the engine compartment hatch

10 BOOT LID AEROFOIL

A Removal Procedure

Special Tools

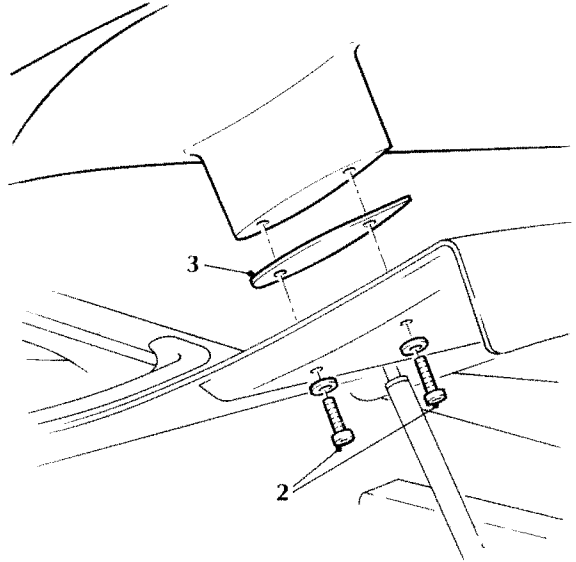
The following special tools will be required

Torque wrench 0 to 65Nm (48ftlb)

NOTES:

- 1 Ensure absolute cleanliness of work area and tools.
- 2 Observe correct use of tools.

- (1) Open the boot lid.
- (2) Support the aerofoil and remove the screws, washers and the height shims securing the aerofoil.
- (3) Remove the aerofoil and gasket from the boot lid.



B Installation Procedure

- (1) Fit a new gasket to the aerofoil.
- (2) Fit the aerofoil to the boot lid, seal the area around the fixings with sealant (Pt. No. 9507).
- (3) Hold the aerofoil in place and fit washers and bolts to secure the aerofoil. Shim the aerofoil pedestals to achieve the correct body gap clearances.
- (4) Torque load nuts 8 to 9.5Nm (6 to 7ftlb).
- (5) Fit bumpstops to the underside of the aerofoil tips (one per side).
- (6) Close the boot lid.



B Installation Preparation

- (1) Assemble the driveline to transaxle as described in Section 5.
- (2) Assemble the clutch as described in Section 8.
- (3) Assemble the transaxle to the engine as described in Section 7.
- (4) Assemble the ridgetent and rear suspension as described in Section 4.
- (5) Assemble the exhaust system as described in Section 9

C Installation Procedure

- (1) Correctly position vehicle on lifting apparatus as described in Jacking and Lifting (Refer to Section 1.1).
- (2) Raise vehicle to suitable working height.
- (3) Fit additional supports to base of rear of the vehicle.
- (4) Position the engine and transaxle beneath the vehicle.
- (5) Fit gear selector rod over transaxle spigot and secure with spring pin.
- (6) Move the gear lever fully forward to prevent the engine fouling on the selector rod during installation.
- (7) Operate the engine trolley hydraulic pump to raise the engine until it is just entering the engine compartment.

CAUTION:

During lowering operations conduct a detailed inspection to ensure no services are snagging and being damaged.

- (8) Feed the throttle cable through the auxiliary drive belt and the fuel return pipe (if removed) through the plenums.
- (9) Remove the upright supports and slowly lower the vehicle making sure the engine bay intermediate plate and rear plate align with the corresponding mountings on the vehicle.
- (10) Refit upright supports.

CAUTION:

Ensure the bolts attaching the engine intermediate and rear plates to the vehicle body mountings can be fitted with thumb pressure. Do not force the bolts through. Adjust the height of the engine trolley or the vehicle to obtain perfect alignment.

- (11) Align the attachment holes in the engine bay intermediate and rear plate with the holes in the corresponding mountings and fit the M6 bolts, washers and nuts. Torque to 12-14Nm (9-10.5lbft)

12 BOOT LID

A Removal Procedure

Special Tools

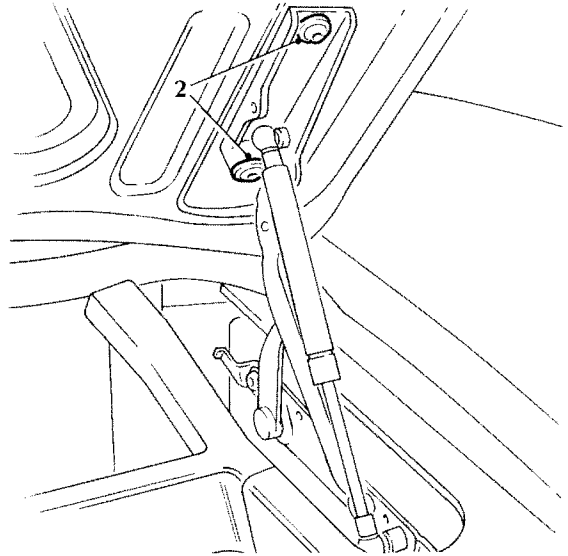
The following special tools will be required

Torque wrench 0 to 65Nm (48ftlb)

NOTES:

- 1 Ensure absolute cleanliness of work area and tools.
- 2 Observe correct use of tools.

- (1) Press boot release and open boot lid.
- (2) Support the boot lid and remove the button head screws securing the hinges to the underside of the boot lid.
- (3) Remove the boot lid.



B Installation Procedure

- (1) Position the boot lid on the hinges.
- (2) Support the boot lid and secure with cap head screws.

NOTE

Do not torque the screws at this stage.

- (3) Close and open boot lid to ensure smooth and unrestricted operation and the lid is faired with the rear quarters. Adjust the boot lid position if necessary.
- (4) Torque the boot lid attachment screws 19.5 to 21.5Nm (14.5 to 16ftlb)

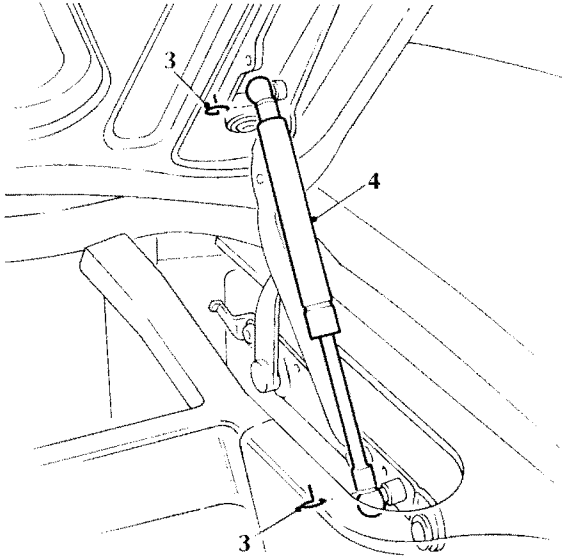
13 BOOT LID GAS STRUTS

A Removal Procedure

NOTES:

- 1 Ensure absolute cleanliness of work area and tools.
- 2 Observe correct use of tools.

- (1) Press the boot release and open the boot lid.
- (2) Remove the RH and LH boot hinge trim panels. (Refer to Section 2.14).
- (3) Remove the clips from the gas strut upper and lower attachments.
- (4) Support the boot lid and pull the gas struts from the ball fittings.



B Installation Procedure

- (1) Support the boot lid and push the gas struts onto the ball fittings as shown.
- (2) Fit the strut clips.
- (3) Fit the RH and LH boot hinge trim panels. (Refer to Section 2.14)
- (4) Close and open boot lid to ensure smooth and unrestricted operation, and satisfactory operation of gas struts.

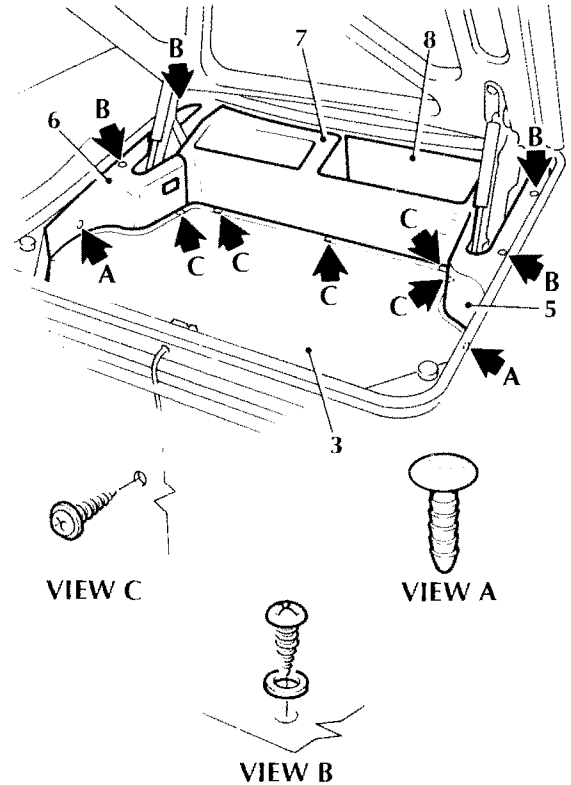
14 BOOT TRIM

A Removal Procedure

NOTES:

- 1 Ensure absolute cleanliness of work area and tools.
- 2 Observe correct use of tools.

- (1) Open the boot lid.
- (2) Turn the main isolator switch to the off position.
- (3) Remove the carpet from the boot floor.
- (4) Remove the tool kit, first aid kit, warning triangle and wheel nut from the boot.
- (5) Remove the trim clips and self-tapping screw attaching the RH trim panel and remove the panel.
- (6) Remove the trim clips and self-tapping screw attaching the LH trim panel. Ease the panel away from the structure, disconnect the boot lamp electrical connectors and remove the panel.
- (7) Remove the self-tapping screws attaching the auto-changer cover and remove the cover.
- (8) Remove the millboard, retained with velcro strip.



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B Installation Procedure

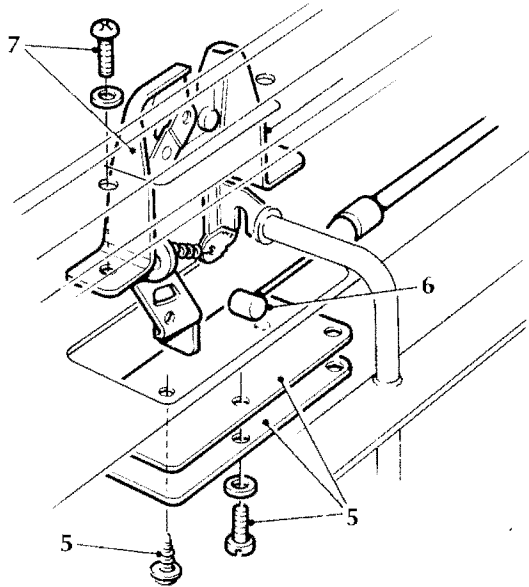
- (1) Fit the auto-changer cover and secure with self-tapping screws.
- (2) fit the millboard and secure with velcro strip.
- (3) Connect the boot lamp electrical connectors and fit the LH trim panel. Secure the panel with trim clips and self-tapping screw.
- (4) Fit the RH trim panel and secure with trim clips and self-tapping screw.
- (5) Fit the boot carpet.
- (6) Fit the tool kit, first aid kit, warning triangle and wheel nut into the boot.
- (7) Turn the main isolator switch to the on position and ensure the boot lamp functions correctly.
- (8) Close the boot lid.

15 BOOT LATCH MECHANISM

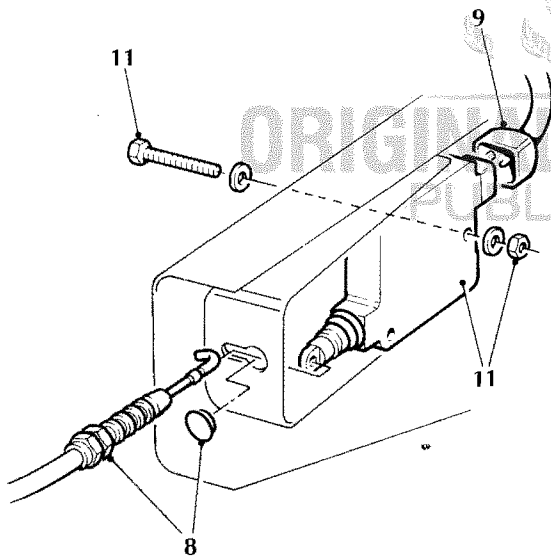
A Removal Procedure

NOTES:

- 1 Ensure absolute cleanliness of work area and tools.
- 2 Observe correct use of tools.



- (1) Press the boot release switch and open the boot lid.
- (2) Turn the main isolator switch to the off position.
- (3) Remove the rear grille. (Refer to Section 2.11).
- (4) Remove the RH rear lamp assembly. (Refer to Section 15).
- (5) Remove the screws securing the boot latch plate, remove the plate and gasket.
- (6) Manually operate the boot latch and remove the cable from the latch.
- (7) Remove the screws attaching the boot latch and remove the latch from the vehicle.
- (8) Disconnect the operating cable from the solenoid, remove the grommet and withdraw the cable.
- (9) Disconnect the electrical connections to the solenoid.
- (10) Lift the carpet and remove the boot RH side trim.
- (11) Remove the two screws, nuts and washers attaching the solenoid and remove the solenoid.



B Installation Procedure

NOTES:

- 1 *Ensure absolute cleanliness of work area and tools.*
- 2 *Observe correct use of tools.*

- (1) Fit the boot latch solenoid to the vehicle and secure using screws, washers and nuts.
- (2) Connect the solenoid electrical connections.
- (3) Feed the operating cable through the rear structure and connect the cable to the solenoid.

NOTE:

Clip the rearmost groove on the cable outer to the bracket.

- (4) Fit the boot latch to the vehicle and secure using two button head screws and washers.
- (5) Manually operate the boot latch and fit the operating cable. Adjust the outer cable position to achieve correct latch release action and fit the grommet.
- (6) Close the boot lid.
- (7) Turn the main isolator switch to the on position.
- (8) Operate the boot release switch several times to ensure correct function of the latch mechanism.
- (9) Fit the gasket and latch cover plate, secure the plate using two button head screws and washers and the self-tapping screws.
- (10) Fit the RH rear lamp assembly. (Refer to Section 15).
- (11) Fit the rear grille. (Refer to Section 2.11).
- (12) Fit the boot carpet and RH boot trim. (Refer to Section 2.14).

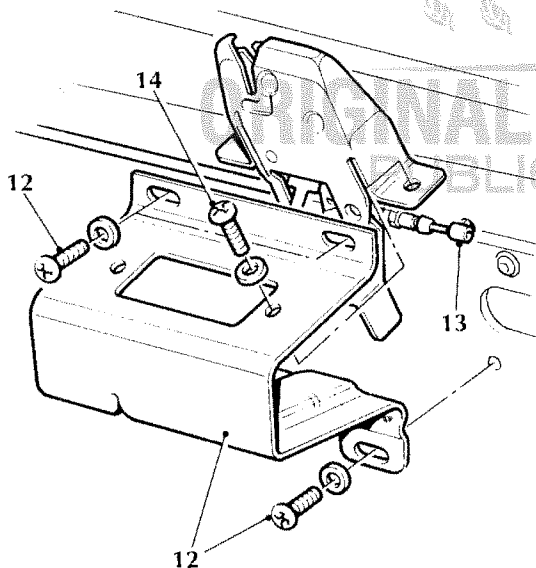
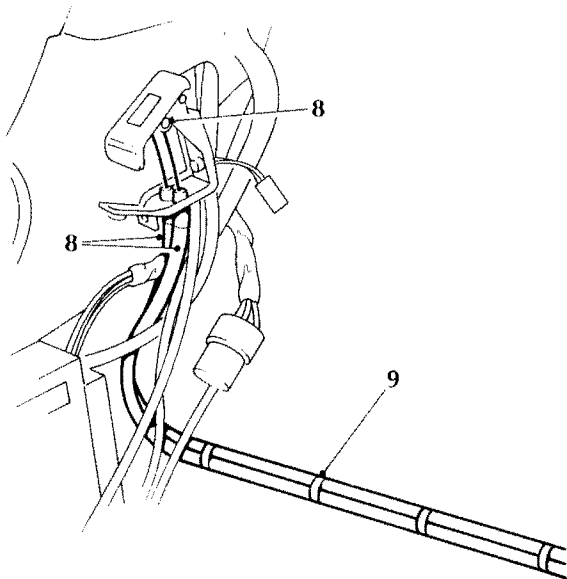
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16 TAILGATE MECHANISM

A Removal Procedure

NOTES:

- 1 Ensure absolute cleanliness of work area and tools.
- 2 Observe correct use of tools.



- (1) Open the tailgate and fit JaguarSport covers.
- (2) Fit JaguarSport sill covers.
- (3) Remove the RH seat (Refer to Section 2.30, Seats and Slides)
- (4) Remove the tunnel trim panel. (Refer to Section 2.31, Tunnel Trim).
- (5) Remove the parcel shelf trim. (Refer to Section 2.27, Rear Trim).
- (6) Remove the ECU cover. (Refer to Section 2.29, Side Trim).
- (7) Remove the RH rear carpet. (Refer to Section 2.25, Carpets).
- (8) Remove the outer cable from the bracket and remove the cable barrel from the release handle.
- (9) Remove the cable ties securing the cables to the bulkhead.
- (10) Remove the tunnel grommet and slide the grommet off the cables.
- (11) To aid refitting of the cables, tie a sufficient length of thin string to the cable barrel ends.
- (12) Remove the three screws and washers securing the latch bracket and remove the bracket.
- (13) Manually operate the latch mechanism and remove the cable from the latch.
- (14) Remove the two screws and washers attaching the latch to the bracket and remove the latch.
- (15) Remove the cable ties attaching the cables to the vehicle body.
- (16) Pull the cables through the tunnel into the engine compartment, untie the string and remove the cables.

B Installation Procedure

NOTES:

- 1 Ensure absolute cleanliness of work area and tools.
- 2 Observe correct use of tools.

NOTE:

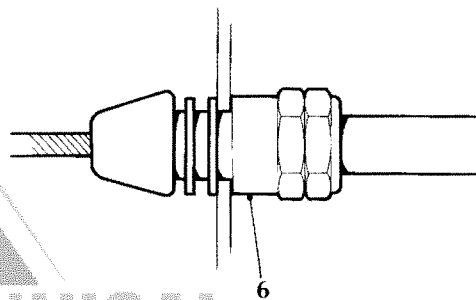
Prior to installing the cables check for free and unrestricted movement of the inner cable

- (1) Tie the string to the barrel ends of the cable.
- (2) Feed the cables through the engine compartment whilst at the same time pulling gently on the string to draw the cables into the passenger compartment.
- (3) Fit the latch to the mounting bracket and secure using two washers and screws.
- (4) Slide the grommet over the cables in the passenger compartment and fit the grommet to the tunnel.
- (5) Remove the string and fit the cables to the bracket and release handle.

NOTE:

Ensure the third groove in the outer cable is used to locate the cable in the bracket.

- (6) Fit the bracket and latch assembly to the vehicle and secure using three washers and screws.
- (7) Secure the cables in the engine compartment using tyrap and ensure a smooth cable lay with no kinks.
- (8) Operate the tailgate mechanism several times to ensure free and unrestricted movement. If necessary adjust the cable outer position at the handle to achieve correct latch release action.
- (9) Fit the RH rear carpet. (Refer to Section 2.25, Carpets)
- (10) Fit the ECU cover. (Refer to Section 2.29, Side Trim)
- (11) Fit the parcel shelf. (Refer to Section 2.27, Rear Trim).
- (12) Fit the tunnel trim. (Refer to Section 2.31, Tunnel Trim).
- (13) Fit the RH seat. (Refer to Section 2.30, Seats and Slides).
- (14) Remove the protective covers and close the tailgate.



17 WINDSCREEN WIPERS

A Removal Procedure

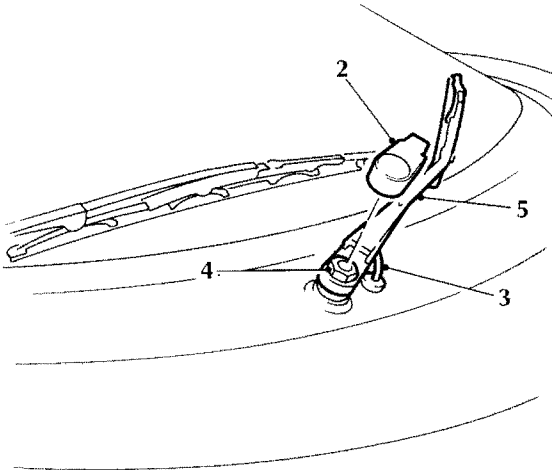
Special Tools

The following special tools will be required

Torque wrench 0 to 23Nm (0 to 17ftlb)

NOTES:

- 1 Ensure absolute cleanliness of work area and tools.
- 2 Observe correct use of tools.



- (1) Fit JaguarSport protective wing covers.
- (2) Remove the cap from the wiper arm.
- (3) Remove the windscreen washer tube from the wiper arm.
- (4) Remove the wiper arm retaining nut.
- (5) Remove the wiper arm and wiper blade.

B Installation Procedure

- (1) With the switch in the 'park position, fit windscreen wiper arm to splined attachment post with align the wiper blade with the base of the windscreen.
- (2) Fit windscreen washer tube to wiper arm.
- (3) Secure wiper arm and wiper blade with washer and nut.
- (4) Torque load nut 20 to 23Nm (15 to 17ftlb).
- (5) Carry out a functional check of the wiper/washer system and ensure correct operation and park positions.

18 WINDSCREEN WIPER MOTOR

A Removal Procedure

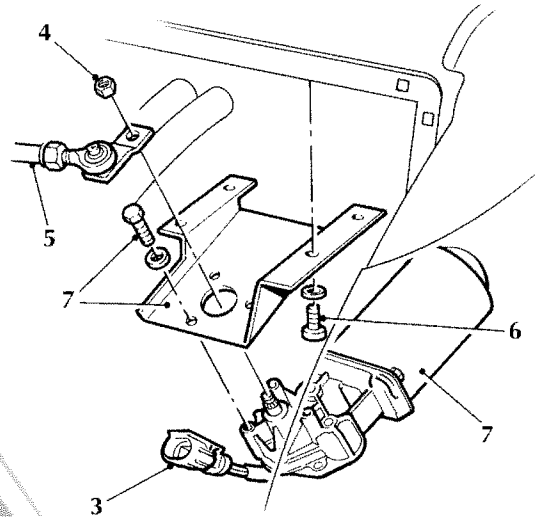
Special Tools

The following special tools will be required

Torque wrench 0 to 65Nm (48ftlb)

NOTES:

- 1 Ensure absolute cleanliness of work area and tools.
- 2 Observe correct use of tools.
- (1) Remove the relevant wheel and wheel arch liner (Refer to Section 4.1 and 2.4) and radiator duct access cover.
- (2) Turn the main isolator switch to the off position.
- (3) Disconnect wiper motor electrical plug.
- (4) Remove the nut securing the wiper arm mechanism to the motor drive shaft.
- (5) Remove wiper arm mechanism from motor.
- (6) Remove the four button head screws attaching the motor bracket to the honeycomb and remove wiper motor and bracket.
- (7) Remove the three screws attaching the motor to the bracket and separate the motor and bracket.



B Installation Procedure

- (1) Fit the wiper motor on the bracket and secure with button head screws.
- (2) Torque load screws 8 to 9.5Nm (6 to 7ftlb).
- (3) Locate the wiper motor assembly and secure with button head screws.
- (4) Torque the screws 8 to 9.5Nm (6 to 7ftlb).
- (5) Fit the wiper mechanism to the wiper motor drive shaft. Check that the wipers are in the 'parked' position.
- (6) Secure the wiper mechanism with nut.
- (7) Torque load nut 20 to 23Nm (15 to 17ftlb).
- (8) Connect wiper motor electrical plug.
- (9) Turn the main isolator switch to the on position

CAUTION:

To prevent the windscreen being scratched do not operate the wipers on a dry windscreen

- (10) Wet the windscreen and switch on the windscreen wipers.
- (11) Ensure wipers have complete sweep of windscreen.
- (12) Fit the radiator duct access cover, wheel arch liner and wheel.

19 WINDSCREEN WIPER MECHANISM

A Removal Procedure

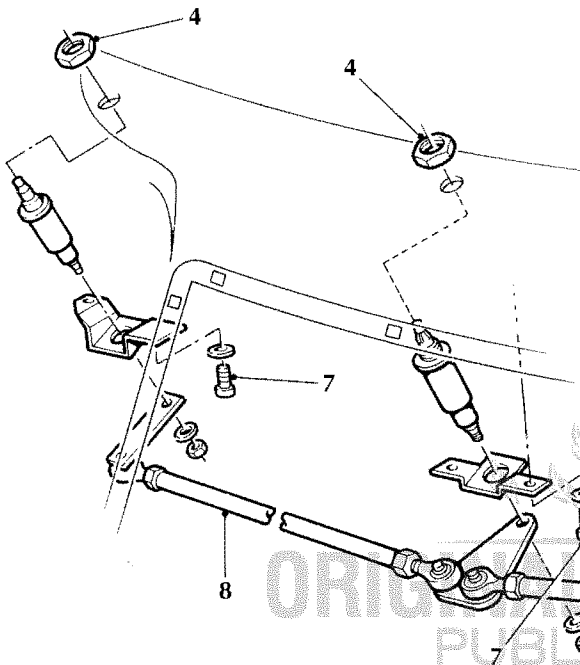
Special Tools

The following special tools will be required

Torque wrench 65Nm (48ftlb)

NOTES:

- 1 Ensure absolute cleanliness of work area and tools.
- 2 Observe correct use of tools.



- (1) Turn the main isolator switch to the off position.
- (2) Remove windscreen wiper arm and wiper blade. (Refer to Section 2.17).
- (3) Remove the HVAC unit (Refer to Section 12).
- (4) Remove the nuts and washers securing the spindles through the vehicle body.
- (5) Remove the front wheels and wheel arch liners. (Refer to Section 4.1 and 2.4).
- (6) Disconnect the wiper motor from the windscreen wiper mechanism (Refer to Section 2.18).
- (7) Remove screws and washers attaching the spindle brackets to vehicle body.
- (8) Remove the windscreen wiper mechanism.

B Installation Procedure

- (1) Fit the mechanism to the vehicle and secure the spindle brackets washers and button head screws.
- (2) Torque the screws 8 to 9.5Nm (6 to 7ftlb).
- (3) Fit the mechanism to the wiper motor and secure the nut.
- (4) Fit the washers and nuts to the spindles and hand torque tighten the nuts.
- (5) Fit wiper arms with wiper blades. (Refer to Section 2.17).
- (6) Move the main isolator switch to the on position.

CAUTION:

To prevent the windscreen being scratched do not operate the wipers on a dry windscreen.

- (7) Wet the windscreen and switch on the windscreen wipers.
- (8) Ensure the wipers have complete sweep of the windscreen and the mechanism has unrestricted movement. Adjust the arm radial position and/or link lengths to achieve correct sweep.
- (9) Fit the wheel arch liners and wheels (Refer to Section 4.1 and 2.4).

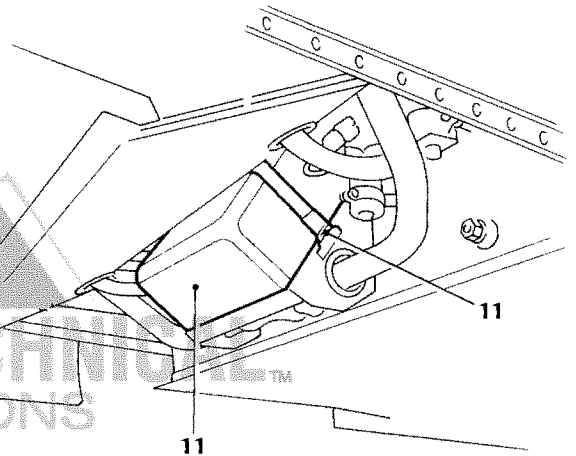
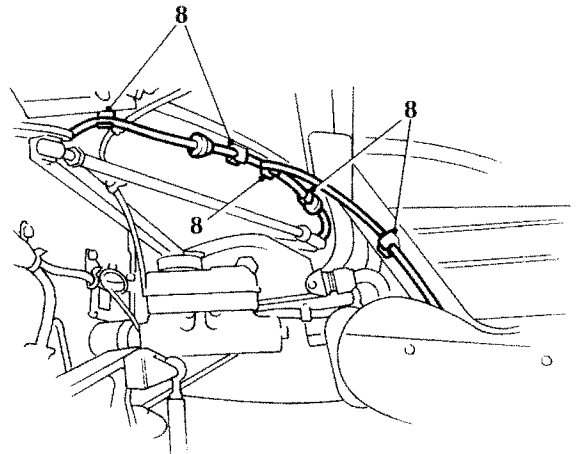
20 SCREENWASH SYSTEM

A Removal Procedure

NOTES:

- 1 Ensure absolute cleanliness of work area and tools.
- 2 Observe correct use of tools.

- (1) Open the bonnet and fit JaguarSport protective wing covers.
- (2) Remove the radiator duct access panel.
- (3) Move the main isolator switch to the off position.
- (4) Drain the washer system.
- (5) Remove the washer jet from the windscreen wiper
- (6) Pull the jet from the washer tubing. To aid refitting of the tubing, tie a sufficient length of string to the tubing ends.
- (7) Feed the washer tubing down through the grommet in the windscreen base.
- (8) Release the tubing from all body clips.
- (9) Remove the front splitter. (Refer to Section 2.3).
- (10) Disconnect the tubing from the washer pump, untie the string, and remove the washer tubing from the vehicle.
- (11) Unhook the washer bottle retaining strap and lift out the bottle.
- (12) Disconnect the washer pump electrical connections.
- (13) Lift the washer pump from the bottle.



B Installation Procedure

- (1) Route the washer tubes from the wiper arms to the water bottle using string ties.
- (2) Push the washer tubing into the retaining clips, and check the clearance to the wiper mechanism.
- (3) Fit the washer jet to the tubing and secure the jet to the wiper arm.
- (4) Fit screenwash pump to water bottle. Ensure the pump is securely located in the grommet.
- (5) Connect washer tube and electrical plug to screenwash pump.
- (6) Fit water bottle in position and retain with rubber strap.
- (7) Fill water bottle and fit filler cap.
- (8) Turn the main isolator switch to the on position.
- (9) Test screenwash system by activating screenwash switch on steering column.
- (10) Fit the radiator duct access panel.
- (11) Remove the protective wing covers and close the bonnet.

21 DOORS

A Removal Procedure

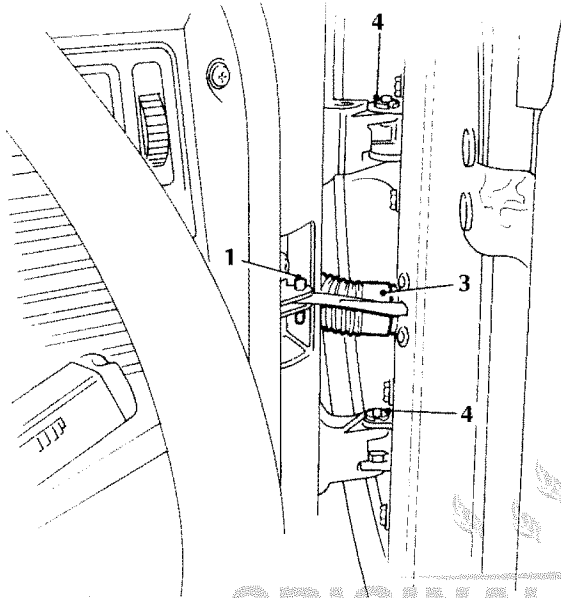
Special Tools

The following special tools will be required.

Jaguar approved work stand for door

NOTES:

- 1 Ensure absolute cleanliness of work area and tools.
- 2 Observe correct use of tools.



- (1) Open the door and remove the check strap pivot pin.
- (2) Turn the main isolator switch to the off position.
- (3) Disconnect the electrical plug.

CAUTION:

Do not loosen or remove the bolts securing the hinges to the vehicle body or the door.

- (4) Remove the 'C' clips from each door hinge pin and remove the door.
- (5) Fit the door to a Jaguar approved work stand, alternatively place door trim side down on a suitable padded work surface to protect the door from damage.

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B Installation Procedure

- (1) Remove door from work stand or work surface.
- (2) Hang the door on the hinge posts.
- (3) Fit new hinge pin spring clips and connect the electrical plug.
- (4) Fit the check strap pivot pin.
- (5) Turn the main isolator switch to the on position.
- (6) Close the door, lock it and check the door is securely locked.
- (7) Check the gap around the door is equal all the way round and that the door is faired with the vehicle body.
- (8) Unlock the door and open it.
- (9) Ensure unrestricted movement of the door through the full arc of travel.
- (10) Ensure the door courtesy lights are operative.
- (11) Close the door.

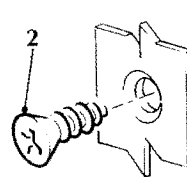
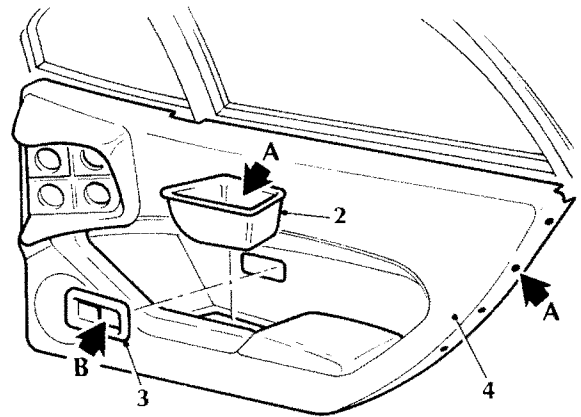
22 DOOR TRIM

A Removal Procedure

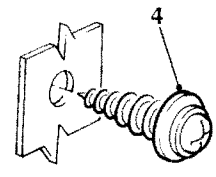
NOTES:

- 1 Ensure absolute cleanliness of work area and tools.
- 2 Observe correct use of tools.

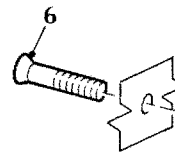
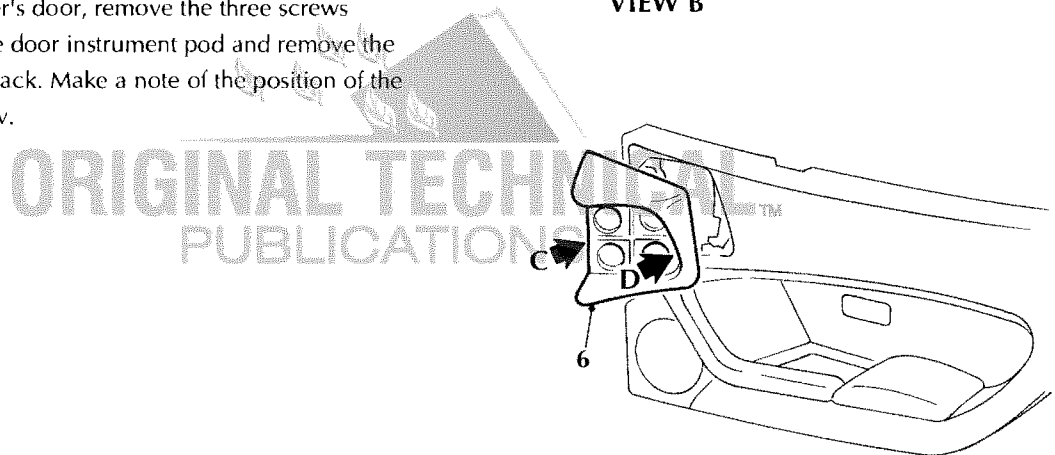
- (1) Open the relevant door and fit JaguarSport protective sill covers, or remove the door and mount on a door stand (Refer to Section 2.21).
- (2) Remove the two self-tapping screws securing the door pocket and remove the pocket.
- (3) Remove the countersunk screw attaching the interior door handle escutcheon and remove the escutcheon.
- (4) Remove the twelve screws attaching the door casing.
- (5) Ease the casing away from the door and disconnect the speaker electrical connections. On the driver's door, disconnect the door pod electrical connectors.
- (6) On the driver's door, remove the three screws attaching the door instrument pod and remove the instrument pack. Make a note of the position of the shorter screw.



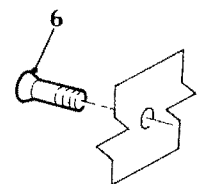
VIEW B



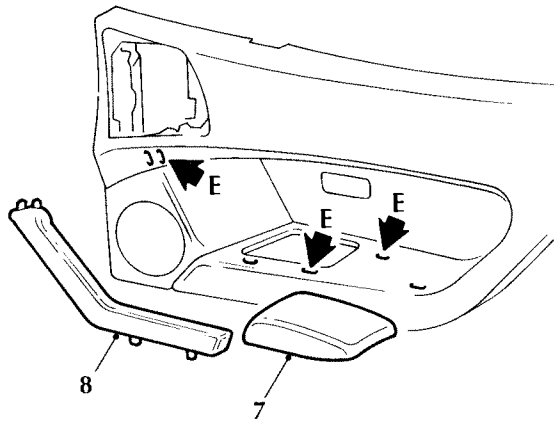
VIEW A



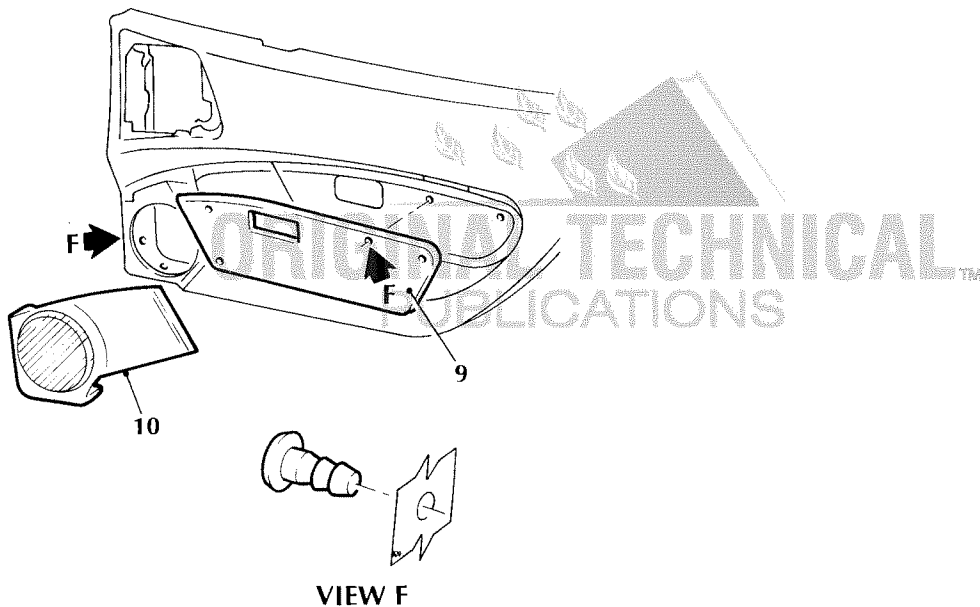
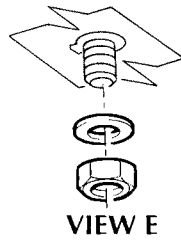
VIEW C



VIEW D



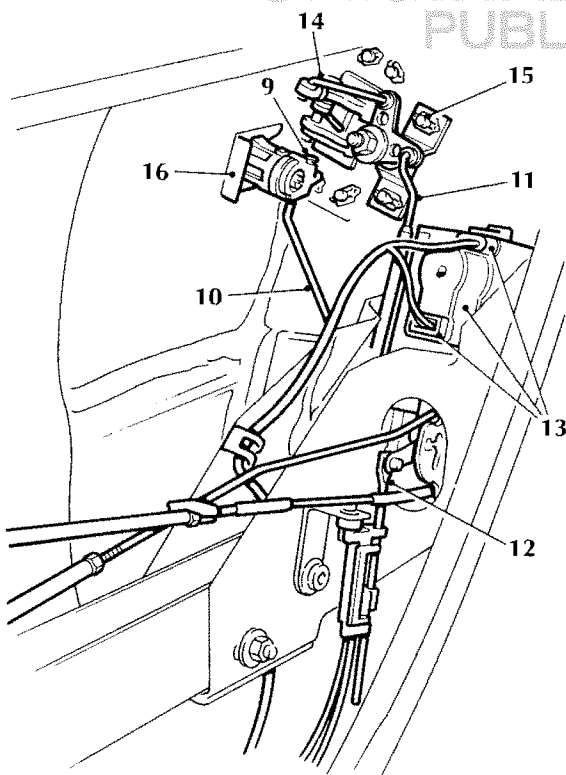
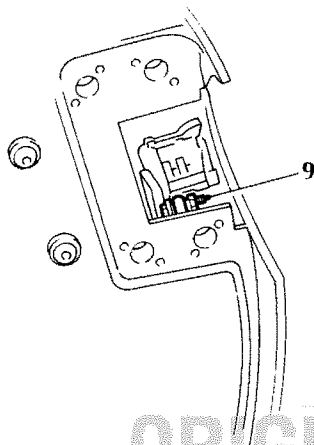
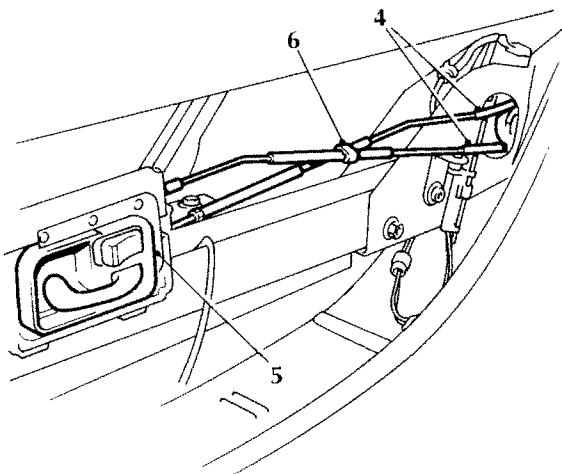
- (7) Remove the four nuts and washers attaching the arm rest to the casing and remove the arm rest.
- (8) Remove the four nuts and washers attaching the door pull and remove from the casing.
- (9) Prise off the trim insert and remove from the casing.
- (10) Remove the fixings and prise off the speaker cover panel and remove from the casing.



B Installation Procedure

NOTES:

- 1 *Ensure absolute cleanliness of work area and tools.*
- 2 *Observe correct use of tools.*
- (1) Fit the speaker cover panel and press in the trim clips.
- (2) Fit the trim insert and press firmly to locate the trim clips.
- (3) Fit the arm rest to the casing and secure with four washers and nuts.
- (4) Fit the door pull to the casing and secure with four washers and nuts.
- (5) On the driver's door, fit the minor instrument pod to the casing and secure with three screws.
- (6) Align the casing with the door and connect the door pod and speaker electrical connectors.
- (7) Secure the casing to the door with twelve screws.
- (8) Fit the door, if removed (Refer to Section 2.21).
- (9) Check the door trim alignment with the front fascia trim and door gap clearance.
- (10) Fit the interior release handle to the casing and secure with a countersunk screw.
- (11) Fit the map pocket to the casing and secure with two self-tapping screws.
- (12) Remove the protective sill covers, if fitted, and close the door.



23 DOOR LOCK MECHANISM

A Removal Procedure

NOTES:

- 1 Ensure absolute cleanliness of work area and tools.
- 2 Observe correct use of tools.

- (1) Remove the door. (Refer to Section 2.21, Doors).
- (2) Remove the door trim. Refer to Section 2.22, Door Trim).
- (3) Peel the water shedder away from the adhesive tape on the underside of the lower door beam.
- (4) Remove the interior door handle rods and clips from the latch.
- (5) Remove the screw attaching the interior door handle to the structure.
- (6) Remove the anti-rattle clip from the handle rods and remove the interior handle complete with rods.
- (7) Remove the window cassette (Refer to Section 2, 24).
- (8) Remove the four bolts and packers securing the exterior door handle to the door.
- (9) Remove the nut securing key cylinder rod to the latch.
- (10) Remove the clip and key cylinder rod connection to the latch.
- (11) Remove the clip and rod from the bellcrank to the latch.
- (12) Unclip the rod from the lock reminder switch to the latch.
- (13) Remove the three torx head screws attaching the latch to the door, disconnect the electrical plug and remove the latch.
- (14) Unclip the rod from the bellcrank to the exterior handle.
- (15) Remove the two screws attaching the bell crank and withdraw the bell crank.
- (16) Remove the retainer securing the lock barrel/key cylinder, remove the key cylinder and collect the pad.

B Installation Procedure

- (1) Fit the door latch and secure with three screws. Torque the screws 8 to 9.5Nm (6 to 7ftlb).
- (2) Fit the bellcrank assembly to the door and secure with two screws. Torque the screws 4 to 5.5Nm (3 to 4ftlb).
- (3) Fit the lock barrel and pad to the door and secure with the retaining clip.
- (4) Fit the rod to the lock barrel with ball joint assembly and nut and clip to the latch mechanism.
- (5) Fit the exterior door release handle and packers to the door, secure with four screws. Torque screws by hand as required.
- (6) Fit the rod and clip from the exterior door release to the bellcrank assembly.
- (7) Fit the rod and clips from the bellcrank assembly to the latch mechanism.
- (8) Fit the rod and clip from the lock reminder switch to the latch lever.
- (9) Fit the interior door release handle, rods and clips.
- (10) Fit the interior door release rod and clip to the latch mechanism.
- (11) Fit the interior lock rod and clip to the latch. Attach the anti-rattle clip. Secure the door handle to the door with a self-tapping screw.
- (12) Carry out an interior door handle latch actuation check as follows;
 - (a) Check that the link rod is securely attached to the interior door handle and latch ends.
 - (b) Close the latch pawl using a suitable screw driver.
 - (c) Open the interior door release handle and check that the latch pawl opens at mid-point of door handle opening travel.
 - (d) Adjust the link rod lengths to achieve latch pawl opening at mid-point of door handle travel (if required) and re-tighten the locknuts.
- (13) Carry out an exterior door handle/latch actuation check as follows;
 - (a) Check the link rods are securely attached to the exterior handle, bell crank and latch.
 - (b) Close the latch pawl using a suitable screw driver.
 - (c) Open the exterior door release handle and check that the latch pawl opens at mid-point of door handle opening travel.
 - (d) Adjust the link rod length to achieve latch pawl opening at mid-point of door handle travel (if required) and re-tighten the locknuts.
- (14) Carry out door key/latch actuation as follows;
 - (a) Check that the link rod is securely attached to the key cylinder lever and latch ends.
 - (b) Close the latch pawl using a suitable screw driver.
 - (c) Insert the key into the door lock barrel and rotate the key (anti-clockwise direction on RHS door and clockwise direction on LHS door) to latch the lock.
 - (d) Adjust the link rod lengths to achieve latch locking at mid-point of key travel (if required) and re-tighten the locknuts.
 - (e) Operate the exterior and interior door handles and check that the latch pawl does not open.
 - (f) Rotate the door lock barrel key (clockwise direction on RHS door and anti-clockwise direction on LHS door) to unlock the latch.
 - (g) Operate the interior door handle and check that the latch pawl releases.
 - (h) Reset the latch pawl to the closed position, operate the exterior door handle and check that the latch pawl releases.
- (15) Carry out an interior door lock knob/latch actuation check as follows;
 - (a) Check that the link rod is securely attached to the lock knob and latch ends.
 - (b) Adjust the rod length to achieve latch locking at mid-point of lock knob travel and re-tighten the locknuts.
 - (c) Close the latch pawl.
 - (d) Rotate the lock knob towards the front of the vehicle to the lock position (orange flash visible).

- (e) Operate the interior and exterior door handles separately, resetting the latch pawl to the closed position each time. Check that the latch pawl does not open.
 - (f) Rotate the lock knob towards the rear of the vehicle to the unlock position (orange flash not visible).
 - (g) Operate the interior and exterior door handles separately, resetting the latch pawl each time. Check that the latch pawl opens.
- (16) Initially assemble the window cassette to the door (Refer to Section 2.24).

NOTE

Do not tighten the bolts at this stage

- (17) Fit the door to the vehicle. (Refer to Section 2.21).
- (18) Finally fit the window cassette (Refer to Section 2.24).
- (19) Connect the electrical connections to the latch mechanism and window lift motor.
- (20) Carry out Vehicle Electrical Test System (Refer to Section 14).
- (21) Connect the battery and ensure the window has full and unrestricted movement throughout.
- (22) Re-adhere water shedder to the underside of the lower door beam.
- (23) Fit the door trim. (Refer to Section 2.22).

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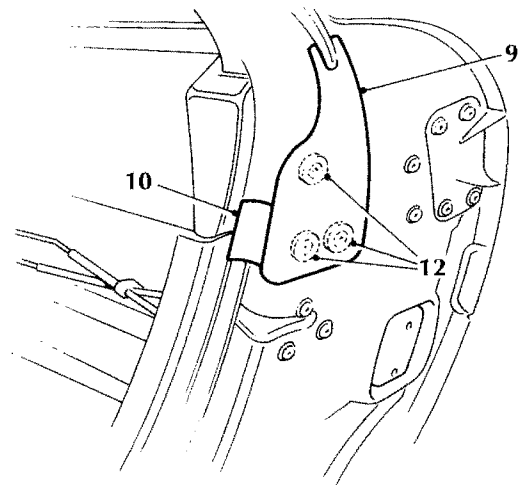
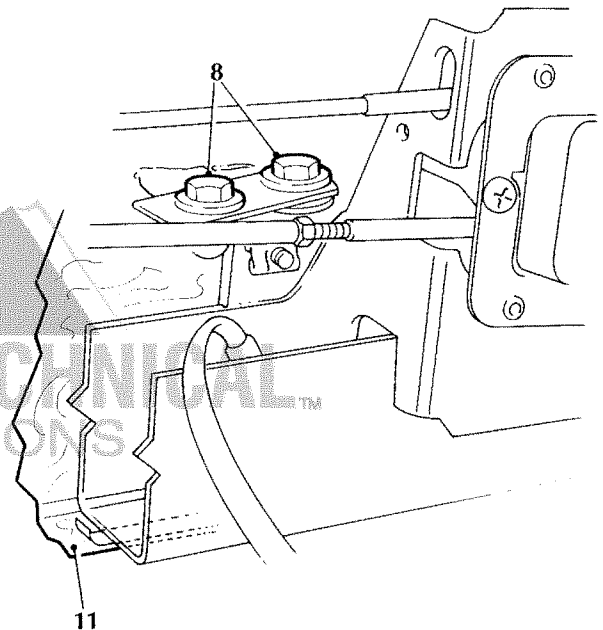
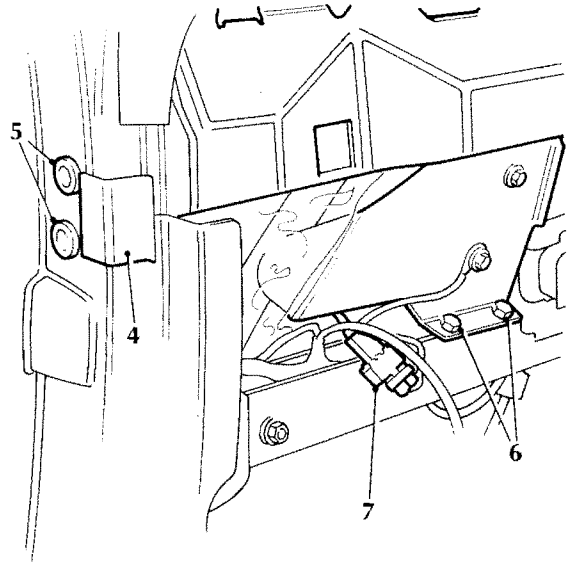
24 DOOR CASSETTE AND WINDOW MECHANISM

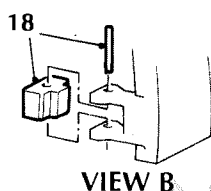
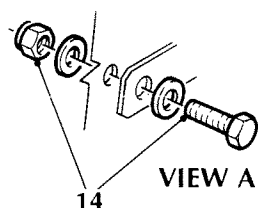
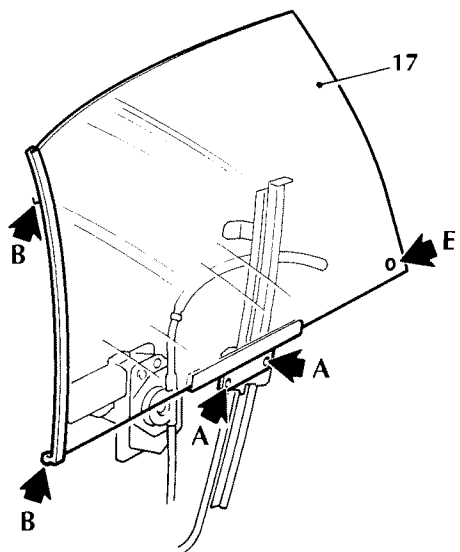
A Removal Procedure

NOTES:

- 1 Ensure absolute cleanliness of work area and tools.
- 2 Observe correct use of tools.

- (1) Remove the door (Refer to Section 2.21).
- (2) Remove the door trim (Refer to Section 2.22).
- (3) Measure the window glass to door gap and record the measurement
- (4) Remove the front door/cassette joint patch.
- (5) Remove the two screws attaching the window cassette to the front of the door.
- (6) Remove the two screws attaching the cassette to the lower door beam.
- (7) Disconnect the lift motor electrical connector.
- (8) Remove the screws securing the window lift mechanism bridging plate and remove the plate.
- (9) Remove the rear glass capping.
- (10) Remove the rear door/cassette joint patch.
- (11) Peel the water shedder away from the adhesive tape on the underside of the lower door beam.
- (12) Remove the three countersunk screws attaching the cassette to the rear of the door and collect the fixings and washers.
- (13) Lift the window cassette assembly from the door.

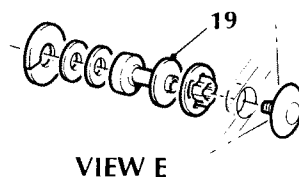
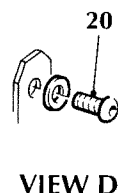
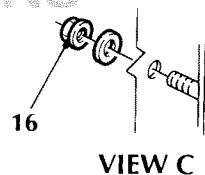
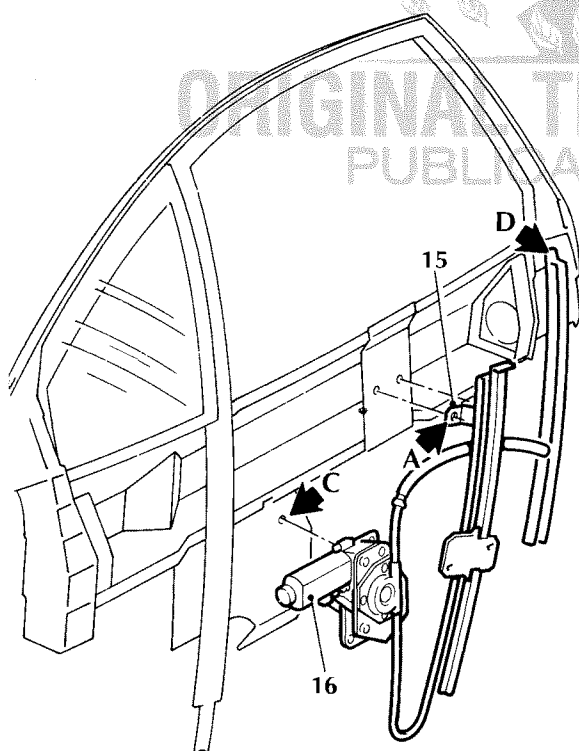




- (14) Remove the two bolts, washers and nuts attaching the window to the mechanism and remove the window glass from the frame.
- (15) Remove the bolts attaching the winder mechanism to the upper mounting plate.
- (16) Remove the two nuts and washers attaching the motor to the frame plate and remove the winder mechanism.
- (17) Slide the drop glass, complete with front retainer out of the door assembly.
- (18) Remove the two spring clips and bearings from the front retainer and remove the retainer from the drop glass.
- (19) Remove the nylon washer, rubber washers, pin collar and cap from the drop glass.
- (20) Remove the two screws and washers from the rear glass retainer and remove the retainer.

NOTE:

Check position and size of any shims fitted to the window mechanism fixings.



B Installation Procedure

- (1) Fit the rear glass retainer to the door and secure with two washers and screws.
- (2) Assemble the collar, pin, rubber washers and nylon washers using Loctite 648. Insert the sub-assembly through the glass and secure using cap.
- (3) Fit the front glass retainer to the glass. Fit the two bearings and secure with spring pins.
- (4) Slide the drop glass assembly into the relevant channels in the door frame.
- (5) Fit the winder mechanism to the upper mounting plate and secure with two screws.
- (6) Fit the two screws attaching the winder mechanism motor to the frame.
- (7) Fit the two bolts, washers, shims and nuts attaching the window glass to the mechanism.
- (8) Fit the window cassette to the door.

NOTE

Do not tighten the screws at this stage.

- (9) Re-adhere the water shedder to the underside of the lower door beam.
- (10) Fit the door to the vehicle.
- (11) Set the glass to door gap/height (using adjustment screw), as measured during removal, and torque tighten the cassette mounting screws 6.8 to 8.8Nm (5 to 6.5ftlb).
- (12) Connect the lift motor electrical connectors.
- (13) Connect the battery and ensure the window has full and unrestricted movement throughout.
- (14) Check the door closing operation and adjust the striker pin on 'B' post as required to achieve correct closing operation.
- (15) Fit the door/cassette joint patches to the front and rear of the door, plus a new rear glass capping.
- (16) Fit the door trim. (Refer to Section 2.22).

25 CARPETS

A Removal Procedure

NOTES:

- 1 Ensure absolute cleanliness of work area and tools.
- 2 Observe correct use of tools.
- (1) Remove the front carpets from the floor mounted clips.
- (2) Remove the seats (Refer to Section 2.30).
- (3) Remove the tunnel saddle trim (Refer to Section 2.31).
- (4) Remove the roll top trim panel (Refer to Section 2.27).
- (5) Pull the carpets from the floor and rear bulkhead.

NOTE

The carpets are glued to the floor and tunnel. It may be necessary to use a solvent to aid the carpet removal.

WARNING:

SOLVENTS ARE TOXIC AND MUST ONLY BE USED IN WELL VENTILATED WORK AREAS. AVOID INHALING THE SOLVENT FUMES AND WEAR PROTECTIVE CLOTHING.

- (6) Pull the carpet from the tunnel. Ensure the sound insulation is not damaged during carpet removal.
- (7) Remove all traces of adhesive from the floor, bulkhead and tunnel areas with a proprietary cleaner.

B Installation Procedure

- (1) Degrease the floor and tunnel areas.
- (2) Apply adhesive to the tunnel carpet and tunnel insulation mating surfaces. Allow the adhesive to dry to the touch.

NOTE

The adhesive used is a contact type. Ensure the carpets are placed correctly at the first attempt.

- (3) Fit the carpets to the tunnel and press firmly to ensure adhesion
- (4) Apply adhesive to the rear carpets and floor/bulkhead mating surfaces. Allow the adhesive to dry to the touch.

NOTE

The adhesive used is a contact type. Ensure the carpets are placed correctly at the first attempt.

- (5) Fit the carpets to the floor and bulkhead and press firmly to ensure adhesion.
- (6) Fit the carpets to the front floor area and secure the clips.
- (7) Fit the roll top trim panel. (Refer to Section 2.27).
- (8) Fit the tunnel saddle trim. (Refer to Section 2.31).
- (9) Fit the seats (Refer to Section 2.30).

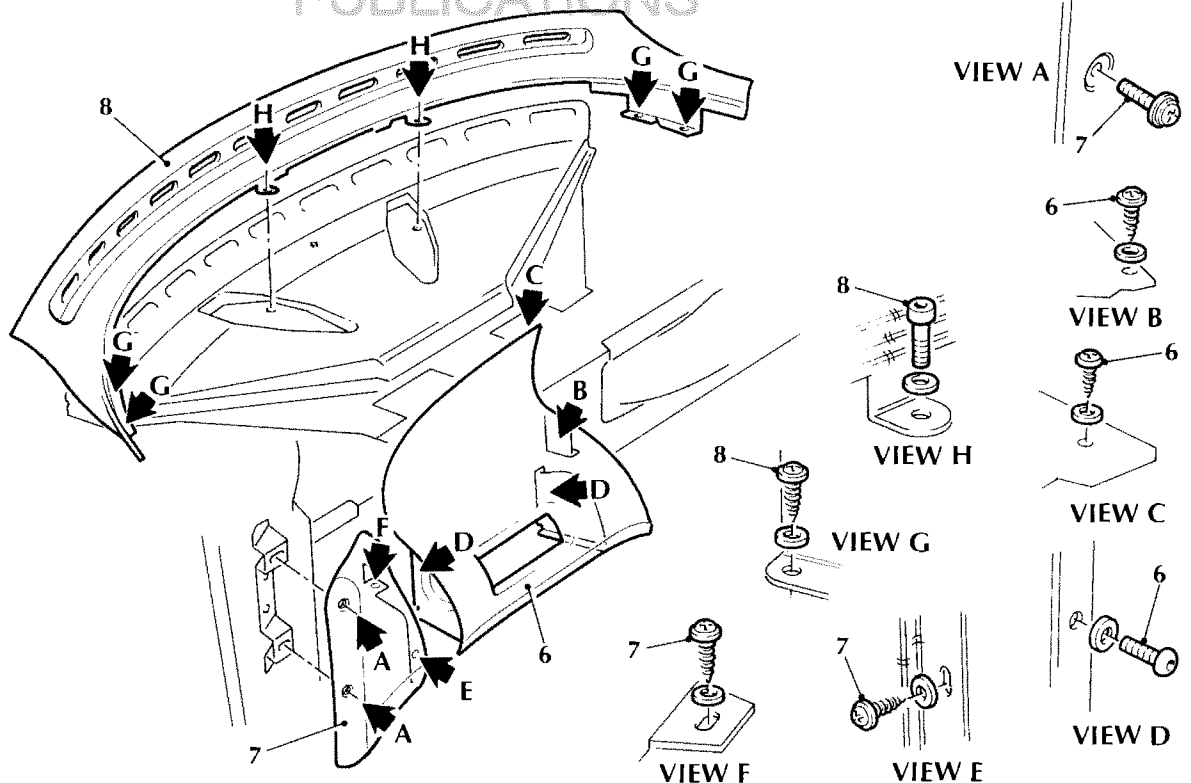
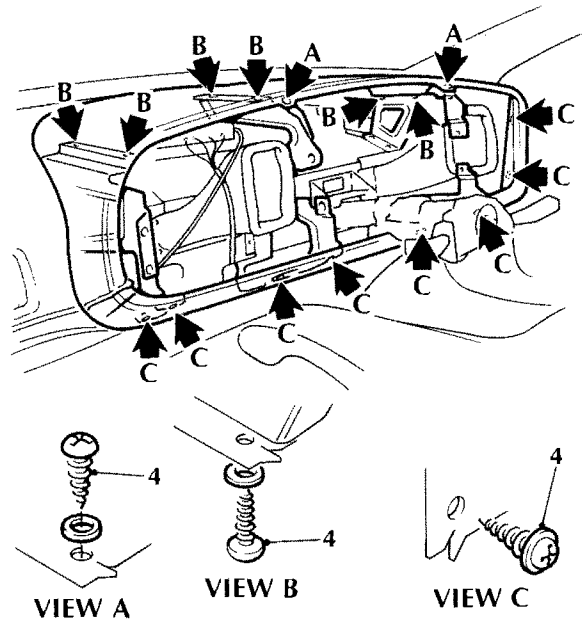
26 FRONT TRIM

A Removal Procedure

NOTES:

- 1 Ensure absolute cleanliness of work area and tools.
- 2 Observe correct use of tools.

- (1) Remove the facia panel assembly (Refer to Section 13).
- (2) Remove the driver and passenger underscuttle and disconnect the footwell lamp connectors.
- (3) Remove the steering column switch cover. (Refer to Section 11).
- (4) Remove the screws attaching the binnacle cover to the brackets and remove the cover.
- (5) Remove the passenger air vent.
- (6) Remove the screws attaching the passenger vent panel to the honeycomb structure.
- (7) Remove the end cap screws and remove the end cap.
- (8) Remove the cap head and self-tapping screws and washers attaching the windscreen base panel.
- (9) Remove the radio amplifier. (Refer to Section 14).
- (10) Remove the screws and washers securing the dash surround brackets to the honeycomb structure and remove the brackets.
- (11) Remove the defrost spyder (Refer to Section 12).



B Installation Procedure

- (1) Fit the air defrost spyder (Refer to Section 12).
- (2) Fit the dash surround brackets and secure to the structure with washers and screws. Tighten the screws to 4 to 5Nm (3 to 4ftlb).
- (3) Fit the radio amplifier. (Refer to Section 14).
- (4) Fit the windscreen base panel to the plenum and secure with cap head and self-tapping screws and washers.
- (5) Torque the cap head screws 5.5 to 7Nm (4 to 5ftlb).
- (6) Fit the end cap trimmed panel and secure with self-tapping screws and washers.
- (7) Fit the passenger vent panel and secure with button head screws, self-tapping screws, washers and 'U' spring nuts.
- (8) Torque the passenger vent panel button head screws 4 to 5.5Nm (3 to 4ftlb).
- (9) Fit the passenger air vent.
- (10) Locate the binnacle cover on the brackets and secure with button head screws, self-tapping screws and washers.
- (11) Torque the button head screws 7.5 to 9Nm (5.5 to 6.5ftlb).
- (12) Fit the steering column cover and secure the four screws.
- (13) Fit the driver and passenger underscuttle and connect the footwell lamp connectors.
- (14) Fit the facia panel assembly. (Refer to Section 13).

27 REAR TRIM

A Removal Procedure

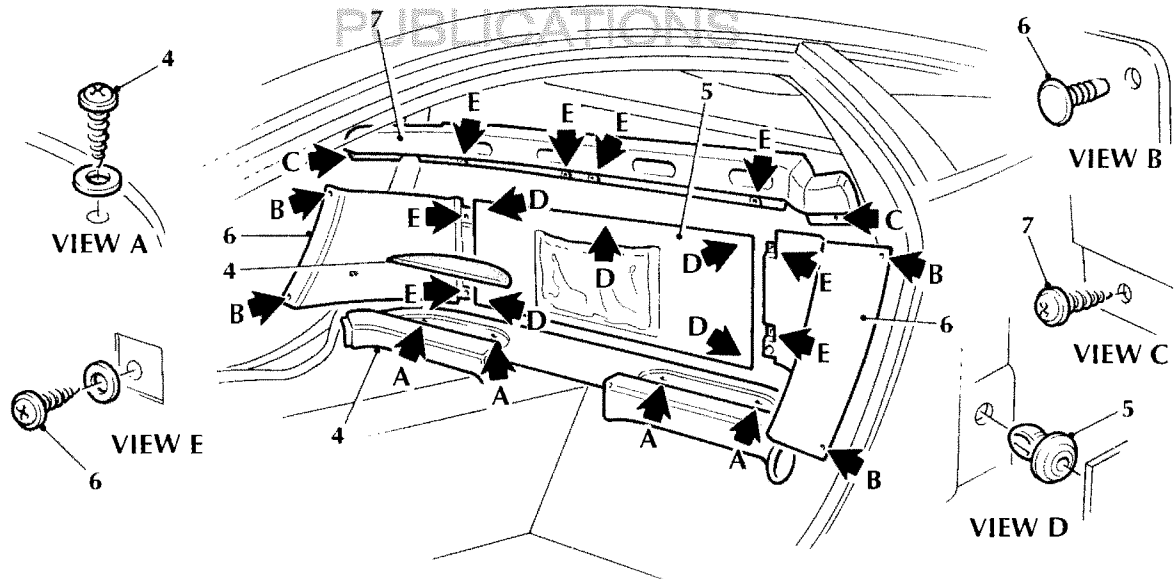
NOTES:

- 1 Ensure absolute cleanliness of work area and tools.
- 2 Observe correct use of tools.

- (1) Remove the seats (Refer to Section 2.30).
- (2) Remove the tunnel trim (Refer to Section 2.31).
- (3) Remove the ECU cover (RHD)/map pocket panel (LHD). (Refer to Section 2.29)
- (4) Remove the mats covering the screws securing the rear stowage tray, remove the screws and remove the tray.
- (5) Insert a screw driver under the backboard panel next to the trim clips and lever off the panel.
- (6) Remove the two self-tapping screws attaching the rear quarter panels; insert a screw driver under the panel next to the trim clips and lever off the panels.
- (7) Remove the screws attaching the backboard roll-top to the seat belt beam, feed the seat belts through the roll-top and remove the panel and sound insulation.

B Installation Procedure

- (1) Feed the seat belt lower attachment through the roll-top and secure the roll-top to the seat belt beam with self-tapping screws and washers.
- (2) Fit the rear quarter panels and secure with self-tapping screws and trim clips.
- (3) Align the backboard trim clips with the rear bulkhead and press the back board to fit.
- (4) Fit the rear stowage tray and secure with screws.
- (5) Re-adhere the stowage tray mats to stowage tray.
- (6) Fit the ECU cover (RHD)/map pocket panel (LHD). (Refer to Section 2.29).
- (7) Fit the tunnel saddle trim (Refer to Section 2.31).
- (8) Fit the seats (Refer to Section 2.30).



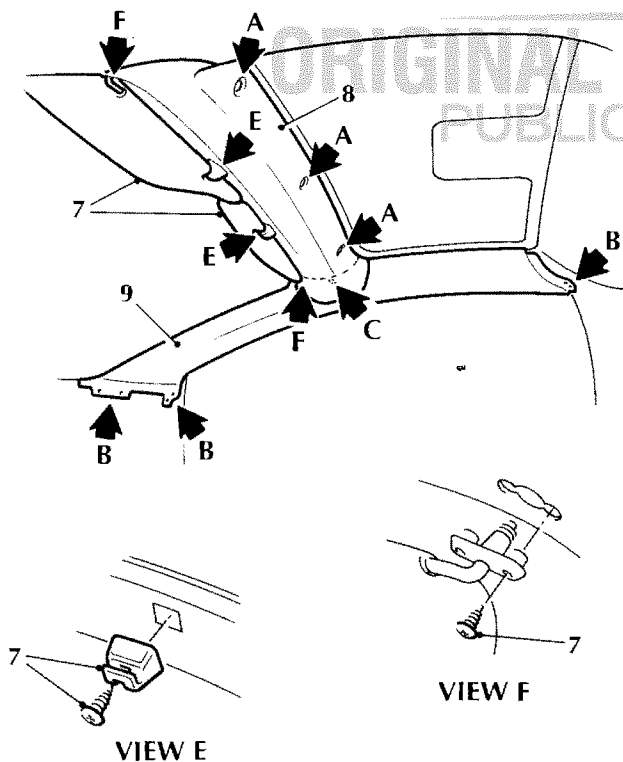
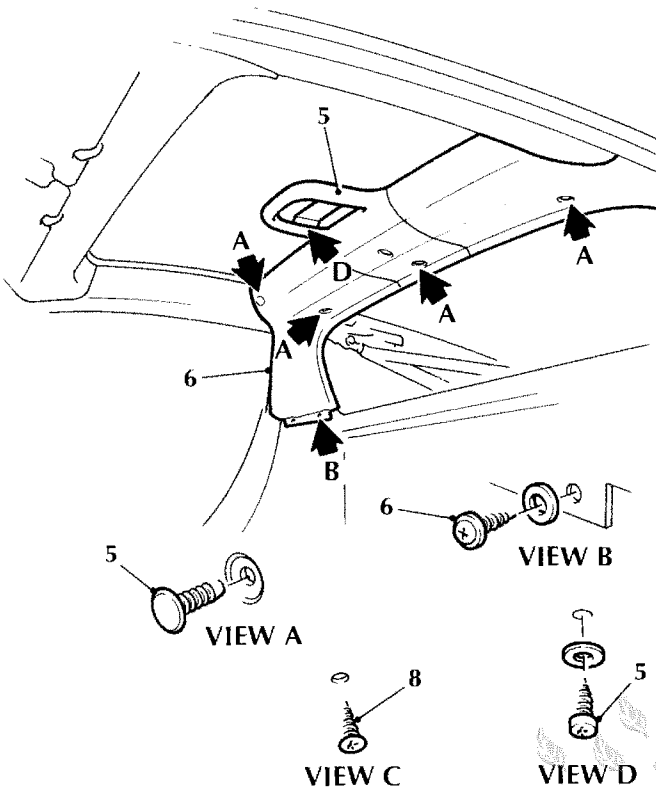
28 ROOF TRIM

A Removal Procedure

NOTES:

- 1 Ensure absolute cleanliness of work area and tools.
- 2 Observe correct use of tools.

- (1) Remove the rear trim (Refer to Section 2.27).
- (2) Remove the front trim (Refer to Section 2.26).
- (3) Remove the interior lamp. (Refer to Section 13).
- (4) Remove the interior lamp mounting bracket screws.
- (5) Insert a screw driver under rear header panel next to the trim clips and lever off the panel.
- (6) Remove the self-tapping screws from the 'B' post covers, insert a screw driver under the panel next to the trim clips and lever off the panel.
- (7) Remove the screws securing the sun visor stowage and mounting brackets and remove the sun visors.
- (8) Remove the front header trim clips and remove the header.
- (9) Remove the self-tapping screws attaching the 'A' post covers and remove the cover.



B Installation Procedure

- (1) Locate the 'A' post covers and secure using self-tapping screws.
- (2) Locate the front header and fit the trim clips.
- (3) Fit the sun visors and secure the mounting and stowage brackets.
- (4) Fit the 'B' post covers and secure with self-tapping screws and trim clips.
- (5) Fit the rear header cover panel and secure with trim clips.
- (6) Fit the interior lamp. (Refer to Section 13).
- (7) Fit the front trim. (Refer to Section 2.26).
- (8) Fit the rear trim. (Refer to Section 2.27).

29 SIDE TRIM

A Removal Procedure

NOTES:

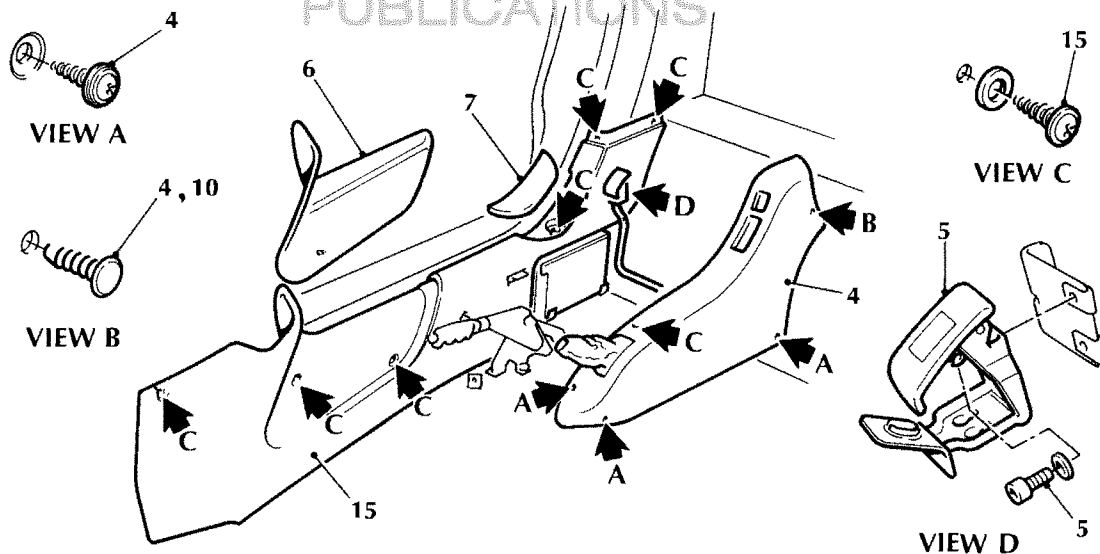
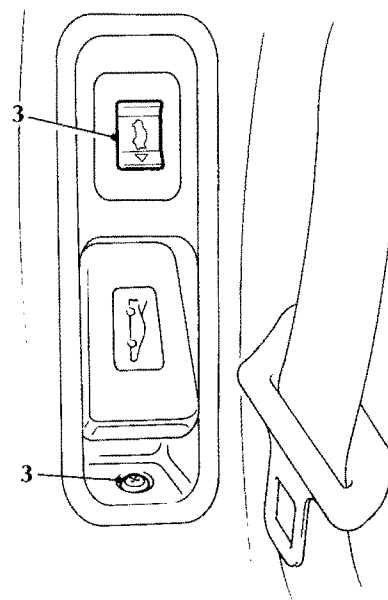
- 1 Ensure absolute cleanliness of work area and tools.
- 2 Observe correct use of tools.

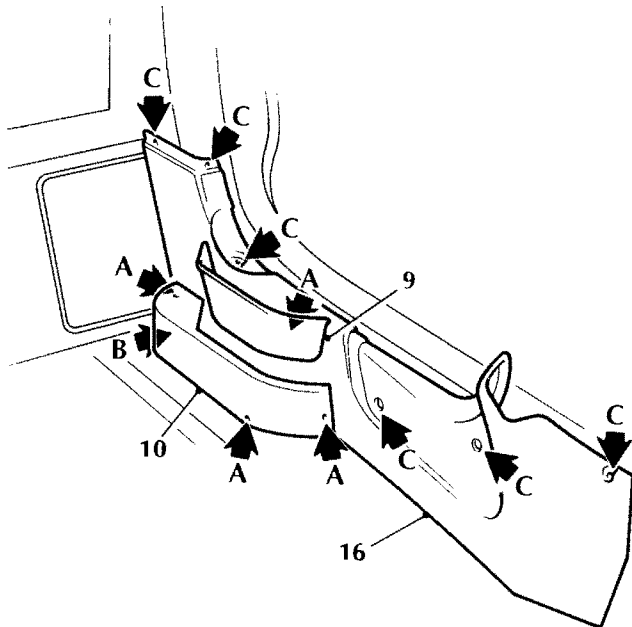
- (1) Remove the seats to gain greater access. (Refer to Section 2.30).
- (2) Turn the main isolator switch to the off position.
- (3) Remove the boot release lever escutcheon and disconnect the boot release electrical connector.
- (4) Remove the screws attaching the ECU cover (RHD) or map pocket panel (LHD) and remove the cover.

NOTE:

To ease removal of the cover, raise the hand brake lever.

- (5) Remove the two screws attaching the tailgate release handle to the bracket and remove the handle.
- (6) Remove the kick panels by inserting a screw driver under the panel and releasing the velcro fastening.
- (7) Remove the hand pads by inserting a screw driver under the pads and releasing the velcro fastening.
- (8) Remove the rear stowage tray and rear quarter panels. (Refer to Section 2.27)



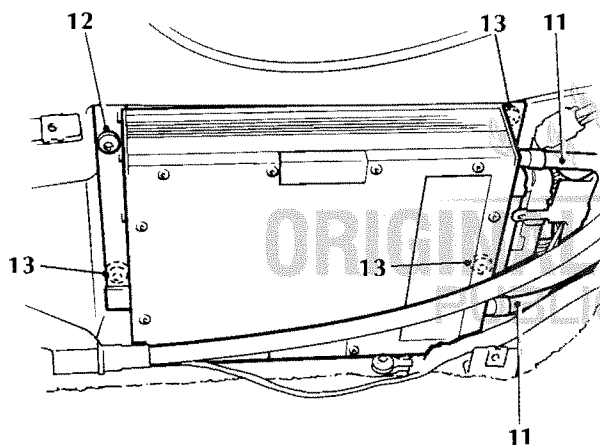


- (9) On RHD vehicles LH side, remove the two screws attaching the map pocket and lift out the pocket.
- (10) On RHD vehicles LH side, remove the self-tapping screws securing the map pocket panel and remove the panel.
- (10a) On LHD vehicles LH side, remove the handbrake cover.
- (10b) On LHD vehicles RH side, remove the ECU cover insert and remove the ECU cover.
- (11) Remove the two boost sensor hoses from the ECU.

NOTE:

Make a note of the fitment of the sensor hoses.

- (12) Remove the button head screw attaching the ECU to the sill panel.
- (13) Remove the three screws attaching the ECU brackets to the sill panel and remove the brackets.
- (14) Remove the screws securing fuse box 'C' to the sill panel.
- (15) Remove the self-tapping screws attaching the RH sill panel and remove the panel.
- (16) Remove the self-tapping screws attaching the LH sill panel.



B Installation Procedure

NOTES:

- 1 Ensure absolute cleanliness of work area and tools.
- 2 Observe correct use of tools.

- (1) Fit the sill panels and secure with self-tapping screws.

NOTE

On the RH sill panel fit the ECU hoses, tailgate release bracket and handbrake cable through the aperture at the rear of the panel.

- (2) On RHD vehicles LH side, fit the map pocket panel and secure with self-tapping screws.
- (3) Fit the map pocket to the panel and secure with self-tapping screws.
- (3a) On LHD vehicles LH side, fit the handbrake cover.
- (4) Fit the hand pads to the sill panels and press the velcro strips together.
- (5) Fit the kick panels to the sill panels and press the velcro strips together.
- (6) Fit the ECU brackets to the sill panel and secure using button head screws.
- (7) Fit the ECU to the bracket and secure with a button head screw.

NOTE

Ensure the ECU is securely located at the bottom

- (8) Fit the boost sensor hoses to the ECU.
- (9) Fit and secure fuse box 'C' and relay module to the sill panel bracket.
- (10) Fit the tailgate release handle to the bracket and secure using two screws.
- (11) Fit the ECU cover (and map pocket panel, LHD) to RH side and secure using self-tapping screws.
- (12) Connect the boot release electrical connectors and fit the boot/tailgate release lever escutcheon.
- (13) Fit the parcel shelf and rear quarter panels and secure with self-tapping screws.
- (14) Move the main isolator switch to the on position and check the operation of the boot release.
- (15) Fit the seats (Refer to Section 2.30).

30 SEATS AND SEAT SLIDES

A Removal Procedure

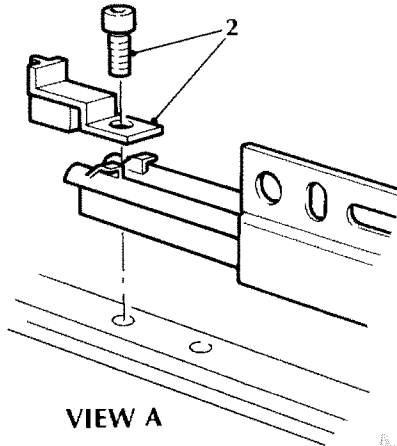
Special Tools

The following special tools will be required.

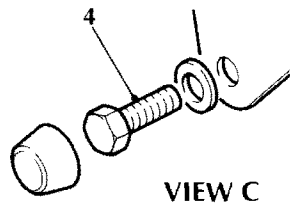
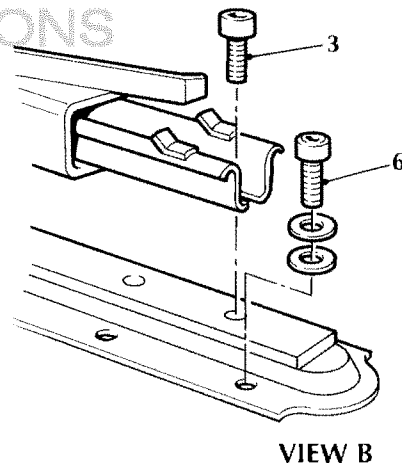
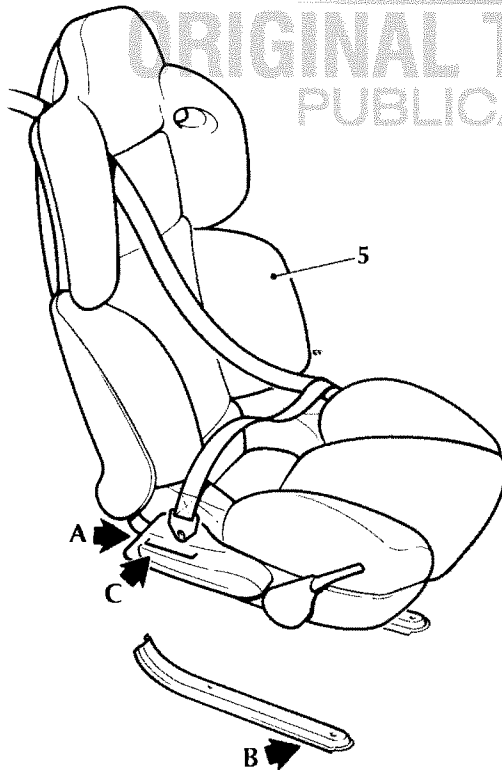
Torque wrench (25ftlb)

NOTES:

- 1 Ensure absolute cleanliness of work area and tools.
- 2 Observe correct use of tools.

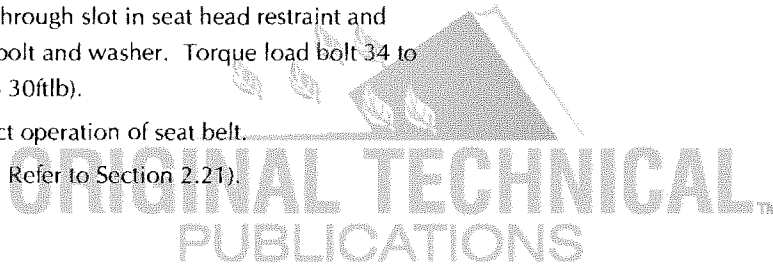


- (1) Remove the door (Refer to Section 2.21).
- (2) Slide the seat forward and remove the screw attaching the seat stop.
- (3) Remove the cap head screws securing the seat rail to seat rail mountings.
- (4) Remove bolt and washer attaching the seat belt to the seat. Feed the belt through the seat head restraint aperture.
- (5) Move seat rearward and up to remove, and place on a clean surface padded to protect the seat rails from damage.
- (6) Remove the cap head screws and washers securing the seat rail assembly (if required).



B Installation Procedure

- (1) Fit the seat rail assembly (if removed).
- (2) Secure seat rail assembly with cap head screws (if removed).
- (3) Torque load cap head screws 8.14 to 9.5Nm (6 to 7ftlb).
- (4) Position seat at the rear end of the seat rail assembly.
- (5) Move the seat slides to the rearmost seat rail assembly fixing.
- (6) Fit the rear seat slide fixings and adjustment stops.
- (7) Slide the seat rearwards and fit the front fixings.
- (8) Torque load cap head screws 20 to 23Nm (15 to 17ftlb).
- (9) Operate the seat adjustment lever to ensure the seat travels forward and rearward without restriction and that the seat position locks when the lever is released..
- (10) Fit seat belt through slot in seat head restraint and secure with bolt and washer. Torque load bolt 34 to 40Nm (25 to 30ftlb).
- (11) Check correct operation of seat belt.
- (12) Fit the door (Refer to Section 2.21).

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31 TUNNEL TRIM

A Removal Procedure

Special Tools

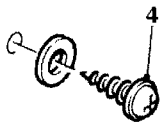
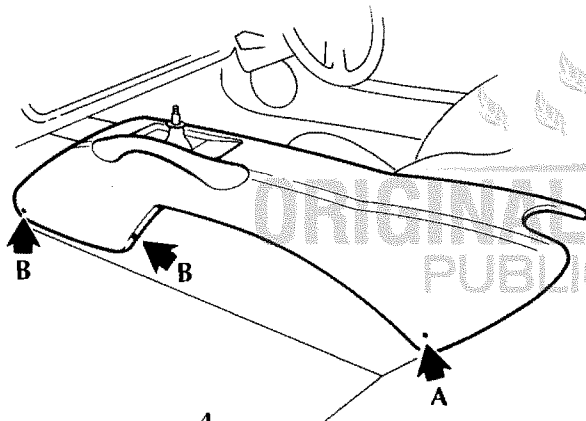
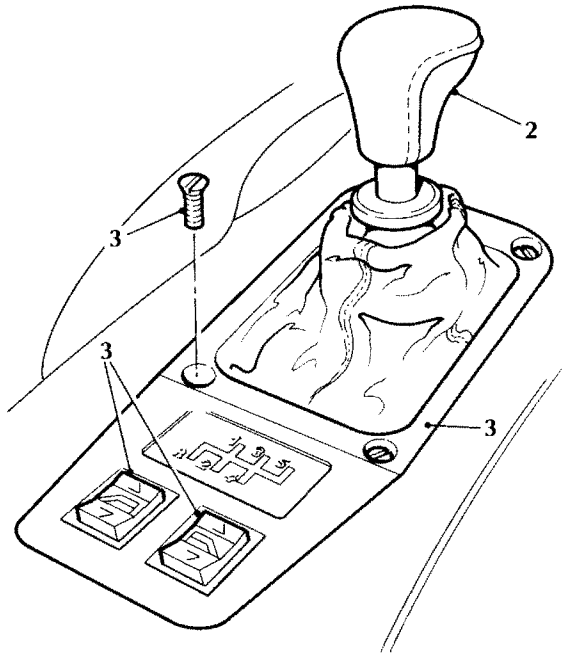
The following special tools will be required.

Torque wrench 7Nm (5ftlb)

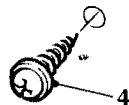
NOTES:

- 1 Ensure absolute cleanliness of work area and tools.
- 2 Observe correct use of tools.

- (1) Remove seats (Refer to Section 2.30).
- (2) Remove the gear knob.
- (3) Remove gear shift surround and disconnect the window switches electrical plugs.
- (4) Remove the six screws securing the tunnel saddle trim and place it on a suitable clean work surface to prevent damage.



VIEW A



VIEW B

B Installation Procedure

- (1) Fit tunnel saddle trim and secure with screws.
- (2) Torque load screws 5.5 to 7Nm (4 to 5ftlb).
- (3) Fit the gear shift surround and connect electrical plugs for window up and down operation.
- (4) Secure the gear shift surround to the tunnel with screws.
- (5) Fit the gear shift knob.
- (6) Fit the seats. (Refer to Section 2.30)
- (7) Switch ignition ON and test window up and down operation.

NOTE:

If window up and down operation is not satisfactory refer to the relevant wiring diagram in the Electrical Systems, Section 14.

32 TUNNEL TOP

A Removal Procedure

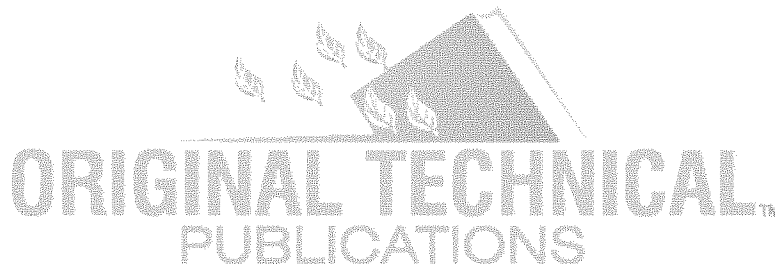
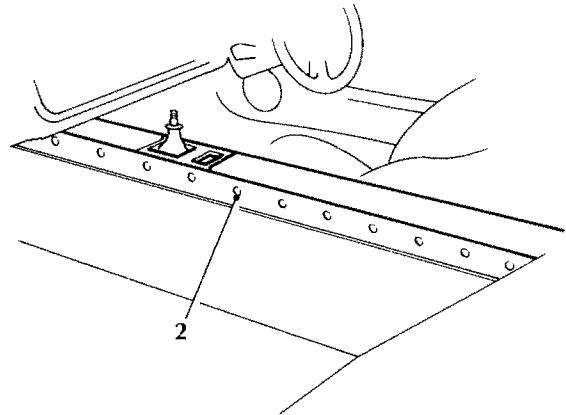
NOTES:

- 1 Ensure absolute cleanliness of work area and tools.
- 2 Observe correct use of tools.
- (1) Lift the carpet away from sides of tunnel.
- (2) Remove the screws securing the tunnel top.

CAUTION:

The tunnel top is secured by a further two screws at the front of the tunnel top. These are inaccessible with the underscuttles fitted. If only minimal access to the tunnel top is required the tunnel top may be gently raised and propped open by suitable means. If the tunnel top is to be removed the underscuttle fittings must be removed.

- (3) Remove the driver and passenger underscuttles. (Refer to Section 2.26, Front Trim).
- (4) Remove two screws at front of tunnel top and remove tunnel top.



B Installation Procedure

- (1) Ensure no tools or other loose articles are in the tunnel.
- (2) Fit tunnel top and fit two screws securing front of tunnel top.
- (3) Torque load screws 5.5 to 7Nm (4 to 5ftlb).
- (4) Fit the driver and passenger underscuttle. (Refer to Section 2.26, Front Trim).

NOTE:

Cover the opening around the gear stick with a suitable material to prevent loose articles falling into the tunnel.

- (5) Fit screws securing tunnel top to sides of tunnel.
- (6) Torque load screws 5.5 to 7Nm (4 to 5ftlb).
- (7) Re-adhere the carpet to sides of tunnel.

33 WINDSCREEN

A Removal Procedure

Materials

The following materials will be required:

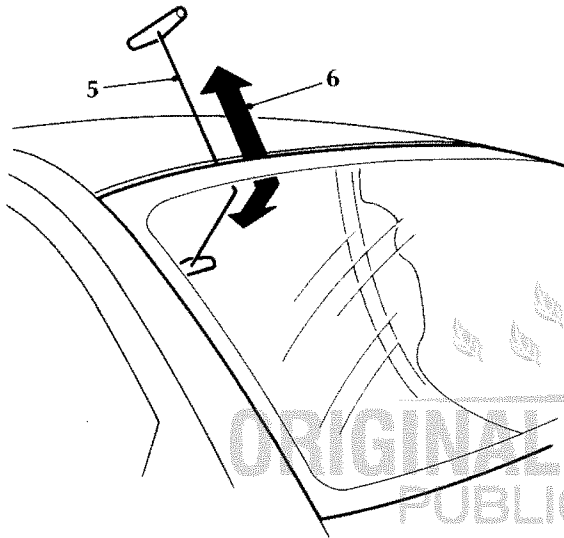
Glass removal kit

NOTES:

- 1 Ensure absolute cleanliness of work area and tools.
- 2 Observe correct use of tools.

CAUTION:

During removal operations extreme care must be taken to avoid damage to the surrounding paintwork and trim.



- (1) Fit JaguarSport protective covers to the front fenders.
- (2) Remove the header trim and loosen the attachment screws on the 'A' post trim. (Refer to Section 2.28)
- (3) Remove the interior lamp assembly and rear header cover (Refer to Section 2.28).
- (4) Insert a length of wire (as shown) between the windscreen glass and the fixing adhesive.
- (5) Using a sawing motion, cut through the adhesive around the windscreen and remove the screen.
- (6) Use a plastic scraper to remove the excess adhesive from the glass and body surfaces.

B Installation Procedure.

Materials

The following materials will be required:

Detergent cleaner (Pt. No. 9553)

Activator, glass/ceramics (Pt. No. 9551)

Primer, glass/ceramics (Pt. No. 9555)

Primer, plastics/paint (Pt. No. 9560)

Adhesive, Polyurethane (Pt. No. 9558)

Adhesive (Pt. No. 9575)

NOTES:

- 1 Ensure absolute cleanliness of work area and tools.
- 2 Observe correct use of tools.

CAUTION:

- 1 *During installation operations extreme care must be taken to avoid damage to the surrounding paintwork and trim.*
- 2 *Extreme care must be taken when handling windscreen glass.*

- (1) Fit new windscreen finisher to the screen using adhesive (9575).
- (2) Remove the lip of the windscreen finisher at the rear of the screen.
- (3) Dry fit the windscreen and adjust the profile of the glass using blocks of cured adhesive (9558). Cut the blocks to suit and lay the windscreen in the aperture to check fit.

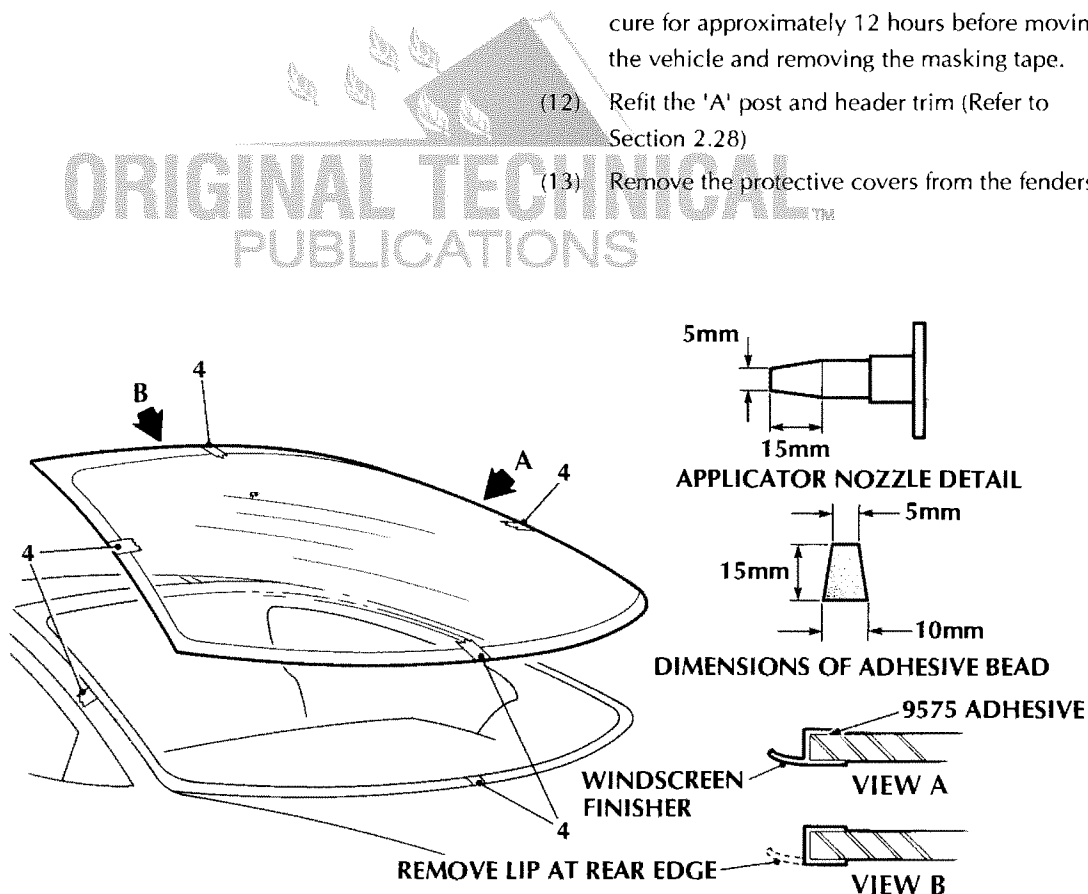
NOTE:

Ensure the windscreen is not touching the body inside the vehicle.

- (4) Centralise the windscreen in the aperture apply strips of masking tape across the glass/body, draw a location line through the tape and cut the masking tape parallel to the glass edge.
- (5) Remove the windscreen and lay it on a soft surface to prevent scratches/damage.
- (6) Degrease the windscreen aperture and the glass screen with JaguarSport recommended detergent (9553).

WARNING:
PRIMERS AND DETERGENTS ARE TOXIC AND MUST ONLY BE USED IN WELL VENTILATED AREAS. AVOID INHALING THE FUMES. USE BARRIER CREAM AND WEAR PROTECTIVE CLOTHING WHEN USING PRIMERS AND DETERGENTS.
REFER TO THE MANUFACTURERS' DATA SHEET FOR APPLICATION AND STORAGE INSTRUCTIONS.

- (7) Treat the vehicle windscreen aperture with JaguarSport plastic/paint primer (9560).
- (8) Treat the windscreen glass with JaguarSport activator (9551) and primer (9555).
- (9) Cut the adhesive applicator nozzle and apply a continuous bead of adhesive (9558) 15mm high to the vehicle windscreen aperture (Refer to illustration).
- (10) Fit the windscreen, aligning the screen with the previously fitted masking tape.
- (11) Apply additional masking tape to control the windscreen location. Allow the adhesive to cure for approximately 12 hours before moving the vehicle and removing the masking tape.
- (12) Refit the 'A' post and header trim (Refer to Section 2.28)
- (13) Remove the protective covers from the fenders.



34 ROOF GLASS

A Removal Procedure

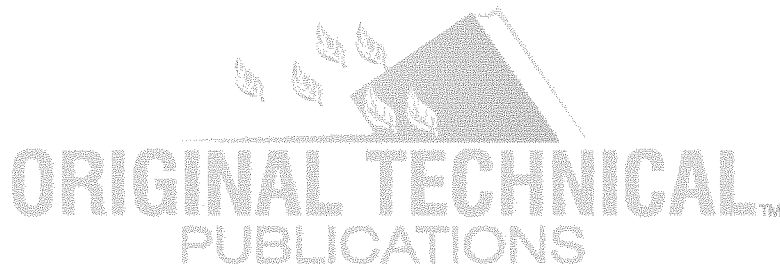
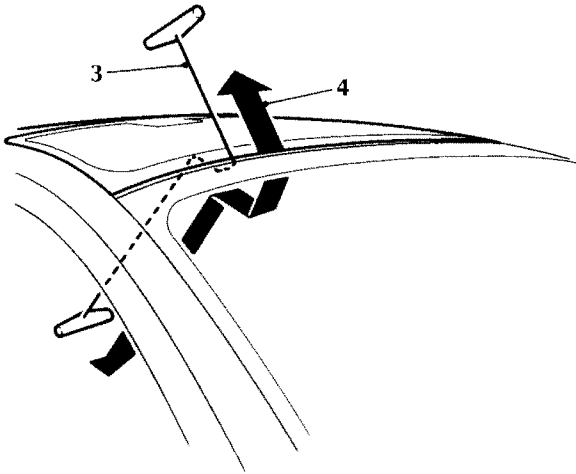
Materials

The following materials will be required:

Glass removal kit

NOTES

- 1 *Ensure absolute cleanliness of work area and tools.*
- 2 *Observe correct use of tools.*
- (1) Fit JaguarSport sill and interior protective covers.
- (2) Remove the rear header (Refer to Section 2.28).
- (3) Insert a length of wire (as illustrated) between the glass and the fixing adhesive.
- (4) Using a sawing motion, cut through the adhesive around the roof glass and remove the glass.
- (5) Use a plastic scraper to remove excess adhesive from glass and body surfaces.



B Installation Procedure

Materials

The following materials will be required:

Detergent cleaner (Pt. No. 9553)

Activator, glass/ceramics (Pt. No. 9551)

Primer, glass/ceramics (Pt. No. 9555)

Primer, plastics/paint (Pt. No. 9560)

Adhesive, Polyurethane (Pt. No. 9558)

Adhesive (Pt. No 9575)

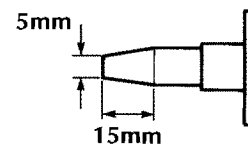
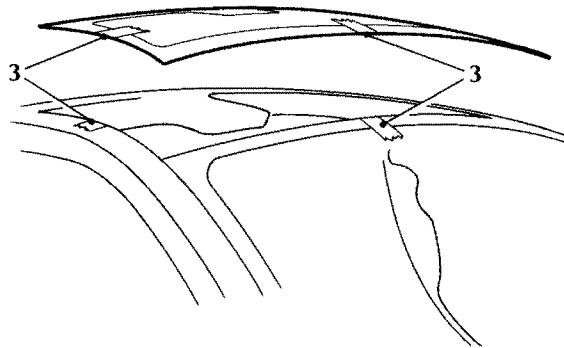
NOTES:

- 1 Ensure absolute cleanliness of work area and tools.
- 2 Observe correct use of tools.
- (1) Fit the finisher to the roof glass using adhesive (9575) and remove the lip of the finisher at the rear of the roof glass.
- (2) Dry fit the roof glass and adjust the profile of the glass using blocks of cured adhesive (9558). Cut the blocks to suit and lay the roof glass in the aperture to check fit.

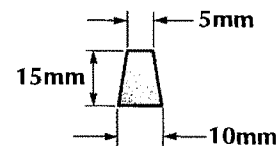
NOTE:

Ensure the glass is not touching the body inside the vehicle.

- (3) Centralise the glass in the aperture and apply strips of masking tape across the glass/body, draw a location line across the tape. Cut the tape parallel to the glass edge.
- (4) Remove the roof light glass and lay it on a soft surface to prevent scratches/damage.
- (5) Degrease the vehicle aperture and the glass with JaguarSport recommended detergent (9553).
- (6) Treat the vehicle aperture with JaguarSport plastic/paint primer (9560).
- (7) Treat the roof glass with JaguarSport activator (9551) and primer (9555).
- (8) Cut the applicator nozzle and apply a continuous bead of adhesive (9558) 15mm high to the vehicle aperture.
- (9) Fit the roof glass, aligning the glass with the previously fitted masking tape.
- (10) Apply additional masking tape to control the roof glass location. Allow the adhesive to cure for approximately 12 hours before moving the vehicle and the masking tape.
- (11) Attach the lamp mounting bracket to the rear header and apply adhesive 9558 to the top surface of the bracket.
- (12) Fit the rear header (Refer to Section 2.28)
- (13) Use masking tape to hold the rear header cover in the correct location until adhesive has cured.
- (14) Fit the interior lamp (Refer to Section 15).
- (15) Remove the sill and interior protective covers.



APPLICATOR NOZZLE DETAIL

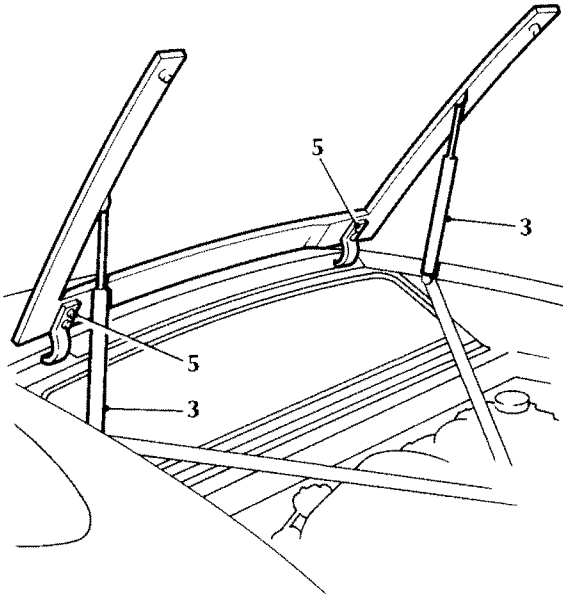


DIMENSIONS OF ADHESIVE BEAD

WARNING:

PRIMERS AND DETERGENTS ARE TOXIC AND MUST ONLY BE USED IN WELL VENTILATED AREAS. AVOID INHALING THE FUMES. USE BARRIER CREAM AND WEAR PROTECTIVE CLOTHING WHEN USING PRIMERS AND DETERGENTS.

REFER TO THE MANUFACTURERS' DATA SHEETS FOR APPLICATIONS AND STORAGE INSTRUCTIONS



35 TAILGATE FRAME

A Removal Procedure

- (1) Open the tailgate and fit JaguarSport protective covers.
- (2) Remove the tailgate glass, if required, (Refer to Section 2.38).
- (3) Disconnect the tailgate gas struts.
- (4) Remove the rear cabin glass sound insulation.
- (5) Support the tailgate, remove the screws and washers securing the tailgate to the hinges and remove from the vehicle.

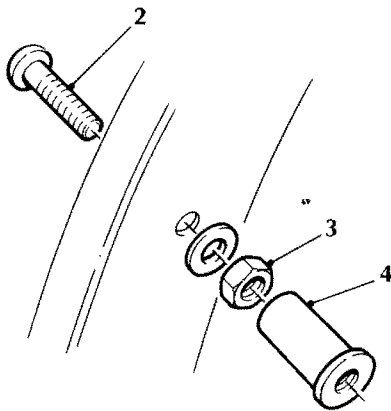
B Installation Procedure

Materials

The following materials will be required.

Studlock 270

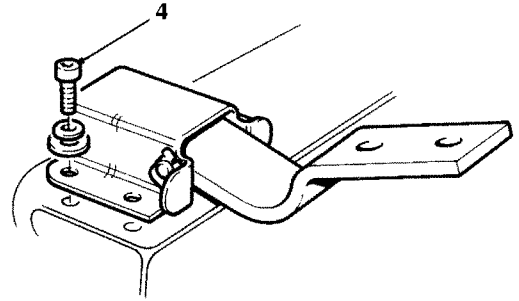
- (1) Fit the striker assembly to the RH and LH legs of the frame and secure using screws and washers. Tighten the screws 7.5 to 9Nm (5.5 to 6.5ftlb).
- (2) Fit the bumpstop screw to the tailgate frame.
- (3) Apply Studlock 270 to the screw threads and fit the washer and nut.
- (4) Apply Studlock 270 to the rubber bump stop threads and fit the bumpstop to the screw.
- (5) Fit the tailgate frame to the hinges and fit the washers and screws finger tight.
- (6) Adjust the tailgate profile/gaping to the body, louvres and latches, and tighten the hinge screws 7.5 to 9Nm (5.5 to 6.5ftlb).
- (7) Fit the tailgate gas struts
- (8) Fit the tailgate glass, if removed, (Refer to Section 2.38).
- (9) Remove the protective covers.
- (10) Operate the tailgate several times through the full range and ensure unobstructed movement, re-adjust the latch striker and bumpstops if necessary.
- (11) Refit the rear cabin glass sound insulation.
- (12) Shut the tailgate.



36 TAILGATE HINGES

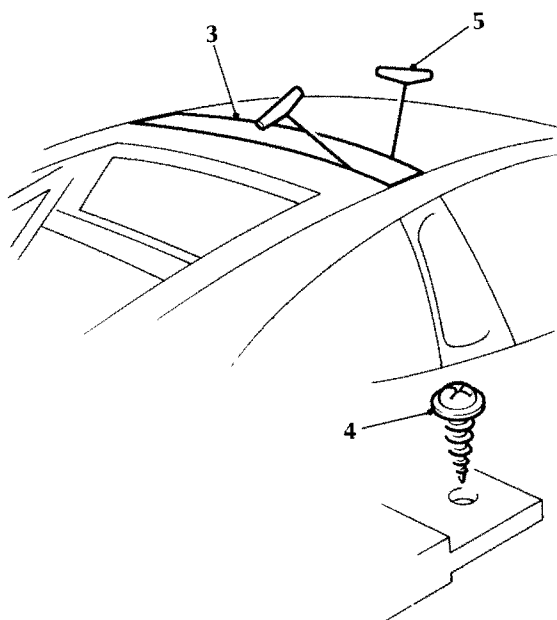
A Removal Procedure

- (1) Open the tailgate and fit JaguarSport protective covers.
- (2) Remove the tailgate frame complete with glass. (Refer to Section 2.35).
- (3) Remove the roof finisher panel (Refer to Section 2.37).
- (4) Remove the screws and washers securing the hinges and remove the hinges.



B Installation Procedure

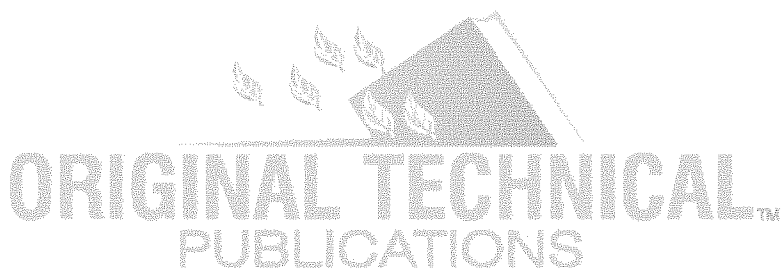
- (1) Locate the hinges and secure to the body using washers and screws. Tighten the screws 7.5 to 9Nm (5.5 to 6.5ftlb).
- (2) Fit the roof finisher panel. (Refer to Section 2.37).
- (3) Fit the tailgate frame and glass assembly. (Refer to Section 2.35).
- (4) Remove the protective covers.



37 ROOF FINISHER PANEL

A Removal Procedure

- (1) Open the tailgate and fit JaguarSport protective covers.
- (2) Remove the rear cabin glass sound insulation.
- (3) Remove the roof glass. (Refer to Section 2.34).
- (4) Remove the two self-tapping screws from the front edge of the roof panel.
- (5) Using a sharp knife, blade or wire, cut through the adhesive securing the roof panel to the tailgate hinge brackets and remove the panel.
- (6) Remove all remaining adhesive and clean the hinge brackets with detergent (9553).



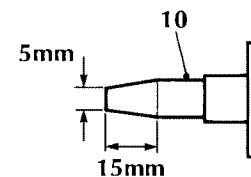
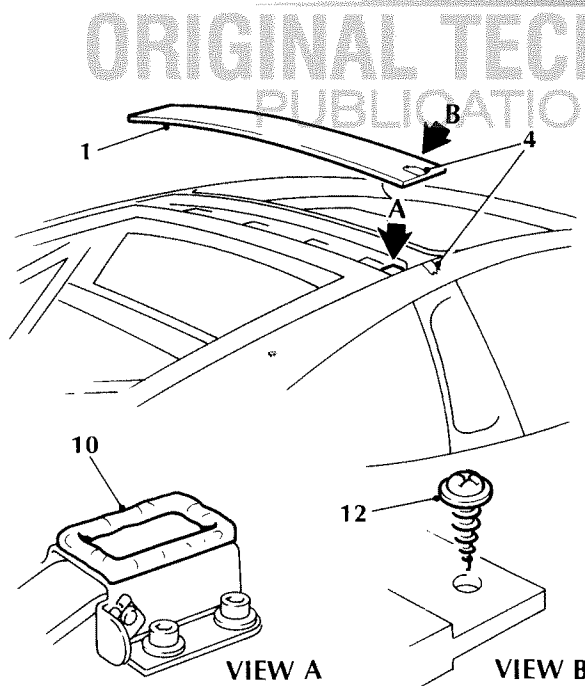
B Installation Procedure

- (1) Locate the roof finisher panel over the tailgate hinges.
- (2) Dry fit the roof glass (complete with edge finisher) and ensure even gaps at both sides and a snug fit to the windscreen.
- (3) Position the roof finisher panel to achieve even gaps at each side and a snug fit to the roof glass.
- (4) Once correct alignment has been achieved apply strips of masking tape across the roof finisher panel, roof glass and body. Open the tailgate glass and check the clearance. Cut the tape parallel to joint lines.
- (5) Carefully remove the roof glass and place on a glazing stand taking care not to disturb the finisher.
- (6) If replacing the roof finisher panel, back-drill the existing holes in the body into the roof panel.
- (7) Remove the panel from the vehicle and deburr the drilled holes.
- (8) Clean the roof finisher panel and body mating surfaces with detergent (9553) and wipe dry.
- (9) Apply plastic primer (9560) to the roof finisher panel underside and tailgate hinge brackets.
- (10) Cut the applicator nozzle and apply a continuous bead of adhesive (9558) 20mm high to the tailgate hinge brackets.
- (11) Carefully locate and position the roof finisher panel to rear header, aligning the marker tapes and ensuring an even gap at each side.
- (12) Secure the roof finisher panel at the front edge with two self-tapping screws.
- (13) Fit the roof glass. (Refer to Section 2.34).
- (14) Fit the rear cabin glass sound insulation.
- (15) Remove the protective covers and close the tailgate.

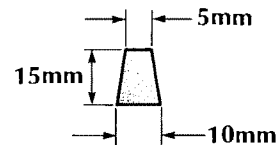
WARNING:

PRIMERS AND DETERGENTS ARE TOXIC AND MUST ONLY BE USED IN WELL VENTILATED AREAS. AVOID INHALING THE FUMES. USE BARRIER CREAM AND WEAR PROTECTIVE CLOTHING WHEN USING PRIMERS AND DETERGENTS.

REFER TO THE MANUFACTURERS' DATA SHEETS FOR APPLICATIONS AND STORAGE INSTRUCTIONS .



APPLICATOR NOZZLE DETAIL



DIMENSIONS OF ADHESIVE BEAD

38 TAILGATE GLASS

Materials

The following materials will be required:

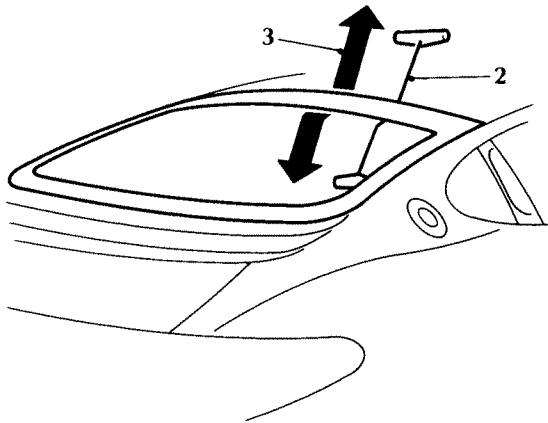
Glass removal kit

NOTES:

- 1 Ensure absolute cleanliness of tools and work area.
- 2 Observe the correct use of tools.

A Removal Procedure

- (1) Fit JaguarSport rear protective covers.
- (2) Insert a length of wire between the glass and the adhesive.
- (3) Using a sawing motion, cut through the adhesive around the glass and remove the glass.
- (4) Use a plastic scraper to remove excess adhesive from the glass and frame.



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B Installation Procedure

Materials

The following materials will be required:

Detergent cleaner (Pt. No. 9553)

Activator, glass/ceramics (Pt. No. 9551)

Primer, glass/ceramics (Pt. No. 9555)

Primer, plastics/paint (Pt. No. 9560)

Adhesive, Polyurethane (Pt. No. 9558)

NOTES:

- 1 Ensure absolute cleanliness of tools and work area.
- 2 Observe the correct use of tools.
- 3 Do not lean on rear quarter body panels.

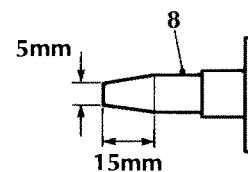
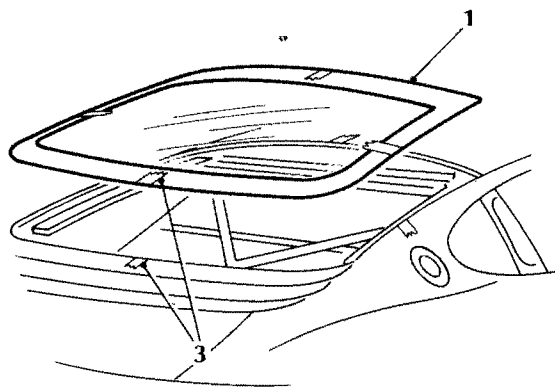
- (1) Dry fit the tail gate glass to the closed tailgate frame to achieve an even gap at the sides and a flush condition along the front edge.
- (2) Adjust the profile of the glass using cured blocks of adhesive (9558). Cut the blocks to suit and lay the tailgate glass in the aperture to check fit.
- (3) Centralise the glass in the aperture and apply masking tape strips across the glass/body, draw a location line through the tape and cut the masking tape parallel to the glass edge.
- (4) Remove the tailgate glass and place on a glazing stand with the interior side uppermost.

- (5) Degrease the frame with JaguarSport detergent (9553).

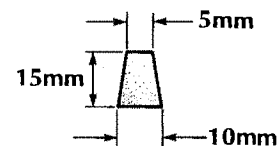
WARNING:

PRIMERS AND DETERGENTS ARE TOXIC AND MUST ONLY BE USED IN WELL VENTILATED AREAS. AVOID INHALING THE FUMES. USE BARRIER CREAM AND WEAR PROTECTIVE CLOTHING WHEN USING PRIMERS AND DETERGENTS. REFER TO MANUFACTURERS' DATA SHEETS FOR APPLICATION AND STORAGE INSTRUCTIONS.

- (6) Treat the frame with JaguarSport plastic/paint primer (9560).
- (7) Treat the glass with JaguarSport activator (9551) and primer (9555).
- (8) Cut the adhesive applicator nozzle (refer to illustration) and apply a continuous bead of adhesive (9558) 15mm high and 5mm from the edge of the frame sides and to the frame top face.
- (9) Fit the glass, aligning the masking tape and ensuring an even gap at the sides.
- (10) Hold the glass in position using tape across the glass and body joint. Allow the adhesive to cure for approximately 12 hours before moving the vehicle and removing the masking tape.
- (11) Remove the rear protective covers.
- (12) Remove the protective covers



APPLICATOR NOZZLE DETAIL



DIMENSIONS OF ADHESIVE BEAD

39 REAR QUARTER LIGHT GLASS

A Removal Procedure

Materials

The following materials will be required:

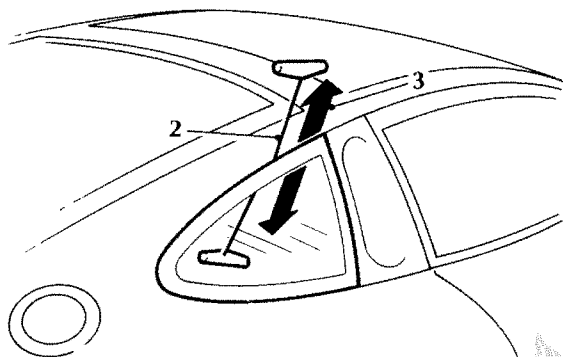
Glass removal kit.

NOTES:

- 1 Ensure absolute cleanliness of tools and work area.
- 2 Observe the correct use of tools.

CAUTION:

During removal operations extreme care must be taken to void damage to the surrounding paintwork.



- (1) Fit JaguarSport rear protective covers.
- (2) Insert a length of wire between the glass and the adhesive.
- (3) Using a sawing motion, cut through the adhesive around the glass and remove the glass.
- (4) Use a plastic scraper to achieve a flat surface on the remaining adhesive.

B Installation Procedure

Materials

The following materials will be required:

Detergent cleaner (Pt. No. 9553)

Activator, glass/ceramics (Pt. No. 9551)

Primer, glass/ceramics (Pt. No. 9555)

Primer, plastics/paint (Pt. No. 9560)

Adhesive, Polyurethane (Pt. No. 9558)

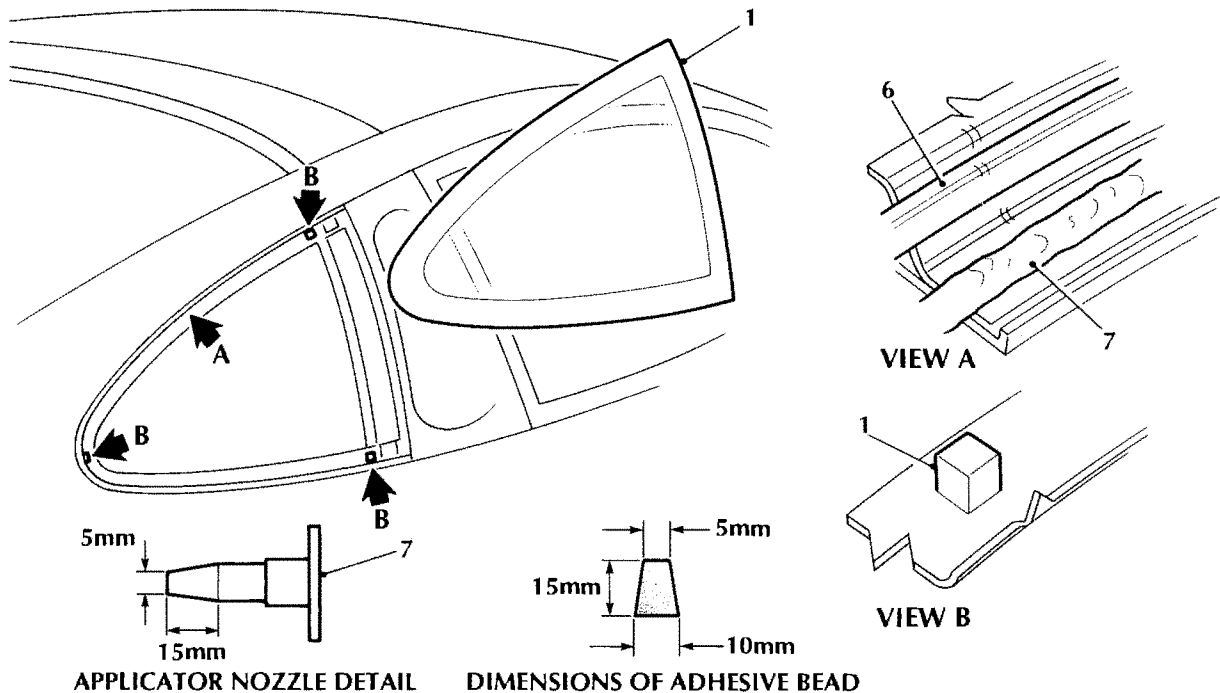
NOTES:

- 1 Ensure absolute cleanliness of tools and work area.
- 2 Observe the correct use of tools.

CAUTION:

- 1 *During installation operations extreme care must be taken to avoid damage to the surrounding paintwork.*
- 2 *Extreme care must be taken when handling glass. Refer to manufacturers' data sheets.*

- (1) Dry fit the glass and adjust the profile using cured blocks of adhesive. Attach the adhesive blocks to the aperture flange using adhesive (9558).
- (2) Remove the glass and place on a workbench interior face uppermost.



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- (3) Degrease the body with JaguarSport detergent (9553).

WARNING:

PRIMERS AND DETERGENTS ARE TOXIC AND MUST ONLY BE USED IN WELL VENTILATED AREAS. AVOID INHALING THE FUMES. USE BARRIER CREAM AND WEAR PROTECTIVE CLOTHING WHEN HANDLING.

WARNING:

REFER TO MANUFACTURERS' DATA SHEETS FOR APPLICATION AND STORAGE INSTRUCTIONS.

- (4) Treat the body with JaguarSport plastic/paint primer (9560).
- (5) Treat the glass with JaguarSport activator (9551) and primer (9555).
- (6) Remove the backing and fit the secondary seal to the outer edge of the aperture, from the top rear corner of the D light intake duct to the bottom rear corner, trimming off excess seal length.

CAUTION:

Ensure the seal section does not collapse during fitting.

- (7) Cut the applicator nozzle and apply a continuous bead of adhesive (9558) 15mm high to the centre of the aperture flange and rear lip of the D light intake duct.
- (8) Carefully fit the glass to the aperture aligning the edge of the glass to the 'D' light intake duct and ensuring an even gap between the body and glass.
- (9) Hold the glass in position using masking tape across the body joints.
- (10) Allow the adhesive to cure for approximately 12 hours before moving the vehicle and removing the masking tape.
- (11) Remove the rear protective covers.

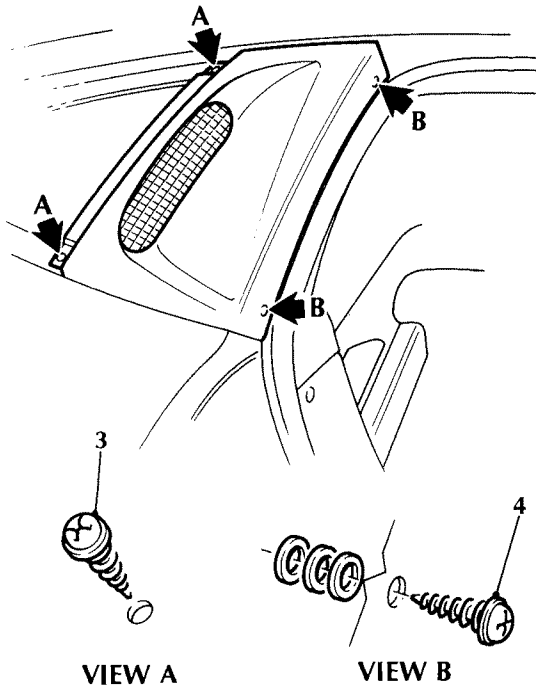
40 D LIGHT INTAKE DUCT

A Removal Procedure

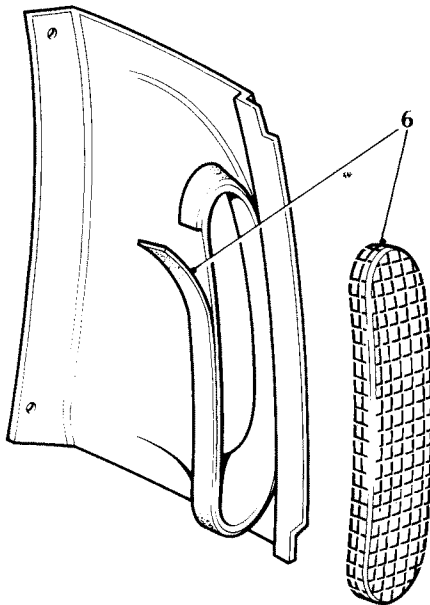
NOTES:

- 1 Ensure absolute cleanliness of tools and work area.
- 2 Observe the correct use of tools.

- (1) Fit JaguarSport protective body covers
- (2) Remove the rear quarter light glass (Refer to Section 2.39).
- (3) Remove the screws securing the rear of the D light intake duct.
- (4) Open the relevant door, and remove the two screws securing the front of the D light. Take care not to damage the door aperture seal.
- (5) Remove the D light intake duct and collect the packers from behind the duct.
- (6) Remove the butyl adhesive tape and remove the D light duct grille (if required).



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B Installation Procedure

- (1) Fit the D light intake duct mesh to the aperture and secure with butyl adhesive tape.
- (2) Align the packers and fit the D light intake duct to the vehicle. Check the alignment of the D light glass and duct with the body and door glass and secure the duct with self-tapping screws.
- (3) Fit the rear quarter light glass (Refer to Section 2.39).
- (4) Remove the JaguarSport protective body covers.

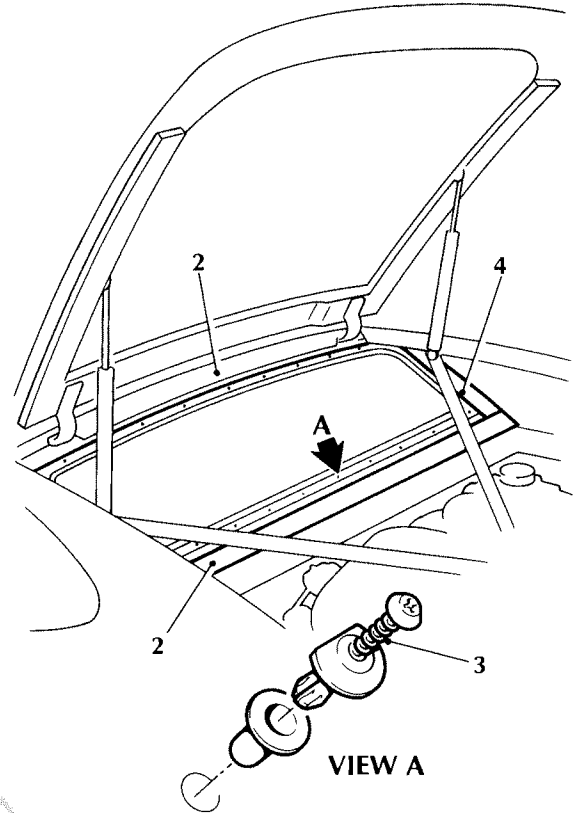
41 REAR BULKHEAD GLASS

A Removal Procedure

NOTES:

- 1 Ensure absolute cleanliness of tools and work area.
- 2 Observe the correct use of tools.

- (1) Open the tail gate and fit JaguarSport protective covers.
- (2) Carefully remove the sound insulation from around the bulkhead glass assembly.
- (3) Remove the scrivenets securing the bulkhead glass and remove the glass from the vehicle.
- (4) Remove the remaining silicone sealant from the body using a plastic scraper.



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B Installation Procedure

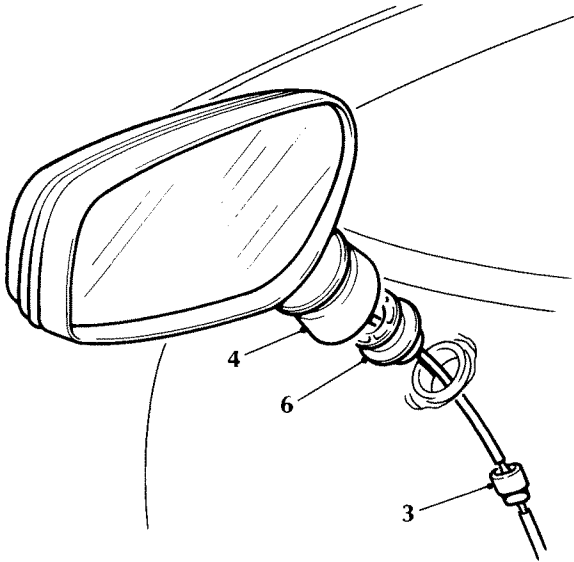
- (1) Fit the bulkhead glass assembly to the rear bulkhead aperture, from the engine compartment side, with the foam seal side of the assembly to the bulkhead.
- (2) Secure the assembly to the body using snapsacs and scrivenets.
- (3) Apply a bead of silicone sealant around the outer edge of the assembly to ensure a water tight seal.
- (4) Refit the sound insulation and replace the velcro attachment strips if necessary.

42 REAR VIEW MIRRORS

A Removal Procedure

NOTES:

- 1 Ensure absolute cleanliness of work area and tools.
- 2 Observe correct use of tools.



Exterior Rear View Mirrors

- (1) Wing-mounted Mirror
Remove wheel arch liner (Refer to Section 2.4).
- (1A) Door-mounted Mirror
Remove door trim (Refer to Section 2.22).
- (2) Turn the main isolator switch to the off position.
- (3) Disconnect the mirror electrical plug.
- (4) Raise the rubber escutcheon to reveal the grub screw.
- (5) Loosen the grub screw and remove the mirror and electrical plug.
- (6) If required, unscrew the mirror mounting boss and remove.

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Interior Rear View Mirror

Special Tools

The following special tools will be required,
Perspex scraper

- (1) Fit JaguarSport interior protective covers.
- (2) Place a perspex scraper at one corner of the mirror base.
- (3) Carefully prise the mirror base off the windscreen.

B Installation Procedure

Materials

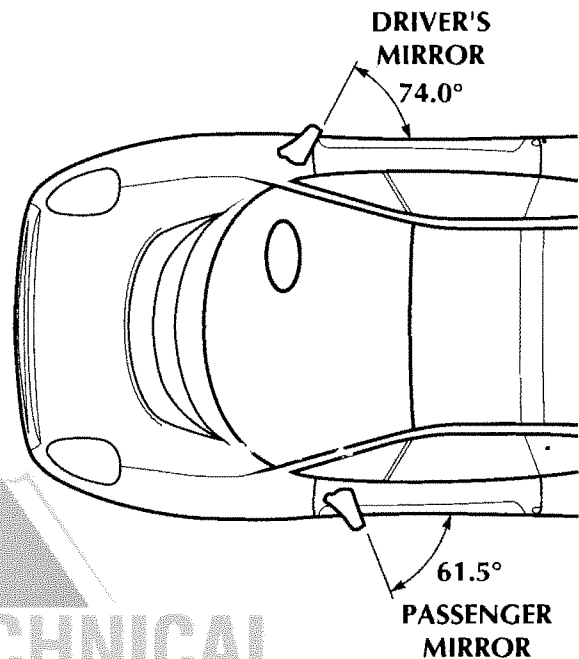
The following materials will be required.

Lint free cloth

Methyl Ethyl Ketone (MEK)

Exterior Mirrors

- (1) If removed, fit the mirror mounting boss to the wing/door.
- (2) Route the mirror electrical harness through the mounting boss and connect to the main harness.
- (3) Fit the mirror to the mounting boss and secure with the grub screw.
- (4) Adjust the mirror head position to the legal field of vision angles and tighten the grub screw.
- (5) Pull down the rubber escutcheon to cover the base.
- (6) Turn the main isolator switch to the on position.
- (7) Carry out a functional check of the mirrors and rectify any faults as necessary.
- (8) Wing Mirror
Fit the wheel arch liner (Refer to Section 2.4).
- (8A) Door Mirror
Fit the door trim (Refer to Section 2.22).



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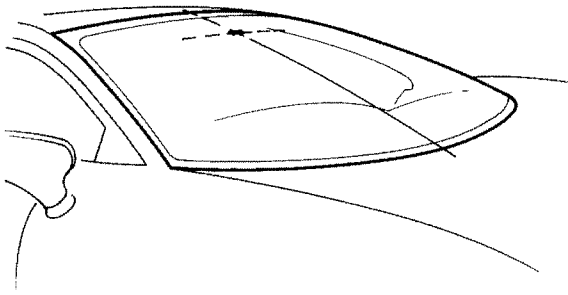
Interior Rear View Mirror

Materials

The following materials will be required.

Lint free cloth

Methyl Ethyl Ketone (MEK)

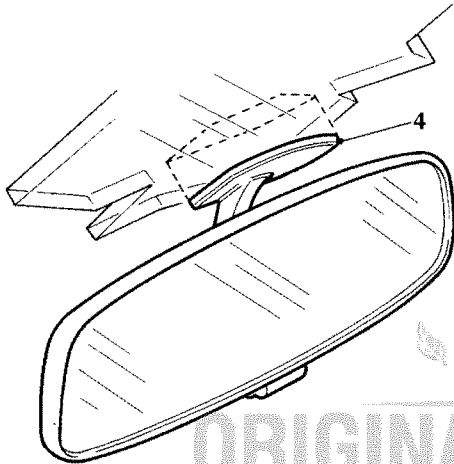


- (1) Thoroughly degrease the area of the windshield around the mirror obscuration zone using MEK.

WARNING:

SOLVENTS ARE TOXIC AND MUST ONLY BE USED IN WELL VENTILATED WORK AREAS. AVOID INHALING THE SOLVENT FUMES AND WEAR PROTECTIVE CLOTHING. REFER TO MANUFACTURER'S DATA SHEET FOR APPLICATION AND STORAGE INSTRUCTIONS.

- (2) Dry and pre-warm the glass surface with a warm air blower.
- (3) Remove the backing from the mirror adhesive pad
- (4) Align the mirror adhesive pad with the mirror obscuration, press the mirror firmly onto the windshield and apply pressure for a minimum of 30 seconds.



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SECTION 03.00 ENGINE

DESCRIPTION

1. INTRODUCTION

The engine, transaxle and driveline have been designed to produce the optimum power to weight ratio. The engine via a twin-plate clutch to a 5-speed fully synchro-mesh transaxle. The transaxle is provided with oil cooling and transmits drive direct to the rear wheels. The exhaust system incorporates catalytic converters and the whole power unit is designed to comply with strict EEC emission regulations. Maximum engine speed is 7,400rpm.

2. ENGINE

The engine is a water cooled, 90° V6 with 4 camshafts, 4 valves-per-cylinder, twin turbochargers, intercooling and dry sump lubrication. Fuelling is by a multi-point sequential injection system with two injectors per cylinder. Ignition is by a fully mapped electronic system driving a single centrally sited spark plug in each cylinder.

The whole unit is constructed from lightweight aluminium alloy utilising the latest thin wall casting techniques. All moving components are designed for light weight and maximum efficiency.

A Cylinder Heads

The heads are cross flow with two inlet and two exhaust valves per cylinder. Separate camshafts are used to operate the inlet and exhaust valves. The valves are operated directly from the camshafts via solid bucket type tappets.

The combustion chambers in the heads are of unique proportions designed to allow for high efficiency, fast burn. The chambers are complemented by high efficiency inlet and exhaust ports designed for maximum gas flow to promote controlled, high velocity in-cylinder gas motion.

B Cylinder Block

The aluminium cylinder block has cast iron liners for the cylinder bores. The block is deep skirted with 4 cross bolted main bearings for the crankshaft. The main bearing caps are steel and the crankshaft runs in conventional steel backed bearing shells.

The crankshaft has three throws, each providing attachment for two connecting rods. This coupled with the 90° Vee angle provides uneven firing intervals of 90° and 150°.

The pistons are forged aluminium of bowl-in-piston design with three rings. The pistons have short skirts on the thrust/anti-thrust sides only. The skirt profile is precisely controlled to minimise friction and control piston motion. The gudgeon pin is fully floating. Oil jets mounted at the base of the cylinders provide both piston cooling and small end lubrication.

The dry sump mounts to the block skirt and the main bearing caps forming a structural member to increase block stiffness. The front engine mounts are incorporated into the sump casing, which also has provision for a locking pin used to lock the crankshaft during camshaft timing operations.

C Cooling System

Each bank is independently cooled by means of its own gear driven high velocity water pump. The outlets from each bank join at the thermostat housing, located between the plenum chambers at the rear of the engine. A full flow by-pass is incorporated into the cooling system which is designed to have minimum volume to promote rapid warm up.

The coolant passes through the transaxle oil heat exchanger after the thermostat and then into the radiator at the front of the vehicle.

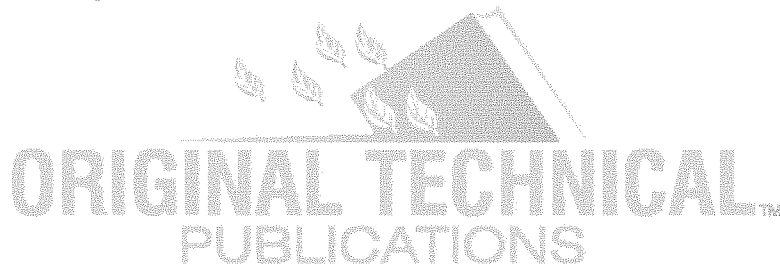
D Lubrication System

The engine has a dry sump lubrication system which allows for constant oil pressure under all operating conditions. It also provides for a lower engine giving a lower centre of gravity for improved road holding.

The oil is pumped into the engine by means of a pressure pump. This pump circulates the oil under pressure to all the internal bearings, plus the piston cooling system. The oil is allowed to drop into the sump where the action of the crankshaft causes it to be thrown into a specially designed gallery.

A scavenge pump sucks the oil from this gallery through a serviceable mesh filter and pumps it back into the oil tank via a vehicle mounted oil cooler.

The two pumps are mounted directly on the nose of the crankshaft in a common housing.



iv

1 INTRODUCTION

This section details the removal instructions for the engine, transaxle and driveline, and preparation for transit.

2 ENGINE, TRANSAXLE AND DRIVELINE

A Removal Procedure

WARNINGS:

- 1 **ERECT NO SMOKING SIGNS AROUND THE VEHICLE AND KEEP A SERVICEABLE HALON FIRE EXTINGUISHER ON HAND.**
- 2 **ENSURE ALL LIFTING EQUIPMENT IS SERVICEABLE AND SATISFIES LOCAL HEALTH AND SAFETY REGULATIONS.**
- 3 **THE BATTERY EARTH LEAD MUST BE DISCONNECTED. MOVE BATTERY MASTER SWITCH TO THE OFF POSITION.**
- 4 **THE ENGINE SHOULD ONLY BE REMOVED WHEN COLD TO ENSURE SAFETY OF PERSONNEL.**

NOTES:

- 1 *Fit JaguarSport body covers prior to commencing work on the vehicle.*
- 2 *Inspect components as they are removed and label for possible repair or rejection.*
- 3 *Left hand (LH) and right hand (RH) are denoted as viewed from the rear of the vehicle FACING FORWARD.*

Special Tools

The following special tools will be required:

Air conditioning system recovery unit.

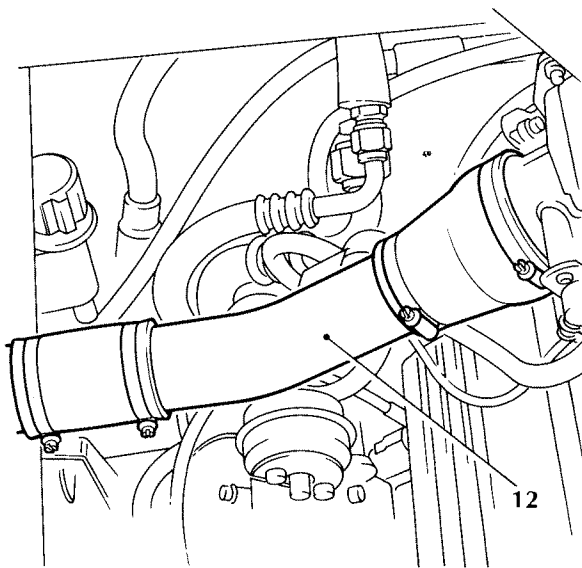
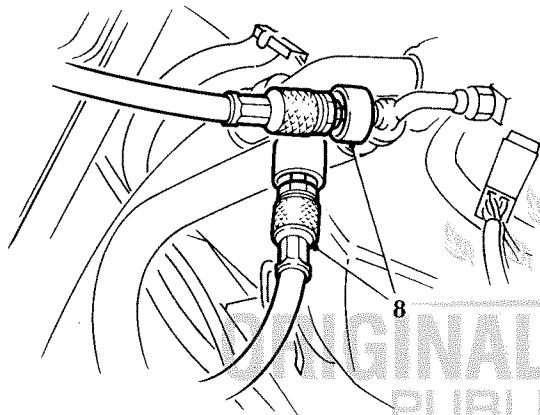
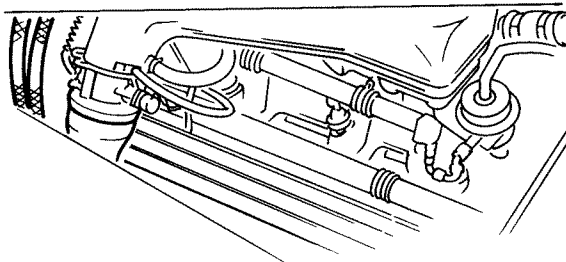
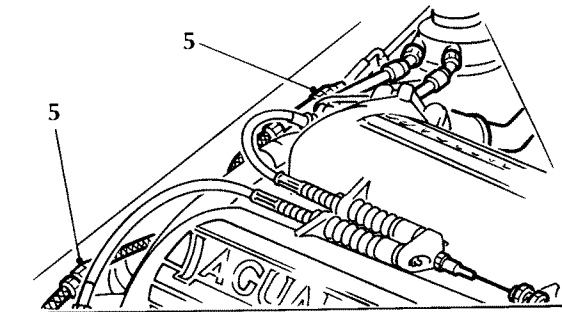
Electro-hydraulic vehicle lifting apparatus.

Hydraulic-mechanical lifting/lowering engine trolley.

JaguarSport approved body covers.

Support for rear axle uprights.

- (1) Correctly position the vehicle on the lifting apparatus as described in Jacking and Lifting-Section 1.1.
- (2) Disconnect the battery.
- (3) Select reverse gear to facilitate the removal of the gear selector rod bolt.
- (4) Disconnect the throttle cable from the connector between the plenum chambers.

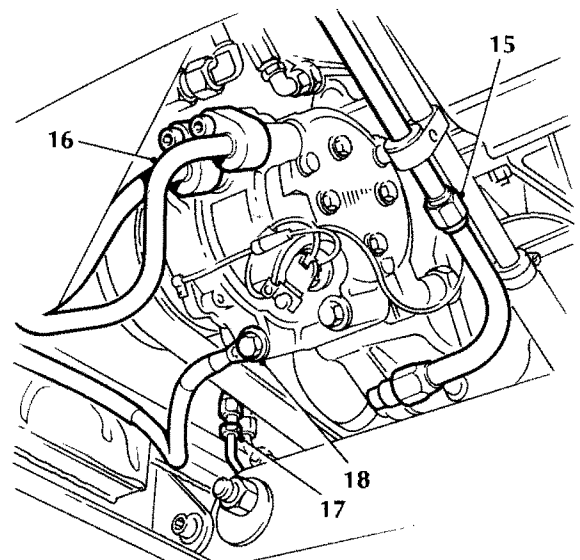
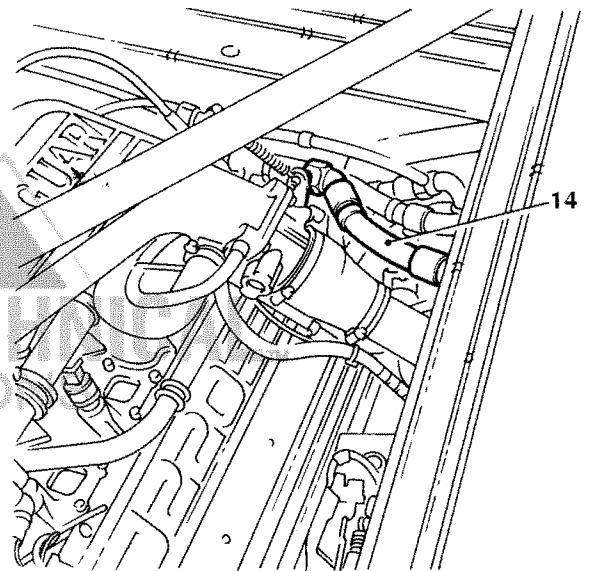
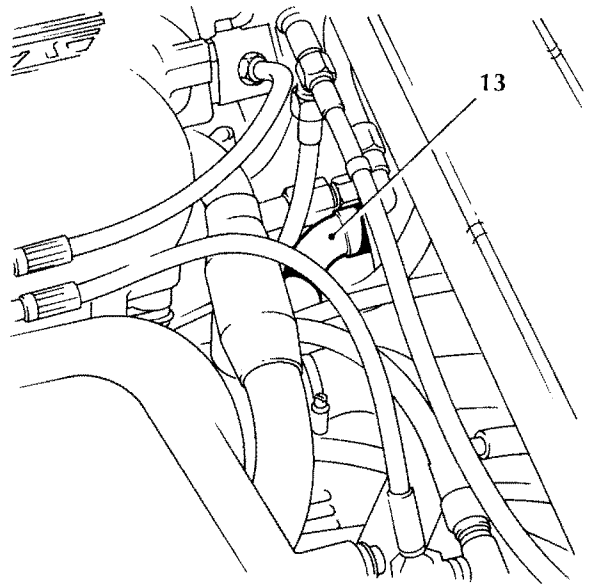


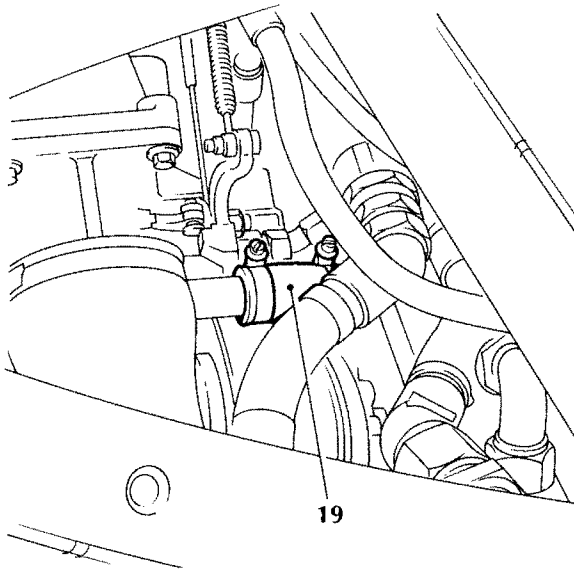
WARNING

THE FUEL INJECTION SYSTEM IS PRESSURISED . DURING DISCONNECTION OF ANY FUEL SYSTEM COMPONENT PLACE COTTON WASTE AROUND THE UNION AND LOOSEN THE UNION SLIGHTLY TO ALLOW THE PRESSURE TO DROP.

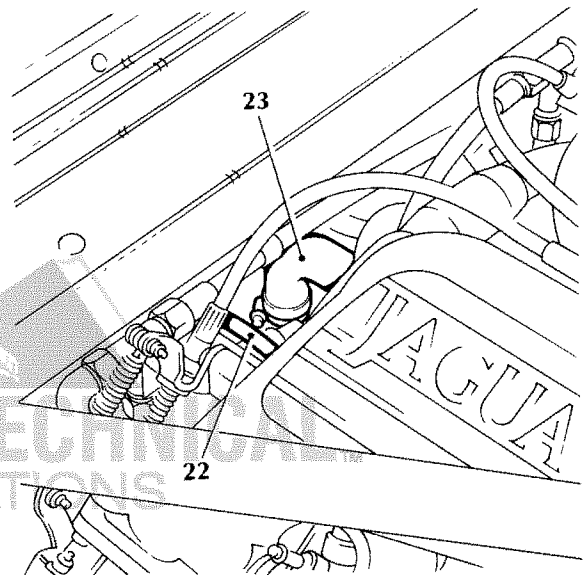
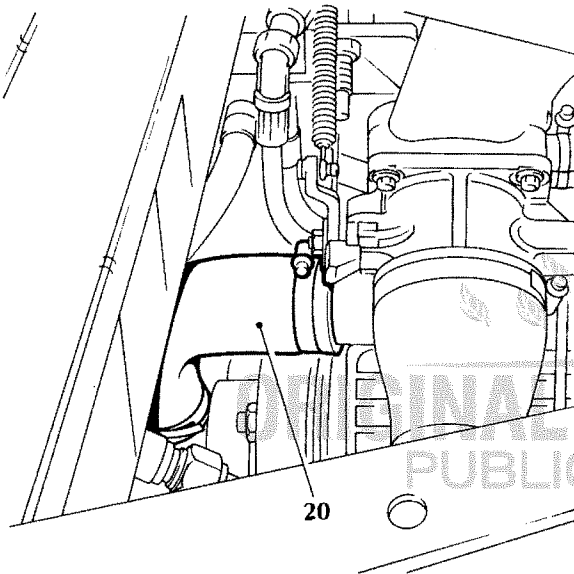
- (5) Disconnect the 'A' bank and 'B' bank fuel inlet pipes at the tee pieces adjacent to the throttle bodies and fit blanking caps to pipe open ends.
- (6) Open the bonnet and remove the quick release fasteners securing the radiator duct access panel and remove panel.
- (7) Remove the blanking caps from the air conditioning system charge pipes.
- (8) Connect the air conditioning system service unit to the vehicle in the following manner;
 - a) Red connector to the fluid filled pipe.
 - b) Blue connector to the gas filled pipe.
- (9) Start the air conditioning system service unit and evacuate the air conditioning system.
- (10) Support the weight of the vehicle with the lifting apparatus, remove the rear wheels (Refer to Section 4.1) and disconnect the handbrake cables at the calipers. Remove the tyrap attaching the handbrake cable to the rear plate.
- (11) Remove the rear grille to allow access to the rear bumper nuts.
- (12) Remove the hose clips from the intercooler pipe and throttle bodies and remove the complete pipe/hose assembly. Plug the open hose ends.

- (13) Disconnect the oil cooler pipe from the front of the engine and fit blanking caps to pipe open ends.
- (14) Remove the -10 breather pipe from the engine to the oil tank.
- (15) Disconnect the turbo charger coolant pipe on both sides of the engine and drain the engine coolant into a suitable container. Reconnect the pipes when the system has been drained.
- (16) Disconnect the inlet and outlet pipes from the air conditioning pump. Fit blanking plugs to pipe open ends.
- (17) Disconnect the brake pipe on the engine bulkhead and fit blanking caps to pipe open ends.
- (18) Disconnect the engine to body earth lead on the air conditioning compressor mounting bolt.

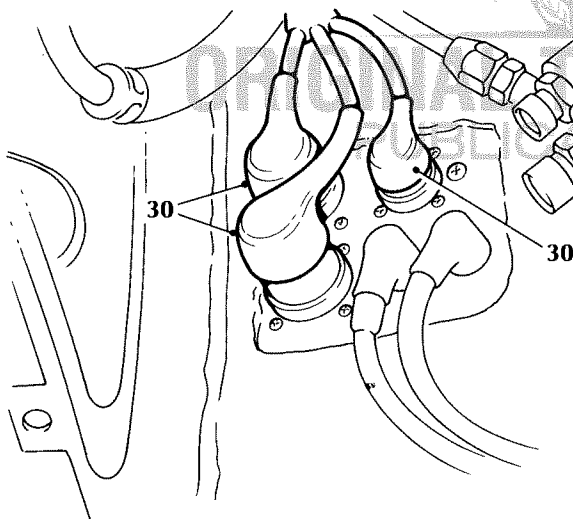
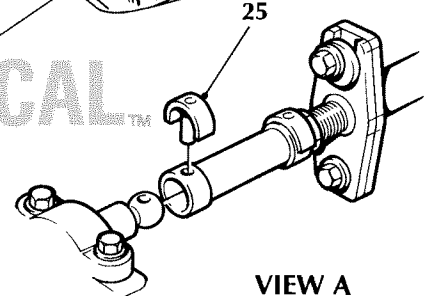
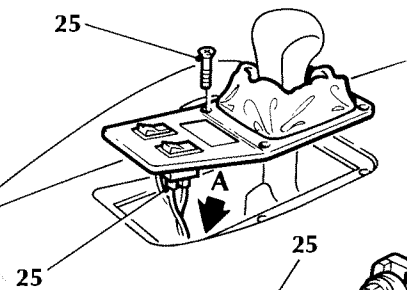
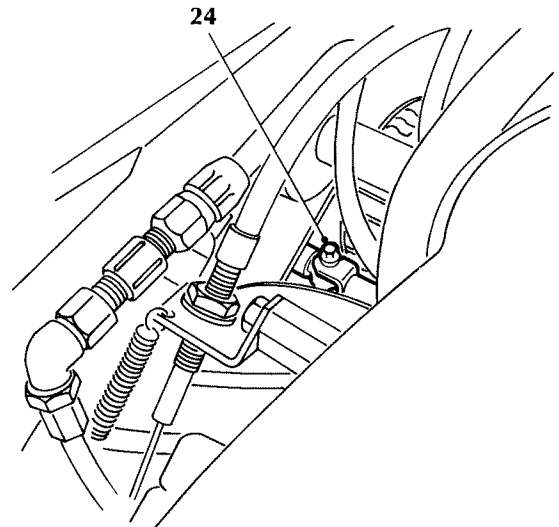


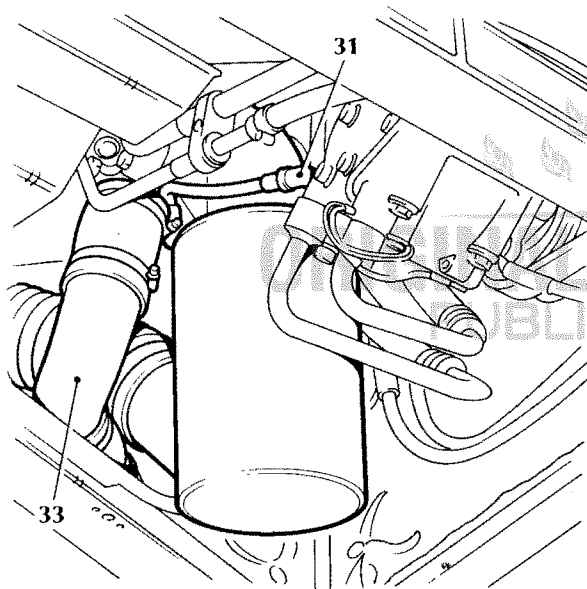
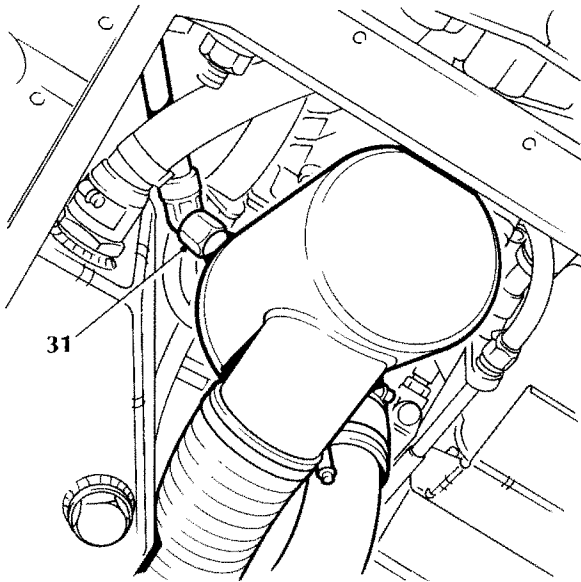


- (19) Disconnect the water by-pass pipe at the front of the engine.
- (20) Disconnect the water pipe over the 'B' bank cam cover at the front of the engine.
- (21) Disconnect the fuel purge hose from the 'Y' piece between the plenum chambers.
- (22) Disconnect the vacuum hoses at the front of 'A' and 'B' bank vacuum rails.
- (23) Disconnect the heater feed pipe at the front of the 'A' bank cam cover.

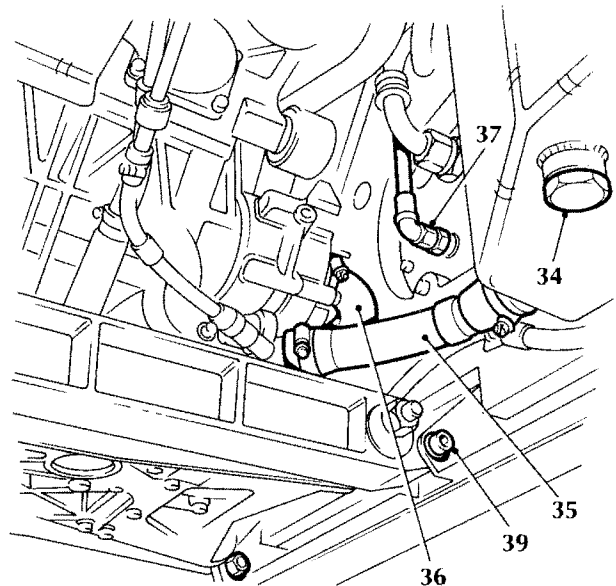


- (24) Remove the M8 bolt on the gear selector rod at the front of the engine and disconnect the rods. After disconnecting the rods, move the gear lever as far forward as possible.
- (25) Remove the four bolts securing the gear lever cover, and disconnect the window switches. Remove the spring clip and move the gear lever forward section clear of the auxiliary drive belt.
- (26) Raise the vehicle to a comfortable working height and fit supports beneath the rear uprights.
- (27) Remove the 5mm and 6mm screws and washers securing the venturi panel and remove the panel. Refer to Fig 220-WM-001 for location of fixings.
- (28) Remove the nuts and washers securing the rear bumper. Lower the bumper to allow access to the rear number plate light plug. Disconnect the plug and remove the bumper (Refer to Section 2).
- (29) Disconnect the electrical cable to the starter motor and the 'P' clips on the intermediate plate.
- (30) Disconnect the engine electrical harness plugs at the engine bulkhead.





- (31) Loosen the Goodridge fittings and remove the air filter breather pipes (RH -10) (LH-6).
- (32) Remove the hose clips at the turbo charger and intake duct then remove the air filters complete with elbow hose.
- (33) Remove the hose clips and remove the intercooler to turbo charger hoses.
- (34) Remove the engine oil tank drain plug and copper washer and drain the oil into a suitable container. Fit new washer, replace and tighten the plug.
- (35) Disconnect the oil feed pipe from the engine and fit blanking caps to the pipe open ends.
- (36) Disconnect the water pipes at the elbow connections at the front of the engine and remove the elbows.
- (37) If the level in the fuel tank is below that of the return line disconnect the fuel return at the fuel tank access cover. Alternatively the fuel return pipe can be disconnected at 'A' bank fuel regulator ensuring the pipe does not lower allowing fuel to drain. Fit blanking caps to pipe open ends.
- (38) Remove the two M8 screws securing the clutch slave cylinder and remove the cylinder. It is permitted to leave the slave cylinder suspended on the hydraulic flexible pipe.
- (39) Remove the nuts and washers securing the engine support beams to the bulkhead. DO NOT REMOVE THE BOLTS AT THIS TIME.



CAUTION

Leave one lower bolt in each side of the intermediate and rear plates.

- (40) Remove the M6 bolts nuts and washers securing the intermediate and rear plates to the vehicle body.
- (41) Remove the two M10 bolts, nuts and washers securing the engine crossbrace to the rear plate.
- (42) Remove the two M8 bolts securing the exhaust silencer assembly to the rubber hangers.

WARNING

ENSURE THE ENGINE TROLLEY AND VEHICLE ARE PARALLEL. CHECK BY LOOKING ALONG THE ENGINE SUPPORT BEAMS AND TOP SURFACE OF THE TROLLEY.

NOTE

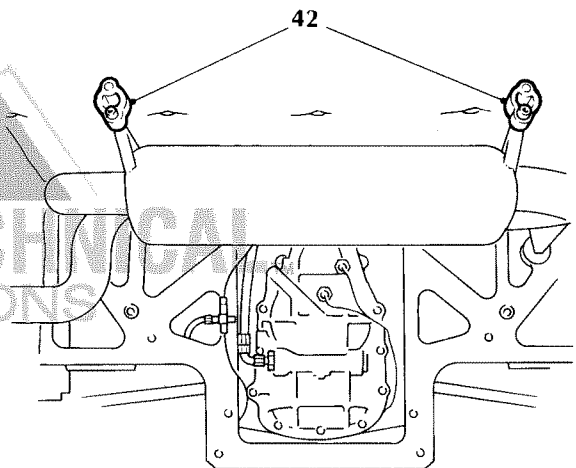
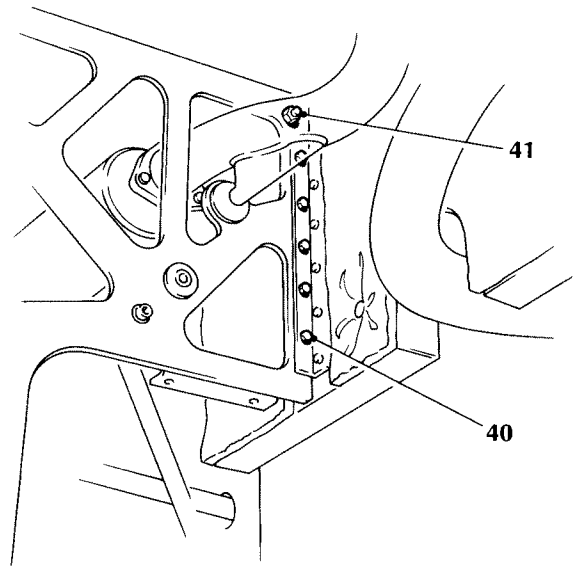
If the engine and transaxle are to be separated when removed from the vehicle, it is recommended that the two bottom bolts attaching the transaxle to the engine be removed at this stage.

- (43) Position the engine trolley directly beneath the engine and transaxle.
- (44) Raise the trolley sufficiently to take the weight of the engine but not raise the vehicle.
- (45) Remove the remaining M6 bolts, nuts and washers securing the intermediate and rear plates.
- (46) Remove the two M10 engine beam attachment bolts.

CAUTION

During the following operations conduct a detailed inspection to ensure no services are still attached to the engine and transaxle assembly.

- (47) Slowly raise the vehicle on the lifting apparatus to clear the top of the plenum chambers.
- (48) Slowly lower the engine trolley to minimum height and remove from under the vehicle.



B Preparation for Transit

This section details the requirements for preparing the engine for transit.

Special Tools

The following special tools will be required:

Hoisting equipment - 500Kg (1100lbs)

Webbing sling.

Materials

The following special tools will be required:

JaguarSport approved container/pallet

Blanking plugs

Cushioning material

Desiccant

Identification labels.

NOTE

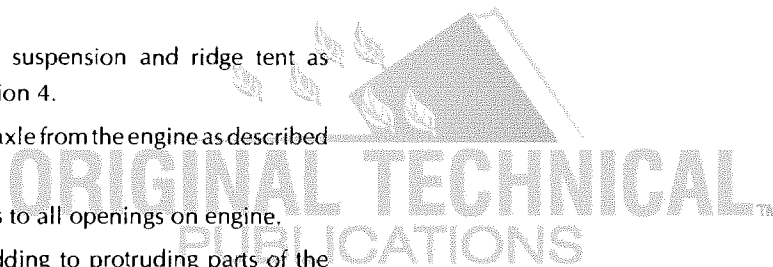
The engine must be returned with all components attached to facilitate hot testing following repair.

- (1) Remove the exhaust system as described in Section 9.
- (2) Remove the rear suspension and ridge tent as described in Section 4.
- (3) Remove the transaxle from the engine as described in Section 7.
- (4) Fit blanking plugs to all openings on engine.
- (5) Fit protective padding to protruding parts of the engine.
- (6) Cushioning material and bracing must be provided in the container to prevent accidental damage to the engine.
- (7) An identification label must be affixed to the engine, and an additional identification label must be affixed to the outside of the container.
- (8) The identification labels must contain all relevant details, including date of removal from vehicle and date of packaging.
- (9) Fit webbing sling under 'A' and 'B' bank exhaust manifolds and attach the sling to the hoist.
- (10) Inspect the webbing sling routing to ensure no external component is being damaged.

WARNING:

**DO NOT STAND BENEATH THE ENGINE
DURING HOISTING OPERATIONS.**

- (11) Ensure that the engine is balanced in the sling, hoist the engine into the packaging container/pallet and remove the webbing sling.
- (12) Ensure the engine is securely bolted onto the packaging container/pallet. Fit more bracing if required.
- (13) Place bags of desiccant around engine and close the packaging container/pallet.



3 ENGINE, TRANSAXLE AND DRIVELINE

A Unpacking

- (1) Check the packaging container/pallet for evidence of external damage caused during transit.
- (2) Remove the cover and check the details on the external identification label are the same as the details on the identification label attached to the engine.
- (3) Check the serial number on the identification label is the same as the serial number on the engine crankcase.
- (4) Fit webbing sling around the exhaust manifolds and attach the sling to the hoist.
- (5) Remove the bolts securing the engine to the container/pallet.
- (6) Inspect the webbing sling routing to ensure no external component is being damaged.

WARNING:

DO NOT STAND BENEATH THE ENGINE DURING HOISTING OPERATIONS.

- (7) Ensure the engine is balanced in the sling, hoist the engine from the packaging container/pallet and lower onto the engine trolley.

CAUTION:

Do not remove the blanking plugs fitted to the engine unless to fit an engine component or connect a service.

- (8) Remove identification label and any protective cushioning material from the engine.

WARNING:

WEAR SAFETY GOGGLES WHEN USING AN AIR JET AND WARN OTHER PERSONNEL TO KEEP CLEAR OF THE AREA.

- (9) Use compressed air at 193kPa (28psi) to remove small particles of cushioning material from inaccessible areas.
- (10) Prepare engine for installation in accordance with the following instructions.

B Installation Preparation

- (1) Assemble the driveline to transaxle as described in Section 5.
- (2) Assemble the clutch as described in Section 8.
- (3) Assemble the transaxle to the engine as described in Section 7.
- (4) Assemble the ridge tent and rear suspension as described in Section 4.
- (5) Assemble the exhaust system as described in Section 9

C Installation Procedure

- (1) Correctly position vehicle on lifting apparatus as described in Jacking and Lifting (Refer to Section 1.1).
- (2) Raise vehicle to suitable working height.
- (3) Fit additional supports to base of rear of the vehicle.
- (4) Position the engine and transaxle beneath the vehicle.
- (5) Fit gear selector rod over transaxle spigot and secure with spring pin.
- (6) Move the gear lever fully forward to prevent the engine fouling on the selector rod during installation.
- (7) Operate the engine trolley hydraulic pump to raise the engine until it is just entering the engine compartment.

CAUTION:

During lowering operations conduct a detailed inspection to ensure no services are snagging and being damaged.

- (8) Feed the throttle cable through the auxiliary drive belt and the fuel return pipe (if removed) through the plenums.
- (9) Remove the upright supports and slowly lower the vehicle making sure the engine bay intermediate plate and rear plate align with the corresponding mountings on the vehicle.
- (10) Refit upright supports.

CAUTION:

Ensure the bolts attaching the engine intermediate and rear plates to the vehicle body mountings can be fitted with thumb pressure. Do not force the bolts through. Adjust the height of the engine trolley or the vehicle to obtain perfect alignment.

- (11) Align the attachment holes in the engine bay intermediate and rear plate with the holes in the corresponding mountings and fit the M6 bolts, washers and nuts. Torque to 12-14Nm (9-10.5lbf ft)

- (12) Fit the two engine support beam to bulkhead M10 bolts, nuts and washers. Torque to 57-65Nm (42-48 ftlb).

NOTE

The M10 bolts should be fitted with the nut closest to the centre-line of the vehicle.

- (13) Lower the engine trolley and remove from beneath the vehicle.
- (14) Remove blanking caps and refit the oil tank feed pipes.
- (15) Fit intercooler to turbo hoses secure with hose clips.
- (16) Fit the water pipes to the elbow connections and secure with hose clips.
- (17) Remove the blanking caps and fit the fuel return pipe (if removed) to the fuel tank access cover.
- (18) Remove the blanking caps and fit the brake pipe at the engine bulkhead.
- (19) Fit air canister charge pipes and fit air canisters complete with elbow hoses to turbos and secure with hose clips.
- (20) Reconnect the turbo charger coolant pipe on both sides of engine.
- (21) Fit clutch slave cylinder using two M8 bolts and torque bolts to 20-23Nm (15-17 ftlb)
- (22) Reconnect electrical lead to starter motor.
- (23) Reconnect the body to engine earth lead.
- (24) Remove the upright supports and lower the car on the lifting apparatus.
- (25) Move the gear lever rearward to align the holes through the gear selector rod fork end and the engine mounted gear selector rod eye end.
- (26) Fit the M8 bolt to the selector rods and torque the bolt 20-30Nm (15-17 ftlb).
- (27) Remove the blanking caps from the oil cooler pipe and fit oil cooler pipe to the engine.
- (28) Remove the blanking cap and refit the oil tank breather pipe.
- (29) Fit the pipe/hose assembly from the intercooler to throttle bodies and secure with hose clips.
- (30) Fit the water by-pass pipe at the front of the engine.
- (31) Remove the blanking caps and fit the 'A' and 'B' bank fuel inlet pipes to the T piece.
- (32) Reconnect the fuel purge system to Y piece at the front of the engine.
- (33) Connect the engine electrical supply harness to the plugs on the bulkhead.

- (34) Refit the heater feed pipe and water pipe over the can cover.
- (35) Ensure the handbrake cables are routed correctly and refit to the calipers. Secure the cables to the rear engine plate using Tyrap.
- (36) Connect the air conditioning service unit and recharge the air conditioning system. Ensure the service unit pipes are connected as follows;
 - a) The red connector to the fluid filled pipe
 - b) The blue connector to the gas filled pipe.
- (37) Replenish engine oil using JaguarSport recommended oil. (See Engine Data)
- (38) Replenish the coolant system with 50% water/ Castrol antifreeze mix (see Engine Data) and bleed the coolant system.
- (39) Replenish brake fluid using Castrol SRF and bleed the brakes as described in Section 6.
- (40) Refit the radiator duct access panel and close the bonnet.
- (41) Check transaxle oil level and replenish if necessary using Castrol Racing XH21.
- (42) Fit road wheels and nuts and tighten nuts.
- (43) Lower the car to the ground and torque the wheel nuts to 520Nm (383 ftlb) and refit the security clips.
- (44) Reconnect the battery and switch the battery master switch to the ON position.
- (45) Reconnect the number plate light plug on the bumper and fit the bumper and venturi panels using M6 screws and M5 nuts and washers.
- (46) Refit the rear grille.
- (47) Road test the car and check for leaks in the fuel, lubrication, coolant and brake systems. Rectify all faults as necessary.

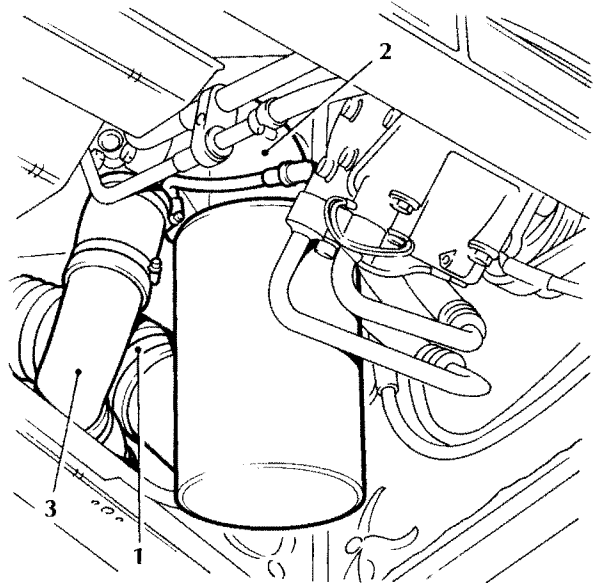
4 AIR INDUCTION HOSES

A Removal Procedure

CAUTION:

Suitably blank off open ports as components are removed.

- (1) Loosen hose clips and remove the hose from air inlet to air filter canister.
- (2) Loosen hose clips and remove air hose from air filter canister to turbo charger.
- (3) Loosen hose clips and remove air hose from turbo charger to intercooler.
- (4) Loosen hose clips and remove pipe/hose assembly from intercooler to throttle body. Remove the small bore hose from the idle air control valve to 'A' bank air inlet pipe.



B Installation Procedure

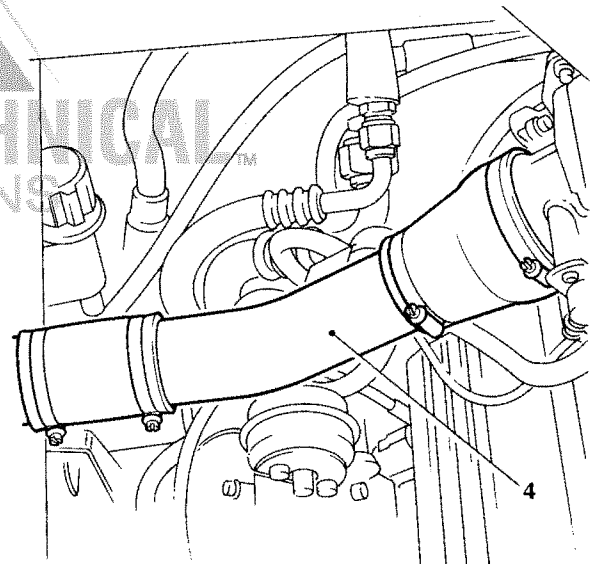
CAUTION:

Remove all blanks prior to assembly.

NOTE:

Before returning turbo charger hoses and hose clips to service, check the hoses and hose clips are undamaged. The hoses and hose clips must be renewed if damaged.

- (1) Fit pipe/hose assembly between throttle body and intercooler and secure with hose clips. Refit the small bore hoses to 'A' bank air inlet pipe to idle air control valve..
- (2) Fit air hose between intercooler and turbo charger and secure with hose clips.
- (3) Fit air hose between turbo charger and air filter canister and secure with hose clips.
- (4) Fit air hose between air filter canister and air inlet and secure with hose clips.



5 AIR CANISTER

A Removal Procedure

- (1) Remove inlet hose from air canister.
- (2) Remove the air canister breather pipe and 90° hose at the turbo charger.
- (3) Remove the air canister complete with 90° hose.
- (4) Remove circlip and remove top cover of the air canister.
- (5) Remove clip securing the air filter to the cover, remove air filter element and discard.
- (6) Wipe clean inside and outside surface of air canister with clean lint free cloth.

NOTE:

Replace air canister if cracked or distorted.

B Installation Procedure

- (1) Check new filter element is oil impregnated.

NOTE:

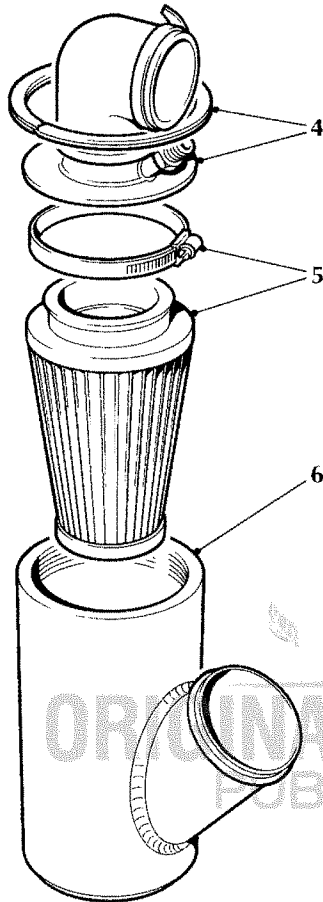
Oil impregnated air filters are red in colour.

- (2) Fit filter element to air canister and secure with a clip..
- (3) Fit top cover and secure with circlip.

CAUTION:

If the air canister is not to be fitted directly to the vehicle, suitably blank off the open connections to prevent filter element contamination.

- (4) Fit air filter canister and 90° hose to turbo charger and secure hose clips, ensuring that the base of the air filter will not foul the venturi floor when fitted.
- (5) Fit air canister breather pipe and inlet hose, secure hose with clip.



6 OIL TANK

A Removal Procedure

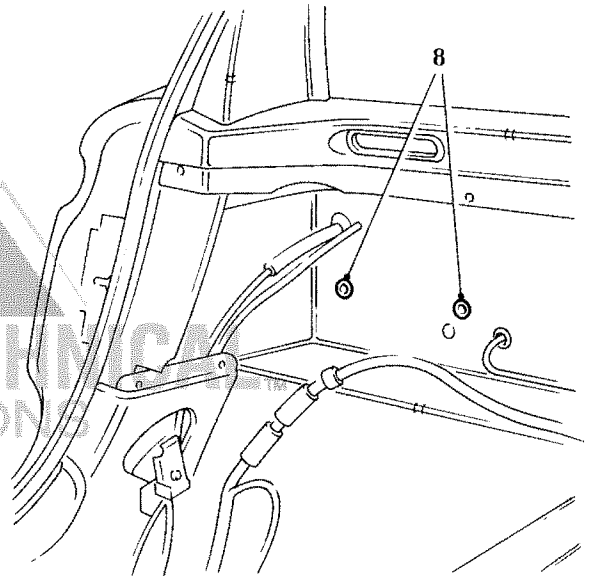
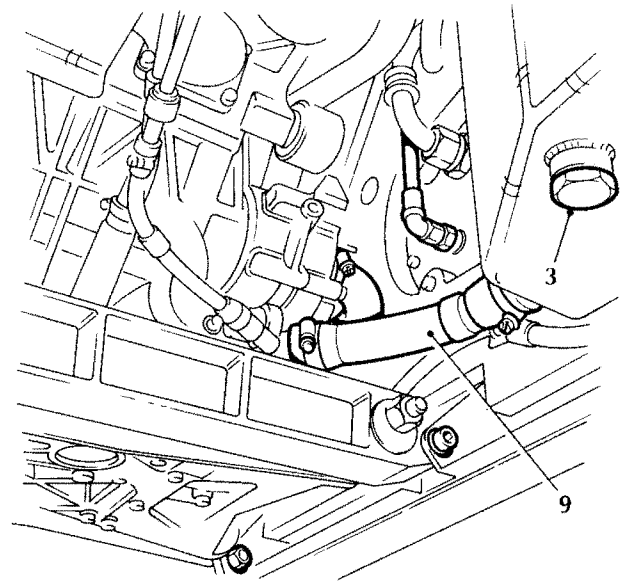
CAUTION:

Suitably blank off all open ports as components are removed.

NOTE

The following procedure deals with the removal of the oil tank with the engine in situ. If the engine has been removed certain operations may be disregarded.

- (1) Position the car on lifting apparatus as described in Lifting and Jacking-Section 1.1.
- (2) Raise the car to a suitable height and remove the venturi panel.
- (3) Remove the oil tank drain plug and drain the oil into a suitable container. Replace the copper washer, refit the drain plug and tighten
- (4) Remove the air canister as described in Section 3.5.
- (5) Remove the intercooler to throttle body pipe/hose assembly as described in Section 3.4.
- (6) Slide and tilt the RH seat fully forwards to gain access to the rear bulkhead trim.
- (7) Remove the screws securing the top roll panel, RH rear quarter panel and the backboard and remove the trim panels (Refer to Section 2).
- (8) Remove the two M8 screws securing the oil tank to the bulkhead.
- (9) Disconnect the oil pipes from the top and bottom of the oil tank. Fit blanking caps to pipe open ends.
- (10) Lift the tank sufficiently to disengage the oil tank locating peg from the bulkhead-mounted bracket and remove the tank from underneath the vehicle.

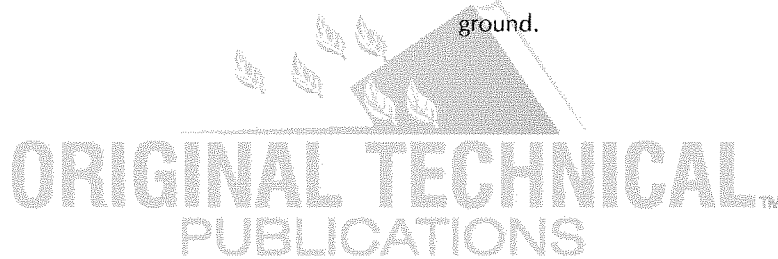


B Installation Procedure

NOTE

The following procedure deals with the installation of the oil tank with the engine in situ. If the engine has been removed from the vehicle certain operations may be disregarded.

- (1) Locate the oil tank in the bulkhead-mounted bracket.
- (2) Fit the two M8 bolts through the bulkhead and secure the tank. Torque the bolt to 30 to 34Nm (22 to 25 ftlb).
- (3) Remove the blanking caps and reconnect the oil pipes to the top and bottom of the oil tank.
- (4) Refit the air canister as described in Section 3.5.
- (5) Refit the intercooler to throttle body pipe/hose assembly as described in Section 3.4.
- (6) Refit the RH quarter panel, backboard and top roll trim panels.
- (7) Fill the oil tank with the recommended quantity and specification of oil. See General Specification-Section 1.
- (8) Run the engine and check for oil leaks. Rectify as necessary.
- (9) Refit the venturi panel and lower the vehicle to the ground.



7 OIL PUMP

A Removal Procedure

CAUTION:

Suitably blank off all open ports as components are removed.

NOTE

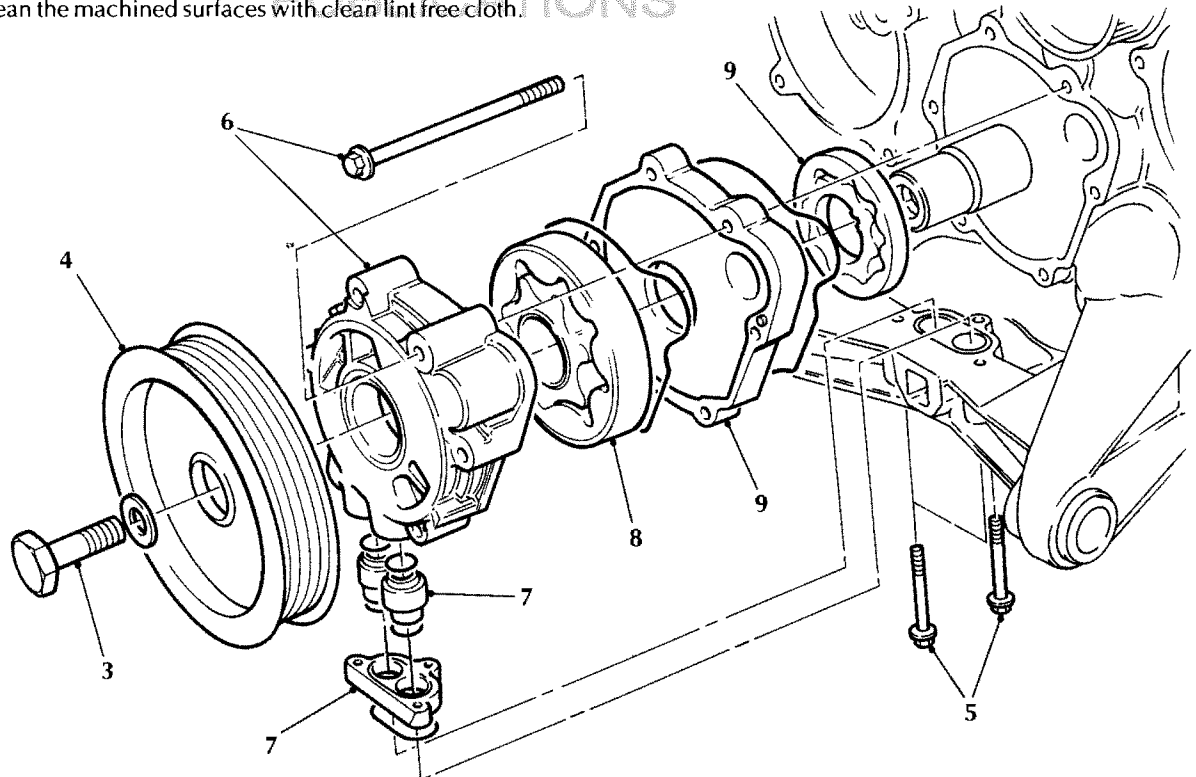
The following procedure must be carried out with the engine out, drained of oil and the transaxle removed.

The engine should be mounted on a stand or table to allow access to the sump retaining bolts.

- (1) Fit special tool to rear of the crankshaft.
- (2) Remove the auxiliary drive belt as described in Section 3.12.
- (3) Hold the crankshaft with a socket spanner on the special tool, and loosen the crankshaft pulley bolt.
- (4) Remove crankshaft pulley.
- (5) Remove the three M5 bolts securing the pump to the sump.
- (6) Remove bolts from oil pump housing and remove oil pump housing.
- (7) Remove adapter plate and connector tubes and discard the 'O' rings.
- (8) Remove scavenge pump and separate pump rotor from pump housing.
- (9) Remove sandwich plate, discard 'O' ring and remove pressure pump rotor.
- (10) Clean the machined surfaces with clean lint free cloth.

B Installation Procedure

- (1) Fit pressure pump rotor to sandwich plate.
- (2) Fit new 'O' ring seal to sandwich plate and fit sandwich plate to engine block.
- (3) Fit pump rotor to scavenge pump housing.
- (4) Fit new 'O' ring seals to connector tubes and adapter plate..
- (5) Fit adapter plate and connector tube assembly to the pump housing.
- (6) Fit new 'O' ring seal to the pump housing and fit housing to front of engine.
- (7) Secure the oil pump to the front cover with M6 bolts and through the bottom of the sump with 3 M5 bolts. Torque M6 bolts 11Nm (8ftlb) and M5 bolts 6Nm(4.5ftlb).
- (8) Fit crankshaft pulley, apply Loctite 270 to the threads and fit the pulley bolt.
- (9) Hold the special tool on the rear of the crankshaft with a socket spanner and tighten the crankshaft pulley bolt. Torque bolt 300Nm (221 ftlb)
- (10) Remove special tool from rear of the crankshaft.
- (11) Refit auxiliary drive belt as described in Section 3.12.

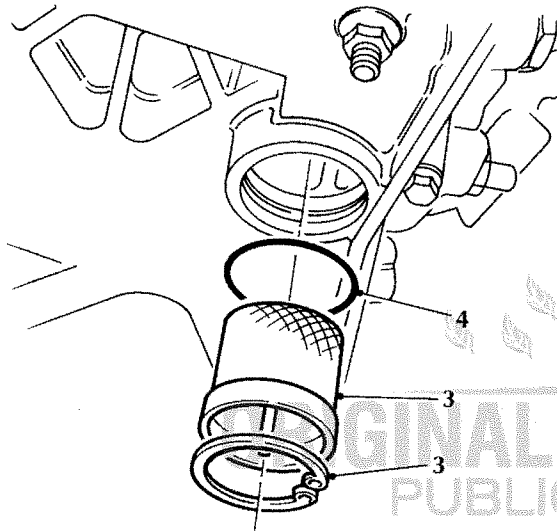


8 OIL SUMP FILTER

A Removal Procedure

WARNINGS:

- 1 **SOLVENTS ARE TOXIC AND MUST ONLY BE USED IN WELL VENTILATED WORK AREAS. AVOID INHALING THE SOLVENT FUMES AND WEAR PROTECTIVE CLOTHING.**
- 2 **WHEN USING AN AIR JET WEAR GOGGLES AND WARN OTHER PERSONNEL TO KEEP CLEAR OF THE AREA.**



- (1) Position the vehicle on lifting apparatus and raise the vehicle to a suitable working height. (Refer to Jacking and Lifting, Section 1.1)
- (2) Remove the venturi panel. (Refer to Section 2)
- (3) Remove circlip and remove oil filter. Allow the oil to drain into a suitable container
- (4) Remove and discard 'O' ring seal.
- (5) Wash gauze filter in approved cleaning solvent.
- (6) Dry clean the filter with an air jet.

B Installation Procedure

CAUTION:

Ensure the O ring seal does not fall out of the machined grooves and is not twisted on installation.

- (1) Fit new 'O' ring seal to sump casing.

CAUTION:

Fit the circlip with the bevelled edge into the engine.

- (2) Re-fit gauze filter and secure with circlip.
- (3) Fit the venturi panel. (Refer to Section 2)
- (4) Lower the vehicle to the ground and remove from the lifting apparatus.

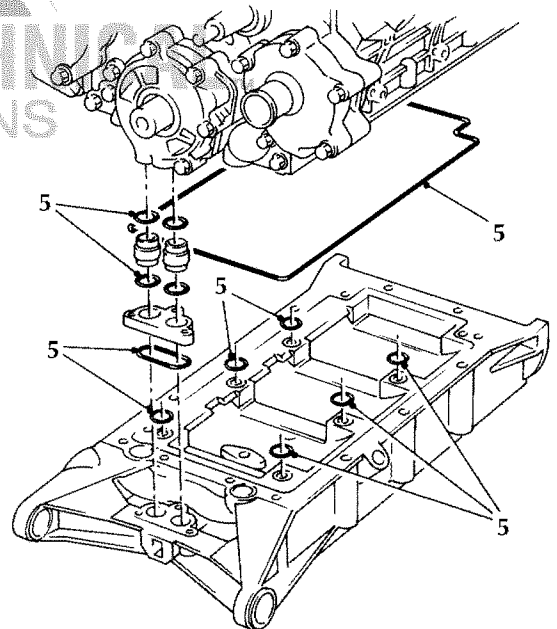
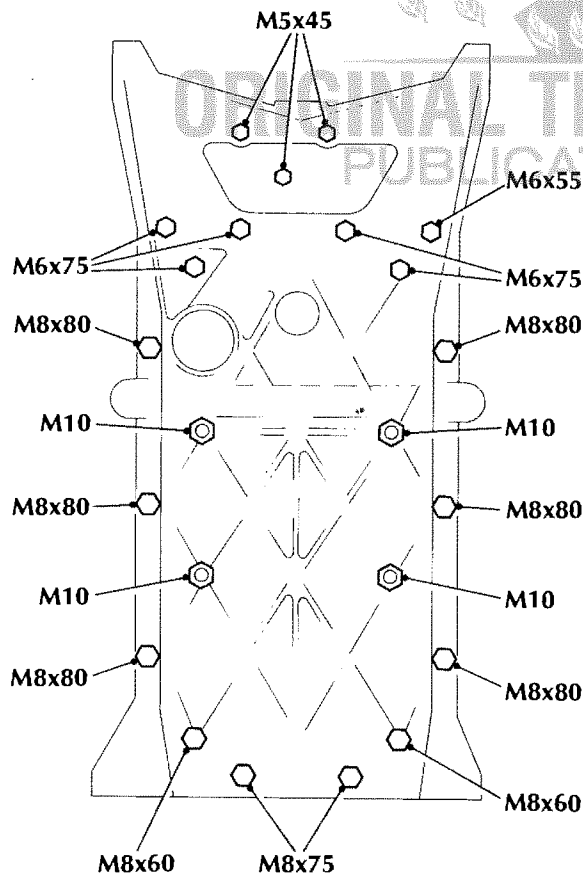
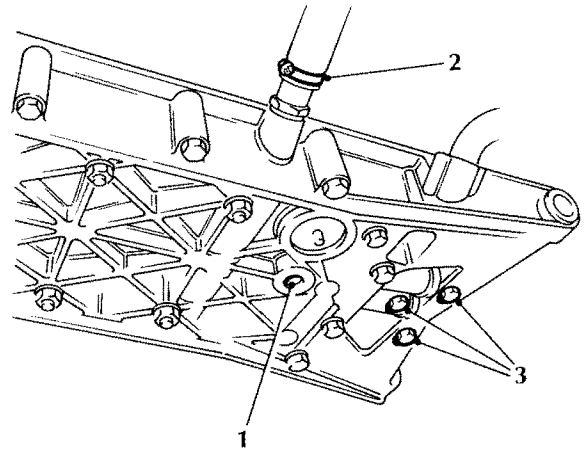
9 OIL SUMP

A Removal Procedure

NOTE

The following procedure should be carried out with the engine removed and on a suitable stand to allow access to the sump fastenings.

- (1) Remove the sump drain plug and drain the sump oil into a suitable container.
- (2) Disconnect the turbo charger oil drain hoses at the sump union.
- (3) Remove the three M5 bolts securing the oil pump adaptor to the sump.
- (4) Remove the combination of M6 bolts, M8 bolts and M10 nuts and washers securing the sump to the engine block. Refer to illustration for locations of fixings.
- (5) Remove the sump making a note of the position of the 'O' ring seals.
- (6) Fit a blanking plate to the oil pump adapter.



B Installation Procedure

CAUTION:

Ensure the O ring seals do not fall out of the machined grooves, and are not twisted on installation.

NOTE

If new sump is to be fitted, fit a new oil fitting and 'O' ring, sump filter, turbo charger oil unions and bonded seals to the sump prior to fitting.

- (1) Fit new 'O' ring seals to sump casing.
- (2) Remove the blanking plate from the oil pump adapter flange. Ensure 'O' ring seal is in place.
- (3) Fit sump casing to engine block and secure with flanged bolts, and flanged nuts with washers. Ensure fastenings are fitted at the correct location. (Refer to illustration)
- (4) Torque load M6 bolts 9.5 to 12 Nm (7 to 9 ftlb), M8 bolts 23 to 27 Nm (17 to 20 ftlb) and M10 nuts 50 Nm (38ftlb).
- (5) Fit the three M5 bolts securing the oil pump adapter and torque bolts 5 to 7Nm (3.5 to 5 ftlb).
- (6) Fit the two M16 bolts, washers and nuts securing sump to engine mounting beams. Torque nuts 127 to 145Nm (93.5 to 107 ftlb).
- (7) Refit turbo charger oil hoses to sump and secure with clips.

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10 ENGINE OIL FILTER (IN-SITU)

A Removal Procedure

Special Tools

The following special tool will be required:

Strap Wrench.

JaguarSport body covers

Lint-free cotton waste

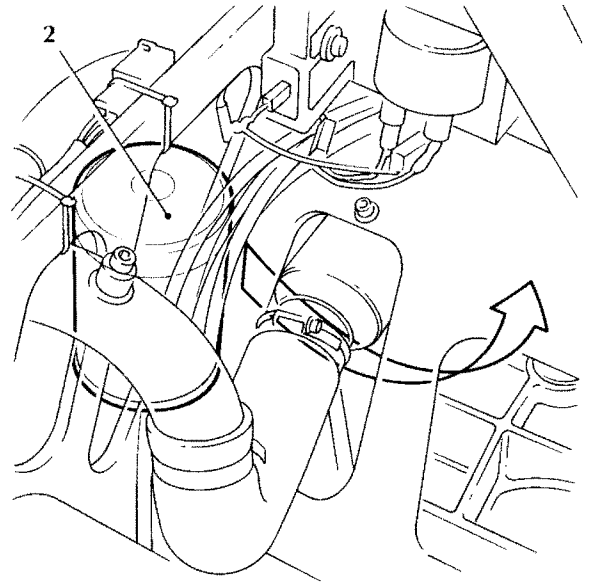
CAUTION:

Ensure JaguarSport approved body covers are fitted before starting work.

- (1) Fit strap wrench around oil filter body and turn oil filter counter-clockwise.

NOTES:

- 1 *It is permissible to loosen the oil filter by tapping the rimmed base with a screw driver, take care to avoid damaging other components.*
- 2 *The oil filter is fitted with non return valves, but it is recommended to place lint-free cotton waste around base of the oil filter to absorb any oil that does leak during removal.*
- (2) Remove oil filter from mounting and pass under engine bay intermediate frame and out through lightening holes in engine bay bridge panel.
- (3) Discard oil filter.

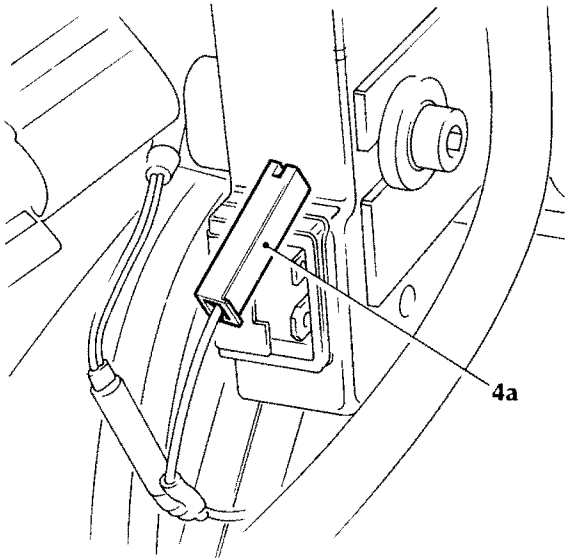


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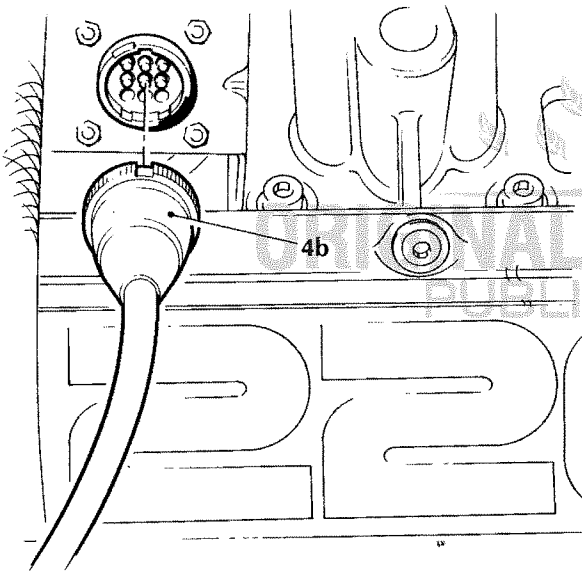
B Installation Procedure

CAUTION:

Ensure JaguarSport approved body covers are fitted before starting work.



- (1) Apply a thin film of clean engine oil to the oil filter seal, then feed the filter through the engine bay bridge panel and intermediate plate into position.
- (2) Screw the oil filter clockwise onto the threaded mounting until the seal makes contact with the sealing surface of the cylinder block.
- (3) Tighten oil filter a further 1/2 turn.
- (4) Before attempting to start the engine observe the following procedure;



- (a) Disconnect lead from the ignition amplifier on the engine bay bridge panel.
 - (b) Disconnect speed sensor loom on RH side of the engine.
 - (c) Remove spark plug covers, HT leads and spark plugs.
 - (d) Crank the engine on the ignition to build up oil pressure.
 - (e) Switch ignition to OFF, fit spark plugs, HT leads and spark plug covers.
 - (f) Connect lead to ignition amplifier.
 - (g) Connect speed sensor loom on RH side of the engine.
- (5) Start the engine and check there are no oil leaks between the oil filter seal and cylinder block.

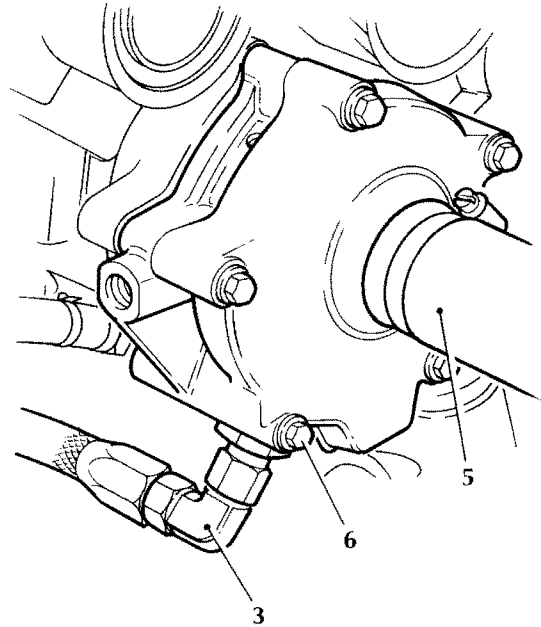
11 WATER PUMP ASSEMBLY

A Removal Procedure

CAUTION:

Suitably blank off open ports as components are removed.

- (1) Position the car on lifting apparatus as described in Jacking and Lifting-Section 1.1.
- (2) Remove the screws and remove the venturi panel.
- (3) Disconnect the turbo charger water feed pipes and allow the coolant to drain into a suitable container. Fit blanking caps to the open ends prevent the ingress of dirt.
- (4) LH water pump only, remove the air conditioning compressor lower front mounting bolt and spacer.
- (5) Loosen the hose clips and remove the elbow hose at the front of the water pump.
- (6) Remove the five M6 flange head bolts securing water pump to engine block.
- (7) Remove water pump.
- (8) Remove and discard 'O' ring seal from T piece in outlet pipe.



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B Installation Procedure

CAUTIONS:

- 1** *Ensure the O ring seal is not twisted on installation.*
 - 2** *Remove all blanks prior to assembly*
- (1) Fit 'O' ring seal to T piece in outlet pipe.

CAUTION:

When installing the water pump ensure the O ring seal is not pinched.

- (2) Fit water pump to engine block and secure with five M6 flange head bolts. Ensure drive gears are located correctly.
- (3) Torque load bolts 9.5 to 12 Nm (7 to 9ftlb).
- (4) LH water pump only, fit the air conditioning compressor lower front mounting bolt and spacer.
- (5) Fit elbow hose to front of water pump.
- (6) Remove the blanking caps and fit the turbo charger water feed pipes to the pump.
- (7) Replenish the coolant system.
- (8) Run the engine to bleed and pressure check the cooling system.
- (9) Fit venturi panel and lower the car to the ground.

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12 AIR CONDITIONING COMPRESSOR

A Removal Procedure

Special Tools

JaguarSport approved body covers

Electro-hydraulic vehicle lifting apparatus

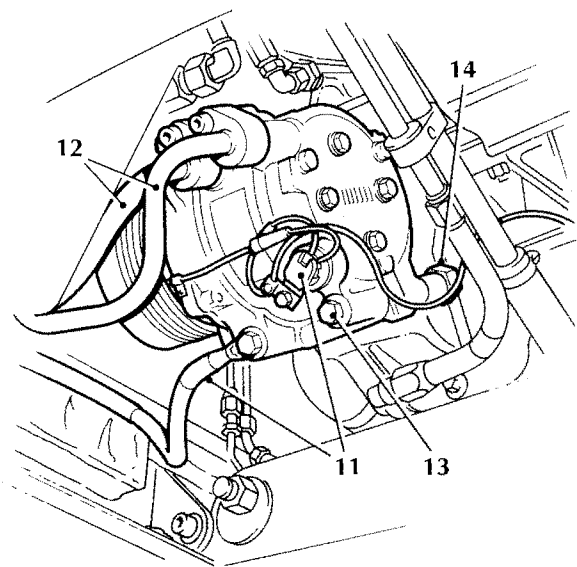
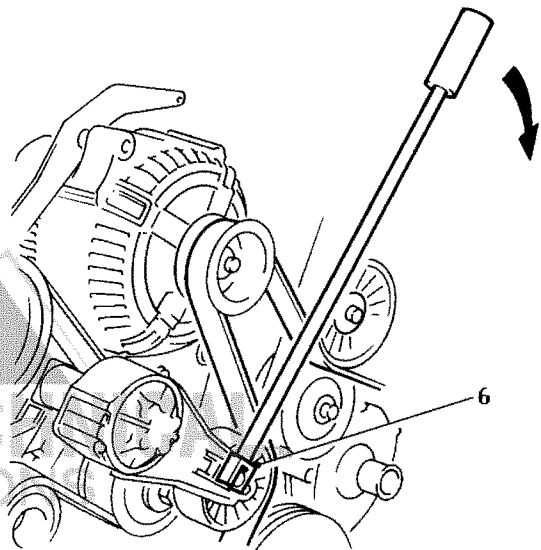
Supports for the rear axle uprights

Air Conditioning service unit

NOTE

The following procedure deals with the removal of the air conditioning compressor with the engine installed. If the engine has been removed certain procedures may be disregarded.

- (1) Position the vehicle on lifting apparatus. (Refer to section 1.1).
- (2) Fit JaguarSport body covers
- (3) Disconnect the battery.
- (4) Open the bonnet and remove the radiator duct access panel.
- (5) Connect the air conditioning service unit to the car and evacuate the air conditioning system.
- (6) Fit 1/2in. square drive wrench in socket hole of dynamic tensioner.
- (7) Move the wrench to oppose spring tension in dynamic tensioner until the auxiliary drive belt is loose enough to be easily removed.
- (8) Gradually reduce pressure on wrench to allow dynamic tensioner to return to neutral position and remove wrench.
- (9) Inspect drive belt for serviceability. If in doubt renew the drive belt.
- (10) Raise the vehicle to a comfortable working height and remove the venturi panel. (Refer to Section 2).
- (11) Disconnect the air conditioning compressor clutch electrical leads and the engine earth lead.
- (12) Remove the manifold pipes from the compressor and fit blanking caps to the pipe open ends.
- (13) Support the compressor and remove the three M10 bolts securing air conditioning compressor to engine block and water pump.
- (14) Remove air conditioning compressor from the engine and retrieve the spacers (early engines have 4 spacers, later engines have only 1). Note the position of the spacers.



B Installation Procedure

- (1) Remove any blanking caps fitted to the compressor.
- (2) Fit the air conditioning compressor to engine. Position the spacers and secure the compressor with the M10 flange head bolts. Torque load bolts 49 to 54 Nm (36 to 40 ftlb).
- (3) Remove the blanking caps and fit the supply pipes to the compressor.
- (4) Connect air conditioning compressor clutch electrical leads and engine earth lead.
- (5) Fit 1/2in square drive wrench in socket hole of dynamic tensioner.
- (6) Move wrench in direction to oppose spring pressure.
- (7) Fit the drive belt.
- (8) Gradually reduce pressure on wrench to allow dynamic tensioner to take up slack in drive belt.
- (9) Remove the wrench.
- (10) Connect the air conditioning service unit and charge the air conditioning system.
- (11) Connect the battery.
- (12) Run the engine and check the air conditioning system for correct function and the pump for leaks. Rectify any faults as necessary.
- (13) Fit the venturi panel.
- (14) Fit the radiator duct access panel and close the bonnet.
- (15) Remove the vehicle from the lifting apparatus and remove the body covers.

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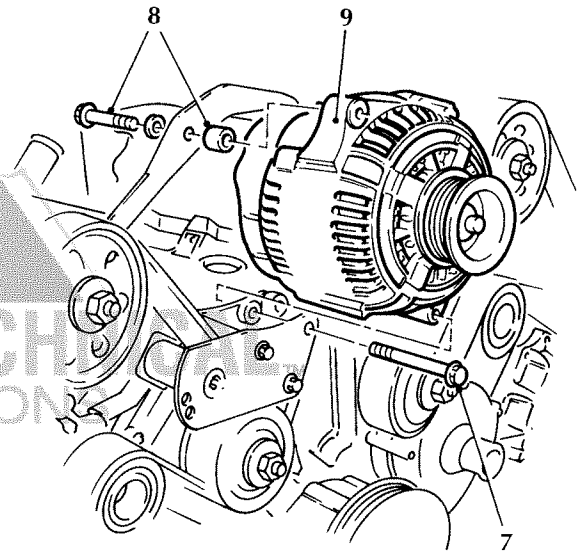
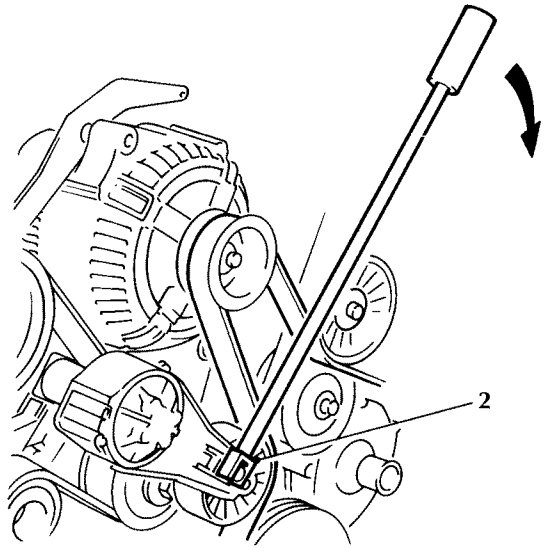
13 ALTERNATOR

A Removal Procedure

NOTE

The following procedure deals with the removal of the alternator with the engine installed. If the engine has been removed certain procedures may be disregarded.

- (1) Remove the right hand plenum and injector housing. (Refer to Section 3.20)
- (2) Fit 1/2in. square drive wrench in socket hole of dynamic tensioner.
- (3) Move the wrench to oppose spring tension in dynamic tensioner until the auxiliary drive belt is loose enough to be easily removed.
- (4) Gradually reduce pressure on wrench to allow dynamic tensioner to return to neutral position and remove wrench.
- (5) Inspect drive belt for serviceability. If in doubt discard drive belt.
- (6) Disconnect electrical plug at the back of the alternator.
- (7) Remove the M10 bolt and spacer from alternator lower attachment.
- (8) Support alternator and remove the M8 bolt and spacer attaching the alternator to the link.
- (9) Remove alternator.



B Installation Procedure

- (1) Position alternator at front of engine.
- (2) Locate the spacer and fit the M10 bolt through alternator lower attachment and secure.
- (3) Locate the spacer and fit the M8 bolt alternator upper attachment and secure.
- (4) Torque load M10 bolts 49 to 54 Nm (36 to 40 ftlb) and M8 bolts 23 to 27 Nm (17 to 20 ftlb).
- (5) Connect the electrical plug at the back of alternator.
- (6) Fit 1/2 in. square drive wrench in socket hole of dynamic tensioner.
- (7) Move wrench in direction to oppose spring pressure.
- (8) Fit drive belt.
- (9) Gradually reduce pressure on wrench to allow dynamic tensioner to take up slack in drive belt.
- (10) Remove wrench.
- (11) Refit the plenum and injector housing. (Refer to Section 3.20)



14 CAM COVERS

A Removal Procedure

NOTE

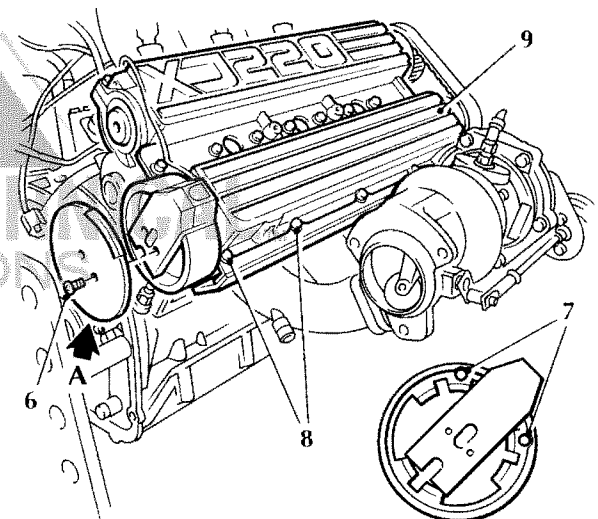
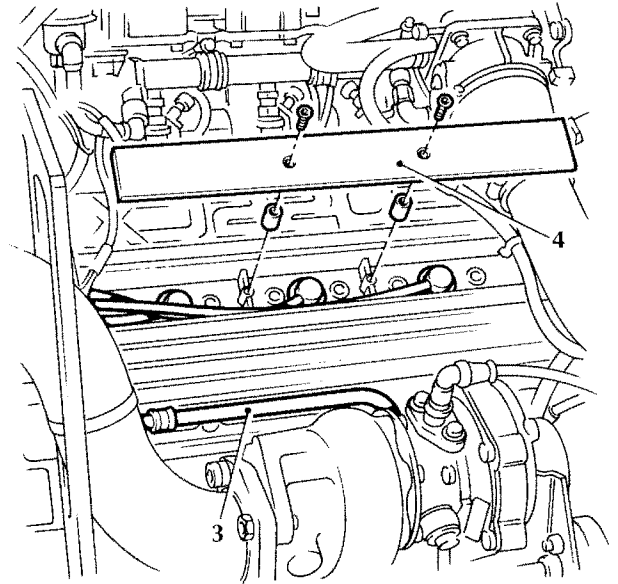
The following procedure deals with the removal of the cam covers with the engine installed. If the engine has been removed certain procedures may be disregarded.

- (1) Open the engine tailgate and fit JaguarSport body covers.
- (2) Disconnect the battery.
- (3) Disconnect the turbo charger water return pipe at the turbo and allow the coolant to drain. Remove the pipe from the thermostat housing, remove the 'P' clips and remove the pipe. Fit blanking caps to the open ends.
- (4) Remove the two M5 screws securing the spark plug lead cover and remove the cover.
- (5) Disconnect the leads from the spark plugs
- (6) On 'A' bank remove the two M4 screws and remove cover from synchroniser.
- (7) Remove top two M5 bolts in synchroniser casing.

NOTE:

If top two bolts are not accessible, push the car along in gear to align bolts with slots in synchroniser disc.

- (8) Remove the eighteen button head screws securing the cam cover and collect the heatshield.
- (9) Remove cam cover.
- (10) Remove and discard cam cover inner and outer gaskets.



VIEW A

B Installation Procedure

- (1) Fit cam cover inner gasket to the cylinder head. Ensure gaskets are correctly seated and not twisted.
- (2) Fit the outer gasket to the cover. Ensure the gaskets are positioned correctly and are not twisted.
- (3) Fit cam cover and correctly position heat shield.
- (4) Fit cam cover M6 screws and torque load 9.5 to 12 Nm (7 to 9 ftlb).
- (5) Fit two top bolts in synchroniser casing.

NOTE:

If top two bolts are not accessible, push the car along in gear to align bolts with slots in synchroniser disc.

- (6) Fit the synchroniser cover and secure with two M4 screws.
- (7) Reconnect the spark plug leads and fit the cover.
- (8) Remove the blanking caps, fit turbo charger water return pipes and secure the 'P' clips.
- (9) Replenish and bleed the cooling system.
- (10) Connect the battery.
- (11) Run the engine and check for coolant and engine oil leaks. Rectify any faults as necessary.
- (12) Remove the body covers and close the tailgate.

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15 CAM SHAFT DRIVE BELTS

A Removal Procedure

Special Tools

The following special tool will be required.

Thin spanner.

Cam pulley locking tool

CAUTION:

Ensure the controlling cylinder for the affected bank is in the valve overlap position. LH 'B' bank controlling cylinder is No. 1. RH 'A' bank controlling cylinder is no. 6. In this configuration all the remaining valves on the affected bank will be closed.

NOTES:

- 1 Ensure absolute cleanliness of work area and tools.
- 2 Observe correct use of tools.

NOTE

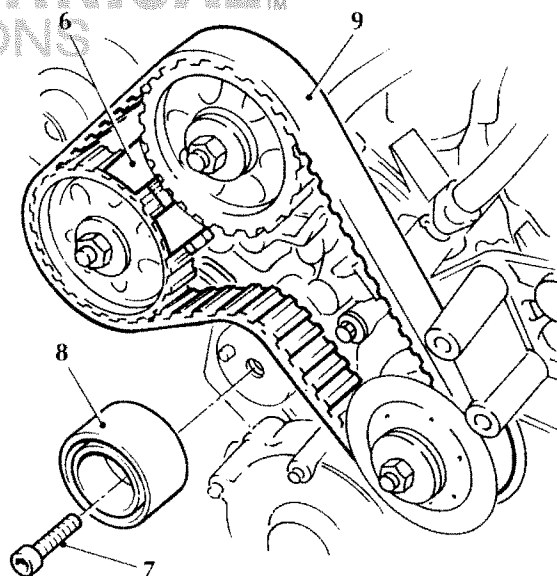
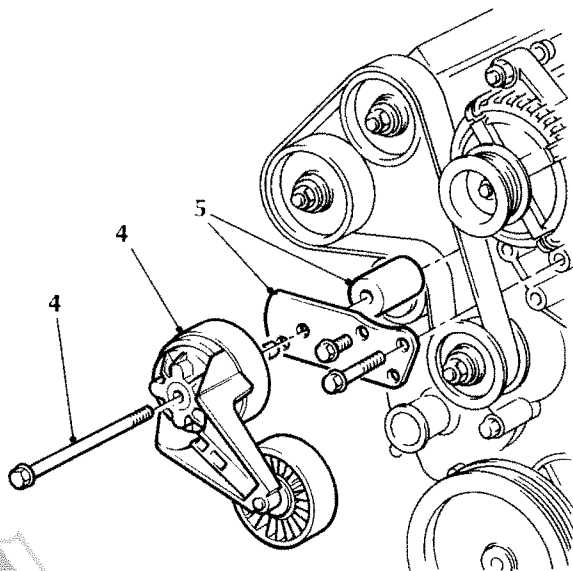
The following procedure should be carried out with the engine removed and placed on a suitable stand

- (1) Remove the water pump ('A' bank only - See Section 3.11) to facilitate removal of the camshaft drive belt
- (2) Fit 1/2in. square drive wrench in socket hole of dynamic tensioner.
- (3) Move the wrench to oppose spring tension in dynamic tensioner until auxiliary drive belt is no longer under load and remove belt.
- (4) Remove M10 bolt and remove dynamic tensioner from mounting plate.
- (5) Remove the bolts securing the mounting plate to the block, remove the plate and lift off the concentric spacer.
- (6) Lock cam pulleys with the locking tool.
- (7) Insert 10mm allen key into bolt in centre of cam belt tensioner.

CAUTION:

The tensioner has a sealed bearing. Do not wash the bearing. Wipe clean with clean lint-free cloth.

- (8) Loosen bolt and remove cam belt tensioner.
- (9) Remove cam belt.



B Installation Procedure

Special Tools

The following special tools will be required:

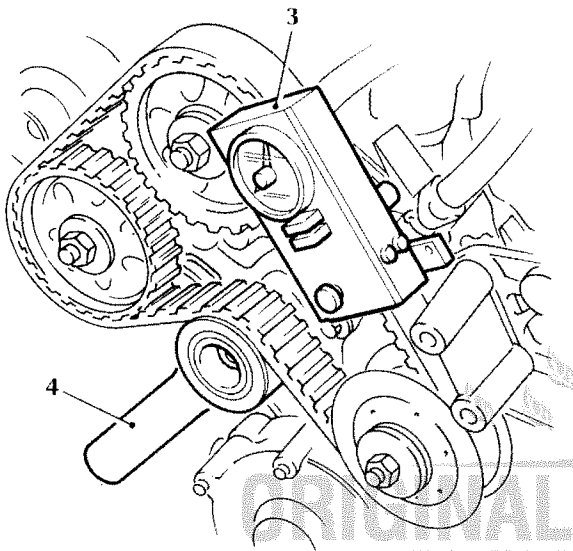
Thin spanner

Cam pulley locking tool

Staegger gauge.

CAUTION:

If refitting the original cam shaft drive belt ensure the belt is free from damage. If in doubt discard the belt and fit new.



- (1) Fit cam belt tensioner and secure loosely using 10mm allen key.
- (2) Fit cam belt over cam pulleys.
- (3) Fit a Staegger gauge to the longest run of the belt.
- (4) Using the thin spanner, turn the hexagon nut on the tensioner wheel to take up the slack in the belt. Nip up the tensioner fixing bolt and remove the pulley locking tool. Turn the hexagon on the tensioner until a belt tension of approximately 7 is achieved.
- (5) Tighten the tensioner wheel M10 bolt and torque 49 to 54Nm (36 to 40ftlb).
- (6) Remove the Staegger gauge.
- (7) Turn the engine manually in forward direction 2 revolutions to settle the belt.
- (8) Turn the engine in reverse direction until the point of lowest tension is found on the longest span of the belt.
- (9) Re-set the belt tension to 4.5 on the Staegger gauge on this span.
- (10) Fit concentric spacer and mounting plate.
- (11) Secure dynamic tensioner to the mounting plate with M10 flange head bolt and torque tighten 49 to 54Nm (36 to 40 ftlb).
- (12) Fit 1/2in. square drive wrench in socket hole of dynamic tensioner.
- (13) Move the wrench to oppose spring tension in dynamic tensioner and refit auxiliary drive belt.
- (14) Slowly allow the dynamic tensioner to take up the slack in the alternator drive belt.
- (15) Refit the water pump, see Section 3.11.

16 CAM SHAFT DRIVE PULLEY

A Removal Procedure

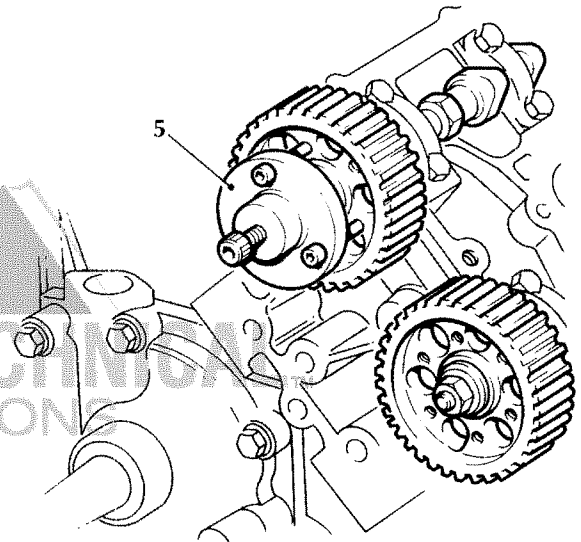
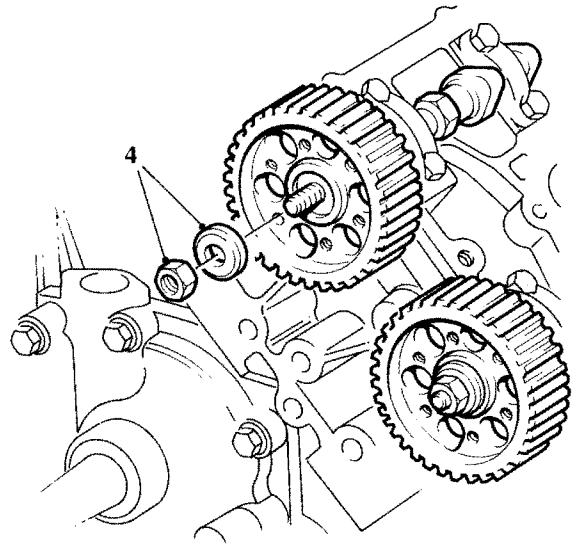
Special Tools

The following special tools will be required:

Wheel Puller.

NOTES:

- 1 Ensure absolute cleanliness of work area and tools.
 - 2 Observe correct use of tools.
-
- (1) For 'A' bank set No. 6 cylinder to TDC (top dead centre) non-firing; for 'B' bank set No. 1 cylinder to TDC non-firing.
 - (2) Remove the cam shaft drive belt. (Refer to Section 3.15)
 - (3) Remove the cam cover (Refer to Section 3.14)
 - (4) Using a spanner on the cam shaft hexagon to prevent rotation, remove the nut and washer securing the cam shaft drive pulley.
 - (5) Fit the wheel puller and remove the camshaft drive pulley



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B Installation Procedure

Special Tools

Dial Test Indicator (DTI)

Timing Disc and Pointer

Materials

The following materials will be required:

Fine grade grinding paste.

NOTE

When rotating the cam shafts or crankshaft with the drive belts removed, take care to ensure piston/valve contact does not occur.

- (1) Clean tapered end of camshaft and ensure no thread damage.
- (2) Lap in the camshaft drive pulley on the camshaft using fine grade grinding paste.
- (3) Thoroughly clean and degrease the camshaft before fitting the pulley.
- (4) Fit camshaft drive pulley to camshaft.
- (5) Apply Loctite 270 to the cam shaft pulley stud and fit the washer and nut finger tight to secure cam shaft drive pulley.
- (6) Set No. 6 piston to TDC for 'A' bank and No. 1 piston for 'B' bank.
- (7) Set 'cylinder at TDC' cams to overlap such that the cams just touch the followers but the valves are not open.
- (8) Fit the camshaft drive belt. (Refer to Section 3.15)
- (9) Tighten the cam pulley sufficiently to lock the pulley to the taper.
- (10) Rotate the engine to 108° ATDC (after top dead centre) for inlet cam and 108° BTDC (before top dead centre) for exhaust cam (non-firing stroke).
- (11) Using the DTI (dial test indicator) on the relevant cam follower, (i.e. No. 1 or No. 6 inlet or exhaust) slacken the cam pulley nut sufficiently to allow the cam shaft to be rotated to achieve the point of maximum lift.
- (12) Tighten the cam pulley nut to 120Nm (88.5ftlb) using the cam shaft hexagon.
- (13) Repeat operations 9 to 11 for the second cam shaft of the relevant bank.

17 CAM SHAFT

A Removal Procedure

NOTES:

- 1 Ensure absolute cleanliness of work area and tools.
- 2 Observe correct use of tools.

- (1) Remove cam shaft cover and cam belt. (Refer to Sections 3.14 and 3.15).
- (2) Remove the camshaft pulleys. (Refer to Section 3.16)

CAUTION:

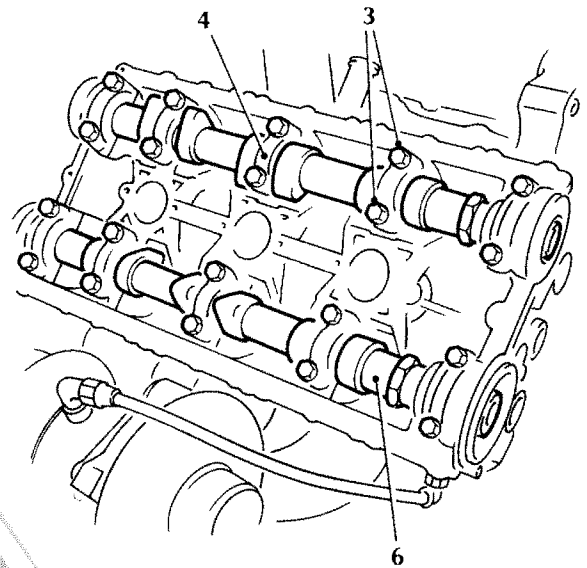
Ensure all valves on the affected bank are closed before attempting to loosen the bolts securing the cam shaft caps.

- (3) Loosen cam shaft bearing cap retaining bolts in pairs.
- (4) Remove cam shaft bearing cap retaining bolts and camshaft bearing caps.
- (5) Inspect the bearing caps for excessive wear and mechanical damage on bearing surfaces.

NOTE

Bearing caps are matched to the cylinder head. If wear is unacceptable return the engine to JaguarSport. NEVER REPLACE BEARING CAPS SINGLY.

- (6) Remove camshaft.
- (7) Remove the front and rear seals from the camshaft.



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B Installation Procedure

Special Tools

The following special tools will be required:

Dial Test Indicator (DTI)

Timing Disc and Pointer

- (1) Apply a thin film of clean engine oil to the bearing surfaces of the bearing caps.

CAUTION:

Fit the cam shaft so that when the cam shaft bearing caps are fitted and tightened down the valves remain closed.

- (2) Fit the cam shaft.
- (3) Fit the centre three cam shaft bearing caps.
- (4) Fit cam shaft bearing cap retaining bolts finger tight only.
- (5) Apply Loctite to the outside diameter of the new bearing cap seals and a smear of engine oil to the inner diameter, fit the seals.
- (6) Apply Loctite to the front and rear bearing cap joint faces.
- (7) Fit the outer bearing caps and retaining bolts.
- (8) Tighten the cam shaft bearing cap retaining bolts in pairs starting in the middle.
- (9) Torque load cam shaft retaining bolts 23 to 27 Nm (17 to 20 ftlb) .
- (10) Fit the cam pulleys and cam belt. (Refer to Sections 3.14 and 3.15)
- (11) Fit the cam shaft cover.

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18 PLENUM CHAMBERS

A Removal Procedure

Special Tools

JaguarSport recommended body covers

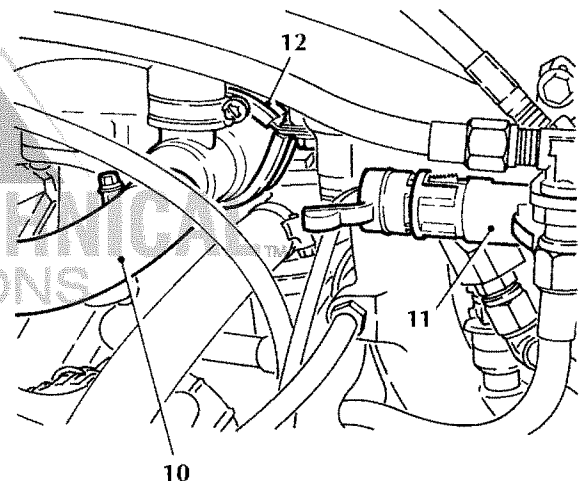
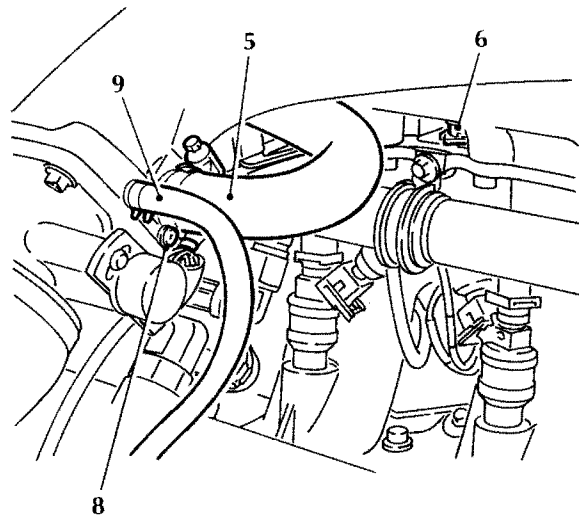
NOTES:

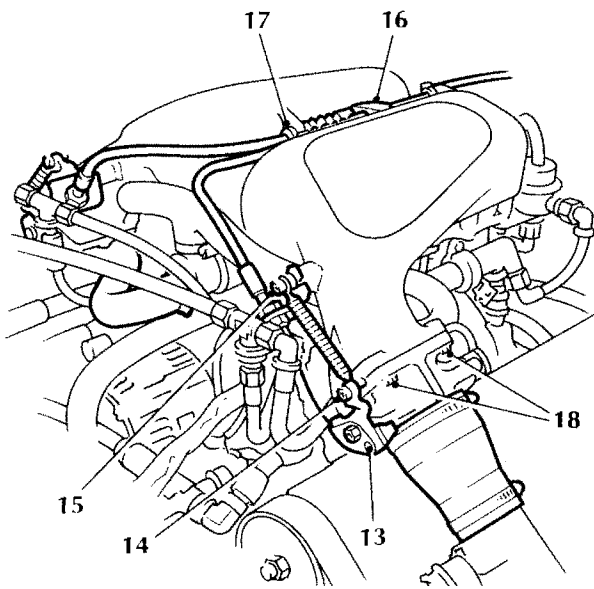
- 1 Ensure absolute cleanliness of work area and tools.
- 2 Observe correct use of tools.

NOTE

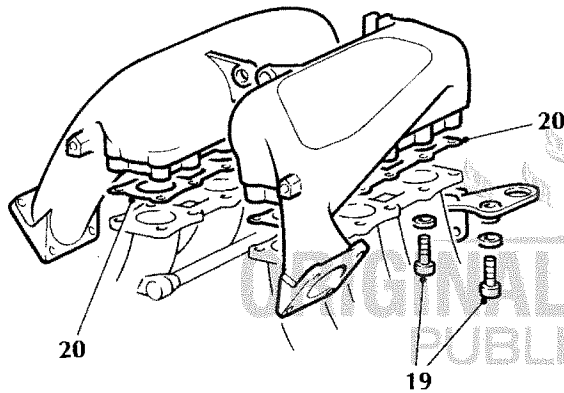
The following procedure deals with the removal of the plenums with the engine installed in the car. If the engine has been removed certain procedures may be disregarded.

- (1) Open the engine tailgate.
- (2) Fit JaguarSport body covers.
- (3) Disconnect the battery.
- (4) Loosen the hose clips and remove the hose/pipe assembly from the intercooler and throttle body.
- (5) Loosen the clips and remove the hose from the idle speed control valve to the plenum.
- (6) Remove the upper fuel rails (Refer to Sub-section 3.21).
- (7) Disconnect the electrical connections to the air thermister.
- (8) Remove the 'P' clip securing the boost control solenoid electrical lead to the plenum.
- (9) Loosen the hose clip and remove the SEM valve hose at the throttle body.
- (10) Loosen the hose clip at the idle speed control valve and remove the hose from the charge pipe to the idle speed control valve.
- (11) On 'B' bank only, lift the throttle position sensor socket off the bracket lug and disconnect the throttle position sensor connections.
- (12) On 'B' bank, remove the bolt retaining the idle speed control valve 'P' clip.





- (13) Rotate the throttle lever by hand and release the cable and drum from the slot.
- (14) Remove the bolt and nut securing the throttle return spring to the link.
- (15) Remove the bolt securing the throttle cable abutment bracket to the plenum.
- (16) Remove the locknuts and release the throttle cable from the cable junction block between the plenums.
- (17) Loosen the locknuts and remove the throttle cable from the plenum lugs.
- (18) Remove the screws securing the throttle body to the plenum, remove the throttle body and discard the gasket.
- (19) Remove the bolts securing the plenum to the injector housing, remove the plenum and collect the throttle position socket bracket spacers.
- (20) Discard the plenum gasket and fit blanking plugs to the injector housing ports.



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B Installation Procedure

Special Tools

Torque wrench 27Nm (20 ftlb)

NOTES:

- 1 Ensure absolute cleanliness of work area and tools.
- 2 Observe correct use of tools.

- (1) Remove the blanking plugs from the injector housing and fit a new plenum gasket.

NOTE

If a new plenum is to be fitted, fit the air thermister to the plenum base.

- (2) Fit the plenum to the injector housing and secure with M6 screws. On 'B' bank only, fit the throttle position socket bracket and spacers to plenum and secure with M6 screws. Torque M6 screws 9.5 to 12.2Nm (7 to 9 ftlb).
- (3) Fit a new gasket and fit the throttle body to the plenum, secure with M6 screws. Torque screws 9.5 to 12.2Nm (7 to 9 ftlb).
- (4) Re-connect the throttle position sensor electrically.
- (5) Fit the 'P' clip securing the boost control solenoid electrical lead to the plenum.
- (6) Re-connect the electrical leads to the air thermister.
- (7) Fit the upper fuel rail (Refer to Sub-section 3.21).
- (8) Fit the SEM valve hose to the throttle body
- (9) Fit the idle speed control valve to plenum hose and secure with hose clips.
- (10) On 'B' bank, fit the bolt securing the idle speed control valve to the plenum.

- (11) Locate the throttle cable through the plenum lugs, fit and tighten locknuts.
- (12) Fit throttle cable abutment bracket, complete with cable, to the plenum and secure with M8 screw. Torque screw 23 to 27Nm (17 to 20 ftlb).
- (13) Fit the throttle cable to the cable junction block between the plenums, fit and tighten the locknuts.
- (14) Refit the cable by rotating the throttle lever by hand and sliding the cable and drum into the slots.
- (15) Fit the cable return spring to the link with M5 bolt and nut.
- (16) Fit the charge pipe assembly to the throttle body and intercooler and secure the hose clips.
- (17) On 'A' bank, fit the charge pipe to idle speed control valve hose and secure the hose clips.
- (18) Reconnect the battery.
- (19) Run the engine, adjust the throttle mechanism and check for induction system leaks. Rectify as necessary.
- (20) Remove the body covers and close the engine tailgate.

19 THROTTLE BODY

A Removal Procedure

Special Tools

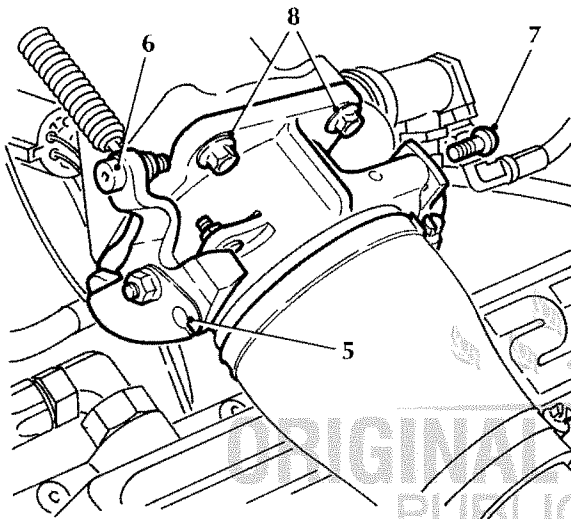
JaguarSport recommended body covers

NOTES:

- 1 Ensure absolute cleanliness of work area and tools.
- 2 Observe correct use of tools.

NOTE

The following procedure deals with the removal of the throttle bodies with the engine installed in the car. If the engine has been removed certain procedures may be disregarded.



- (1) Open the engine tailgate.
- (2) Fit JaguarSport body covers.
- (3) Disconnect the battery.
- (4) Loosen the hose clips and remove the charge pipe assembly from the intercooler to throttle body.
- (5) Rotate the throttle lever by hand and release the cable and drum from the slots.
- (6) Remove the bolt and nut securing the throttle return spring to the throttle link.
- (7) On 'B' bank only, remove the two screws securing the throttle position sensor to the throttle body.
- (8) Remove the four bolts securing the throttle body to the plenum and remove the throttle body.
- (9) Discard the throttle body gasket and fit blanking plugs to the plenum

B Installation Procedure

Special Tools

Torque wrench 12.2Nm (9ftlb)

NOTES:

- 1 Ensure absolute cleanliness of work area and tools.
- 2 Observe correct use of tools.

- (1) Remove the blanking plugs from the plenum and fit the throttle body using a new gasket.
- (2) Secure the throttle body with four M6 bolts and torque bolts 9.5 to 12.2Nm (7 to 9 ftlb).
- (3) Fit and secure the throttle position sensor to 'B' bank throttle body using two M4 screws. Torque screws 3Nm (2ftlb)
- (4) Fit the M5 bolt and nut securing the throttle return spring to the link.
- (5) Fit the throttle cable to the link
- (6) Refit the cable by rotating the throttle lever by hand and sliding the cable and drum into the slots
- (7) Fit the charge pipe assembly to throttle body and intercooler and secure with hose clips.
- (8) Reconnect the battery.
- (9) Check the setting of the throttle position sensor. (Refer to Section 1 - Engine Management).
- (10) Run the engine and check for induction system leaks and correct throttle operation. Rectify as necessary.
- (11) Remove the body covers and close the engine tailgate.

20 INJECTOR HOUSING

A Removal Procedure

Special Tools

JaguarSport recommended body covers

NOTES:

- 1 Ensure absolute cleanliness of work area and tools.
- 2 Observe correct use of tools.

NOTE

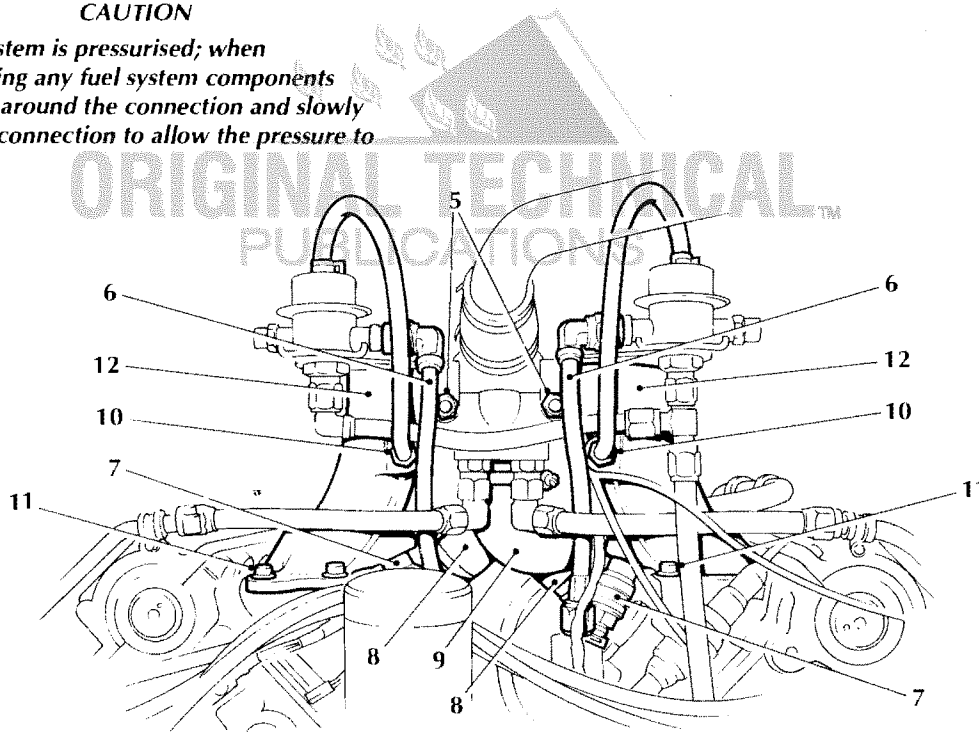
The following procedure deals with the removal of the injector housing with the engine installed. If the engine has been removed certain procedures may be disregarded

- (1) Open the engine tailgate.
- (2) Fit JaguarSport body covers.
- (3) Disconnect the battery.
- (4) Remove the plenum and throttle body (Refer to Section 3.19).
- (5) Remove the nut securing the thermostat housing to the fuel regulator bracket.

CAUTION

The fuel system is pressurised; when disconnecting any fuel system components wrap a rag around the connection and slowly loosen the connection to allow the pressure to drop.

- (6) Disconnect the front and rear fuel pipes to the lower fuel rail. Fit blanks to the open ends.
- (7) Disconnect the electrical leads to the primary and secondary injectors, remove the cable ties.
- (8) Remove the hose clip and disconnect the hose at the water rail stub pipe. Fit blanks to the open ends.
- (9) Remove the hose clip and remove the water by-pass pipe at the thermostat housing. Fit blanks to the open ends.
- (10) Remove the hose clips at the front and rear vacuum rail connections and disconnect the hoses. Fit blanks to the open ends.
- (11) Remove the bolts securing the injector housing to the cylinder head. Collect the speed/synchroniser connector bracket spacers. ('A' bank only).
- (12) Lift the housing, complete with fuel rails and injectors, clear of the cylinder head and retrieve the gaskets.
- (13) Fit blanking caps to the cylinder head ports.



B Installation Procedure

Special Tools

Torque wrench 12.2Nm (9 ftlb)

NOTES:

- 1 *Ensure absolute cleanliness of work area and tools.*
- 2 *Observe correct use of tools.*

- (1) Remove the blanking caps from the cylinder head ports and fit new gaskets. Ensure gasket orientation is correct.
- (2) Fit the injector housing assembly to the cylinder head. Fit the speed synchroniser bracket spacers and secure bracket and housing with M6 screws. Torque screws 9.5 to 12.2Nm (7 to 9 ftlb).
- (3) Remove the blanks and fit the front and rear hoses to the vacuum rail and secure with hose clips.
- (4) Remove the blanks and fit the heater feed pipe to the thermostat housing and secure the hose clips.
- (5) Remove the blanks and fit the hose from the thermostat housing to the water rail stub pipe and secure the hose clips.
- (6) Connect the electrical leads to the primary and secondary injectors. Secure the cable ties.
- (7) Remove the blanking caps and fit the front and rear fuel pipes to the lower fuel rail.
- (8) Fit the thermostat housing to the fuel regulator bracket and secure with M6 nut.
- (9) Fit the plenum and throttle body. (Refer to Section 3.19)
- (10) Connect the battery.
- (11) Replenish the cooling system as necessary.
- (12) Run the engine and check for fuel, cooling and induction system leaks. Rectify any faults as necessary.
- (13) Remove the body covers and close the engine tailgate.

21 PRIMARY FUEL INJECTORS

A Removal Procedure

Special Tools

JaguarSport recommended body covers

NOTES:

- 1 Ensure absolute cleanliness of work area and tools.
- 2 Observe correct use of tools.

NOTE

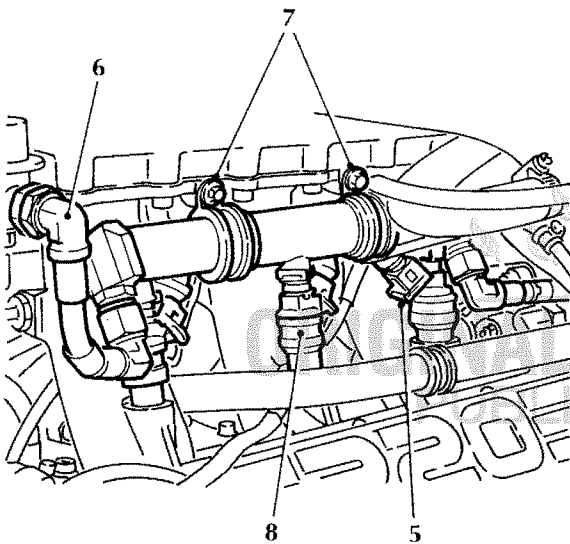
The following procedure deals with the removal of the fuel injectors with the engine installed. If the engine has been removed certain procedures may be disregarded.

- (1) Open the engine tailgate.
- (2) Fit JaguarSport body covers.
- (3) Disconnect the battery.
- (4) Disconnect the injector electrical connections.
- (5) Remove the injector retaining clips.

CAUTION

The fuel system is pressurised; when disconnecting any fuel system components wrap a rag around the connection and slowly loosen the connection to allow the pressure to drop.

- (6) Remove the front and rear fuel pipes to the upper fuel rail. Fit blanks to the open ends.
- (7) Remove the bolts securing the fuel rail 'P' clips and remove the fuel rail.
- (8) Remove fuel injectors from the housing.
- (9) Fit blanks to the injector housing ports.



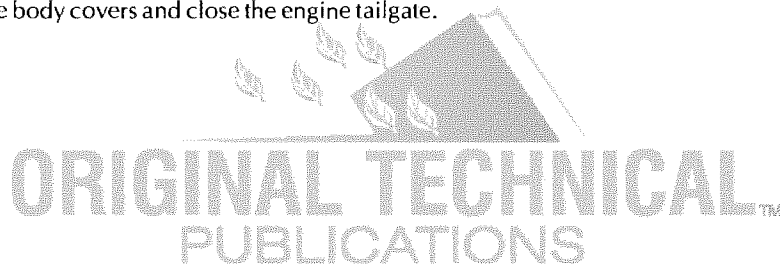
B Installation Procedure

Special Tools

Torque wrench 7Nm (5.5 ftlb)

NOTES:

- 1 *Ensure absolute cleanliness of work area and tools.*
 - 2 *Observe correct use of tools.*
-
- (1) Remove the blanks from the injector housing ports and fit fuel injectors.
 - (2) Fit the upper fuel rail to fuel injectors and secure the injector retaining clips
 - (3) Secure P clips to fuel rail with M6 bolts. Torque bolts 5 to 7Nm (3.5 to 5 ftlb)
 - (4) Fit injector electrical connections.
 - (5) Remove the blanking caps and fit the front and rear fuel pipes to the fuel rail.
 - (6) Connect the battery.
 - (7) Run the engine and check the fuel system for leaks. Rectify any faults as necessary.
 - (8) Remove the body covers and close the engine tailgate.



22 SECONDARY FUEL INJECTORS

A Removal Procedure

Special Tools

JaguarSport recommended body covers

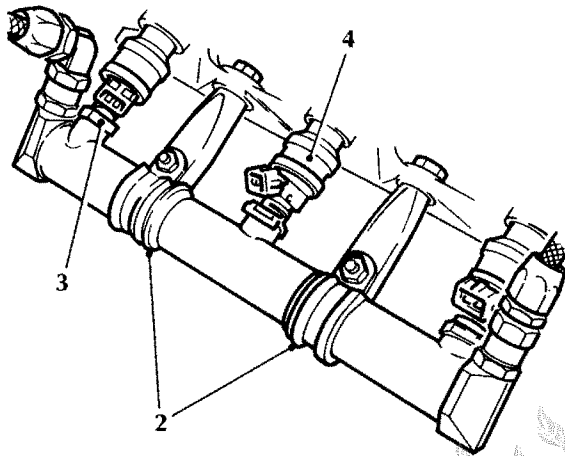
NOTES:

- 1 Ensure absolute cleanliness of work area and tools.
- 2 Observe correct use of tools.

NOTE

The following procedure deals with the removal of the fuel injectors with the engine installed. If the engine has been removed certain procedures may be disregarded.

- (1) Remove the relevant injector housing. (Refer to Section 3.20).
- (2) Remove the 'P' clips securing the lower fuel rail to the spacers.
- (3) Remove the injector retaining clips and remove the fuel rail.
- (4) Remove the injectors from the housing.



B Installation Procedure

NOTES:

- 1 Ensure absolute cleanliness of work area and tools.
- 2 Observe correct use of tools.

- (1) Fit injectors into the fuel rail and fit the injector retaining clips.
- (2) Fit the injectors into the injector housing.
- (3) Fit the 'P' clips to the fuel rail and secure the rail to the spacers with M6 bolts and nuts.
- (4) Fit the injector housing. (Refer to Section 3.20).

23 UPPER FUEL RAILS

A Removal Procedure

Special Tools

JaguarSport recommended body covers

NOTES:

- 1 Ensure absolute cleanliness of work area and tools.
- 2 Observe correct use of tools.

NOTE

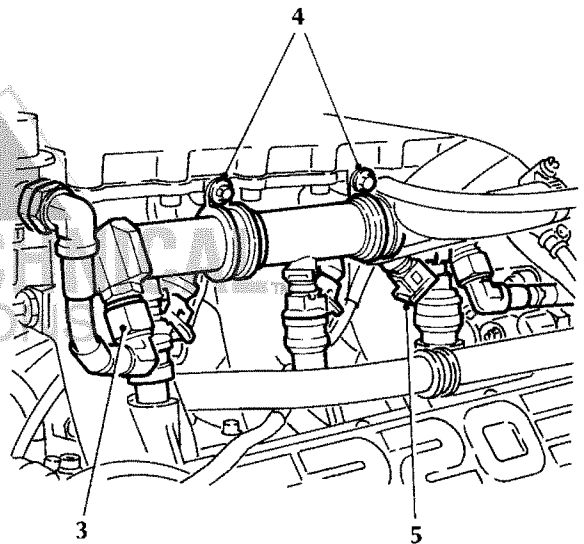
The following procedure deals with the removal of the fuel injectors with the engine installed. If the engine has been removed certain procedures may be disregarded.

- (1) Fit JaguarSport body covers.
- (2) Disconnect the battery.

CAUTION

The fuel system is pressurised; when disconnecting any fuel system components wrap a rag around the connection and slowly loosen the connection to allow the pressure to drop.

- (3) Disconnect the front and rear fuel connections to the rail. Fit blanking caps to the open ends.
- (4) Remove the bolts securing the 'P' clips to the injector housing.
- (5) Remove the injector retaining clips and remove the fuel rail.
- (6) Fit blanking caps to the fuel rail and injectors.



B Installation Procedure

Special Tools

Torque wrench 7Nm (5.5 ftlb)

NOTES:

- 1 Ensure absolute cleanliness of work area and tools.
- 2 Observe correct use of tools.
- (1) Remove the blanking caps, fit the fuel rail to the injectors and secure the injector retaining clips.
- (2) Fit the 'P' clips to the fuel rail and secure with M6 screws. Torque screws 5 to 7Nm (3.5 to 5 ftlb).
- (3) Remove the blanking caps and connect the front and rear fuel pipes to the fuel rail.
- (4) Connect the battery.
- (5) Run the engine and check the fuel rails and injectors for leaks. Rectify any faults as necessary.
- (6) Remove the body covers and close the engine tailgate.

24 LOWER FUEL RAILS

A Removal Procedure

Special Tools

JaguarSport recommended body covers

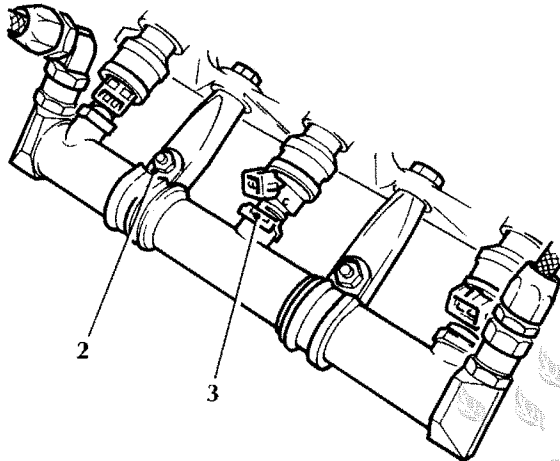
NOTES:

- 1 Ensure absolute cleanliness of work area and tools.
- 2 Observe correct use of tools.

NOTE

The following procedure deals with the removal of the fuel injectors with the engine installed. If the engine has been removed certain procedures may be disregarded.

- (1) Remove the relevant injector housing. (Refer to Section 3.20)
- (2) Remove the bolts and nuts securing the fuel rail 'P' clips
- (3) Remove the injector retaining clips and remove the fuel rail. Fit blanking caps to the fuel rail and injectors.



B Installation Procedure

NOTES:

- 1 Ensure absolute cleanliness of work area and tools.
- 2 Observe correct use of tools.

- (1) Remove the blanking caps and fit the fuel rail to the injectors.
- (2) Fit and secure the injector retaining clips.
- (3) Fit the 'P' clips to the fuel rail and secure to the spacers using M6 bolts and nuts.
- (4) Fit the injector housing. (Refer to Section 3.20).

25 ENGINE BAY BRIDGE PANEL

A Removal Procedure

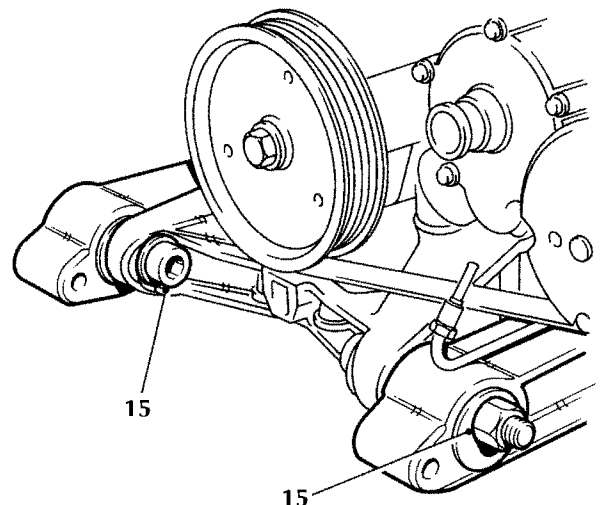
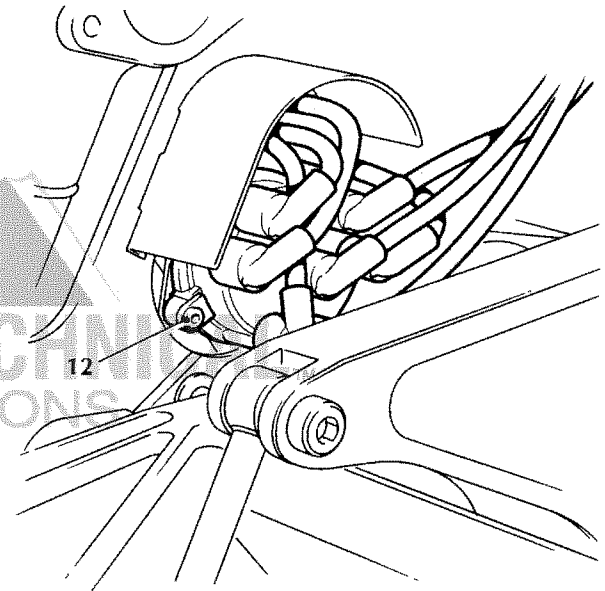
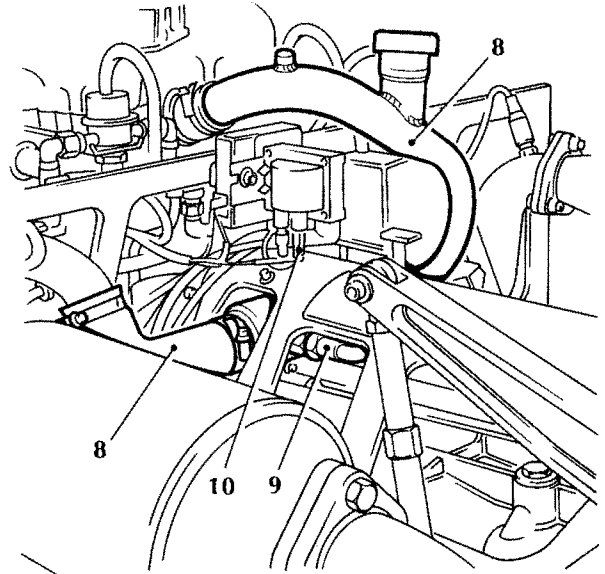
Special Tools

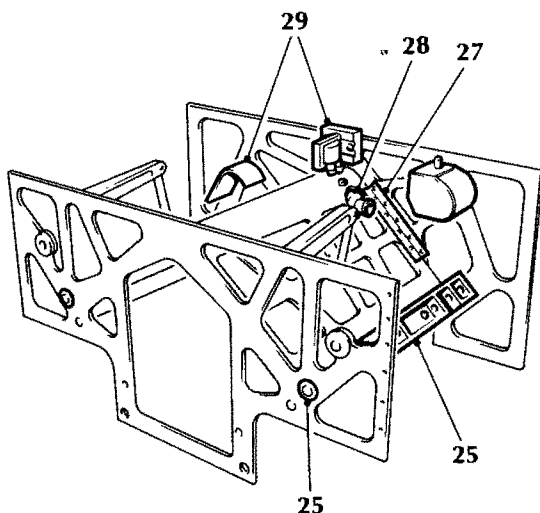
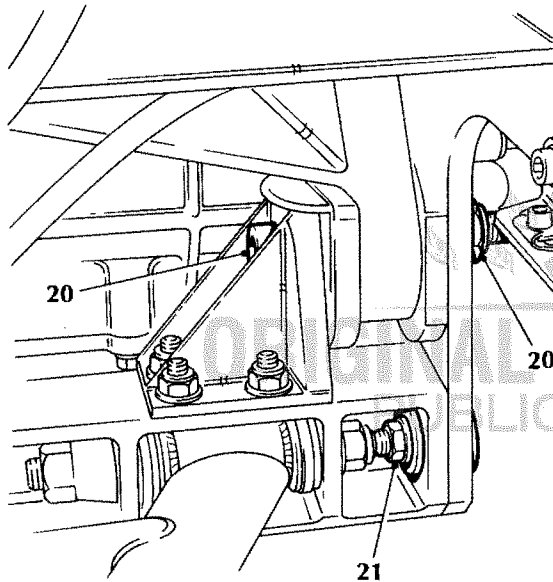
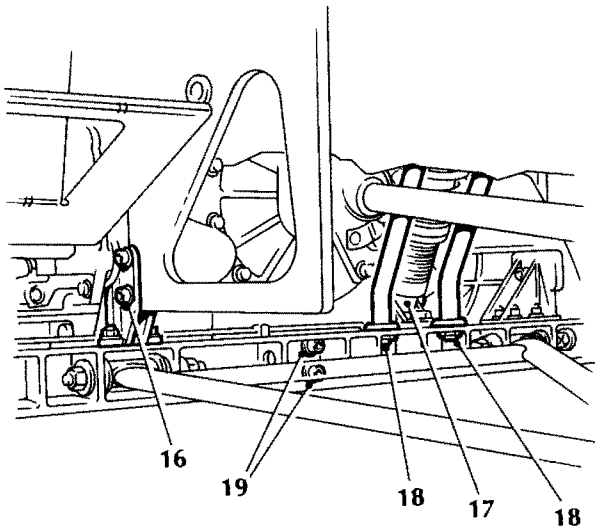
Suitable engine/transaxle stand/table or wooden blocks

NOTES:

- 1 Ensure absolute cleanliness of work area and tools.
- 2 Observe correct use of tools.

- (1) Remove the engine and transaxle unit. (Refer to Section 3.1).
- (2) Place the engine on a stand or table ensuring the engine beams are free.
- (3) Remove the transaxle sump drain plug and drain the oil. (Refer to Section 7)
- (4) Remove the exhaust system. (Refer to Section 9).
- (5) Remove the drive line. (Refer to Section 5).
- (6) Remove the oil inlet pipe from the transaxle rear casing and fit blanking caps. (Refer to Section 7)
- (7) Remove the oil outlet pipe from the transaxle and fit blanking caps. (Refer to Section 7).
- (8) Loosen the hose clips and remove the water inlet and outlet pipes from the transaxle oil heat exchanger.
- (9) Remove the oil inlet and outlet pipes from the heat exchanger and fit blanking caps.
- (10) Disconnect the ignition module electrical connections.
- (11) Remove the spark plug covers and disconnect the spark plug leads.
- (12) Remove the 2 screws securing the distributor cap, remove the cap and the rotor arm.
- (13) Remove the cable ties securing the transaxle rear casing sensor wiring to the engine bay bridge panel.
- (14) Disconnect the brake rigid pipe at the T piece on the rear plate. (Refer to Section 6).
- (15) Remove the engine mounting bolts, nuts and washers.

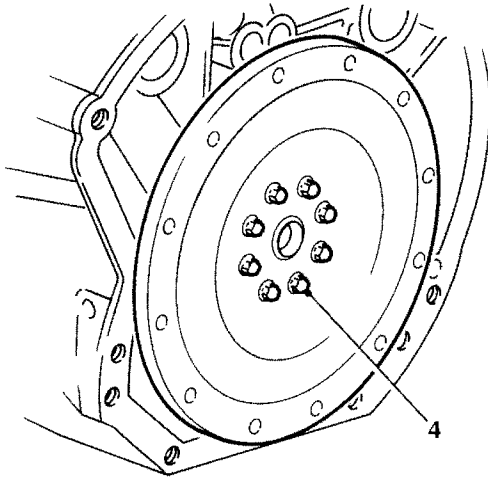




- (16) Remove the bolts, nuts and washers securing the intermediate plate to the mounting bracket.
- (17) Remove the damper lower attachment bolt, nut and washer.
- (18) Remove the bolts and washers securing the steady assemblies to the engine beams.
- (19) Remove the bolts securing the engine brace to the beams.
- (20) Ensure the transaxle is stable on the stand and remove the transaxle mounting bolts, nuts, washers and sleeves.
- (21) Support the engine bay bridge panel and remove the bolts nuts and washers securing the rear plate to the engine beams.
- (22) Remove the engine beams and lower wishbone assemblies.
- (23) Carefully lift the engine bay bridge panel and upper suspension assemblies away from the engine and transaxle ensuring the speed synchroniser electrical cables do not foul on the engine bay bridge panel.
- (24) Remove the upper wishbones and suspension assemblies. (Refer to Section 4).
- (25) Remove the bolts, nuts and washers securing the upper wishbone support to the engine plates and bridge plate and remove the support.
- (26) Remove the 'P' clips securing the flexible brake hoses to the bridge plate; disconnect the hoses at the T piece and calipers and remove the hoses. (Refer to Section 6).
- (27) Remove the bolts, nuts and washers from the cleats attaching the engine plates to the bridge plate and separate the plates.
- (28) Remove the bolts securing the transaxle oil heat exchanger to the bridge plate.
- (29) Remove the coolant expansion tank, ignition module and distributor heat shield from the intermediate plate.

B Installation Procedure

- (1) Fit the bridge plate to the rear and intermediate plates and secure with M6 bolts washers and nuts. Torque nuts 8 to 9.5Nm (6 to 7ftlb)
- (2) Fit the transaxle oil heat exchanger to the bridge plate and secure with M6 bolts and washers. Torque the screws 8 to 9.5Nm (6 to 7ftlb).
- (3) Fit the coolant expansion tank, ignition module and distributor heat shield to the intermediate plate.
- (4) Fit the brake flexible hoses to the bridge plate P clips and secure with M6 bolts, washers and nuts.
- (5) Fit the upper wishbone support to the bridge plate, intermediate and rear plates and secure with M8 bolts, washers and nuts. Torque nuts to 30 to 34Nm (22 to 25 ftlb).
- (6) Fit the engine bay bridge panel over the engine and transaxle and support the engine bay bridge panel.
- (7) Fit the engine beams and lower wishbone assemblies to the engine bay bridge panel. Secure the engine beams to the intermediate and rear plates with M8 bolts, washers and nuts. Torque the nuts 30 to 34Nm (22 to 25 ftlb).
- (8) Fit the transaxle mounting sleeves, bolts washers and nuts. Torque the nuts 102 to 117Nm (75 to 86ftlb).
- (9) Fit the engine mounting bolts, washers and nuts. Torque nuts 57 to 65Nm (42 to 48ftlb).
- (10) Fit the upper wishbones and suspension assemblies. (Refer to Section 4).
- (11) Fit the engine brace to the beams and secure with M8 bolts and washers. Torque bolts 20 to 23Nm (15 to 17ftlb).
- (12) Fit the steady assemblies to the beams and secure using M8 bolts and washers. Torque bolts 30 to 34Nm (22 to 25ftlb).
- (13) Fit the damper lower attachment bolt, washer and nut. Torque the nut 30 to 34Nm (22 to 25ftlb).
- (14) Connect the brake flexible hoses and rigid pipe to the T piece on the rear plate.
- (15) Remove the blanking caps and fit the transaxle oil pipes to the heat exchanger and transaxle.
- (16) Fit the water inlet and outlet pipes to the heat exchanger and secure the hose clips.
- (17) Fit the rotor arm and distributor cap and secure the cap with M5 screws.
- (18) Fit the spark plug leads to the plugs and secure the cover.
- (19) Connect the ignition module electrical wiring.
- (20) Fit the drive line. (Refer to Section 5)
- (21) Fit the exhaust system. (Refer to Section 9).
- (22) Fit the engine and transaxle unit to the vehicle. (Refer to Section 3.1).



26 FLYWHEEL

A Removal Procedure

NOTE

The following procedure deals with removal of the flywheel with the engine installed. If the engine has been removed certain procedures may be disregarded.

- (1) Remove the engine. (Refer to Section 3.1)
- (2) Remove the transaxle from the engine. (Refer to Section 7)
- (3) Remove the clutch. (Refer to Section 8)
- (4) Remove the eight bolts securing the flywheel to the crankshaft and remove the flywheel.

B Installation Procedure

- (1) Fit flywheel to crankshaft and secure with eight M10 flanged head bolts..
- (2) Torque tighten bolts to 95Nm (70 ftlb).
- (3) Fit clutch. (Refer to Section 8)
- (4) Fit transaxle to engine. (Refer to Section 7)
- (5) Fit the engine to the vehicle. (Refer to Section 3.2)

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27 ENGINE SPEED AND POSITION SENSOR TIMING

NOTES:

- 1 Ensure absolute cleanliness of work area and tools.
- 2 Observe correct use of tools.

Special tools

The following special tools will be required

JaguarSport service tool STM.90.013

Procedure

- (1) Fit the JaguarSport service tool M.90.013 and turn flywheel to align 40 degree BTDC timing mark on flywheel and timing mark on crankcase with number one cylinder firing.

NOTE:

No. 1 cylinder firing can be determined from the rotor arm position relative to the distributor cap.

- (2) Loosen centre bolt retaining the sensing disc.
- (3) Move sensing disc in the direction of correct camshaft rotation to achieve correct setting.

NOTE:

Correct camshaft rotation is counter-clockwise as viewed from the rear.

- (4) The status of the speed and synch sensor outputs is shown by the service diagnostic software. Refer to the appropriate page of the diagnostic aid and proceed as follows:

- (a) Rotate disc until synch sensor status changes from OFF to ON indication;

NOTE:

"ON" indication identifies position of the disc slot at the synch sensor position.

- (b) Rotate sensing disc until speed sensor output signal changes from OFF to ON;

NOTE:

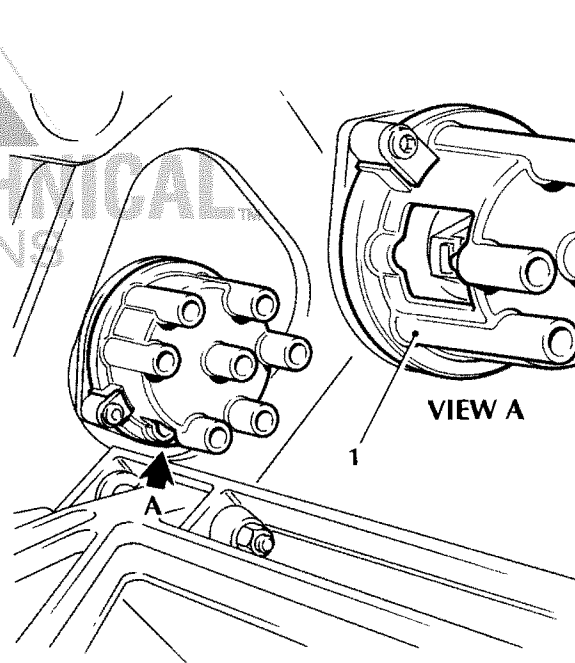
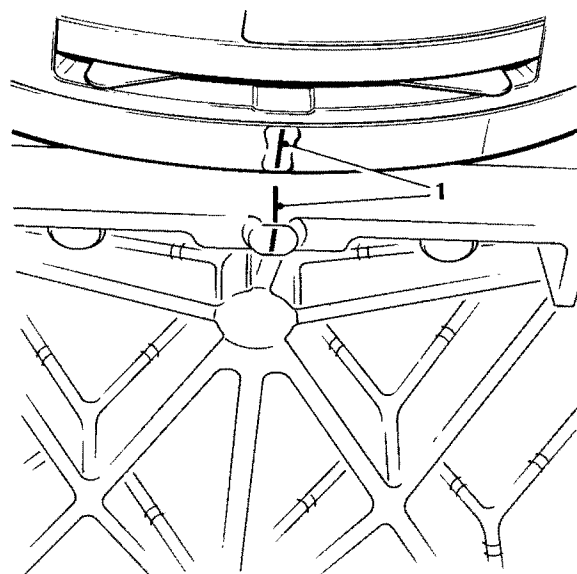
This exact sequence of events can only occur once in a camshaft rotation.

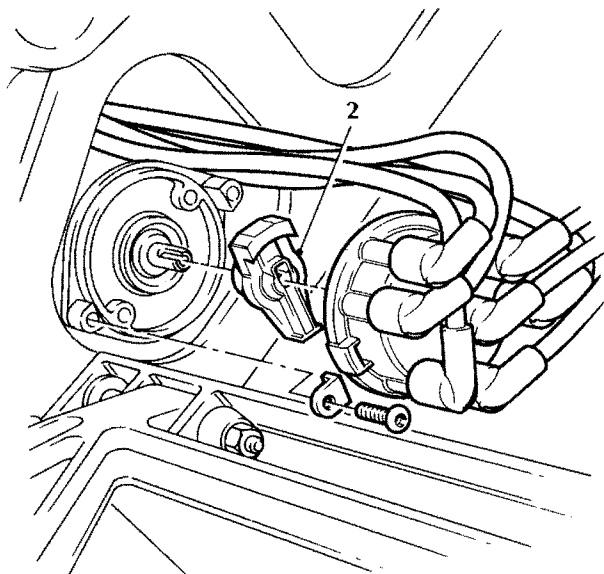
- (c) Fit centre bolt to sensing disc.

NOTE:

Ensure the sensing disc does not move as the centre bolt is tightened.

- (5) Check the ignition timing by rotating the engine in the correct direction of rotation to 40 degrees BTDC and confirm the sequence of synch "ON" followed by speed "ON" at 40 degrees $\pm$ one degree.
- (6) Drill camshaft adaptor and secure with pin.
- (7) Remove the service tool and fit the distributor cap.





28 DISTRIBUTOR TIMING

Procedure

- (1) Turn flywheel to align 20 degrees BTDC timing mark on flywheel and timing mark on crankcase with number one cylinder firing.

NOTE:

No. 1 cylinder firing can be determine by the alignment of the timing marks on the drive pulleys of "B" bank camshafts.

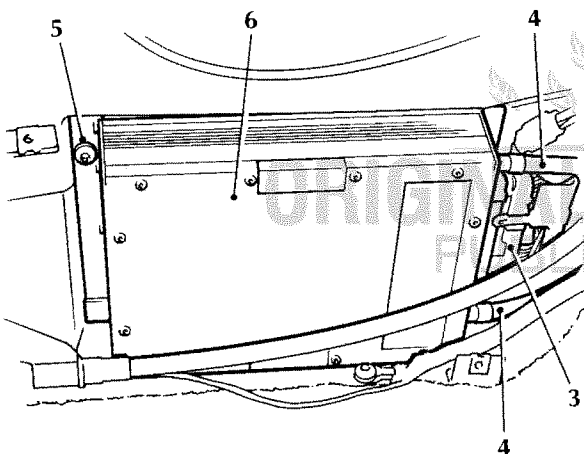
- (2) Fit rotor arm to distributor using Loctite 405 and fit distributor cap.

29 ELECTRONIC CONTROL UNIT (ECU)

A Removal Procedure

NOTES:

- 1 Ensure absolute cleanliness of work area and tools.
- 2 Observe correct use of tools.



- (1) Move the battery master switch to the OFF position.
- (2) Remove the trim surround from the ECU.
- (3) Remove the electrical plugs from the ECU.
- (4) Remove the ECU sensor hoses
- (5) Remove the screws securing the ECU.
- (6) Remove the ECU.

B Installation Procedure

Special Tools

The following special tools will be required.

Torque wrench 0 to 10.5 Nm (0 to 14 ftlb)

WARNING;

ENSURE BATTERY MASTER SWITCH IS OFF.

NOTES:

- 1 Ensure absolute cleanliness of work area and tools.
- 2 Observe correct use of tools.

- (1) Fit ECU and secure with screws.
- (2) Torque load screws 9 to 10.5Nm (12 to 14ftlb).
- (3) Connect the electrical plugs to ECU.
- (4) Fit the ECU sensor hoses.
- (5) Fit trim surround.
- (6) Move battery master switch to ON position.
- (7) Run the engine and check the ECU functions correctly.

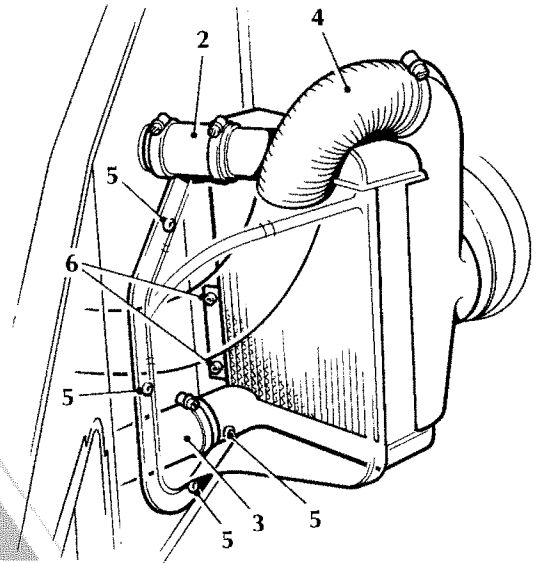
30 INTERCOOLER ASSEMBLY

A Removal Procedure

NOTES:

- 1 Ensure absolute cleanliness of work area and tools.
- 2 Observe correct use of tools.
- 3 For intercooler maintenance the vehicle must be supported on a wheel free ramp with the respective wheel removed.

- (1) Remove wheel arch liner. (Refer to Section 2)
- (2) Remove intercooler to body side panel pipe assembly.
- (3) Remove hose clips and remove the intercooler to turbo hoses. Fit blanking caps to open connections.
- (4) Remove the hose clips and disconnect the air intake hose from the front intercooler duct.
- (5) Remove the screws securing the intercooler rear ducting to the chassis. Note the position of the button head screw. Feed the air intake hose through the duct and remove the duct.
- (6) Remove the two screws securing the intercooler to the chassis and remove the intercooler.



B Installation Procedure

Special Tools

Torque wrench 0 to 10 Nm (0 to 14 ftlb).

NOTES:

- 1 Ensure absolute cleanliness of work area and tools.
- 2 Observe correct use of tools.

- (1) Fit intercooler and secure with two M6 screws.
- (2) Torque load screws 2 to 2.5 Nm (1.5 to 2 ftlb).
- (3) Feed the air intake hose through the rear duct; fit the duct and secure with four M6 screws. Torque the screws 2 to 2.5Nm (1.5 to 2 ftlb).
- (4) Fit the air intake hose to the front duct and secure the hose clip.
- (5) Remove the blanks, fit the intercooler to turbo hoses and secure with hose clips.
- (6) Remove the blanks, fit intercooler body side panel pipe assembly and secure the hose clips.
- (6) Fit wheel arch liner and secure with screws. (Refer to Section 2)
- (7) Torque load screws 7.5 to 9 Nm (5.5 to 6.5 ftlb).

- (11) Fit washers and nuts to the top pins and torque load nuts 57 to 65Nm (42 to 48ftlb).
- (12) Fit wheel (Refer to Section 4.1).
- (13) Lower lifting apparatus and check wheel alignment.
- (14) If wheel alignment is satisfactory remove vehicle from lifting apparatus. If wheel alignment is unsatisfactory repeat this procedure.

28 RIDE HEIGHT TRIM

A Check

Special Tools

The following special tools will be required:

JaguarSport protective body covers

NOTES:

1. Ensure absolute cleanliness of work area and tools.
2. Ensure correct use of tools.
3. Ensure the vehicle is at the correct kerb weight.
4. Ensure all tyre pressures are correct.
5. Ensure the vehicle is on a flat and level surface.
6. Fit JaguarSport protective body covers in immediate work area.

Kerb Weight

The kerb weight of the vehicle is defined as the complete vehicle plus all fluids. This includes a full tank of petrol.

Front Axle	LH Wheel	288kg (635.040lbs)
	RH Wheel	309kg (681.345lbs)

Rear Axle	LH Wheel	445kg (981.225lbs)
	RH Wheel	428kg(943.740lbs)
		873kg (1924.965lbs)

Totals

Front Axle	=	597kg (1316.385lbs)
		+

Rear Axle	=	873kg (1925.965lbs)
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Kerb Weight	=	1470kg.(3241.35lbs)
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32 OIL COOLER

A Removal Procedure

NOTES:

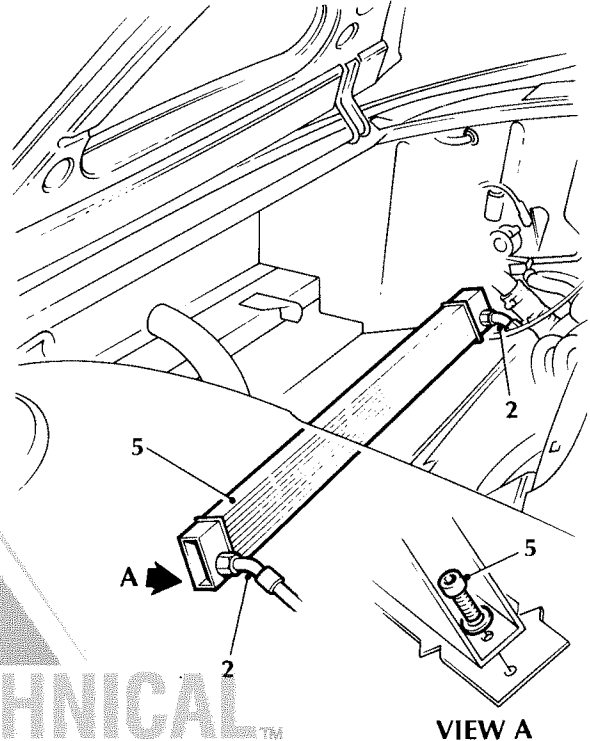
- 1 Ensure absolute cleanliness of work area and tools.
- 2 Observe correct use of tools.

- (1) Remove the radiator from vehicle. (Refer to Section 3.31)

WARNING:

BEFORE DISCONNECTING OIL PIPES FROM OIL COOLER ALLOW OIL TO COOL DOWN.

- (2) Remove the oil pipes from the oil cooler.
- (3) Fit blanking caps to the open connections.
- (4) Wipe away any spillage.
- (5) Remove the screws and remove the oil cooler from the vehicle.
- (6) Drain the oil from the oil cooler.



B Installation Procedure

Special Tools

The following special tools will be required.

Torque wrench 0 to 41Nm (0 to 30 ftlb)

NOTES:

- 1 Ensure absolute cleanliness of work area and tools.
- 2 Observe correct use of tools.

- (1) Remove blanking caps.
- (2) Fit the oil cooler in the vehicle and secure with screws.
- (3) Torque load screws 8 to 9.5 Nm (6 to 7 ftlb).
- (4) Connect the oil pipes to the oil cooler. Torque pipe unions 35 to 41Nm (26 to 30 ftlb).
- (5) Fit the radiator. (Refer to Section 3.31)
- (6) Check the engine oil level and replenish as necessary.
- (7) Run the engine and check for oil leaks. Rectify any faults as necessary.

33 THROTTLE CABLE AND PEDAL

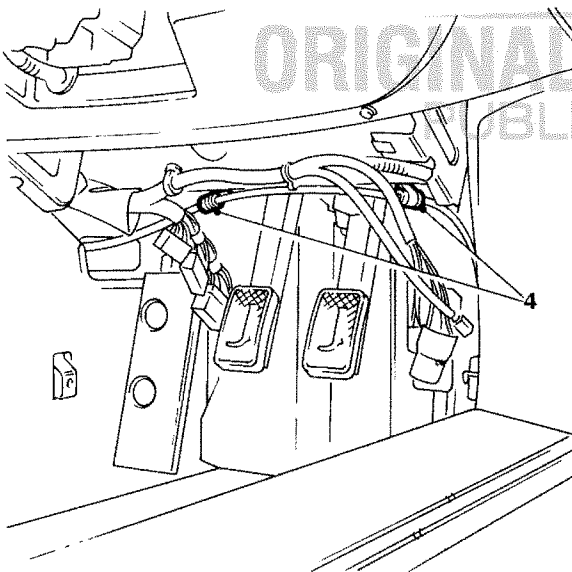
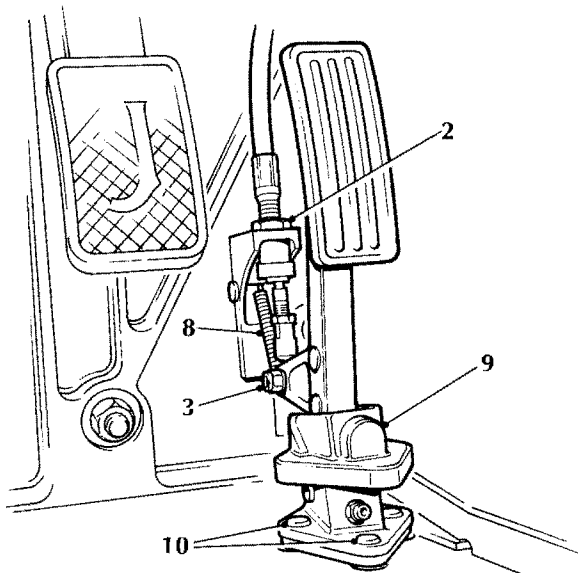
A Removal Procedure

WARNING:

WHEN HANDLING CABLE ASSEMBLIES BEWARE OF FRAYED AND BROKEN STRANDS. WEAR PROTECTIVE GLOVES.

NOTES:

- 1 *Ensure absolute cleanliness of work area and tools.*
- 2 *Observe correct use of tools.*



- (1) Remove the nut and washer from the cable adjuster on the plenum chambers and remove the throttle cable from the cable block.
- (2) Loosen the nut and washer on the adjustment bracket fixed to the bulkhead in the driver's footwell.
- (3) Remove the screw, washers and nut and remove the throttle cable from the pedal.
- (4) Remove the P clips on the underside of dashboard securing the throttle cable.
- (5) Remove the tunnel trim and top cover. (Refer to Section 2)
- (6) Remove the throttle cable.
- (7) Remove carpets from around pedal.
- (8) Remove the spring anchor pin and safety fastener from the throttle pedal lever.
- (9) Push the rubber gaiter as far as possible up the pedal.
- (10) Remove the screws and washers attaching the pedal assembly to the vehicle floor and remove the pedal.

B Installation Procedure

WARNING:

WHEN HANDLING CABLE ASSEMBLIES BEWARE OF FRAYED AND BROKEN STRANDS. WEAR PROTECTIVE GLOVES.

NOTES:

- 1 Ensure absolute cleanliness of work area and tools.
- 2 Observe correct use of tools.

NOTE

Prior to installing the throttle pedal inspect the condition of the gaiter. Replace if necessary.

- (1) Locate the throttle pedal and secure to the vehicle floor with screws and washers. Torque tighten the screws 8 to 9.5Nm (6 to 7ftlb).
- (2) Locate the return spring in the lever and secure with anchor pin and safety fastener.
- (3) Pull down the rubber gaiter.
- (4) Fit the carpet around the pedal.
- (5) Route the cable from the plenum chambers through the rear bulkhead and tunnel, through the adjuster bracket on the front bulkhead to the pedal.

NOTE:

Fit nut and washer onto throttle cable before connecting throttle cable to pedal.

- (6) Locate the throttle cable in the cable block and fit to the pedal using the screw, nut and washers.
- (7) Fit the shouldered nut and washer to the threaded portion of the throttle cable at the adjuster bracket on the front bulkhead.
- (8) Secure the throttle cable to the underside of the dashboard with P clips.
- (9) Set up the throttle cable at the engine, (Refer to Engine Tuning) and adjust the cable at the pedal to ensure full throttle travel. Run the engine and ensure correct throttle operation. Ensure that the throttle stop is correctly adjusted in the pivot block to prevent any strain in the cable when the pedal is depressed. Rectify any faults as necessary.

CAUTION:

Before fitting the tunnel cover and trim, inspect the tunnel to ensure there are no loose articles.

- (10) Fit the tunnel top cover and trim (Refer to Section 2).

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INTRODUCTION

The vehicle is fitted front and rear with four wheel independent unequal length wishbone suspension with inboard concentric coil springs and dampers, and anti-roll bars. The suspension has been designed to provide optimum road handling and ride comfort.

DESCRIPTION

The LH and RH front and rear suspension each consists of an upper and lower wishbone, a pushrod, a nodder, a damper, a coil spring, a drop link attaching an anti-roll bar and an upright.

Front Suspension

The upper wishbone is pivoted at the inner end using rubber bushes and thrust washers, and is fitted at the outer end with a spherical bearing which is attached to the upright. The spherical bearing is fitted into a separate cap which is bolted onto the wishbone and uses shims to adjust the camber.

The lower wishbone is pivoted at the inner end in the lower suspension beam, using rubber bushes and thrust washers and is attached to the upright with a ball joint. The pin of the ball joint bolts onto the upright through a steel bush.

The cast front upright houses the wheel bearings and oil seals and provides the location for the brake caliper.

The pushrod attaches with a rubber bush to the lower wishbone and with a spherical joint to the nodder, the nodder is pivoted on a spindle and operates the spring/damper unit. The anti-roll bar is operated via a drop link attached to the nodder.

Rear Suspension

The upper wishbone pivots at the inboard end using rubber bushes and thrust washers and is attached to the upright using two spherical bearings. One spherical bearing is housed in a cap which bolts onto the wishbone and uses shims to provide toe adjustment.

The lower wishbone also pivots on rubber bushes and uses thrust washers at the inboard end. The outer end is attached

to a pin on the upright using a spherical bearing. The bearing is housed in a bolt-on cap which provides, with shims, the camber adjustment.

The pushrod attaches to the upright with a spherical joint and to the nodder with a rubber bush. The nodder pivots on a spindle and operates the spring/damper unit.

The cast rear upright houses the wheel bearings, oil seals and the stub axle and provides location for the hand and service brake calipers. The anti-roll bar is operated by a drop link attached to the nodder.

Front and Rear Anti-roll Bars

The drop link attaches to the lever arm. The other end of the lever arm and the anti-roll bar are splined. A nut and bolt clamp the lever arm onto the anti-rollbar.

OPERATION

Wheel loads are transmitted through the pushrod to the nodder, which pivots on its bushes and transmits the loads to the spring/damper unit.

The drop links actuate the anti-roll bar via the splined lever arms to provide roll stiffness. Road noise and vibration is reduced by the use of rubber bushes in the suspension system.

INTRODUCTION

This section details the procedures for replacement of suspension system components. Replacement of components or disassembly instructions not covered in this section require the component be returned to the JaguarSport.

1 ROAD WHEELS

A Removal Procedure

Special Tools

The following special tools will be required:

JaguarSport protective body covers

Low line trolley jack

Socket spanner 90mm (3.5in)

Torque wrench 488Nm (0-360 ftlb)

Wheel chocks

Materials

The following materials will be required.

High temperature resistant anti-corrosive copper grease.

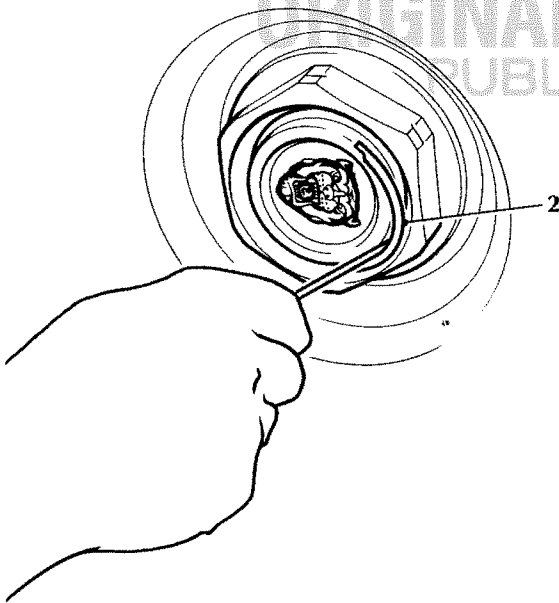
NOTES:

- 1 Ensure absolute cleanliness of work area and tools.
- 2 Observe correct use of tools.
- 3 Check serviceability of components as they are removed. If in doubt reject component.
- 4 Clean all parts where applicable before reassembly using a soluble proprietary engine cleaner.
- 5 Fit JaguarSport protective body covers in immediate work area.
- 6 When using wheel free lifting apparatus follow steps 1 to 5. When using manual jacking procedures follow steps 1, 2, and 6 to 10

- (1) Insert a small screwdriver in stub axle undercut and lever out retaining clip.
- (2) Slide the screwdriver in a clockwise direction to remove retaining clip.

CAUTION:

The wheel nuts on the RH side of the vehicle are LH thread and the nuts on the LH side of the vehicle are RH thread.



Wheel Free Lifting Apparatus

- (3) Correctly position the vehicle on wheel free lifting apparatus.
- (4) Chock the wheels, apply the foot brake and loosen wheel nut using socket spanner and a suitable wrench.
- (5) Raise the lifting apparatus to a suitable working height and remove the wheel nut and wheel.

Manual Jacking

- (6) Ensure the handbrake is off and the transmission is in neutral selection.
- (7) Ensure the jack lift plate is free from debris.

NOTES:

1. The jack lift plate should be flat to avoid underbody damage.
2. During jacking operations the vehicle will move toward the jack.

- (8) For front end jacking

Position jack on the jacking point precisely under zero bulkhead with the centreline of the jack aligned with the centreline of the vehicle and lift the vehicle and lift the vehicle.

- (8A) For rear end jacking

Position jack precisely under the rear engine bay plate with the centreline of the jack aligned with the centreline of the vehicle, and lift the vehicle.

- (9) Apply handbrake, select any gear and chock the wheels.
- (10) Remove wheel nut and remove wheel.

B Installation Procedure

CAUTION:

When a wheel is re-fitted, it must be fitted on the same axle it was removed from.

- (1) Apply high temperature resistant anti-corrosive copper grease to stub axle threads.
- (2) Align wheel location pegs with the wheel and fit the wheel.
- (3) Fit wheel nut hand tight.

Wheel Free Lifting Apparatus

- (4) Lower lifting apparatus, fit wheel chocks and apply footbrake.

Manual Jacking

- (4A) Release the handbrake and select neutral gear
- (4B) Lower the vehicle, remove jack, fit wheel chocks and apply foot brake.
- (5) Torque load wheel nut to 461 to 488Nm (340 to 360ftlb) using socket spanner and torque wrench.
- (6) Fit retaining clip.
- (7) Remove wheel chocks and apply hand brake.

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2 FRONT LOWER WISHBONE

A Removal Procedure

Special Tools

The following special tools will be required.

JaguarSport protective body covers

JaguarSport service tools ST0022 and ST0023

Torque wrench 0 - 68Nm (50ftlb)

NOTES:

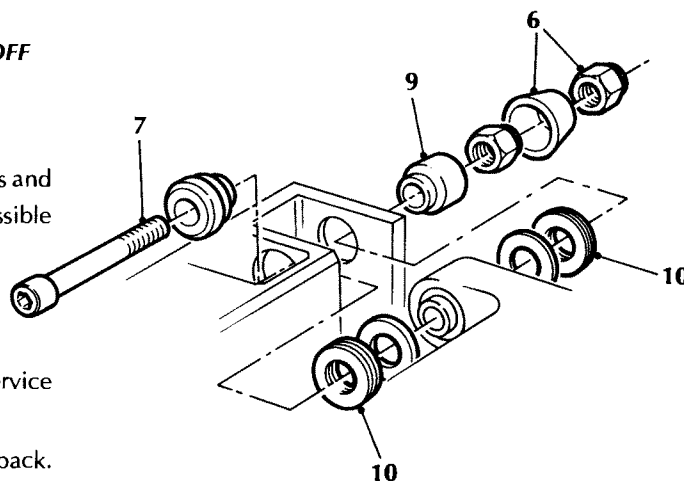
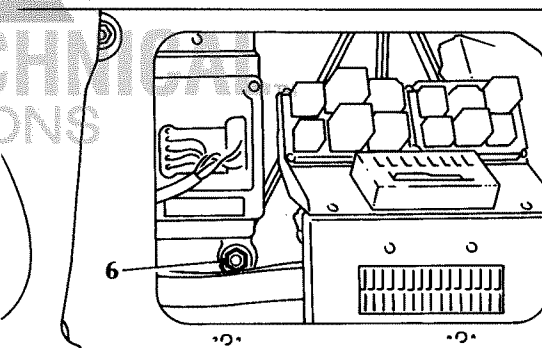
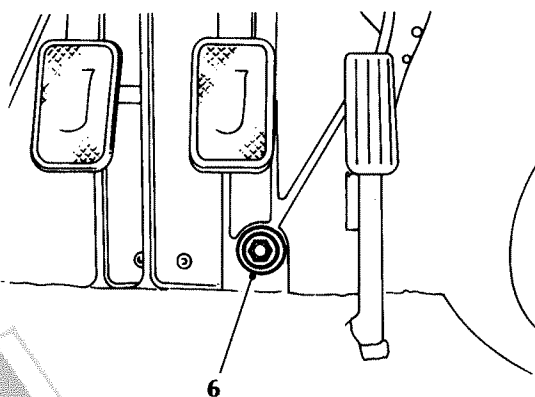
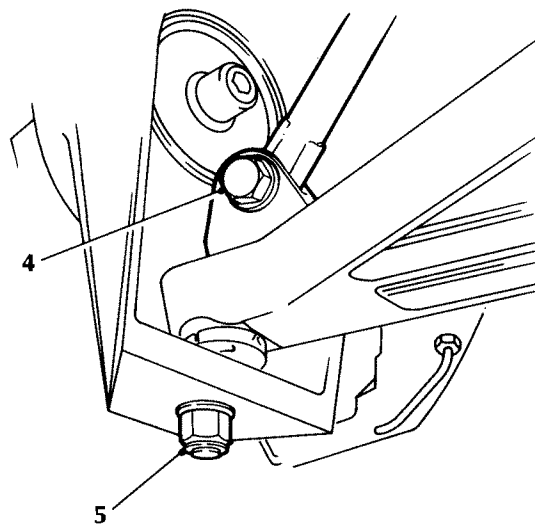
- 1 Ensure absolute cleanliness of work area and tools.
- 2 Ensure correct use of tools.
- 3 Check serviceability of components as they are removed. If in doubt reject component.
- 4 Clean all parts where applicable before reassembly using a soluble proprietary engine cleaner.
- 5 Fit JaguarSport protective body covers in immediate work area.
- 6 JaguarSport service tools must be used to prevent damage to the components and to ensure correct assembly.

- (1) Correctly position vehicle on lifting apparatus (Refer to Section 1.1), and remove the wheel. (Refer to Section 4.1).
- (2) Raise the vehicle to a suitable working height, remove screws and washers and remove wheel arch liner.
- (3) When removing the passenger's side components, remove the footwell panel to gain access to the rear locking caps and nuts.
- (4) Support the upright and remove bolt, washer and nut attaching push rod to wishbone.
- (5) Remove nut and washer from ball pin. To prevent the ball pin from rotating it will be necessary to use an allen key in the end of the ball pin.

WARNING

ENSURE THE BATTERY MASTER SWITCH IS OFF PRIOR TO WORKING IN THE PASSENGER'S FOOTWELL PANEL.

- (6) Remove front and rear wishbone attachment nuts and locking caps. The rear attachment fixings are accessible from the driver/passenger footwell.
- (7) Remove spindle nuts and withdraw spindles.
- (8) Lift ball pin clear of upright.
- (9) Withdraw wishbone pivot bushes slightly using service tools ST0022 and ST0023, if necessary.
- (10) Remove wishbone and collect the thrust washer pack.



B Installation Procedure

- (1) Fit wishbone in position, locate the thrust washer pack and push in the wishbone pivot bushes.
- (2) Fit spindles and spindle nuts.
- (3) Torque load nuts 41 to 47Nm (30 to 35ftlb).
- (4) Fit front and rear wishbone attachment nuts and locking caps. The rear attachments are accessible from inside passenger compartment.
- (5) Torque load nuts 30 to 34Nm (22 to 25ftlb).
- (6) Locate the ball pin through the upright.
- (7) Fit washer and nut to ball pin.
- (8) Torque load nut 61 to 68Nm (45 to 50ftlb). It may be necessary to hold the ball pin with an allen key to prevent rotation during tightening.
- (9) Fit push rod to wishbone and secure with bolt, washer and nut.
- (10) Torque load nut 41 to 45Nm (30 to 33ftlb)
- (11) Fit passenger footwell panel and secure with quick release fasteners, if removed.
- (12) Fit the wheel arch liners and secure with screws and washers.
- (13) Fit wheels (Refer to Section 4.1).
- (14) Lower the lifting apparatus and remove the vehicle.

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3 FRONT LOWER WISHBONE BUSHINGS

A Removal Procedure

Special Tools

The following special tools will be required.

Hydraulic press

JaguarSport service tools ST0008, ST0006 and ST0015

JaguarSport protective body covers

Materials

The following materials will be required.

Approved degreaser

Soluble proprietary engine cleaner

Lint free cloth

NOTES:

1. Ensure absolute cleanliness of work area and tools.
2. Observe correct use of tools.
3. Check serviceability of components as they are removed. If in doubt reject component.
4. Clean all parts where applicable before reassembly.
5. Fit JaguarSport protective body covers in immediate work area.
6. JaguarSport service tools must be used to prevent damage to the components and to ensure correct assembly.

- (1) Remove wheel and wishbone (Refer to Section 4.2).
- (2) Place wishbone in hydraulic press.
- (3) Carefully press out rubber bushes using service tools ST0008, ST0006 and ST0015.

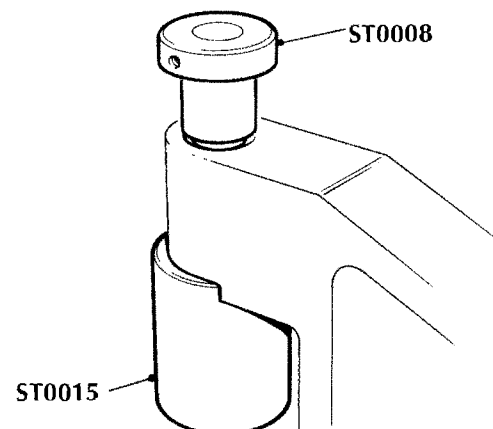
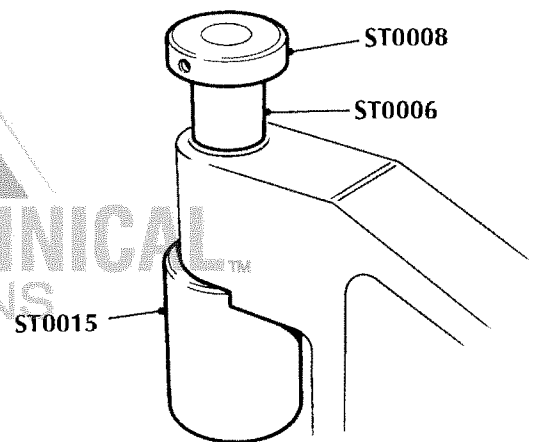
WARNING:

SOLVENTS ARE TOXIC AND MUST ONLY BE USED IN WELL VENTILATED WORK AREAS. AVOID INHALING THE SOLVENT FUMES.

- (4) Clean the inside surface of the wishbone bush bore using lint free cloth moistened with approved degreaser.

B Installation Procedure

- (1) Place wishbone in hydraulic press.
- (2) Press in the rubber bushes using service tools ST0008 and ST0015, taking care to ensure the bush does not tip to one side.
- (3) Fit wishbone (Refer to Section 4.2).



4 LOWER BALL PIN

A Removal Procedure

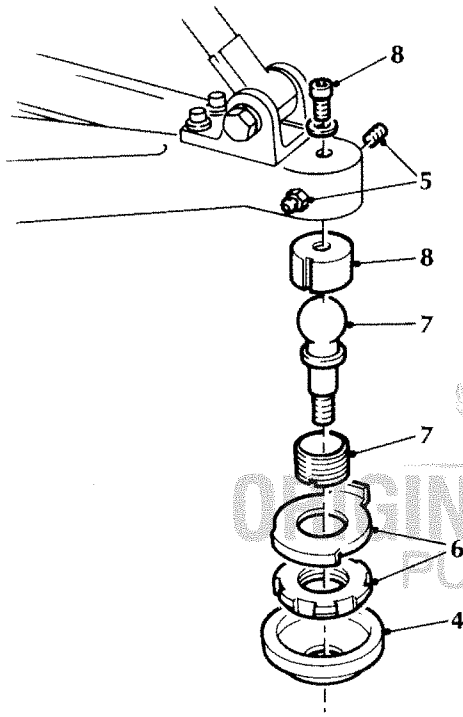
Special Tools

The following special tools will be required.

JaguarSport protective covers

JaguarSport service tool ST0001

Torque wrench 0 - 46Nm (35ftlb)



NOTES:

- 1 Ensure absolute cleanliness of work area and tools.
- 2 Ensure correct use of tools.
- 3 Clean all parts where applicable before reassembly using a soluble proprietary engine cleaner.
- 4 Fit JaguarSport protective body covers in immediate work area.
- 5 JaguarSport service tools must be used to prevent damage to the components and to ensure correct assembly.

- (1) Correctly position vehicle on lifting apparatus (Refer to Section 1.1), and remove wheels (Refer to Section 4.1).
- (2) Raise the vehicle to a suitable working height.
- (3) Remove the upright (Refer to Section 4.10)
- (4) Remove ball pin seal.
- (5) Remove the grub screw and grease nipple.
- (6) Straighten the locking tab washer and remove ball pin locking ring with service tool ST0001.
- (7) Remove ball pin cap with service tool ST0001 and remove the ball pin.
- (8) Remove the cap head screw and remove the ball pin seat from wishbone. If necessary, gently tap the seat through the hole with a thin flat-ended punch.

NOTE:

If any components require replacement, then the ball pin, seat and cap must be fitted as a new matched assembly.

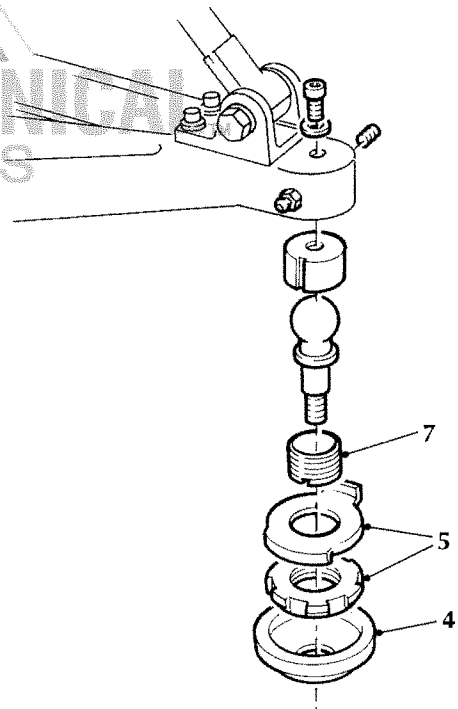
B Installation Procedure

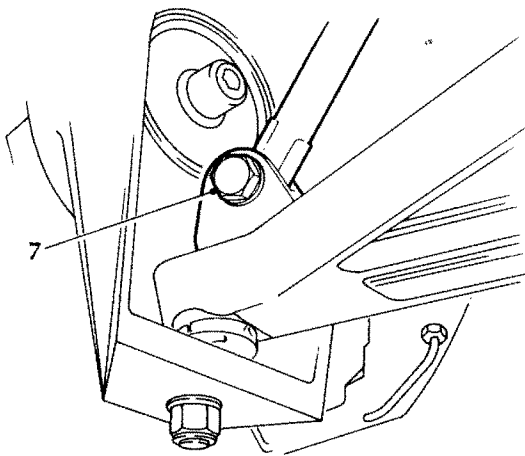
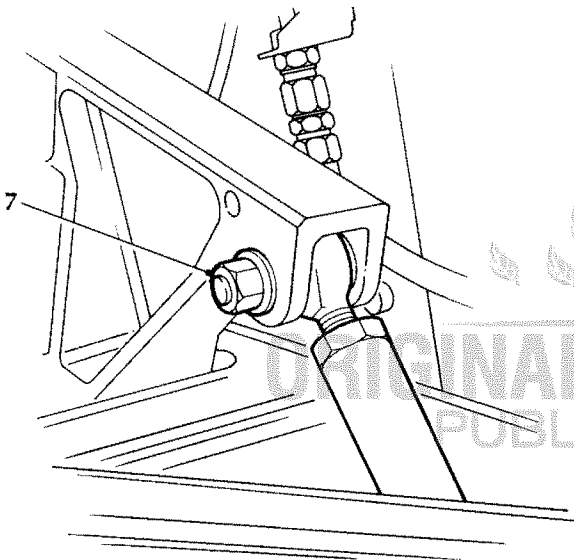
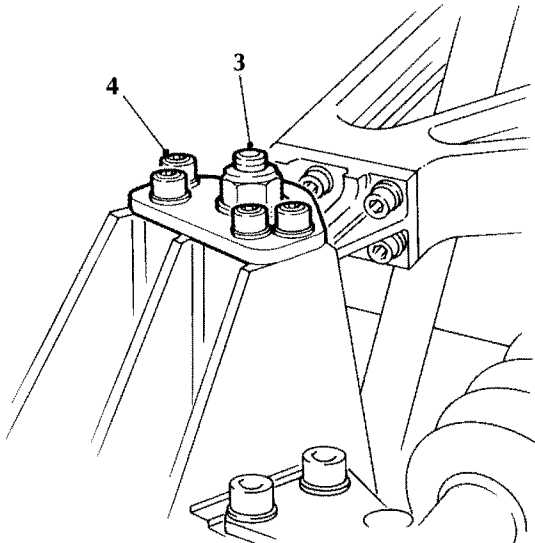
- (1) Fit the grub screw loosely
- (2) Fit ball pin seat to wishbone, ensuring the groove locates with the grubscrew.
- (3) Apply grease Castrol LMX to ball end of ball pin, ball pin seat and ball pin cap.
- (4) Fit ball pin to ball pin seat.
- (5) Fit ball pin cap finger tight.

- (6) Tighten ball pin cap with service tool ST0001 until a rotating torque of 5.5 to 8Nm (4 to 6ftlb) is achieved on ball pin .
- (7) Fit new tab washer and fit ball pin locking ring.
- (8) Torque load ball pin locking ring 41 to 46Nm (30 to 34.5ftlb) whilst ensuring the ball pin cap does not rotate, using service tool ST0001.
- (9) Lock the tab washer by neatly staking into the locking ring with a round nosed punch
- (10) Fit and tighten the grubscrew, cap head screw and grease nipple.
- (11) Grease the locking ring and new ball pin seal liberally with Castrol LMX.
- (12) Fit new ball pin seal.
- (13) Fit upright (Refer to Section 4.10).
- (14) Fit wheel (Refer to Section 4.1).
- (15) Lower the lifting apparatus and remove the vehicle.

Ball Pin Adjustment

- (1) Correctly position vehicle on lifting apparatus (Refer to Section 1.1), and remove wheels (Refer to Section 4.1).
- (2) Raise the vehicle to a suitable working height.
- (3) Remove the upright (Refer to Section 4.10)
- (4) Remove ball pin seal.
- (5) Straighten tabs on tab washer and remove the locking ring
- (6) Fit a new locking tab washer.
- (7) Tighten ball pin cap with service tool ST0001 whilst rotating ball pin until a rotating torque of 5.5 to 8Nm (4 to 6ftlb) is achieved on ball pin .
- (8) Torque load ball pin locking ring 41 to 46Nm (30 to 34.5ft lb), whilst ensuring that the ball pin cap does not rotate, using service tool ST0001.
- (9) Lock the tab washer by neatly staking into the locking ring with a round nosed punch.
- (10) Grease the locking ring, grease and fit new ball pin seal.
- (11) Fit upright (Refer to Section 4.10).
- (12) Check for play in ball pin and readjust if necessary.
- (13) Fit wheel (Refer to Section 4.1).
- (14) Lower the lifting apparatus and remove the vehicle.





5 FRONT UPPER WISHBONE

A Removal Procedure

Special Tools

The following special tools will be required.

JaguarSport protective covers

JaguarSport service tools ST0022 and ST0023

Torque wrench 0 - 47Nm (35ftlb)

NOTES:

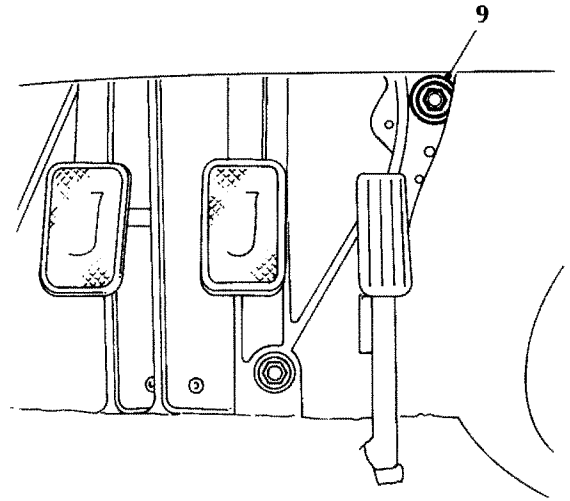
- 1 Ensure absolute cleanliness of work area and tools.
- 2 Ensure correct use of tools.
- 3 Check serviceability of components as they are removed. If in doubt reject component.
- 4 Clean all parts where applicable before reassembly using a soluble proprietary engine cleaner.
- 5 Fit JaguarSport protective body covers in immediate work area.
- 6 JaguarSport service tools must be used to prevent damage to the components and to ensure correct assembly.

- (1) Correctly position vehicle on lifting apparatus (Refer to Section 1.1), and remove wheel. (Refer to Section 4.1).
- (2) Raise the vehicle to a suitable working height, remove screws and washers and remove wheel arch liner.
- (3) Remove the nut attaching wishbone to upright.
- (4) Remove cap head screws securing top plate to upright, and remove top plate.
- (5) Carefully release flexible brake hose from the wishbone
- (6) Support the upright.
- (7) Remove upper and lower push rod attachment screws, washers and nuts and remove push rod.
- (8) Open the bonnet and remove the radiator duct access panel to gain access to the front locking caps and nuts.
- (9) When removing passenger's side component, remove the footwell panel to gain access to the rear locking caps and nuts.

WARNING

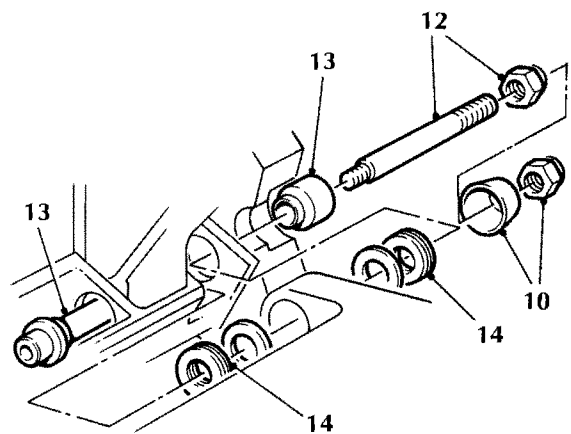
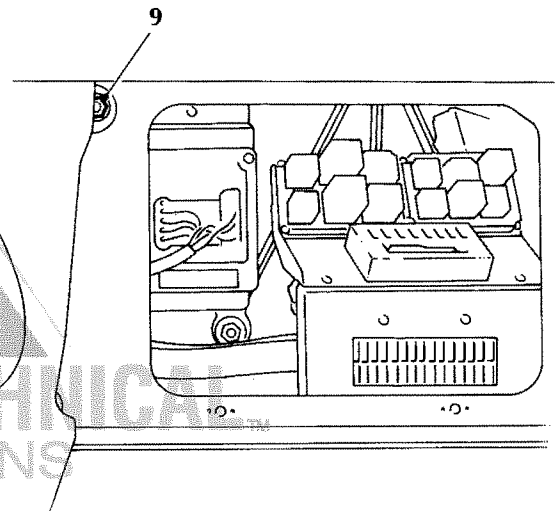
ENSURE THE BATTERY MASTER SWITCH IS OFF PRIOR TO WORKING IN THE PASSENGER'S FOOTWELL PANEL.

- (10) Remove wishbone nuts and locking caps.
- (11) Lift the outer joint off the upright pin.
- (12) Remove spindle nuts and remove spindles.
- (13) Withdraw wishbone front pivot bushes slightly (using service tools ST0022 and ST0023, if necessary).
- (14) Remove wishbone and collect thrust washer pack.



B Installation Procedure

- (1) Support the upright, fit wishbone in position, locate the thrust washer packs and push in the wishbone pivot bushes.
- (2) Fit spindles and spindle nuts.
- (3) Torque load wishbone/nodder spindle 20Nm (15ftlb) and spindle nuts 41 to 47Nm (30 to 35ftlb).
- (4) Fit wishbone locking caps and secure with nuts.
- (5) Torque load nuts 30 to 34Nm (22 to 25ftlb).
- (6) Fit push rod and secure with screws, washers and nuts.
- (7) Torque load lower attachment nut 41 to 45Nm (30 to 33ftlb) and upper attachment nut 35 to 41Nm (26 to 30ftlb).
- (8) Fit the wishbone to the top pin on the upright.
- (9) Fit the insert and top plate and secure with cap head screws.
- (10) Torque load cap head screws 8 to 9.5Nm (6 to 7ftlb).
- (11) Fit the nut to the pin and torque load to 41 to 47Nm (30 to 34.5ftlb).
- (12) Attach the flexible brake hose to the wishbone using new tyrap.
- (13) Fit radiator duct panel and secure screws.
- (14) Fit the footwell panel (if applicable).
- (15) Fit wheel arch liners and secure with screws and washers.
- (16) Fit wheels (Refer to Section 4.1).
- (17) Lower the lifting apparatus and remove the vehicle.



6 FRONT UPPER WISHBONE BUSHINGS

WISHBONE BUSHES

A Removal Procedure

Special Tools

The following special tools will be required.

JaguarSport protective body covers

JaguarSport service tools ST0006, ST0008, ST0009 and ST0015

Hydraulic press

Materials

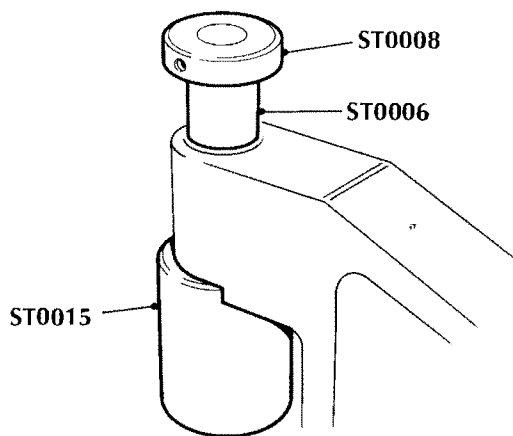
The following materials will be required.

Approved degreaser

Lint free cloth

NOTES:

- 1 Ensure absolute cleanliness of work area and tools.
- 2 Ensure correct use of tools.
- 3 Check serviceability of components as they are removed. If in doubt reject component.
- 4 Clean all parts where applicable before reassembly using a soluble proprietary engine cleaner.
- 5 Fit JaguarSport protective body covers in immediate work area.
- 6 JaguarSport service tools must be used to prevent damage to the components and to ensure correct assembly.



- (1) Remove the wheel and wishbone. (Refer to Section 4.5).
- (2) Place wishbone in hydraulic press.
- (3) Carefully press out rubber bushes using service tools ST0006, ST0015 and ST0008.

WARNING:

SOLVENTS ARE TOXIC AND MUST ONLY BE USED IN WELL VENTILATED WORK AREAS. AVOID INHALING THE SOLVENT FUMES.

- (4) Clean inside surface of wishbone bush bore using lint free cloth moistened with approved degreaser.

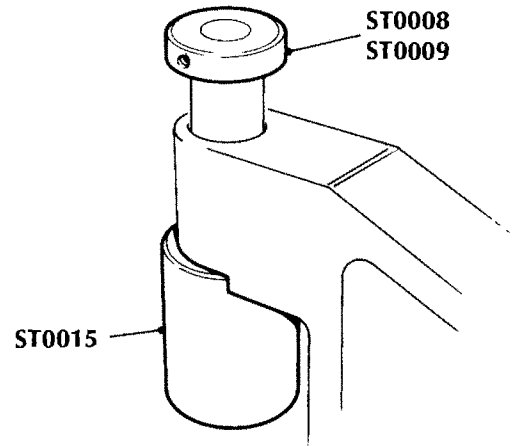
B Installation Procedure

- (1) Place wishbone in hydraulic press.
- (2) Press in rubber bushes using service tools ST0015, ST0009 (for the narrow wishbone leg) and ST0008 (for the wide wishbone leg), taking care the bush does not tip to one side during installation.

NOTE:

The service tools will correctly position the bushes in the wishbone.

- (3) Fit wishbone and wheel (Refer to Section 4.5).




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END CAP SPHERICAL JOINT

A Removal Procedure

Special Tools

The following special tools will be required;

JaguarSport protective body cover

JaguarSport service tools ST0003, ST0010, ST0011, ST0012 and ST0013

Hydraulic press

Protective gloves

Materials

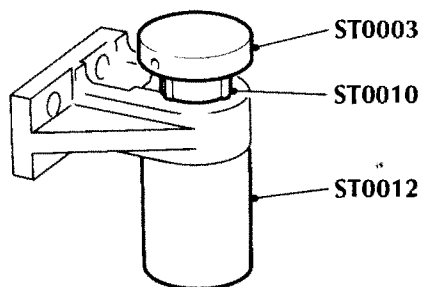
The following materials will be required;

Approved degreaser

Lint free cloth

NOTES:

- 1 Ensure absolute cleanliness of work area and tools.
- 2 Ensure correct use of tools.
- 3 Check serviceability of components as they are removed. If in doubt reject component.
- 4 Clean all parts where applicable before reassembly using a soluble proprietary engine cleaner.
- 5 Fit JaguarSport protective body covers in immediate work area.
- 6 JaguarSport service tools must be used to prevent damage to the components and to ensure correct assembly.



- (1) Remove the upright (Refer to Section 4.10).
- (2) Remove the 4 cap head screws and washers securing the end cap to the wishbone and remove the end cap and shims
- (3) Remove the spirolox clip.
- (4) Place the end cap in a hydraulic press.
- (5) Carefully press out the end cap spherical joint using service tools ST0003, ST0010 and ST0012.

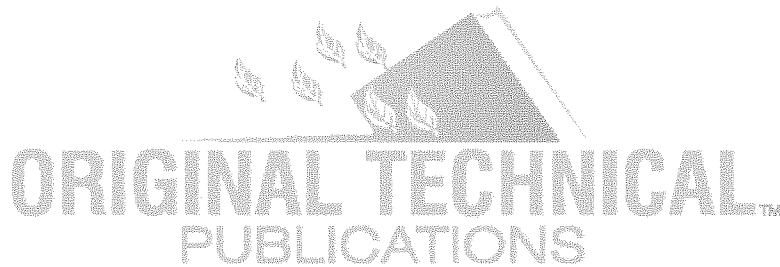
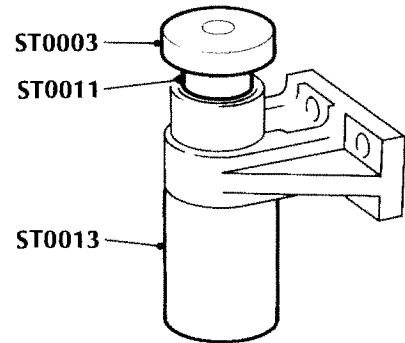
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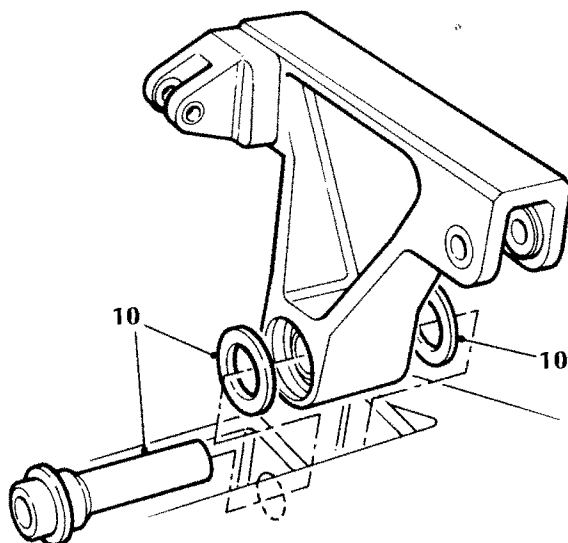
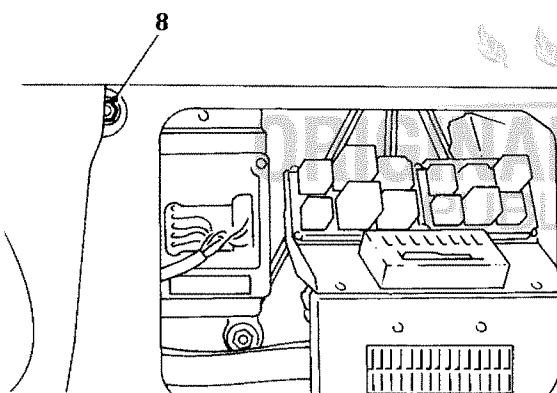
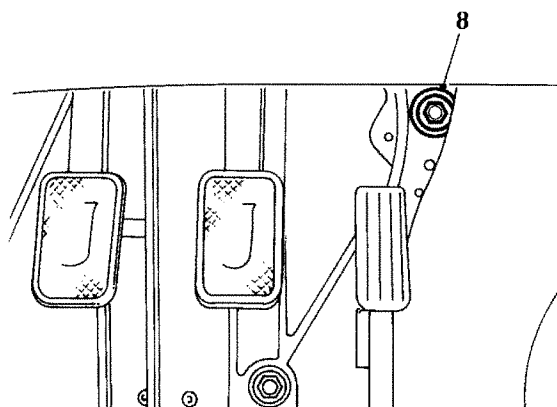
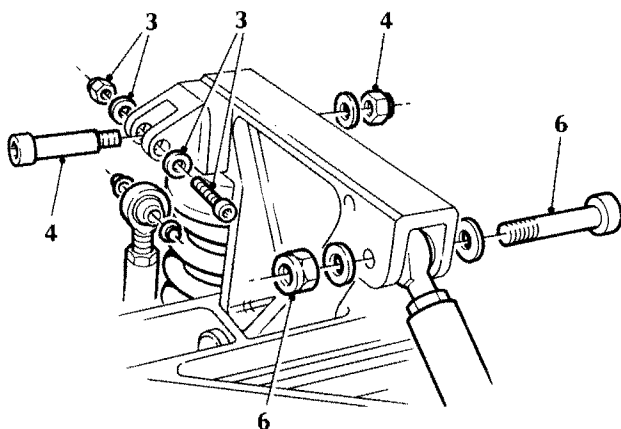
SOLVENTS ARE TOXIC AND MUST ONLY BE USED IN WELL VENTILATED WORK AREAS. AVOID INHALING SOLVENT FUMES.

- (5) Clean the inside surface of the end cap using lint free cloth moistened with approved degreaser.

B Installation Procedure

- (1) Place the spherical joint in a freezer at -20 degrees C (21.6°F).
- (2) Place the end cap in a hydraulic press.
- (3) Using protective gloves remove the spherical joint from the freezer.
- (4) Carefully press the spherical joint into the end cap using service tools ST0003, ST0011 and ST0013.
- (5) Fit the spirolox clip.
- (6) Fit the end cap and shims to the wishbone and torque the screws 8 to 9.5Nm (6 to 7ftlb).
- (7) Fit the upright (Refer to Section 4.10).
- (8) On completion, check the front camber (Refer to Section 4.26).





7 FRONT SUSPENSION NODDERS

A Removal Procedure

Special Tools

The following special tools will be required:

JaguarSport protective body covers

Torque wrench 47Nm (35ftlb)

NOTES:

- 1 Ensure absolute cleanliness of work area and tools.
- 2 Ensure correct use of tools.
- 3 Check serviceability of components as they are removed. If in doubt reject component.
- 4 Clean all parts where applicable before reassembly using a soluble proprietary engine cleaner.
- 5 Fit JaguarSport protective body covers in immediate work area.

- (1) Correctly position vehicle on lifting apparatus (Refer to Section 1.1) and raise vehicle to a suitable working height.
- (2) Remove wheel (Refer to Section 4.1) and remove wheel arch liner.
- (3) Remove bolt, washers and nut securing drop link to nodder bracket.
- (4) Remove bolt, washers and nut securing coil spring and damper to nodder.
- (5) Support the upright.
- (6) Remove bolt, washers and nut securing push rod to nodder.
- (7) When removing passenger's side components, remove the footwell panel to gain access to the locking caps and nuts.

WARNING

ENSURE THE BATTERY MASTER SWITCH IS OFF PRIOR TO WORKING IN THE PASSENGER'S FOOTWELL PANEL.

- (8) Remove nut and locking cap on nodder spindle from inside the vehicle.
- (9) Remove spindle nut and undo the spindle from the nodder pivot. Withdraw the spindle sufficiently to allow removal of the nodder without disrupting the wishbone.

NOTE:

During removal of nodder spindle ensure thrust washers packing wishbone rear upper attachment do not fall out of the assembly.

- (10) Remove nodder pivot, collect thrust washers and withdraw nodder.

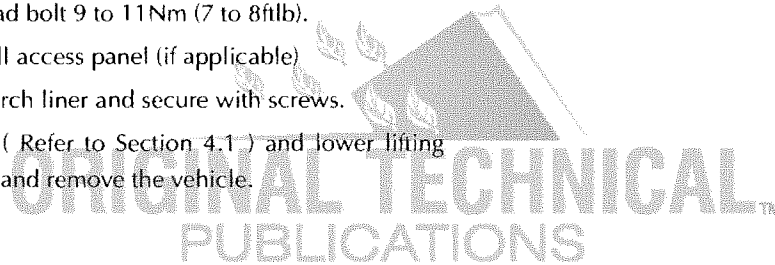
B Installation Procedure

- (1) Position nodder with thrust washers in upper front suspension beam.
- (2) Fit nodder pivot and nodder spindle. Torque load spindle 20Nm (15ftlb).
- (3) Fit nut and torque load to 41 to 47Nm (30 to 34.5ftlb).
- (4) Fit the locking cap and nut. Torque load nut 30 to 34Nm (22 to 25ftlb).
- (5) Fit push rod to nodder and secure with bolt, washers and nut.

NOTE:

The nut must be fitted to the front of the vehicle.

- (6) Torque load bolt 35 to 41Nm (26 to 30ftlb).
- (7) Fit coil spring and damper to nodder, fit bolt, washer and nut. .
- (8) Torque load nut 20 to 23Nm (15 to 17ftlb).
- (9) Fit drop link to nodder and secure bolt, washer and nut.
- (10) Torque load bolt 9 to 11Nm (7 to 8ftlb).
- (11) Fit footwell access panel (if applicable)
- (12) Fit wheelarch liner and secure with screws.
- (13) Fit wheel (Refer to Section 4.1) and lower lifting apparatus and remove the vehicle.



8 FRONT NODDER SPHERICAL JOINTS

A Removal Procedure

Special Tools

The following special tools will be required:

JaguarSport protective body covers

JaguarSport service tools ST0031 and ST0032

Hydraulic press

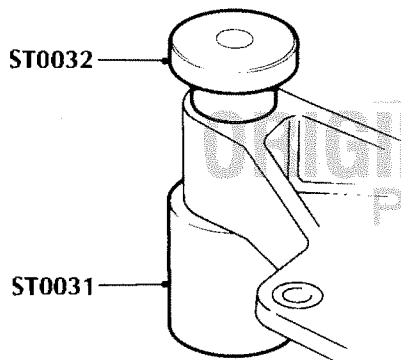
Materials

Approved degreaser

Lint free cloth

NOTES:

- 1 Ensure absolute cleanliness of work area and tools.
- 2 Ensure correct use of tools.
- 3 Check serviceability of components as they are removed. If in doubt reject component.
- 4 Clean all parts where applicable before reassembly using a soluble proprietary engine cleaner.
- 5 Fit JaguarSport protective body covers in immediate work area.
- 6 JaguarSport service tools must be used to prevent damage to the components and to ensure correct assembly.

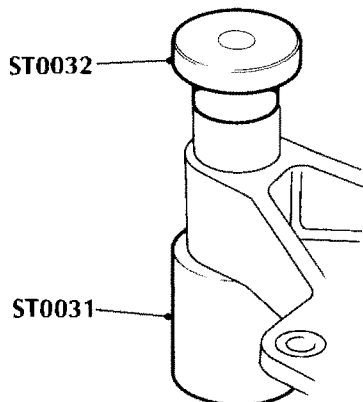


- (1) Remove the front nodder (Refer to Section 4.7).
- (2) Place the nodder in hydraulic press.
- (3) Carefully press out the nodder spherical joints using service tools ST0031 and ST0032.

WARNING

SOLVENTS ARE TOXIC AND MUST ONLY BE USED IN WELL VENTILATED AREAS. AVOID INHALING THE SOLVENT FUMES.

- (4) Clean the inside surface of the nodder bore using lint free cloth moistened with approved degreaser.



B Installation Procedure

- (1) Place the nodder in hydraulic press.
- (2) Press in the spherical joints using service tools ST0031 and ST0032 taking care the joints do not tip to one side during installation.
- (3) Fit the nodder (Refer to Section 4.7).

9 PUSHROD BUSHES

A Removal Procedure

Special Tools

The following special tools will be required:

JaguarSport protective body covers

JaguarSport service tools ST0006, ST0009 and ST0016

Hydraulic press

Materials

Approved degreaser

Lint free cloth

NOTES:

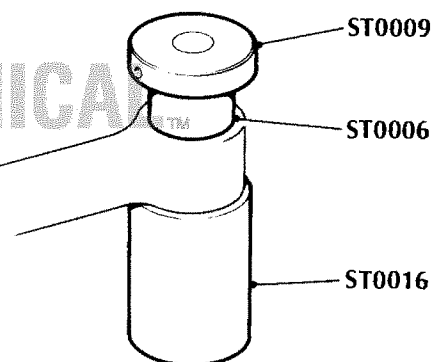
- 1 Ensure absolute cleanliness of work area and tools.
- 2 Ensure correct use of tools.
- 3 Check serviceability of components as they are removed. If in doubt reject component.
- 4 Clean all parts where applicable before reassembly using a soluble proprietary engine cleaner.
- 5 Fit JaguarSport protective body covers in immediate work area.
- 6 JaguarSport service tools must be used to prevent damage to the components and to ensure correct assembly.

- (1) Remove the relevant pushrod.
- (2) Place the pushrod in hydraulic press.
- (3) Carefully press out the pushrod rubber bush using service tools ST0006, ST0009 and ST0016.

WARNING

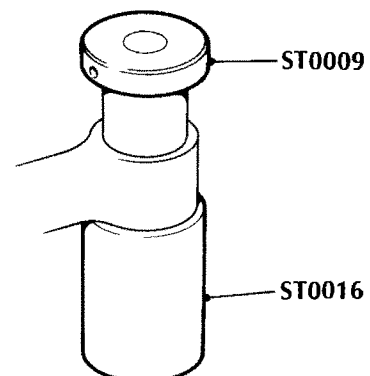
SOLVENTS ARE TOXIC AND MUST ONLY BE USED IN WELL VENTILATED AREAS. AVOID INHALING THE SOLVENT FUMES.

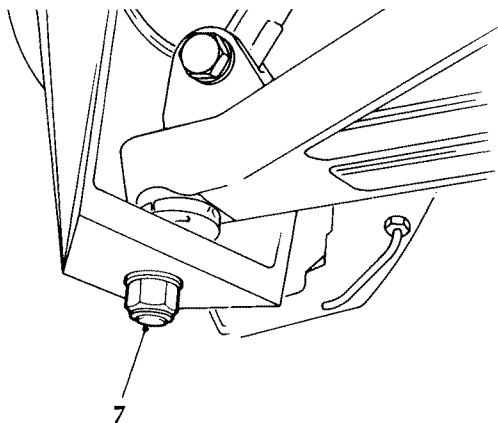
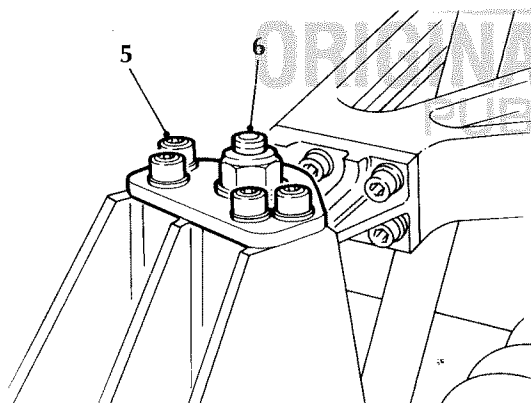
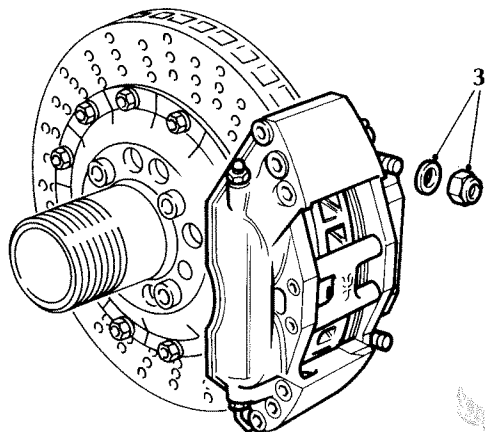
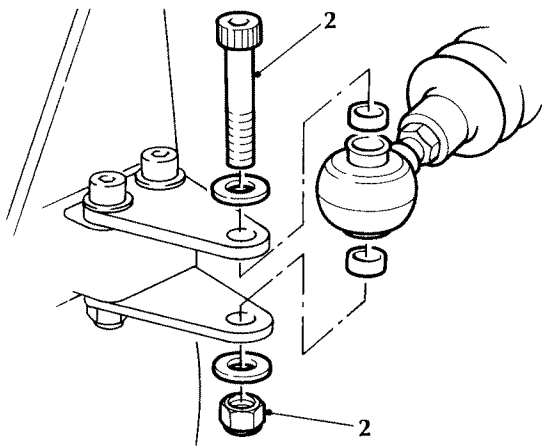
- (4) Clean the inside surface of the pushrod bore using lint free cloth moistened with approved degreaser.



B Installation Procedure

- (1) Place the pushrod in hydraulic press.
- (2) Press in the rubber bush using service tools ST0016 and ST0009 taking care the bush does not tip to one side during installation.
- (3) Fit the pushrod.





10 FRONT SUSPENSION UPRIGHT

A Removal Procedure

Special Tools

The following special tools will be required:

JaguarSport protective body covers

Torque wrench 0 - 68Nm (50ftlb)

NOTES:

- 1 Ensure absolute cleanliness of work area and tools.
- 2 Ensure correct use of tools.
- 3 Check serviceability of components as they are removed. If in doubt reject component.
- 4 Clean all parts where applicable before reassembly using a soluble proprietary engine cleaner.
- 5 Fit JaguarSport protective body covers in immediate work area.

- (1) Correctly position vehicle on lifting apparatus, raise vehicle to suitable working height and remove the wheel (Refer to Section 4.1).
- (2) Remove bolt, washer, spacers and nut and disconnect track rod from steering bracket. Note the position and fitment of spacers.
- (3) Remove nuts and remove brake caliper.

CAUTION:

Whilst the caliper is disconnected, ensure the brake pedal is not operated.

- (4) Attach brake caliper to top wishbone with tyrap or similar to prevent placing any strain on the flexible hose.

CAUTION:

Support the upright during the removal of the pushrod and wishbone.

- (5) Remove screws and washers securing top plate to upright.
- (6) Remove nut and washer securing upright to upper wishbone and remove the plate.
- (7) Remove ball pin nut and washer and remove upright.

B Installation Procedure

- (1) Fit upright in position and attach to lower wishbone ball pin with washer and nut.
- (2) Fit upper wishbone over attachment stud on upright.
- (3) Fit top plate, washer and nut.
- (4) Fit screws and washers to secure top plate.
- (5) Fit tie rod with spacers to steering bracket on side of upright and secure with bolt washers and nut. Ensure correct fitment of spacers.
- (6) Fit brake caliper and secure with washers and nuts.
- (7) Torque tighten the ball pin nut 61 to 68Nm (45 to 50ftlb).
- (8) Torque tighten the upright stud nut to 41 to 46Nm (30 to 34.5ftlb).
- (9) Torque tighten the top plate nuts securing the top plate to the upright to 8 to 9.5Nm (6 to 7ftlb).
- (10) Torque tighten the tie rod attachment nut 35 to 41 Nm (26 to 30ftlb).
- (11) Torque tighten the brake caliper nuts 57 to 65Nm (42 to 48ftlb).
- (12) Secure the brake pipe to wishbone with new cable ties.
- (13) Fit the road wheels (Refer to Section 4.1), and lower the lifting apparatus.
- (14) Remove the vehicle.

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11 FRONT WHEEL BEARINGS

A Removal Procedure

Special Tools

The following special tools will be required.

Hydraulic press (or suitable alternative)

JaguarSport service tools ST0002 and ST0030

JaguarSport protective body covers

Torque wrench 0 - 380Nm (280ftlb) and 0 - 65Nm (48ftlb).

Materials

The following materials will be required.

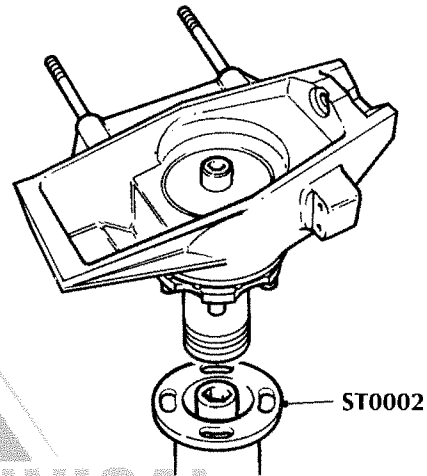
Lint free cloth

Protective gauntlets

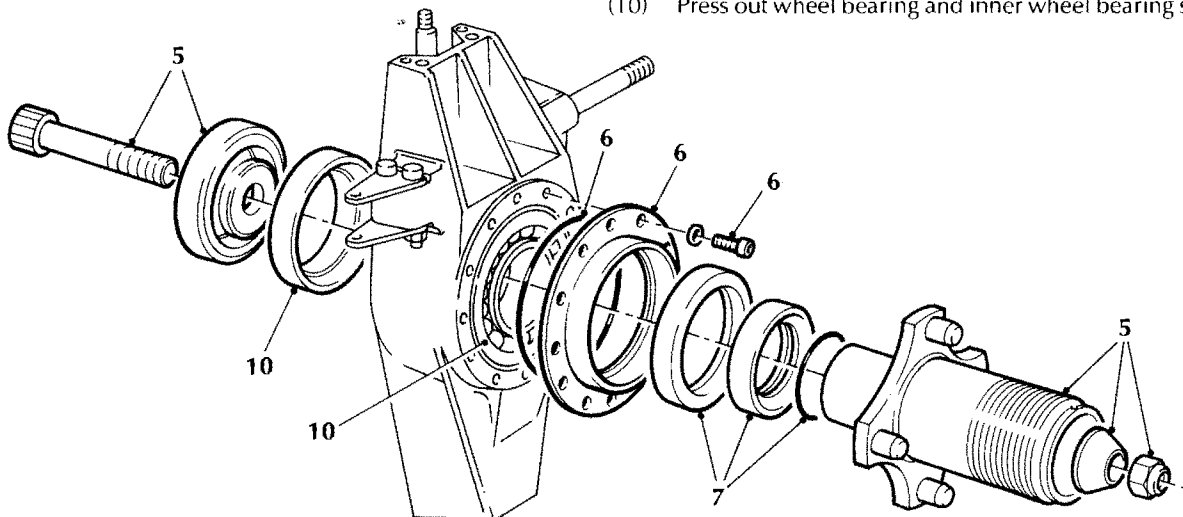
NOTES:

- 1 Ensure absolute cleanliness of work area and tools.
- 2 Ensure correct use of tools.
- 3 Check serviceability of components as they are removed. If in doubt reject component.
- 4 Clean all parts where applicable before reassembly using a soluble proprietary engine cleaner.
- 5 Fit JaguarSport protective body covers in immediate work area.
- 6 JaguarSport service tools must be used to prevent damage to the components and to ensure correct assembly.

- (1) Correctly position vehicle on lifting apparatus (Refer to Section 1.1) and remove road wheel (Refer to Section 4.1) and raise vehicle to suitable working height.
- (2) Remove the upright (Refer to Section 4.10).
- (3) Remove logo cap from end of stub axle (Refer to Section 11.5)
- (4) Remove countersink screws and remove brake disc with brake bell. (Refer to Section 6).
- (5) Place the upright in the service tool ST0002 and remove bolt and nut with load spreader and inner axle collar, and remove stub axle.



- (6) Remove 12 screws with washers and remove bearing retainer and O ring seal.
- (7) Remove collar, O ring and wheel bearing seal from bearing retainer.
- (8) Carefully apply heat to upright bearing housing as evenly as possible.
- (9) Place upright in hydraulic press.
- (10) Press out wheel bearing and inner wheel bearing seal.



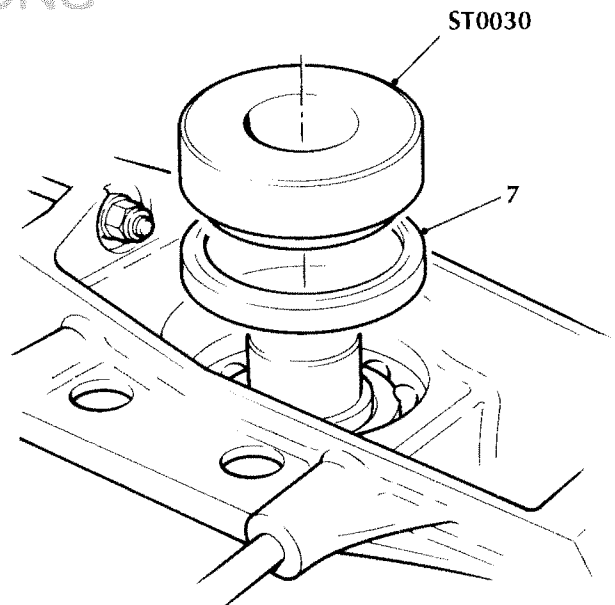
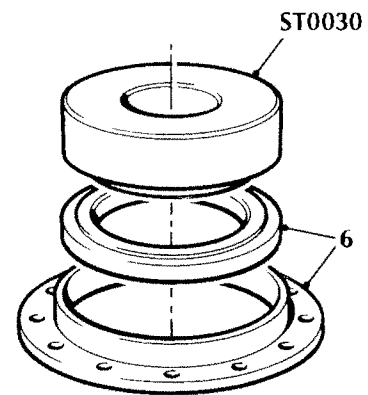
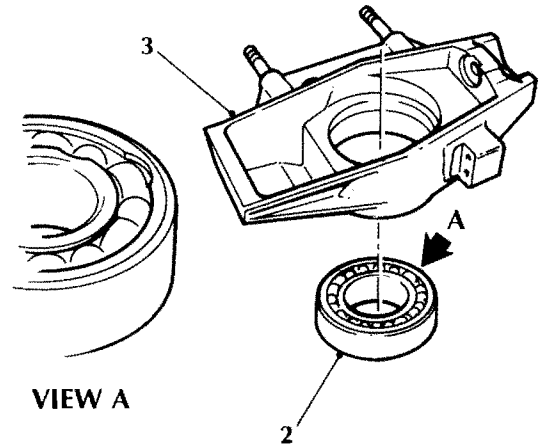
B Installation Procedure

- (1) Place upright in oven for 6 to 8 hours at 120 degrees C (273 degrees F). Alternatively, heat the upright evenly with a propane torch.
- (2) Pack wheel bearing with 50cc Castrol LMX grease and place wheel bearing in freezer at -20 degrees C (21.6 degrees F).
- (3) Using protective gloves, remove the upright from the oven
- (4) Using protective gloves, remove the bearing from the freezer and fit the bearing to the upright, ensuring the bearing filling slots are towards the vehicle centreline.

NOTE:

No force should be used to fit the bearing.

- (5) Leave the upright in position to cool.
- (6) Press wheel bearing seal into bearing retainer using service tool ST0030.
- (7) Press inner wheel bearing seal into upright using service tool ST0030.
- (8) Fit new O ring to the bearing retainer.
- (9) Secure bearing retainer to upright with 12 screws.
- (10) Torque load screws 8 to 9Nm (6 to 7ftlb).
- (11) Fit new O ring to collar and fit collar to stub axle.
- (12) Smear Castrol LMX grease onto the wheel bearing seal faces.
- (13) Fit stub axle to upright and secure with inner axle collar, bolt, load spreader and nut using service tool ST0002.
- (14) Torque load nut 366 to 380Nm (270 to 280ftlb).
- (15) Fit upright (Refer to Section 4.10).
- (16) Fit brake disc with brake bell and secure with countersink screws.
- (17) Torque load countersink screws 7 to 8Nm (5 to 6ftlb).
- (18) Fit the brake caliper and secure nuts. Torque nuts 57 to 65Nm (42 to 48ftlb)
- (19) Fit cap to end of stub axle, and tighten the grub screw.
- (20) Fit road wheel (Refer to Section 4.1) and lower lifting apparatus.
- (21) Remove vehicle from lifting apparatus.



12 FRONT ANTI-ROLL BAR

A Removal Procedure

Special Tools

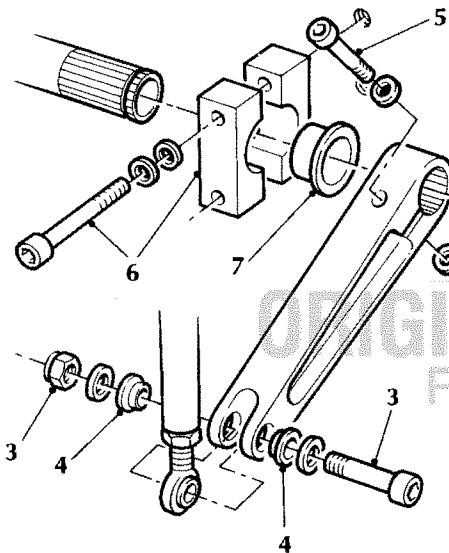
The following special tools will be required:

JaguarSport protective body covers

Torque wrench 34Nm (25ftlb)

NOTES:

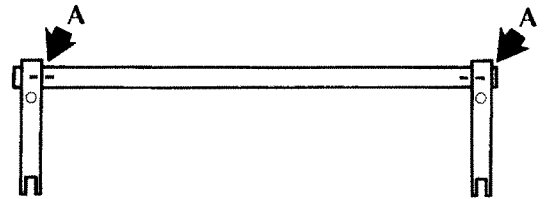
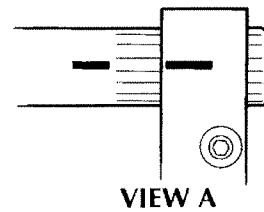
- 1 Ensure absolute cleanliness of work area and tools.
- 2 Ensure correct use of tools.
- 3 Check serviceability of components as they are removed. If in doubt reject component.
- 4 Clean all parts where applicable before reassembly using a soluble proprietary engine cleaner.
- 5 Fit JaguarSport protective body covers in immediate work area.



- (1) Open vehicle bonnet, remove quick release fasteners and remove radiator duct access cover.
- (2) Correctly position vehicle on lifting apparatus (Refer to Section 1.1), and remove wheels (Refer to Section 4.1).
- (3) Remove screws, washers and nuts and disconnect drop links from anti-roll bar levers.
- (4) Remove the inserts from the levers.
- (5) Remove screws, washers and nuts and retaining clips, and remove levers from anti-roll bar.
- (6) Remove screws and washers and remove clamps.
- (7) Remove bushes.
- (8) Tip the anti-roll bar and remove from under the RH side crush tube.

B Installation Procedure

- (1) Before fitting the anti-roll bar to the vehicle proceed as follows:
 - Position the anti-roll bar on a level work surface ;
 - Fit the levers and ensure they are perfectly aligned;
 - Mark a line on the levers and a corresponding line on the anti-roll bar ;
 - Remove levers from the anti-roll bar.
- (2) Fit anti-roll bar and position clamps and bushes at each end.
- (3) Secure anti-roll bar and clamps with screws and washers.
- (4) Torque load screws 8 to 9.5Nm (6 to 7ftlb).
- (5) Fit levers to anti-roll bar aligning the marked lines on the anti-roll bar with those on the levers and secure with screws, washers and nuts.
- (6) Fit the retaining clips.
- (7) Torque load bolts 30 to 34Nm (22 to 25ftlb).
- (8) Fit drop links to levers and secure with screws, washers and nuts.
- (9) Torque load screws 12 to 14Nm (9 to 10.5ftlb).
- (10) Fit the wheels (Refer to Section 4.1) lower the lifting apparatus and remove the vehicle .
- (11) Fit radiator duct access cover and secure with quick release fasteners.
- (12) Close vehicle bonnet.



13 FRONT SUSPENSION COIL SPRING AND DAMPER ASSEMBLY

A Removal Procedure

Special Tools

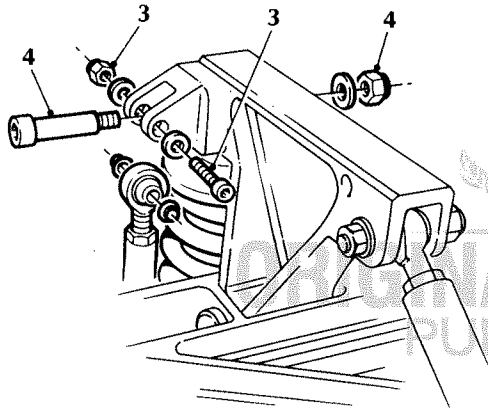
The following special tools will be required:

JaguarSport protective body covers

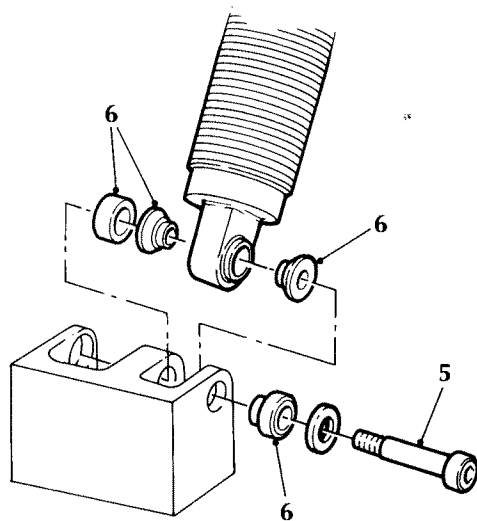
Torque wrench 0 - 23Nm (17ftlb)

NOTES

1. Ensure absolute cleanliness of work area and tools.
2. Observe correct use of tools.
3. Check serviceability of components as they are removed. If in doubt reject component.
4. Clean all parts where applicable before reassembly using a soluble proprietary engine cleaner.
5. Fit JaguarSport protective body covers in immediate work area.



- (1) Correctly position vehicle on lifting apparatus (Refer to Section 1.1), and remove wheels (Refer to Section 4.1).
- (2) Raise the vehicle to a suitable working height, remove screws and washers and remove wheel arch liner.
- (3) Remove the screw securing the drop link to the nodder bracket, disconnect the drop link and collect the bushes.
- (4) Support the upright and remove cap head screw, washer and nut securing the coil spring and damper to the nodder.



- (5) Remove the lower cap head screw and remove coil spring and damper from lower support.
- (6) Remove the coil spring and damper and collect inserts.

B Installation Procedure

- (1) Fit inserts to eye ends of coil spring and damper.
- (2) Fit coil spring and damper to lower attachment fitting.
- (3) Fit M8 cap head screw to secure coil spring and damper to lower attachment.
- (4) Torque screw 20 to 23Nm (15 to 17 ftlb).
- (5) Fit 3/8in cap head screw, washer and nut to secure coil spring and damper to nodder.
- (6) Torque load cap head screw 20 to 23Nm (15 to 17 ftlb).
- (7) Fit the bushes and drop link to the nodder bracket and secure with 1/4in special screw.
- (8) Torque the screw 9.5 to 11Nm (7 to 8 ftlb).
- (9) Fit wheel arch liner and secure with screws and washers.
- (10) Fit wheel (Refer to Section 4.1), lower the lifting apparatus and remove the vehicle.



14 UPPER WISHBONE SUPPORT BEAM

A Removal Procedure

Special Tools

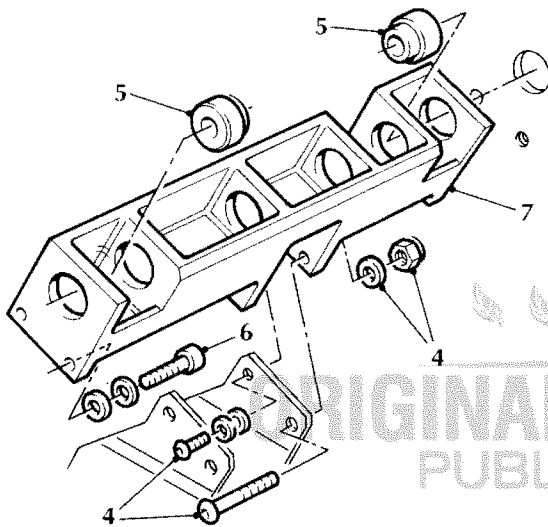
The following special tools will be required:

JaguarSport protective body covers

Torque wrench 0 - 23Nm (17ftlb)

NOTES

1. Ensure absolute cleanliness of work area and tools.
2. Observe correct use of tools.
3. Check serviceability of components as they are removed. If in doubt reject component.
4. Clean all parts where applicable before reassembly using a soluble proprietary engine cleaner.
5. Fit JaguarSport protective body covers in immediate work area.



- (1) Correctly position vehicle on lifting apparatus (Refer to Section 1.1), and remove wheels (Refer to Section 4.1).
- (2) Raise the vehicle to a suitable working height, remove screws and washers and remove wheel arch liner.
- (3) Remove upper wishbone (Refer to Section 4.5).
- (4) Remove bolts, washers and nuts attaching steering rack bracket to wishbone upper support beam.
- (5) Extract the front and rear support to bulkhead bushes.
- (6) Remove bolts securing upper wishbone support beam to zero bulkhead and suspension bulkhead.
- (7) Remove upper wishbone support beam.

B Installation Procedure

- (1) Fit wishbone upper support beam and insert the front and rear support to bulkhead bushes.
- (2) Secure wishbone upper support beam to zero bulkhead and suspension bulkhead with bolts.
- (3) Torque load bolts 20 to 23Nm (15 to 17ftlb).
- (4) Attach steering rack bracket to wishbone upper support beam with screws.
- (5) Torque M6 screws 7.5 to 9Nm (5.5 to 6.5ftlb) and M5 screw 5 to 7Nm (3.5 to 4.5ftlb).
- (6) Fit upper wishbone (Refer to Section 4.5).
- (7) Fit wheel arch liners and secure with screws and washers.
- (8) Fit wheels (Refer to Section 4.1).
- (9) Lower the lifting apparatus and remove the vehicle.

15 REAR LOWER WISHBONE

A Removal Procedure

Special Tools

The following special tools will be required.

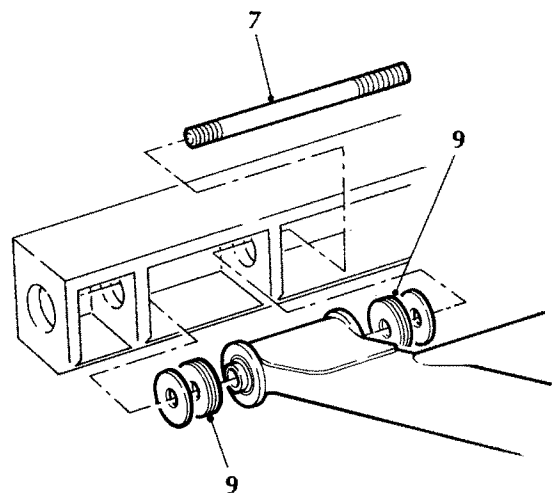
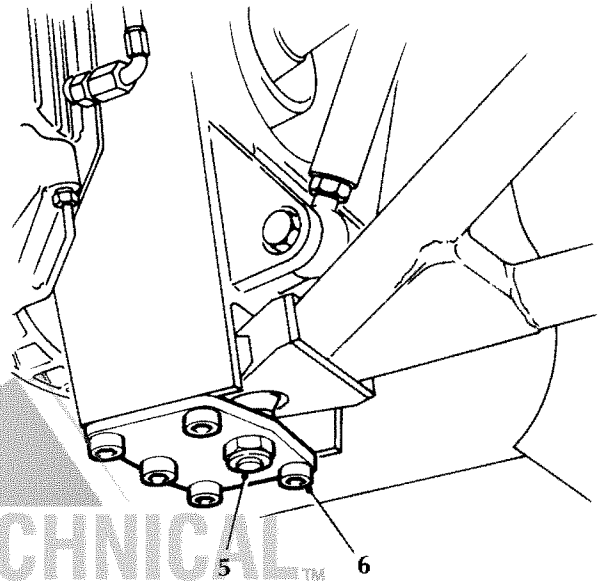
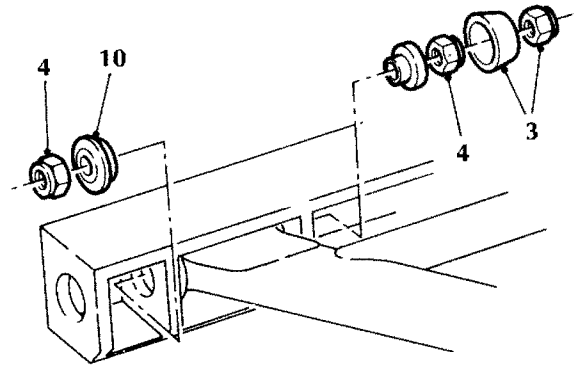
JaguarSport protective body covers

Torque wrench 0 - 65Nm (48ftlb)

NOTES:

1. *Ensure absolute cleanliness of work area and tools.*
2. *Observe correct use of tools.*
3. *Check serviceability of components as they are removed. If in doubt reject component.*
4. *Clean all parts where applicable before reassembly using a soluble proprietary engine cleaner.*
5. *Fit JaguarSport protective body covers in immediate work area.*

- (1) Correctly position vehicle on lifting apparatus (Refer to Section 1.1), and remove wheels (Refer to Section 4.1).
- (2) Raise the vehicle to a suitable working height, remove screws and washers and remove venturi and bumper. (Refer to Section 2).
- (3) Remove nuts and collect locking caps off each leg of the wishbone.
- (4) Remove spindle nuts.
- (5) Remove nut and washer securing upright bottom pin to the lower plate.
- (6) Remove 5 screws and washers securing plate to upright and remove the plate.
- (7) Remove wishbone pivot spindles.
- (8) Remove wishbone.
- (9) Collect thrust washer packs and washer assemblies.
- (10) Remove wishbone pivot bushes.



B Installation Procedure

- (1) Fit pivot bushes if removed.
- (2) Fit wishbone and thrust washer packs to engine support beam.
- (3) Fit wishbone pivot spindles.
- (4) Fit nuts to pivot spindles and torque load 41 to 47Nm (30 to 35ftlb).
- (5) Fit wishbone locking caps and nuts to secure wishbone to rear suspension lower beam.

NOTE:

Do not tighten nuts securing locking caps at this stage.

- (6) Fit wishbone on to pin on bottom of upright.
- (7) Fit bottom plate to upright and secure with 5 bolts and washers.
- (8) Torque load screws 20 to 23Nm (15 to 17ftlb).
- (9) Fit nut and washer securing upright bottom pin to lower plate.
- (10) Torque load nut 57 to 65Nm (42 to 48ftlb).
- (11) Torque load nuts securing locking caps 30 to 34Nm (22 to 25ftlb).
- (12) Fit venturi and bumper and secure with screws and washers.
- (13) Fit the wheel (Refer to Section 4.1).
- (14) Lower lifting apparatus and remove vehicle.

ORIGINAL TECHNICAL
PUBLICATIONS

16 REAR LOWER WISHBONE BUSHINGS

WISHBONE BUSHES

A Removal Procedure

Special Tools

The following special tools will be required.

Hydraulic press

JaguarSport protective body covers

JaguarSport service tools ST0004, ST0005, ST0006 and ST0007

Materials

The following materials will be required.

Approved degreaser

Lint free cloth

NOTES:

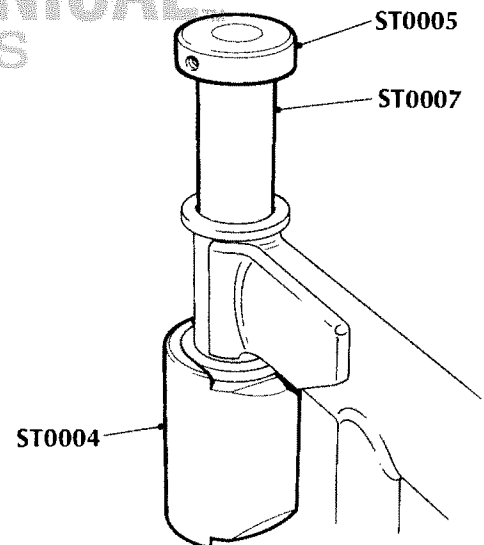
- 1 *Ensure absolute cleanliness of work area and tools.*
- 2 *Observe correct use of tools.*
- 3 *Check serviceability of components as they are removed. If in doubt reject component.*
- 4 *Clean all parts where applicable before reassembly using a soluble proprietary engine cleaner.*
- 5 *Fit JaguarSport protective body covers in immediate work area.*
- 6 *JaguarSport service tools must be used to prevent damage to the components and to ensure correct assembly.*

- (1) Remove wheel and wishbone (Refer to Section 4.15).
- (2) Place wishbone in hydraulic press.
- (3) Press out rubber bushes using service tools ST0004, ST0007 and ST0005.
- (4) Collect spacer.

WARNING:

SOLVENTS ARE TOXIC AND MUST ONLY BE USED IN WELL VENTILATED WORK AREAS. AVOID INHALING THE SOLVENT FUMES.

- (5) Clean the inside surface of the wishbone bush bore using lint free cloth moistened with approved degreaser.



B Installation Procedure

- (1) Place wishbone in hydraulic press.

CAUTION:

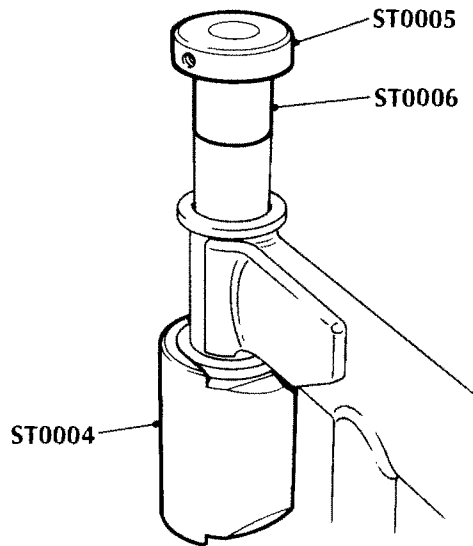
Ensure the bush does not tip to one side during installation

- (2) Press in one rubber bush using service tools ST0004, ST0005 and ST0006, taking care to ensure the bush does not tip to one side, and fit spacer.
- (3) Press in second rubber bush using service tools ST0004 and ST0005.

NOTE:

The service tools will correctly position the bushes in the wishbone.

- (4) Fit the wishbone (Refer to Section 4.15).



ORIGINAL TECHNICAL
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END CAP SPHERICAL JOINT

A Removal Procedure

Special Tools

The following special tools will be required;

JaguarSport protective body cover

JaguarSport service tools ST0017, ST0018, ST0019 and ST0021

Hydraulic press

Protective gloves

Materials

The following materials will be required;

Approved degreaser

Lint free cloth

NOTES:

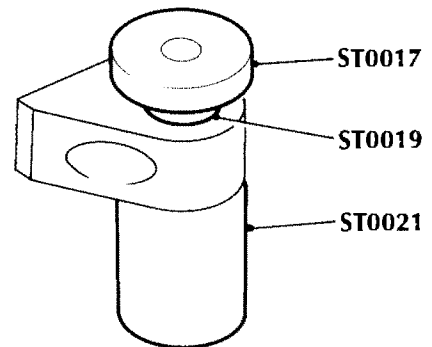
- 1 Ensure absolute cleanliness of work area and tools.
- 2 Ensure correct use of tools.
- 3 Check serviceability of components as they are removed. If in doubt reject component.
- 4 Clean all parts where applicable before reassembly using a soluble proprietary engine cleaner.
- 5 Fit JaguarSport protective body covers in immediate work area.
- 6 JaguarSport service tools must be used to prevent damage to the components and to ensure correct assembly.

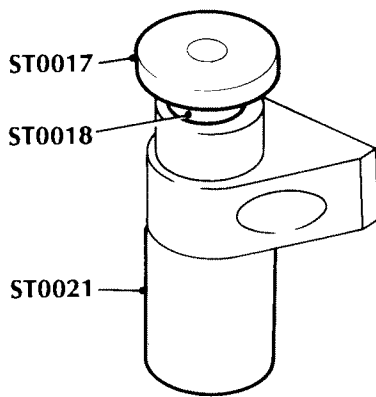
- (1) Remove the upright (Refer to Section 4.19).
- (2) Remove the 2 nuts and washers securing the end cap to the wishbone and remove the end cap and shims
- (3) Remove the spirolox clip.
- (4) Place the end cap in a hydraulic press.
- (5) Carefully press out the end cap spherical joint using service tools ST0017, ST0019 and ST0021.

WARNING:

SOLVENTS ARE TOXIC AND MUST ONLY BE USED IN WELL VENTILATED WORK AREAS. AVOID INHALING SOLVENT FUMES.

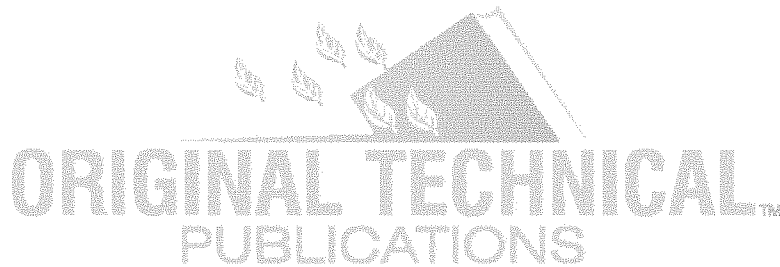
- (5) Clean the inside surface of the end cap using lint free cloth moistened with approved degreaser.





B Installation Procedure

- (1) Place the spherical joint in a freezer at -20 degrees C (21.6°F).
- (2) Place the end cap in a hydraulic press.
- (3) Using protective gloves remove the spherical joint from the freezer.
- (4) Carefully press the spherical joint into the end cap using service tools ST0017, ST0018 and ST0021.
- (5) Fit the spirolox clip.
- (6) Fit the end cap and shims to the wishbone and torque the nuts 57 to 65Nm (42 to 48ftlb).
- (7) Fit the upright (Refer to Section 4.19).



17 REAR UPPER WISHBONE

A Removal Procedure

Special Tools

The following special tools will be required:

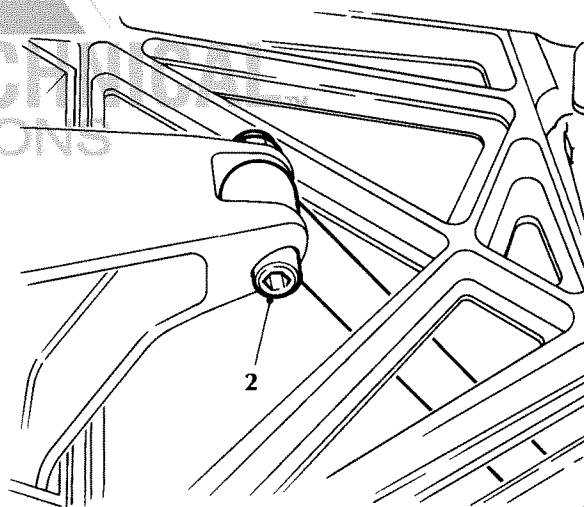
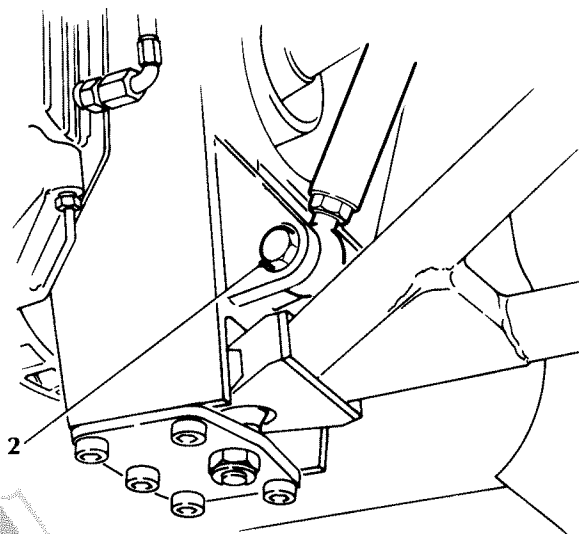
JaguarSport protective body covers

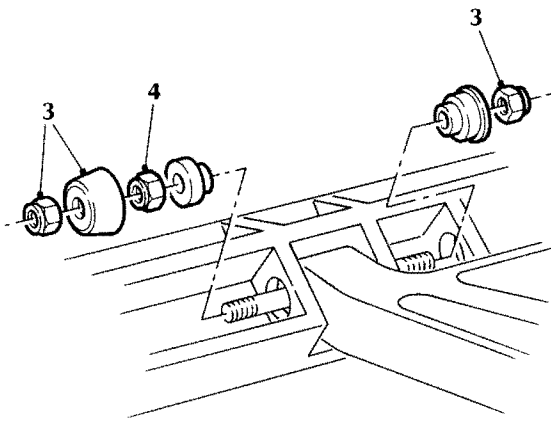
Torque wrench 0 - 118Nm (87ftlb)

NOTES:

- 1 Ensure absolute cleanliness of work area and tools.
- 2 Ensure correct use of tools.
- 3 Check serviceability of components as they are removed. If in doubt reject component.
- 4 Clean all parts where applicable before reassembly using a soluble proprietary engine cleaner.
- 5 Fit JaguarSport protective body covers in immediate work area.

- (1) Correctly position vehicle on lifting apparatus (Refer to Section1.1), and remove wheel (Refer to Section 4.1).
- (2) Support the upright and remove nuts, washers and bolts securing the push rod to the upright and nodder. Remove the push rod.

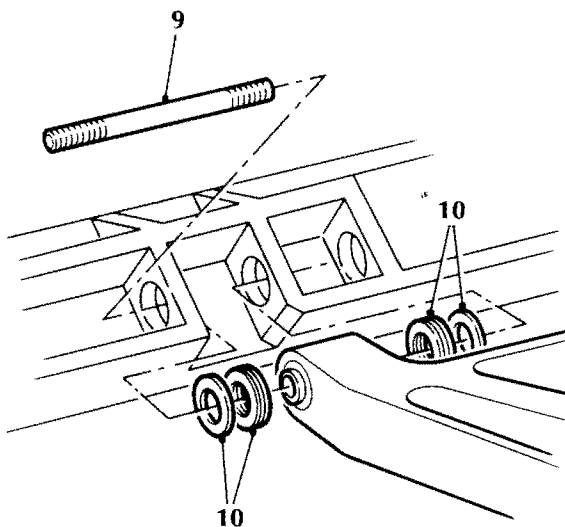
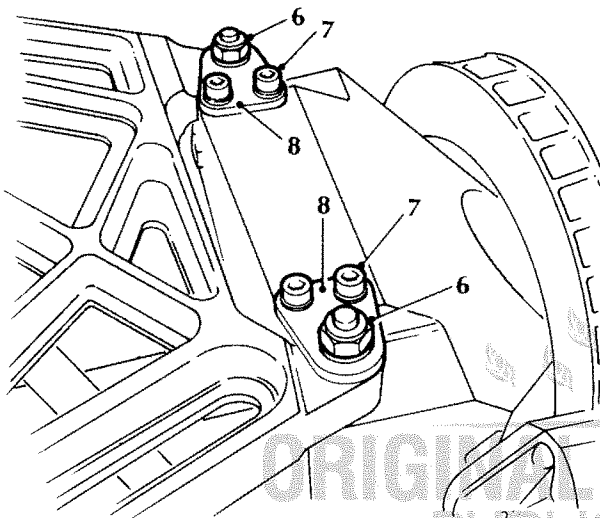




NOTE:

It is advisable to remove the catalytic converter to gain greater access to the spindle nuts. (Refer to Section 9).

- (3) Remove the wishbone locking nuts and collect wishbone locking caps.
- (4) Remove spindle nuts.
- (5) Remove clips securing brake hose to wishbone.
- (6) Remove nuts and washers securing upright to wishbone.
- (7) Remove the bolts securing plate to upright.
- (8) Remove plate from upright.
- (9) Remove wishbone pivot spindles.
- (10) Remove wishbone and collect wishbone thrust washers and washer assemblies.



B Installation Procedure

- (1) Fit wishbone to support and locate the thrust washer packs.
- (2) Fit wishbone pivot spindles.
- (3) Fit nuts to pivot spindles and torque load 41 to 47Nm (30 to 35ftlb).
- (4) Fit wishbone locking caps and secure with nuts.

NOTE:

Do not tighten nuts securing locking caps at this stage.

- (5) Fit wishbone over pins on upright.
- (6) Fit the upright top plate and secure with screws and washers.
- (7) Torque load screws 20 to 23Nm (15 to 17ftlb).
- (8) Fit the wishbone attachment nut and washer and torque nut 57 to 65Nm (42 to 48ftlb).
- (9) Torque load nuts securing locking caps 30 to 34Nm (22 to 25ftlb).
- (10) Attach the flexible brake hose to the inside edge of the wishbone using new cable ties.
- (11) Fit push rod to upright using special bolt, washer and nut and torque load bolt 57 to 65Nm (42 to 48ftlb).
- (12) Fit push rod to nodder using M12 bolt, washer and nut, and torque load bolt 68 to 74.5Nm (50 to 55ftlb).
- (13) Fit the catalytic converter (if removed). (Refer to Section 9).
- (14) Fit road wheel (Refer to Section 4.1), lower the lifting apparatus and remove vehicle.

18 REAR UPPER WISHBONE BUSHINGS

WISHBONE BUSHES

A Removal Procedure

Special Tools

The following special tools will be required.

JaguarSport service tools ST0006, ST0008 and ST0014

JaguarSport protective body covers

Hydraulic press

Materials

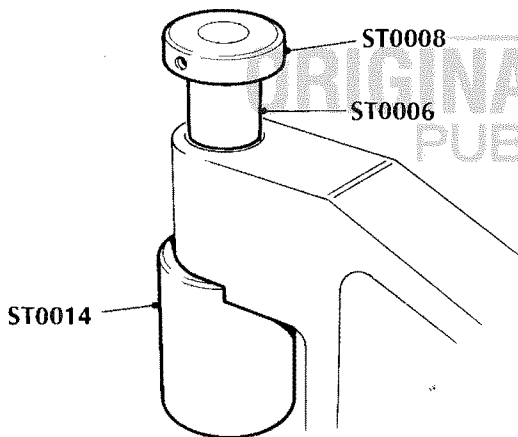
The following materials will be required.

Approved degreaser

Lint free cloth

NOTES:

- 1 Ensure absolute cleanliness of work area and tools.
- 2 Ensure correct use of tools.
- 3 Check serviceability of components as they are removed. If in doubt reject component.
- 4 Clean all parts where applicable before reassembly using a soluble proprietary engine cleaner.
- 5 Fit JaguarSport protective body covers in immediate work area.
- 6 JaguarSport service tools must be used to prevent damage to the components and to ensure correct assembly.



- (1) Remove the wishbone. (Refer to Section 4.17).
- (2) Place wishbone in hydraulic press.
- (3) Press out rubber bushes using service tools ST0006, ST0008 and ST0014.

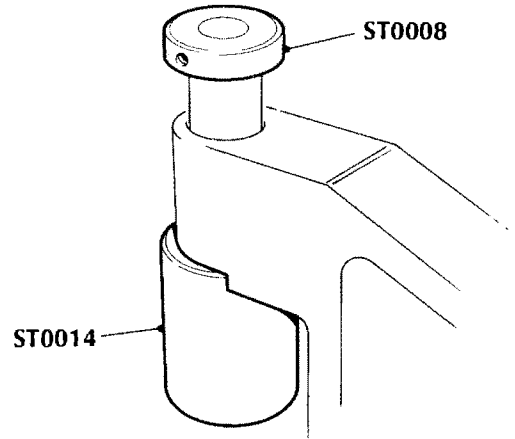
WARNING:

SOLVENTS ARE TOXIC AND MUST ONLY BE USED IN WELL VENTILATED WORK AREAS. AVOID INHALING THE SOLVENT FUMES.

- (4) Clean inside surface of wishbone bush bore using lint free cloth moistened with approved degreaser.

B Installation Procedure

- (1) Place wishbone in hydraulic press.
- (2) Carefully press in rubber bushes using service tools ST0008 and ST0014, taking care to ensure the bush does not tip to one side during installation.
- (3) Check the tube through the centre of the bush has not moved or been damaged during installation.
- (4) Fit wishbone and wheel (Refer to Section 4.17).




ORIGINAL TECHNICAL™
PUBLICATIONS

END CAP SPHERICAL JOINTS

A Removal Procedure

Special Tools

The following special tools will be required.

JaguarSport service tools ST0003, ST0010, ST0011, ST0012 and ST0013

JaguarSport protective body covers

Hydraulic press

Materials

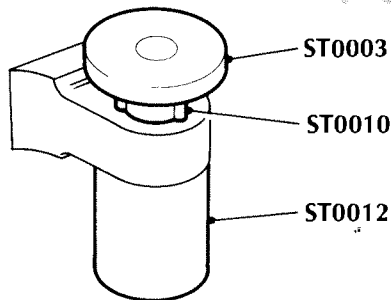
The following materials will be required.

Approved degreaser

Lint free cloth

NOTES:

- 1 Ensure absolute cleanliness of work area and tools.
- 2 Ensure correct use of tools.
- 3 Check serviceability of components as they are removed. If in doubt reject component.
- 4 Clean all parts where applicable before reassembly using a soluble proprietary engine cleaner.
- 5 Fit JaguarSport protective body covers in immediate work area.
- 6 JaguarSport service tools must be used to prevent damage to the components and to ensure correct assembly.



- (1) Remove the upright (Refer to Section 4.19).
- (2) Remove the 4 screws and washers securing the end cap to the wishbone and remove the end cap and shims
- (3) Remove the spirolox clip.
- (4) Place the end cap in a hydraulic press.
- (5) Carefully press out the end cap spherical joint using service tools ST0003, ST0010 and ST0012.

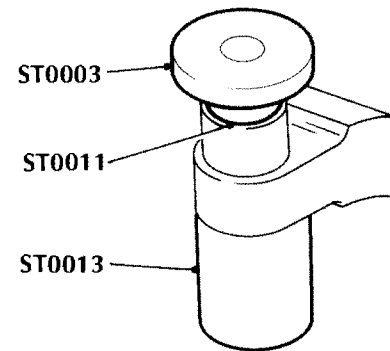
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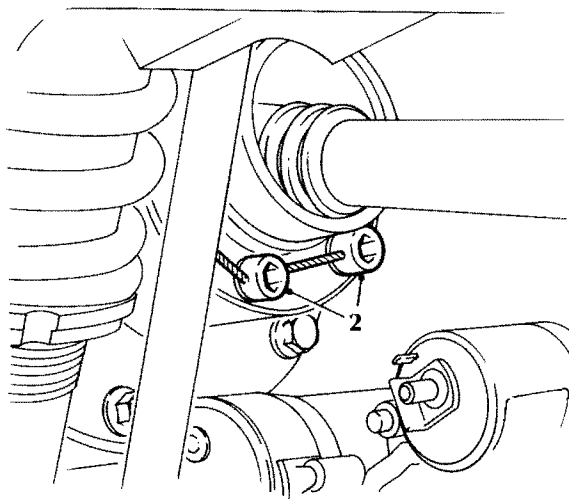
SOLVENTS ARE TOXIC AND MUST ONLY BE USED IN WELL VENTILATED WORK AREAS. AVOID INHALING SOLVENT FUMES.

- (6) Clean the inside surface of the end cap using lint free cloth moistened with approved degreaser.

B Installation Procedure

- (1) Place the spherical joint in a freezer at -20 degrees C (21.6°F).
- (2) Place the end cap in a hydraulic press.
- (3) Using protective gloves remove the spherical joint from the freezer.
- (4) Carefully press the spherical joint into the end cap using service tools ST0003, ST0010 and ST0013.
- (5) Fit the spirolox clip.
- (6) Fit the end cap and shims to the wishbone and torque the screws 8 to 9.5Nm (6 to 7ftlb).
- (7) Fit the upright (Refer to Section 4.19).





19 REAR SUSPENSION UPRIGHT

A Removal Procedure

Special Tools

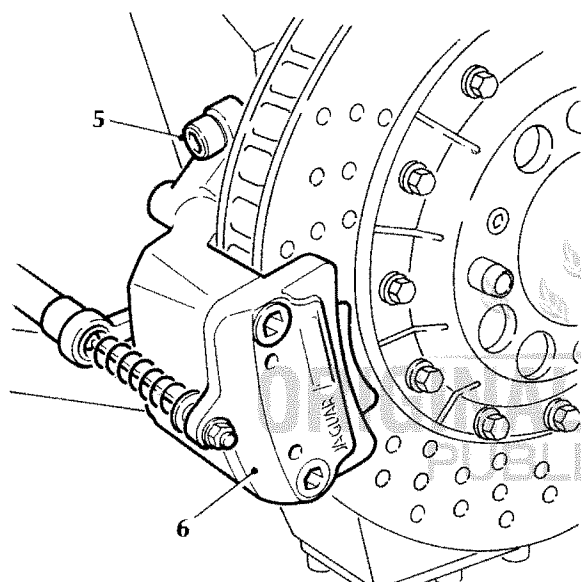
The following special tools will be required:

JaguarSport protective body covers

Torque wrench 0 - 118Nm (87ftlb)

NOTES:

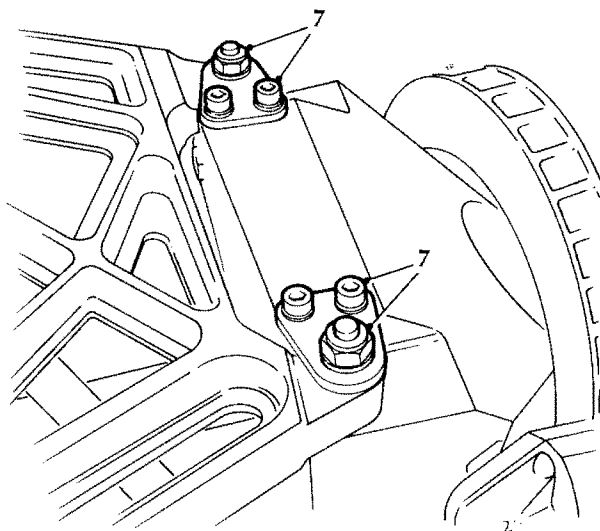
- 1 Ensure absolute cleanliness of work area and tools.
- 2 Ensure correct use of tools.
- 3 Check serviceability of components as they are removed. If in doubt reject component.
- 4 Clean all parts where applicable before reassembly using a soluble proprietary engine cleaner.
- 5 Fit JaguarSport protective body covers in immediate work area.



- (1) Correctly position vehicle on lifting apparatus (Refer to Section 1.1), remove wheel (Refer to Section 4.1) and raise vehicle to a suitable working height.
- (2) Cut and remove the locking wire from the driveshaft bolts.
- (3) Remove the six bolts securing driveshaft to transaxle output flange.
- (4) Remove nuts securing the main brake caliper to upright.
- (5) Remove brake caliper from upright and secure to upper wishbone with tyrap or similar to ensure no strain is placed on the flexible brake hose.

CAUTION

Whilst the caliper is removed ensure the brakes are not operated.



- (5) Remove bolts securing handbrake caliper to upright.
- (6) Remove handbrake caliper from upright and secure to upper wishbone with tyrap or similar.
- (7) Remove bolts, nuts and washers securing retaining plates to top wishbone and upright and remove the plate.

- (8) Remove 5 bolts and nut securing retaining plate on bottom wishbone and upright and remove the plate.
- (9) Remove nut and washer securing push rod to upright.

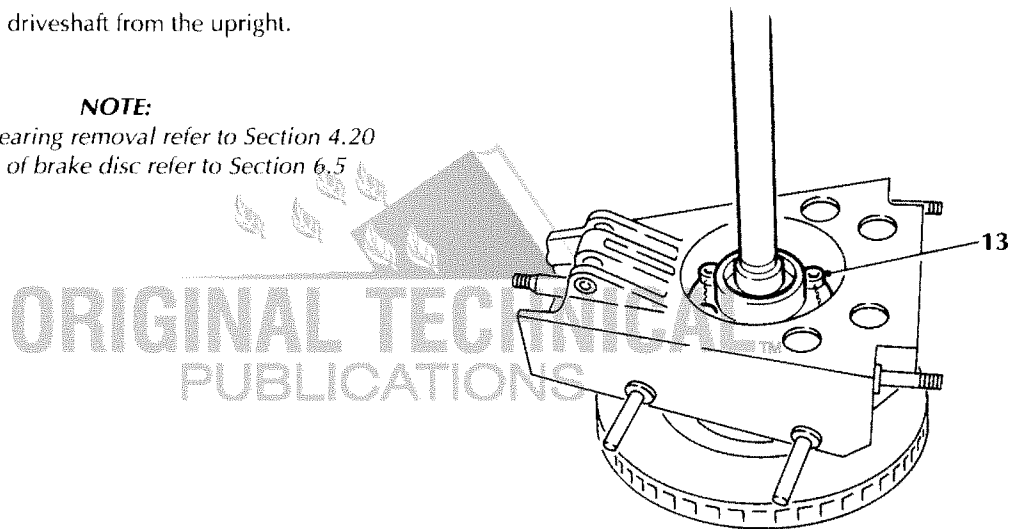
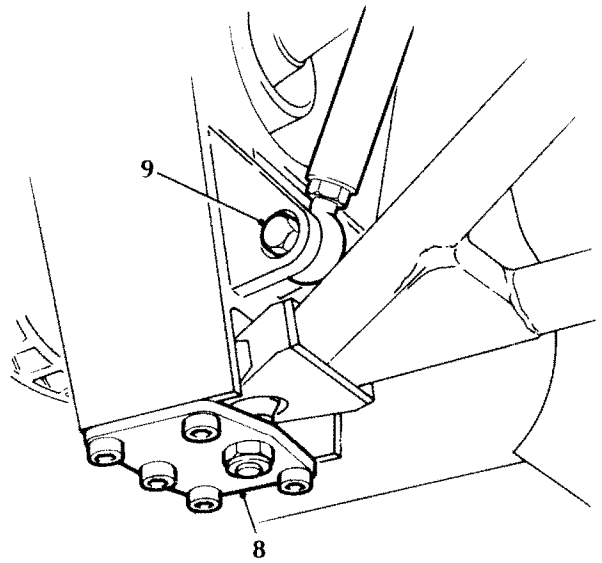
NOTE:

Do not remove the push rod bolt at this stage.

- (10) Support the upright.
- (11) Raise top wishbone and lower bottom wishbone, remove push rod bolt and withdraw driveshaft with upright and brake disc.
- (12) Cover exposed end of CV joint to prevent ingress of foreign matter.
- (13) Securely clamp upright in padded vice with driveshaft vertical.
- (14) Remove locking wire and remove the six bolts securing outer end of driveshaft to axle output flange mounted in upright.
- (15) Remove the driveshaft from the upright.

NOTE:

- 1 For wheel bearing removal refer to Section 4.20
- 2 For removal of brake disc refer to Section 6.5



B Installation Procedure

- (1) Securely clamp upright in padded vice.
- (2) Fit drive shaft to upright and secure outer end of driveshaft to axle output flange with the six bolts.
- (3) Tighten the bolts evenly and torque load 57 to 65Nm (42 to 48ftlbs).
- (4) Wire lock the bolts using a single piece of 22swg locking wire in a single pass through each bolt head.
- (5) Raise top wishbone and lower bottom wishbone and fit upright with driveshaft until inner end of driveshaft mates with transaxle drive flange.
- (6) Fit push rod to upright using special bolt, washer and nut and torque load bolt 57 to 65Nm (42 to 48ftlb).
- (7) Secure top wishbone to upright with retaining plate, nuts and washers, and cap head screws fitted with washers.
- (8) Torque load nut 57 to 65Nm (42 to 48ftlb) and cap head screws 20 to 23Nm (15 to 17ftlb).
- (9) Secure bottom wishbone to upright with retaining plate, nut and washer, and bolts fitted with washers.
- (10) Torque load nut 57 to 65Nm (42 to 48ftlb) and cap head screws 20 to 23Nm (15 to 17ftlb).
- (11) Fit the main brake caliper to upright and secure with nuts and washers.
- (12) Torque load nuts 57 to 65Nm (42 to 48ftlb).
- (13) Fit handbrake caliper to upright and secure with bolts.
- (14) Torque load bolts 40.5 to 46.5Nm (30 to 34.5ftlb).
- (15) Secure flanged end of transaxle to gear box output drive with bolts.
- (16) Torque load bolts to 57 to 65Nm (42 to 48ftlb) and wire lock using three lengths of 24 swg locking wire , wiring the bolts together in adjacent pairs.
- (17) Fit wheel (Refer to Section 4.1), lower the lifting apparatus and remove the vehicle.

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20 REAR WHEEL BEARINGS

A Removal Procedure

Special Tools

The following special tools will be required.

Hydraulic press (or suitable alternative)

JaguarSport service tools ST0002 and ST0030

JaguarSport protective body covers

Torque wrench 0 - 380Nm (280ftlb) and 0 - 9.5Nm (7ftlb)

Materials

The following materials will be required.

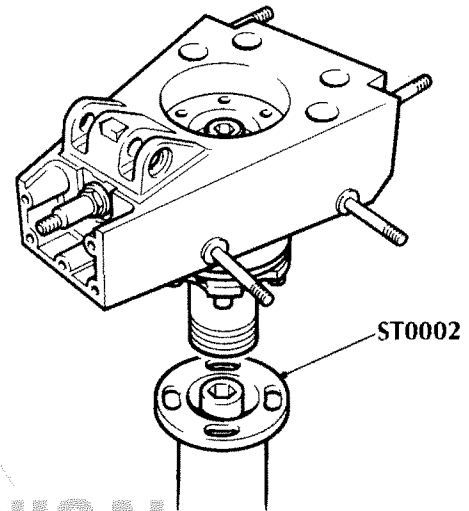
Lint free cloth

Protective gauntlets

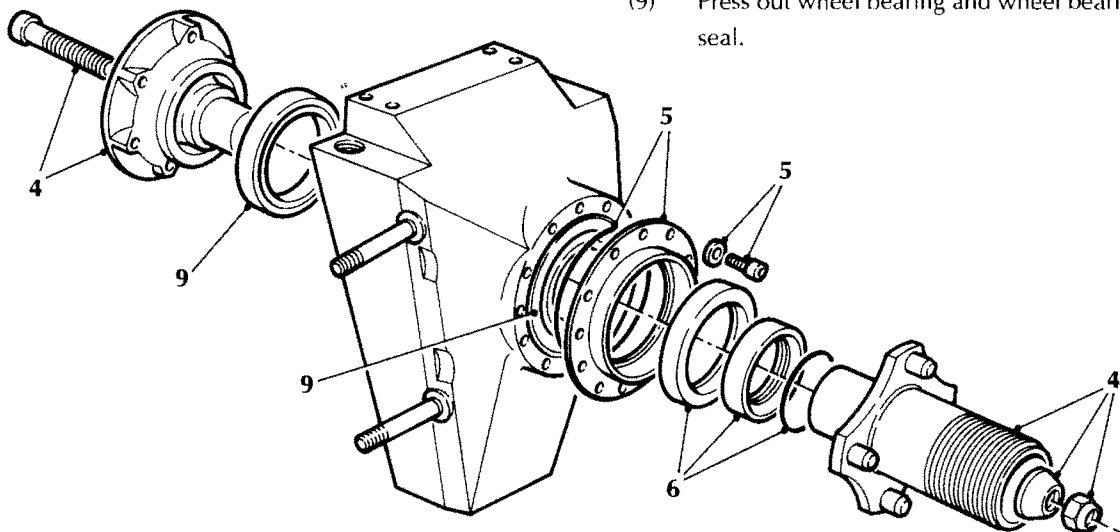
NOTES:

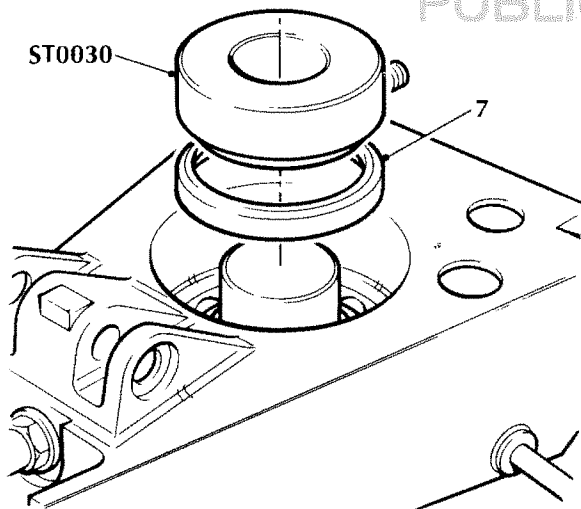
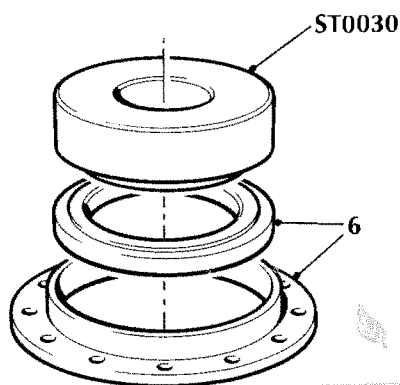
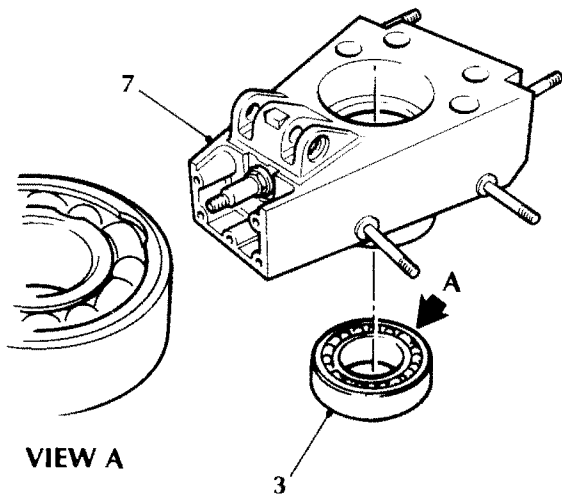
- 1 Ensure absolute cleanliness of work area and tools.
- 2 Ensure correct use of tools.
- 3 Check serviceability of components as they are removed. If in doubt reject component.
- 4 Clean all parts where applicable before reassembly using a soluble proprietary engine cleaner.
- 5 Fit JaguarSport protective body covers in immediate work area.
- 6 JaguarSport service tools must be used to prevent damage to the components and to ensure correct assembly.

- (1) Remove the upright (Refer to Section 4.19).
- (2) Remove logo cap from end of stub axle (Refer to Section 11.5)
- (3) Remove countersink screws and remove brake disc with brake bell. (Refer to Section 6)
- (4) Place upright in the service tool ST0002 and remove the central bolt and nut with load spreader and stub flange, and remove stub axle.
- (5) Remove 12 screws with washers and remove bearing retainer and O ring seal.



- (6) Remove collar, O-ring and wheel bearing seal from bearing retainer.
- (7) Apply local heat to upright bearing housing as evenly as possible.
- (8) Place upright in hydraulic press.
- (9) Press out wheel bearing and wheel bearing inner seal.





B Installation Procedure

- (1) Place upright in oven for 6 to 8 hours at 120 degrees C (273 degrees F). Alternatively, heat the upright evenly using a propane torch.
- (2) Pack wheel bearing with 50cc Castrol LMX grease and place wheel bearing in freezer at -20 degrees C (-21.6 degrees F).
- (3) Using protective gloves, remove bearing from freezer.
- (4) Using protective gloves, remove the upright from the oven and fit the bearing to the upright, ensuring the bearing filling slots are towards the vehicle centreline.
- (5) Leave the upright in position to permit the upright to cool.
- (6) Press wheel bearing seal into bearing retainer using service tool ST0030.
- (7) Press inner wheel bearing seal into upright using service tool ST0030
- (8) Fit new O ring to the bearing retainer.
- (9) Secure bearing retainer to upright with 12 screws.
- (10) Torque load screws 8 to 9.5Nm (6 to 7ftlb).
- (11) Fit new O ring to collar and fit collar to stub axle.
- (12) Smear Castrol LMX grease onto the wheel bearing seal faces.
- (13) Fit stub axle to upright and secure with stub flange, bolt, load spreader and nut.
- (14) Torque load nut 366 to 380Nm (270 to 280ftlb).
- (15) Fit upright (Refer to Section 4.17).
- (16) Fit brake disc with brake bell and secure with countersink screws.
- (17) Torque load countersink screws 7 to 8Nm (5 to 6ftlb).
- (18) Fit the main brake caliper and handbrake caliper. (Refer to Section 4.19).
- (19) Fit cap to end of stub axle and tighten the grubscREW.
- (20) Fit wheel (Refer to Section 4.1), lower lifting apparatus and remove vehicle.

21 REAR ANTI-ROLL BAR

A Removal Procedure

Special Tools

The following special tools will be required.

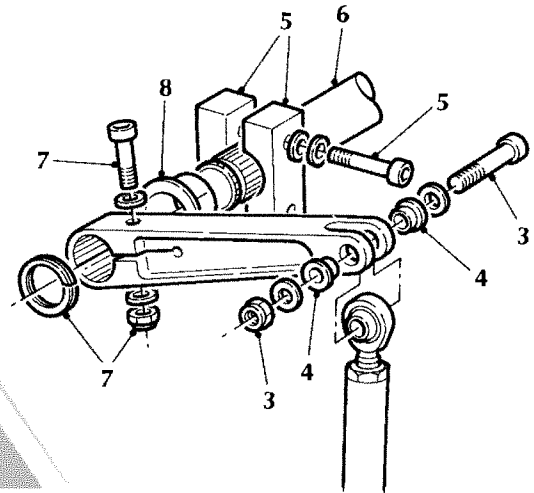
JaguarSport protective body covers

Torque wrench 0 - 34Nm (25ftlb).

NOTES:

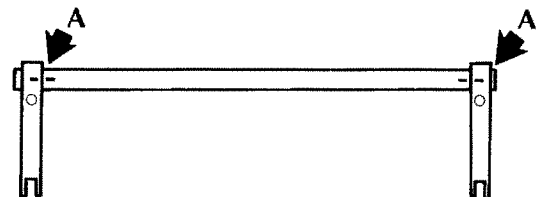
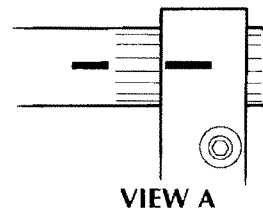
- 1 Ensure absolute cleanliness of work area and tools.
- 2 Ensure correct use of tools.
- 3 Check serviceability of components as they are removed. If in doubt reject component.

- (1) Fit JaguarSport protective body covers.
- (2) Open engine bay hatch.
- (3) Remove bolts, washers and nuts and disconnect drop links from levers.
- (4) Remove the inserts from the levers.
- (5) Remove bolts and washers and remove clamps.
- (6) Remove anti-roll bar.
- (7) Remove bolts, washers and nuts and retaining clips, and remove levers from anti-roll bar.
- (8) Remove bushes.



B Installation Procedure

- (1) Position the anti-roll bar on a level work surface.
- (2) Fit bushes to anti-roll bar.
- (3) Fit the levers to the anti-roll bar and ensure they are perfectly aligned.
- (4) Secure levers to anti-roll bar with bolts, washers and nuts.
- (5) Torque load screws 30 to 34Nm (22 to 25ftlb).
- (6) Fit retaining clips.
- (7) Fit anti-roll bar in vehicle and position clamps at each end.
- (8) Secure each anti-roll bar clamp with 2 bolts and washers.
- (9) Torque load bolts 8 to 9.5Nm (6 to 7ftlb).
- (10) Fit the inserts to the levers.
- (11) Fit drop links to levers and secure with screws, washers and nuts.
- (12) Torque load screws 9 to 10Nm (6.5 to 7.5ftlb).
- (13) Close engine bay hatch.



22 REAR SUSPENSION COIL SPRING AND DAMPER

A Removal Procedure

Special Tools

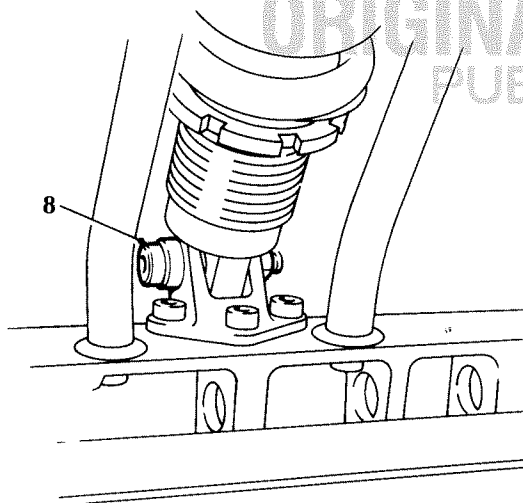
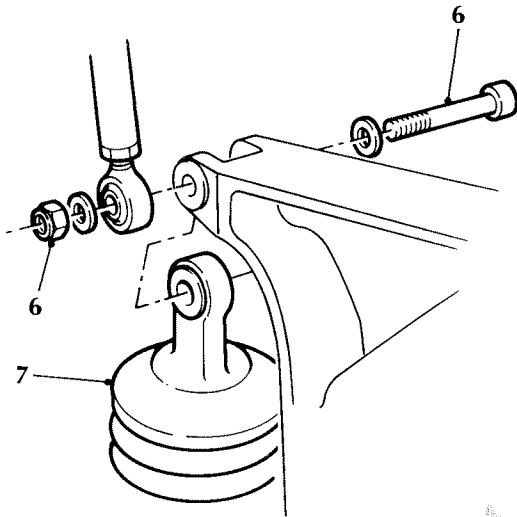
The following special tools will be required.

JaguarSport protective covers.

Torque wrench 0 - 34Nm (25ftlb)

NOTES:

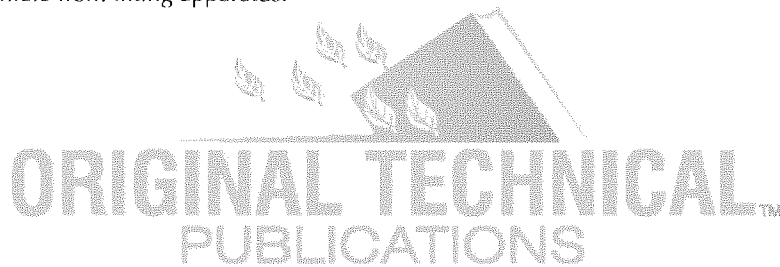
- 1 Ensure absolute cleanliness of work area and tools.
- 2 Ensure correct use of tools.
- 3 Check serviceability of components as they are removed. If in doubt reject component.
- 4 Clean all parts where applicable before reassembly using a soluble proprietary engine cleaner.



- (1) Fit JaguarSport protective body covers.
- (2) Correctly position vehicle on lifting apparatus (Refer to Section 1.1) , remove the wheels (Refer to Section 4.1) and raise vehicle to a suitable working height.
- (3) Remove the rear grille
- (4) Remove the M5 and M6 screws, nuts and washers securing the rear bumper. Lower the rear bumper to allow access to the rear number plate light plug. Disconnect plug and remove the bumper. (Refer to Section 2).
- (5) Fit support to upright.
- (6) Remove nut, washer and screw sufficiently to remove drop link.
- (7) Support coil spring and damper in nodder and remove screw.
- (8) Remove nut, washer and screw attaching damper lower bracket.
- (9) Remove the coil spring and damper and collect inserts from eye ends.

B Installation Procedure

- (1) Fit inserts to damper eye ends.
- (2) Fit coil spring and damper to nodder and damper block.
- (3) Fit M8 screw, washer and nut to secure the damper to the damper mounting block.
- (4) Fit 5/16in screw to support coil spring and damper in nodder, and fit drop link.
- (5) Secure damper and drop link with washer and nut.
- (6) Torque load 5/16in screw 24 to 28Nm (17.5 to 20.5ftlb).
- (7) Torque load M8 screw 30 to 34Nm (22 to 25ftlb).
- (8) Fit bumper and secure with screws. (Refer to Section 2).
- (9) Fit rear grille.
- (10) Fit wheel (Refer to Section 4.1).
- (11) Remove support from upright, lower lifting apparatus and remove body covers.
- (12) Remove vehicle from lifting apparatus.



23 REAR SUSPENSION NODDER

A Removal Procedure

Special Tools

The following special tools will be required.

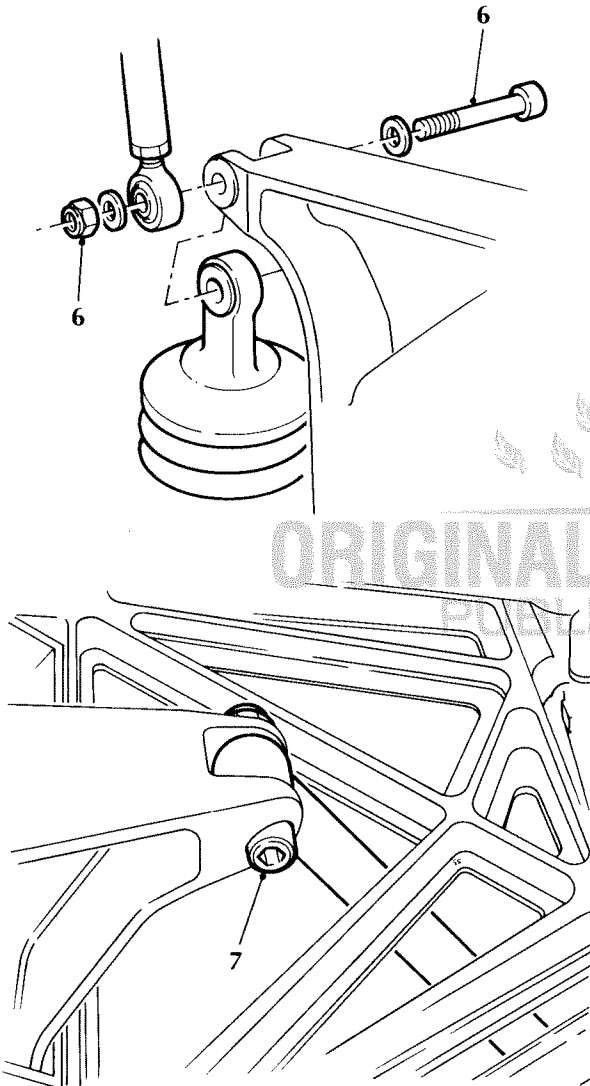
JaguarSport protective covers

Torque wrench 0 - 118Nm (87ftlb).

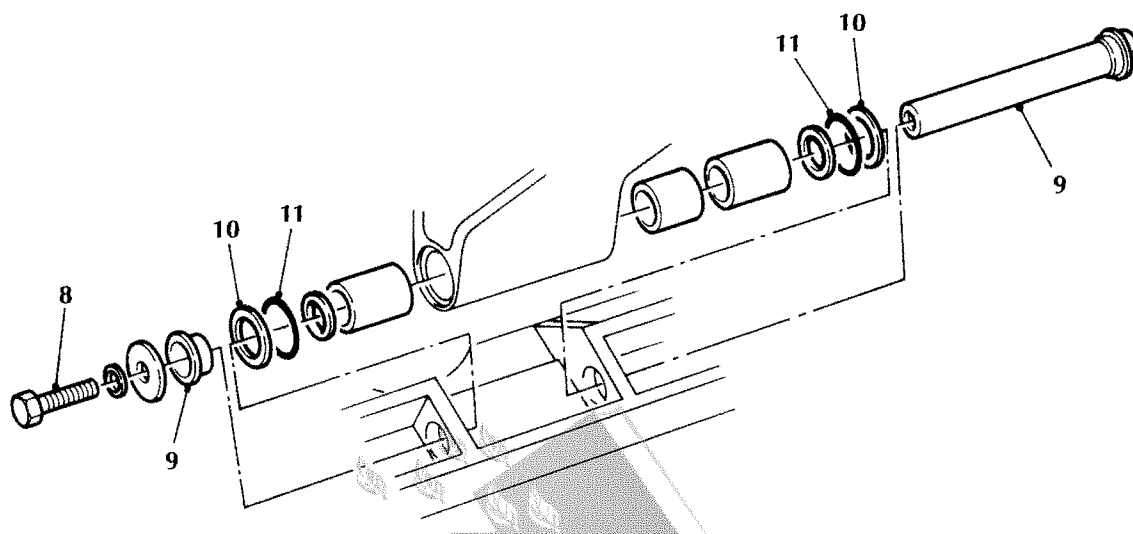
NOTES:

- 1 Ensure absolute cleanliness of work area and tools.
- 2 Ensure correct use of tools.
- 3 Check serviceability of components as they are removed. If in doubt reject component.
- 4 Clean all parts where applicable before reassembly using a soluble proprietary engine cleaner.

- (1) Fit JaguarSport protective covers and open engine bay hatch.
- (2) Apply penetrating oil to the screws and nuts attaching exhaust catalytic converter to S pipe and rear exhaust system.
- (3) Remove the exhaust catalytic converter. (Refer to Section 9).
- (4) Correctly position vehicle on lifting apparatus (Refer to Section 1.1) and raise vehicle to a suitable working height.
- (5) Support the upright.
- (6) Remove screw, washer and nut, securing drop link and coil spring and damper upper attachment.
- (7) Remove screw, washer and nut and disconnect the push rod at the nodder.



- (8) Remove the bolt and washers from the nodder spindle.
- (9) Support the nodder and remove the nodder spindle and nodder pivot.
- (10) Remove nodder from the upper suspension support and collect washers.
- (11) Remove and discard the nodder O ring seals.



B Installation Procedure

- (1) Apply a thin smear of Castrol rubber grease to the new O rings.
- (2) Fit O rings and thrust washers to the nodder.
- (3) Position nodder between mountings on upper wishbone support.
- (4) Apply a thin film of SMIC 1020B grease to the nodder spindle, fit the spindle and secure with washer and bolt.
- (5) Torque load bolt 40.5 to 46.5Nm (30 to 34.5ftlb).
- (6) Fit push rod to nodder using M12 bolt, washer and nut, and torque load bolt 68 to 74.5Nm (50 to 55ftlb).
- (7) Fit coil spring and damper upper attachment to nodder and fit bolt.
- (8) Fit drop link to bolt, and secure with washer and nut.
- (9) Torque load nut 24 to 28Nm (17.5 to 20.5ftlb).
- (10) Remove support from upright and lower lifting apparatus.
- (11) Fit the exhaust catalyst converter with new gaskets and secure to the S pipe and rear exhaust system with bolts, washers and nuts.

NOTE:

The mating surfaces of the S pipe, exhaust catalyst converter and rear exhaust system must be absolutely clean prior to assembly.

- (12) Torque load screws 20.5 to 23Nm (15 to 17ftlb).
- (13) Close engine bay hatch, remove protective covers and remove vehicle from lifting apparatus.
- (14) Run the engine and check for exhaust gas leaks

24 REAR NODDER BUSHES

A Removal Procedure

Special Tools

The following special tools will be required:

JaguarSport protective body covers

JaguarSport service tools ST0026, ST0027, ST0028 and ST0029

Hydraulic press

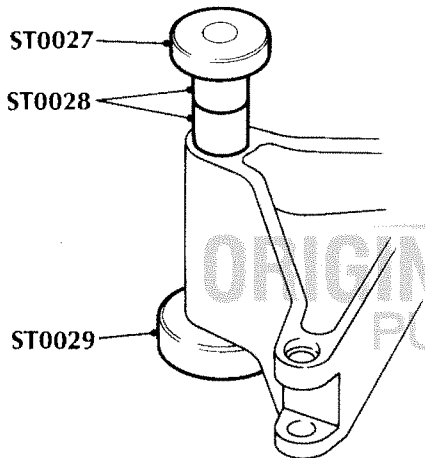
Materials

Approved degreaser

Lint free cloth

NOTES:

- 1 Ensure absolute cleanliness of work area and tools.
- 2 Ensure correct use of tools.
- 3 Check serviceability of components as they are removed. If in doubt reject component.
- 4 Clean all parts where applicable before reassembly using a soluble proprietary engine cleaner.
- 5 Fit JaguarSport protective body covers in immediate work area.
- 6 JaguarSport service tools must be used to prevent damage to the components and to ensure correct assembly.

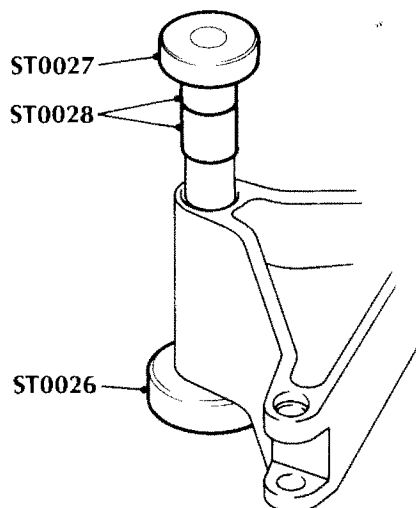


- (1) Remove the rear nodder (Refer to Section 4.23).
- (2) Place the nodder in hydraulic press.
- (3) Carefully press out the nodder bushes using service tools ST0027, ST0028 and ST0029.

WARNING

SOLVENTS ARE TOXIC AND MUST ONLY BE USED IN WELL VENTILATED AREAS. AVOID INHALING THE SOLVENT FUMES.

- (4) Clean the inside surface of the nodder bore using lint free cloth moistened with approved degreaser.



B Installation Procedure

- (1) Place the nodder in hydraulic press.
- (2) Carefully press in the centre bush using ST0027, ST0028 and ST0026.
- (3) Carefully press in the end bushes using ST0027 and ST0026
- (4) Fit the nodder (Refer to Section 4.23).

25 WHEELS AND TYRES

A Wheels

Special Tools

The following special tools will be required

Magnifying glass-10X magnification

NOTES:

- 1 Ensure absolute cleanliness of work area and tools.
- 2 Ensure correct use of tools.
- 3 For this inspection to be successfully completed will require the tyre be removed.

Inspection

- (1) Roll the wheel along a level surface and check there is no evidence of distortion or buckling. Evidence of distortion or buckling requires the wheel be rejected.
- (2) Check there is no evidence of surface corrosion. Corrosion on aluminium is identified by grey powder deposits. Surface corrosion may be removed using Scotchbrite or fine grade abrasive paper. Treat the affected area with a de-oxidizing fluid and restore paint finish. Check wheel balance.
- (3) Check the wheel for cracks using a torch and 10X magnifying glass. Alternatively, use the Ardrex crack detection method. Reject the wheel if it is cracked.

B Tyres

CAUTION:

It is not permitted to repair punctures.

Inspection

- (1) Correctly position vehicle on lifting apparatus (Refer to Section 1.1), and raise the vehicle to a suitable working height.
- (2) Check the tyre is fitted to the wheel with the direction of rotation indicator arrows pointing to forward rotation on the outside wall of the tyre.
- (3) Inspect the tyre walls for evidence of damage due to scuffing, bulging, cuts or cracking. Reject the tyre if any damage is found.
- (4) Inspect the tread for even wear pattern, and that the remaining tread satisfies local safety regulations. If wear is uneven check the severity of wear to determine whether the tyre is serviceable. If in doubt reject the tyre. If wear satisfies local safety regulations correct the fault by adjusting wheel camber and/or alignment.
- (5) Inspect the tread to ensure there are no cuts or cracks. Cuts in the tread that penetrate deeper than 3mm (0.118 in.) will require the tyre to be rejected.
- (6) Reject the tyre if there are any flat spots.
- (7) Reject the tyre if there is evidence of chemical contamination.

26 WHEEL CAMBER

Special Tools

The following special tools will be required.

Camber gauge

JaguarSport protective body covers

Torque wrench 0 - 65Nm (48ftlb)

NOTES:

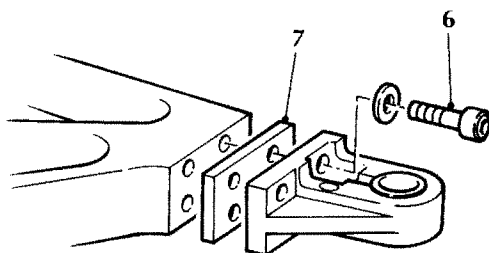
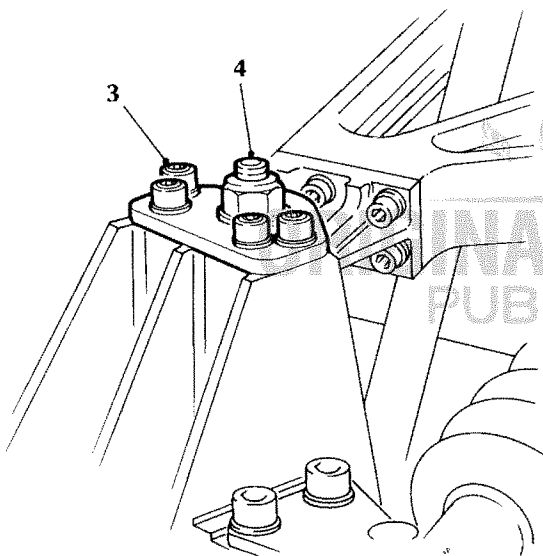
- 1 Ensure absolute cleanliness of work area and tools.
- 2 Ensure correct use of tools.
- 3 Ensure the vehicle is set to the correct ride height
- 4 After correcting the camber it will be necessary to check/adjust the wheel alignment.
- 5 Fit JaguarSport protective body covers in immediate work area.

Checking

- (1) Measure the camber using the camber gauge.
- (2) If camber setting is unsatisfactory note the displacement angle in degrees and minutes and proceed as follows.

A Adjustment of Front Wheel Camber

- (1) Correctly position vehicle on lifting apparatus (Refer to Section 1.1), and remove wheels (Refer to Section 4.1).
- (2) Raise the vehicle to a suitable working height.
- (3) Remove cap head bolts and washers securing top plate to upright.
- (4) Remove nut and washer securing upright to upper wishbone.
- (5) Support the upright and lift wishbone clear.
- (6) Remove the 4 cap head bolts and remove cap from end of upper wishbone.
- (7) Remove shims and measure shim thickness.
- (8) To adjust the camber by ± 0 degrees 13 minutes, or multiples thereof, use shim thickness 0.91mm, 20 SWG (0.36in.) P/No.5689AA.
- (8A) To adjust the camber by ± 0 degrees 28 minutes, or multiples thereof, use shim thickness 2.0mm, 14 SWG (0.80in.) P/No.5690AA.
- (9) Add or remove shims as necessary.
- (10) Record new shim thickness in case further adjustment is necessary.



- (11) Fit shims to the wishbone and secure cap to wishbone with 4 cap head bolts.

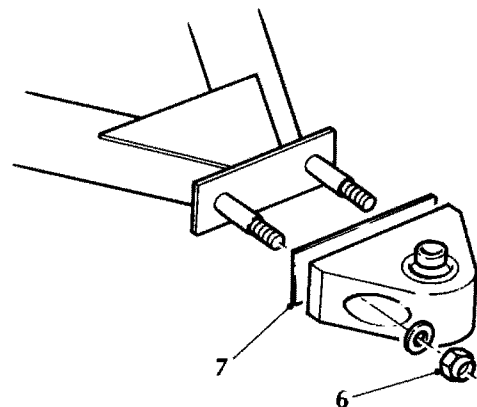
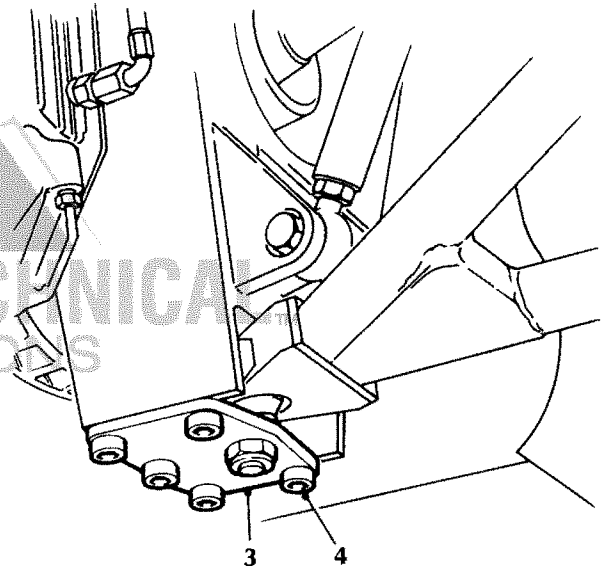
CAUTION

Ensure cap is refitted with the circlip groove to the underside.

- (12) Torque load screws 8 to 9.5Nm (6 to 7ftlb).
- (13) Fit wishbone to upright top pin.
- (14) Fit top plate to upright and secure with cap head screws and washers.
- (15) Torque load cap head screws 8 to 9.5Nm (6 to 7ftlb).
- (16) Fit wishbone attachment nut and washer and torque nut 41 to 47Nm (30 to 34.5ftlb)
- (17) Fit wheel (Refer to Section 4.1) and measure the camber. Repeat process if camber is incorrect.
- (18) Lower lifting apparatus and remove the vehicle.
- (19) Check wheel alignment.
- (12) Torque load the nuts 50 to 55.5Nm (37 to 41ftlb).
- (13) Fit wishbone to upright.
- (14) Fit plate to upright and secure with cap head screws and washers.
- (15) Torque load cap head screws 20 to 23Nm (15 to 17ftlb).
- (16) Fit wishbone attachment nut and washer and torque nut 57 to 65Nm (42 to 48ftlb).
- (17) Fit wheel (Refer to Section 4.1) and measure the camber. Repeat process if camber is incorrect.
- (18) Lower lifting apparatus and remove the vehicle.
- (19) Check wheel alignment.

B Adjustment of Rear Wheel Camber

- (1) Correctly position vehicle on lifting apparatus (Refer to Section 1.1), and remove wheels (Refer to Section 4.1).
- (2) Raise the vehicle to a suitable working height.
- (3) Remove nut and washer securing lower wishbone to upright.
- (4) Remove 5 bolts and washers securing plate to upright.
- (5) Support the upright and move lower wishbone away from the upright.
- (6) Remove two locking nuts and remove cap from end of lower wishbone.
- (7) Remove shims and measure shim thickness.
- (8) To adjust the camber by +/- 0 degrees 11 minutes, or multiples thereof, use shim thickness 0.91mm, 20 SWG (0.36in.) P/No.5686AA.
- (8A) To adjust the camber by +/- 0 degrees 20 minutes, or multiples thereof, use shim thickness 1.63mm, 16 SWG (0.64in.) P/No. 5685AA.
- (9) Add or remove shims as necessary.
- (10) Record new shim thickness in case further adjustment is necessary.
- (11) Fit shims to the wishbone and secure cap to wishbone with 2 washers and locking nuts.



27 WHEEL ALIGNMENT

Special Tools

The following special tools will be required.

JaguarSport protective body covers

Torque wrench 0 - 65Nm (48ftlb)

Wheel alignment equipment

NOTES:

- 1 *Ensure absolute cleanliness of work area and tools.*
- 2 *Ensure correct use of tools.*
- 3 *Ensure the vehicle is at the correct kerb weight.(Refer to Section 4.24)*
- 4 *Ensure all tyre pressures are correct.*
- 5 *Ensure the vehicle is on a flat and level surface and is at the correct ride height at each corner.*
- 6 *Fit JaguarSport protective body covers in immediate work area.*

- (1) Position vehicle and set up wheel alignment checking equipment.
- (2) If wheel alignment is unsatisfactory note the displacement toe-in/toe-out angle in degrees and minutes and proceed as follows.

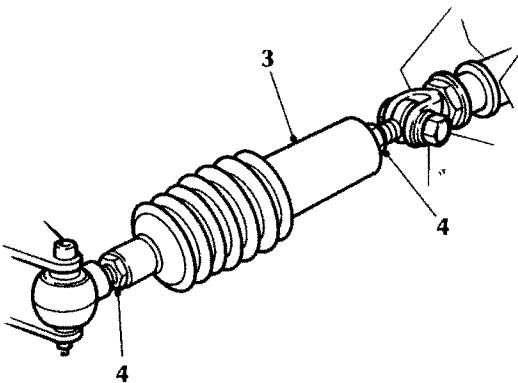
A Adjustment of Front Wheel Alignment

- (1) Correctly position the vehicle on lifting apparatus (Refer to Section 1) and remove the wheels (Refer to Section 4.1).
- (2) Raise the vehicle to a suitable working height.
- (3) Remove the cable ties from the inner end of the steering rack gaiter and pull the gaiter outboard to reveal the inner steering tie rod joint.

CAUTION:

The ends of the tie rods attaching to the steering rack have left hand threads

- (4) Loosen the inboard and outboard lock nuts on the tie rods.
- (5) Centralise the steering rack.
- (6) Refit the wheels and lower the lifting equipment.
- (7) Without disturbing the wheels, rotate the steering tie rods to shorten (increase toe-in) or lengthen (decrease toe-in) the tie rod length.
- (8) Tighten lock nuts and check wheels are still aligned.



- (9) Roll vehicle back sufficiently to allow wheels to fully rotate.
- (10) Reposition vehicle and check wheel alignment.
- (11) Check steering wheel spoke alignment.
- (12) On completion of wheel alignment adjustments torque the steering tie-rod locknuts 28.5 to 33Nm (21 to 24ftlb).
- (13) Raise the vehicle to a suitable working height, remove the wheels (Refer to Section 4.1) and fit the gaiter onto the steering rack using new cable ties.
- (14) Fit the wheels (Refer to Section 4.1), lower the lifting apparatus and remove the vehicle.

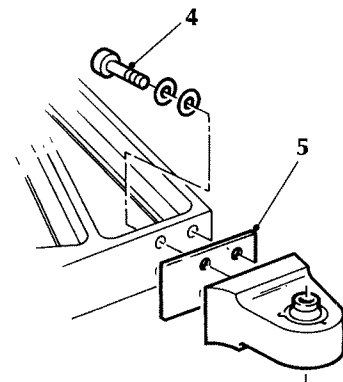
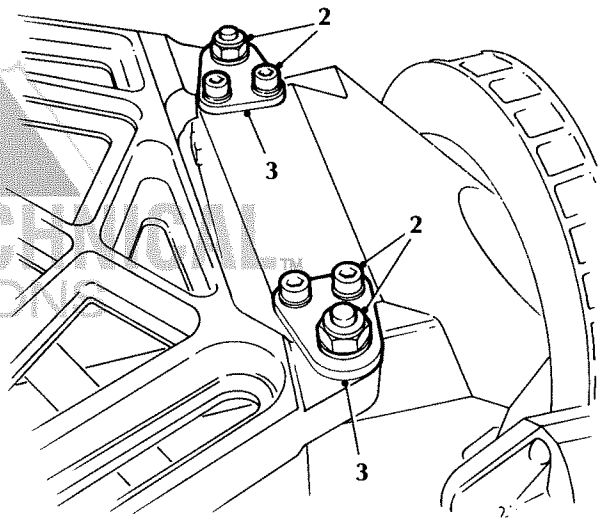
B Adjustment of Rear Wheel Alignment

- (1) Correctly position vehicle on lifting apparatus (Refer to Section 1.1), remove wheel (Refer to Section 4.1) and raise vehicle to a suitable working height.
- (2) Support rear upright and remove cap head screws and washers, and nut and washer securing wishbone to upright..
- (3) Collect plates and lift wishbone off upright.
- (4) Remove the wishbone to spherical bearing cap screws and remove cap and shims.
- (5) Measure and record shim thickness.
- (6) To adjust the toe-in by +/- 0 degrees 15 minutes, or multiples thereof, add shim thickness 0.91mm, 20 SWG (0.36in.) P/No.5688AA.
- (6A) To adjust the toe-in by +/- 0 degrees 28 minutes, or multiples thereof, add shim thickness 1.63mm, 16 SWG (0.64in.) P/No. 5687AA.
- (7) Fit shims to cap and secure cap to wishbone with cap head screws.

CAUTION

Ensure the cap is refitted with the circlip groove on the underside.

- (8) Torque load screws 8 to 9.5Nm (6 to 7ftlb).
- (9) Fit wishbone to upright and fit plates.
- (10) Secure plates to upright with cap head screws torque loaded 20 to 23Nm (15 to 17ftlb).



- (11) Fit washers and nuts to the top pins and torque load nuts 57 to 65Nm (42 to 48ftlb).
- (12) Fit wheel (Refer to Section 4.1).
- (13) Lower lifting apparatus and check wheel alignment.
- (14) If wheel alignment is satisfactory remove vehicle from lifting apparatus. If wheel alignment is unsatisfactory repeat this procedure.

28 RIDE HEIGHT TRIM

A Check

Special Tools

The following special tools will be required:

JaguarSport protective body covers

NOTES:

- 1 Ensure absolute cleanliness of work area and tools.
- 2 Ensure correct use of tools.
- 3 Ensure the vehicle is at the correct kerb weight.
- 4 Ensure all tyre pressures are correct.
- 5 Ensure the vehicle is on a flat and level surface.
- 6 Fit JaguarSport protective body covers in immediate work area.

Kerb Weight

The kerb weight of the vehicle is defined as the complete vehicle plus all fluids. This includes a full tank of petrol.

Front Axle	LH Wheel	288kg (635.040lbs)
	RH Wheel	309kg (681.345lbs)
Rear Axle	LH Wheel	445kg (981.225lbs)
	RH Wheel	428kg(943.740lbs)
		873kg (1924.965lbs)

Totals

Front Axle	=	597kg (1316.385lbs)
		+
Rear Axle	=	873kg (1925.965lbs)
Kerb Weight	=	1470kg.(3241.35lbs)

Tyre Pressure

The tyre pressure must be checked with the tyres at ambient temperature.

Front Tyres 2.5 bar-0+.1bar
(37psi -0+1.5psi)

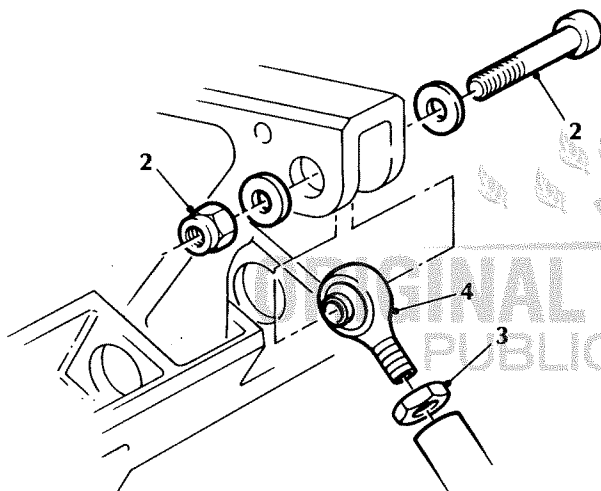
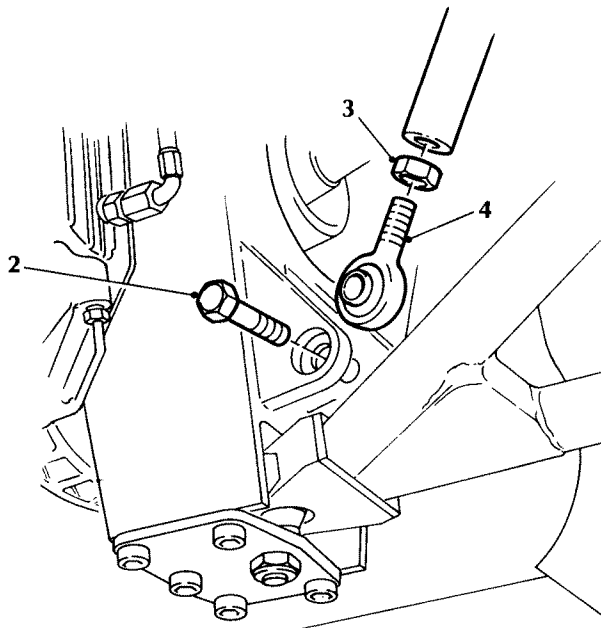
Rear Tyres 1.8 bar-0+.1bar
(26psi -0+1.5psi)

Procedure

CAUTION:

Do not mark the underside of the vehicle with an implement which will penetrate the anti-corrosion protective finish.

- (1) Locate the longitudinal centre line of the vehicle.
- (2) Mark the longitudinal centre line between the front wheels and rear wheels.
- (3) Locate the horizontal centre line between the axles.
- (4) Mark where the horizontal lines cross the longitudinal lines.
- (5) Measure from the flat surface to the points where the longitudinal and horizontal lines cross.
- (6) Check the measurement is between 126mm and 128mm.



B Adjustment

- (1) Remove the front and rear wheels (Refer to Section 4.1).
- (2) Support the uprights and remove the front and rear pushrod bolts, washers and nuts.
- (3) Loosen the locknut on the push rods.
- (4) To raise the ride height, screw the pushrod eye-end out. To lower the ride height, screw the pushrod eye-end in.
- (5) Tighten the pushrod locknut and torque 27 to 31Nm (20 to 23ftlb).
- (6) The front pushrods should be secured to the nodder with bolt, washers and nut. Torque load the nut to 35 to 41Nm (26 to 30ftlb).
- (6A) The rear pushrod should be secured to the upright using special bolt, washer and nut. Torque load the nut 57 to 65Nm (42 to 48ftlb)
- (7) Fit the front and rear wheels (Refer to Section 4.1)
- (8) Check ride height again and adjust if necessary.

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DESCRIPTION

The driveline transfers the transaxle drive to the rear wheels and consists of a driveshaft fitted at each end with a constant velocity joint. The inboard and outboard constant velocity joints are each housed in an attachment flange. The flanges are bolted to the transaxle output flange and the stub axle inside the rear upright.

1 INTRODUCTION

This section details the procedures for removal and installation of driveline components. Removal of components or disassembly instructions not covered in this section require the driveline be returned to JaguarSport.

2 DRIVESHAFT

A Removal Procedure

Special Tools

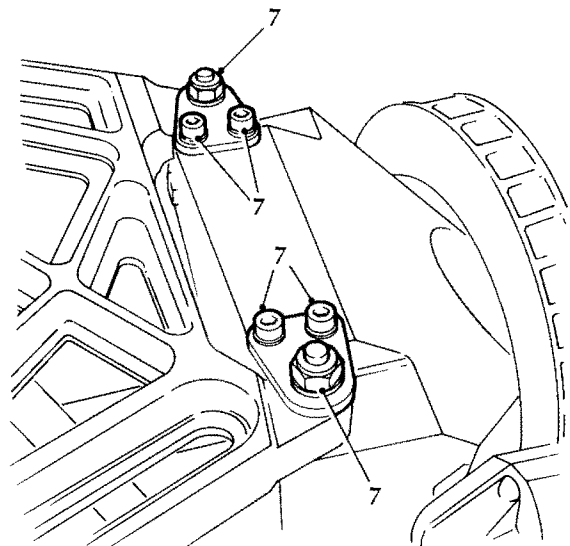
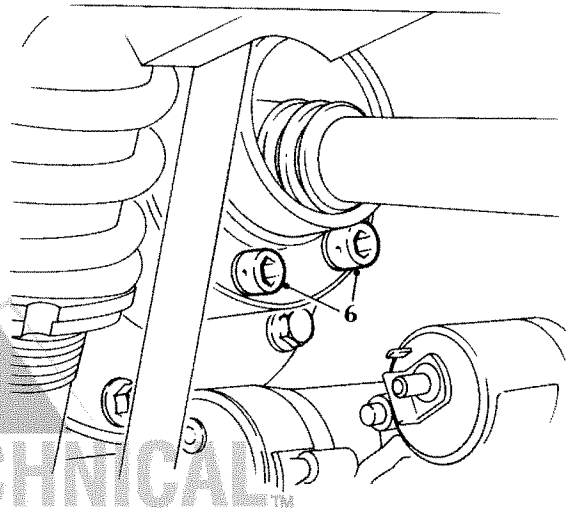
The following special tools will be required:

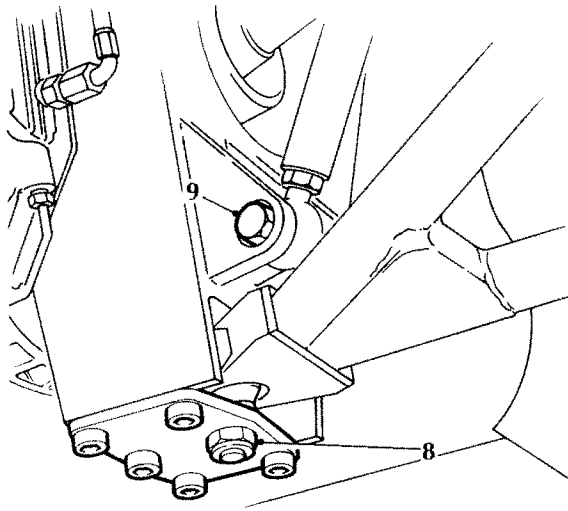
Torque wrench 65Nm (48ftlb).

NOTES:

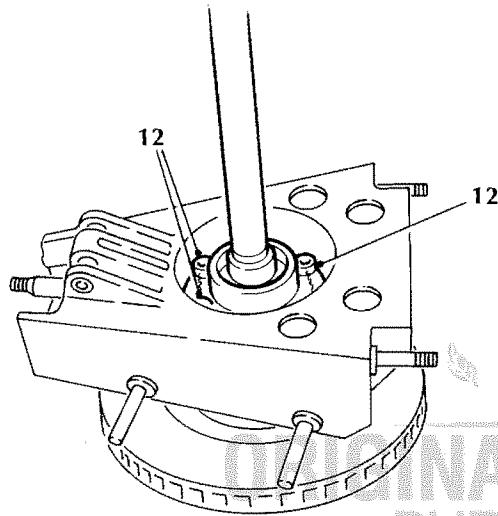
- 1 Ensure absolute cleanliness of work area and tools.
- 2 Observe correct use of tools.
- 3 Isolate battery by moving battery master switch to OFF position.

- (1) Correctly position vehicle on lifting apparatus (Refer to Section 1) and raise vehicle to a suitable working height.
- (2) Remove the rear wheel (Refer to Section 4).
- (3) Remove screws and remove the venturi and bumper. (Refer to Section 2).
- (4) Remove nuts securing brake caliper and handbrake caliper to upright. (Refer to Section 6).
- (5) Remove brake calipers from the upright and secure to upper wishbone with tyrap or similar to avoid any tension on the brake pipe.
- (6) Remove locking wire and remove the six bolts securing driveshaft to transaxle output flange.
- (7) Remove bolts, nuts and washers securing retaining plates to the upright.





- (8) Remove bolts and nut securing retaining plate on bottom wishbone and upright.
- (9) Remove bolt, nut and washer securing pushrod to upright.
- (10) Raise top wishbone and lower bottom wishbone, remove push rod and withdraw driveshaft with upright and brake disc.
- (11) Cover the exposed end of the CV joint to prevent ingress of dust/dirt and securely clamp upright in padded vice with driveshaft vertical.
- (12) Remove locking wire and remove the six bolts securing outer end of driveshaft to the stub flange.
- (13) Remove driveshaft.



B Installation Procedure

- (1) Cover exposed end of CV joint and securely clamp upright in padded vice.
- (2) Fit drive shaft to upright and secure outer end of driveshaft to the stub flange with the six bolts.
- (3) Tighten the bolts evenly and torque load 57 to 65Nm (42 to 48ftlb).
- (4) Wire lock the bolts using 0.032in locking wire using a single pass through each screw head in a continuous ring.
- (5) Raise top wishbone and lower bottom wishbone and fit upright with driveshaft until inner end of driveshaft mates with transaxle drive flange.
- (6) Fit push rod to attachment lug on upright and secure with screw, washer and nut.
- (7) Tighten the nut to 57 to 65Nm (42 to 48ftlb)
- (8) Fit wishbones to upright.
- (9) Secure top wishbone to upright with retaining plates, nuts and washers, and cap head screws fitted with washers.
- (10) Tighten the nuts to 57 to 65Nm (42 to 48ftlb) and cap head screws 20 to 23Nm (15 to 17ftlb).
- (11) Secure bottom wishbone to upright with retaining plate, nut and washer, and bolts fitted with washers.
- (12) Tighten the nuts to 57 to 65Nm (42 to 48ftlb) and cap head screws 20 to 23Nm (15 to 17ftlb).
- (13) Fit brake caliper to upright and secure with nuts and washers.

- (14) Torque load nuts 57 to 65Nm (42 to 48ftlb).
- (15) Fit handbrake caliper to upright and secure with washers and screws.
- (16) Torque load screw to 40.5 to 46.5Nm (30 to 34.5ftlb).
- (17) Secure the inner driveshaft flange to the transaxle output flange with six bolts.
- (18) Tighten the bolts to 57 to 65Nm (42 to 48ftlb) and wirelock using three lengths of 0.024in wire, wiring the screws in adjacent pairs.
- (19) Fit venturi and bumper and secure with screws.
- (20) Lower the lifting apparatus and remove the vehicle.



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DESCRIPTION

The XJ220 is fitted with disc brakes front and rear which are operated by hydraulic pressure. The hand brake is a mechanical system which operates independently on the rear brake discs only.

The foot brake system is hydraulically operated and consists of a master cylinder and brake pedal with hydraulic pipes connecting the master cylinder to brake calipers.

The handbrake mechanism consists of a manually operated lever connected by cables to handbrake calipers.

The foot brake pedal attaches the piston rod of the master cylinder. Two rigid pipes attach the master cylinder; the rear rigid pipe connects the master cylinder to the front brake calipers, the front rigid pipe connects the master cylinder to the rear brake calipers.

The brake calipers, which are attached to the suspension uprights, are each fitted with 4 pistons. The pistons operate non-asbestos brake pads against the ventilated brake discs. The front discs are 330.2mm (13in) diameter. The rear discs are 300mm (11in) diameter.

The handbrake lever is floor mounted on the door side of the driver's seat. The primary cable connects the handbrake lever to a mechanical lever system in the engine compartment. The mechanical lever attaches to the two secondary cables. One cable attaches the RH handbrake caliper, and the other cable attaches the LH handbrake caliper.

The handbrake caliper is a mechanical unit attached to the rear suspension upright and the handbrake pads operate against the service brake disc.

OPERATION

Pressure on the brake pedal operates the piston in the master cylinder, which applies pressure to the brake fluid forcing the caliper pistons against the brake pads which apply pressure to the brake discs.

When the pressure is removed from the brake pedal the brake pads move away from the brake disc.

Upward movement of the handbrake lever applies tension to the handbrake primary cable. The tension is transmitted to the mechanical lever system which transfers the movement to the two secondary handbrake cables. The secondary cables operate the handbrake calipers to apply pressure to the brake discs.

When the handbrake cable tension is removed the handbrake pads move away from the brake disc.

1 INTRODUCTION

This section details the procedures for replacement of brake system components. Replacement of components or disassembly instructions not covered in this section require the component be returned to JaguarSport.

2 BRAKE PAD REPLACEMENT

A Preparation

Special Tools

The following special tools will be required:

Caliper piston lever

Jaguar XJ220 protective body covers

Jaguar XJ220 jack

Socket spanner 90mm (3.5in)

Torque wrench 488Nm (0–360ftlb).

Materials

The following materials will be required:

Brake cleaning fluid

Castrol synthetic racing brake fluid

Face mask

High temperature anti-corrosive copper grease

Lint free cloth.

NOTES:

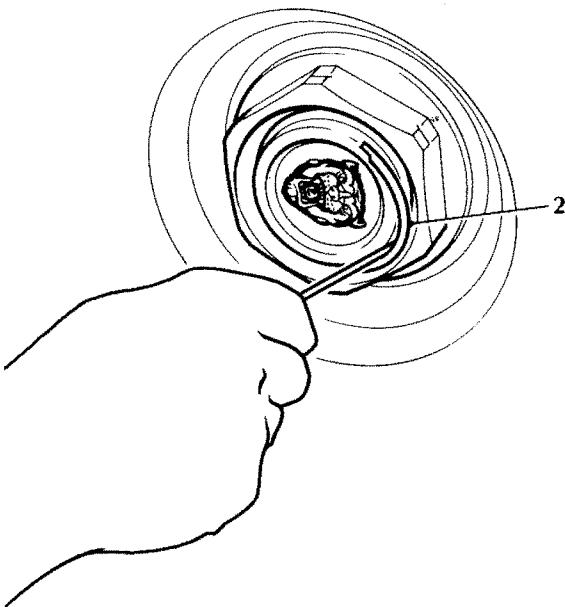
- 1 Ensure absolute cleanliness of work area and tools.
- 2 Observe correct use of tools.

B Procedure

CAUTION:

The wheel nuts on the RH side of the vehicle are LH thread. The wheel nuts on the LH side of the vehicle are RH thread.

- (1) Insert a small screwdriver in stub axle undercut and lever out retaining clip.
- (2) Slide the screwdriver in a clockwise direction to remove retaining clip.
- (3) Apply the brakes and loosen wheel nut using socket spanner and a suitable wrench.



CAUTIONS:

1. *Do not attempt to jack up the vehicle unless the handbrake is off and the transaxle is in neutral.*
2. *Ensure the jack lift plate is free from debris which could damage the underside of the vehicle.*

NOTE:

During jacking operations the vehicle will move toward the jack.

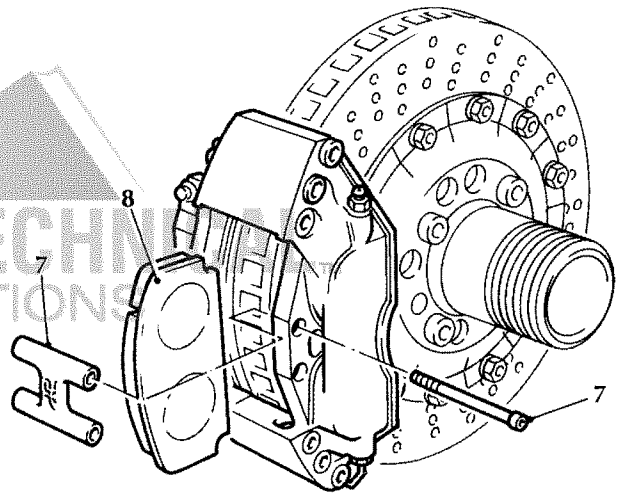
- (4) For front end jacking;
Position jack on the jacking point directly under zero bulkhead and lift the vehicle.
- (4A) For rear end jacking;
Position the jack on the jacking point directly under engine rear plate and lift the vehicle.
- (5) Apply handbrake and select any gear.
- (6) Remove wheel nut and remove wheel.

WARNING:

THE BRAKE PADS ARE CERAMIC. DO NOT USE COMPRESSED AIR TO REMOVE ACCUMULATIONS OF BRAKE PAD DUST. USE CLEAN LINT FREE CLOTH MOISTENED WITH BRAKE CLEANING FLUID. WEAR PROTECTIVE GLOVES AND FACE MASK.

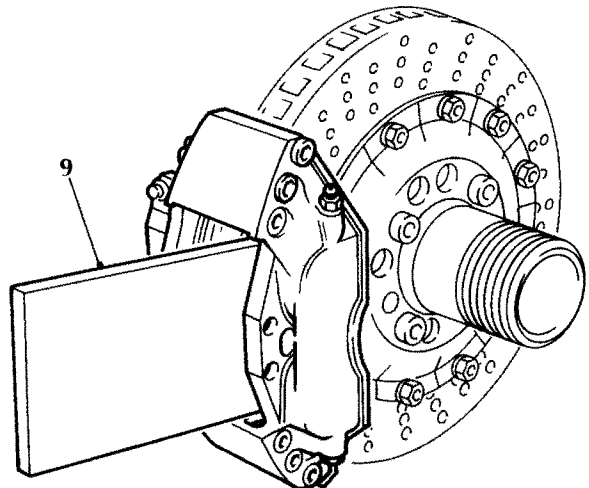
NOTES:

- 1 Check the condition of the tyre, especially the inner wall which is not normally visible.
- 2 Check the wheel is free from damage.
- 3 Check the brake disc is free from damage such as grooves, cracks, excessive wear and burn marks.
- 4 Check there is no brake fluid leakage.
- 5 Check the nuts and bolts securing the disc bell are all fitted.
- (7) Remove bolts securing caliper bridge using 5mm allen key, and remove the caliper bridge.



CAUTIONS:

- 1 *Check the brake fluid level in the master cylinder. If the master cylinder is full take precautions to prevent spillage.*
2. *Remove only one brake pad at a time, starting with the outer pad. Do not remove the inner pad until the new outer pad has been fitted.*
- (8) Remove outer brake pad and discard.
- (9) Depress brake caliper pistons using caliper piston lever.



- (10) Fit new brake pad.
- (11) Remove inner brake pad and discard.
- (12) Repeat steps (9) and (10).
- (13) Fit caliper bridge and secure with bolts and torque loaded to 0.225Nm (2inlb).
- (14) Rotate brake disc to ensure unrestricted rotation.
- (15) Apply high temperature anti-corrosive copper grease to stub axle threads.
- (16) Align wheel location pegs with the wheel and fit the wheel.
- (17) Fit wheel nut hand tight.

CAUTION:

Release the handbrake and select neutral gear.

- (18) For front end jacking;
Lower the vehicle and remove the jack.
- (18A) For rear end jacking;
Lower the vehicle, remove jack and apply handbrake.
- (19) Torque load wheel nut to 461 to 488Nm (340 to 360ftlb) using socket spanner and torque wrench.
- (20) Fit retaining clip and check brake fluid level.

CAUTIONS:

1 ***Ensure JaguarSport recommended protective covers are fitted before work is started.***

2 ***Wipe away any spillage using clean lint free cloth.***

- (21) Replenish brake fluid if necessary using a syringe fitted with plastic tube.
- (22) Fit master cylinder cap and fit electrical plug.
- (22) Remove protective covers and wheel chocks.
- (23) See Brake Bedding-In procedures.

3 HANDBRAKE PAD REPLACEMENT

A Removal Procedure

Special Tools

The following special tools will be required.

Jaguar XJ220 jack

Socket spanner 90mm (3.5in)

Torque wrench 488Nm (360ftlb)

Wheel chocks

Materials

The following materials will be required.

Brake cleaning fluid

Face mask

Lint free cloth

Stiff bristle brush

CAUTION:

The wheel nuts on the RH side of the vehicle are LH thread. The wheel nuts on the LH side of the vehicle are RH threads.

- (1) Position the vehicle on lifting apparatus. (Refer to Introduction, Lifting and Jacking).
- (2) Remove the rear wheels.

WARNING:

THE BRAKE PADS ARE CERAMIC. DO NOT USE COMPRESSED AIR TO REMOVE ACCUMULATIONS OF BRAKE PAD DUST. USE STIFF BRISTLE BRUSH AND LINT FREE CLOTH MOISTENED WITH BRAKE CLEANING FLUID. WEAR FACE MASK.

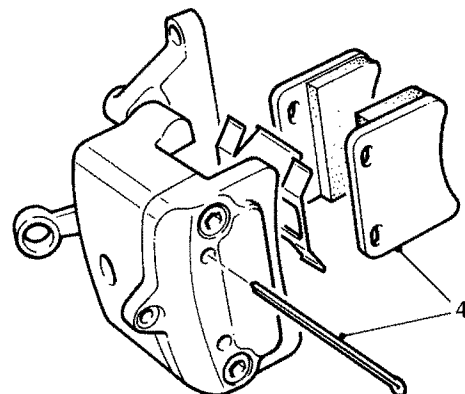
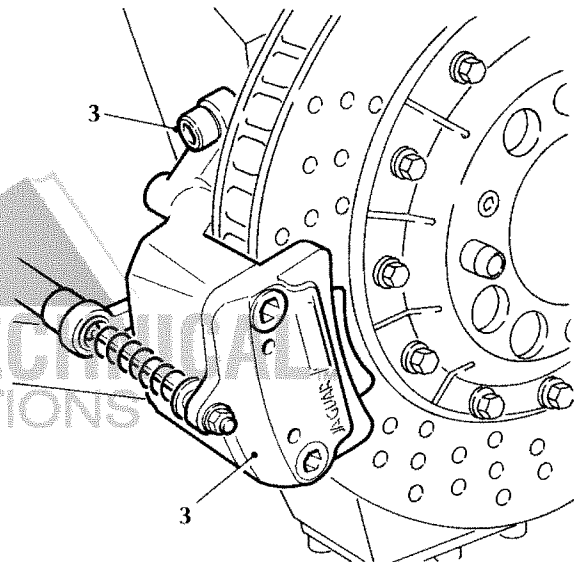
NOTES:

- 1 Check the condition of the tyre, especially the inner wall which is not normally visible.
- 2 Check the wheel is free from damage.
- 3 Check the brake disc is free from damage such as grooves, cracks, excessive wear and burn marks.
- 4 Check there is no brake fluid leakage.
- 5 Check the nuts and bolts securing the disc bell are all fitted and the nuts are in safety.

- (3) Remove bolts securing handbrake calipers to upright and remove the calipers.
- (4) Remove and discard split pins, and remove handbrake pads.

NOTE:

Do not re-use split pins.



- (5) Clean the caliper units using lint free cloth moistened with brake cleaning fluid. Remove stubborn accumulations of foreign matter with the stiff bristle brush.
- (6) Fit new handbrake pads and secure with new split pins.
- (7) Secure caliper units to upright and torque load nuts 57 to 65Nm (42 to 48ftlb).
- (8) Connect and adjust the handbrake operating cables (Refer to Section 6.12).
- (9) Fit wheels and lower the vehicle (Refer to Section 4).

4 BRAKE BEDDING-IN

This is a general procedure which can only be accomplished by either driving the vehicle, or by simulated driving conditions on a rolling road.

Drive the vehicle according to road conditions using the brakes whenever necessary. Avoid harsh braking. Allow an extra two (2) car lengths braking distance for at least the first 100 applications.

It is anticipated that the brakes will be bedded-in satisfactorily after 100 brake applications. However, it is recommended that care is taken for a further 50 applications.

If in doubt refer to manufacturer.

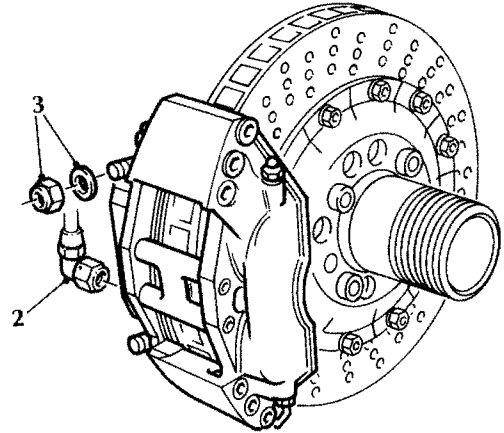
5 BRAKE CALIPERS

A Removal Procedure

WARNING:
ENSURE BRAKES HAVE COOLED DOWN BEFORE STARTING WORK.

NOTES:

- 1 Ensure absolute cleanliness of work area and tools.
 - 2 Observe correct use of tools.
-
- (1) Raise the vehicle and remove the respective wheel (Refer to Section 4).
 - (2) Remove brake hose from brake caliper and fit blanking plugs.
 - (3) Remove nuts and washers securing brake caliper to upright.
 - (4) Remove brake caliper from brake disc.



B Installation Procedure

- NOTES:**
- 1 Ensure absolute cleanliness of work area and tools.
 - 2 Observe correct use of tools.

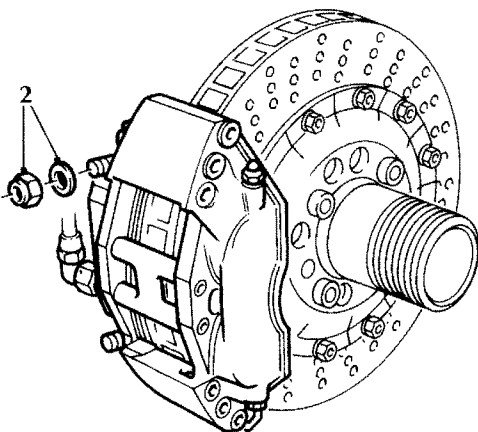
- (1) Fit brake caliper to brake disc.
- (2) Secure brake caliper to upright with nuts and washers.
- (3) Torque load nuts 57 to 65Nm (42 to 48ftlb).
- (4) Remove blanking caps and connect brake hose to brake caliper. Torque the hose union 12 to 16Nm (9 to 12ftlb).
- (5) Bleed the brake system (Refer, to Section 10).
- (6) Fit the wheel and lower the vehicle (Refer to Section 4).

6 BRAKE DISC

A Removal Procedure

NOTES:

- 1 Ensure absolute cleanliness of work area and tools.
- 2 Observe correct use of tools.
- 3 Check serviceability of components as they are removed. If in doubt reject component.

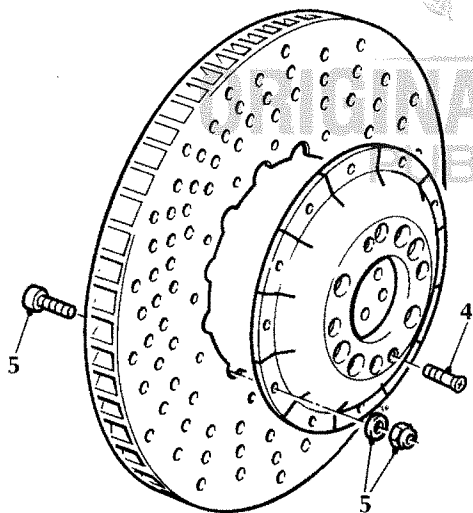


- (1) Remove the relevant road wheel. (Refer to Road Wheels, Section 4).
- (2) Remove nuts and washers securing brake caliper to the upright and secure brake caliper to the wishbone with tyrap to avoid any tension on the flexible brake hose.

CAUTION

Take care not to kink the flexible brake hose.

- (3) On the rear brakes, remove the bolts securing the hand brake caliper to the upright and secure the caliper assembly to the wishbone with tyrap.
- (4) Remove the two countersunk screws, and remove brake disc and bell from stub axle.
- (5) Remove screws, washers and nuts securing the bell to the disc.



B Installation Procedure

Special Tools

The following special tools will be required

Torque wrench 65Nm (48ftlb).

NOTES:

- 1 Ensure absolute cleanliness of work area and tools.
- 2 Observe correct use of tools.
- (1) Ensure all mating surfaces are clean.
- (2) Ensure the disc is correctly orientated and fit the bell to the brake disc with screws, washers and nuts. Torque load nuts 13.5 to 16Nm (10 to 12ftlb).
- (3) Fit brake disc with brake bell to stub axle and secure with countersunk screws. Torque load screws 6.5 to 8Nm (5 to 6ftlb).
- (4) Fit brake caliper to studs and secure with washers and nuts.
- (5) Torque load nuts 57 to 65Nm (42 to 48).
- (6) On the rear discs, fit the handbrake caliper and secure with screws and washers. Torque the screws 41 to 47Nm (30 to 34.5ftlb).

7 BRAKE PIPES

REAR BRAKE PIPES

A Removal Procedure

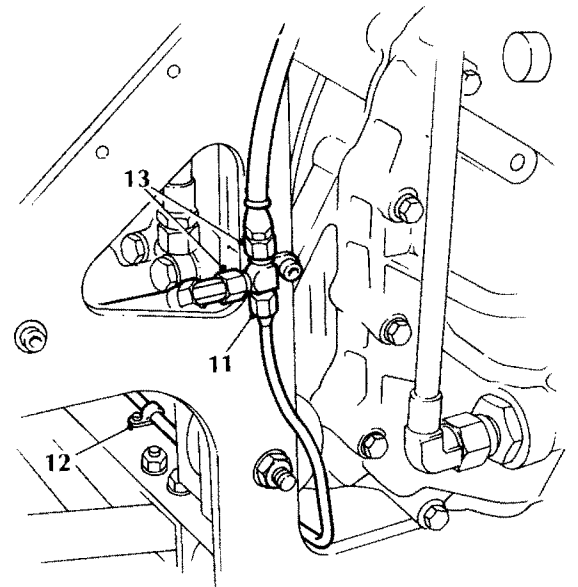
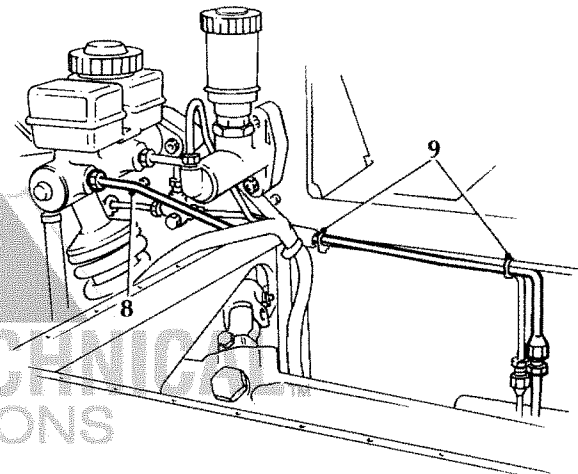
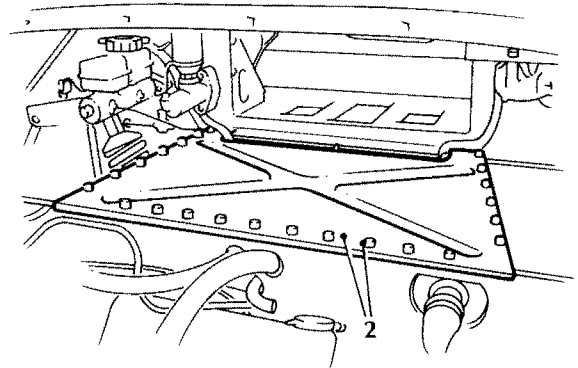
WARNING:
BRAKE FLUID CAN HAVE A HARMFUL EFFECT ON THE SKIN. AVOID CONTACT WITH THE SKIN AND USE A BARRIER CREAM. WEAR EYE PROTECTION.

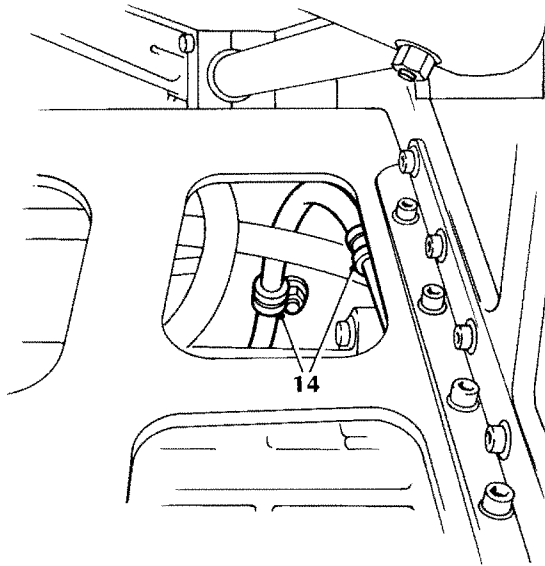
CAUTION:
Brake fluid can have a damaging effect on paint surfaces. Avoid brake fluid spillage.

NOTES:

- 1 Ensure absolute cleanliness of work area and tools.
- 2 Observe correct use of tools.

- (1) If the tunnel pipe is to be removed, remove the engine and transaxle. (Refer to Section 3).
- (2) Remove the screws securing the structural hatch and remove the hatch.
- (3) Remove the air conditioning unit. (Refer to Section 12).
- (4) Remove the driver and passenger underscuttle. (Refer to Section 2).
- (5) Remove the tunnel trim and carpet. (Refer to Section 2).
- (6) Remove the screws securing the tunnel top cover and remove the cover.
- (7) Drain the fluid from the master cylinder.
- (8) Disconnect the pipe from the master cylinder to the front of the tunnel, fit blanking caps to the master cylinder.
- (9) Remove the cable clips and remove the pipe.
- (10) Remove the tunnel and rear bulkhead pipe clips, disconnect the tunnel pipe from the engine beam pipe and remove the pipe from the tunnel through the engine compartment. Ensure the pipe does not foul on adjacent pipework or structure.
- (11) Disconnect the rear brake pipe to flexible hose connection at the T piece on the engine rear plate.
- (12) Remove the P clips securing the pipe to the engine beam and remove the pipe.
- (13) Disconnect the flexible pipes at the rear plate T piece and brake calipers. Fit blanking caps to the calipers.





- (14) Remove the P clips securing the pipes to the bridge cleats and remove the pipes.

B Installation Procedure

NOTES:

- 1 *Ensure absolute cleanliness of work area and tools.*
 - 2 *Observe correct use of tools.*
- (1) Remove blanking caps and connect flexible brake pipes to brake caliper and T piece on engine rear plate. Torque pipe unions 12 to 16Nm (9 to 12ftlb).
 - (2) Fit 'P' clips to the flexible pipes and secure to bridge plate cleats.
 - (3) Attach the flexible pipes to the inside edge of the rear wishbones using tyraps.
 - (4) Remove blanking caps and connect brake pipe to T piece and route along the engine beam to rear bulkhead. Torque the pipe union 9 to 11Nm (6.5 to 8ftlb).
 - (5) Secure brake pipe to engine beam with retaining clips.
 - (6) Route the brake pipe through the tunnel, ensure the pipe does not foul on adjacent pipework or structure, and secure with cable ties.
 - (7) Remove blanks and fit brake pipe to connector at front bulkhead and to master cylinder. Torque the pipe unions 9 to 11Nm (6.5 to 8ftlb).
 - (8) Fit retaining clips.
 - (9) Refit tunnel access cover. (Refer to Section 2).
 - (10) Fit the tunnel carpet and trim (Refer to Section 2)
 - (11) Fit driver and passenger underscuttle. (Refer to Section 2).
 - (12) Fit the air conditioning unit. (Refer to Section 12).
 - (13) Fit structural hatch and torque screws 12 to 14Nm (9 to 10.5ftlb).
 - (14) Fit the engine and transaxle. (Refer to Section 3).

ORIGINAL TECHNICAL
PUBLICATIONS

FRONT BRAKE PIPES

A Removal Procedure

WARNING:

BRAKE FLUID CAN HAVE A HARMFUL EFFECT ON THE SKIN. AVOID CONTACT WITH THE SKIN AND USE A BARRIER CREAM. WEAR EYE PROTECTION.

CAUTION:

Brake fluid can have a damaging effect on paint surfaces. Avoid brake fluid spillage.

NOTES:

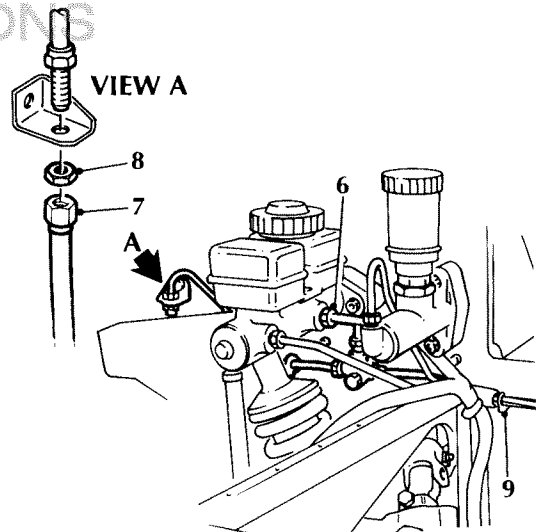
- 1 Ensure absolute cleanliness of work area and tools.
- 2 Observe correct use of tools.

- (1) Position the vehicle on the lifting apparatus (Refer to Introduction, Lifting and Jacking) and raise to a suitable working height.
- (2) Remove the wheels and wheel arch liners. (Refer to Road Wheels, Section 4).
- (3) Remove the radiator duct access panel.
- (4) Remove the air conditioning unit. (Refer to Section 12).
- (5) Drain the fluid from the master cylinder.

CAUTION:

Ensure the brakes are not operated while the system is disconnected.

- (6) Remove brake pipe from master cylinder to T piece. Fit blanking caps to open connections.
- (7) Remove flexible hoses from brake calipers and connections to rigid brake pipes at the brackets. Fit blanking caps to open connections.
- (8) Remove the locknuts securing the brake pipes to the brackets and disconnect the pipes at the T piece. Fit blanking caps to the open connections.
- (9) Remove retaining clips securing brake pipes and remove the pipes.
- (10) Lower the lifting apparatus .



B Installation Procedure

NOTES:

- 1 *Ensure absolute cleanliness of work area and tools.*
- 2 *Observe correct use of tools.*

- (1) Remove blanking caps and fit brake pipe from master cylinder to T piece. Torque unions 9 to 11Nm (6.5 to 8ftlb).
- (2) Remove blanking caps and fit the brake pipes to the T piece and to brackets.

NOTE

Do not tighten connections at this stage.

- (3) Fit retaining clips to brake pipe.
- (4) Check the route of the brake pipes to ensure the brake pipe cannot chafe against adjacent structure. Torque tighten the pipe unions 9 to 11Nm (6.5 to 8ftlb).
- (5) Raise the vehicle on the lifting apparatus (Refer to Introduction, Lifting and Jacking) and fit lock nut to brake pipe through bracket. Torque nut 17.5 to 20Nm (13 to 15ftlb).
- (6) Connect flexible brake pipe from rigid brake pipe to brake caliper. Torque tighten unions 12 to 16Nm (9 to 12ftlb) at the caliper and 9 to 11Nm (6.5 to 8ftlb) to the rigid pipes.
- (7) Check route of brake pipe to ensure the brake pipe cannot chafe against adjacent components and secure to wishbone with tyrap.
- (8) Bleed the brake system (Refer to Section 6.10).

CAUTION:

Check there are no leaks from the master cylinder or any joint in the brake system.

- (9) Fit the air conditioning unit. (Refer to Section 12).
- (10) Fit the radiator duct access panel.
- (11) Fit the wheel arch liners and wheels. (Refer to Road Wheels, Section 4).
- (12) Lower the lifting apparatus and remove the vehicle.
- (13) Test the brakes (Refer to Section 6.4).

8 MASTER CYLINDER

A Removal Procedure

Materials

The following materials will be required:

Barrier cream

Brake fluid

Lint free cloth

Container for brake fluid.

WARNING:

BRAKE FLUID CAN HAVE A HARMFUL EFFECT ON THE SKIN. AVOID PROLONGED CONTACT WITH THE SKIN AND USE A BARRIER CREAM. WEAR EYE PROTECTION.

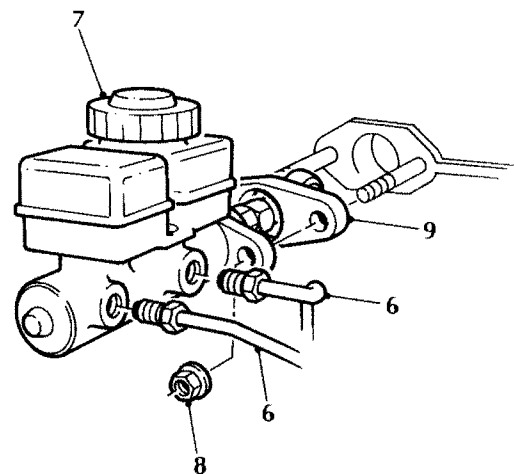
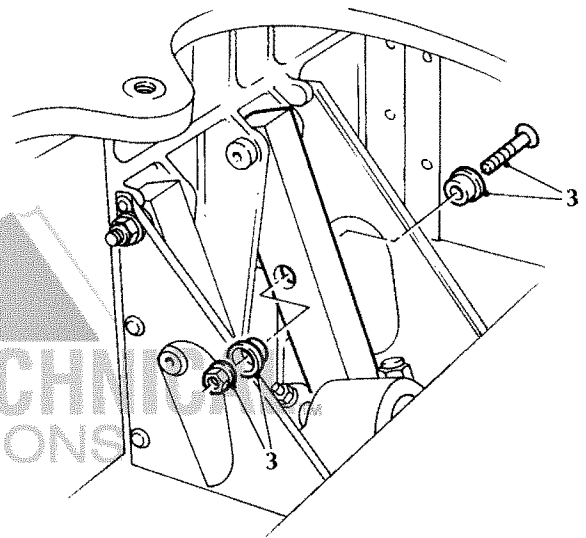
CAUTION:

Brake fluid can have a damaging effect on paint surfaces. Avoid brake fluid spillage.

NOTES:

- 1 Ensure absolute cleanliness of work area and tools.
- 2 Observe correct use of tools.

- (1) Remove the RH wheel and wheel arch liner. (Refer to Road Wheels, Section 4).
- (2) Remove interior front end trim (Refer to Section 2).
- (3) Remove the bolt and nut securing the master cylinder push rod to the brake pedal and retrieve the pedal inserts.
- (4) Remove the radiator duct access panel.
- (5) Drain brake fluid from master cylinder into a container. Refit the master cylinder cap.
- (6) Remove the brake pipes from the master cylinder, fit blanking caps to the open connections.
- (7) Disconnect the electrical cables to the master cylinder cap.
- (8) Remove the nuts attaching master cylinder to zero bulkhead.
- (9) Remove the master cylinder and the collect spacer.



B Installation Procedure

Special Tools

The following special tools will be required.

Large syringe and plastic tubing

Materials

The following materials will be required:

Barrier cream

Brake fluid

Lint free cloth.

WARNING:

BRAKE FLUID CAN HAVE A HARMFUL EFFECT ON THE SKIN. AVOID PROLONGED CONTACT WITH THE SKIN AND USE A BARRIER CREAM. WEAR EYE PROTECTION.

CAUTION:

Brake fluid can have a damaging effect on paint surfaces. Avoid brake fluid spillage.

NOTES:

1 Ensure absolute cleanliness of work area and tools.

2 Observe correct use of tools.

(1) Fit master cylinder and spacer to zero bulkhead and secure with nuts. Torque the nuts 30 to 34Nm (22 to 25ftlb).

(2) Fit the inserts to the pedal and connect the master cylinder push rod to the pedal using bolt and Kaylock nut. Torque nut 8.5 to 10Nm (6.5 to 7.5ftlb).

(3) Remove blanking caps and connect the brake pipes to master cylinder.

(4) Torque load brake pipe unions 9 to 11Nm (6.5 to 8ftlb).

(5) Fill master cylinder to the bottom of the filler neck with recommended brake fluid using suitable syringe and plastic tube.

(6) Connect the electrical cables to the master cylinder cap.

WARNING:

DO NOT ATTEMPT TO MOVE THE VEHICLE UNTIL THE BRAKE SYSTEM HAS BEEN BLED TO REMOVE AIR FROM THE SYSTEM.

(7) Bleed the brake system (Refer to Section 10).

(8) Fit the interior front end trim (Refer to Section 2).

(9) Fit the radiator duct access panel.

(10) Fit the wheel and wheel arch liner. (Refer to Road Wheels, Section 4).

(11) Test the brakes. (Refer to Section 6.4).

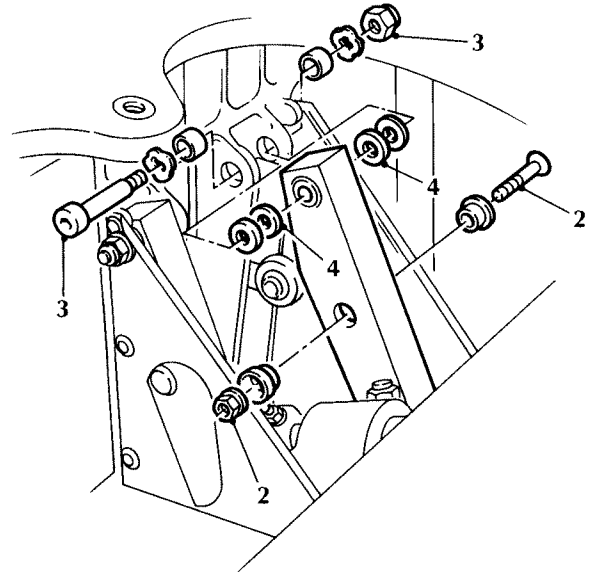
9 BRAKE PEDAL

A Removal Procedure

NOTES:

- 1 Ensure absolute cleanliness of work area and tools.
- 2 Observe correct use of tools.

- (1) Remove interior front end trim. (Refer to Section 2).
- (2) Disconnect the master cylinder push rod from the brake pedal.
- (3) Remove bolt and nut securing brake pedal to pedal box. Note the position and number of wavey washers fitted to the bolt.
- (4) Remove brake pedal and collect the shims and thrust washers. Note the number of shims fitted.

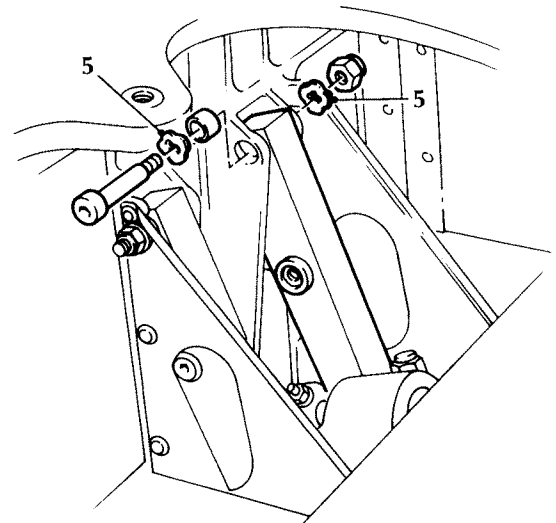


B Installation Procedure

NOTES:

- 1 Ensure absolute cleanliness of work area and tools.
- 2 Observe correct use of tools.

- (1) When fitting a new pedal, press new pivot bushes into the pedal.
- (2) Locate a thrust washer (coated face against the pedal) and shim either side of the pedal.
- (3) Fit the pedal between the lugs and add shims as removed.
- (4) Fit the pedal attachment bolt together with a plain M8 nut and measure the bolt end float.
- (5) To adjust end float proceed as follows;
 - a) For end float greater than 1mm insert one wavey washer (Pt. No. 8490) under the head of the bolt and one under nut.
 - b) For end float between 0.5mm and 1mm insert a wavey washer under the head of the bolt.
 - c) For an end float of less than 0.5mm no washers are required.
- (6) Replace the M8 plain nut with an M8 Philidas and torque tighten the nut 20 to 23Nm (15 to 17ftlb).
- (7) Fit inserts to the pedal and secure the master cylinder push rod to the pedal with a bolt and Kaylock nut. Torque the nut 8.5 to 10Nm (6.5 to 7.5ftlb).
- (8) Fit the interior front end trim. (Refer to Section 2).



10 BRAKE SYSTEM BLEEDING

Special Tools

The following special tools will be required:

Brake system bleeding equipment.

Materials

The following materials will be required:

Barrier cream

Brake fluid

Lint free cloth.

Procedure

WARNING:

BRAKE FLUID CAN HAVE A HARMFUL EFFECT ON THE SKIN. AVOID CONTACT WITH THE SKIN AND USE A BARRIER CREAM. WEAR EYE PROTECTION.

CAUTION:

Brake fluid can have a damaging effect on paint surfaces. Avoid brake fluid spillage.

NOTES:

- 1 Ensure absolute cleanliness of work area and tools.
- 2 Observe correct use of tools.
- 3 The road wheels must be removed in order to bleed the brakes. (Refer to Section 4).
- 4 It is recommended that the front and rear brakes on each side of the vehicle are bled simultaneously, for example, the LH front and rear brakes are bled before the RH front and rear brakes.

- (1) Remove filler cap from master cylinder.

CAUTION:

Always use clean brake fluid when topping up the reservoir. Never use fluid which has been bled from the system.

NOTE:

Start the bleeding operation on LH side for RHD and RH side for LHD.

- (2) Ensure brake fluid in the master cylinder is level with the bottom of the filler neck.
- (3) Loosen outboard bleed nipple on front and rear brake caliper.
- (4) Fit brake bleeding equipment to loosened caliper bleed nipples.

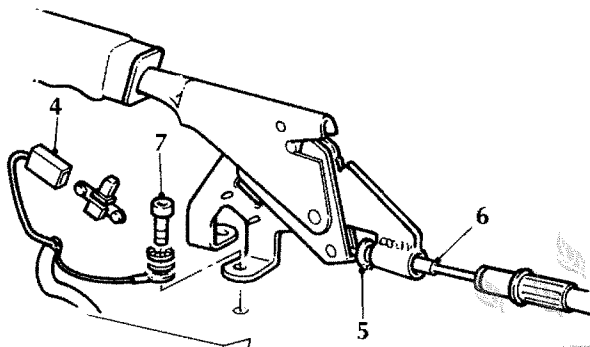
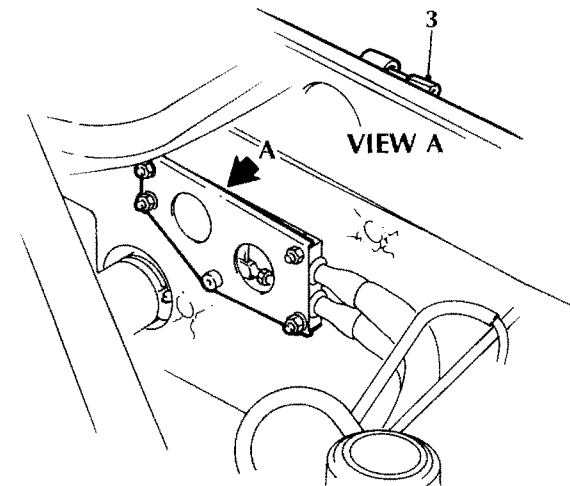
- (5) Operate brake pedal several times to remove air from the system.

NOTES:

- 1 *Check the brake fluid level in the master cylinder at regular intervals.*
 - 2 *It is recommended that the brake pedal is held at the full brakes on position while the bleed nipples are tightened to prevent air being drawn back into the system.*
- (6) Allow brake pedal to return to brakes OFF position.
- (7) Loosen inboard bleed nipple on front and rear brake caliper.
- (8) Repeat steps (3), (4), (5), (6) and (7) on the other side of vehicle.
- (9) Inspect the brake system as far as possible to ensure there are no leaks.
- (10) Refit road wheels (Refer to Section 4).
- (11) When brake system bleeding has been successfully completed position the vehicle on a 'rolling road' brake test rig.
- (12) Test the braking efficiency of the vehicle in accordance with 'rolling road' test rig operating instructions.



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11 HANDBRAKE LEVER ASSEMBLY

A Removal Procedure

- (1) Move the battery master switch to the off position.
- (2) Remove the handbrake lever trimmed cover.
- (3) In the engine compartment, remove the nut attaching the primary cable to the link trunnion.
- (4) Disconnect the electrical connections to the handbrake warning switch.
- (5) Remove the circlip attaching the cable outer to the handbrake lever.
- (6) Unscrew the cable from the handbrake lever.
- (7) Remove the two screws and washers attaching the lever assembly to the floor and remove the lever assembly.

B Installation Procedure

- (1) Fit the lever assembly to the vehicle and secure with washers and screws. Torque screws 20 to 23Nm (15 to 17ftlb).
- (2) Fit the primary cable to the lever and secure with circlip.
- (3) Screw the primary cable inner fully into the trunnion leaving a maximum of one thread protruding from the front side.
- (4) Fit the electrical connector to the handbrake warning switch.
- (5) In the engine compartment, fit the cable to the trunnion with the sleeved nut and adjust the nut to remove all slack from the cables.
- (6) Fit the handbrake lever trimmed cover. (Refer to Section 2).
- (7) Move the battery master switch to the on position.
- (8) Check the handbrake setting and adjust if necessary. (Refer to Section 6.13)

12 HANDBRAKE CABLES

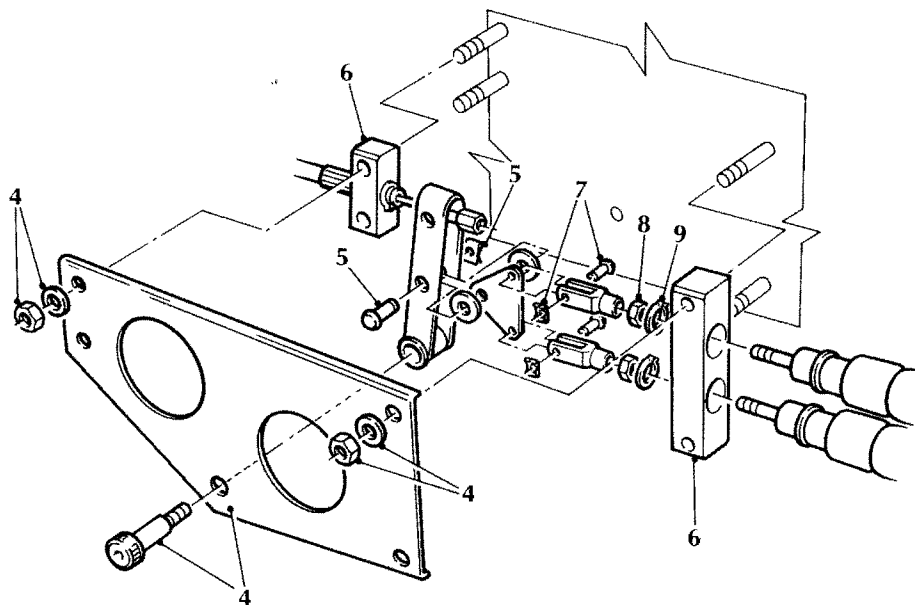
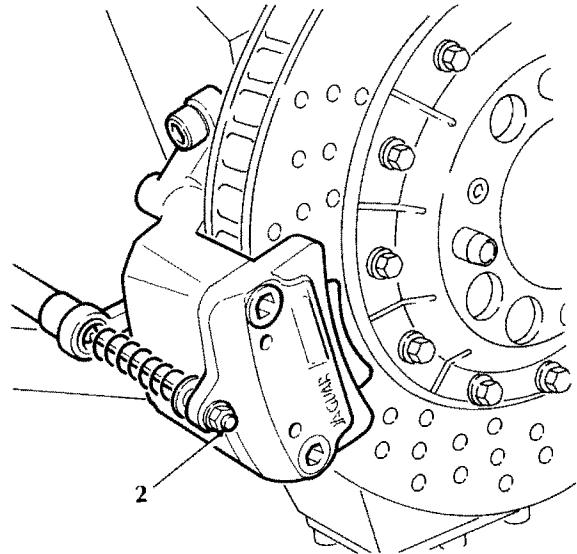
SECONDARY CABLES

A Removal Procedure

NOTES:

- 1 Ensure absolute cleanliness of work area and tools.
- 2 Observe correct use of tools.

- (1) Remove rear wheel and raise vehicle. (Refer to Section 4).
- (2) Remove nuts and disconnect handbrake cables from brake caliper.
- (3) Collect washer and spring.
- (4) Remove nuts, washers and bolt and remove handbrake mechanism cover plate and booster sensor bracket (LHD only).
- (5) Remove the spring clip and clevis pin securing the equalising yoke to the link, withdraw the yoke from the link and collect the spacers. Note the position of the spacers.
- (6) Lift the cable abutment off the studs, remove the cable ties on the engine rear plate and remove the cables from the vehicle.
- (7) Remove the spring clips and clevis pins attaching the cables to the equalising yoke.
- (8) Loosen the clevis locknut and unscrew the clevis and locknut from the cables.
- (9) Remove the circlip from the cable outer cover and remove the cable from the abutment.



B Installation Procedure

NOTES:

- 1 *Ensure absolute cleanliness of work area and tools.*
- 2 *Observe correct use of tools.*

- (1) Fit handbrake cables to the abutment and secure with circlips.
- (2) Fit the locknut and screw clevis as far as possible onto the cable. Torque tighten locknut 4 to 4.5Nm (3 to 3.5ftlb).
- (3) Fit the heatshields to approximately 50mm (1.96in) from the cable abutment.
- (4) Fit the firesleeves over the heatshields and overlap equally at each end.
- (5) Apply high performance adhesive around the ends of the firesleeves and locate the heatshrink ends half covering the cable and firesleeves.
- (6) Heat the heatshrink evenly using a hot air gun until heatshrink grips the cable/firesleeve.
- (7) Fit clevis ends to the equalising yoke and secure with clevis pins and spring clips.
- (8) Fit cable abutment over studs.
- (9) Locate the yoke and spacers in the link and secure using clevis pin and spring clip.
- (10) Locate the spacer between the link and the vehicle body, fit the cover plate and boost sensor (LHD only) and secure with washers, nuts and bolt.
- (11) Torque the nuts 30 to 34Nm (22 to 25ftlb) and the bolt 20 to 23Nm (15 to 17ftlb).
- (12) Inspect the spring, replace if unserviceable.
- (13) Fit spring and washer to handbrake cable, attach handbrake cable to brake caliper and secure the nut.
- (14) Secure the cables to the bridge plate 'P' clips.
- (15) Set and adjust the handbrake mechanism. (Refer to Section 6.13).
- (16) Fit the wheels. (Refer to Road Wheels, Section 4).

PRIMARY CABLE

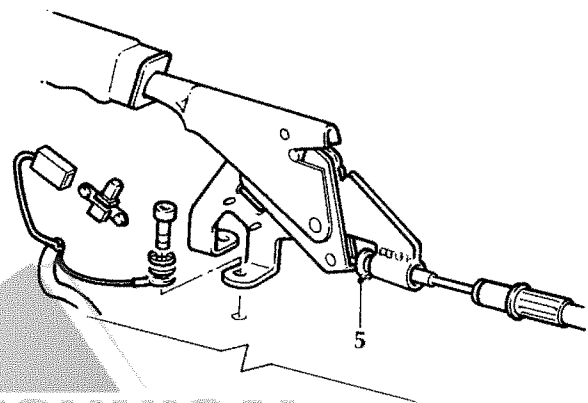
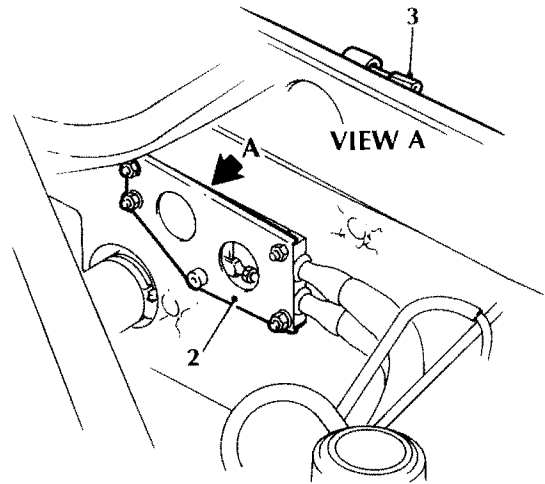
A Removal Procedure

- (1) Remove the handbrake trimmed cover, backboard and RH corner trim panel. (Refer to Section 2).
- (2) Remove the handbrake mechanism cover plate and booster sensor (LHD only).

NOTE:

When removing the cover plate take care not to lose spacer from behind lever.

- (3) Remove the sleeved nut securing the primary cable to the trunnion in the lever.
- (4) Remove the circlip holding the cable into the abutment and withdraw the cable from the abutment.
- (5) At the handbrake lever, remove the circlip securing the cable outer to the lever and unscrew the cable inner from the trunnion.
- (6) Pull the cable through the engine bulkhead and out of the vehicle.



B Installation Procedure

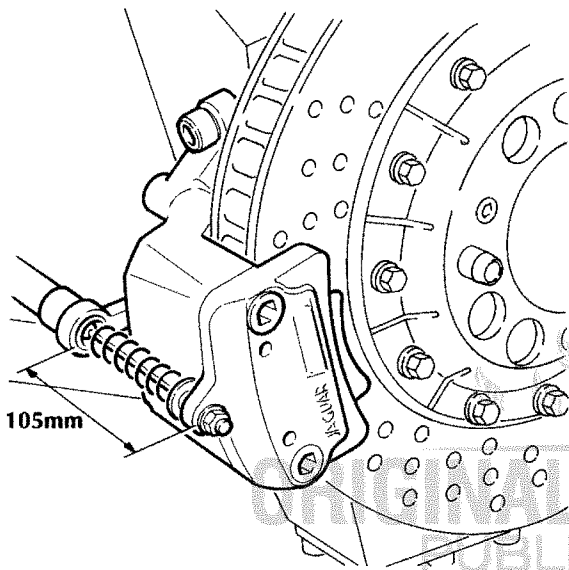
- (1) Route the cable through the engine bulkhead.
- (2) With the handbrake lever fully down, screw the cable inner fully into the trunnion, leaving a maximum of one thread protruding from the front side.
- (3) Fit the circlip securing the cable outer to the handbrake lever.
- (4) Fit the cable through the abutment in the engine compartment and secure with a circlip.
- (5) Fit the cable through the link trunnion and fit the sleeved nut.
- (6) Adjust the nut to remove all slack from all cables.
- (7) Locate the spacer between the link and the body and fit the cover plate and boost sensor (LHD only). Secure with washers, nuts and bolt.
- (8) Torque the nuts 30 to 34Nm (22 to 25ftlb) and the bolt 20 to 23Nm (15 to 17ftlb).
- (9) Set and adjust the handbrake mechanism. (Refer to Section 6.13).
- (10) Fit the RH corner trimmed panel, backboard and handbrake cover.

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13 HANDBRAKE SETTING AND ADJUSTMENT

A Procedure

- (1) Remove the rear wheels. (Refer to Road Wheels, Section 4).
- (2) Remove the handbrake lever trimmed cover. (Refer to Section 2).
- (3) In the engine compartment, remove the handbrake mechanism cover plate.
- (4) Remove the spring clip and clevis pin attaching the equalising yoke to the link. Collect yoke spacers and note their position.
- (5) Turn the adjuster screws on the handbrake caliper until the pads contact the disc but the disc can still be easily rotated by hand.
- (6) At the rear calipers, adjust the inner cable nut to achieve an overall dimension of 105mm (4.13in) between the actuating lever and the caliper when the other ends of the inner cable are fully retracted into the outer cables.
- (7) Locate the spacers and fit the yoke to the link, secure with clevis pin and spring clip.
- (8) Loosely fit the pivot bolt (without the cover plate) through the link into the vehicle body.
- (9) Ensure the handbrake lever is in the off position and check the primary cable is fully screwed into the trunnion with a maximum of one thread protruding from the front side.
- (10) Adjust the sleeved nut on the primary cable at the link to remove all slack from all cables. Ensure that the discs still rotate easily by hand.
- (11) Remove the pivot bolt, fit the cover plate and torque tighten the cover plate nuts 30 to 34Nm (22 to 25ftlb) and the bolt 20 to 23Nm (15 to 17ftlb).
- (12) Fit the handbrake lever trimmed cover.
- (13) Adjust the caliper adjusting screws to lock the discs against hand pressure then release the screws by 1/4 turn and lock the locknut.
- (14) The discs should not rotate under hand pressure when the handbrake lever is pulled up 2 'clicks' on the ratchet. Readjust after road test if necessary as in operation 13.
- (13) Fit the wheels. (Refer to Road Wheels, Section 4).



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DESCRIPTION

The transaxle is a five speed unit incorporating a viscous coupling limited slip differential, an oil pump and a heat exchanger. First and second gear are each fitted with a triple cone synchroniser.

Characteristics of the transaxle are listed below.

Weight (dry without clutch)	94kg (207lbs)
Oil capacity (less heat exchanger)	4 litres (0,88 gals)
Oil specification	Castrol XH21
Transmission ratios	
First gear	3.000:1
Second gear	1.955:1
Third gear	1.423:1
Fourth gear	1.097:1
Fifth gear	0.853:1
Reverse gear	2.579:1
Axle ratio	3.364:1
Oil pump gear	1.368:1

The main case provides fittings for bearings which support the input shaft, the lay shaft and the pinion shaft.

The forward end of the input shaft is splined to engage with the clutch assembly fitted to the engine. An input shaft bearing carrier is attached to the front section of the main case. The input shaft bearing carrier supports the clutch release bearing carrier and clutch release bearing which are operated by the clutch release lever. An oil seal and a bearing are fitted inside the bearing carrier. The oil seal prevents leakage from the main case, and the bearing provides the forward support for the input shaft.

The input shaft passes through the differential section of the transaxle casing and is splined into the lay shaft. The lay shaft is supported at the front by a bearing fitted into the main case, and at the rear by a bearing fitted into the sandwich plate. Each bearing is retained in place by a circlip. First, second and reverse input gears are an integral part of the lay shaft. Third, fourth and fifth input gears are splined onto the lay shaft. A locknut secures the lay shaft and input gears in the transaxle casing. The input gears mesh with the gear assemblies.

The pinion transfers drive to the differential and is supported by a roller bearing at the rear end and twin opposed taper roller bearings at the forward end. Oil channels are drilled through the pinion shaft into the oil gallery which is machined longitudinally through the centre of the pinion. The forward end of the pinion is plugged. Oil feed into the oil gallery is through a metering jet fitted to the rear cover.

The output gear assemblies and the synchromesh assemblies are fitted to the pinion shaft directly over the oil channels. The output gear assemblies rotate on bearings that are splined onto the pinion shaft. Gear selection is accomplished by the selector rod mounted above the pinion shaft.

The rear casing provides mounting for the speed sensor, reverse light switch and the return connection with filter for the oil pipe from the heat exchanger. Oil to the heat exchanger is tapped from the main case. The sump is fitted with flaps and a temperature sensor. The flaps reduce oil surge in the sump.

The limited slip differential is housed inside the main case and consists of a crown wheel, side gears, four bevels a viscous coupling and 1/4 shafts. The crown wheel meshes with the pinion.

The crown wheel is bolted to the differential case assembly. The differential case assembly houses the bevels and the side gears. The RH side gear is splined onto the drive sleeve, the LH side gear is splined onto the LH drive shaft. The drive sleeve is splined into the viscous coupling which attaches the RH drive shaft. The RH and LH drive shafts are supported in bearings. The bearings are fitted in the RH and LH side covers.

The gear lever unit assembly consists of the gear lever and a shift trunnion mounted in a machined gear shift housing. The shift trunnion connects to an adjuster tube assembly.

The gear lever pivots about a bearing and is mounted in the shift trunnion on a bolt. The bearing permits lateral movement of 30 degrees. A tie rod assembly is fitted between the lower end of the gear lever and the gear shift housing.

The adjuster tube assembly consists of an internally threaded barrel fitted with a flanged threaded rod and a pin. This is used for adjusting the overall length of the front and rear tube assemblies. There is a cam adjustment to set the vertical position of the gear lever. The rear tube is supported in a plastic bush at the front of the engine and is connected to the selector rail at the transaxle end. The selector rail is supported by a linear bearing in the sandwich plate, and a bush in the rear casing, and fitted with a selector finger which engages the selector rod. An interlock ensures movement of only one selector rail at any time.

LUBRICATION

The oil is drawn from the sump by the oil pump and directed to the heat exchanger. From the heat exchanger the oil is returned via the filter in the rear casing. The oil passes through the filter and is then directed to the oil gallery in the pinion, a jet to the pinion mesh and a spray bar. The drillings in the pinion direct the oil to the bearings which support the output gear assemblies. The spray bar sprays lubricating oil onto the gears. The oil then returns to the sump.

The lay shaft and the differential crown wheel are partially immersed in the sump oil. Rotation of these components splash feeds oil to the drive gear assemblies, the pinion and the differential component parts. The CWP mesh is lubricated by its own jet.

OPERATION

Engine output drives the transaxle input shaft. The transaxle input shaft is splined into the lay shaft. The layshaft gears mesh with the associated drive gear assemblies. The drive gear assemblies transmit rotational force to the pinion when a gear is selected. The gear selectors are attached to the selector rod which is operated by the selector finger fitted onto the selector rail. The pinion drives the viscous coupling limited slip differential which transmits the drive force to the road wheels.

When reverse gear is selected the selector rail moves rearward and depresses the plunger of the reverse light switch. This completes the electrical circuit and the reverse light illuminates.



1 INTRODUCTION

This section details the procedures for removal and installation of the transaxle and transaxle components. Removal of components or disassembly instructions not covered in this section require the transaxle or transaxle component to be returned to JaguarSport.

2 TRANSAXLE

A Removal Procedure

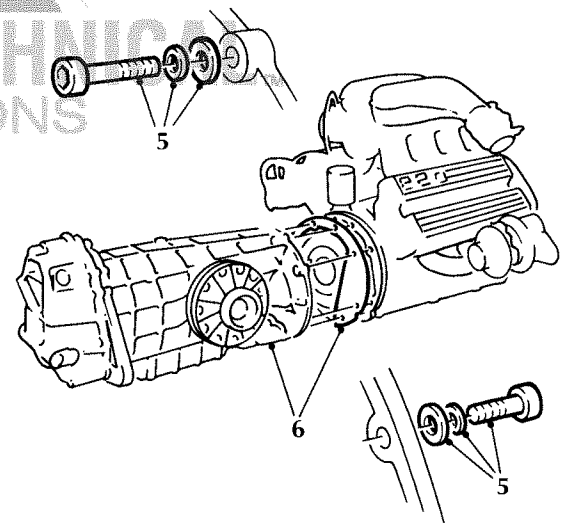
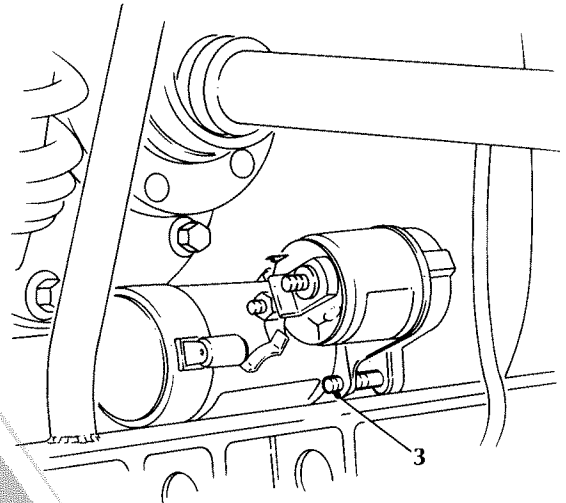
CAUTION:

When removing the transaxle, support the engine to prevent it tipping

NOTES:

- 1 Ensure absolute cleanliness of work area and tools.
- 2 Observe correct use of tools.
- 3 If the transaxle has developed a fault during service take a sample from the oil being drained for analysis by JaguarSport.

- (1) Remove the engine and transaxle unit from the vehicle. (Refer to Section 3)
- (2) Remove the ridge tent assembly. (Refer to Section 4).
- (3) Remove the bolts securing the starter motor and remove the starter from the transaxle.
- (4) Remove transaxle temperature sensor and drain the oil from the transaxle.
- (5) Remove bolts securing transaxle to engine.
- (6) Remove transaxle from engine and collect spacer plate.



3 PREPARATION FOR TRANSIT

Special Tools

The following special tools will be required:

Hoisting equipment

Webbing sling.

Materials

The following materials will be required:

Transit Container

Cushioning material

Desiccant

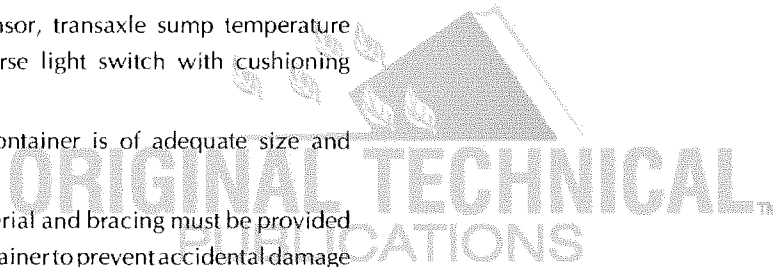
Identification labels.

- (11) Attach hoisting equipment to webbing sling and take the weight.
- (12) Inspect the webbing sling routing to ensure no external component is being damaged.

WARNING:

DO NOT STAND BENEATH THE TRANSAXLE DURING HOISTING OPERATIONS.

- (13) Ensure the transaxle is balanced on the hoist and lift the transaxle into the transit container and remove webbing sling.
 - (14) Ensure the transaxle is stable in the transit container. Fit more bracing if required.
 - (15) Place bags of desiccant around the transaxle and fit lid to transit container.
- (1) Remove the temperature sensor and drain transaxle oil.
 - (2) Refit the temperature sensor.
 - (3) Fit blanking plugs to all openings on the transaxle.
 - (4) Protect end of selector rail with cushioning material.
 - (5) Protect RPM sensor, transaxle sump temperature sensor and reverse light switch with cushioning material.
 - (6) Ensure transit container is of adequate size and strength.
 - (7) Cushioning material and bracing must be provided in the transit container to prevent accidental damage to the transaxle.
 - (8) An identification label must be affixed to the transaxle, and an additional identification label must be affixed to the outside of the transit container.
 - (9) The identification labels must contain all relevant details, including date of removal from vehicle and date of packaging.
 - (10) Fit the webbing sling around the transaxle to distribute the weight as evenly as possible.



4 UNPACKING AND PREPARATION FOR INSTALLATION

WARNING:

- 1 **DO NOT STAND BENEATH THE TRANSAXLE DURING HOISTING OPERATIONS.**
- 2 **WEAR SAFETY GOGGLES WHEN USING AN AIR JET AND WARN OTHER PERSONNEL TO KEEP CLEAR OF THE AREA.**

CAUTIONS:

- 1 **Ensure the transaxle is balanced in the webbing sling.**
- 2 **Do not remove the blanking plugs fitted to the transaxle unless to fit a component part or to connect a service.**

- (1) Check the transit container for evidence of external damage caused during transit.
- (2) Remove lid and check the details on the external identification label are the same as the details on the identification label attached to the transaxle.
- (3) Check the serial number on the identification label is the same as the serial number on the transaxle.
- (4) Fit the webbing sling around the transaxle to distribute the weight as evenly as possible.
- (5) Attach hoisting equipment to webbing sling and take the weight.
- (6) Remove bracing from transit container as required.
- (7) Hoist the transaxle from the transit container onto a suitable clean surface, padded to prevent damage to the transaxle.
- (8) Remove identification label and any protective cushioning material from the transaxle.
- (9) Use compressed air at approx. 193kPa (28 psi) to remove small particles of cushioning material from inaccessible areas.
- (10) Prepare the transaxle for installation in accordance with the following instructions.

5 INSTALLATION PROCEDURE

Materials

The following material will be required:

Anti-seize compound.

- (1) Fit spacer plate to rear of engine and check correct alignment of spacer plate bolt holes with corresponding holes in engine block and transaxle.
- (2) Apply anti-seize compound to the splines of the transaxle input shaft and the rubbing surfaces of the clutch release housing and sleeve.
- (3) Fit the transaxle to engine. Support transaxle and secure with bolts and washers finger tight only.
- (4) Torque tighten bolts to 54Nm (40ftlb) in a diagonal sequence.
- (5) Fit starter motor and secure with bolts and washers. Torque bolts 20 to 23Nm 15 to 17ftlb).
- (6) Fit the ridge tent assembly. (Refer to Section 4).
- (7) Fit the engine and transaxle unit to the vehicle. (Refer to Section 3).
- (8) Road test the vehicle and ensure correct function of the transaxle.

6 OUTPUT SHAFT OIL SEAL AND BEARING (RH)

A Removal Procedure

Special Tools

The following special tools will be required:

Torque wrench 55Nm (40ftlb).

Materials

The following materials will be required:

Approved cleaning solvent

Barrier cream.

WARNINGS:

- 1 **SOLVENTS ARE TOXIC AND MUST ONLY BE USED IN WELL VENTILATED WORK AREAS. AVOID INHALING SOLVENT FUMES.**
- 2 **USE BARRIER CREAM AND WEAR PROTECTIVE CLOTHING WHEN WASHING COMPONENT PARTS IN CLEANING SOLVENT.**

NOTES:

- 1 It is not necessary to drain the transaxle in order to change the output shaft oil seal.
- 2 LH and RH are as viewed from the rear of the transaxle.
- 3 Ensure absolute cleanliness of work area and tools.
- 4 Observe correct use of tools.

NOTE

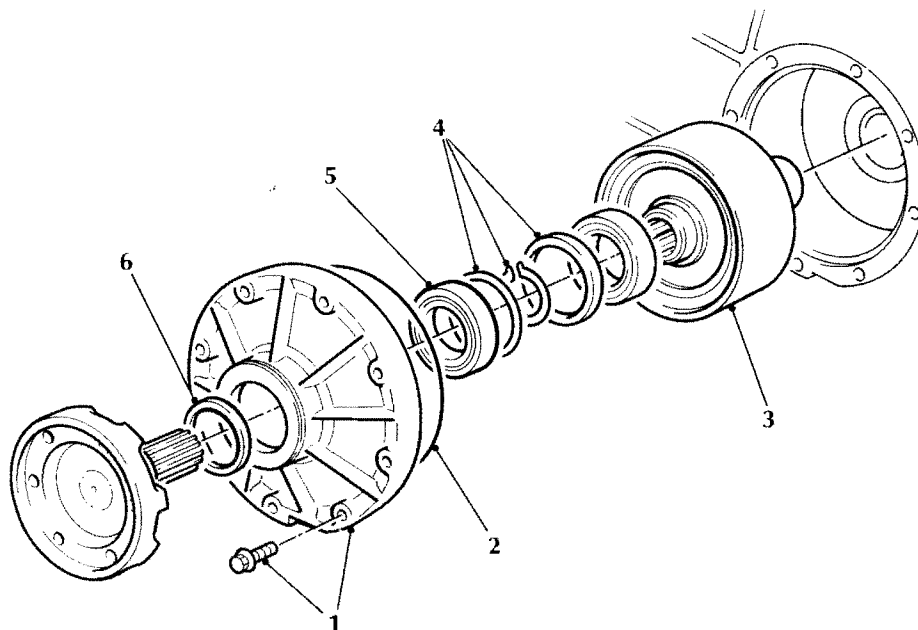
The following procedure should be carried out with the transaxle removed from the vehicle

- (1) Remove bolts and remove side cover with drive shaft.
- (2) Remove and discard side cover 'O' ring.
- (3) Remove viscous coupling.
- (4) Remove spacer and 2 circlips, and remove drive shaft from side cover.

CAUTION:

To remove the bearing from the side cover will require the application of heat. To prevent damage to the side cover, apply even heat around the bearing housing in the side cover sufficiently to allow the bearing to be removed without the use of force.

- (5) Remove the bearing from the side cover.
- (6) Remove and discard output shaft oil seal.



B Installation Procedure

NOTES:

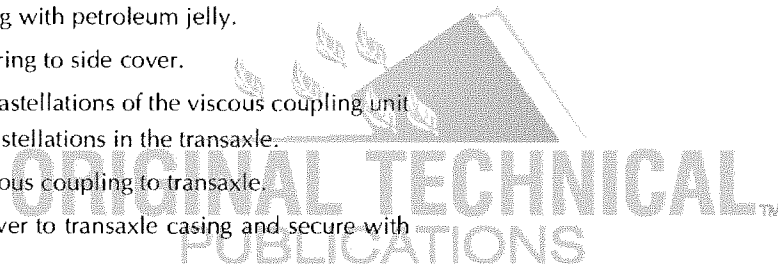
- 1 Wash all other disassembled component parts in a suitable cleaning solvent to remove all traces of sealant, oil and grease.
- 2 Inspect the drive shaft to ensure there is no damage to the splines or the machined groove for the circlip.

- (1) Press the new output shaft oil seal into position flush with the face of the side cover.
- (2) Evenly warm the side cover to approximately 70°C.
- (3) Press bearing into side cover and fit circlip.
- (4) Fit drive shaft through bearing in side cover and secure with circlip.
- (5) Fit spacer.

CAUTION:

Ensure the new O ring is seated correctly and is not twisted when fitted to the machined groove in the side cover.

- (6) Treat O ring with petroleum jelly.
- (7) Fit new O ring to side cover.
- (8) Align the castellations of the viscous coupling unit with the castellations in the transaxle.
- (9) Fit the viscous coupling to transaxle.
- (10) Fit side cover to transaxle casing and secure with bolts.
- (11) Torque tighten bolts to 55Nm (40ftlb).
- (12) Check and correct the transaxle oil level.



7 OUTPUT SHAFT OIL SEAL AND BEARING (LH)

A Removal Procedure

NOTES:

- 1 LH and RH are as viewed from the rear of the transaxle.
- 2 Ensure absolute cleanliness of work area and tools.
- 3 Observe correct use of tools.

NOTE

The following procedure should be carried out with the transaxle removed from the vehicle

- (1) Remove the oil temperature sensor and drain the oil from the transaxle.
- (2) Remove bolts and remove side cover with drive shaft.

(3) Remove the circlip.

(3a) Remove the taper bearing outer race.

CAUTION:

To remove the bearing and taper bearing outer race from the side cover will require the application of heat. To prevent damage to the side cover, apply the heat evenly around the bearing housing in the side cover sufficiently to allow the bearing to be removed without the use of force.

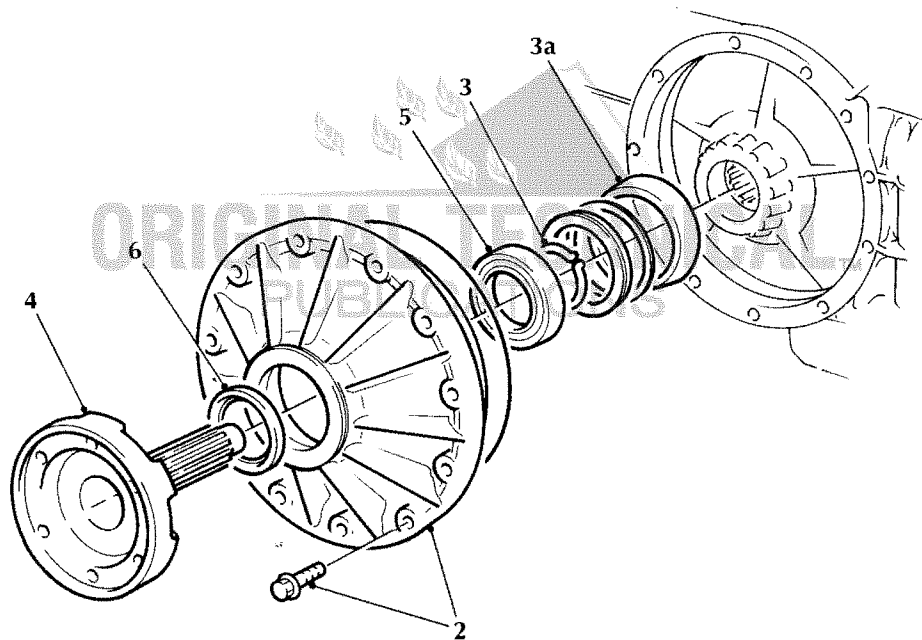
(4) Remove the drive shaft from side cover.

NOTE:

There are shims between the spacer and the differential case taper roller bearing. Take care not to disturb or lose the shims.

(5) Remove bearing from side cover.

(6) Remove and discard output shaft oil seal.



B Installation Procedure (LH)

WARNINGS:

- 1 ***SOLVENTS ARE TOXIC AND MUST ONLY BE USED IN WELL VENTILATED WORK AREAS. AVOID INHALING SOLVENT FUMES.***
- 2 ***USE BARRIER CREAM AND WEAR PROTECTIVE CLOTHING WHEN WASHING COMPONENT PARTS IN CLEANING SOLVENT.***

CAUTION:

The bearings are lubricated with oil during manufacture. Do not immerse the bearings in cleaning solvent.

NOTES:

- 1 *Wash all other disassembled component parts in a suitable cleaning solvent to remove all traces of sealant, oil and grease.*
- 2 *Inspect the drive shaft to ensure there is no damage to the splines or the machined groove for the circlip.*

Special Tools

The following special tools will be required:

Torque wrench 55Nm (40ftlb)

Materials

The following materials will be required:

Approved cleaning solvent

Barrier cream

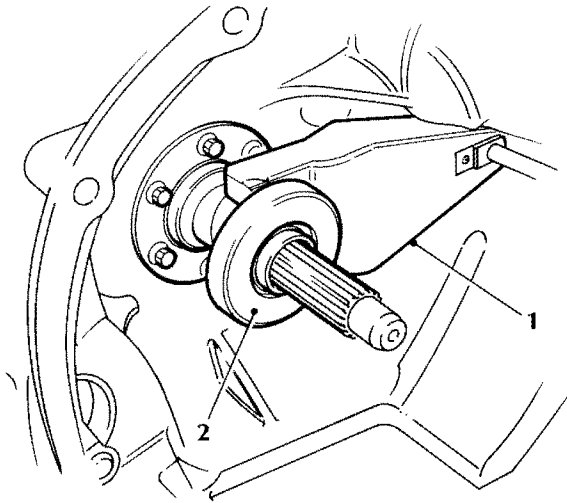
Petroleum Jelly

- (1) Press new output shaft oil seal flush with the face of the side cover.
- (2) Evenly warm the side cover to approximately 70°C.
- (3) Fit bearing and taper bearing outer race into side cover, replacing the spacer and shims.
- (4) Fit drive shaft through bearing in side cover and secure with circlip.

CAUTIONS:

- 1 ***Ensure the shims are not damaged and are in the correct position.***
 - 2 ***Ensure the new O ring seated correctly and is not twisted when fitted to the machined groove in the side cover.***
 - 3 ***The taper bearing may not be renewed without adjusting the shim pack.***
- (5) Treat new O ring with Petroleum Jelly and fit to side cover.

- (6) Fit side cover to transaxle casing and secure with bolts.
- (7) Torque tighten bolts to 55Nm (40ftlb).
- (8) Replenish the transaxle oil.



8 CLUTCH ACTUATION COMPONENTS

A Removal Procedure

NOTES:

- 1 Ensure absolute cleanliness of work area and tools.
- 2 Observe correct use of tools.

NOTE

The following procedure should be carried out with the transaxle removed from the vehicle

- (1) Remove clutch release lever.
- (2) Remove clutch release bearing and bearing carrier.

B Installation Procedure

NOTES:

- 1 Ensure absolute cleanliness of work area and tools.
- 2 Observe correct use of tools.

WARNING:

WEAR PROTECTIVE GLOVES WHEN HANDLING COPPER SLIP GREASE.

Materials

The following materials will be required:

Copper slip grease

Disposable protective gloves.

- (1) Apply an even coat of Easyrun 100 grease to the mating surface of transaxle input shaft, bearing carrier, and clutch release bearing carrier.
- (2) Fit clutch release bearing and bearing carrier on transaxle input shaft.
- (3) Apply an even coat of Easyrun 100 grease to the release lever pivot and fork pads.
- (4) Fit clutch release lever.

9 INPUT SHAFT OIL SEAL AND BEARING

A Removal Procedure

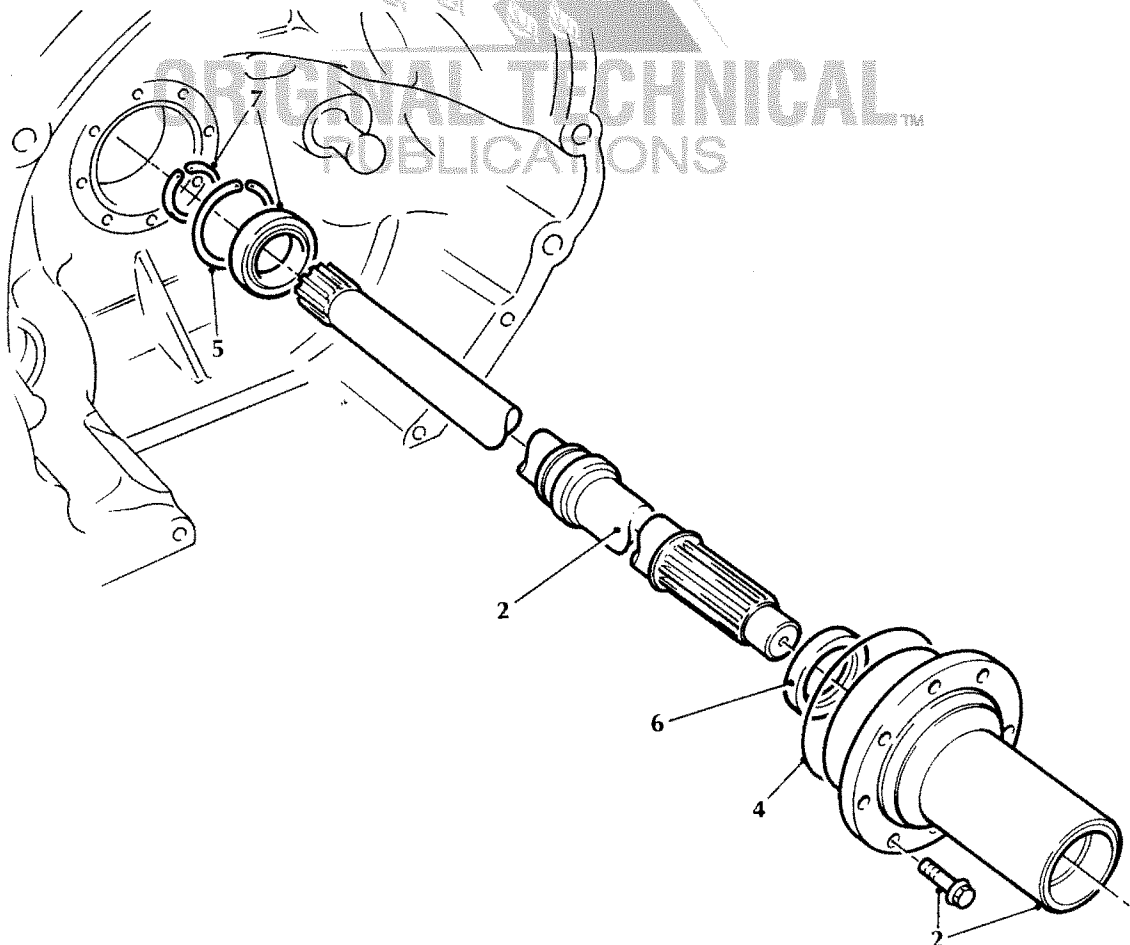
NOTES:

- 1 If transaxle oil has not been drained raise front end of transaxle a few inches to prevent oil spillage when bearing support is removed.
- 2 Ensure absolute cleanliness of work area and tools.
- 3 Observe correct use of tools.

NOTE

The following procedure should be carried out with the transaxle removed from the vehicle

- (1) Remove clutch actuation components.
- (2) Remove screws and remove bearing support and input shaft from transaxle.
- (3) Fit blanking plug to the transaxle to prevent ingress of dust.
- (4) Remove O ring seal from bearing support.
- (5) Remove circlip and remove bearing support.
- (6) Remove input shaft oil seal from bearing support.
- (7) Remove circlip and remove bearing from input shaft.



B Installation Procedure

WARNINGS:

- 1 SOLVENTS ARE TOXIC AND MUST ONLY BE USED IN WELL VENTILATED WORK AREAS. AVOID INHALING SOLVENT FUMES.**
- 2 USE BARRIER CREAM AND WEAR PROTECTIVE CLOTHING WHEN WASHING COMPONENT PARTS IN CLEANING SOLVENT.**

CAUTION:

The bearings are lubricated with oil during manufacture. Do not immerse bearings in cleaning solvents.

NOTES:

- 1 Wash all other disassembled component parts in a suitable cleaning solvent to remove all traces of sealant and grease.*
- 2 Inspect input drive shaft to ensure there is no damage to the spines.*

Special Tools

The following special tools will be required:

Torque wrench 10Nm (7ftlb)

Materials

The following materials will be required:

Approved cleaning solvent

Barrier cream

NOTES:

- 1 Ensure absolute cleanliness of work area and tools.*
 - 2 Observe correct use of tools.*
-
- (1) Fit bearing to input shaft and fit circlip.
 - (2) Press new input shaft oil seal into position in the bearing support.
 - (3) Fit bearing support over input drive shaft bearing and secure with circlip.
 - (4) Fit new 'O' ring seal to the bearing support.

CAUTION:

Ensure the O ring seal is not twisted when fitted to the bearing support.

- (5) Remove blanking plug and fit input shaft into transaxle and secure bearing support with screws.
- (6) Torque load screws 10Nm (7ftlb).
- (7) Fit clutch actuation components.
- (8) Lower front end of transaxle.

10 SELECTOR RAIL OIL SEAL

A Removal Procedure

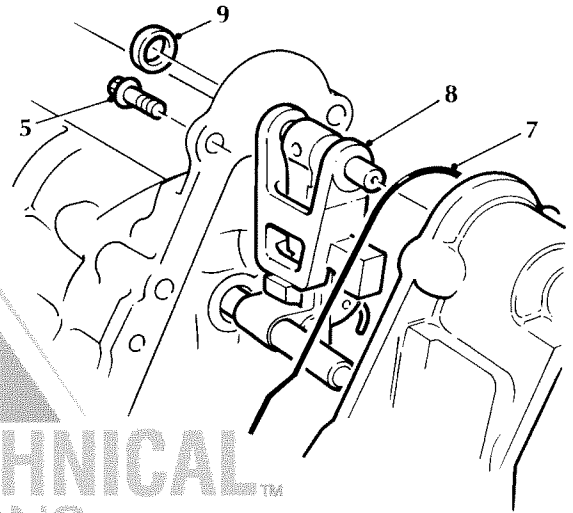
NOTES:

- 1 The gear change selector rod must be disconnected from selector rail before disassembly begins.
- 2 Ensure absolute cleanliness of work area and tools.
- 3 Observe correct use of tools.

NOTE

The following procedure deals with removal while the transaxle is in-situ. If the transaxle has been removed certain procedures may be disregarded.

- (1) Position the vehicle on lifting apparatus. (Refer to Introduction, Lifting and Jacking)
- (2) Raise the vehicle to a comfortable working height and remove the bumper and venturi panels. (Refer to Section 2).
- (3) Remove the exhaust silencer. (Refer to Section 9).
- (4) Drain transaxle oil into a suitable container.
- (5) Remove bolts securing transaxle rear casing. Note the position of the casing bolts.
- (6) Remove rear casing.
- (7) Remove and discard O ring cord seal from rear transaxle casing.
- (8) Remove interlock and selector rail.
- (9) Remove and discard selector rail oil seal.



B Installation Procedure

Special Tools

The following special tools will be required:

Torque wrench 25Nm (18.4ftlb)

Materials

The following materials will be required:

Lint free cloth

Oil, Castrol XH21, 4 litres (0,88 gal).

NOTES:

- 1 *Ensure absolute cleanliness of work area and tools.*
- 2 *Observe correct use of tools.*

- (1) Fit new seal to transaxle oil temperature sensor and refit the sensor.
- (2) Fit new selector rail oil seal to transaxle sandwich plate.
- (3) Remove all traces of oil from mating surfaces of transaxle rear casing and transaxle casing using clean lint free cloth.

CAUTION:

Ensure the new O ring seal is seated correctly in the machined groove in the rear cover.

- (4) Fit new O ring seal to groove in transaxle rear casing.
- (5) Fit interlock and selector rail.

NOTE

Check the selector finger and interlock attached to the selector rail are engaged in the machined slots in the gear selector rods. The selector finger must be in the slot of 3rd and 4th gear selector rod.

- (6) Fit transaxle rear casing to transaxle casing and secure with bolts.
- (7) Torque load bolts to 25Nm (18.4ftlb).
- (8) Replenish transaxle oil as necessary with Castrol XH21 oil.
- (9) Fit the exhaust silencer. (Refer to Section 9).
- (10) Fit the bumper and venturi panels. (Refer to Section 2).
- (11) Lower the vehicle and remove from the lifting apparatus.

11 HEAT EXCHANGER RETURN FITTING 'O' RING SEAL

A Removal Procedure

Materials

The following materials will be required:

Approved solvent

Stiff Bristle Brush.

NOTES:

- 1 Ensure absolute cleanliness of work area and tools.
- 2 Observe correct use of tools.

NOTE

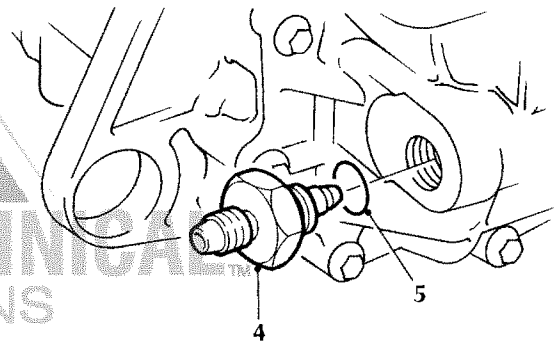
The following procedure deals with the removal while the transaxle is in-situ. If the transaxle has been removed certain operations may be disregarded.

- (1) Position the vehicle on lifting apparatus. (Refer to Introduction, Lifting and Jacking).
- (2) Raise the vehicle to a comfortable working height and remove the rear bumper and venturi panels. (Refer to Section 2).
- (3) Clean the area around the lubrication fitting with stiff bristle brush and approved solvent.
- (4) Remove adapter for attaching heat exchanger return pipe.

NOTE:

Fit blanking plugs while adapter is removed from transaxle casing.

- (5) Remove and discard O ring seal.



B Installation Procedure

NOTES:

- 1 Ensure absolute cleanliness of work area and tools.
- 2 Observe correct use of tools.

CAUTION:

Ensure the O ring seal is seated correctly on the adaptor.

- (1) Fit new O ring seal to adaptor.
- (2) Remove the blanking plugs and fit adaptor to transaxle casing. Tighten the adaptor to 30Nm (22ftlb).
- (3) Fit the rear bumper and venturi panels. (Refer to Section 2).
- (4) Lower the vehicle and remove from the lifting apparatus.
- (5) Check transaxle oil level and replenish as necessary.

12 HEAT EXCHANGER INLET BANJO FITTING CRUSH WASHERS

A Removal Procedure

Materials

The following materials will be required:

Approved solvent

Stiff Bristle Brush.

NOTES:

- 1 Ensure absolute cleanliness of work area and tools.
- 2 Observe correct use of tools.

NOTE

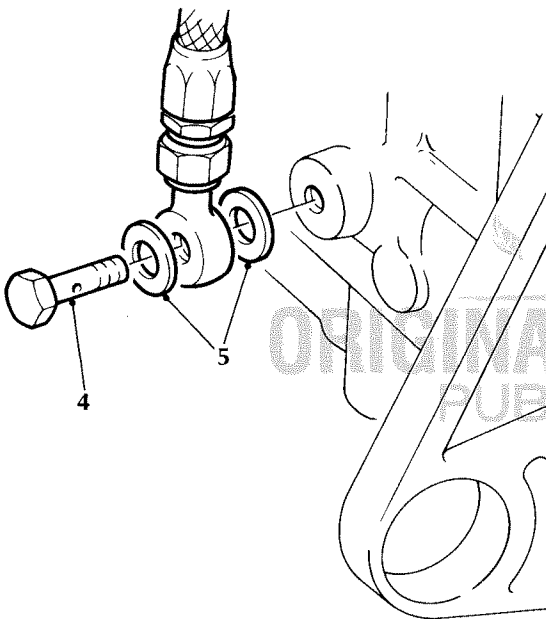
The following procedure deals with removal while the transaxle is in-situ. If the transaxle has been removed certain operations may be disregarded.

- (1) Position the vehicle on lifting apparatus. (Refer to Introduction, Lifting and Jacking).
- (2) Raise the vehicle to a comfortable working height and remove the rear bumper and venturi panels. (Refer to Section 2).
- (3) Clean the area around the banjo fitting with stiff bristle brush and approved solvent.
- (4) Remove bolt attaching banjo to transaxle casing.

NOTE:

Fit blanking plugs while banjo is removed from transaxle casing.

- (5) Remove and discard crush washers.



B Installation Procedure

NOTES:

- 1 Ensure absolute cleanliness of work area and tools.
- 2 Observe correct use of tools.

- (1) Fit the banjo and new crush washers to the bolt.
- (2) Remove the blanking plugs and fit banjo to transaxle casing and tighten bolt to 24 to 27Nm (18 to 20ftlb).
- (3) Fit the rear bumper and venturi panels. (Refer to Section 2).
- (4) Lower the vehicle and remove from the lifting apparatus.
- (5) Check the transaxle oil level and replenish as necessary.

13 OIL FILTER

A Removal Procedure

Special Tools

The following special tools will be required:

Compressed air jet at 193kPa (28 psi) max.

Materials

The following materials will be required:

Approved cleaning solvent

Lint free cloth.

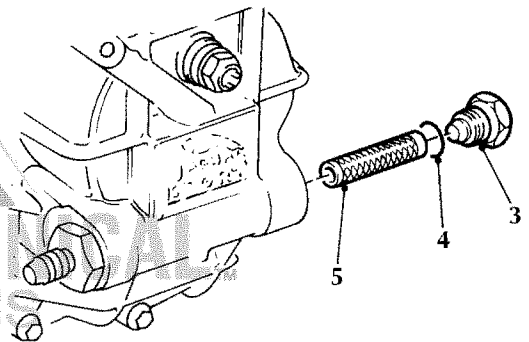
NOTES:

- 1 Ensure absolute cleanliness of work area and tools.
- 2 Observe correct use of tools.

NOTE

The following procedure deals with removal while the transaxle is in-situ. If the transaxle has been removed certain operations may be disregarded.

- (1) Position the vehicle on lifting apparatus. (Refer to Introduction, Lifting and Jacking).
- (2) Raise the vehicle to a comfortable working height and remove the rear bumper and venturi panels. (Refer to Section 2).
- (3) Remove the filter end cap.
- (4) Remove and discard O ring from end cap.
- (5) Pull the filter body assembly from the housing.
- (6) Fit a blanking plug to the filter housing.



B Cleaning

WARNING:

SOLVENTS ARE TOXIC AND MUST ONLY BE USED IN WELL VENTILATED WORK AREAS. AVOID INHALING THE SOLVENT FUMES AND WEAR PROTECTIVE CLOTHING.

- (1) Wash the filter body in approved cleaning solvent.

WARNING:

WHEN USING AN AIR JET WEAR GOGGLES AND WARN OTHER PERSONNEL TO KEEP CLEAR OF THE AREA.

- (2) Dry filter body with an air jet at approximately 193kPa (28 psi).
- (3) Ensure the filter gauze is free from damage.
- (4) Clean filter body assembly housing with clean lint free cloth moistened with approved cleaning solvent.

C Installation Procedure

Special Tools

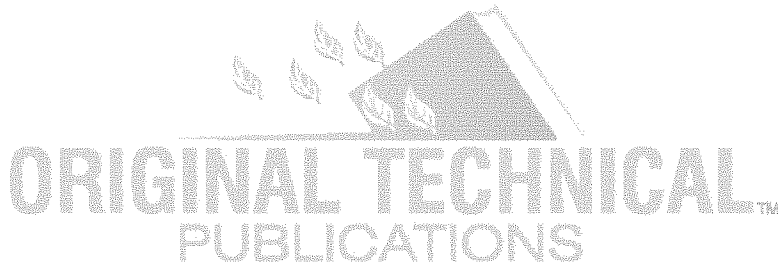
The following special tools will be required:

Torque wrench 30Nm (22ftlb)

NOTES:

- 1 *Ensure absolute cleanliness of work area and tools.*
- 2 *Observe correct use of tools.*

- (1) Fit new O ring seal to the end cap.
- (2) Remove the blanking plug and fit filter body assembly in the housing.
- (3) Fit cap and torque load to 30Nm (22ftlb).
- (4) Fit the rear bumper and venturi panels. (Refer to Section 2).
- (5) Lower the vehicle and remove from the lifting apparatus.
- (6) Check the transaxle oil level and replenish as necessary.



14 ROAD SPEED SENSOR

A Removal Procedure

Materials

The following materials will be required.

Loctite 542

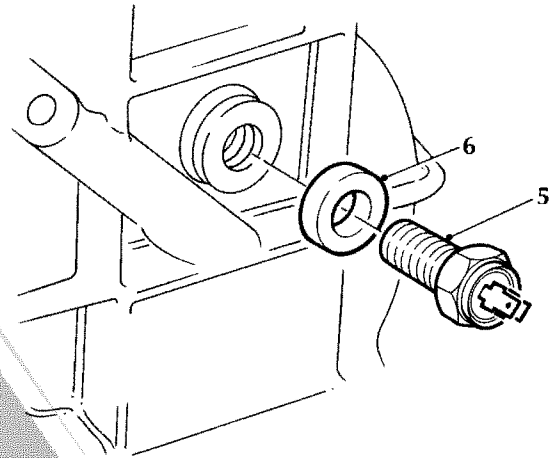
NOTES:

- 1 Ensure absolute cleanliness of work area and tools.
- 2 Observe correct use of tools.

NOTE

The following procedure deals with the removal with the transaxle in-situ. If the transaxle has been removed certain operations may be disregarded.

- (1) Position the vehicle on lifting apparatus. (Refer to Introduction, Lifting and Jacking).
- (2) Disconnect the battery.
- (3) Raise the vehicle to a comfortable working height and remove the rear bumper. (Refer to Section 2).
- (4) Remove the speed sensor electrical plug.
- (5) Remove speed sensor from transaxle rear casing.
- (6) Remove the shims and make a note of the pack thickness.



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B Installation Procedure

NOTES:

- 1 Ensure absolute cleanliness of work area and tools.
- 2 Observe correct use of tools.

- (1) Fit shims to speed sensor.
- (2) Treat threaded section of speed sensor with Loctite 542 and fit speed sensor to transaxle rear casing.
- (3) Fit the sensor electrical plug.
- (4) Fit the rear bumper. (Refer to Section 2).
- (5) Lower the vehicle and remove from the lifting apparatus.
- (6) Connect the battery.

15 REVERSE LIGHT SWITCH

A Removal Procedure

NOTES:

- 1 Ensure absolute cleanliness of work area and tools.
- 2 Observe correct use of tools.

NOTE

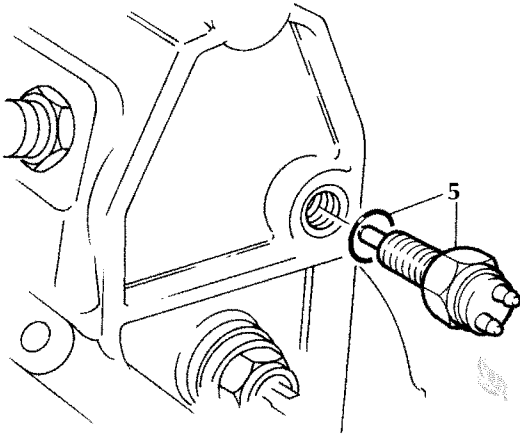
The following procedure deals with the removal with the transaxle in-situ. If the transaxle has been removed certain operations may be disregarded.

- (1) Position the vehicle on lifting apparatus. (Refer to, Lifting and Jacking, Section 1).
- (2) Disconnect the battery.
- (3) Raise the vehicle to a comfortable working height and remove the rear bumper. (Refer to Section 2).
- (4) Disconnect the electrical cables to the switch.

NOTE:

Clean the area around the reverse light switch before removal.

- (5) Remove the switch and discard the O ring seal.



B Installation Procedure

NOTES:

- 1 Ensure absolute cleanliness of work area and tools.
- 2 Observe correct use of tools.

- (1) Fit new O ring seal.
- (2) Fit reverse light switch.
- (3) Fit electrical connections to the switch.
- (4) Fit the rear bumper. (Refer to Section 2).
- (5) Lower the vehicle and remove from the lifting apparatus.
- (6) Connect the battery.

Testing

- (1) Switch ignition to ON position.
- (2) Select reverse gear.
- (3) Visually check reverse light is lit.
- (4) Select neutral gear and switch ignition to OFF position.

16 TRANSAXLE OIL TEMPERATURE SENSOR

A Removal Procedure

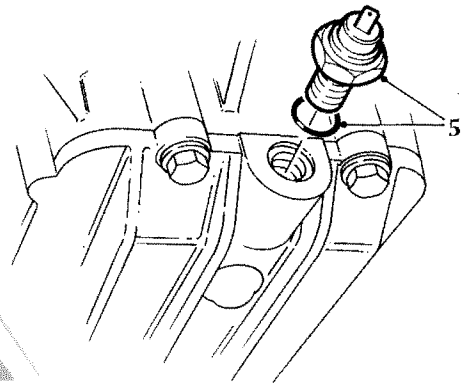
NOTES:

- 1 Ensure absolute cleanliness of work area and tools.
- 2 Observe correct use of tools.

NOTE

The following procedure deals with removal with the transaxle in-situ. If the transaxle has been removed certain operations may be disregarded.

- (1) Position the vehicle on lifting apparatus. (Refer to Introduction, Lifting and Jacking).
- (2) Disconnect the battery.
- (3) Raise the vehicle to a comfortable working height and remove the rear bumper. (Refer to Section 2).
- (4) Disconnect the electrical cables from the temperature sensor.
- (5) Remove temperature sensor and discard the O ring seal.
- (6) Allow oil to drain from sump.



B Installation Procedure

NOTES:

- 1 Ensure absolute cleanliness of work area and tools.
- 2 Observe correct use of tools.

- (1) Fit new O ring seal to the temperature sensor.
- (2) Fit the temperature sensor to the sump casing.
- (3) Fit the electrical cables to the sensor.
- (4) Fit the rear bumper. (Refer to Section 2).
- (5) Lower the vehicle and remove from the lifting apparatus.
- (6) Replenish the transaxle oil.
- (7) Road test the vehicle and ensure that all transaxle temperatures are correct.

17 SUMP O RING SEAL

A Removal Procedure

WARNING:

SOLVENTS ARE TOXIC AND MUST ONLY BE USED IN WELL VENTILATED WORK AREAS. AVOID INHALING THE SOLVENT FUMES AND WEAR PROTECTIVE CLOTHING.

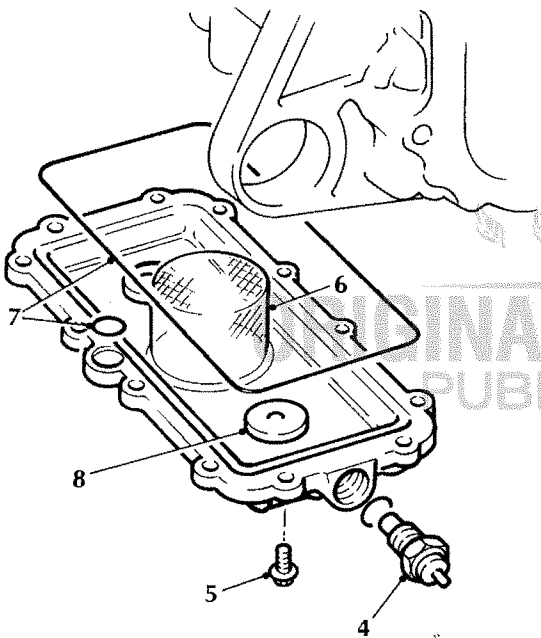
NOTES:

- 1 Ensure absolute cleanliness of work area and tools.
- 2 Observe correct use of tools.

NOTE

The following procedure deals with removal with the transaxle in-situ. If the transaxle has been removed certain operations may be disregarded.

- (1) Position the vehicle on lifting apparatus. (Refer to Introduction, Lifting and Jacking).
- (2) Disconnect the battery.
- (3) Raise the vehicle to a comfortable working height and remove the rear bumper and venturi panels. (Refer to Section 2).
- (4) Disconnect the electrical cables and remove temperature sensor; allow oil to drain from sump into a suitable container.
- (5) Remove bolts securing sump to the transaxle casing.
- (6) Remove sump and gauze filter.
- (7) Remove and discard sump seal and oil pressure relief valve seal.
- (8) Inspect magnets in sump and remove all traces of magnetically attracted particles.
- (9) Wash sump in approved solvent to remove all traces of sludge.
- (10) Wash the gauze filter in approved solvent and inspect filter for damage. Replace if unserviceable.



B Installation Procedure

Special Tools

The following special tools will be required:

Torque wrench 25Nm (18.4ftlb)

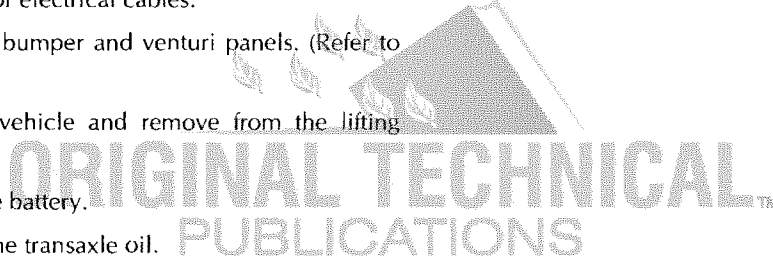
NOTES:

- 1 *Ensure absolute cleanliness of work area and tools.*
- 2 *Observe correct use of tools.*

CAUTION:

Ensure the O ring seals are not twisted when fitted.

- (1) Treat all 'O' ring seals with Petroleum Jelly.
- (2) Fit new oil pressure relief valve seal and sump seal.
- (3) Fit sump and gauze filter to transaxle casing and fit bolts.
- (4) Torque load the bolts to 25Nm (18.4ftlb) in a diagonal sequence to ensure even loading.
- (5) Fit a new oil temperature sensor 'O' ring seal and fit the sensor.
- (6) Fit the sensor electrical cables.
- (7) Fit the rear bumper and venturi panels. (Refer to Section 2).
- (8) Lower the vehicle and remove from the lifting apparatus.
- (9) Connect the battery.
- (10) Replenish the transaxle oil.
- (11) Road test the vehicle and ensure all transaxle functions and temperatures are correct.



18 CENTRALISING CARTRIDGE ASSEMBLY

A Removal Procedure

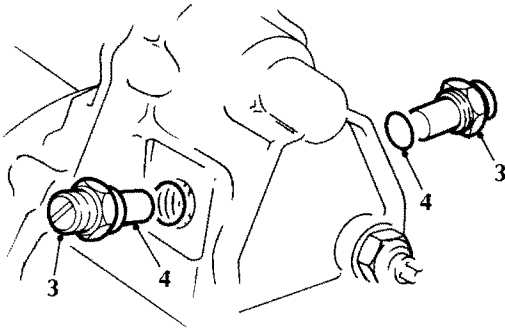
NOTES:

- 1 Ensure absolute cleanliness of work area and tools.
- 2 Observe correct use of tools.

NOTE

The following procedure deals with removal with the transaxle in-situ. If the transaxle has been removed certain operations may be disregarded.

- (1) Position the vehicle on lifting apparatus. (Refer to Introduction, Lifting and Jacking).
- (2) Raise the vehicle to a comfortable working height and remove the rear bumper and venturi panels. (Refer to Section 2).
- (3) Remove centralising cartridge assembly from rear transaxle casing.
- (4) Remove and discard O ring seal.



B Installation Procedure

NOTES:

- 1 Ensure absolute cleanliness of work area and tools.
- 2 Observe correct use of tools.

- (1) Fit new 'O' ring seal.
- (2) Fit centralising cartridge assembly to rear transaxle casing.
- (3) Fit the rear bumper and venturi panels. (Refer to Section 2).
- (4) Lower the vehicle and remove from the lifting apparatus.

19 GEAR LINKAGE ADJUSTMENT

NOTES:

- 1 Ensure absolute cleanliness of work area and tools.
- 2 Observe correct use of tools.

Procedure

- (1) Remove the tunnel trim and top cover. (Refer to Section 2).
- (2) Measure the distance from centre of the forward spring pin to the aft face of the welded attachment flange.

NOTES:

- 1 The dimension should be $101\text{mm} \pm 0.5\text{mm}$. To achieve this dimension proceed as follows.
- 2 If lateral pivot movement of the gear stick is less than 20 degrees to the right of centre position and less than 10 degrees to the left of centre position, proceed to step 9.

- (3) Loosen thin nut and remove rear spring pin.
- (4) Remove bolts and nuts attaching adjuster tube assembly to front tube sub assembly.
- (5) Turn flanged end of adjuster tube assembly to obtain a dimension of 101mm (3.9in).

NOTE:

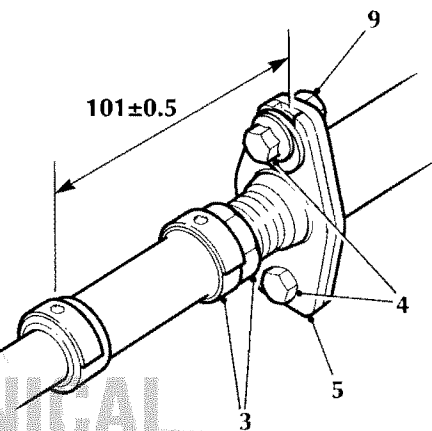
The dimension given is nominal and may differ between vehicles due to alignment of the holes for the spring pin.

- (6) Align attachment holes for the spring pin, and the holes through flanges of adjuster tube assembly and front tube sub assembly.
- (7) Fit spring pin and attach welded flanges with bolts and nuts.
- (8) Torque load thin nut to 30Nm (22ftlb).

NOTE:

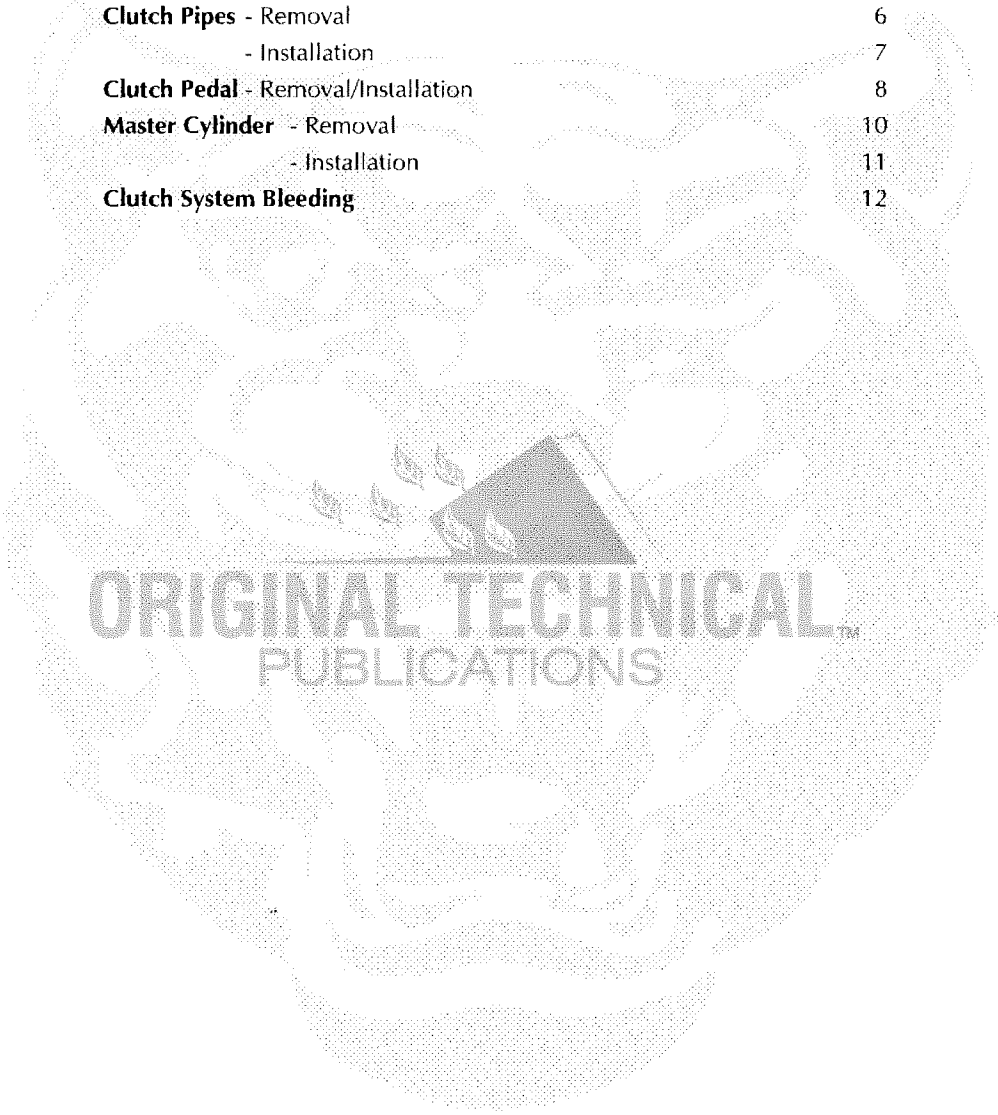
If lateral movement of gear stick is incorrect leave bolts and nuts attaching welded flanges finger tight only.

- (9) Adjust the gear lever movement by rotating the eccentric adjuster bolt.
- (10) Tighten bolts and nuts to secure welded flanges assemblies.
- (11) Refit the tunnel top and trim. (Refer to Section 2).



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DESCRIPTION

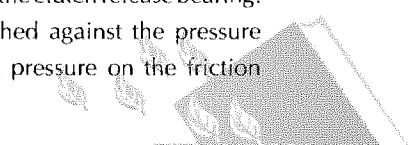
The clutch assembly fitted to the XJ220 is a twin-plate hydraulically actuated design. Hydraulic fluid is stored in the master cylinder reservoir. The master cylinder is connected to the clutch pedal with rigid pipes connecting the master cylinder to the slave cylinder.

The slave cylinder is mounted on the transaxle casing and attached to the clutch actuation components.

The clutch assembly is attached by bolts to the flywheel and consists of twin friction and pressure plates with a ring gear attached to its periphery, to mesh with the starter motor.

OPERATION

Pressure on the clutch pedal actuates the piston in the master cylinder forcing fluid through the pipes to actuate the slave cylinder. The slave cylinder movement is transmitted through the clutch actuation components to the clutch release bearing. The clutch release bearing is pushed against the pressure plate diaphragm fingers, relieving pressure on the friction surfaces.



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1 INTRODUCTION

This section details the procedures for replacement of clutch system components. Replacement of components or disassembly instructions not covered in this section require the component to be returned to JaguarSport.

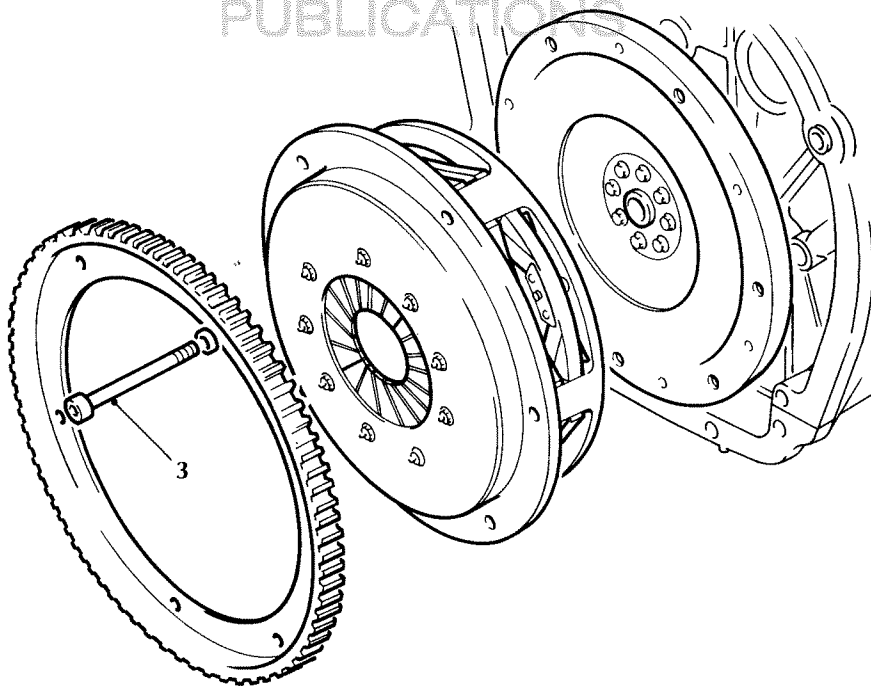
2 CLUTCH ASSEMBLY

A Removal Procedure

NOTES:

- 1 *Ensure absolute cleanliness of work area and tools.*
- 2 *Observe correct use of tools.*
- 3 *Check serviceability of components as they are removed. If in doubt reject the component.*

- (1) Remove the engine and transaxle. (Refer to Section 3).
- (2) Remove the transaxle from the engine. (Refer to Section 7).
- (3) Remove bolts securing clutch assembly to fly wheel.
- (4) Remove the ring gear.
- (5) Remove the clutch assembly.



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B Installation Procedure

NOTES:

- 1 Ensure absolute cleanliness of work area and tools.
- 2 Observe correct use of tools.
- 3 Clean all parts where applicable before reassembly using a soluble propriety engine cleaner.

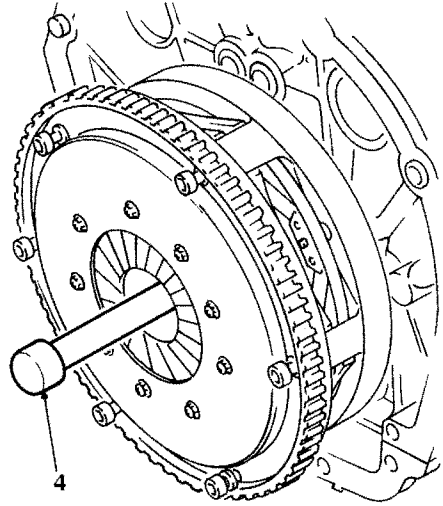
Special Tools

The following special tools will be required:

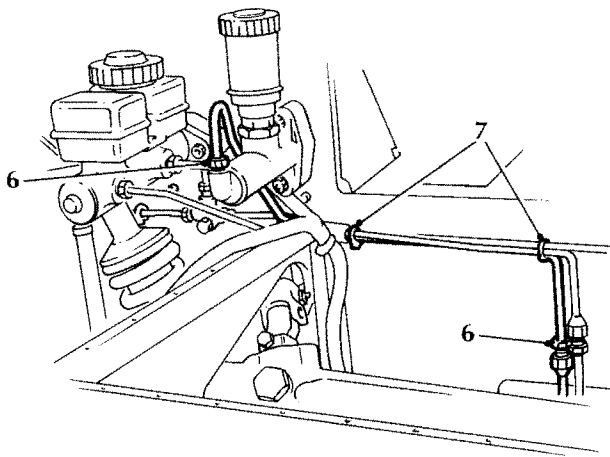
Clutch alignment tool

Torque wrench.

- (1) Locate the ring gear on the clutch assembly.
- (2) Fit clutch assembly to engine fly wheel and fit bolts finger tight.
- (3) Tighten bolts evenly finger tight.
- (4) Fit alignment tool to ensure correct alignment.
- (5) Torque tighten bolts to 33.9Nm (25 ftlb) in a diagonal sequence.
- (6) Remove clutch alignment tool.
- (7) Fit transaxle to engine. (Refer to Section 7).
- (8) Fit engine and transaxle to vehicle. (Refer to Section 3).



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3 CLUTCH PIPES

A Removal Procedure

WARNING:

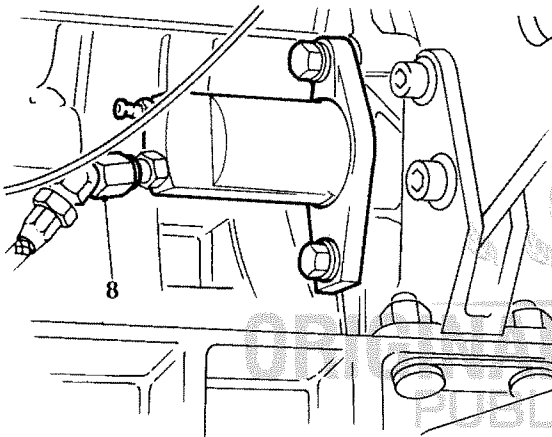
CLUTCH FLUID CAN HAVE A HARMFUL EFFECT ON THE SKIN. AVOID PROLONGED CONTACT WITH THE SKIN AND USE A BARRIER CREAM. WEAR EYE PROTECTION.

CAUTION:

Clutch fluid can have a damaging effect on paintwork. Avoid spillage of clutch fluid.

NOTES:

- 1 Ensure absolute cleanliness of work area and tools.
- 2 Observe correct use of tools.
- 3 Check serviceability of components as they are removed. If in doubt reject the component.



- (1) If the tunnel pipe is to be removed, remove the engine. (Refer to Section 3).
- (2) Remove the structural hatch. (Refer to Section 11).
- (3) Remove the air HVAC unit. (Refer to Section 12).
- (4) Remove driver and passenger underscuttle. (Refer to Section 2).
- (5) Remove tunnel trim and top cover. (Refer to Section 2).
- (6) Disconnect the pipe from master cylinder to front of tunnel and fit blanking caps.
- (7) Remove the cable ties and remove the pipe.
- (8) Remove the flexible pipe from bulkhead to slave cylinder and fit blanking caps.
- (9) Remove clutch pipe cable ties from the tunnel and rear bulkhead.
- (10) Remove clutch pipe from the tunnel through the engine compartment. Ensure the pipe does not foul on adjacent pipework or structure.

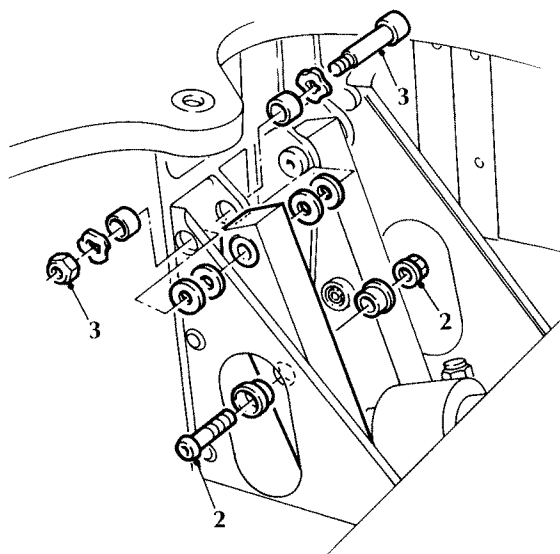
B Installation Procedure

NOTES:

- 1 *Ensure absolute cleanliness of work area and tools.*
- 2 *Observe correct use of tools.*
- 3 *Clean all parts where applicable before reassembly using a soluble propriety engine cleaner.*

- (1) Fit the clutch pipe through the tunnel from the engine compartment. Ensure the pipe does not foul on adjacent pipework or structure.
- (2) Fit cable ties in tunnel and on rear bulkhead.
- (3) Remove blanking caps and fit flexible pipe from rear bulkhead to slave cylinder. Torque pipe unions 12 to 16Nm (9 to 12ftlb).
- (4) Remove the blanking caps and fit the pipe from the master cylinder to the front of the tunnel and secure the pipe with cable ties. Torque the pipe unions 9 to 11Nm (6.5 to 8ftlb).
- (5) Bleed the clutch system. (Refer to Section 8.8).
- (6) Fit the air conditioning unit. (Refer to Section 12).
- (7) Fit the structural hatch. (Refer to Section 11).
- (8) Fit the engine and transaxle, if removed. (Refer to Section 3).

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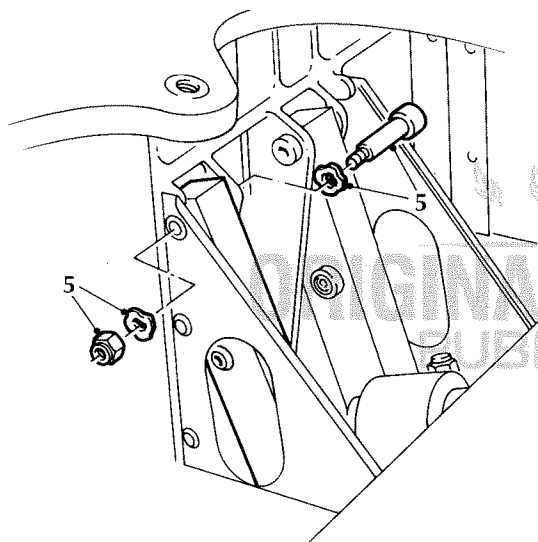
5 CLUTCH PEDAL

A Removal Procedure

NOTES:

- 1 Ensure absolute cleanliness of work area and tools.
- 2 Observe correct use of tools.

- (1) Remove interior front end trim (Refer to Section 2).
- (2) Disconnect the master cylinder push rod from the clutch pedal.
- (3) Remove bolt and nut securing clutch pedal to pedal box. Note the position and number of wavey washers fitted to the bolt.
- (4) Remove clutch pedal and collect the shims and thrust washers. Note the number of shims fitted.



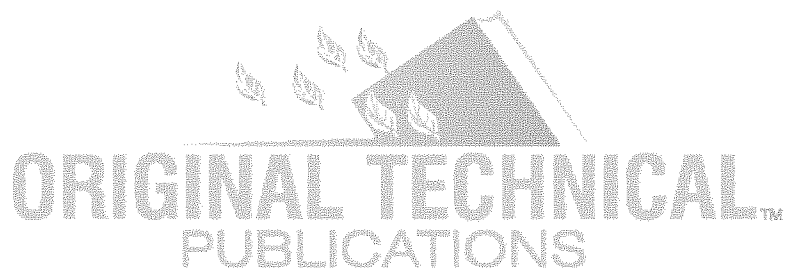
B Installation Procedure

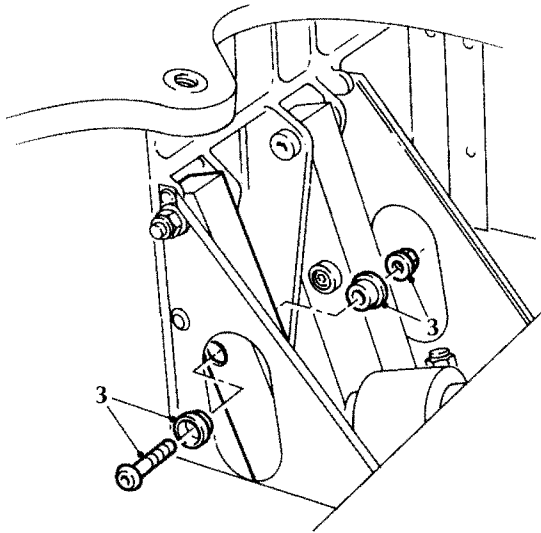
NOTES:

- 1 Ensure absolute cleanliness of work area and tools.
- 2 Observe correct use of tools.

- (1) When fitting a new pedal, press new pivot bushes into the pedal.
- (2) Locate a thrust washer (coated face against the pedal) either side of the pedal.
- (3) Fit the pedal between the lugs and using feeler gauges measure the gap between the outer face of the thrust washer and the pedal box. Fit the appropriate number of shims to ensure the gap is between 0 and 0.09in.
- (4) Fit the pedal attachment bolt together with a plain M8 nut and measure the bolt end float.
- (5) To adjust end float proceed as follows;
 - a) For end float greater than 1mm (0.04in) insert one wavey washer (Pt. No. 8490) under the head of the bolt and one under nut.
 - b) For end float between 0.5mm (0.02in) and 1mm (0.04in) insert a wavey washer under the head of the bolt.
 - c) For an end float of less than 0.5mm (0.02in) no washers are required.

- (6) Replace the M8 plain nut with an M8 Philidas and torque tighten the nut 20 to 23Nm (15 to 17ftlb).
- (7) Fit inserts to the pedal and secure the master cylinder push rod to the pedal with a bolt and Kayloc nut. Ensure there is 0.5 to 1.5mm clearance between the rear face of the pedal and the steering column lower bracket. If not, adjust the position of the rod end on the master cylinder push rod by loosening the lock nut and screwing the rod end in or out as required. Torque the rod end attachment nut 8.5 to 10Nm (6.5 to 7.5ftlb).
- (8) Fit the interior front end trim. (Refer to Section 2).





6 MASTER CYLINDER

A Removal Procedure

Materials

The following materials will be required:

Barrier cream

Clutch fluid

Lint free cloth

Container for clutch fluid.

WARNING:

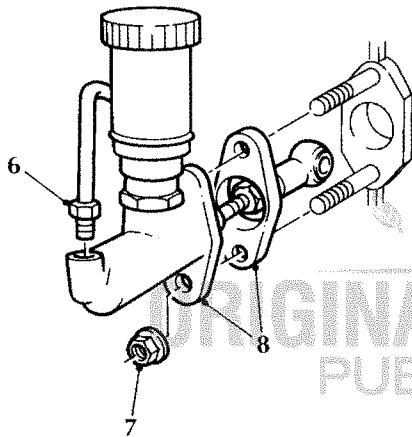
CLUTCH FLUID CAN HAVE A HARMFUL EFFECT ON THE SKIN. AVOID PROLONGED CONTACT WITH THE SKIN AND USE A BARRIER CREAM. WEAR EYE PROTECTION.

CAUTION:

Clutch fluid can have a damaging effect on paintwork. Avoid spillage of clutch fluid.

NOTES:

- 1 Ensure absolute cleanliness of work area and tools.
- 2 Observe correct use of tools.



- (1) Remove the RH wheel. (Refer to Section 4, Road Wheels).
- (2) Remove interior front end trim. (Refer to Section 2).
- (3) Remove the bolt and nut securing the master cylinder push rod to the clutch pedal and retrieve the pedal inserts.
- (4) Remove the radiator duct access panel.
- (5) Drain clutch fluid from master cylinder into container.
- (6) Remove the pipe from the master cylinder, fit blanking caps to the open connections.
- (7) Remove nuts attaching master cylinder to front bulkhead.
- (8) Remove master cylinder and collect the spacer.

B Installation Procedure

Special Tools

The following special tools will be required.

Large syringe and plastic tubing

Materials

The following materials will be required:

Barrier cream

Clutch fluid

Lint free cloth.

WARNING:

CLUTCH FLUID CAN HAVE A HARMFUL EFFECT ON THE SKIN. AVOID PROLONGED CONTACT WITH THE SKIN AND USE A BARRIER CREAM. WEAR EYE PROTECTION.

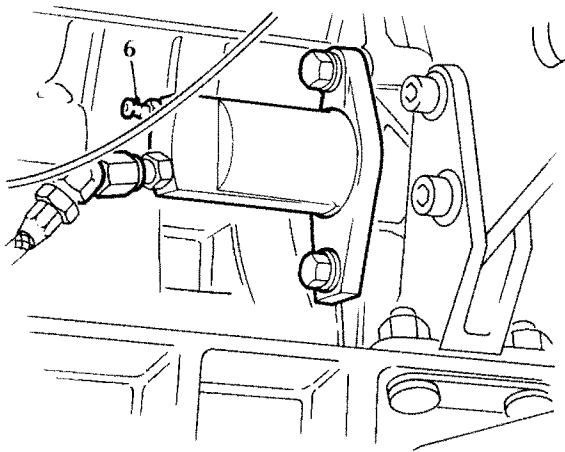
CAUTION:

Clutch fluid can have a damaging effect on paintwork. Avoid spillage of clutch fluid.

NOTES:

- 1 Ensure absolute cleanliness of work area and tools.
- 2 Observe correct use of tools.

- (1) Fit master cylinder and spacer to front bulkhead and secure with nuts. Torque the nuts 30 to 34Nm (22 to 25ftlb).
- (2) Fit the inserts to the pedal and connect the master cylinder push rod to the pedal using bolt and kaylock nut. Ensure there is 0.5 to 1.5mm clearance between the rear face of the pedal and the steering column lower bracket. If not, adjust the position of the rod end on the master cylinder push rod by loosening the lock nut and screwing the rod end in or out as required. Torque the rod end attachment nut 8.5 to 10Nm (6.5 to 7.5ftlb).
- (3) Remove blanking caps and connect the clutch pipes to master cylinder.
- (4) Tighten the pipe unions to 9 to 11Nm (6.5 to 8ftlb).
- (5) Fill master cylinder to the bottom of the filler neck with recommended clutch fluid using suitable syringe and plastic tube.
- (6) Bleed the clutch system. (Refer to Section 8.8).
- (7) Fit the interior front end trim. (Refer to Section 2).
- (8) Fit the radiator duct access panel.
- (9) Fit the wheel. (Refer to Section 4, Road Wheels).
- (10) Road test the car and ensure the clutch functions correctly.



7 CLUTCH SYSTEM BLEEDING

Special Tools

The following special tools will be required:

Clutch system bleeding equipment.

Materials

The following materials will be required:

Barrier cream

Clutch fluid

Lint free cloth.

Procedure

WARNING:

CLUTCH FLUID CAN HAVE A HARMFUL EFFECT ON THE SKIN. AVOID CONTACT WITH THE SKIN AND USE A BARRIER CREAM. WEAR EYE PROTECTION.

CAUTION:

Clutch fluid can have a damaging effect on paintwork. Avoid spillage of clutch fluid.

NOTES:

- 1 Ensure absolute cleanliness of work area and tools.
- 2 Observe correct use of tools.
- 3 Check serviceability of components as they are removed. If in doubt reject the component.

- (1) Position the vehicle on lifting apparatus. (Refer to Introduction, Lifting and Jacking).
- (2) Raise the vehicle to a comfortable working height and remove the venturi panel. (Refer to Section 2).
- (3) Remove the radiator duct access panel.
- (4) Remove filler cap from master cylinder.
- (5) Ensure clutch fluid in master cylinder is level with the bottom of the filler neck.
- (6) Loosen bleed nipple on slave cylinder.
- (7) Fit clutch bleeding equipment to loosened slave cylinder bleed nipple.

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- (8) Operate clutch pedal slowly several times until all the air is removed from the system.

CAUTION:

Ensure the master cylinder fluid is maintained at the correct level. Always use new, clean fluid. Never re-use fluid.

- (9) Hold the clutch pedal fully down and tighten bleed nipple.

NOTE:

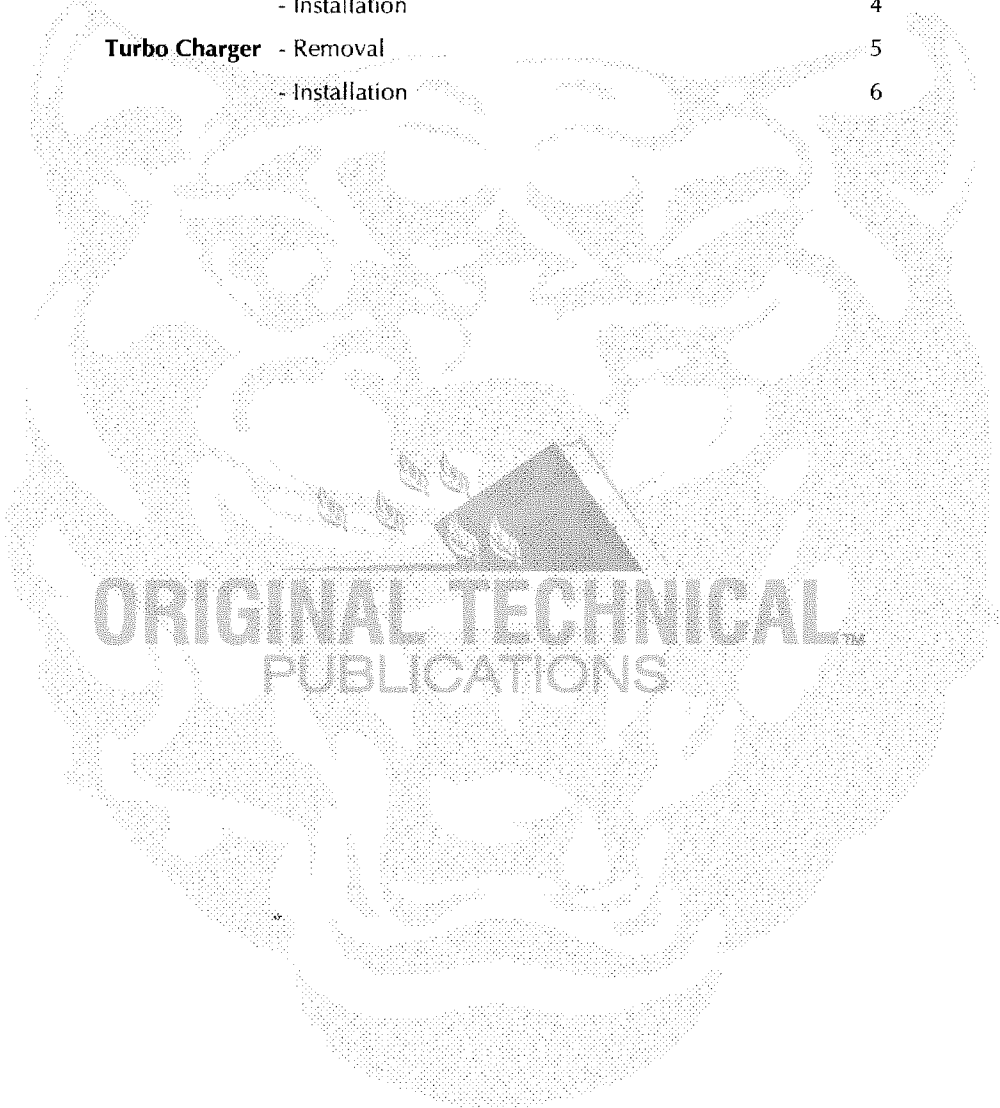
It is recommended that the clutch pedal is held at maximum travel while the bleed nipple is tightened to prevent air being drawn back into the system.

- (10) Allow clutch pedal to return to fully up position.
- (11) Operate the clutch through several cycles and check there are no leaks.
- (12) Fit the venturi panel (Refer to Section 2), and remove the vehicle from the lifting apparatus.
- (13) Top up the master cylinder and refit the radiator duct access panel.
- (14) Road test the vehicle to ensure smooth gear selection.



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1 EXHAUST SYSTEM

A Removal Procedure

NOTES:

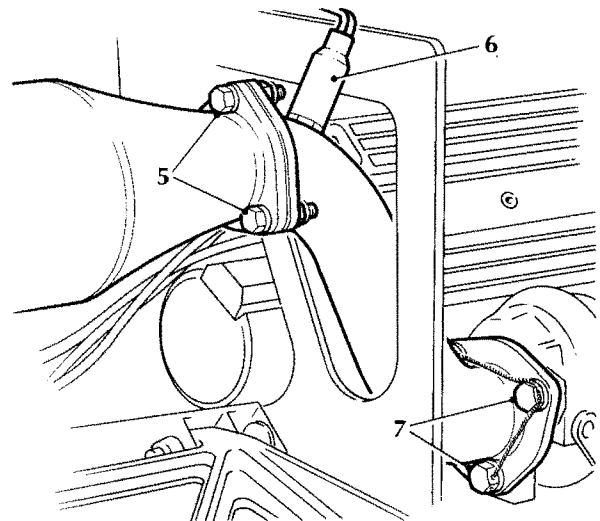
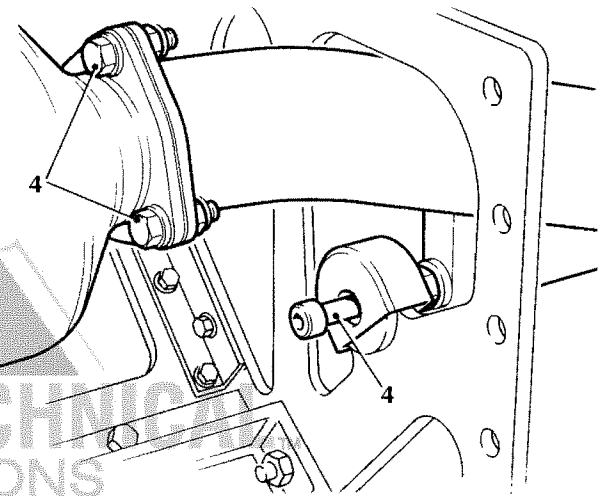
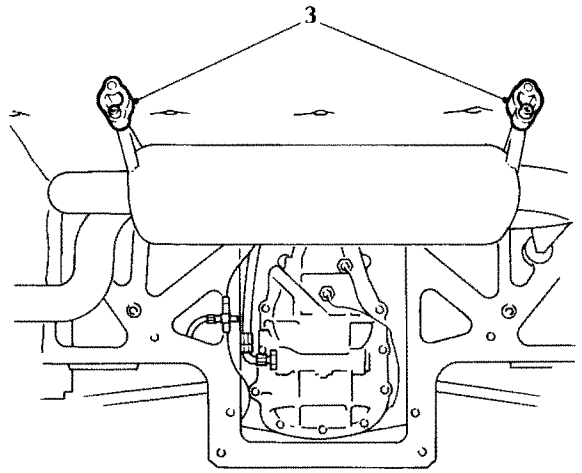
- 1 Ensure absolute cleanliness of work area and tools.
- 2 Observe correct use of tools.

Materials

The following materials will be required

Penetrating oil.

- (1) Apply penetrating oil to the bolts and nuts securing the exhaust segment to the turbo chargers and catalytic converters, and catalytic converters to silencer system.
- (2) Remove the rear bumper. (Refer to Section 2).
- (3) Support the silencer system and remove the bolts securing silencer system to the rubber mountings fitted to underside of the boot.
- (4) Remove bolts and nuts securing the silencer system to the catalytic converters and the engine bay rear plate, remove silencer system and discard the gaskets.
- (5) Remove bolts and nuts securing catalytic converters to the 'S' pipes and remove catalytic converters, discard the gaskets.
- (6) Disconnect the electrical connections to the lambda sensors.
- (7) Undo the locking wire and remove the bolts securing 'S' pipes to the turbo chargers, and remove the 'S' pipes.
- (8) Fit blanking plugs to turbo charger outlets.



B Installation Procedure

NOTES:

- 1 Ensure absolute cleanliness of work area and tools.
- 2 Observe correct use of tools.
- 3 Prior to installation check the condition of all rubber components. Replace if split or oil contaminated.

- (1) Remove blanking plugs from turbo charger outlets.

NOTE:

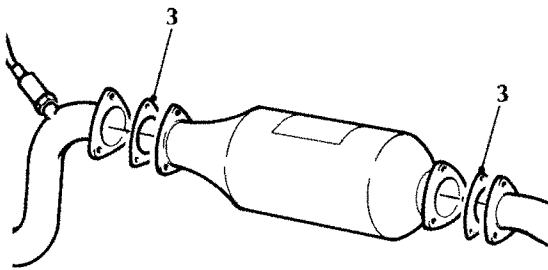
It is recommended to fit new bolts and nuts when fitting the exhaust system.

- (2) Fit 'S' pipes to turbo chargers and secure with bolts. Torque bolts 41 to 47Nm (30 to 34.5ftlb) and wirelock.
- (3) Fit new gaskets and fit the catalytic converters to 'S' pipes and secure with bolts and nuts. Torque nuts 20 to 23Nm (15 to 17ftlb).

CAUTION:

Ensure the catalytic converters are fitted with the can biased to the rear of the car.

- (4) Connect electrical leads to lambda sensors.
- (5) Fit silencer system to rubber mountings fitted on underside of boot and on rear engine plate, secure with bolts and locknuts. Torque locknuts 18 to 20Nm (13 to 15ftlb).
- (6) Secure silencer system to catalytic converters with bolts and nuts. Torque nuts 20 to 23Nm (15 to 17ftlb).
- (7) Start the engine and check for exhaust system leaks, rectify as necessary.
- (8) Fit the rear bumper. (Refer to Section 2).



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2 TURBO CHARGER

A Removal Procedure

Materials

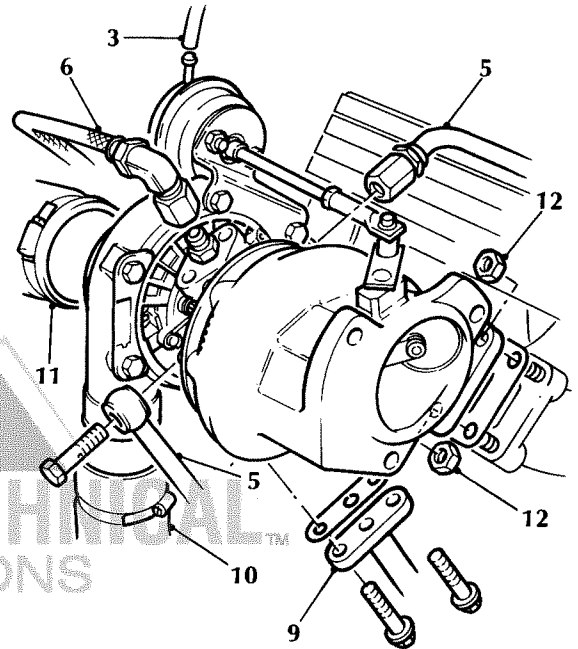
The following materials will be required:

Penetrating oil.

NOTES:

- 1 Apply penetrating oil to all threaded connections and allow time for penetration to be effective.
- 2 Ensure absolute cleanliness of work area and tools.
- 3 Observe correct use of tools.

- (1) Position the vehicle on lifting apparatus. (Refer to Introduction, Lifting and Jacking)).
- (2) Fit protective body covers.
- (3) Release the hose clips and remove the hose from the boost control valve. Fit blanks to open ends.
- (4) Undo the locking wire and remove bolts, nuts and washers securing 'S' pipe to the turbo charger. (Refer to sub-section 1 of this section).
- (5) Remove the turbo charger water inlet and return pipes. Fit blanking caps to open ends.
- (6) Remove turbo charger oil inlet pipe. Fit blanking cap to the open ends.
- (7) Raise the vehicle to a comfortable working height.
- (8) Remove the venturi panel. (Refer to Section 2).
- (9) Remove the turbo charger oil return pipe and discard the gasket. Fit blanks to open ends.
- (10) Loosen the hose clips and remove the turbo charger to intercooler hose. Fit blanks to open ends.
- (11) Loosen the hose clips and remove the turbo to air canister hose. Fit blanks to open ends.
- (12) Remove nuts and washers and remove turbo charger from engine exhaust manifold. Discard the gasket and collect the heatshield.



B Installation Procedure

NOTES:

- 1 Ensure absolute cleanliness of work area and tools.
- 2 Observe correct use of tools.
- 3 It is recommended that components associated with the exhaust system are assembled with new bolts, washers and nuts.

NOTE

The following procedure deals with turbo charger installation on a vehicle already on lifting apparatus.

- (1) Fit new exhaust manifold gasket and secure turbo charger and heatshield to the manifold with M10 nuts and washers. Torque nuts 49 to 54Nm (36 to 40ftlb).
- (2) Remove the pipe blanks and using a new gasket fit oil return pipe to turbo.
- (3) Remove the blanks and fit the turbo to intercooler hose and secure the pipe clips.
- (4) Remove the blanks and fit the turbo to air canister hose and secure the pipe clips.
- (5) Remove the blanks and fit the turbo oil inlet pipe.
- (6) Remove the blanks and fit the turbo water inlet and return pipes.
- (7) Fit the 'S' pipe to the turbo. Torque 'S' pipe bolts (M10) 41 to 47Nm (30 to 34.5ftlb) and wire lock.
- (8) Remove the blanks and fit the hoses to the boost control valve and turbo and secure the hose clips.
- (9) Check the oil and coolant level and replenish if necessary.
- (10) Fit the venturi panel. (Refer to Section 2).
- (11) Remove the protective body covers.
- (12) Lower the car and remove from the lifting apparatus.
- (13) Road test the car and check the turbo charger function is normal. Also check for leaks on the oil, coolant and exhaust systems.

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INTRODUCTION

The fuel system consists of a bag tank, a collector pot, two low pressure pumps with inlet filters, two high pressure fuel pumps, two full flow filters, upper and lower fuel rails for each cylinder bank, two fuel regulators, supply and return fuel pipes and tank breather pipes.

DESCRIPTION

The fuel system is a ZYTEK electronic multi-point sequential injection system with electronic boost control.

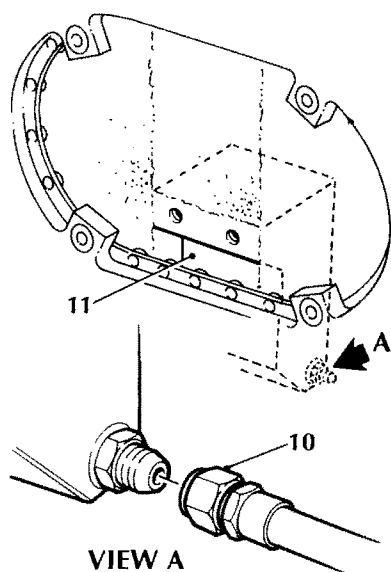
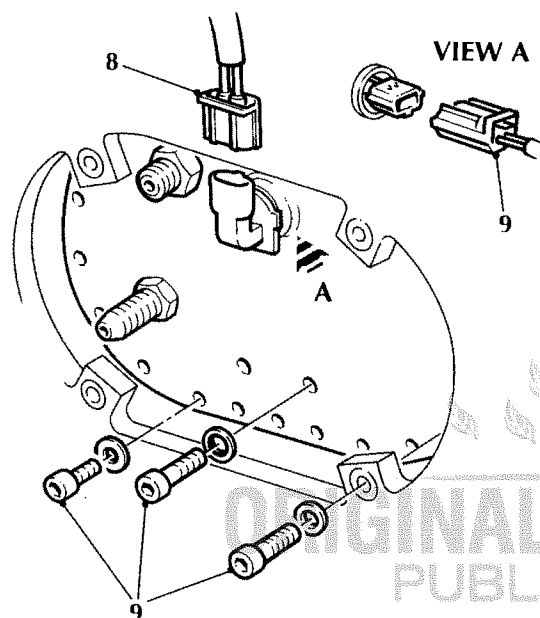
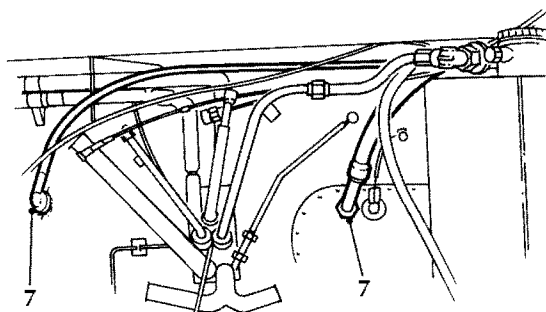
The bag tank, located in a cavity forward of the engine bulkhead, is filled with pre-formed aerated foam to prevent fuel movement under high 'G' forces.

Fuel is drawn into a collector pot (inside the tank) by two low pressure pumps via two bag type filters. From the collector pot the fuel is directed through an in-line filter to two high pressure pumps which direct the fuel through two full flow filters. From the filters the fuel is directed to the upper and lower fuel rails and to the primary and secondary injectors.

Two vacuum controlled fuel regulators (one for each cylinder bank) ensure the fuel pressure remains constant.

A fuel return pipe connected to the fuel regulators returns unused fuel to the tank.

Breather pipes from the LH side of the tank and from the access hatch connect to the fuel filler neck and allow air to escape during filling. A vent pipe leads to atmosphere just behind the RH rear wheel.



1 FUEL TANK

A Removal Procedure

NOTES:

- 1 Ensure absolute cleanliness of work area and tools.
- 2 Observe correct use of tools.

- (1) Remove the engine and transaxle. (Refer to Section 3).
- (2) Remove the oil tank (Refer to Section 3).
- (3) Remove the RH wheel arch liner. (Refer to Section 4)
- (4) Remove the RH intercooler. (Refer to Section 3)
- (5) Drain the fuel tank as follows;

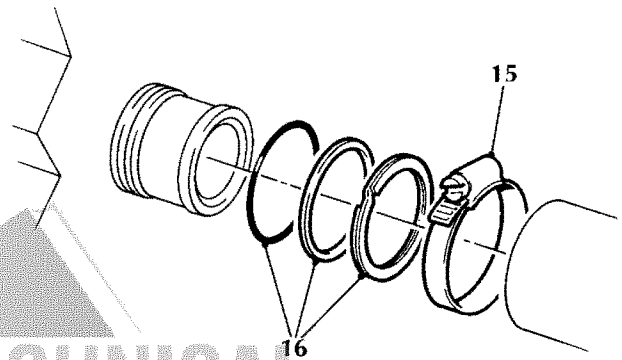
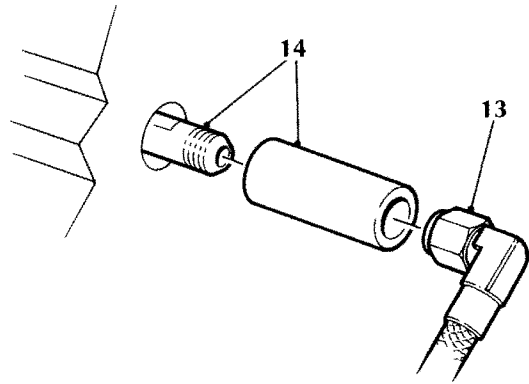
- a) Remove the blanks from the fuel inlet hoses and place the hoses in a container with sufficient capacity to hold the contents of the tank.
- b) Disconnect the engine main electrical feed and insulate the terminal post.
- c) Connect the battery.
- d) Fit service tool ST0034 in place of the fuel pump relay, switch on the ignition and allow the fuel pumps to drain the tank.

CAUTION:

Do not allow the pumps to run dry.

- e) On completion of the draining operation disconnect the battery and fit blanking caps to the fuel inlet pipes.
- (6) Remove the tank sender unit. (Refer to Section 10.6).
- (7) Remove the tank and access panel breather pipes and fit blanks to the pipe open ends.
- (8) Disconnect the electrical connector on the fuel tank access panel.
- (9) Remove the screws securing the tank access panel and collector pot, remove the panel and disconnect the fuel pump harness from the back of the panel. Make a note of the position of the four longer screws.
- (10) Disconnect the supply hose from the collector pot outlet.

- (11) Remove the collector pot from the tank.
- (12) Remove the foam from the tank making a note of the removal sequence.
- (13) Disconnect the supply hose to the high pressure pumps at the fuel tank spigot. Fit blanking caps to the hose.
- (14) Remove the supply spigot insulator and push the spigot into the tank cavity.
- (15) Remove the hose clips and remove the hose from the fuel filler pipe to the tank inlet spigot.
- (16) On the tank inlet spigot, remove the spirolox circlip, load ring and 'O' ring seal and push the spigot into the tank cavity.
- (17) Push the LH breather spigot into the tank cavity.
- (18) Starting in the top RH corner, pull the tank towards the access panel aperture. When sufficiently free, manoeuvre the tank through the aperture and remove completely from the vehicle.
- (19) Remove the fuel supply hose and O ring seal from inside the tank.
- (20) Refit the foam to the tank to ensure the tank retains its shape.

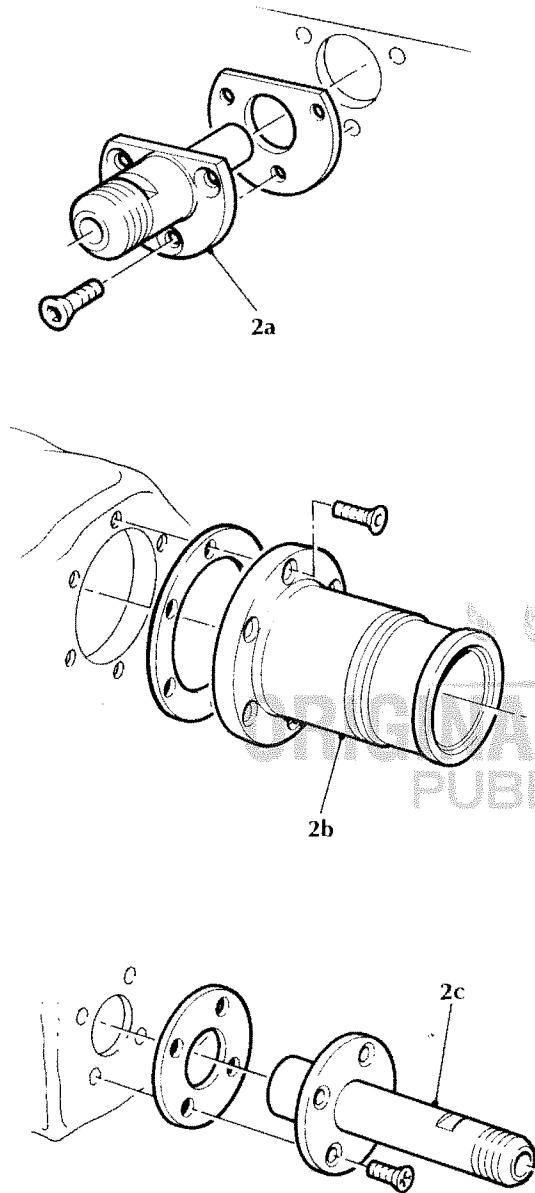


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B Installation Procedure

NOTES:

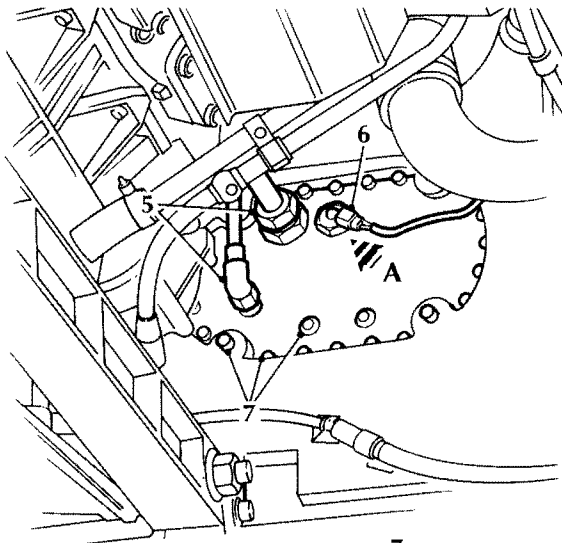
- 1 Ensure absolute cleanliness of work area and tools.
- 2 Observe correct use of tools.



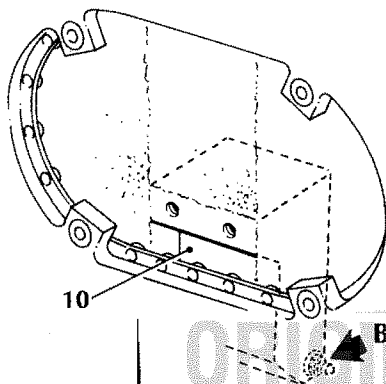
- (1) Remove the foam from the tank making a note of the removal sequence.
- (2) Prior to fitting a new fuel tank, install the following using new gaskets throughout;
 - a) Fit the LH breather spigot and secure to the tank using countersunk screws. Torque screws 1.4 to 1.7Nm (1 to 1.25ftlb).
 - b) Fit the fuel inlet spigot and new 'O' ring and secure to the tank using countersunk screws. Torque screws 1.4 to 1.7Nm (1 to 1.25ftlb).
 - c) Fit the fuel supply spigot and secure to the tank using countersunk screws. Torque screws 1.4 to 1.7Nm (1 to 1.25ftlb).
 - d) Fit the fuel hose to the supply spigot inside the tank. Torque pipe union 20 to 24Nm (15 to 18ftlb).
- (3) Feed the tank through the access panel aperture, taking care not to abrade the tank whilst passing it through the rear bulkhead and push the tank into the corners of the cavity.
- (4) Fit new 'O' ring seal and load ring to the tank inlet spigot and secure with spirolox circlip.
- (5) Fit the tank foam to the left and right of the access panel aperture only.
- (6) Locate the collector pot and connect the fuel supply hose to the collector pot outlet. Torque the hose union 20 to 24Nm (15 to 18ftlb).
- (7) Fit the remaining sections of foam to the tank.
- (8) Fit 4 locating studs evenly around the fuel tank aperture and locate a new gasket on the tank.
- (9) Position the access panel through the aperture and connect the fuel pump harness to the connector on the rear of the panel.
- (10) Loosely secure the access panel to the collector pot using two screws and new crush washers.
- (11) Locate the hatch over the studs and secure with screws and washers.

- (12) Remove the locating studs and replace with screws and washers.
- (13) Torque the screws 1.7 to 2Nm (1.25 to 1.5ftlb).
- (14) Torque tighten the collector pot screws 19.5 to 21.5Nm (14.5 to 16ftlb).
- (15) Fit the fuel supply spigot insulator from outboard side of honeycomb.
- (16) Remove the blanking caps and fit the fuel pump supply hose to the fuel tank spigot. Torque hose union 20 to 24Nm (15 to 18ftlb).
- (17) Fit the hose to the tank inlet spigot and fuel filler pipe and secure with hose clips.
- (18) Remove the blanking caps and fit the fuel tank breather pipes. Torque LH breather union 20 to 24Nm (15 to 18ftlb) and the access panel breather union 35 to 40Nm (26 to 29.5ftlb).
- (19) Fit fuel tank sender unit. (Refer to Section 10.6).
- (20) Fill the fuel tank and carryout a leak check on the fuel system. Rectify any faults as necessary.
- (21) Fit the RH intercooler assembly. (Refer to Section 3).
- (22) Fit the RH wheel arch liner. (Refer to Section 4).
- (23) Fit the oil tank. (Refer to Section 3).
- (24) Fit the engine and transaxle. (Refer to Section 3).

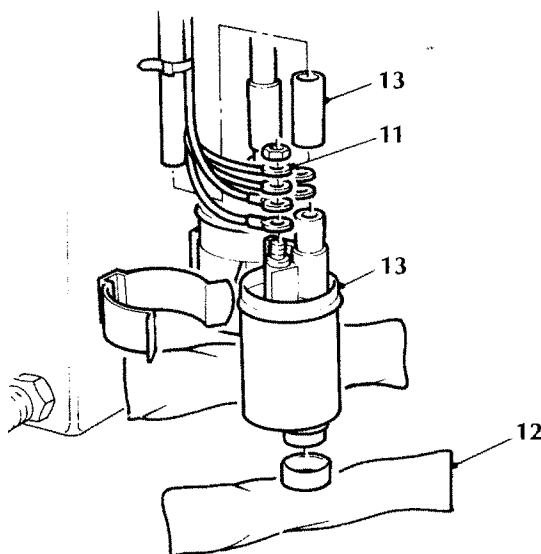
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VIEW A



VIEW B



2 COLLECTOR POT AND LOW PRESSURE FUEL PUMPS

A Removal Procedure

NOTES:

- 1 Ensure absolute cleanliness of work area and tools.
- 2 Observe correct use of tools.

NOTE:

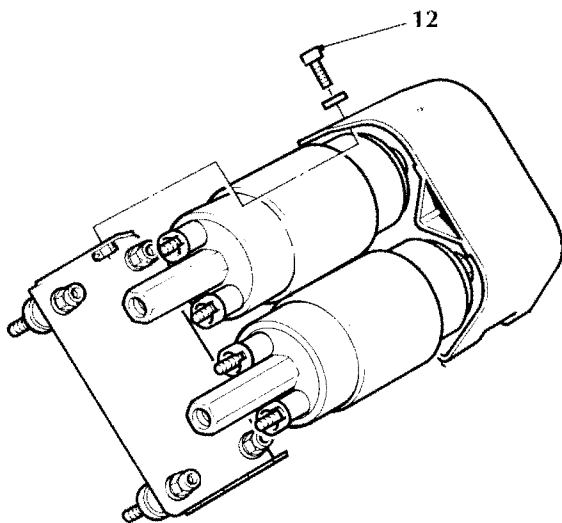
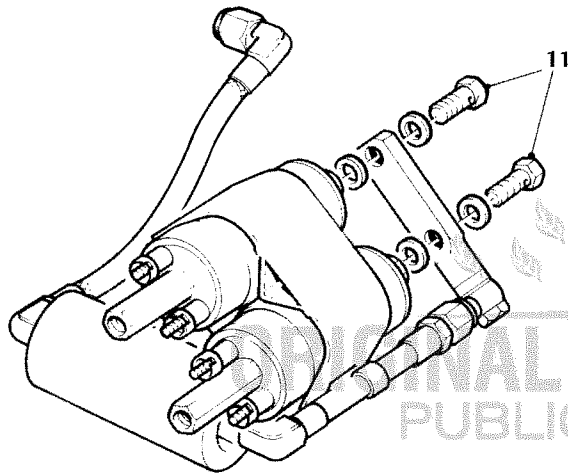
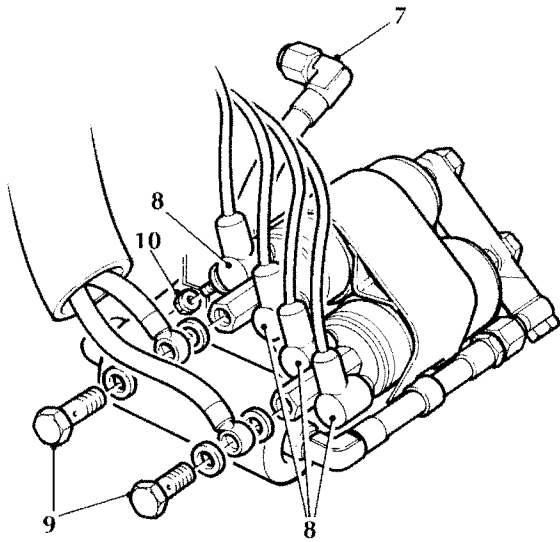
The following procedure deals removal with the engine in-situ. If the engine has been removed certain procedures may be disregarded.

- (1) Remove the venturi panel. (Refer to Section 2).
- (2) Remove the RH air filter canister. (Refer to Section 3).
- (3) Remove the oil tank. (Refer to Section 3).
- (4) Drain the fuel tank. (Refer to Fuel Tank, Section 10.1, operation 5).
- (5) Remove the access panel breather pipe and fuel return pipe. Fit blanking caps to open ends.
- (6) Disconnect the electrical connector on the tank access panel.
- (7) Remove the screws securing the tank access panel and collector pot, remove the panel and disconnect the fuel pump harness from the connector on the rear of the panel.
- (8) Remove the foam sections to the rear of the collector pot only.
- (9) Disconnect the fuel supply hose from the collector pot. Fit blanks to the open ends.
- (10) Remove the collector pot from the tank.
- (11) Disconnect the electrical connections to the pumps.
- (12) Pull the filters from the pumps.
- (13) Pull the pumps from the clip and fuel hose.

B Installation Procedure

NOTES:

- 1 *Ensure absolute cleanliness of work area and tools.*
 - 2 *Observe correct use of tools.*
-
- (1) Connect the fuel pump harness with four eyelets to one pump as follows;
 - a) Two black lead to the negative (-ve) terminals.
 - b) Two white leads to the positive (+ve) terminals.
 - c) Secure the electrical leads using washers and brass nuts.
 - (2) Fit the pump to the fuel hose and clip on the collector pot.
 - (3) Connect the remainder of the harness to the second pump as follows;
 - a) Single white lead to the positive (+ve) terminal and single black lead to the negative (-ve) terminal.
 - b) Secure the leads using washers and brass nuts.
 - (4) Fit the pump to the fuel hose and clip on the collector pot.
 - (5) Push fit the filters to the bottom of each pump with orientation as illustrated.
 - (6) Locate the collector pot in the tank.
 - (7) Remove the blanking caps and connect the fuel supply hose to the bottom of the collector pot. Torque the hose union 20 to 24Nm (15 to 18ftlb).
 - (8) Fit the foam sections to the fuel tank.
 - (9) Fit 4 locating studs evenly around the fuel tank aperture and locate a new gasket on the tank.
 - (10) Position the access panel through the aperture and connect the fuel pump harness to the connector on the rear of the panel.
 - (11) Loosely secure the access panel to the collector pot using two screws and new crush washers.
 - (12) Locate the hatch over the studs and secure with screws and washers.
 - (13) Remove the locating studs and replace with screws and washers.
 - (14) Torque the screws 1.7 to 2Nm (1.25 to 1.5ftlb).
 - (15) Torque tighten the collector pot screws 19.5 to 21.5Nm (14.5 to 16ftlb).
 - (16) Fit the access panel breather pipe. Torque pipe union 35 to 40Nm (26 to 29.5ftlb).
 - (17) Fit the fuel return pipe and tighten the union 13.5 to 16Nm (10 to 12ftlb).
 - (18) Fit the electrical connection to the access panel connector.
 - (19) Fill the fuel tank and carryout a leak check on the fuel system. Rectify any faults as necessary.
 - (20) Fit the oil tank. (Refer to Section 3).
 - (21) Fit the RH air filter canister. (Refer to Section 3).
 - (22) Fit the venturi panel. (Refer to Section 2).



3 HIGH PRESSURE FUEL PUMPS

A Removal Procedure

NOTES:

- 1 Ensure absolute cleanliness of work area and tools.
- 2 Observe correct use of tools.

- (1) Position the vehicle on lifting apparatus. (Refer to Introduction, Lifting and Jacking).
- (2) Raise the vehicle to a comfortable working height and remove the RH rear wheel and wheel arch liner. (Refer to Road Wheels, Section 4).
- (3) Remove the venturi panel. (Refer to Section 2).
- (4) Remove the RH intercooler and ducting. (Refer to Section 3).
- (5) Drain the fuel tank. (Refer to Section 10.1, Fuel Tank, operation 5).
- (6) Disconnect the battery.
- (7) Disconnect the fuel supply hose at the fuel tank spigot. Fit blanking caps to the open ends.
- (8) Disconnect the electrical harness to the fuel pumps.
- (9) Remove the banjo bolts and disconnect the fuel hoses at the pump outlets. Fit blanking caps to the open ends.
- (10) Remove the four Durlock nuts securing the fuel pumps mounting bracket to the vehicle and remove the pumps complete with fuel manifold assembly.

NOTE

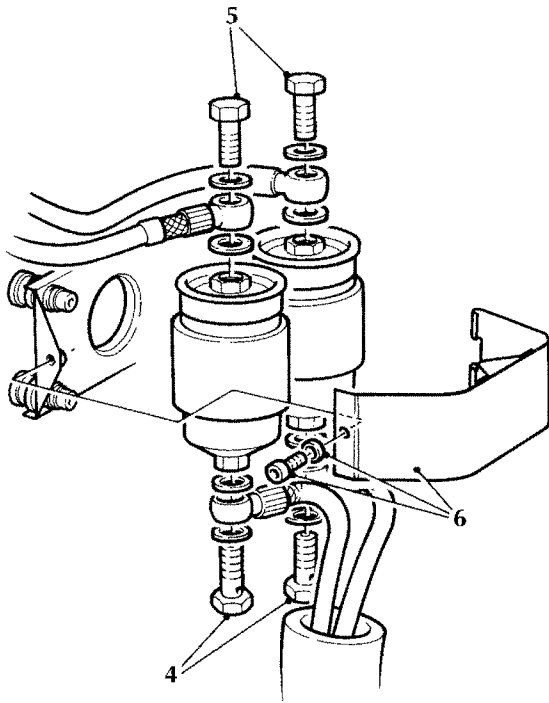
Do not disturb the fuel pump to supply pipe connections.

- (11) Remove the banjo bolts securing the manifold to the pumps and remove the manifold and supply pipe complete.
- (12) Remove the screw and washer securing the mounting bracket strap, release the strap and remove the pumps from the bracket.

B Installation Procedure

NOTES:

- 1 *Ensure absolute cleanliness of work area and tools.*
 - 2 *Observe correct use of tools.*
-
- (1) If fitting new fuel pumps, fit a foam strip around each pump.
 - (2) Locate the pumps in the carrier bracket, fit the strap and loosely secure the strap using an M6 washer and bolt.
 - (3) Orientate the pumps in the carrier as illustrated and torque tighten the strap bolt 8 to 9.5Nm (6 to 7ftlb).
 - (4) Fit the fuel pump manifold and supply hose assembly using banjo bolts and new crush washers. Torque the banjo bolts 16 to 20Nm (12 to 15ftlb).
 - (5) Fit the fuel pump assembly to the vehicle and secure with four Durlock nuts. Torque nuts 12 to 14Nm (9 to 10.5ftlb).
 - (6) Remove the blanking caps and fit the fuel supply hose to the tank spigot. Torque the hose union 20 to 24Nm (15 to 18ftlb).
 - (7) Fit the hoses from the filters to the pump outlets using banjo bolts and new crush washers. Torque the banjo bolts 13.5 to 16Nm (10 to 12ftlb).
 - (8) Connect the pump electrical harness as follows;
 - a) M4 eyelets to the positive (+ve) terminals.
 - b) M5 eyelets to the negative (-ve) terminals.
 - (9) Secure the harness to the pumps using M4 and M5 washers and Nyloc nuts.
 - (10) Torque the M4 nuts 3 to 4Nm (2 to 3ftlb) and the M5 nuts 7 to 8Nm (5 to 6ftlb).
 - (11) Fit harness boots over the pump terminal posts.
 - (12) Fill the fuel tank and ensure the pumps function correctly and carryout a leak check. Rectify any faults as necessary.
 - (13) Fit the intercooler and ducting. (Refer to Section 3).
 - (14) Fit the venturi panel. (Refer to Section 2).
 - (15) Fit the wheel arch liner and wheel. (Refer to Section 4).
 - (16) Connect the battery.
 - (17) Remove the vehicle from the lifting apparatus.



4 FUEL FILTERS

A Removal Procedure

NOTES:

- 1 Ensure absolute cleanliness of work area and tools.
- 2 Observe correct use of tools.

- (1) Position the vehicle on lifting apparatus. (Refer to Introduction, Lifting and Jacking).
- (2) Switch the battery master switch to off.
- (3) Remove the RH wheel and wheel arch liner. (Refer to Section 4).
- (4) Disconnect the hose from the pumps to the filter at the bottom of the filters. Make a note of the hose position and fit blanking caps to the open ends.
- (5) Disconnect the hose from the filters to the side wall at the top of the filters. Make a note of the hose position and fit blanking caps to the open ends.
- (6) Remove the screw and washer securing the filter retaining strap, release the strap and remove the filters.

B Installation Procedure

- (1) If fitting new filters, fit foam strips to both filters.
- (2) Locate the filters in the carrier, fit the retaining strap and secure with a washer and screw. Torque the screw 12 to 14Nm (9 to 10.5ftlb).
- (3) Fit the hoses from the side wall to the filters using banjo bolts and new crush washers. Torque the banjo bolts 13.5 to 16Nm (10 to 12ftlb).
- (4) Fit the hoses from the pumps to the filters using banjo bolts and new crush washers. Torque the banjo bolts 13.5 to 16Nm (10 to 12ftlb).
- (5) Switch on the battery master switch.
- (6) Run the engine and check for fuel leaks. Rectify any faults as necessary.
- (7) Fit the wheel arch liner and wheel. (Refer to Section 4).
- (8) Remove the vehicle from the lifting apparatus.

5 FUEL FILLER ASSEMBLY

A Removal Procedure

NOTES:

- 1 Ensure absolute cleanliness of work area and tools.
- 2 Observe correct use of tools.

NOTE

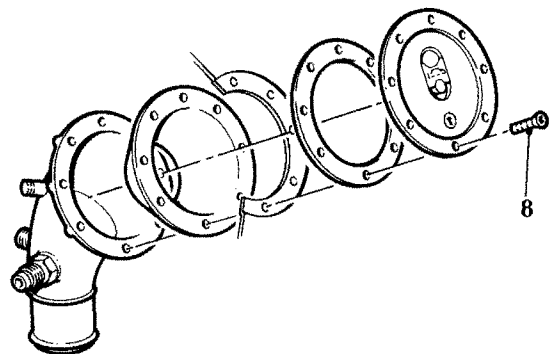
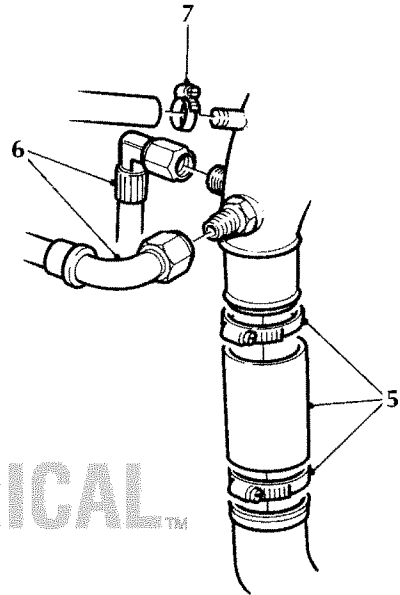
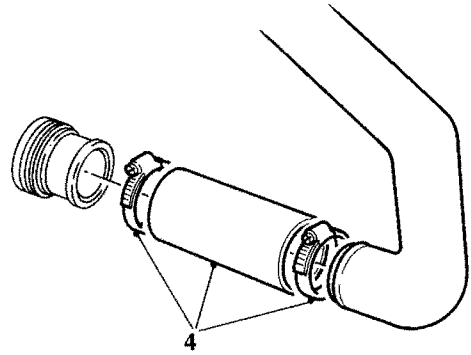
Prior to removing the fuel filler assembly ensure the fuel level in the tank is below that of the inlet spigot.

- (1) Position the vehicle on lifting apparatus. (Refer to Introduction, Lifting and Jacking).
- (2) Remove the RH rear wheel and wheel arch liner. (Refer to Road Wheels, Section 4).
- (3) Remove the RH intercooler and ducting. (Refer to Section 3).
- (4) Loosen the hose clips and remove the tank inlet spigot to filler pipe hose. Fit a blanking cap to the tank spigot.
- (5) Loosen the hose clips, remove the filler neck to filler pipe hose and remove the filler pipe complete with insulation.
- (6) Disconnect the fuel tank breather pipes at the filler neck. Fit blanking caps to the pipes.
- (7) Loosen the hose clip and remove the roll-over valve tubing from the filler neck.

NOTE

Make a note of the filler neck position to ensure correct alignment of filler neck and filler pipe on assembly.

- (8) Remove the eight screws securing the filler neck to the filler cap, remove the filler neck and cap.



B Installation Procedure

NOTES:

- 1 *Ensure absolute cleanliness of work area and tools.*
 - 2 *Observe correct use of tools.*
-
- (1) Locate the filler cap assembly and new gasket in the aperture with the lock vertically.
 - (2) Fit the filler neck and new gasket and secure to the vehicle using eight screws. Torque the screws 5.5 to 8Nm (4 to 6ftlb).
 - (3) Locate the fuel filler pipe in the vehicle.
 - (4) Fit the hose between the filler neck and filler pipe and secure with hose clips.
 - (5) Remove the blanking cap from the tank spigot and fit the spigot to filler pipe hose and secure with hose clips.
 - (6) Fit the breather pipes to the filler neck.
 - (7) Torque tank to filler breather union 20 to 24Nm (15 to 18ftlb) and the tank access panel to filler neck breather union 35 to 40Nm (26 to 30ftlb).
 - (8) Fit the roll-over valve tubing to the filler neck and secure the hose clip.
 - (9) Replenish the fuel tank and check for any filler leaks. Rectify any faults as necessary.
 - (10) Fit the intercooler and ducting. (Refer to Section 3).
 - (11) Fit the venturi panel. (Refer to Section 2).
 - (12) Fit the wheel arch liner and wheel. (Refer to Section 4).
 - (13) Remove the vehicle from the lifting apparatus.

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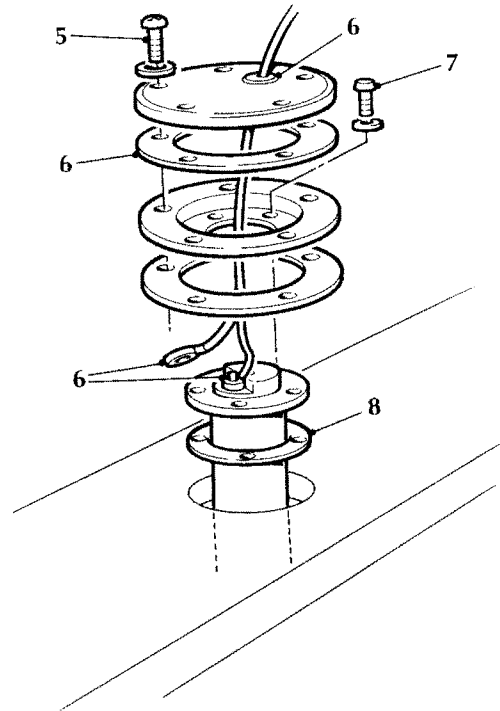
6 FUEL SENDER UNIT

A Removal Procedure

NOTES:

- 1 Ensure absolute cleanliness of work area and tools.
- 2 Observe correct use of tools.

- (1) Move the seats fully forward and move the seatback to the upright position.
- (2) Fit recommended seat covers.
- (3) Disconnect the battery.
- (4) Loosen the tunnel saddle panel, remove the ECU cover and remove the rear stowage tray. (Refer to Section 2).
- (5) Remove the screws securing the sender unit cover.
- (6) Disconnect the electrical connections to the sender unit, feed the leads through the grommet and remove the cover. Discard the cover gasket.
- (7) Remove the bolts securing the sender unit to the fuel tank and withdraw the unit.
- (8) Retrieve the gasket from the top of the fuel tank.



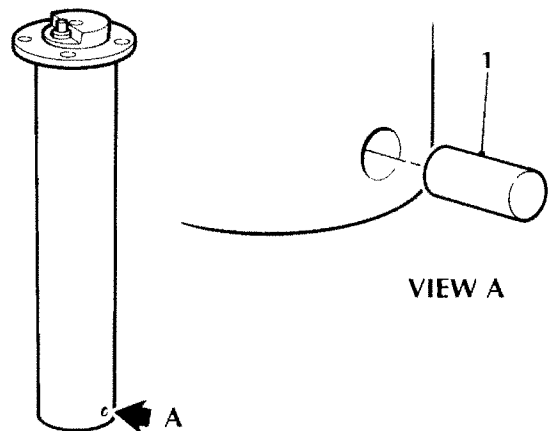
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B Installation Procedure

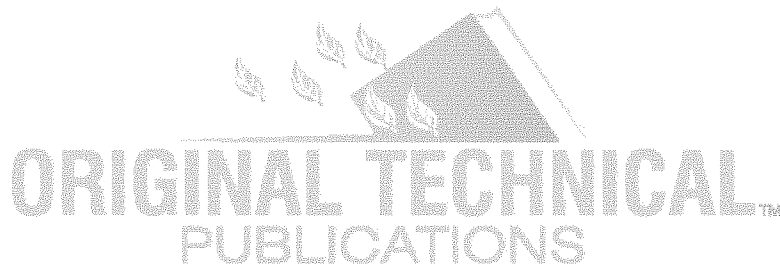
NOTES:

- 1 Ensure absolute cleanliness of work area and tools.
- 2 Observe correct use of tools.

- (1) If fitting a new sender unit, remove the locking pin from the bottom of the dip tube.
- (2) Fit a new gasket to the sender unit and align the sender unit, fit the retaining plate and new retaining plate gasket to the fuel tank.
- (3) Secure the sender unit assembly to the tank using four screws and washers leaving the fifth screw for harness connection. Torque the screws 1.4 to 1.7Nm (1 to 1.25ftlb).
- (4) Locate a new gasket on the top face of the retaining plate.

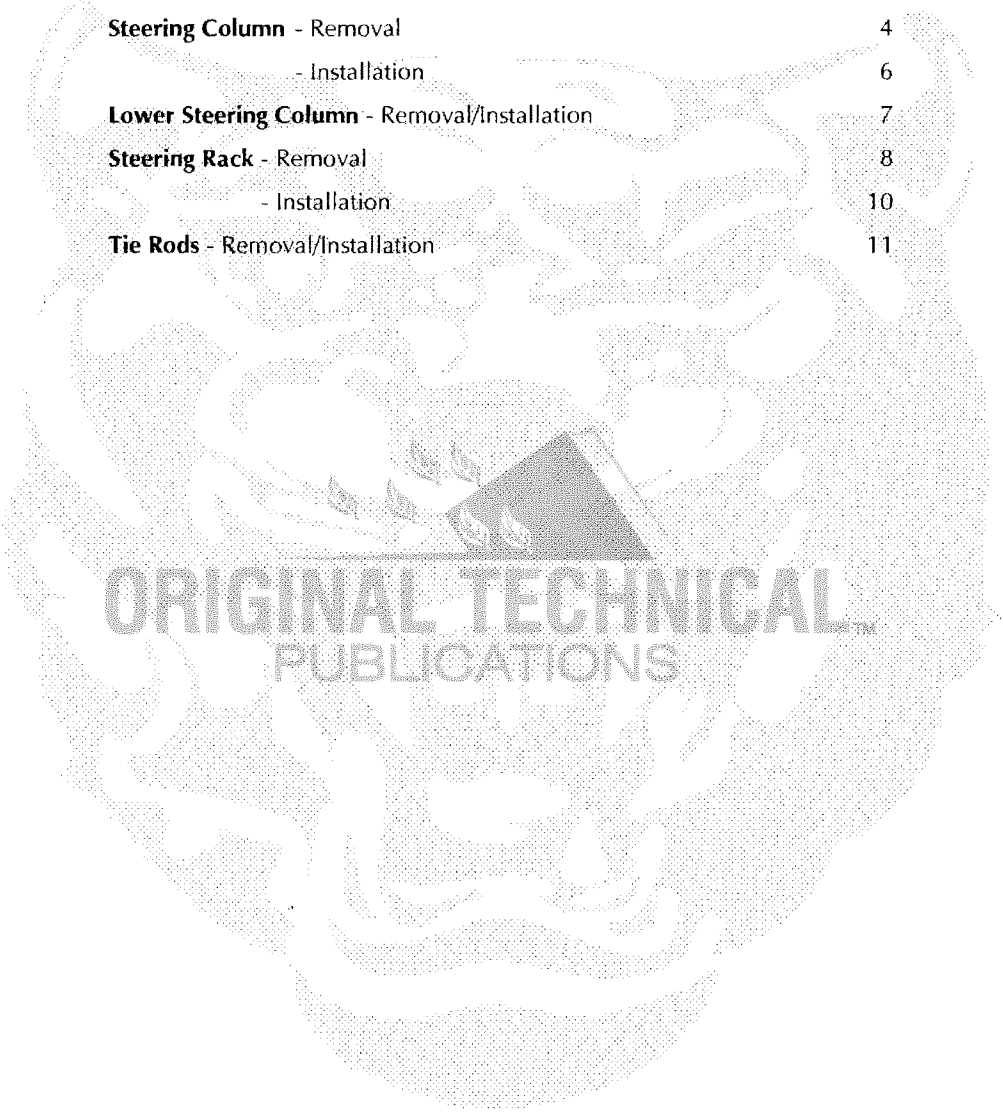


- (5) Feed the unit electrical leads through the cover plate grommet. Fit the positive lead to the sender terminal and the earth lead to the sender unit using the fifth screw and washer. Torque the screw 1.4 to 1.7Nm (1 to 1.25ftlb).
- (6) Fit the sender unit cover plate and secure with five washers and screws. Torque the screws 8 to 9.5Nm (6 to 7ftlb).
- (7) Connect the battery and check the sender functions correctly. Rectify any faults as necessary.
- (8) Fit the rear stowage tray, ECU cover and tunnel saddle. (Refer to Section 2).
- (9) Remove the seat covers and return seats to their original position.



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DESCRIPTION

The steering is a mechanical rack and pinion system consisting of the steering wheel, the upper and lower steering columns, steering rack and tie-rods.

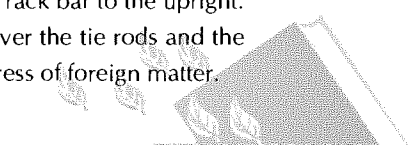
The steering wheel is leather bound and fitted with a horn button in the centre. The wheel is splined onto the upper steering column.

The upper steering column combines the ignition switch, steering lock and steering wheel height/rake adjustment.

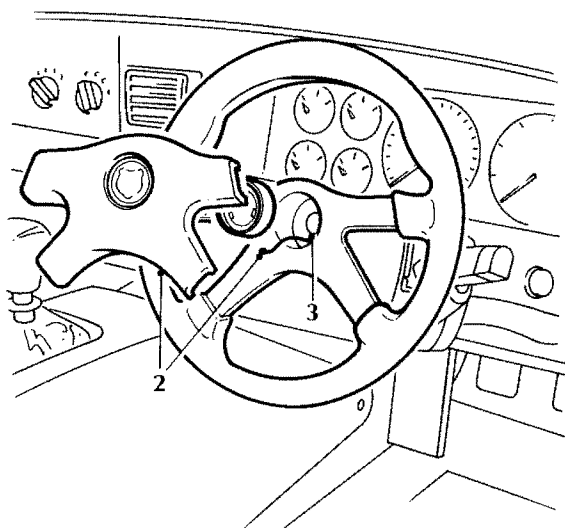
The lower steering column is splined at both ends to attach to the upper and lower universal joints. The upper universal joint attaches between the upper and lower steering columns. The lower universal joint attaches between the steering rack and the lower column.

The steering rack assembly consists of a steering housing which contains the rack and pinion mechanism. Tie rods attach the ends of the rack bar which meshes with the steering pinion in the steering housing.

The tie rods connect the ends of the rack bar to the upright. Rubber steering joint gaiters fitted over the tie rods and the ends of the rack bar prevent the ingress of foreign matter.



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1 INTRODUCTION

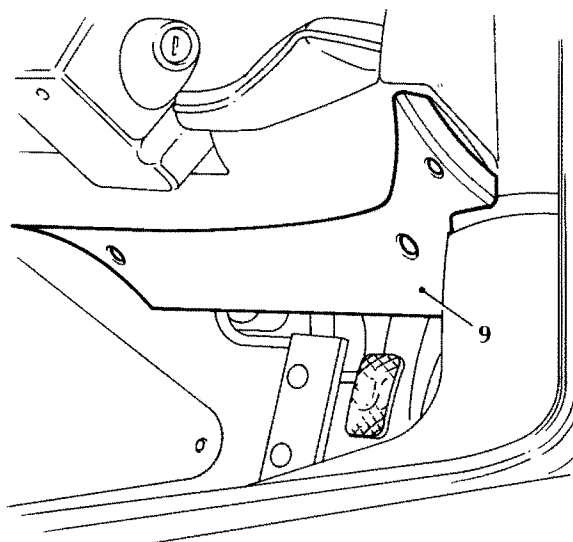
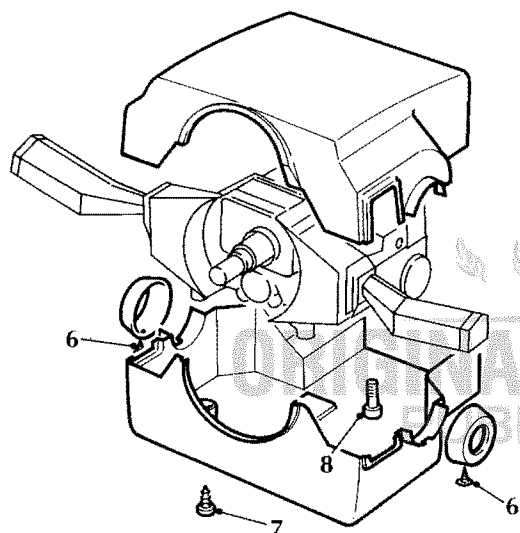
This section details the procedures for replacement of steering system components. Replacement of components or disassembly instructions not covered in this section require the component be returned to JaguarSport.

2 STEERING COLUMN

A Removal Procedure

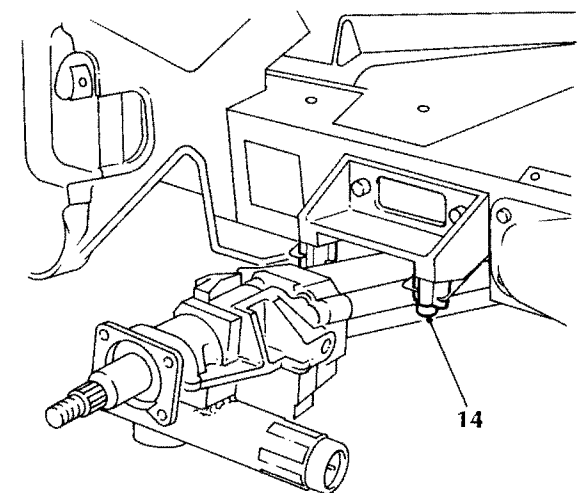
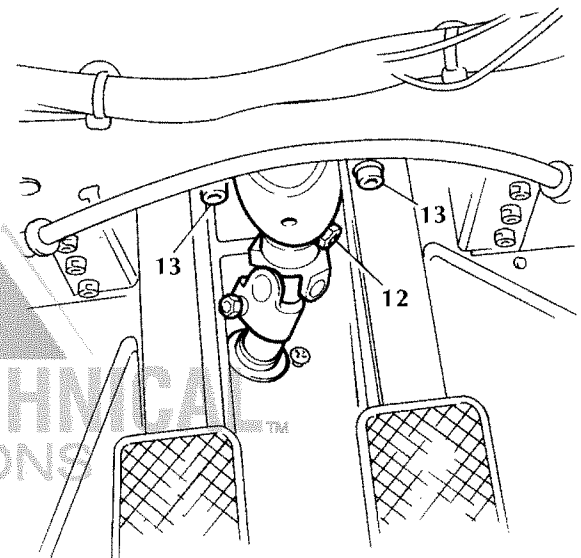
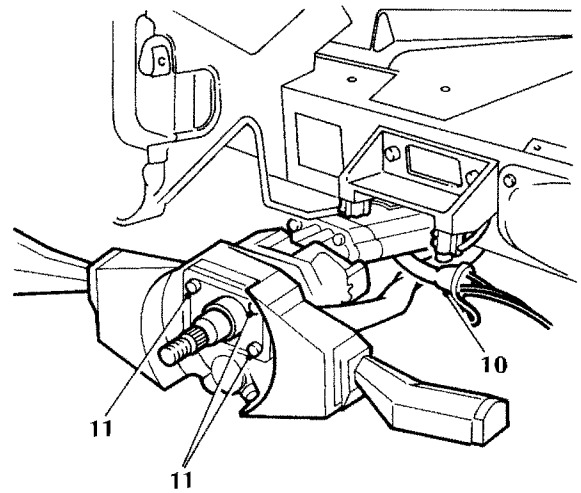
NOTES:

- 1 Ensure absolute cleanliness of work area and tools.
- 2 Observe correct use of tools.



- (1) Switch the battery master switch to the off position.
- (2) Remove the steering wheel centre pad by gently prising off and disconnect the horn electrical connector.
- (3) Remove the steering wheel centre nut.
- (4) Mark across the boss and the end of the upper steering column using permanent pen to ensure correct alignment on re-assembly and remove the steering wheel.
- (5) Remove the instrument panel. (Refer to Section 13).
- (6) Remove the screws securing the ignition switch escutcheons and blank escutcheon. Remove the escutcheons.
- (7) Remove the two screws securing the lower switch cover and remove the cover.
- (8) Remove the three screws securing the upper switch cover and remove the cover.
- (9) Remove the driver's underscuttle to gain access to the column upper mounting bolts (Refer to Section 2).

- (10) Disconnect steering column switch cluster electrical harness and ignition switch electrical harness.
- (11) Remove the four bolts securing the steering column switch cluster to the column and remove the switches.
- (12) Remove bolt, washer and nut securing upper universal joint to steering column situated between clutch and brake pedal in the footwell.
- (13) Remove lower attachment screws and washers securing steering column to pedal box.
- (14) Support steering column and remove upper attachment screws with washers and spacer washers.
- (15) Remove steering column from the vehicle.



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B Installation Procedure

NOTES:

- 1 Ensure absolute cleanliness of work area and tools.
- 2 Observe correct use of tools.

Special Tools

The following special tools will be required

Torque wrench 0 to 47Nm (34.5ftlb)

- (1) Locate the steering column in vehicle and attach universal joint to steering column between clutch and brake pedal.

NOTE:

Do not secure universal joint with bolt, washer and nut at this stage.

- (2) Fit the steering column to the upper attachment points and secure steering column with screws and washers and spacer washers.

NOTE

Do not torque tighten column attachment bolts at this stage.

- (3) Fit the steering column to the lower attachment points and secure steering column with screws and washers.
- (4) Fit bolt, washer and nut and secure upper universal joint to steering column.
- (5) Torque load bolt and nut 20 to 23Nm (15 to 17 ftlb).
- (6) Fit the column switch cluster and secure with four screws. Torque the screws 8 to 9.5Nm (6 to 7ftlb).
- (7) Connect switch cluster electrical harness and ignition switch electrical harness.
- (8) Fit steering column switch covers and secure with screws. Torque upper cover screws 3 to 4Nm (2 to 3ftlb).
- (9) Fit the ignition switch escutcheon and blank escutcheon and secure with screws.
- (10) Torque load the column lower attachment bolts 20 to 23Nm (15 to 17ftlb) and the upper attachment bolts 41 to 47Nm (30 to 34.5ftlb).
- (11) Fit the instrument panel. (Refer to Section 13).
- (12) Fit the steering wheel and boss to the upper column, taking care to align the marks made on removal. Tighten the centre nut to 35 to 41Nm (26 to 30ftlb).
- (13) Connect the horn electrical connector and fit the steering wheel centre pad.
- (14) Connect the battery and ensure all column-mounted switches function correctly.

3 LOWER STEERING COLUMN

A Removal Procedure

NOTES:

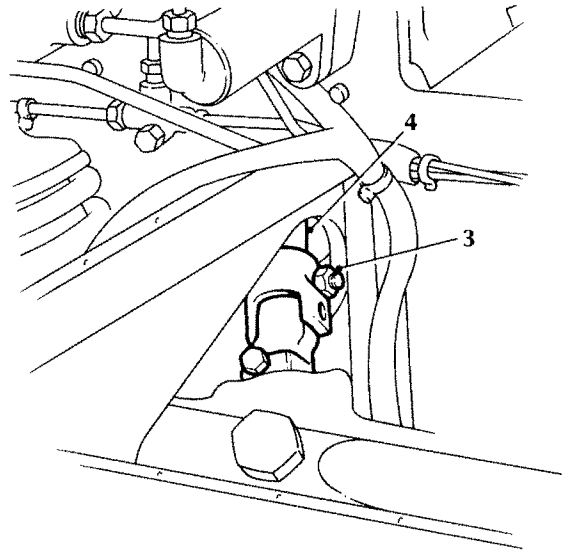
- 1 Ensure absolute cleanliness of work area and tools.
- 2 Observe correct use of tools.

- (1) Remove wheel and remove wheel arch liner. (Refer to Road Wheels, Section 4).
- (2) Remove steering wheel and upper steering column. (Refer to sub section 2 of this section).
- (3) Remove bolt, washer and nut securing lower universal joint to lower steering column.

NOTE:

It is not necessary to remove the universal joint from the steering rack input shaft.

- (4) Remove lower steering column.



B Installation Procedure

NOTES:

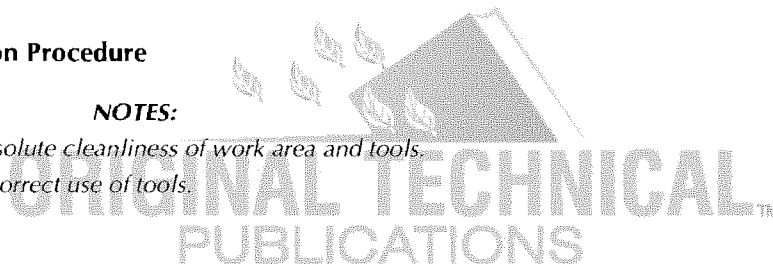
- 1 Ensure absolute cleanliness of work area and tools.
- 2 Observe correct use of tools.

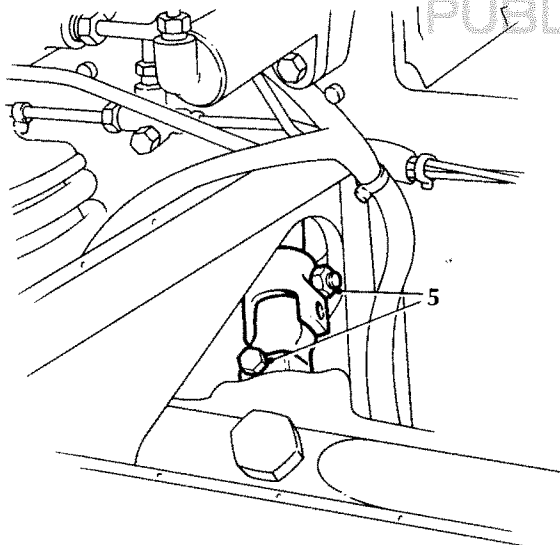
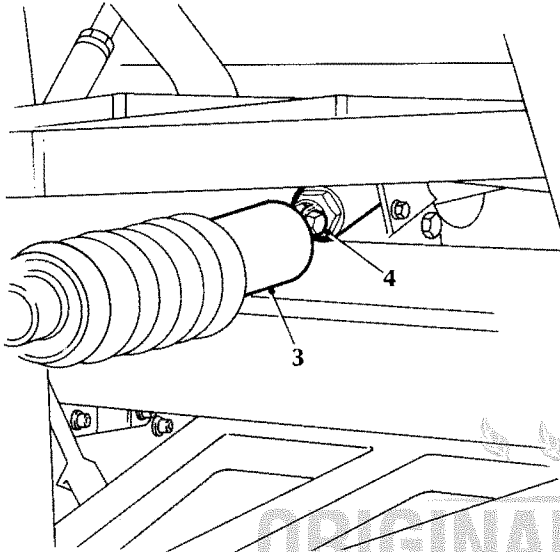
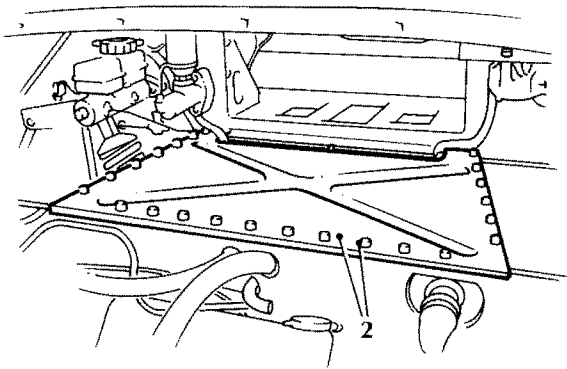
Special Tools

The following special tools will be required

Torque wrench 0 to 23Nm (17ftlb)

- (1) Locate the lower steering column through the seal assembly and secure to the lower universal joint with bolt, washer and nut. Lubricate the seal with red rubber grease.
- (2) Torque load nut 20 to 23 Nm (15 to 17 ftlb).
- (3) Fit the upper steering column and steering wheel.
- (4) Fit wheel arch liner and wheel. (Refer to Road Wheels, Section 4).





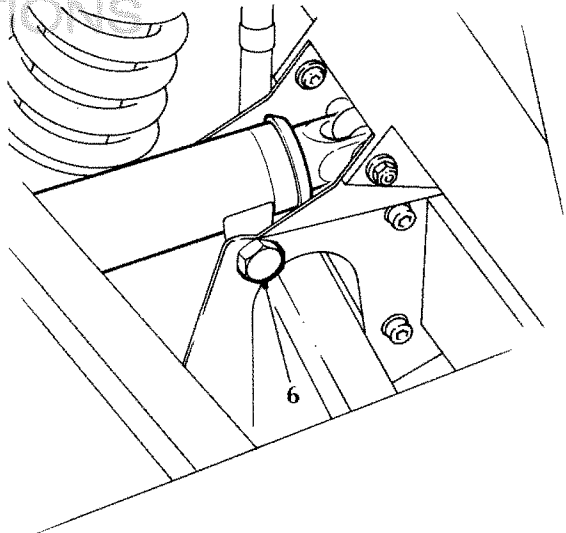
4 STEERING RACK

A Removal Procedure

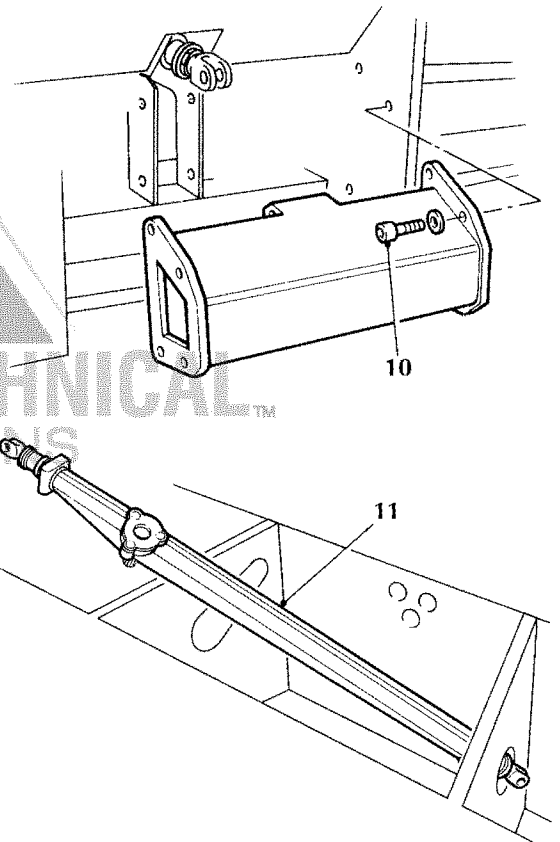
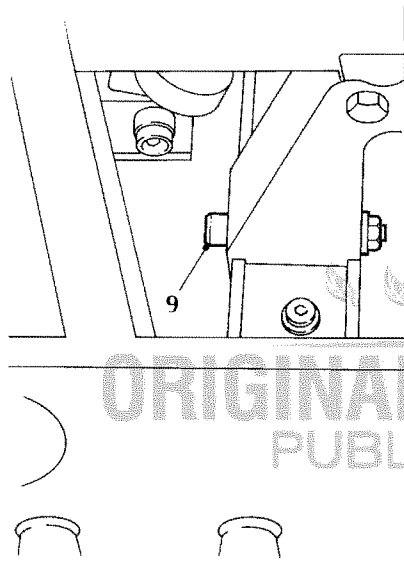
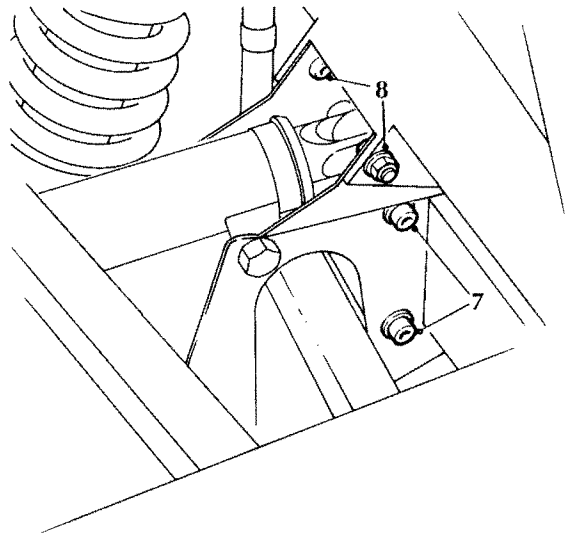
NOTES:

- 1 Ensure absolute cleanliness of work area and tools.
- 2 Observe correct use of tools.
- 3 Fit Jaguar approved protective body covers.

- (1) Remove the front wheels and wheel arch liners (Refer to Road Wheels, Section 4).
- (2) Remove the cap head screws and remove the cover from the structural hatch.
- (3) Release the gaiters from the rack housing and slide back to gain access to the inner rod joint.
- (4) Remove the bolts, washers and nuts and remove the tie rods from the steering rack.
- (5) Remove the bolts, washers and nuts securing the lower universal joint to the steering rack and lower column.
- (6) Remove the bolts, washers and nuts securing the steering rack to the mounting brackets.



- (7) Remove the two bolts, washers and nuts securing the steering rack mounting brackets to the crush tubes.
- (8) Remove the button head screws securing the steering rack mounting brackets to the upper wishbone supports.
- (9) Remove the bolts and washers securing the steering rack brackets to the lower wishbone supports, remove the steering rack mounting brackets and collect the spacers from the wishbone supports.
- (10) Remove cap head screws and remove RH crush tube for LHD or LH crush tube for RHD from between suspension and zero bulkhead.
- (11) Remove steering rack through the structural hatch.



B Installation Procedure

NOTES:

- 1 *Ensure absolute cleanliness of work area and tools.*
- 2 *Observe correct use of tools.*
- 3 *Fit Jaguar approved protective body covers.*

Special Tools

The following special tools will be required

Torque wrench 0 to 65Nm (48ftlb)

- (1) Locate the steering rack in the vehicle.
- (2) Locate the spacer, using an M4 screw fitted into the side of the spacer to aid location, and attach the steering rack bracket to the lower wishbone support. Torque the bolt 20 to 23Nm (15 to 17ftlb).
- (3) Fit the steering rack mounting bracket to the upper wishbone support using M6 screws, washers and nuts and M5 screw.
- (4) Torque load M6 screws 7.5 to 9Nm (5.5 to 6.5ftlb) and M5 screw 5 to 6Nm (3.5 to 4.5ftlb).
- (5) Fit crush tube between suspension and zero bulkhead and secure with cap head screws.
- (6) Torque load cap head screws 8 to 9.5Nm (6 to 7ftlb).
- (7) Secure the steering rack bracket to the crush tube with bolts, washers and nuts.
- (8) Torque load nuts 12 to 14Nm (9 to 10.5ftlb).
- (9) Secure steering rack to mounting bracket with bolts, washers and nuts.
- (10) Torque load bolts and nuts 57 to 65Nm (42 to 48 ftlb).
- (11) Fit universal joint to steering rack and lower steering column and secure with bolts, washers and nuts.
- (12) Torque load bolt and nut 20 to 23Nm (15 to 17ftlb).
- (13) Fit tie rods to steering rack and secure with bolt, washers and nut.
- (14) Torque load bolt nut 35 to 40Nm (26 to 30ftlb) and fit rubber boot to the rack housing.
- (15) Fit structural hatch and secure with cap head screws and washers.
- (16) Torque load cap head screws 12 to 14Nm (9 to 10.5ftlb).
- (17) Fit wheel arch liner and wheel. (Refer to Road Wheels, Section 4).
- (18) Check wheel alignment (Refer to Wheel Alignment and Camber, Section 4).

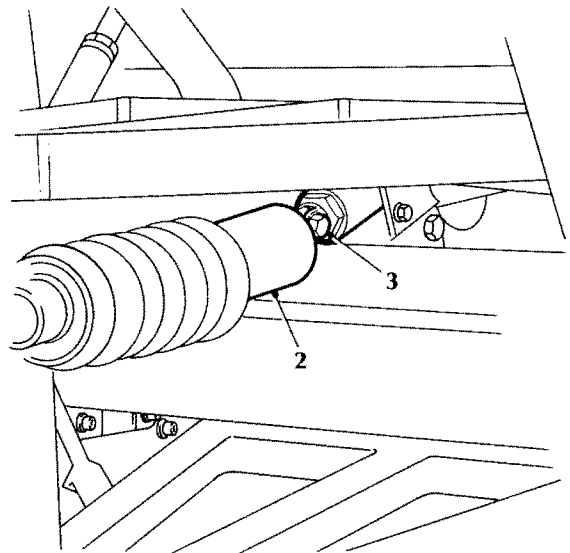
5 TIE RODS

A Removal Procedure

NOTES:

- 1 Ensure absolute cleanliness of work area and tools.
- 2 Observe correct use of tools.
- 3 Fit Jaguar approved protective body covers.

- (1) Remove the relevant wheel and wheel arch liner. (Refer to Road Wheels, Section 4).
- (2) Release the cable tie and pull back the steering rack gaiter to gain access to the tie-rod inboard fixings.
- (3) Remove the bolt, nut and washer securing the tie-rod to the steering rack.
- (4) Remove the bolt, nut and washers securing the tie-rod to the upright bracket.
- (5) Withdraw the tie-rod from the steering rack and upright bracket and collect the spacers from the bracket.



B Installation Procedure

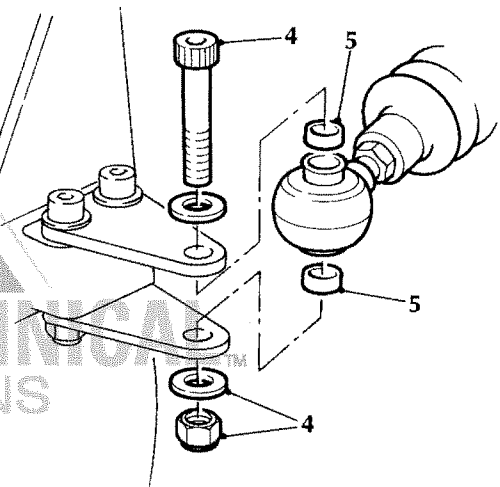
Special Tools

Torque wrench 0 - 41Nm (30ftlb)

NOTE

If a new tie-rod is to be fitted, fit a new gaiter prior to fitting the tie-rod.

- (1) Position the spacers in the upright bracket and fit the tie-rod to the steering rack and upright.
- (2) Secure the tie-rod to upright using 3/8in UNF bolt, washers and nut.
- (3) Torque bolt and nut 35 to 41Nm (26 to 30ftlb).
- (4) Secure the tie-rod to the steering rack using 3/8in UNF bolt, washer and nut.
- (5) Torque bolt and nut 35 to 41Nm (26 to 30ftlb).
- (6) If a new tie-rod has been fitted, or the tie rods have been adjusted, check wheel alignment. (Refer to Wheels and Tyres, Section 4).
- (7) Fit wheel arch liner and road wheel. (Refer to Road Wheels, Section 4).



HEATING AND VENTILATION

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DESCRIPTION

The heating, ventilation and air conditioning (HVAC) functions of the vehicle are formed by two systems.

The air conditioning system comprises of an engine mounted refrigerant compressor, a condenser/receiver drier bottle and an evaporator. The condenser/receiver bottle is located forward of the radiator and the evaporator is located within the HVAC unit. The HVAC unit being located under the facia/dash top panels.

The heating system comprises of an electric water pump, a flow control valve (located in front of the HVAC unit) and a heater matrix (located within the HVAC unit).

Fresh, ambient temperature, air is drawn from the front of the vehicle and passes through both the evaporator and heater matrices and is ducted to the screen, facia or floor depending upon the control. In the 'RECIRC' mode or 'A/C MAX' control mode, only a small quantity of fresh air is drawn into the vehicle, the remainder of the air being drawn from within the occupant area and recirculated through the HVAC unit.

Operating control of both systems is achieved via three facia control selectors; temperature, blower fan speed and mode control switch which controls both air conditioning and air distribution.

ORIGINAL TECHNICAL
PUBLICATIONS

INTRODUCTION

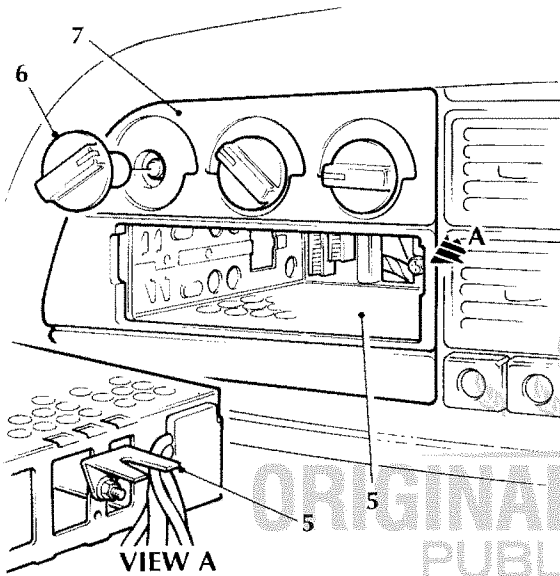
This section details the procedures for replacement of heating and ventilation system components. Replacement of components or disassembly instructions not covered in this section require the component be returned to JaguarSport.

1 HVAC CONTROLS

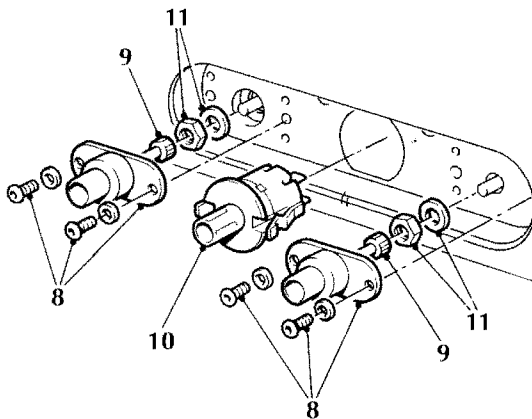
A Removal Procedure

NOTES:

- 1 Ensure absolute cleanliness of work area and tools.
- 2 Observe correct use of tools.

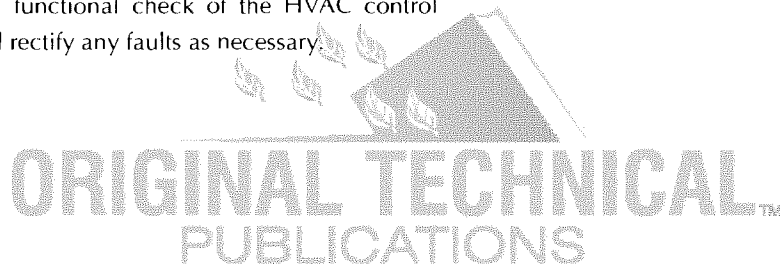


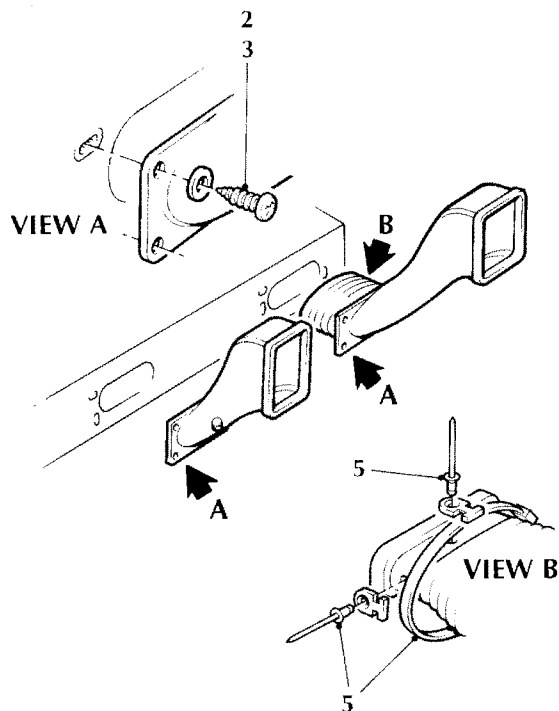
- (1) Fit JaguarSport seat covers.
- (2) Disconnect the battery.
- (3) Remove the passenger side underscuttle (Refer to Section 2).
- (4) Pull out the audio unit.
- (5) Remove the bolt at the rear of the audio unit cage, straighten the front locking tabs and remove the audio unit cage.
- (6) Pull off the three HVAC selector control knobs.
- (7) Insert a screwdriver behind the HVAC fascia panel and carefully prise off the panel.
- (8) Remove the screws securing the mode and temperature control spindle adaptor cages and remove the cages.
- (9) Loosen the grub screws and remove the spindle adaptors.
- (10) Remove the blower speed control and disconnect the electrical connections.
- (11) Remove the nuts and washers securing the HVAC control board to the fascia and remove the board through the audio unit aperture.



B Installation Procedure

- (1) Fit the HVAC control board through the audio unit aperture and locate in the fascia.
- (2) Fit the washers and nuts and secure the board to the fascia.
- (3) Electrically connect the blower speed control and fit to the fascia.
- (4) Fit the spindle adaptors and secure the grub screws.
- (5) Fit the spindle cages and secure to the fascia with screws.
- (6) Apply new adhesive and fit the HVAC fascia panel.
- (7) Fit the selector knobs.
- (8) Fit the audio unit cage and secure the bolt and locking tabs.
- (9) Fit the audio unit.
- (10) Fit the passenger side underscuttle. (Refer to Section 2).
- (11) Connect the battery and remove the seat covers.
- (12) Carryout a functional check of the HVAC control system and rectify any faults as necessary.





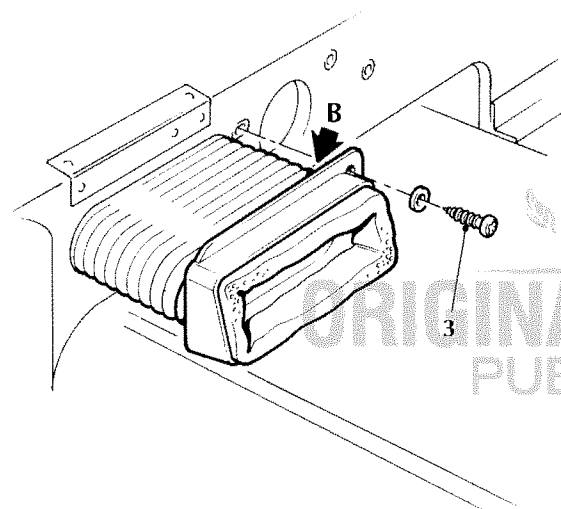
2A FACE LEVEL VENT DUCTS

A Removal Procedure

NOTES:

- 1 Ensure absolute cleanliness of work area and tools.
- 2 Observe correct use of tools.

- (1) Remove the instrument panel, binnacle, driver and passenger underscuttle and passenger vent panel (Refer to Section 2)
- (2) Remove the screws attaching the centre duct to the honeycomb and remove the duct.
- (3) Remove the screws attaching the driver and passenger ducts to the honeycomb.
- (4) Pull the ducts and distribution hoses through the outboard apertures to gain access to the hose attachments.
- (5) Remove the tyrap and carefully drill out the rivets attaching the hoses and cable tie mounting blocks.
- (6) Remove the ducts.



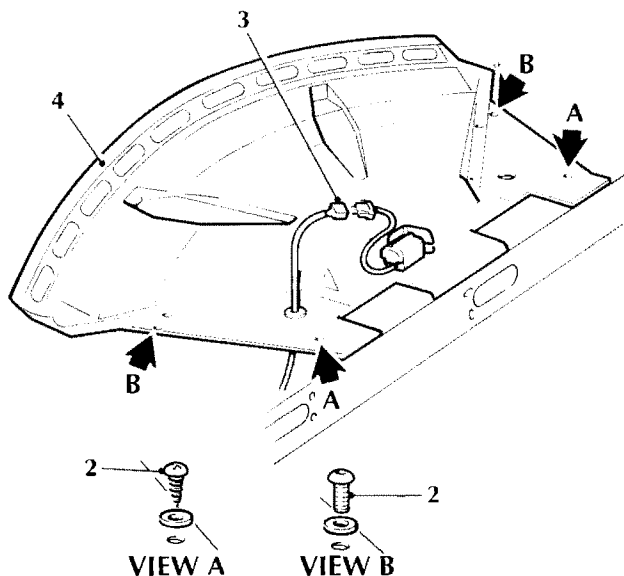
2B DUCT ASSEMBLY - DEFROST SPIDER

A Removal Procedure

NOTES:

- 1 Ensure absolute cleanliness of work area and tools.
- 2 Observe correct use of tools.

- (1) Remove the instrument panel, binnacle, driver and passenger under scuttle, passenger vent panel and windscreen base panel (Refer to Section 2).
- (2) Remove the retaining self tapping and machine screws.
- (3) Disconnect the screen flap electrical connections.
- (4) Remove the defrost spider from the vehicle.



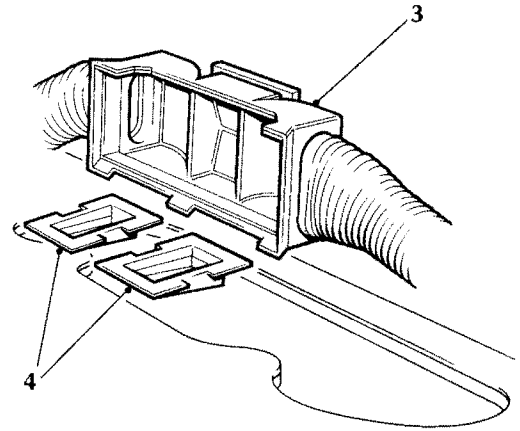
2C AIR DISTRIBUTION DUCT ASSEMBLY AND FOOTWELL DUCTS

A Removal Procedure

NOTES:

- 1 Ensure absolute cleanliness of work area and tools.
- 2 Observe correct use of tools.

- (1) Remove the face level ducts and defrost spider (Refer to Section 12.2A and 12.2B).
- (2) Remove the HVAC unit (Refer to Section 12.3).
- (3) Pull the distribution duct assembly hoses through the apertures and remove the duct assembly.
- (4) Lift out the footwell ducts.



AIR CONDITIONING DUCT SYSTEM

B Installation Procedure

- (1) Check the foam seal around the distribution duct assembly is not twisted or torn. If in doubt renew the foam seal.
- (2) Remove all traces of old adhesive from the footwell duct assemblies and fit new adhesive strips.
- (3) Fit the footwell duct assemblies.
- (4) Fit the distribution duct assembly and route the hoses through the vent apertures.
- (5) Fit the driver and passenger ducts to the hoses.
- (6) Secure the ducts and hoses using rivets and cable tie mounts. Fit and secure the tyrap.
- (7) Fit the ducts to the honeycomb and secure with screws.
- (8) Fit the centre duct to the honeycomb and secure with screws.
- (9) Fit the HVAC unit. (Refer to Section 12.3).
- (10) Fit the defrost spider duct assembly, fasten and reconnect the screen flap electrical connection.
- (11) Ensure the screen flap operates satisfactorily.
- (12) Check that the system is properly sealed with no or minimal air leaks.
- (13) Fit all trimmed dashboard panels (Refer to Section 2).

GENERAL NOTE

When any major air conditioning component is changed/renewed or there has been an obvious oil leak, the following procedures should be carried out after repairs have been made to the affected components.

After following the recommended refrigerant recovery procedures, the compressor should be removed from the vehicle and placed on a clean bench.

Drain the oil from the compressor via the suction and discharge ports. Drain for as long as possible to ensure that the maximum amount of oil is drained from the compressor. Rotate the compressor crankshaft several times whilst draining the oil.

Using a total of 130cc of clean refrigerant oil, Suniso 5gs (P/No.9985), charge 65cc of oil through each of the suction and discharge ports.

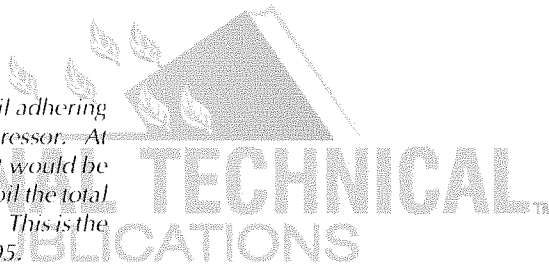
NOTE:

Generally, there will be a certain amount of oil adhering to the internal working surfaces of the compressor. At ambient temperature of 20-25°C, this amount would be approximately 20cc, and by adding 130cc of oil the total amount of oil in the compressor will be 150cc. This is the factory charged oil amount of a typical TR 105.

As the compressor now has a factory oil charge, it is necessary to clean oil from the remaining components before fitting the compressor back to the system. If this is not done there is a possibility of an overcharge of oil which could result in a performance reduction of the air conditioning system.

As a guide to system flushing the following procedure should be followed;

- Recover the refrigerant, disconnect the suction and discharge lines from the compressor.
- Remove the compressor.
- Remove and discard the receiver drier bottle.
- Flush the system with a suitable flushing agent (R11).
- Install a new receiver drier bottle and other components as necessary.
- Complete the oil charge procedure and refit the compressor.
- De-hydrate and charge the system with the 1.08kg of R12.



3 HEATING, VENTILATION AND AIR CONDITIONING UNIT (HVAC)

A Removal Procedure

Special Tools

The following special tools will be required;

Air Conditioning service unit

JaguarSport protective body covers

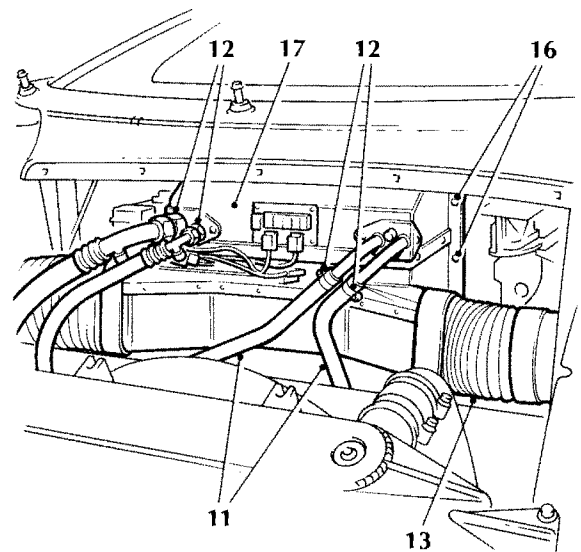
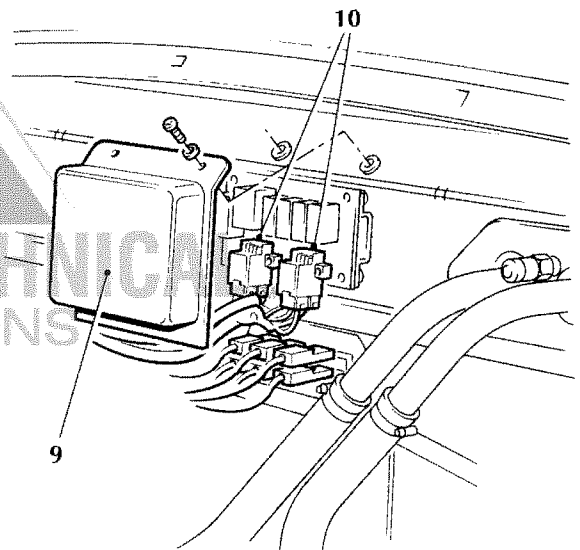
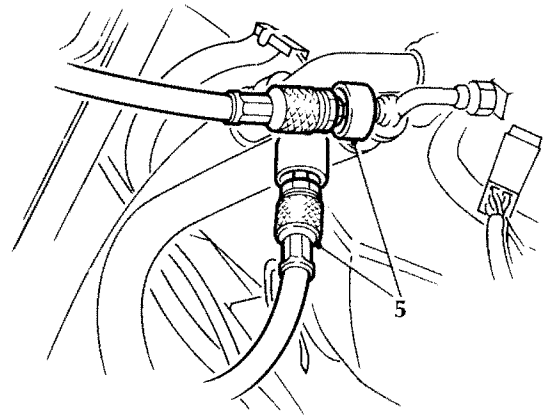
Refrigerant leak detector.

NOTES:

1 Ensure absolute cleanliness of work area and tools.

2 Observe correct use of tools.

- (1) Switch on the ignition and set the HVAC distribution control to face/floor. Switch off the ignition.
- (2) Disconnect the battery.
- (3) Pull bonnet release handle and open bonnet.
- (4) Remove the radiator duct access panel and suspension box structural hatch.
- (5) Connect the air conditioning service unit to the vehicle evaporator piping service valves as follows in order to remove the refrigerant gas;
 - a) Connect the RED connector to the high pressure pipe (small diameter pipe).
 - b) Connect the BLUE connector to the low pressure pipe (large diameter pipe).
- (6) Start the service unit and evacuate the refrigerant system
- (7) Fit blanking caps to the evaporator service connections on HVAC unit.
- (8) Remove the front interior trim panels and air distribution plenum. (Refer to Section 2).
- (9) Remove the HVAC control module cover.
- (10) Disconnect the electrical harness to the HVAC unit.
- (11) Clamp the water hoses if cooling system is not drained.
- (12) Remove the refrigerant hoses and water pipes from the unit. Fit blanking caps to the open connections.
- (13) Release the hose bands and remove the flexible air distribution hoses from the HVAC unit inlets.
- (14) Disconnect the condensate drain tubes from underneath the HVAC.



- (15) Remove the windscreen wiper spindle links (Refer to Section 2).
- (16) Remove the cap head screws with washers securing the HVAC unit to zero bulkhead.
- (17) Carefully remove HVAC unit ensuring the unit casing and flap motor linkages do not foul on adjacent structure, pipework or passenger side windscreen wiper spindle crank.

B Pre-Installation Check

- (1) Connect the tester to the HVAC control module and harness.
- (2) Connect the tester to a 12 volt electrical supply.
- (3) Select low fan speed.
- (4) Check the HVAC unit flap logic as follows;
Turn the mode selector switch to:-
 - a) Screen Check that air is directed to the screen only.
 - b) Screen/floor Check that air is directed to the screen and footwells.
 - c) Face/floor Check that air is directed to face level and footwells.
 - d) A/C Norm Check that air is directed to face level only.
 - e) A/C Max Check that air is directed to face level.
- (5) On completion select face/floor to pre-set the unit for installation.

C Installation Procedure

NOTES:

- 1 Ensure absolute cleanliness of work area and tools.
- 2 Observe correct use of tools.

Special Tools

The following special tools will be required

Torque wrench 41Nm (30ftlb)

CAUTION

To prevent damage during installation ensure the vent flaps are in the face/floor position.

- (1) Locate the HVAC unit in the vehicle and fit the drain hoses to the under side of the unit.
- (2) Align the HVAC unit face level outlet with the air distribution duct assembly. Ensure the foam seal on the distribution duct is not twisted and gently install until the HVAC unit mounting bracket fully contacts the Zero bulkhead.
- (3) Fit the cap head screws and torque tighten 8 to 9Nm (6 to 7ftlb).
- (4) Fit the suspension box structural hatch.
- (5) Fit the air ducting hoses to the HVAC unit inlets and secure the hose bands.
- (6) Fit the refrigerant hoses and torque tighten the connections as follows:
 - condesate/evaporator feed pipes (#6) 16Nm (12.5ftlb).
 - evaporator/compressor suction pipe (#12) 41Nm (30ftlb).
- (7) Fit the water hoses to the unit and secure the hose clips. Release the hose clamps if the cooling system was not drained.
- (8) Replenish the cooling system, if necessary, and carry out a leak check on the disturbed components. Ensure there are no air locks in the HVAC unit by bleeding the Schrader valve located at the HVAC unit heater feed and return pipe manifolds.
- (9) Fit windscreen wiper spindle links (Refer to Section 2).
- (10) Connect electrical harness to the unit.
- (11) Fit the HVAC control module cover.
- (12) Connect the air conditioning servicing unit (as follows) and charge the air conditioning system.
 - a) Connect the RED connector to the small diameter pipe (high pressure).
 - b) Connect the BLUE connector to the large diameter pipe (low pressure).
- (13) When charging is complete remove the air conditioning service unit and fit blanking caps.
- (14) Fit the defrost spider, duct assemblies and interior front end trim panels. (Refer to Section 2).
- (15) Connect the battery.
- (16) Operate the system and ensure there are no leaks.
- (17) Fit the radiator duct access cover, secure the quick release fasteners and close the bonnet.

4 COMPRESSOR

A Removal Procedure

Special Tools

JaguarSport approved body covers

Electro-hydraulic vehicle lifting apparatus

Supports for the rear axle uprights

Air Conditioning service unit

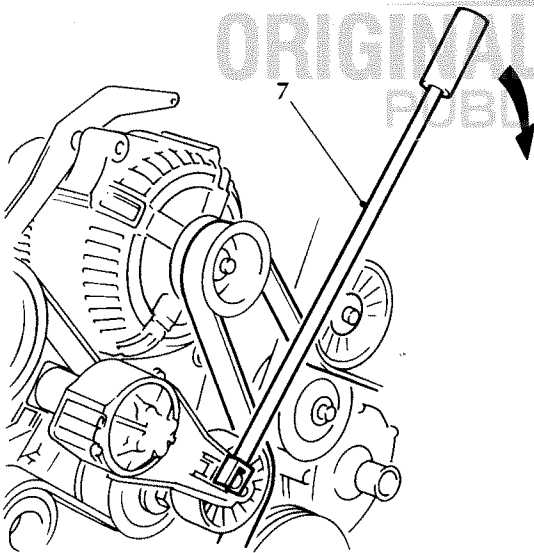
NOTES:

- 1 Ensure absolute cleanliness of work area and tools.
- 2 Observe correct use of tools.

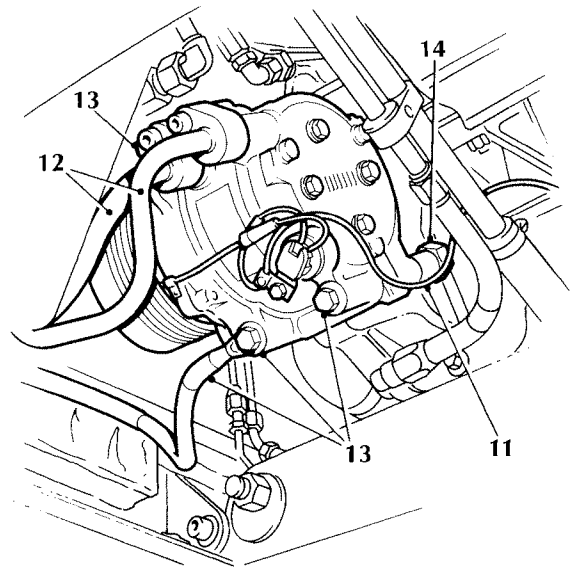
NOTE

The following procedure deals with the removal of the air refrigerant compressor with the engine installed. If the engine has been removed certain procedures may be disregarded.

- (1) Position the vehicle on lifting apparatus. (Refer to Section 1).
- (2) Fit JaguarSport body covers
- (3) Disconnect the battery.
- (4) Open the bonnet and remove the radiator duct access panel.
- (5) Connect the air conditioning service unit to the car and evacuate the air conditioning system (Refer to Section 12.3).
- (6) Raise the vehicle to a comfortable working height and remove the venturi panel (Refer to Section 2).
- (7) Fit 1/2in. square drive wrench in socket hole of dynamic tensioner, if the engine is out of the vehicle. If the engine is installed use a lever to release the spring tension.
- (8) Move the wrench to oppose the spring tension in the dynamic tensioner until the auxiliary drive belt is loose enough to be easily removed.
- (9) Gradually reduce pressure on the wrench to allow the dynamic tensioner to return to a neutral position and remove the wrench.
- (10) Inspect the drive belt for serviceability. If in doubt renew the drive belt.

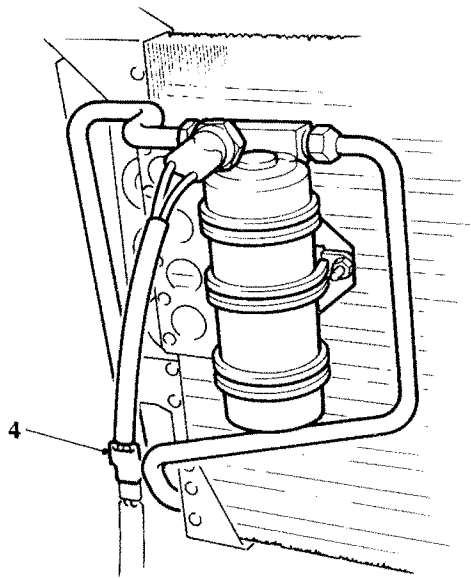


- (11) Disconnect the compressor clutch electrical leads.
- (12) Remove the manifold pipes from the compressor and fit blanking caps to the pipe open ends.
- (13) Support the compressor and remove the three M10 bolts securing the compressor to engine block and water pump and remove the engine earth lead.
- (14) Remove the compressor from the engine and retrieve the three spacers. Note the position of the spacers.



B Installation Procedure

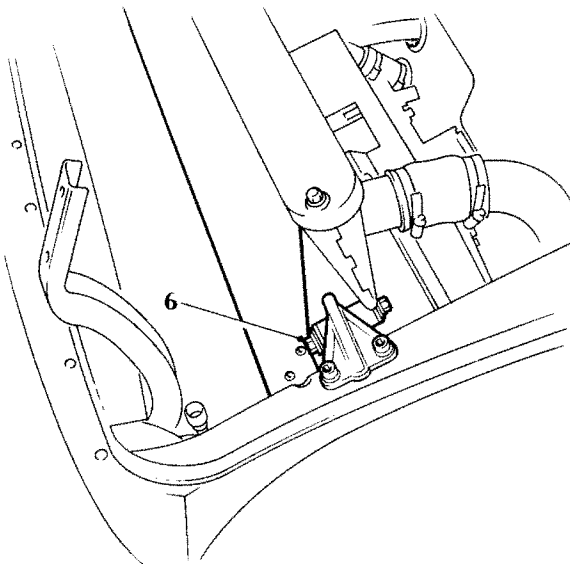
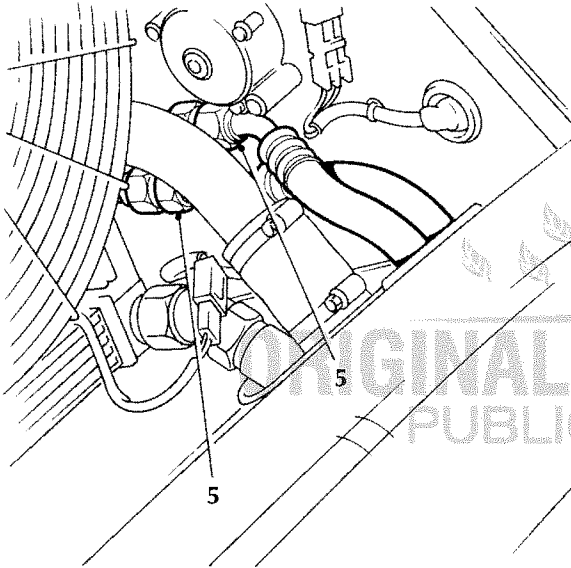
- (1) Remove any blanking caps fitted to the compressor.
- (2) Fit the compressor to engine. Position the spacers and engine earth lead and secure the compressor with the M10 flange head bolts. Torque load bolts 49-54 Nm (36-40 ftlb).
- (3) Remove the blanking caps and lubricate the 'O' ring seals on the manifold with compressor oil (P/No.9985) before fitting the supply pipes to the compressor.
- (4) Connect the compressor clutch electrical leads.
- (5) Fit 1/2in square drive wrench in socket hole of dynamic tensioner or carefully use a lever to relieve spring tension.
- (6) Move wrench in direction to oppose spring pressure.
- (7) Fit drive belt.
- (8) Gradually reduce pressure on wrench to allow dynamic tensioner to take up slack in drive belt.
- (9) Remove wrench.
- (10) Connect the air conditioning service unit and charge the air conditioning system.
- (11) Connect the battery.
- (12) Run the engine and check the air conditioning system for correct function and the compressor for leaks. Rectify any faults as necessary.
- (13) Fit the venturi panel.
- (14) Fit the radiator duct access panel and close the bonnet.
- (15) Remove the vehicle from the lifting apparatus and remove the body covers.



5 CONDENSER

A Removal Procedure

- (1) Remove the bonnet. (Refer to Section 2).
- (2) Disconnect the battery.
- (3) Connect an air conditioning service unit and evacuate the system (Refer to Section 12.3)
- (4) Disconnect the electrical connections to the receiver/drier bottle.
- (5) Disconnect the hoses to the condenser. Fit blanks to the open connections.
- (6) Remove the nuts and washers attaching the condenser to the bracket and carefully lift the condenser from the vehicle.



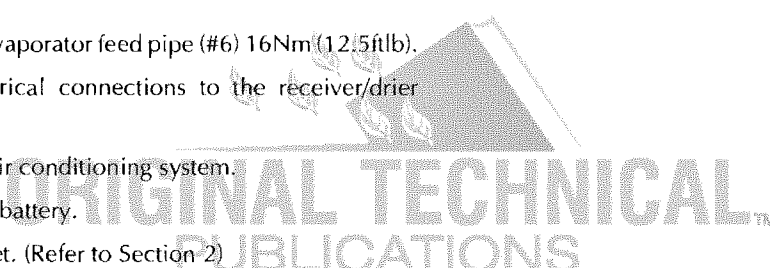
B Installation Procedure

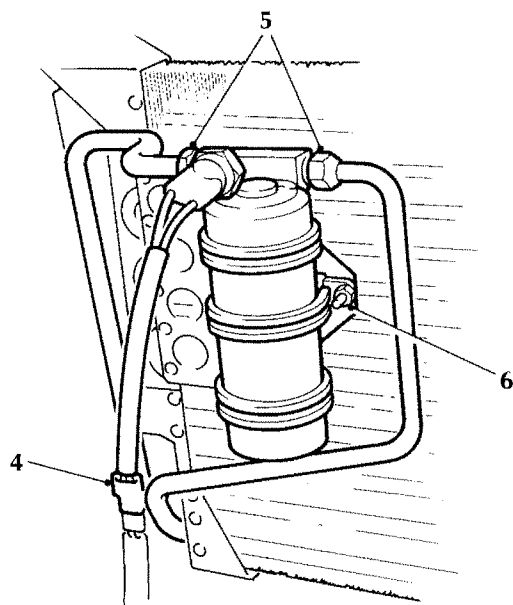
- (1) If fitting a new condenser sub assemble the condenser as follows:
 - a) Locate the RH and LH radiator sealing panels and secure using rivets.
 - b) Locate the top radiator sealing panel and secure using rivets.

CAUTION:

Ensure that the condenser matrix is not damaged during the installation procedure.

- (2) Fit the condenser assembly to the radiator duct.
- (3) Fit the condenser to the mounting bracket and secure with washers and nuts.
- (4) Remove the blanking caps and connect the air conditioning pipes to the condenser and receiver/drier.
- (5) Torque the condenser hose connections as follows:-
compressor/condenser discharge pipe (#8) 24Nm (18ftlb)
condenser/evaporator feed pipe (#6) 16Nm (12.5ftlb).
- (6) Fit the electrical connections to the receiver/drier bottle.
- (7) Charge the air conditioning system.
- (8) Connect the battery.
- (9) Fit the bonnet. (Refer to Section 2)
- (10) Carryout a functional check of the air conditioning system. Rectify any faults as necessary.





6 RECEIVER/DRIER BOTTLE

A Removal Procedure

- (1) Disconnect the battery.
- (2) Connect an air conditioning service unit and evacuate the system.
- (3) Remove the front grille to gain access to the receiver/drier. (Refer to Section 2).
- (4) Disconnect the electrical harness to the receiver/drier.
- (5) Disconnect the pipes from the receiver/drier. Fit blanks to the open connections.
- (6) Loosen the clamp nuts and slide out the receiver drier. Note the direction of refrigerant flow.

NOTE

The direction of refrigerant flow is indicated by an arrow engraved on the top of the receiver/drier bottle

B Installation Procedure

- (1) Fit the receiver/drier to the clamps and tighten the nuts. Ensure the direction of flow is correct.

NOTE

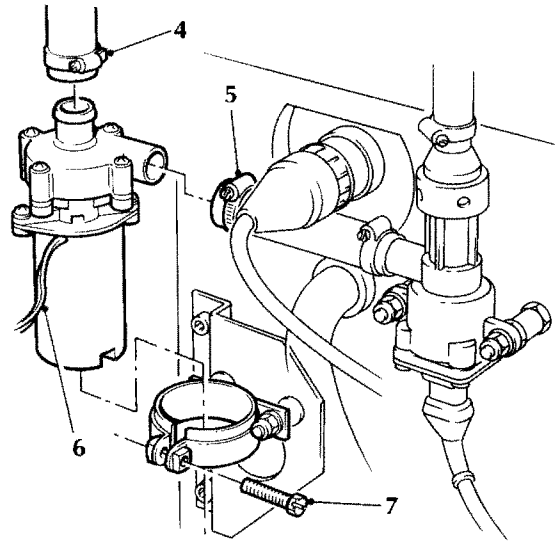
The direction of refrigerant flow is indicated by an arrow engraved on the top of the receiver/drier bottle

- (2) Remove the blanking caps and fit the pipes to the receiver/drier.
- (3) Connect the electrical harness to the receiver/drier.
- (4) Fit the front grille. (Refer to Section 2).
- (5) Charge the air conditioning system.
- (6) Connect the battery.
- (7) Carryout a functional check on the air conditioning system and rectify any faults as necessary.

7 HEATER WATER PUMP

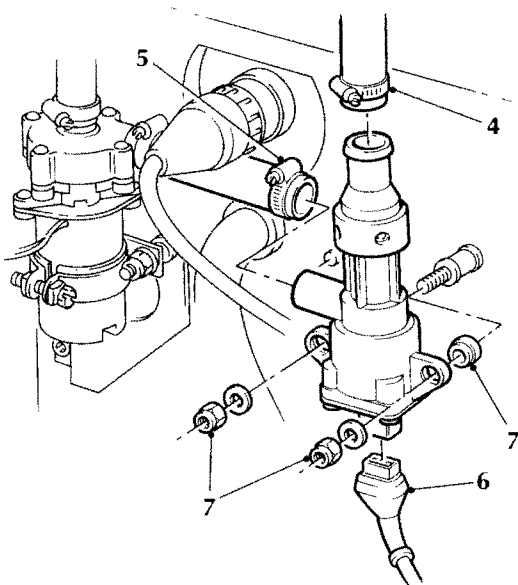
A Removal Procedure

- (1) Open the bonnet and remove the radiator duct access panel.
- (2) Disconnect the battery.
- (3) Drain the cooling system (Refer to Section 3).
- (4) Loosen the hose clip and remove the hose from the top of the pump. Fit blanking caps to the hose to prevent excessive loss of fluid.
- (5) Loosen the hose clip and remove the hose from the pump to the water valve. Fit blanking caps to the open connections.
- (6) Disconnect the electrical connection to the pump.
- (7) Loosen the pump clamp bolt and remove the pump.



B Installation Procedure

- (1) Fit the pump to the clamp and secure the bolt.
- (2) Remove the blanking caps and fit the hose from the water valve to the pump and secure the hose clip.
- (3) Remove the blanking caps and fit the hose to the top of the pump and secure the hose clip.
- (4) Connect the electrical harness to the pump.
- (5) Connect the battery.
- (6) Replenish the coolant system, if necessary, and carry out a leak check on the disturbed components. Ensure there are no air locks in the HVAC unit by bleeding the Schrader valve located at the HVAC unit heater feed and return pipe manifolds.
- (7) Fit the radiator duct access panel and close the bonnet.



8 WATER VALVE

A Removal Procedure

- (1) Open the bonnet and remove the radiator duct access panel.
- (2) Disconnect the battery.
- (3) Drain the cooling system (Refer to Section 3)
- (4) Loosen the hose clip and remove the top hose from the water valve. Fit blanking caps to open connections.
- (5) Loosen the hose clip and remove the hose from the valve to the pump. Fit blanking caps to the open ends.
- (6) Disconnect the electrical connections to the valve.
- (7) Remove the nuts and washers attaching the valve to the rubber mounts, remove the valve and collect the adaptors.

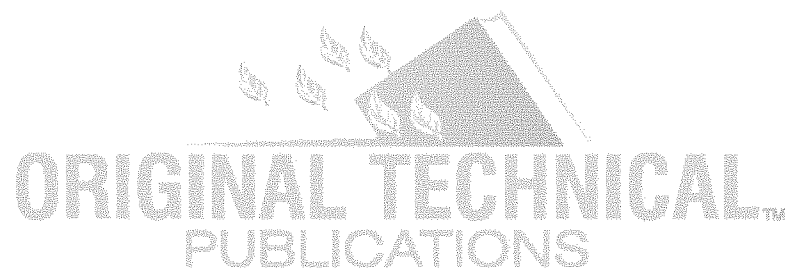
B Installation Procedure

- (1) Fit the adaptors to the valve and fit the valve to the rubber mounts. Secure the pump with washers and nuts.
- (2) Fit the hose from the pump to the valve and secure the hose clip.
- (3) Fit the top hose to the valve and secure the hose clip.
- (4) Connect the electrical harness to the valve.
- (5) Connect the battery.
- (6) Replenish the coolant system, if necessary, and carry out a leak check on the disturbed components. Ensure there are no air locks in the HVAC unit by bleeding the Schrader valve located at the HVAC heater feed and return pipe manifolds.
- (7) Fit the radiator duct access panel and close the bonnet.

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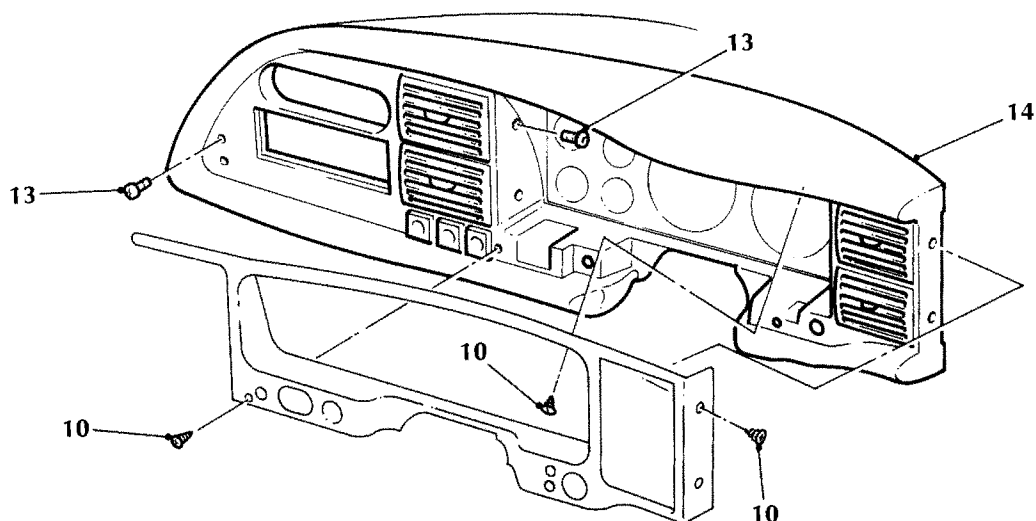
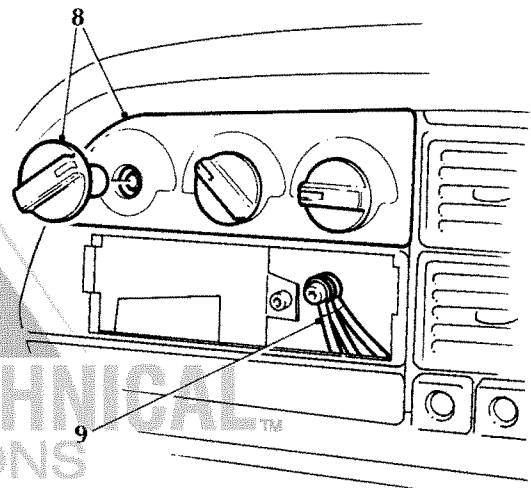
1 INSTRUMENT PANEL ASSEMBLY

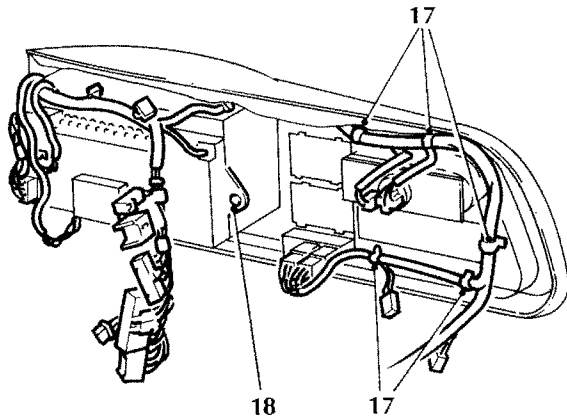
A Removal Procedure

NOTES:

- 1 Ensure absolute cleanliness of work area and tools.
- 2 Observe correct use of tools.

- (1) Fit JaguarSport seat covers.
- (2) Disconnect the battery.
- (3) Pull off the steering wheel centre trim.
- (4) Remove the nut and washer securing the steering and boss assembly boss to the steering column and remove the steering wheel assembly.
- (5) Remove the steering column switch covers.
- (6) Remove the driver and passenger underscuttles. (Refer to Section 2).
- (7) Loosen the steering column upper attachment bolts sufficiently to facilitate removal of the fascia assembly.
- (8) Remove the A/C control knobs and fascia panel. (Refer to Section 12).
- (9) Disconnect the fascia harness earth connections.
- (10) Remove the 6 screws attaching the instrument panel finisher, remove the finisher and lift out the mirror switch.
- (11) Disconnect the exterior mirror switch electrical connections. (Refer to Section 13, sub-section 4).
- (12) Disconnect the alarm LED electrical connections from the harness.
- (13) Remove the six screws attaching the fascia assembly to the dash surround.
- (14) Ease the fascia assembly forwards and disconnect the fascia harness and audio equipment connections.





- (15) Disconnect the air conditioning air thermister harness.
- (16) Carefully lift the facia assembly out of the vehicle.

CAUTION

The instruments are silicone damped. never lay the instrument panel face down, always store the instrument panel upright.

- (17) Remove the facia harness clips.
- (18) Remove the four screws securing the instrument panel to the facia and remove the instrument panel.

B Installation Procedure

NOTES:

- 1 Ensure absolute cleanliness of work area and tools.
- 2 Observe correct use of tools.

- (1) Locate the instrument panel in the facia and secure with screws.
- (2) Secure the harness to the facia using clips.
- (3) Locate the facia assembly in the vehicle and connect the air conditioning air thermister harness.
- (4) Connect the facia/main harness connections.
- (5) Secure the facia assembly to the dash surround with screws.
- (6) Connect the facia harness earth leads.
- (7) Connect and fit the audio cage to the facia.
- (8) Fit the mirror switch to the finisher.
- (9) Connect the mirror switch electrical connectors and alarm LED to facia harness.
- (10) Fit the instrument panel finisher and secure with 6 screws.
- (11) Fit the HVAC controls (Refer to Section 12).
- (12) Torque tighten the steering column upper attachment bolts.
- (13) Fit the driver and passenger underscuttles (Refer to Section 2).
- (14) Fit the steering column switch covers.
- (15) Fit the steering wheel to the boss using countersunk bolts. Torque tighten the bolts and fit the steering wheel centre trim.
- (16) Fit the audio unit.

- (17) Connect the battery.
- (18) Road test the vehicle and ensure all gauges, switches and warnings operate correctly. Rectify any faults as necessary.
- (19) Remove the seat covers.

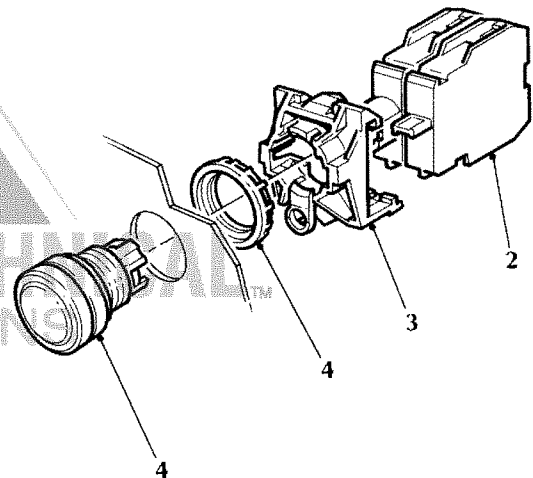
2 STARTER SWITCH ASSEMBLY

A Removal Procedure

NOTES:

- 1 Ensure absolute cleanliness of work area and tools.
- 2 Observe correct use of tools.

- (1) Remove the instrument panel assembly. (Refer to Sub-section 1 of this section).
- (2) Unclip and remove the starter switch contact element.
- (3) Unclip and remove the starter switch fixing adaptor.
- (4) Unscrew the plastic ring attaching the lamp holder to the push button switch and remove the push button.



B Installation Procedure

NOTES:

- 1 Ensure absolute cleanliness of work area and tools.
- 2 Observe correct use of tools.

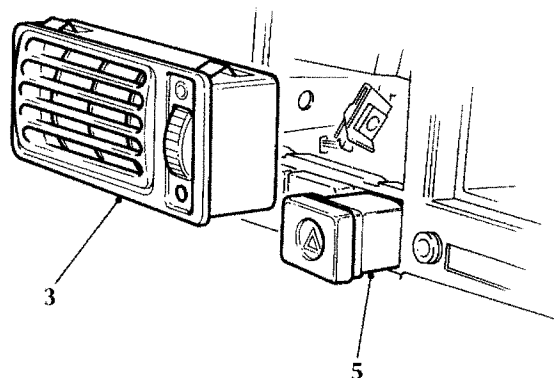
- (1) Fit the push button through the finisher aperture.
- (2) Fit the lamp holder to the push button and secure the plastic ring.
- (3) Clip the switch fixing adaptor to the holder.
- (4) Clip the switch contact element to the adaptor.
- (5) Fit the instrument panel assembly.

3 HAZARD WARNING AND REAR FOG LAMP SWITCHES

A Removal Procedure

NOTES:

- 1 Ensure absolute cleanliness of work area and tools.
- 2 Observe correct use of tools.



- (1) Fit JaguarSport seat covers.
- (2) Disconnect the battery.
- (3) Carefully pull out the facia vent being careful not to damage the facia trim panels.
- (4) Working through the vent apertures, partially push the switches forwards out of their housings.
- (5) Disconnect the harness switch connectors from the switches and fully remove the switches.

B Installation Procedure

NOTES:

- 1 Ensure absolute cleanliness of work area and tools.
- 2 Observe correct use of tools.

- (1) Partially fit the switches to the facia panel.
- (2) Working through the vent apertures, connect the switch harness connectors to the switches.
- (3) Push the switches fully home into the facia panel.
- (4) Connect the battery.
- (5) Ensure the switches function correctly.
- (6) Fit the centre air conditioning vents.

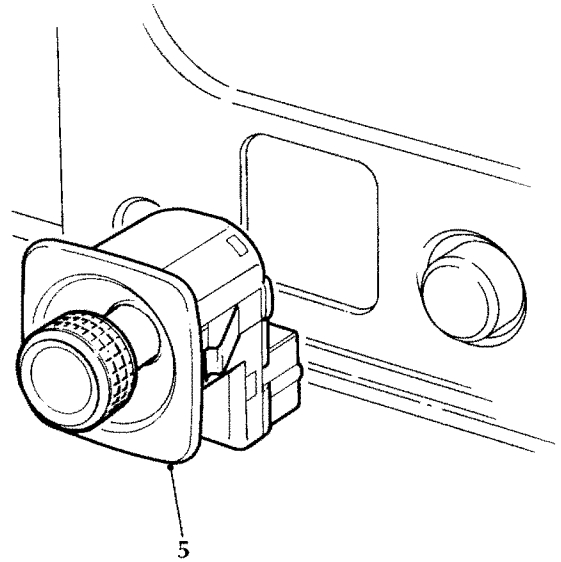
4 EXTERIOR MIRROR SWITCH

A Removal Procedure

NOTES:

- 1 Ensure absolute cleanliness of work area and tools.
- 2 Observe correct use of tools.

- (1) Fit JaguarSport seat covers.
- (2) Disconnect the battery.
- (3) Remove the screws attaching the instrument panel finisher.
- (4) Ease the finisher away from the instrument panel and disconnect the electrical connection to the switch and LED.
- (5) Press the spring retaining clips together and remove the switch from the finisher.



B Installation Procedure

NOTES:

- 1 Ensure absolute cleanliness of work area and tools.
- 2 Observe correct use of tools.

- (1) Locate the switch in the finisher and press fully home.
- (2) Align the finisher to the instrument panel and connect the switch electrical connector.
- (3) Fit the finisher and secure with screws.
- (4) Connect the battery.
- (5) Carryout a functional check of the switch.
- (6) Remove the seat covers.

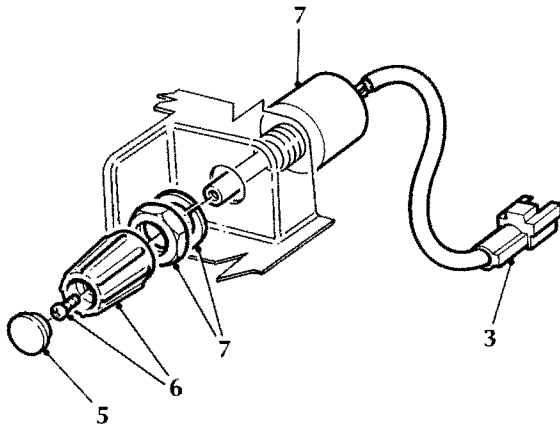
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5 INSTRUMENT PANEL DIMMER SWITCH

A Removal Procedure

NOTES:

- 1 Ensure absolute cleanliness of work area and tools.
- 2 Observe correct use of tools.



- (1) Remove the driver's underscuttle (Refer to Section 2).
- (2) Disconnect the battery.
- (3) Disconnect the switch electrical connection from the harness.
- (4) Remove the instrument panel finisher (Refer to Section 2).
- (5) Insert a small screw driver into the top of the dimmer switch control knob and carefully prise off the top of the control knob.
- (6) Loosen the screw inside the control knob and remove the knob from the spindle.
- (7) Remove the nut and lockwasher attaching the switch to the facia and remove the switch.

B Installation Procedure

NOTES:

- 1 Ensure absolute cleanliness of work area and tools.
- 2 Observe correct use of tools.

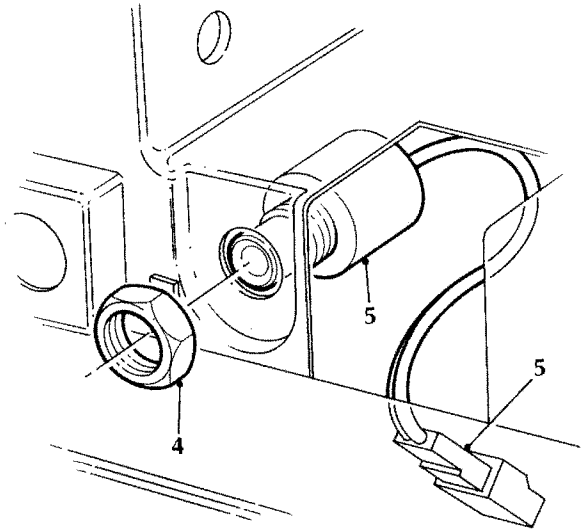
- (1) Insert the switch from the rear of the facia.
- (2) Secure the switch to the facia using a lockwasher and nut.
- (3) Fit the control knob and tighten the screw to secure the knob to the spindle.
- (4) Press the top of the control knob into position.
- (5) Connect the switch electrical connector to the harness.
- (6) Connect the battery.
- (7) Carry out a functional check of the dimmer switch and rectify any faults as necessary.
- (8) Fit the instrument panel finisher (Refer to Section 13.1).
- (9) Fit the driver's underscuttle (Refer to Section 2).

6 ODOMETER TRIP RESET SWITCH

A Removal Procedure

NOTES:

- 1 Ensure absolute cleanliness of work area and tools.
 - 2 Observe correct use of tools.
-
- (1) Disconnect the battery.
 - (2) Remove the driver's underscuttle (Refer to Section 2).
 - (3) Remove the instrument panel finisher (Refer to Section 2).
 - (4) Remove the nut securing the switch to the facia.
 - (5) Remove the switch from the rear of the facia and disconnect the electrical connections.



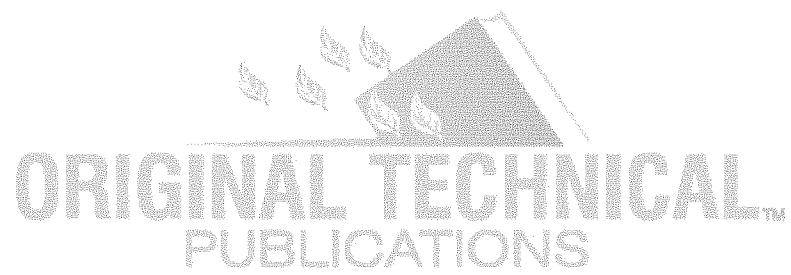
B Installation Procedure

NOTES:

- 1 Ensure absolute cleanliness of work area and tools.
 - 2 Observe correct use of tools.
-
- (1) Fit the electrical connections to the switch.
 - (2) Working through the underscuttle aperture fit the switch to the rear of the facia and secure using a thin nut.
 - (3) Connect the battery.
 - (4) Carryout a functional check of the trip switch and rectify any faults as necessary.
 - (5) Fit the instrument panel finisher (Refer to Section 2).
 - (6) Fit the driver's underscuttle (Refer to Section 2).

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INTRODUCTION

This section details the procedures for removal and installation of electrical components. Removal of components or disassembly instructions not covered in this section require the component be returned to manufacturer.

1 BATTERY

A Removal Procedure

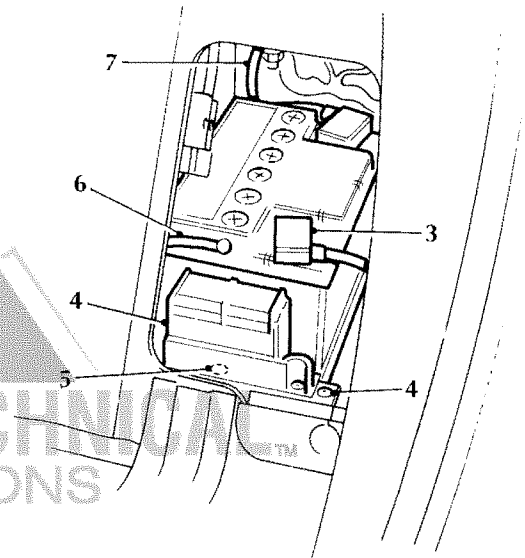
NOTE

- 1 *Ensure absolute cleanliness of work area and tools.*
- 2 *Observe the correct use of tools.*

- (1) Turn the main isolator switch to the off position.
- (2) Remove the battery cover.
- (3) Slacken and remove the negative (black) lead from the battery terminal post.
- (4) Remove the screw securing the maxi fuse box and move the fuse box out of the way to expose the battery clamp.
- (5) Remove the screw from the battery clamp and remove the clamp.
- (6) Remove the battery breather pipe.
- (7) Unscrew and remove the positive (red) lead from the bulkhead terminal post taking care not to loose the retaining nut and washer.
- (8) Carefully lift the battery out of the vehicle.

NOTE:

Be careful when lifting out the battery as it is a tight fit.

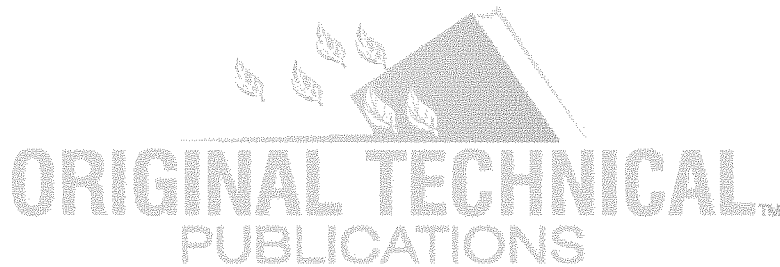


B Installation Procedure

- (1) Ensure the positive (red) lead is secured to the battery. Note the correct position of the clamp/lead.
- (2) Fit the battery into the compartment.
- (3) Connect the battery positive lead to the bulkhead terminal post with the retaining nut and washer.

CAUTION:

- 1. Prior to connecting the battery negative terminal post, check there are no short circuits between cables and body.***
 - 2. Ensure the terminal cover is fitted to the maxi fuse box.***
- (4) Connect the negative (black) lead to the battery terminal post and reconnect the breather pipe.
 - (5) Fit the maxi fuse box and secure with the screw.
 - (6) Attach the battery clamp to the battery and secure with the screw.
 - (7) Fit the battery cover and turn the main isolator switch to the on position.



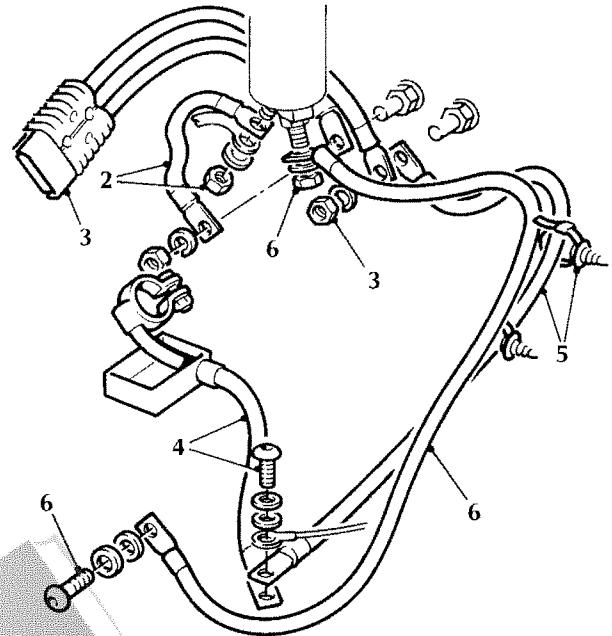
2 BATTERY CABLES

A Removal Procedure

NOTE

- 1 Ensure absolute cleanliness of work area and tools.
- 2 Observe the correct use of tools.

- (1) Remove the battery (Refer to Section 14.1)
- (2) Remove the nut and washer from the isolator switch incoming terminal (battery side) and remove the positive cable and the main harness breakout leads.
- (3) Remove the nut and washer from the negative terminal post and remove the charging cable adapter assembly.
- (4) Remove the main chassis earth point screw and remove the battery negative cable.
- (5) Remove the cable ties securing the negative terminal post to main chassis earth point cable and remove the cable.
- (6) Remove the nut and washer from the isolator switch outgoing terminal, the maxi fuse box positive terminal cover and screw and remove the cable ties. Remove the cable



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B Installation Procedure

- (1) Locate the isolator switch to positive terminal post cable on to the isolator switch incoming terminal and the inboard positive terminal post.
- (2) Fit the breakout leads from the main harness to the isolator switch incoming terminal and loosely secure the nut and washer.
- (3) Fit the isolator switch to maxi fuse box cable onto the outgoing isolator switch terminal and loosely secure with the nut and washer. Secure to body (with main harness) using cable ties.
- (4) Fit one end of the negative terminal post to chassis ground cable on to the outboard negative terminal post. Secure to the body using cable ties.
- (5) Locate the positive lead of the charging adapter on to the inboard positive terminal post and loosely secure with the washer and nut.
- (6) Locate the negative lead of the charging adapter on to the negative terminal post and loosely secure with the washer and nut.
- (7) Fit the battery negative cable, together with the free end of the negative terminal post to chassis ground cable and the main harness earth eyelets, to the main chassis earth point securing with screw and washers.
- (8) Tighten both the incoming and outgoing isolator terminal nuts.
- (9) Tighten the negative outboard terminal post.
- (10) Connect the isolator to maxi fuse box cable to the maxi fuse box using screw and washer. Fit the maxi fuse box terminal cover.
- (11) Refit the battery (Refer to Section 14.1)

CAUTION:

Prior to connecting the battery negative terminal to the battery negative terminal post, check there are no short circuits between cables and body.

3 ALTERNATOR

A Removal Procedure

NOTES:

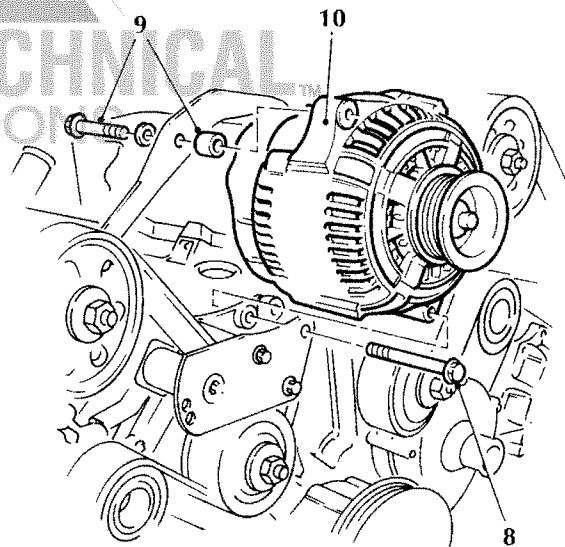
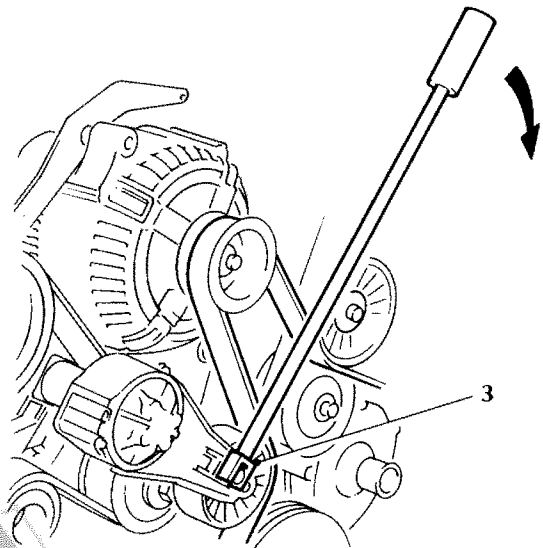
- 1 If drive belt has not been removed proceed as detailed in steps (1) to (4).
- 2 Ensure absolute cleanliness of work area and tools.
- 3 Observe correct use of tools.

- (1) Switch off the main isolator switch,
- (2) Disconnect the battery earth cable from the negative terminal post.

WARNING:

THE MAIN ISOLATOR SWITCH DOES NOT ISOLATE THE STARTER/ALTERNATOR CIRCUIT FROM THE BATTERY.

- (3) Fit 1/2in. square drive wrench in socket hole of dynamic tensioner.
- (4) Move the wrench to oppose spring tension in dynamic tensioner until drive belt is loose enough to be easily removed.
- (5) Gradually reduce pressure on wrench to allow dynamic tensioner to return to neutral position and remove wrench.
- (6) Inspect drive belt for serviceability. If in doubt discard drive belt.
- (7) Disconnect electrical leads from back of alternator.
- (8) Remove bolt from alternator lower attachment.
- (9) Support alternator and remove upper attachment bolt and spacer.
- (10) Remove alternator.



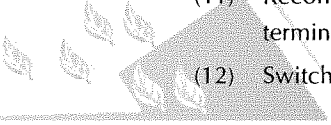
B Installation Procedure

- (1) Support alternator at front of engine.
- (2) Fit bolt through lower alternator attachment and secure.
- (3) Fit bolt and spacer through upper alternator attachment and secure.
- (4) Torque load M10 bolts 40.5 to 46.5Nm (30 to 34.5ftlb) and M8 bolts 20 to 23Nm (15 to 17ftlb).
- (5) Connect electrical leads to back of alternator.

NOTE:

If drive belt is to be fitted here proceed as follows.

- (6) Fit 1/2in. square drive wrench in socket hole of dynamic tensioner.
- (7) Move wrench in direction to oppose spring pressure.
- (8) Fit drive belt.
- (9) Gradually reduce pressure on wrench to allow dynamic tensioner to take up slack in drive belt.
- (10) Remove wrench.
- (11) Reconnect the battery earth cable to negative terminal post.
- (12) Switch on the main isolator switch.



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4 STARTER CABLES

A Removal Procedure

NOTE

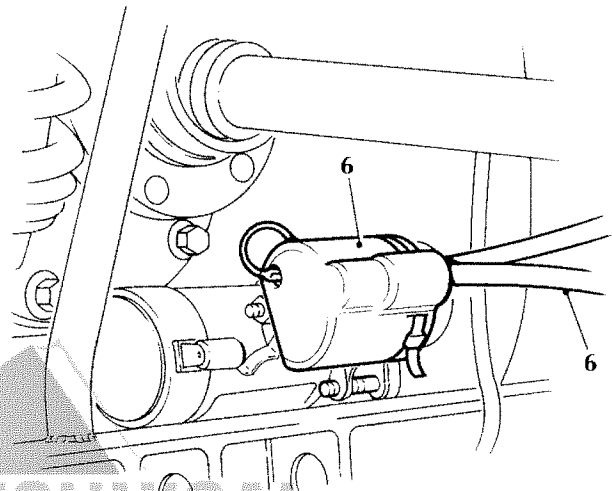
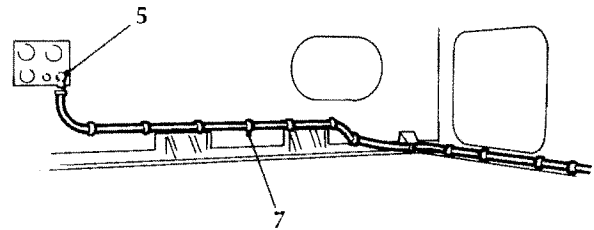
- 1 Ensure absolute cleanliness of work area and tools.
- 2 Observe the correct use of tools.

- (1) Position the vehicle on lifting apparatus. (Refer to Lifting and Jacking, Section 1.1)
- (2) Switch off the main isolator switch.

WARNING:

THE MAIN ISOLATOR SWITCH DOES NOT ISOLATE THE STARTER/ALTERNATOR CIRCUIT FROM THE BATTERY.

- (3) Disconnect the battery earth cable from the negative terminal post
- (4) Remove the venturi panel. (Refer to Section 2)
- (5) Slide back the rubber cover and remove the nut and spring washer securing the cable to the terminal post.
- (6) Remove the starter motor cover and remove the nut and washer securing the cables at the starter motor.
- (7) Remove the cable ties and 'P' clips securing the cable to the structure and oil tank bracket and remove the cable from the vehicle.

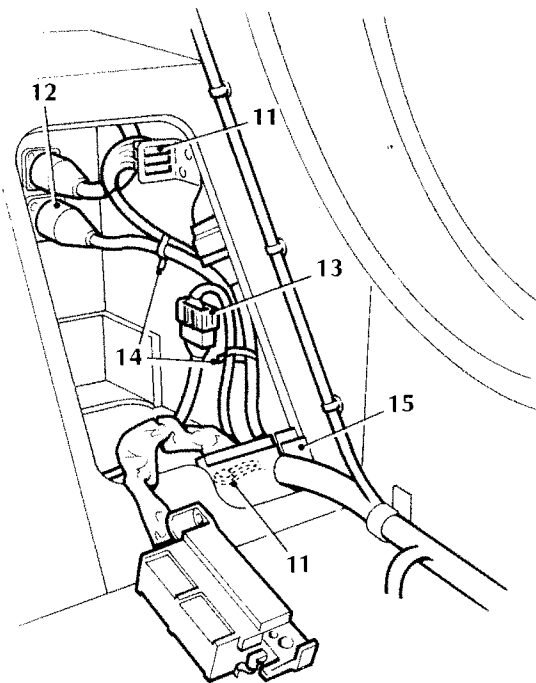
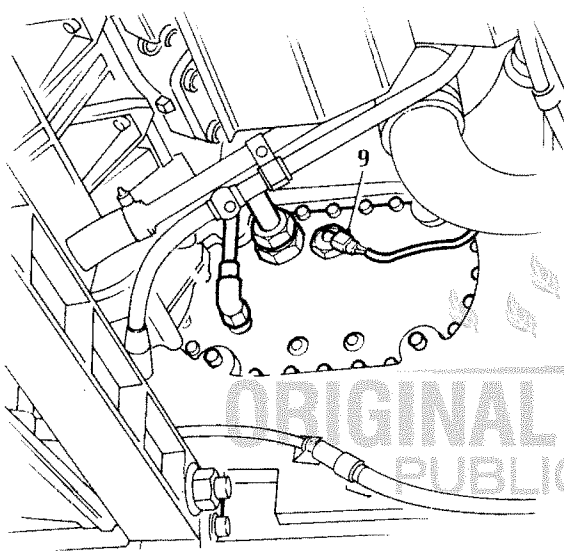
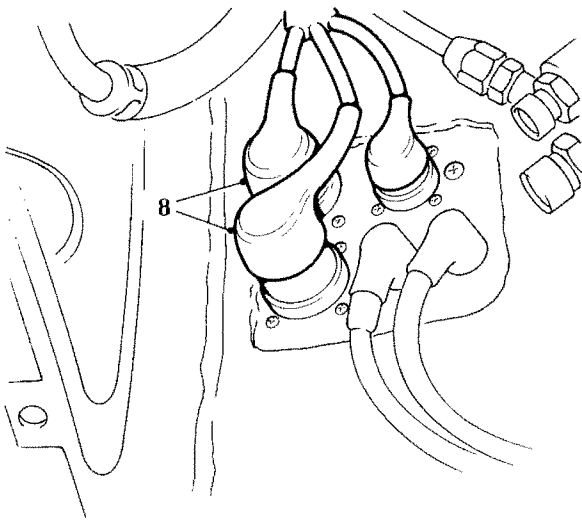


B Installation Procedure

NOTE

- 1 Ensure absolute cleanliness of work area and tools.
- 2 Observe the correct use of tools

- (1) Secure the flat connector end of the cable to the negative terminal post and secure with spring washer and nut.
- (2) Route the cable along the bulkhead and secure to the structure and oil tank bracket with tyrap.
- (3) Route the cable along the side bulkhead and secure with cable tie and 'P' clips.
- (4) Fit the cable, together with the alternator cable, to the starter motor terminal and secure to correct torque with washer and nut.
- (5) Fit the starter motor cover and secure with cable tie.
- (6) Reconnect the battery earth cable to the negative terminal post.
- (7) Fit the venturi panel. (Refer to Section 2)
- (8) Lower the vehicle and remove from the lifting apparatus.
- (9) Switch on the main isolator switch.



5 MAIN HARNESS

Special Tools

The following special tools will be required
JaguarSport protective body covers.

A Removal Procedure

NOTE

- 1 Ensure absolute cleanliness of work area and tools.
- 2 Observe the correct use of tools.

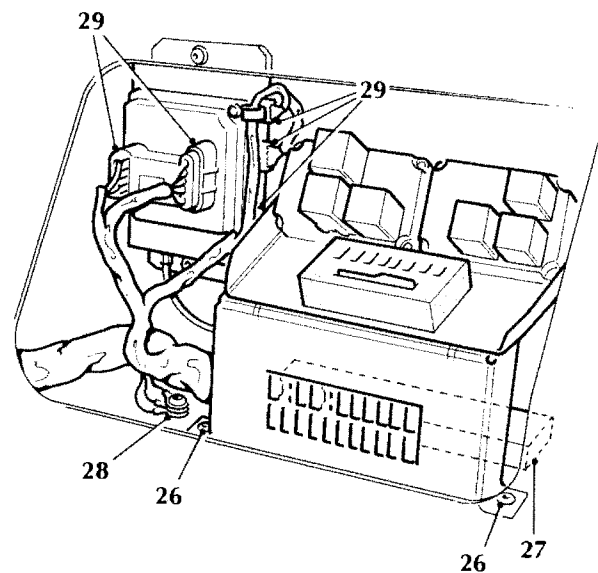
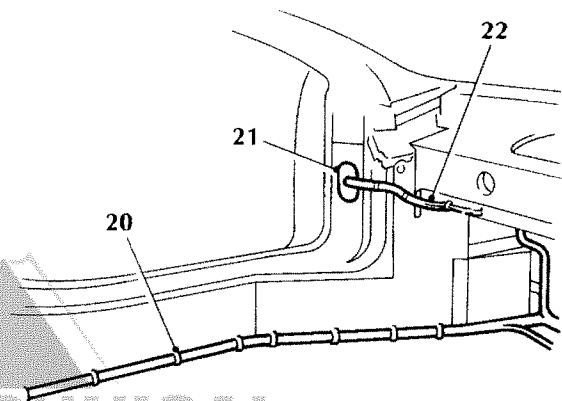
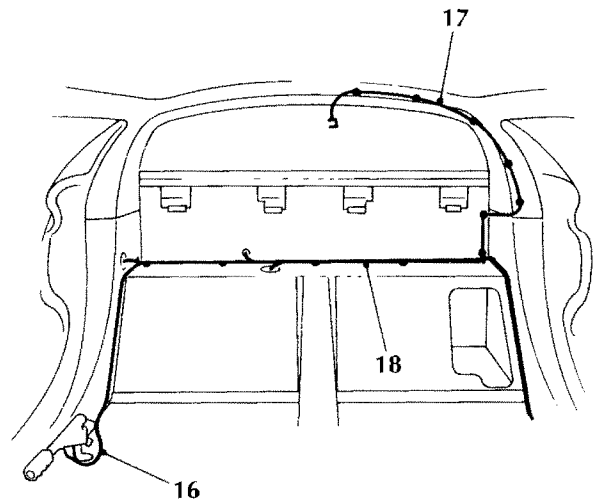
- (1) Remove the interior trim (Refer to Section 2).
- (2) Remove the metal tunnel cover (Refer to Section 2).
- (3) Remove the air conditioning unit (Refer to Section 12).
- (4) Remove the battery (Refer to Section 14.1).
- (5) Disconnect the battery positive lead from the maxi fuse box.
- (6) Remove both front and rear right wheelarch liners (Refer to Section 4).
- (7) Remove the venturi panel (Refer to Section 2).

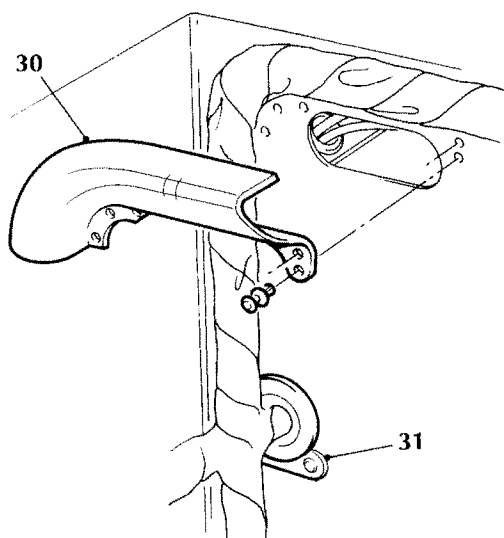
NOTE

If the harness is to be reused, extreme care must be taken to prevent unnecessary damage to the harness during removal.

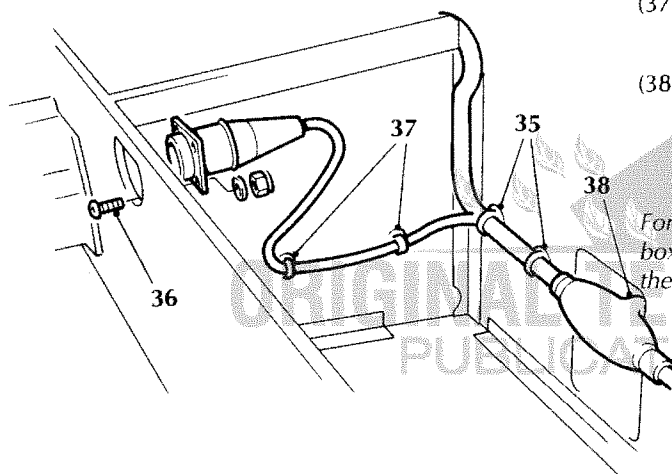
- (8) Disconnect the engine harness to main harness bulkhead connectors in the engine compartment.
- (9) Disconnect the low pressure fuel pump connectors at the tank plate and feed the breakout through the bulkhead into the passenger compartment.
- (10) Disconnect the fuel pump harness and the fuel system purge breakout in the rear right hand wheelarch. Feed the harness through the bulkhead into the passenger compartment.
- (11) Remove fuse box D from the bracket and disconnect the main harness positive eyelets from the master switch.
- (12) Remove the 4 screws, washers and nuts securing the connectors.
- (13) Disconnect the main harness to rear body harness body harness connector.
- (14) Remove the cable ties securing the main harness inside the battery box.
- (15) Release the harness from the battery cover grommet.

- (16) Disconnect the electrical connections at the handbrake.
- (17) Remove the cable ties securing the map light breakout to the rear header and 'B' post.
- (18) Remove the cable ties securing the harness to the fuel tank bulkhead and rear bulkhead.
- (19) Remove the cable ties securing the CD lead to the main harness.
- (20) Release the harness from the plastic clips along the sill.
- (21) Disconnect the left hand door to main harness connectors. Remove the grommet in the 'A' post and feed the harness through into the passenger compartment.
- (22) Remove the cable ties securing the harness to the left hand 'A' post and under knee panel.
- (23) Disconnect the connections to the inertia switch and the stop lamp switch.
- (24) Disconnect the right hand door to main harness connectors. Remove the grommet in the 'A' post and feed the harness through into the passenger compartment.
- (25) Remove the cable ties securing the harness to the right hand 'A' post and under knee panel.
- (26) Remove the screws securing the main fuse box and relay module assembly to the floor.
- (27) Prise off the top of the header retainer and lift out the headers.
- (28) Remove the screw and washers attaching the earth tags to the floor.
- (29) Disconnect the connections to the alarm module and headlamp door module.





- (30) Remove the plastic rivets from the harness shield and remove the shield.
- (31) Remove the rivets securing the tunnel grommet filler plate, remove the plate and remove the grommet from the aperture.
- (32) On LHD, remove the rivets securing the grommet plate.
- (33) Disconnect the electrical connections to the wiper motor, brake fluid level sensor, exterior mirror and brake pad sensors.
- (34) Remove the screws securing the earth tags on the right hand and left hand front zero bulkhead.
- (35) Release the cable ties and clips securing the harness to zero bulkhead.
- (36) Remove the screws, washers and nuts securing the harness CPC's to the suspension bulkhead.
- (37) Remove the cable ties securing the harness to the suspension box.
- (38) Feed the harness from the suspension box through the tunnel and remove the complete main harness from the vehicle.



NOTE

For LHD, remove cover, base and fuses from the maxi fuse box assembly and carefully remove the harness through the tunnel towards the passenger compartment.

B Installation Procedure RHD

NOTE

Prior to installation of the harness check the cable main insulation for splits, the security of all end connectors and serviceability of all grommets. Reject the harness if any faults are found. If new header retainer is being fitted, ensure surface fixing area is clean.

- (1) Prior to installing the harness into the vehicle, sub-assemble the following:
 - Fit mounting bracket to fuse box 'A' using screws and washers.
 - Fit relays to fuse box 'A'.
 - Fit new header retainer if required.
 - Fit mounting bracket to fuse box/relay base 'C'.
 - Fit relays to relay base 'C'.
 - Using screw, washer and nut, fit maxi fuse box mounting bracket to maxi fuse box base.
- (2) Place the main harness in to the front of the passenger footwell.
- (3) Identify the front harness breakout with 2 CPC connectors and foam tape and feed it through the hole in the left hand tunnel side on to the front suspension box. Secure the grommet in the aperture.
- (4) Route the right hand door breakout over the tunnel and along the right hand knee panel. Loosely secure with cable ties.
- (5) Fit the harness shield to the harness, locate to the tunnel side and secure using plastic rivets. Route the harness tight into the corners of the footwell.
- (6) Secure the harness to the floor using cable ties. Locate the headers into the header retainer channel and snap on the retainer lid. If a new header retainer is being fitted, remove the backing from the header retainer assembly and stick the assembly to the floor between the fuse box fixing points.
- (7) Secure the earth tags to the floor using screw and washers.
- (8) Locate the main fuse box and secure to the floor using screws and washers. Fit modules to module mounting bracket and secure assembly to fuse box using screws and washers.
- (9) Connect the harness to the alarm module and headlamp door module. Secure top alarm connector breakout into 'P' clip.
- (10) Fit modules and relays into module bases.
- (11) Route the right hand door breakout from the grommet in the left hand tunnel side along the joint of the tunnel to bulkhead, and under the right hand knee panel. Tighten cable ties.
- (12) Feed the free end of the door breakout through the aperture in the door ring and out through the 'A' post.
- (13) Secure the end of the convoluted tube to the 'A' post and secure the grommet on the harness to the aperture in the door ring. Connect to door.
- (14) Secure the harness to the 'A' post using cable ties ensuring the harness sits tight against the body.
- (15) Route the left hand door breakout along the corner of the floor/zero bulkhead and up along the body side panel to the 'A' post.
- (16) Loosely secure the left hand door breakout together with the alarm aerial lead to the body side panel and 'A' post using cable ties and loosely secure the footwell light breakout.
- (17) Feed the end of the left hand door breakout through the aperture in the door ring and out through the 'A' post.
- (18) Secure the end of the convoluted tube to the 'A' post aperture and secure the grommet on the harness into the door ring. Connect to door.
- (19) Ensure the harness sits tight against the body side and tighten the cable ties.
- (20) Route the harness rearwards from the corner at the left hand body side panel and bulkhead and secure in the open 'P' clips.
- (21) Fit the harness into the battery cover grommet and locate into cut-out.
- (22) Secure the harness inside the battery box using cable ties.
- (23) Using washer and screw, secure the harness earth eyelet, together with the battery negative cable, to the floor. Tighten to 20Nm (14.8ftlb).
- (24) Connect the harness to chassis and rear body harness connectors and secure with cable ties.
- (25) Connect the maxi fuse box battery positive cable to the busbar.
- (26) Fit fuse box 'D' to the bracket in the battery compartment.
- (27) Locate the map light and fuel pump breakouts into the corner between the body side panel and battery box opening. Secure the harness with cable ties.

- (28) Secure the map light breakout to the left hand 'B' post and rear header using cable ties.
- (29) Route the harness across the corner of the rear bulkhead seat back bulkhead and secure with cable ties.
- (30) Connect the alarm microwave sensor.
- (31) Route the low pressure fuel pump breakout through the plate on the rear bulkhead and secure the grommet to the plate.
- (32) Feed the fuel pump/purge system breakout through the large hole in the right hand side honeycomb and secure the grommet on the harness into the hole.
- (33) Route the handbrake breakout down the right hand side honeycomb and secure with cable ties. Connect the harness to the handbrake.
- (34) Feed the purge system breakout through the grommeted aperture and connect the harness to the purge valves.
- (35) Feed the fuel pump breakout down between the roll cage and 'B' post closing panel and out of the aperture at the bottom of the roll cage. Connect the harness to the fuel pumps.
- (36) Fit foam location pad between 'B' post closing panel and roll cage to secure the harness breakout. Ensure that the protective grommet strip is in place along the bottom of the 'B' post closing panel.
- (37) Feed the harness to the H. P. fuel pumps and connect, observing correct polarity.
- (38) Connect the low pressure fuel pump breakout to the connector on the fuel tank access plate.
- (39) Fit the harness CPC connector to the connector plate and secure using washers, screws and nuts. Ensure correct orientation of connector.
- (40) Connect the engine harness connectors to the bulkhead connector plate connectors.
- (41) Route the CD lead through the battery cover grommet along side the main harness and secure to harness with cable ties. Connect to audio interface unit.
- (42) Locate the foam wrapped section of the harness into the centre of the tunnel/zero bulkhead in the front suspension box.
- (43) Route the CPC connector with the centre pins, numbers 8 and 9, towards the right hand aperture in the suspension bulkhead and secure using washers, screws and nuts.
- (44) Secure the harness to the right hand side of the suspension box using cable ties. Connect the right hand front body harness.
- (45) Route the CPC connector without the centre pins, numbers 8 and 9, towards the left hand aperture in the suspension bulkhead and secure using washers, screws and nuts.
- (46) Secure the harness to the left hand side of the suspension box using cable ties. Connect the left hand front body harness.
- (47) Route the harness breakout to the brake fluid level switch and external mirrors up and across the front of the bulkhead. Secure with cable ties and open 'P' clips.
- (48) Route the breakouts to the external mirrors and connect. Secure the harness with cable ties.
- (49) Connect the brake fluid level switch.
- (50) Secure the right hand earth eyelet to the bulkhead using a new ripp washer and screw.
- (51) Route the right hand brake pad harness breakout along the front brake pipe and connect to the brake pad sensor lead. Secure back tightly to the brake pipe using cable ties.
- (52) Route the harness breakout to the wiper motor and connect. Secure the harness back to the wiper motor rear fixing using 'P' clip.
- (53) Secure the left hand harness earth eyelet to the bulkhead using a new ripp washer and screw.
- (54) Route the left hand brake pad harness breakout along the front brake pipe and connect to the brake pad sensor lead. Secure back tightly to the brake pipe using cable ties.
- (55) Fit HVAC unit (Refer to Section 12).
- (56) Fit fascia panel (Refer to Section 13).
- (57) Fit the battery (Refer to Section 14.1). Do not connect battery earth cable.

NOTE

Before connecting the battery earth terminal, it is recommended that the supply to the vehicle is connected first and tested via a test lead with a 20 amp in-line fuse.

- (58) Carry out a full Vehicle Electrical Test. Rectify any faults found.
- (59) Fit the venturi panel (Refer to Section 2).
- (60) Fit the wheelarch liners (Refer to Section 4).
- (61) Fit the tunnel metal cover (Refer to Section 2).
- (62) Fit the interior trim (Refer to Section 2).

C Installation Procedure LHD

NOTE

Prior to installation of the harness check the cable main insulation for splits, the security of all end connectors and serviceability of all grommets. Reject the harness if any faults are found. If new header retainer is being fitted, ensure surface fixing area is clean.

- (1) Prior to installing the harness into the vehicle, sub-assemble the following:
 - Fit mounting bracket to fuse box 'A' using screws and washers.
 - Fit relays to fuse box 'A'.
 - Fit new header retainer if required.
 - Fit mounting bracket to fuse box/relay base 'C'.
 - Fit relays to relay base 'C'.
 - Using screw, washer and nut, fit maxi fuse box mounting bracket to maxi fuse box base.
 - Fit harness grommet plate to second largest tunnel grommet.
- (2) Place the main harness in to the front of the passenger footwell.
- (3) Feed the maxi fuse box assembly and harness through the tunnel up to the first large grommet. Push the large grommet through the right hand tunnel hole.
- (4) Identify the front harness breakout with 2 CPC connectors and foam tape and feed it through the hole in the left hand tunnel side on to the front suspension box. Secure the grommet in the aperture.
- (5) Position the right hand grommet plate to the right hand tunnel side and secure using rivets.
- (6) Locate the left hand grommet filler plate to the tunnel and secure with rivets. Pull through and locate the large main harness grommet into the left hand tunnel hole.
- (7) Route the left hand door breakout over the tunnel and along the right hand knee panel. Loosely secure with cable ties.
- (8) Fit the harness shield to the harness, locate to the tunnel side and secure using plastic rivets. Route the harness tight into the corners of the footwell.
- (9) Secure the harness to the floor using cable ties. Locate the headers into the header retainer channel and snap on the retainer lid. If a new header retainer is being fitted, remove the backing from the header retainer assembly and stick the assembly to the floor between the fuse box fixing points.
- (10) Secure the earth tags to the floor using screw and washers.
- (11) Locate the main fuse box and secure to the floor using screws and washers. Fit modules to module mounting bracket and secure assembly to fuse box using screws and washers.
- (12) Connect the harness to the alarm module and headlamp door module. Secure top alarm connector breakout into 'P' clip.
- (13) Fit modules and relays into module bases.
- (14) Route the left hand door breakout from the grommet in the right hand tunnel side along the joint of the tunnel to bulkhead, and under the left hand knee panel. Tighten cable ties.
- (15) Route inertia switch breakout forward and connect.
- (16) Route stop lamp switch breakout forward and connect.
- (17) Feed the free end of the door breakout through the aperture in the door ring and out through the 'A' post.
- (18) Secure the end of the convoluted tube to the 'A' post and secure the grommet on the harness to the aperture in the door ring. Connect to door.
- (19) Secure the harness to the 'A' post using cable ties ensuring the harness sits tight against the body.
- (20) Route the right hand door breakout along the corner of the floor/front bulkhead and up along the body side panel to the 'A' post.
- (21) Loosely secure the right hand door breakout together with the alarm aerial lead to the body side panel and 'A' post using cable ties and loosely secure the footwell light breakout.
- (22) Feed the end of the left hand door breakout through the aperture in the door ring and out through the 'A' post.
- (23) Secure the end of the convoluted tube to the 'A' post aperture and secure the grommet on the harness into the door ring. Connect to door.
- (24) Ensure the harness sits tight against the body side and tighten the cable ties.
- (25) Lay harness into 'U' channel retainer across driver's footwell area and secure cover. (On later cars locate retention plate over harness and secure to floor using rivets).

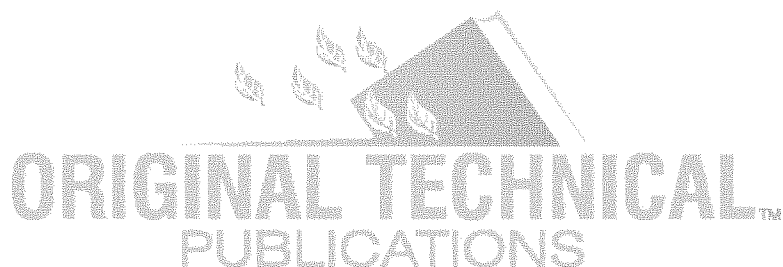
- (26) Route the harness rearwards from the corner at the left hand body side panel and bulkhead and secure in the open 'P' clips.
- (27) Fit the harness into the battery cover grommet and locate into cut-out.
- (28) Secure the harness inside the battery box using cable ties.
- (29) Using washer and screw, secure the harness earth eyelet, together with the battery negative cable, to the floor. Tighten to 20Nm (14.8ftlb).
- (30) Connect the harness to chassis and rear body harness connectors and secure with cable ties.
- (31) Connect the maxi fuse box battery positive cable to the busbar.
- (32) Fit fuse box 'D' to the bracket in the battery compartment.
- (33) Locate the map light and fuel pump breakouts into the corner between the body side panel and battery box opening. Secure the harness with cable ties.
- (34) Secure the map light breakout to the left hand 'B' post and rear header using cable ties.
- (35) Route the harness across the corner of the rear bulkhead seat back bulkhead and secure with cable ties.
- (36) Connect the alarm microwave sensor.
- (37) Route the low pressure fuel pump breakout through the plate on the rear bulkhead and secure the grommet to the plate.
- (38) Feed the fuel pump/purge system breakout through the large hole in the right hand side honeycomb and secure the grommet on the harness into the hole.
- (39) Feed the purge system breakout through the grommeted aperture and connect the harness to the purge valves.
- (40) Feed the fuel pump breakout down between the roll cage and 'B' post closing panel and out of the aperture at the bottom of the roll cage.
- (41) Fit foam location pad between 'B' post closing panel and roll cage to secure the harness breakout. Ensure that the protective grommet strip is in place along the bottom of the 'B' post closing panel.
- (42) Feed the harness to the H. P. fuel pumps and connect, observing correct polarity.
- (43) Connect the low pressure fuel pump breakout to the connector on the fuel tank access plate.
- (44) Fit the harness CPC connector to the connector plate and secure using washers, screws and nuts. Ensure correct orientation of connector.
- (45) Connect the engine harness connectors to the bulkhead connector plate connectors.
- (46) Route the CD lead through the battery cover grommet along side the main harness and secure to harness with cable ties. Connect to audio interface unit.
- (47) Connect flying lead to handbrake.
- (48) Locate the foam wrapped section of the harness into the centre of the tunnel/zero bulkhead in the front suspension box.
- (49) Route the CPC connector with the centre pins, numbers 8 and 9, towards the right hand aperture in the suspension bulkhead and secure using washers, screws and nuts.
- (50) Secure the harness to the right hand side of the suspension box using cable ties. Connect the right hand front body harness.
- (51) Route the CPC connector without the centre pins, numbers 8 and 9, towards the left hand aperture in the suspension bulkhead and secure using washers, screws and nuts.
- (52) Secure the harness to the left hand side of the suspension box using cable ties. Connect the left hand front body harness.
- (53) Route the harness breakout to the brake fluid level switch and external mirrors up and across the front of the bulkhead. Secure with cable ties and open 'P' clips.
- (54) Route the breakouts to the external mirrors and connect. Secure the harness with cable ties.
- (55) Connect the brake fluid level switch.
- (56) Secure the earth eyelet to the bulkhead using a new ripp washer and screw.
- (57) Route the brake pad harness breakout along the front brake pipe and connect to the brake pad sensor lead. Secure back tightly to the brake pipe using cable ties.
- (58) Route the harness breakout to the wiper motor and connect. Secure the harness back to the wiper motor rear fixing using 'P' clip.
- (59) Secure the harness earth eyelet to the bulkhead using a new ripp washer and screw.

- (60) Route the brake pad harness breakout along the front brake pipe and connect to the brake pad sensor lead. Secure back tightly to the brake pipe using cable ties.
- (61) Fit HVAC unit (Refer to Section 12).
- (62) Fit fascia panel (Refer to Section 13).
- (63) Fit the battery (Refer to Section 14.1). Do not connect battery earth cable.

NOTE

Before connecting the battery earth terminal, it is recommended that the supply to the vehicle system is first connected and tested via a test lead containing a 20 amp in-line fuse.

- (64) Carry out a full Vehicle Electrical Test. Rectify any faults found.
- (65) Fit the venturi panel (Refer to Section 2).
- (66) Fit the wheelarch liners (Refer to Section 4).
- (67) Fit the tunnel metal cover (Refer to Section 2).
- (68) Fit the interior trim (Refer to Section 2).



6 FASCIA HARNESS

A Removal Procedure

NOTE

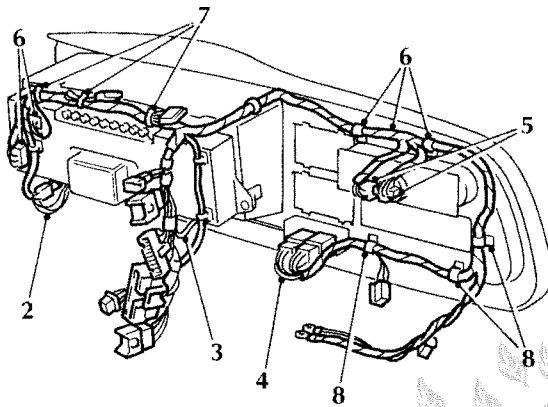
- 1 Ensure absolute cleanliness of work area and tools.
- 2 Observe the correct use of tools

- (1) Remove the instrument panel assembly. (Refer to Section 13).

CAUTION

The instruments are silicone damped. Never lay the instrument panel face down, always store the instrument panel upright.

- (2) Disconnect the starter button electrical connector and the alarm LED.
- (3) Disconnect the instrument panel dimmer, trip reset, mirror switch electrical connectors.
- (4) Disconnect the hazard and rear fog switches electrical connectors.
- (5) Disconnect heater control panel and fan switch connectors.
- (6) Disconnect instrument panel electrical connectors.
- (7) Remove the cable ties securing the harness to the instrument panel.
- (8) Remove the harness from the plastic 'P' clips and remove the harness.



B Installation Procedure

NOTE

Prior to installation of the harness check the wire insulation for splits, the security of all end connectors and serviceability of all grommets. Reject the harness if any faults are found.

- (1) Route the harness along the instrument panel assembly and secure to the binnacle with 'P' clips and to the instrument panel with cable ties.
- (2) Connect the starter button electrical connectors and alarm LED.
- (3) Connect the trip reset, instrument panel dimmer and mirror switch electrical connectors.
- (4) Connect the hazard and rear fog switches electrical connectors.
- (5) Connect the heater control panel and fan switch connectors.
- (6) Connect the three instrument panel connectors.
- (7) Fit the instrument panel. (Refer to Section 13).

7 CHASSIS HARNESS

A Removal Procedure

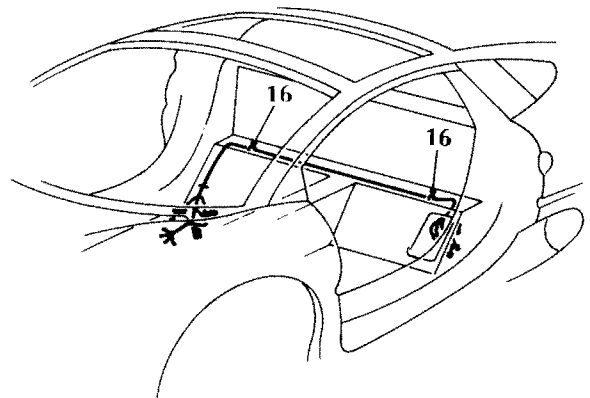
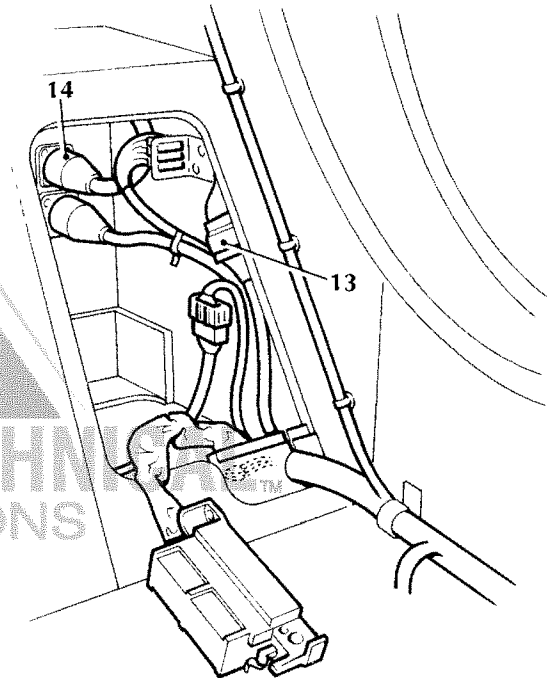
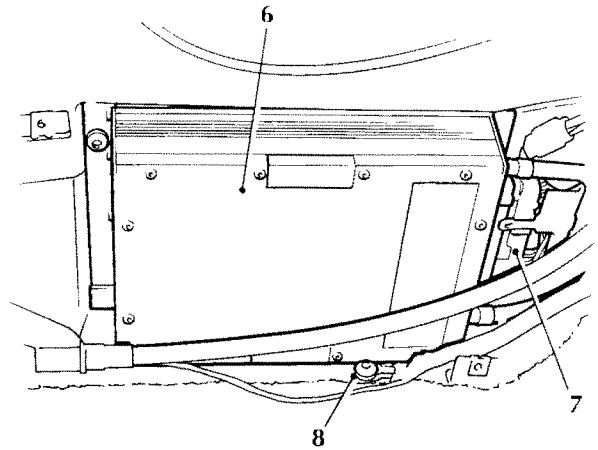
NOTE

- 1 Ensure absolute cleanliness of work area and tools.
- 2 Observe the correct use of tools.

Special Tools

JaguarSport protective seat covers

- (1) Fit JaguarSport protective seat covers
- (2) Slide both seats fully forward.
- (3) Remove the battery (Refer to Section 14.1)
- (4) Remove the rear trim panels (Refer to Section 2).
- (5) Remove the ECU cover (Refer to Section 2).
- (6) Remove the ECU (Refer to Section 3)
- (7) Disconnect the chassis harness to ECU connector.
- (8) Remove the screw, washer and nut attaching the earth leads to the floor and release the earth connections.
- (9) Unclip the diagnostic connector from the support bracket.
- (10) Disconnect the chassis harness from the engine bay bulkhead CPC connectors.
- (11) Disconnect the battery cables from the main isolator switch (Refer to section 14.2)
- (12) Remove the rivets securing the isolator switch housing and remove the main isolator switch and housing assembly.
- (13) Disconnect the chassis harness to main harness connectors.
- (14) Remove the 8 screws, washers and nuts securing the chassis harness CPC bulkhead connectors to the connector plate.
- (15) Feed the CPC connectors through the main isolator switch aperture.
- (16) Remove the harness from the two P clips and remove the harness from the vehicle.



B Installation Procedure

NOTE

Prior to installation of the harness check the wire insulation for splits, the security of all end connectors and serviceability of all grommets. Reject the harness if any faults are found.

- (1) Carefully feed the CPC connectors through the main isolator switch aperture and connect to the connector plate with the 8 screws, washers and nuts. Torque the nuts 4Nm (3ftlb).
- (2) Route the harness along the fuel tank bulkhead and secure to the bulkhead using two P clips.
- (3) Connect the harness to the ECU.
- (4) Locate the earth eyelets on the floor and secure with screw, washers. Torque the screw 22Nm (16ftlb).
- (5) Fit the ECU (Refer to Section 3)
- (6) Fit the ECU cover (Refer to Section 2).
- (7) Connect the harness to the corresponding main harness connectors.
- (8) Fit the main isolator switch and housing and secure with rivets.
- (9) Apply adhesive to chassis harness grommet and locate in main isolator switch housing/plate.
- (10) Connect the battery cables to the main isolator switch (Refer to Section 14.2)
- (11) Connect the engine harness to the engine bay CPC bulkhead connectors
- (12) Fit the battery (Refer to Section 14.1).
- (13) Carry out functional check of vehicle to ensure correct installation.
- (14) Fit the rear trim panels (Refer to Section 2).
- (15) Return the seats to their original position and remove the protective covers.

ORIGINAL TECHNICAL
PUBLICATIONS

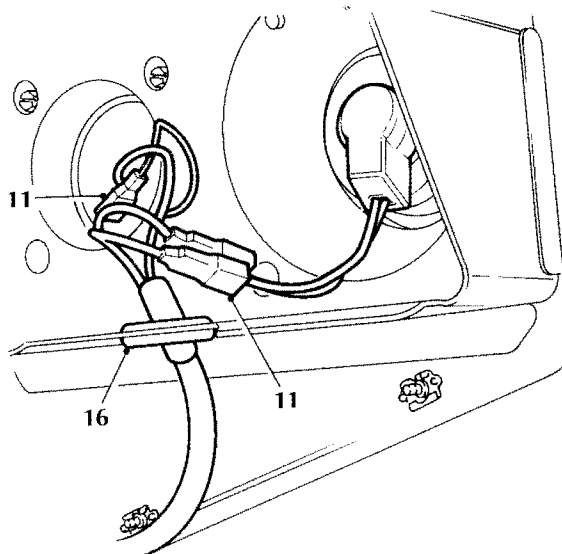
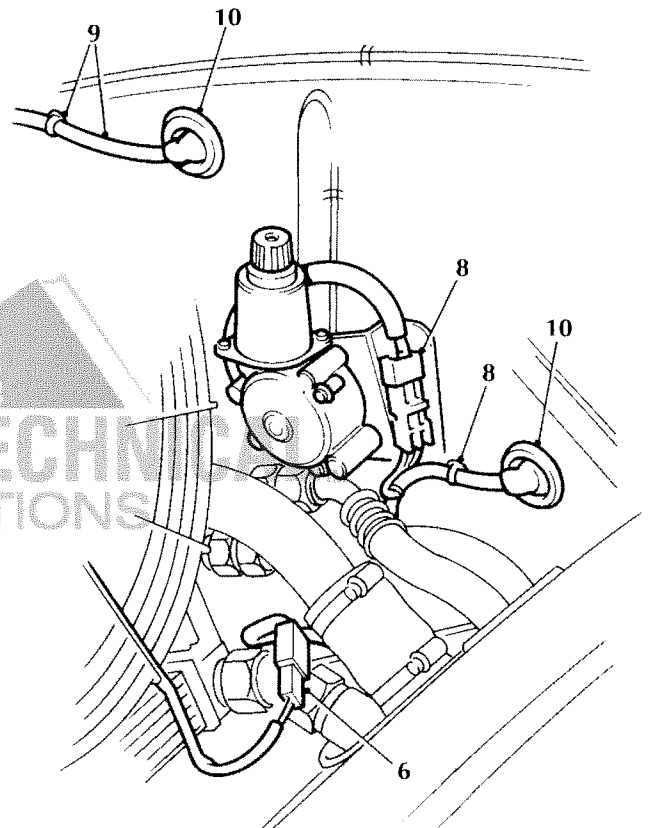
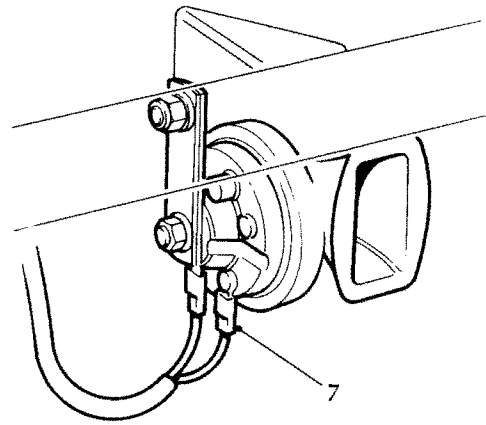
8 FRONT BODY HARNESS RH

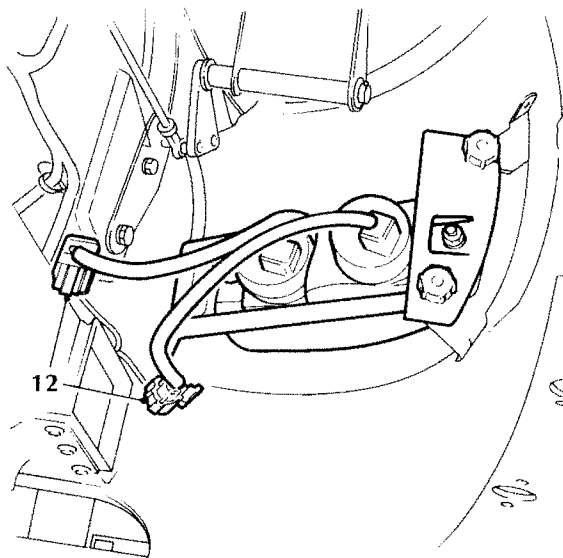
A Removal Procedure

NOTE

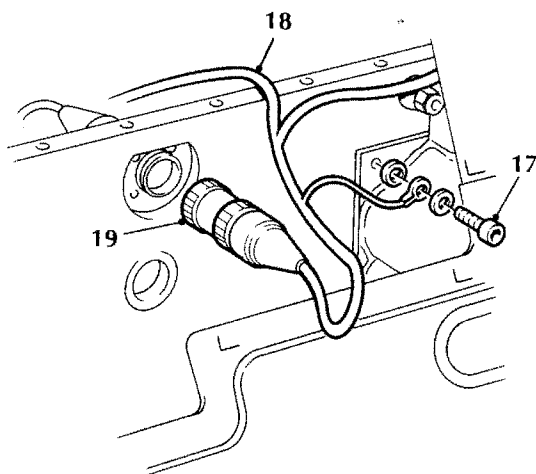
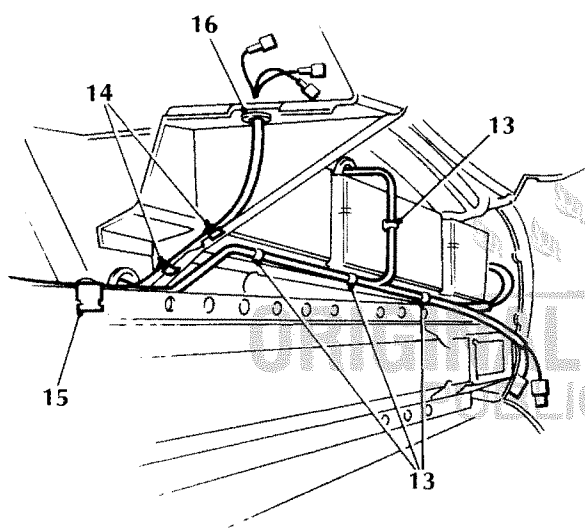
- 1 Ensure absolute cleanliness of work area and tools.
- 2 Observe the correct use of tools

- (1) Fit front JaguarSport protective body covers.
- (2) Position the vehicle on lifting apparatus. (Refer to Lifting and Jacking, Section 1).
- (3) Turn the main isolator switch to the off position.
- (4) Disconnect the battery earth cable.
- (5) Open the bonnet and remove the radiator duct access panel.
- (6) Remove the RH front wheel (Refer to Section 4) and wheel arch liner (Refer to Section 2).
- (7) Disconnect the electrical connectors at the horn
- (8) Disconnect the radiator fan and headlamp cover motor electrical connectors and remove the 'P' clip and cable tie securing the harness to the radiator side duct.
- (9) Disconnect the electrical connector to the receiver/drier bottle and remove the cable tie securing the harness to the radiator side duct.
- (10) Remove the grommets from the side duct and feed the harness through the duct.
- (11) Disconnect the electrical connectors at the RH headlamp assembly.





- (12) Disconnect the electrical connectors at the RH side/indicator lamps.
- (13) Remove the cable ties securing the harness to the radiator side duct.
- (14) Remove the harness from the edge clips on the headlamp bracket.
- (15) Remove the harness from the large edge clip on the top flange of the crush tube.
- (16) Remove the grommet in the headlamp bracket and pull the headlamp breakout through the aperture.
- (17) Remove the top inboard screw on the crush tube flange and remove the harness earth connector. Refit the screw and tighten 9Nm (6.5ftlb).
- (18) Remove washer bottle (Refer to Section 2) and disconnect the screen wash pump.
- (19) Remove the harness main connector at the suspension bulkhead and remove the complete harness from the vehicle.



B Installation Procedure

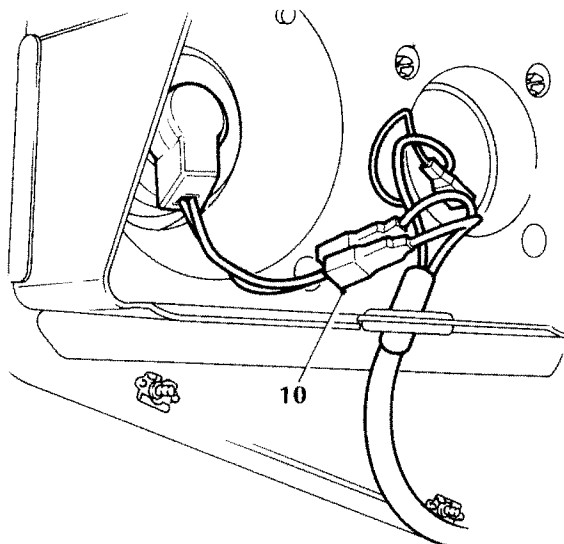
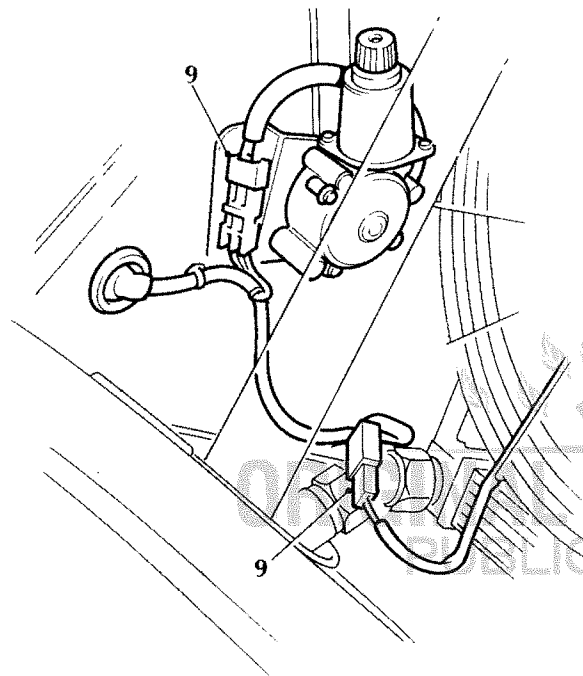
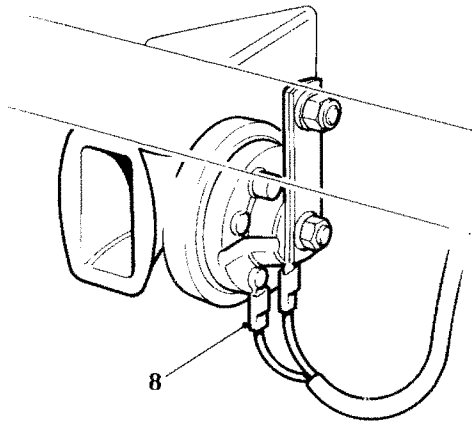
NOTE

- 1 Ensure absolute cleanliness of work area and tools.
- 2 Observe the correct use of tools

NOTE

Prior to installation of the harness check the wire insulation for splits, the security of all end connectors and serviceability of all grommets. Reject the harness if any faults found.

- (1) Secure the harness main connector to the main harness at the suspension bulkhead.
- (2) Remove the top inboard screw and washer on the crush tube flange and refit together with the harness earth lead and new ripp washer. Tighten the screw 9Nm (6.5ftlb).
- (3) Route the harness toward the front of the vehicle and secure to the top flange of the crush tube and headlamp bracket with edge clips.
- (4) Feed the radiator fan and headlamp cover motor breakouts through the radiator side duct and secure the grommet.
- (5) Feed the receiver/drier bottle breakout through the radiator side duct and secure the grommet.
- (6) Feed the headlamp breakout through the headlamp bracket and secure the grommet.
- (7) Secure the harness to the radiator side duct with cable ties.
- (8) Feed the horn breakout through the front structure.
- (9) Connect the harness to the side lamps and headlamps.
- (10) Connect the harness to the horn, radiator fans and headlamp cover motor and secure the harness to the side duct using a cable tie. Fit 'P' clip.
- (11) Connect the harness to the receiver/drier bottle and secure the harness to the side duct using a cable tie.
- (12) Connect the harness to the screen wash pump and fit washer bottle (Refer to Section 2).
- (13) Connect the battery and carry out a functional check on all disturbed systems.
- (14) Fit the wheel arch liner (Refer to Section 2) and the wheel. (Refer to Section 4).
- (15) Fit the radiator duct access panel and close the bonnet.



9 FRONT BODY HARNESS LH

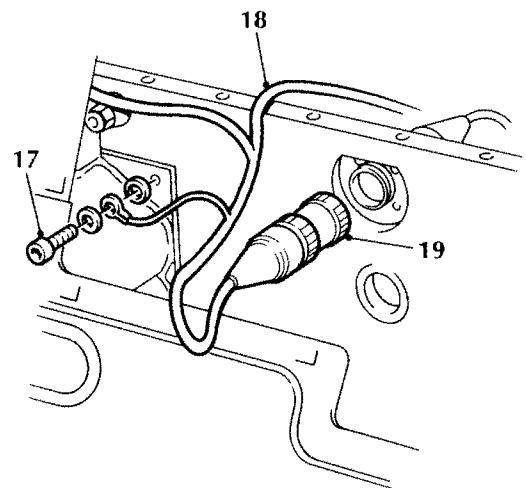
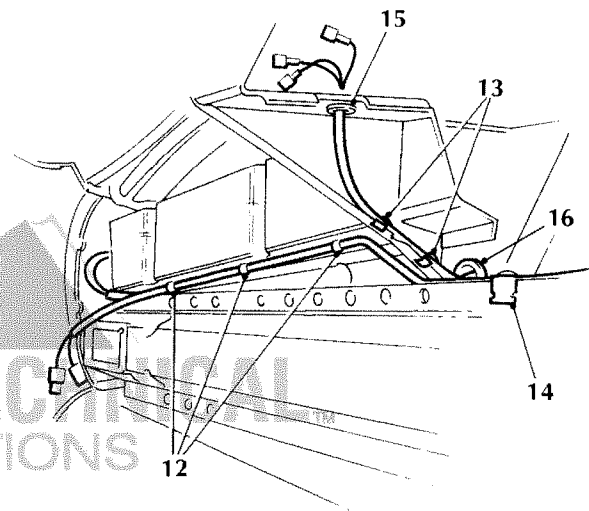
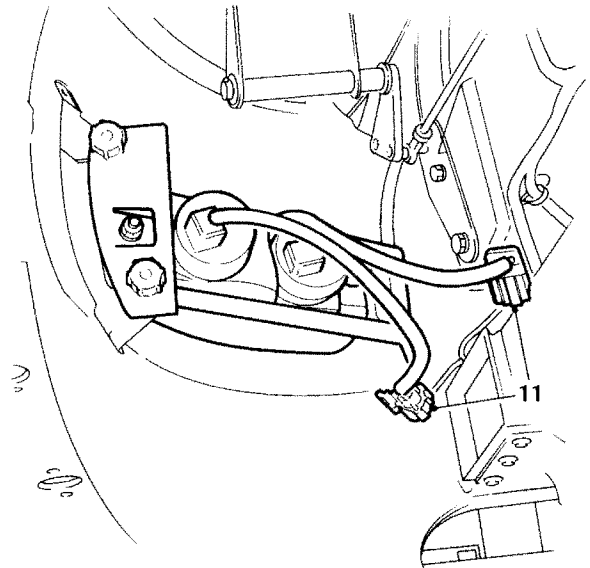
A Removal Procedure

NOTE

- 1 Ensure absolute cleanliness of work area and tools.
- 2 Observe the correct use of tools

- (1) Fit front JaguarSport protective body covers.
- (2) Position the vehicle on lifting apparatus. (Refer to Lifting and Jacking, Section 1).
- (3) Turn the main isolator switch to the off position.
- (4) Disconnect the battery earth cable.
- (5) Open the bonnet and remove the radiator duct access panel.
- (6) Remove the Tyreweld canister.
- (7) Remove the LH front wheel (Refer to Section 4) and wheel arch liner (Refer to Section 2).
- (8) Disconnect the electrical connectors at the horn.
- (9) Disconnect the radiator fan and headlamp cover motor electrical connectors and remove the 'P' clip and cable tie securing the harness to the radiator side duct.
- (10) Disconnect the electrical connectors at the head lamp assembly.

- (11) Disconnect the electrical connectors at the LH side/indicator lamps.
- (12) Remove the cable ties securing the harness to the radiator side duct.
- (13) Remove the harness from the edge clips on the headlamp bracket.
- (14) Remove the harness from the large edge clip on the top flange of the crush tube.
- (15) Remove the grommet in the headlamp bracket and pull the headlamp breakout through the aperture.
- (16) Remove the grommet from the radiator side duct and pull the radiator fan and headlamp motor breakouts through the aperture.
- (17) Remove the top inboard screw on the crush tube flange and remove the harness earth connector. Refit the screw and tighten 9Nm (6.5ftlb).
- (18) Disconnect the electrical connectors from the water pump and control valve.
- (19) Remove the harness main connector at the suspension bulkhead and remove the complete harness from the vehicle.



B Installation Procedure

NOTE

- 1 Ensure absolute cleanliness of work area and tools.
- 2 Observe the correct use of tools

NOTE

Prior to installation of the harness check the wire insulation for splits, the security of all end connectors and serviceability of all grommets. Reject the harness if any faults found.

- (1) Secure the harness main connector to the main harness at the suspension bulkhead.
- (2) Remove the top inboard screw and washer on the crush tube flange and refit together with the harness earth lead and new ripp washer. Tighten the screw 9Nm (6.5ftlb).
- (3) Connect the water pump and control valve.
- (4) Route the harness toward the front of the vehicle and secure to the top flange of the crush tube and headlamp bracket with edge clips.
- (5) Feed the radiator fan and headlamp motor breakouts through the radiator side duct and secure the grommet.
- (6) Feed the headlamp breakout through the headlamp bracket and secure the grommet.
- (7) Secure the harness to the radiator side duct with cable ties.
- (8) Feed the horn breakout through the front structure.
- (9) Connect the harness to the side lamps and headlamps.
- (10) Connect the harness to the horn, radiator fans and headlamp cover motor and secure the harness to the side duct using a cable tie. Fit 'P' clip.
- (11) Connect the battery and carry out a functional check on all disturbed systems.
- (12) Fit the wheel arch liner (Refer to Section 2) and the wheel. (Refer to Section 4).
- (13) Fit the radiator duct access panel.
- (14) Fit the Tyreweld canister and close the bonnet.

10 REAR BODY HARNESS

A Removal Procedure

NOTE

- 1 Ensure absolute cleanliness of work area and tools.
- 2 Observe the correct use of tools

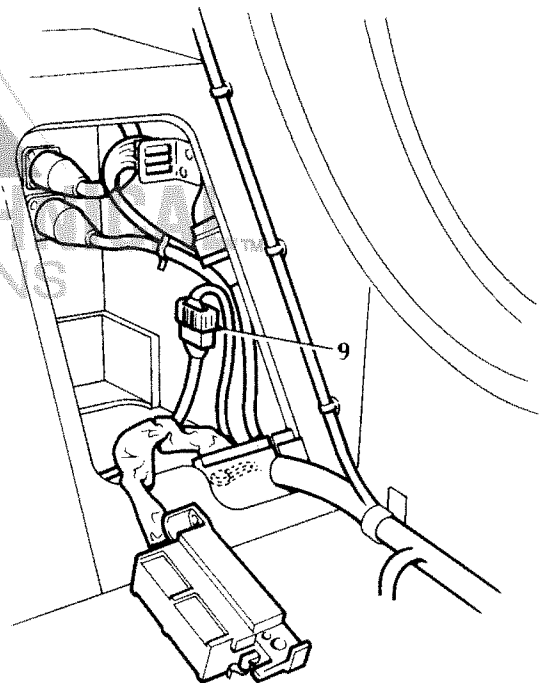
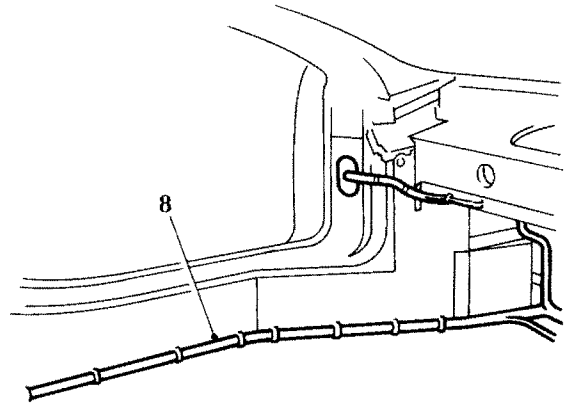
Special Tools

The following special tools will be required

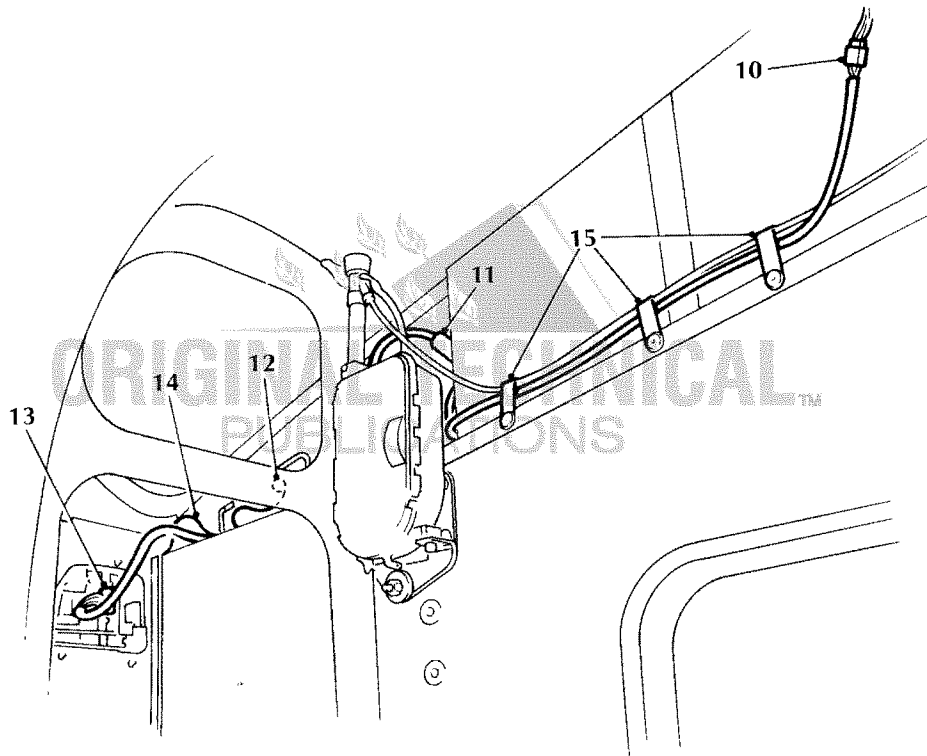
JaguarSport protective body covers

JaguarSport protective seat covers.

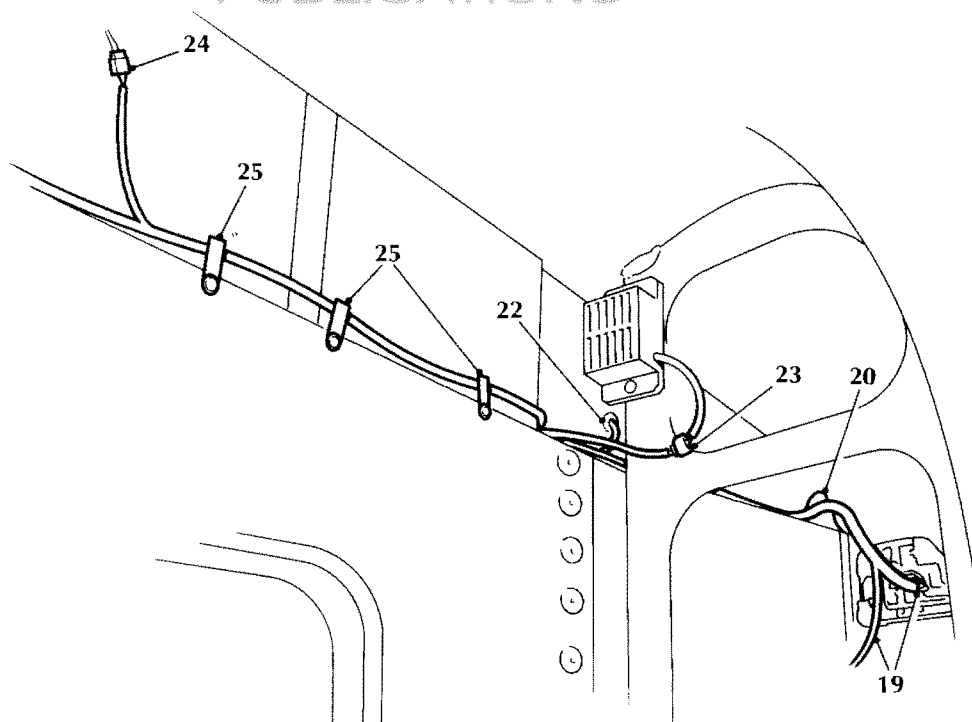
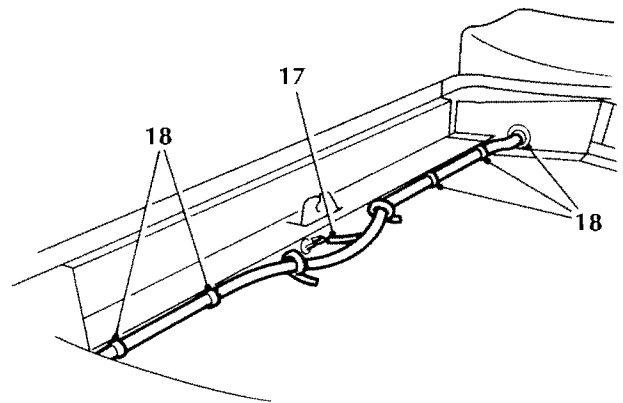
- (1) Fit JaguarSport protective covers to body and RH seat.
- (2) Disconnect battery.
- (3) Remove the RH and LH rear wheelarch liners (Refer to Section 4).
- (4) Remove the boot carpet, LH trim panel and auto-changer cover (Refer to Section 2).
- (5) Remove the LH side trim (Refer to Section 2).
- (6) Remove the RH underscuttle (Refer to Section 2).
- (7) Disconnect the CD lead from audio connectors and feed through the front trim bracketry.
- (8) Remove the cable ties securing the CD lead to the front and side honeycomb structure.
- (9) Disconnect the rear to main harness connectors and remove the cable ties and grommet from the flange.



- (10) Disconnect the RH engine bay lamp connector and RH engine bay lamp switch connector.
- (11) Disconnect the aerial to harness connector.
- (12) Disconnect the harness from the boot release solenoid.
- (13) Remove the harness connector from the RH rear lamp cluster.
- (14) Remove the screw and washers securing the harness earth tags to the structure.
- (15) Release the harness from the clips in the wheelarch, remove the grommet from the RH corner of the boot and feed the harness into the boot area.



- (16) Remove the rear grille (Refer to Section 2).
- (17) Remove the boot latch cover, disconnect the boot lamp switch connections and feed the breakout through into the boot.
- (18) Remove the cable ties securing the harness to the boot floor, remove the grommet from the LH corner of the boot and feed the harness through into the LH wheel arch area.
- (19) Disconnect the number plate harness and LH rear lamp cluster connectors.
- (20) Remove the screw and washers securing the harness earth tags to the structure.
- (21) Disconnect the CD lead from the auto-changer.
- (22) Remove the grommet and feed the boot lamp breakout and CD lead into the LH wheelarch area.
- (23) Disconnect the harness from the alarm sounder unit.
- (24) Disconnect the LH engine bay lamp connector and RH engine bay lamp switch connectors.
- (25) Release the harness from the clips in the wheelarch.
- (26) Remove the grommet at the forward end of the wheelarch.
- (27) Feed the CD lead through from the battery bay and remove the harness from the wheelarch.



B Installation Procedure

NOTE

Prior to installation of the harness check the wire insulation for splits, the security of all end connectors and serviceability of all grommets. Reject the harness if any faults are found.

- (1) Feed the main connector, CD lead and first large grommet through the aperture in the 'B' post closure panel. Locate the second large grommet in the aperture.
- (2) Route the harness through the honeycomb body side panel into the battery compartment through the flange and secure the grommet in the flange.
- (3) Feed the breakout for the LH engine bay lamp through the panel, and secure the grommet to the panel.
- (4) Route the harness rearwards securing it to the clips.
- (5) Feed the CD lead and boot lamp breakout through the boot side panel and secure the grommet to the panel.
- (6) Connect the CD lead to the auto-changer.
- (7) Connect the harness to the alarm sender unit
- (8) Route the harness through the hole at the rear of the LH boot panel and into the boot.
- (9) Secure the harness earth tags to the structure using screw and washers.
- (10) Connect the number plate harness and LH rear lamp cluster plugs.
- (11) Route the harness across the rear of the boot and through the hole at the rear of the RH boot panel.
- (12) Route the boot lamp switch breakout through the panel and secure the grommet. Connect the harness to the boot latch switch and refit the latch cover.
- (13) Secure the harness to the boot floor using cable ties. Cut off excess cable tie length and discard.
- (14) Secure the harness earth tags to the structure using screw and washers.
- (15) Route the harness along the top of the RH honeycomb body side panel and fit to the clips.
- (16) Connect the RH rear lamp cluster and boot release solenoid connections.
- (17) Connect the harness to the aerial and fit the aerial lead to the large P clip.
- (18) Feed the RH engine lamp breakout through the panel, connect to the engine bay lamp and secure the grommet.
- (19) Connect the rear harness to the main harness connectors and secure with cable ties.
- (20) Route the CD lead from the battery compartment along the outer edge of the floor pan parallel to the main harness.
- (21) Secure the lead CD lead to the main harness using cable ties.
- (22) Route the CD lead along the front hinge pillar bulkhead and secure with cable ties.
- (23) Feed the CD lead through the trim bracketry and connect to the audio connectors.
- (24) Connect the battery (Refer to Section 14.?) and ensure all associated electrical equipment functions correctly.
- (25) Fit the rear grille (Refer to Section 2).
- (26) Fit the boot LH trim panel, auto-change cover and boot carpet (Refer to Section 2).
- (27) Fit the rear wheel arch liners (Refer to Section 4).
- (28) Fit the RH side sill panel trim (Refer to Section 2).
- (29) Fit the RH underscuttle (Refer to Section 2).
- (30) Remove the JaguarSport protective body and seat covers.

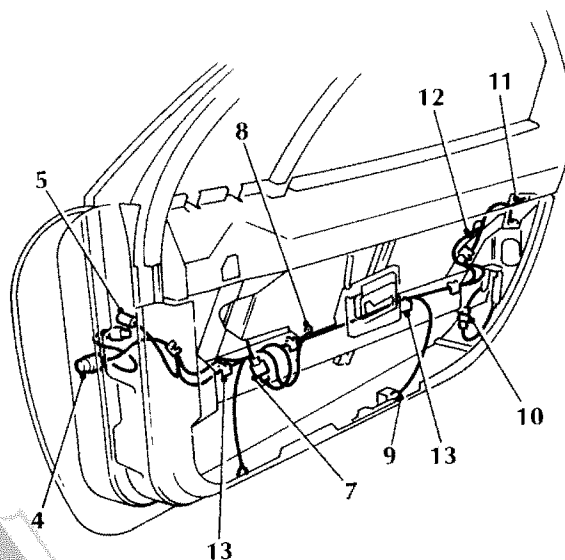
11 DOOR HARNESS

A Removal Procedure

NOTE

- 1 Ensure absolute cleanliness of work area and tools.
- 2 Observe the correct use of tools.

- (1) Move the main isolator switch to the off position and disconnect the battery earth.
- (2) Remove the door trim (refer to Section 2).
- (3) Disconnect the main harness to door harness connector.
- (4) Remove the large connector locking ring and remove the connector from the door aperture.
- (5) Disconnect the side repeater bulb holder assembly.
- (6) Disconnect the passenger door mirror connection.
- (7) Disconnect the window lift motor connections.
- (8) Remove the screw and washer securing the harness earth connections.
- (9) Remove the connections to the kerb lamp.
- (10) Disconnect the connections at the door lock module.
- (11) Disconnect the lock reminder switch connector.
- (12) Disconnect the connections to the door warning lamp.
- (13) Remove the harness from the retaining clips and remove the harness from the door.



B Installation Procedure

NOTE

Prior to installation of the harness check the wire insulation for splits, the security of all end connectors and serviceability of all grommets. Reject the harness if any faults are found.

- (1) Locate the door connector into the aperture and secure with the large connector locking ring.
- (2) Route the harness along the door beam and fit the harness into the retaining clips.
- (3) Connect the harness to the side repeater lamp.
- (4) Connect the harness to the window lift motor.
- (5) Secure the earth tag to the door beam using screw and washer.
- (6) Connect the harness to the kerb lamp.
- (7) Connect the harness to the door lock actuator.
- (8) Connect the harness to the lock reminder switch.
- (9) Connect the harness to the door warning lamp.
- (10) Connect the main harness door connector to the door harness.
- (11) Connect the battery earth.

NOTE

Make sure that the speaker connections are not shorted together or to ground.

- (12) Move the main isolator switch to the on position and ensure all door electrical equipment is functioning correctly.
- (13) Fit the door trim (Refer to Section 2).
- (14) Check function of speaker and door instruments in the driver's door.

12 STEERING COLUMN SWITCH ASSEMBLY

A Removal Procedure

NOTE

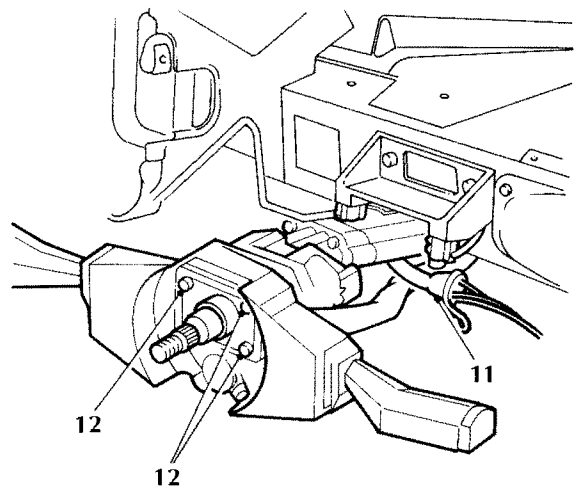
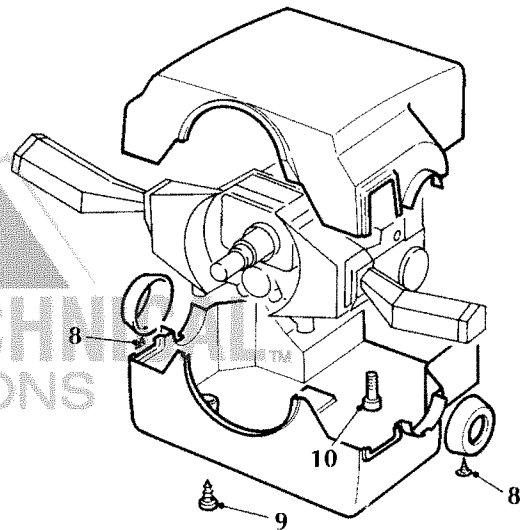
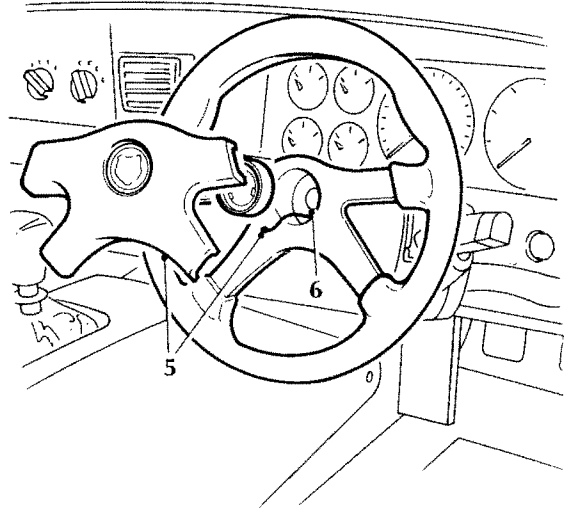
- 1 Ensure absolute cleanliness of work area and tools.
- 2 Observe the correct use of tools.

Special Tools

The following special tools will be required.

JaguarSport protective seat covers

- (1) Fit JaguarSport protective seat covers.
- (2) Switch the main isolator switch to the off position.
- (3) Disconnect the battery earth.
- (4) Remove the driver's underscuttle
- (5) Remove the steering wheel centre pad by gently prising off and disconnect the horn electrical connector.
- (6) Remove the steering wheel centre nut.
- (7) Mark across the boss and the end of the upper steering column using a permanent pen to ensure correct alignment on re-assembly, and remove the steering wheel.
- (8) Remove the screws securing the ignition switch escutcheon and blank escutcheon. Remove escutcheons.
- (9) Remove the two screws securing the lower switch cover and remove the cover.
- (10) Remove the three screws securing the upper switch cover and remove the cover.
- (11) Disconnect the steering column switch assembly electrical harness from the main harness.
- (12) Remove the four screws securing the steering column switch cluster to the column and remove the switches.



B Installation Procedure

Special Tools

The following special tools will be required

Torque wrench 0 to 47Nm (43.5ftlb)

- (1) Fit the steering column switch assembly to the column and secure using four screws. Torque the screws 8 to 9.5Nm (6 to 7ftlb).
- (2) Connect the column switch assembly harness connector to the main harness column switch connector.
- (3) Fit the steering column switch covers and secure, together with the column switch earth lead, with screws. Torque the upper cover screws 3 to 4Nm (2 to 3ftlb).
- (4) Fit the ignition switch blank escutcheon and ignition switch escutcheon and secure with screws.
- (5) Fit the steering wheel and boss to the upper column, taking care to align the marks made on removal. Tighten the centre nut to 35 to 41Nm (26 to 30ftlb).
- (6) Connect the horn electrical connector and fit the steering wheel centre pad.
- (7) Connect the battery earth.
- (8) Switch the main isolator switch to the on position and ensure all column-mounted switches function correctly.
- (9) Remove the JaguarSport protective seat covers.

13 AUDIO UNIT AMPLIFIER

A Removal Procedure

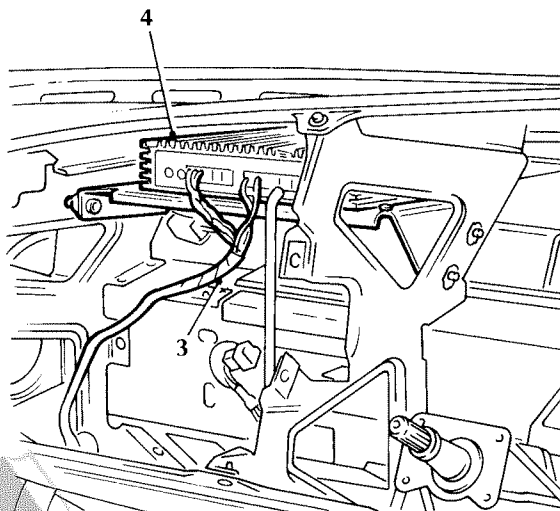
Special Tools

The following special tools will be required
JaguarSport protective seat covers.

NOTE

- 1 Ensure absolute cleanliness of work area and tools.
- 2 Observe the correct use of tools.

- (1) Fit JaguarSport protective seat covers.
- (2) Remove the instrument panel (Refer to Section 13).
- (3) Disconnect the electrical connections to the amplifier.
- (4) Remove the screws and washers securing the amplifier bracket to the dash surround brackets, withdraw the amplifier and collect the spacers.
- (5) Remove the 4 screws, washers and nuts securing the amplifier to the bracket and remove the amplifier.



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B Installation Procedure

- (1) Locate the amplifier on the mounting bracket and secure with 4 screws, washers and nuts.
- (2) Fit the amplifier and bracket to the dash surround brackets, locate the spacers and secure the bracket with 4 screws and washers.
- (3) Secure the main harness to the amplifier as follows;
 - a) Red/Black lead to right negative terminal
 - b) Red lead to right positive terminal
 - c) Green/Black lead to left negative terminal
 - d) Green lead to left positive terminal
 - e) Purple/Yellow lead to "BATT" terminal
 - f) Blue/White lead to "REMOTE" terminal
 - g) Amplifier earth to Ground.
- (4) Fit the instrument panel (Refer to Section 13).
- (5) Remove the JaguarSport protective seat covers.

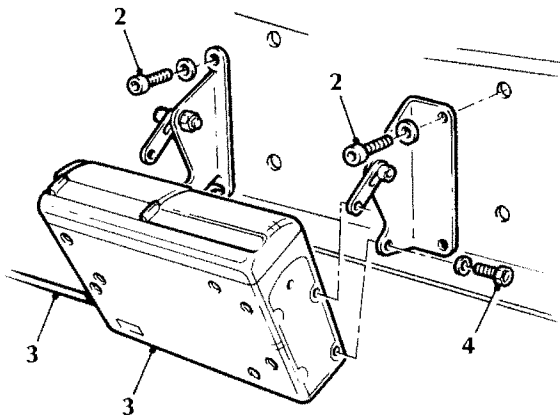
14 CDAUTO-CHANGERAND MOUNTINGS

A Removal Procedure

NOTE

- 1 Ensure absolute cleanliness of work area and tools.
- 2 Observe the correct use of tools.

- (1) Remove the auto-changer trim panel (Refer to Section 2).
- (2) Remove the four screws, washers and packers (if fitted) securing the auto-changer to the bulkhead.
- (3) Disconnect the CD lead from the auto-changer, slide the auto-changer LH bracket from behind the LH boot panel and remove from the vehicle.
- (4) Remove the four bolts and washers securing the mounting brackets to the auto-changer.



B Installation Procedure

- (1) If fitting a new auto-changer, remove the three shipping screws from the auto-changer and cover the holes with the patches provided. Remove both side covers and reposition the springs to notch 4 marked on the guide.
- (2) Fit the two mounting brackets and secure with four bolts and washers.
- (3) Connect the CD lead to the auto-changer.
- (4) Slide the LH bracket lower fixing under the LH boot panel and secure the auto-changer to the bulkhead using screws, washers and packers (previously removed).
- (5) Fit the auto-changer trim panel.
- (6) Ensure the magazine cover on the auto changer operates smoothly in the trim panel aperture. If a fouling condition occurs, pack out the changer brackets to achieve a central position.

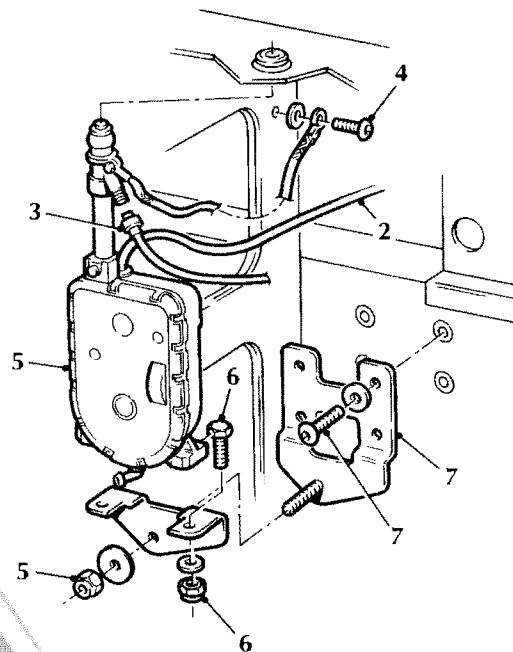
15 AERIAL

A Removal Procedure

NOTE

- 1 Ensure absolute cleanliness of work area and tools.
- 2 Observe the correct use of tools.

- (1) Remove the left hand rear wheel arch (Refer to Section 4).
- (2) Disconnect the aerial lead from the rear harness.
- (3) Remove the aerial cable from the aerial and remove the in-line choke from the open 'P' clip.
- (4) Remove the M5 screw and washer and remove the aerial earth strap.
- (5) Remove the M8 nut and washer securing the aerial bracket to the body; pull the aerial down through the body grommet and remove the aerial from the vehicle.
- (6) Remove the 2 M5 bolts, washers and nuts securing the bracket to the aerial and remove the bracket.
- (7) Remove the 4 M6 screws and washers securing the aerial mounting bracket to the body and remove the bracket from the vehicle.



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B Installation Procedure

Special tools

The following special tools will be required

Torque wrench 0-13Nm (0-10ftlb)

Materials

The following materials will be required

Red rubber grease

- (1) Fit the aerial mounting bracket to the vehicle and secure with 4 M6 screws and washers. Torque tighten screws 2 to 3Nm (1.5 to 2ftlb).
- (2) Secure the aerial to the aerial bracket using 2 M5 screws, washers and nuts.
- (3) Remove the grommet from the aerial and fit to the body using red rubber grease.
- (4) Locate the aerial in the grommet fitted in the body and secure to the body side using M8 nut and special washer. Torque the nut 12 to 13Nm (9 to 10ftlb).
- (5) Connect the aerial cable to the aerial.
- (6) Secure the earth strap to the body using M5 screw and washer. Torque tighten the screw 4 to 5Nm (3 to 4ftlb).
- (7) Connect the aerial lead to the rear harness.
- (8) Fit the in-line choke into the open 'P' clip on body.
- (9) Fit the wheel arch liner (Refer to Section 4).

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16 ALARM MODULE

A Removal Procedure

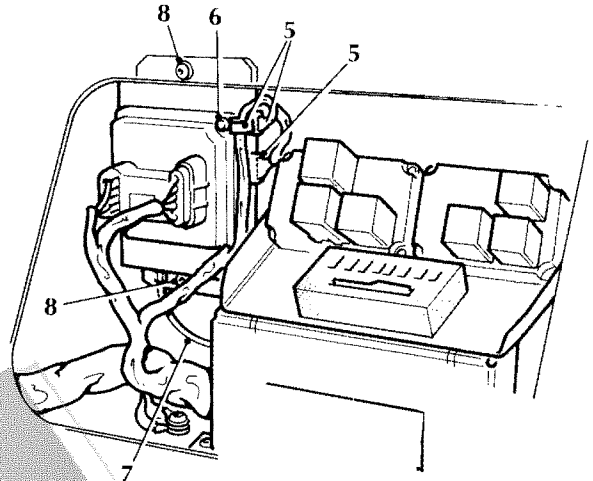
Special Tools

The following special tools will be required

JaguarSport Protective seat covers.

NOTE

- 1 Ensure absolute cleanliness of work area and tools.
- 2 Observe the correct use of tools.
- (1) Fit JaguarSport protective seat covers
- (2) Move the main isolator switch to the off position.
- (3) Disconnect battery earth.
- (4) Release the fasteners securing the passenger footwell panel and remove the panel.
- (5) Disconnect the harness connections to the alarm module and remove the upper connector from the P clip.
- (6) Remove the 2 nuts and washers securing the headlamp door module to the alarm module and remove the headlamp door module. Retrieve the plastic P clip.
- (7) Disconnect the alarm aerial from the module.
- (8) Remove the 2 screws and washers securing the alarm module to the bulkhead and remove the module from the vehicle.



B Installation Procedure

- (1) Fit the alarm module to the bulkhead and secure with 2 screws and washers.
- (2) Connect the alarm aerial to the module.
- (3) Fit the headlamp door module to the alarm module and secure with 2 washers and nuts. Fit plastic P clip to the upper right stud
- (4) Connect the harness connectors to the alarm module; secure the upper harness connector to the P clip.
- (5) Fit the footwell panel and secure the fasteners.

NOTE

When battery earth is connected the CDL alarm system will perform a lock/unlock function. Do not close doors until the CDL alarm system has performed its lock/unlock function.

- (6) Connect battery earth.
- (7) Move the main isolator switch to the on position.
- (8) Remove the JaguarSport protective seat covers.

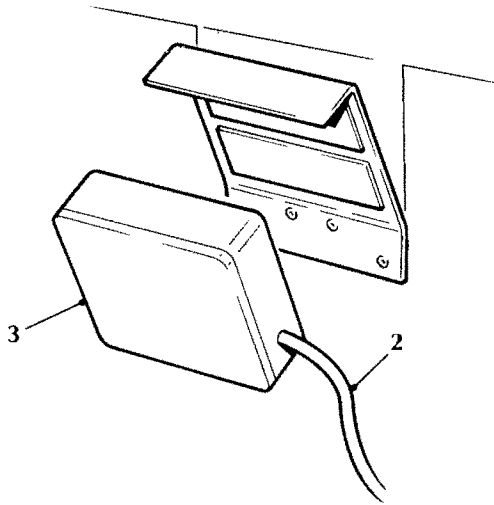
17 ALARM SENSOR

A Removal Procedure

NOTE

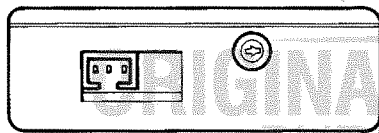
- 1 Ensure absolute cleanliness of work area and tools.
- 2 Observe the correct use of tools.

- (1) Remove the rear trim (Refer to Section 2).
- (2) Disconnect the sensor breakout from the harness.
- (3) Pull the sensor from the bracket.



B Installation Procedure

- (1) If fitting a new sensor, verify that the unit's sensitivity adjustment is set as shown. Thoroughly clean the back of the sensor and stick a velcro strip to the back of the sensor.
- (2) Locate the sensor in the bracket and press firmly to ensure correct contact of the velcro.
- (3) Connect the sensor breakout to the main breakout.
- (4) Fit the rear trim (Refer to Section 2).



18 VETS CHECK

The following check must be carried out after any modification, adjustment or replacement of parts associated with the central door locking or alarm systems.

The check must be carried out in the sequence shown in order to ensure the system is fully checked.

Procedure

NOTE:

The check must be carried out with the windows down to enable access to the internal door handles.

Check remote keys arm/dis-arm, lock/unlock and sensitivity.

Check courtesy lighting off/lock, on/unlock.

Check LED operation (illuminates for approximately 10 seconds, then flashes when armed).

Arm alarm.

Unlock offside door using door button, open offside door (carry out within 10 seconds after the alarm LED flashes).

Check alarm operation, close door.

Reset to armed state.

Unlock nearside door using door button, open nearside door (carry out within 10 seconds after the alarm LED flashes).

Check alarm operation, close door.

Reset to armed state.

Open tailgate, check alarm operation.

Check engine bay lamps, close tailgate.

Reset to armed state.

Open boot, check alarm operation.

Dis-arm alarm, check boot lamp operation and LH engine bay lamp on.

NOTE

LH engine bay lamp will illuminate with boot lamp due to the interconnection of the alarm circuit.

Close boot.

Reset alarm to armed state (wait 60 seconds clear of the vehicle).

NOTE:

Any movement around the vehicle, with the windows open, in the 60 second period will effect the sensitivity of the microwave sensor setting.

Sweep arm across glass areas, check alarm remains untriggered.

Reach into the vehicle, check the alarm operation and microwave sensitivity.

Dis-arm the alarm.

Open both doors.

Arm alarm.

Check door mislock, close both doors.

Dis-arm the alarm.

CDL operation (door key each).

Lock offside door, (1)/(2) check doors locked.

Unlock offside door, (1)/(2) check doors unlocked.

Lock nearside door, (1)/(2) check doors locked.

Unlock offside door, (1)/(2) check doors unlocked.

Open/close door, check interior light delay.

Open door, enter vehicle, lift the inertia switch button, close and lock the doors using door buttons.

Ignition switch on, check door unlock sequence (doors should unlock with ignition on).

Check courtesy lighting off.

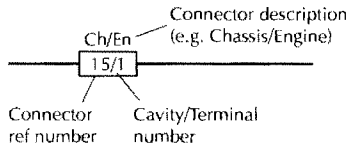
Check 'Engine Check' warning lamp on.

Ignition switch off, inertia switch button down (ECU fault code 32 is set during this test).

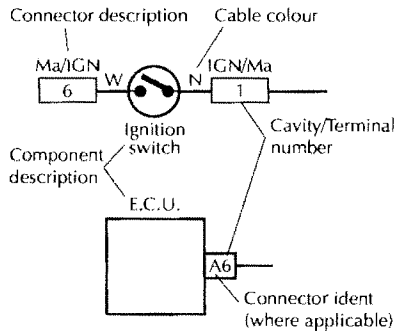
Clear fault code 32 with lap top computer (Refer to Engine Fault Code Analysis in Introduction).

CIRCUIT DIAGRAM KEY

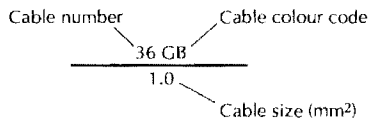
MAIN HARNESS INTERFACE CONNECTION



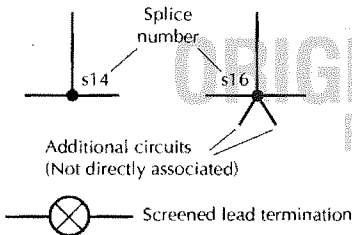
COMPONENT TO HARNESS CONNECTION



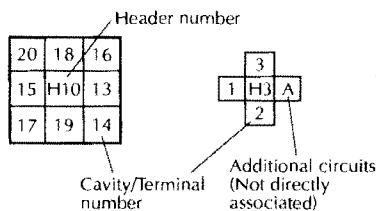
CABLE IDENTIFICATION



INTERNAL SPICE CONNECTION



HEADER CONNECTORS



HARNESS EARTH POINT



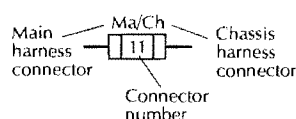
INTEGRAL EARTH POINT



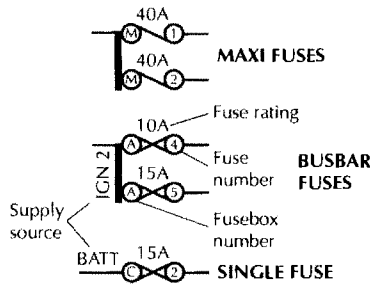
CONNECTOR (General)



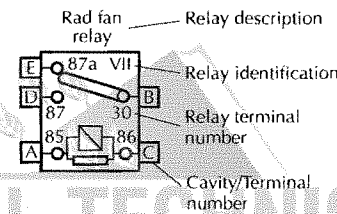
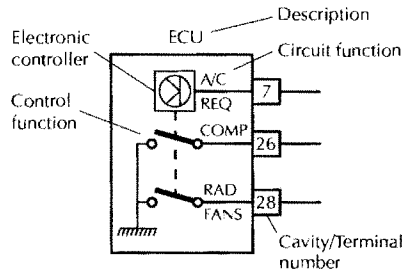
INTER-HARNESS CONNECTOR



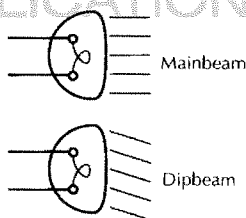
FUSES



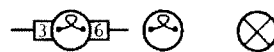
TYPICAL COMPONENT DETAILS



HEADLAMPS



LAMP - TYPICAL PANEL LAMP



LED



Diode D1



MOTOR - TYPICAL

LH Rad fan

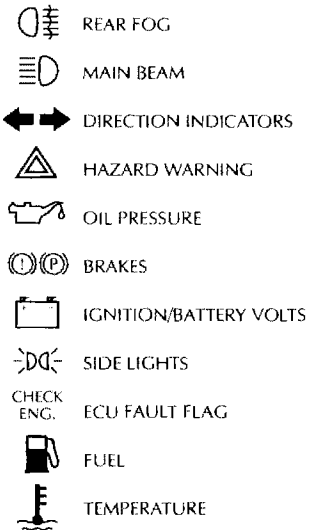


Function symbol

SWITCH - TYPICAL



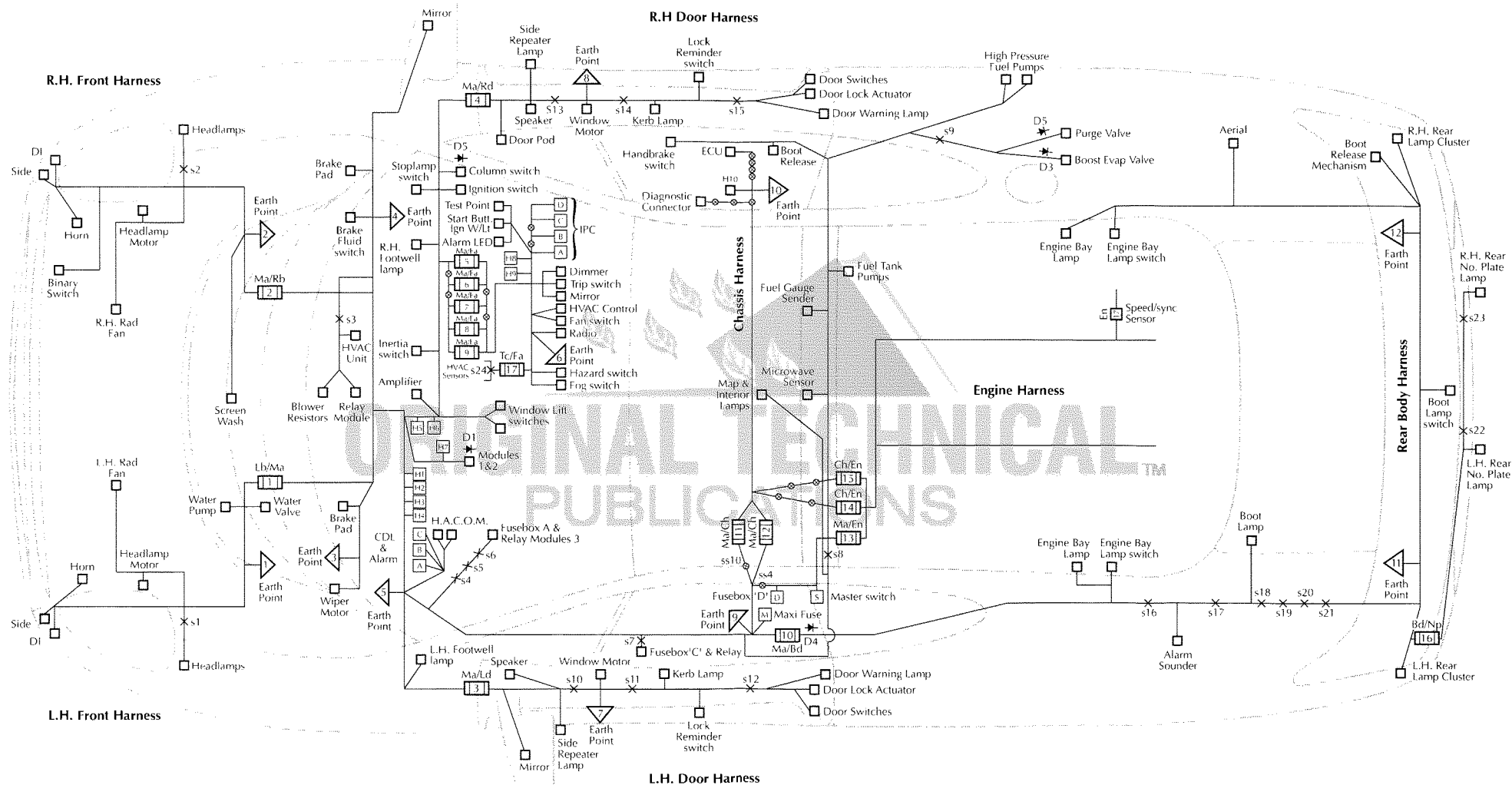
WARNING LAMPS & SYMBOLS

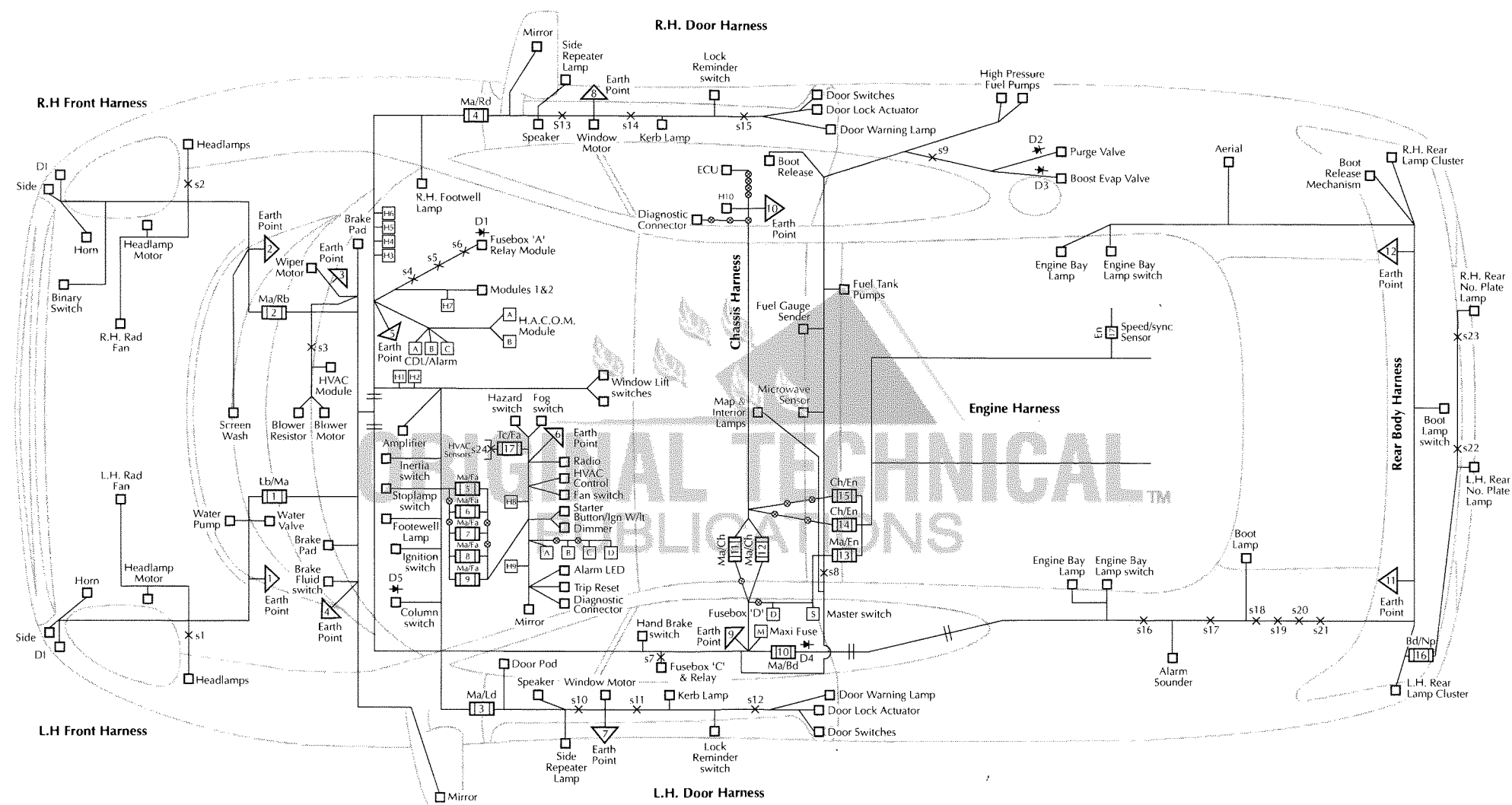


COLOUR CODE	
B	BLACK
N	BROWN
R	RED
O	ORANGE
Y	YELLOW
G	GREEN
U	BLUE
P	PURPLE
S	SLATE
W	WHITE
K	PINK
LG	LIGHT GREEN

ABBREVIATIONS

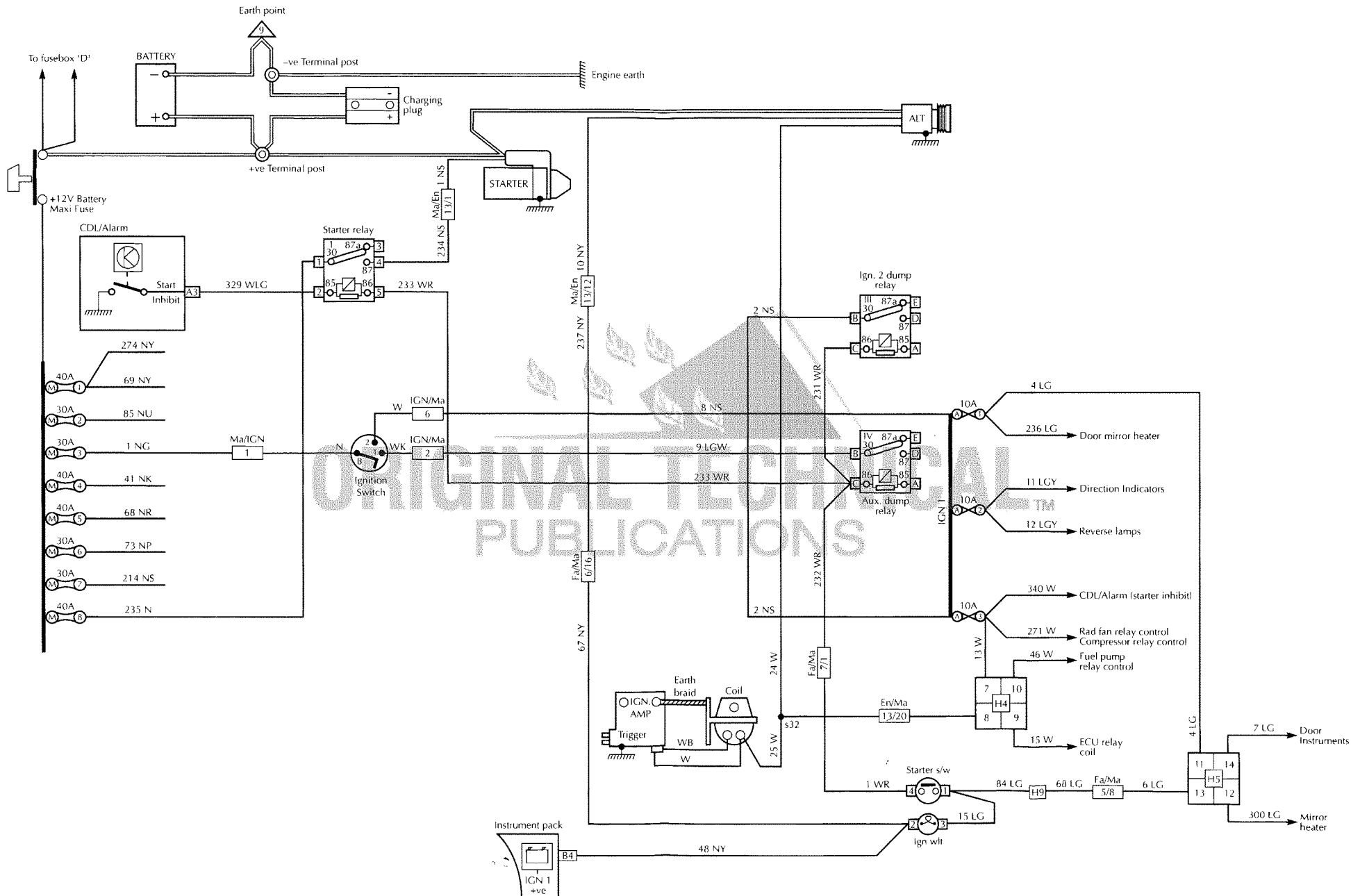
- Lb - Front LH Body Harness
- Rb - Front Right Body Harness
- Fa - Facia Harness Connector
- Ma - Main Harness Connector
- Rd - RH Door Harness Connector
- Ld - LH Door Hand Connector
- Bd - Rear Body Harness Connector
- En - Engine Harness Connector
- Ch - Chassis Harness Connector
- AUX - Ignition switch controlled feed
Key position 1 & 2, Disconnected during cranking
- IGN 1 - Ignition switched controlled feed
Key position 2
- IGN 2 - Ignition switch controlled feed
Key position 2, Disconnected during cranking
- BATT - Battery feed, Controlled via Isolator switch
- BATT (dir) - Direct Battery feed
(i.e. NOT controlled by Master Switch)
- P.W.M. - Pulse Width Modulation
- I.S.C.V. - IDUs Speed Control Unit
- E.C.U. - Engine Control Unit
- H.V.A.C. - Heating Ventilating & Air Conditioning

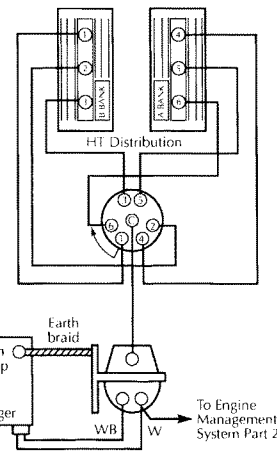


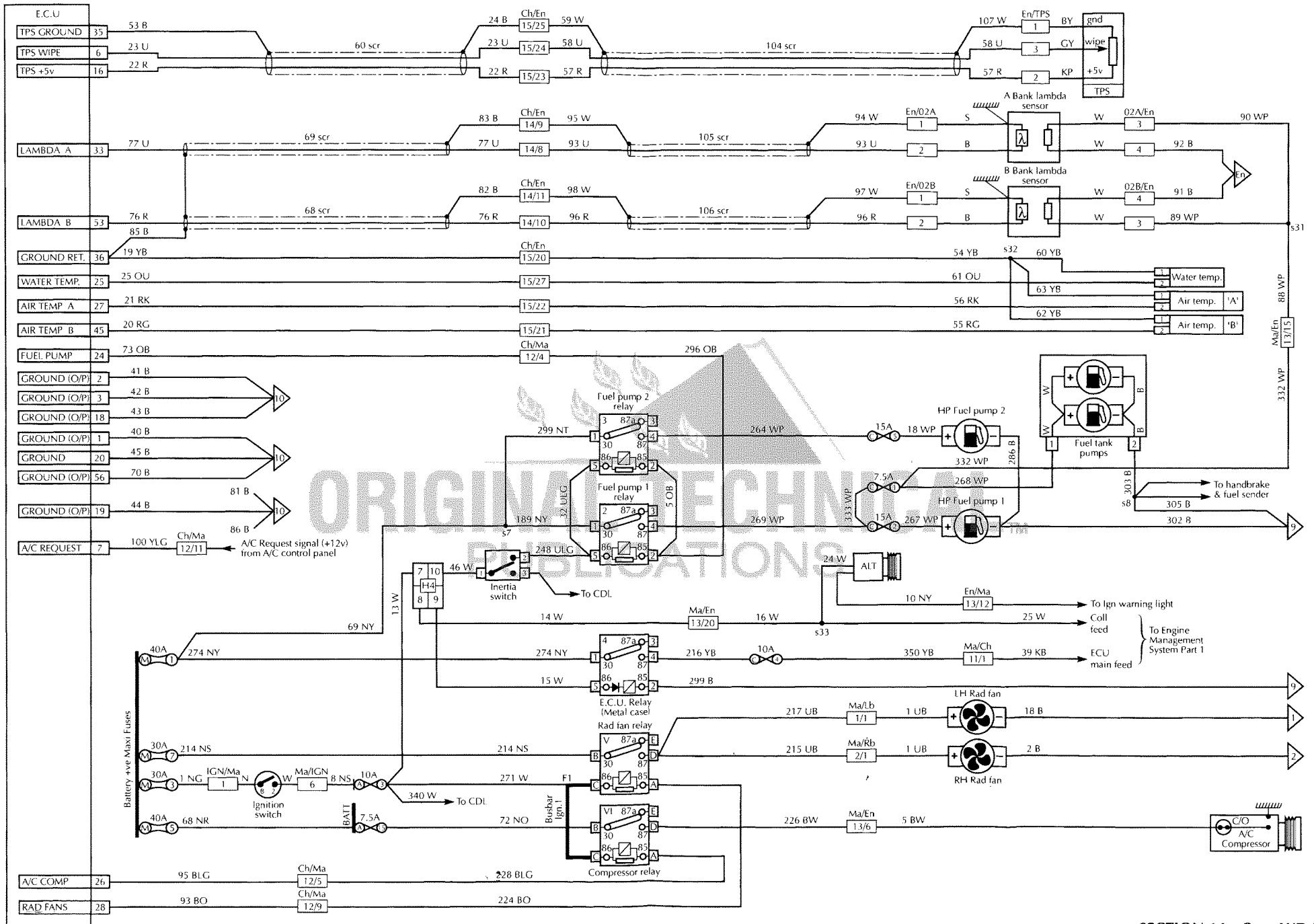


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BATTERY, STARTER & ALTERNATOR SYSTEM

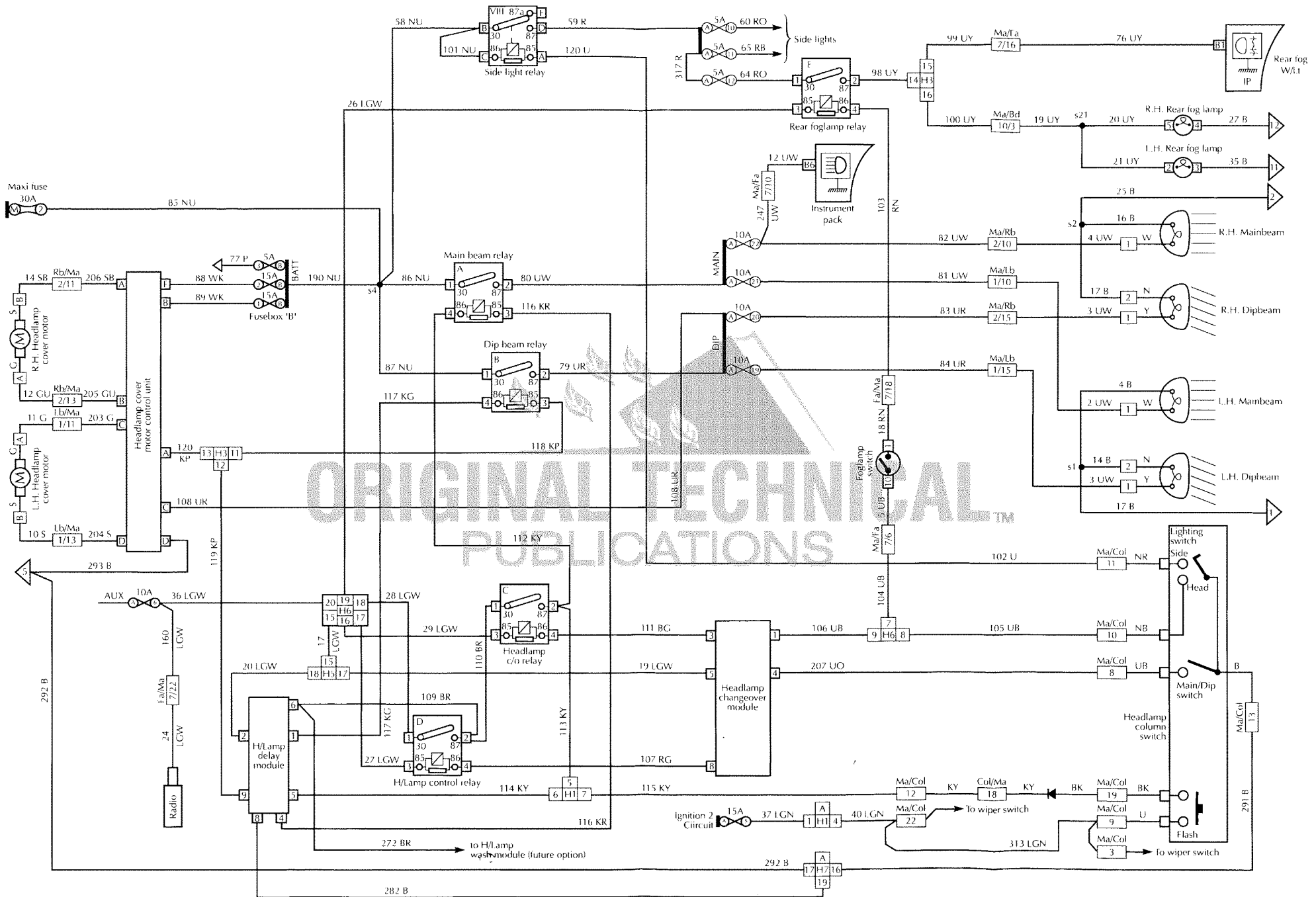




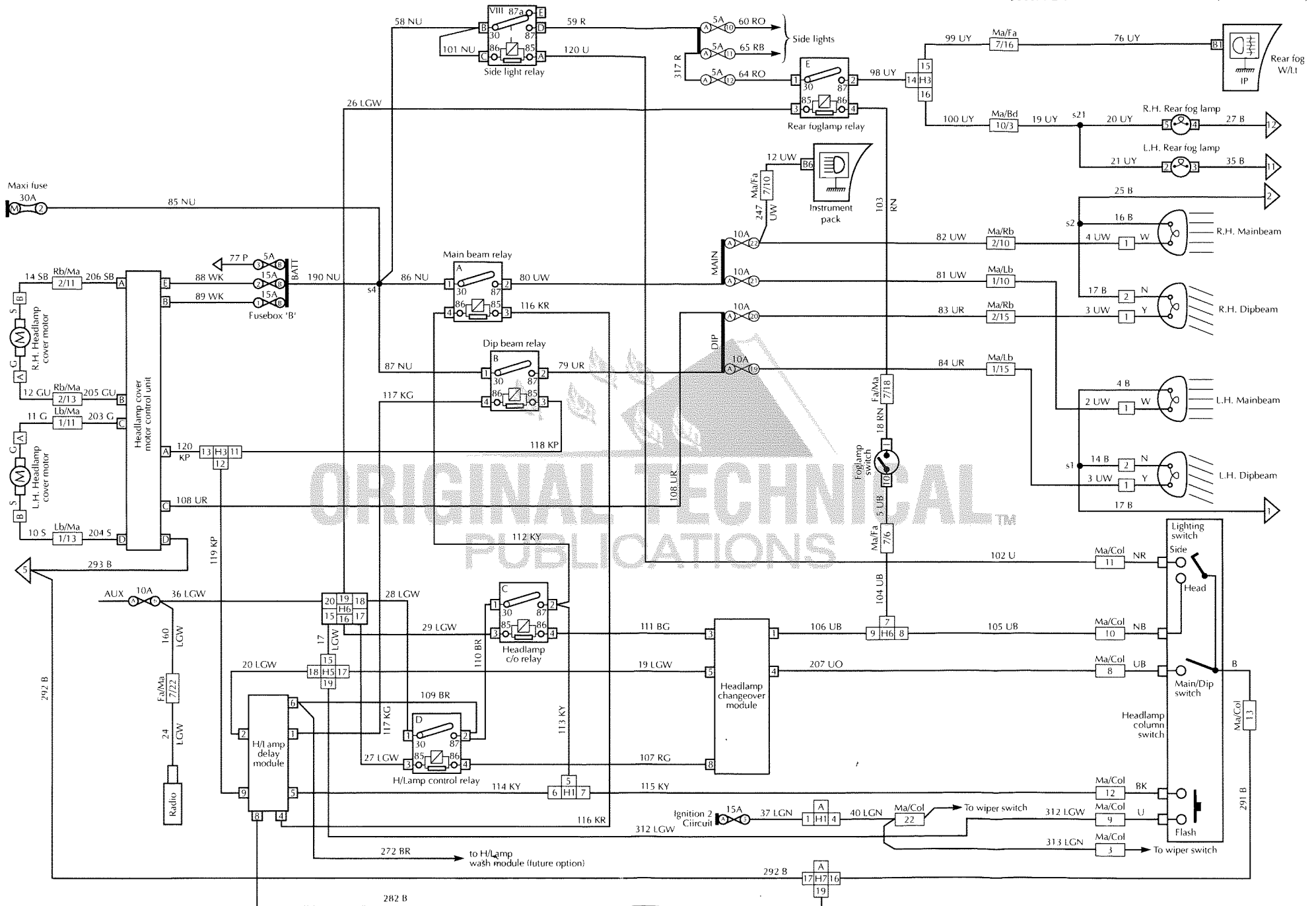


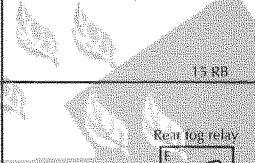


LIGHTING HEADLAMP CIRCUIT (EARLY CARS)

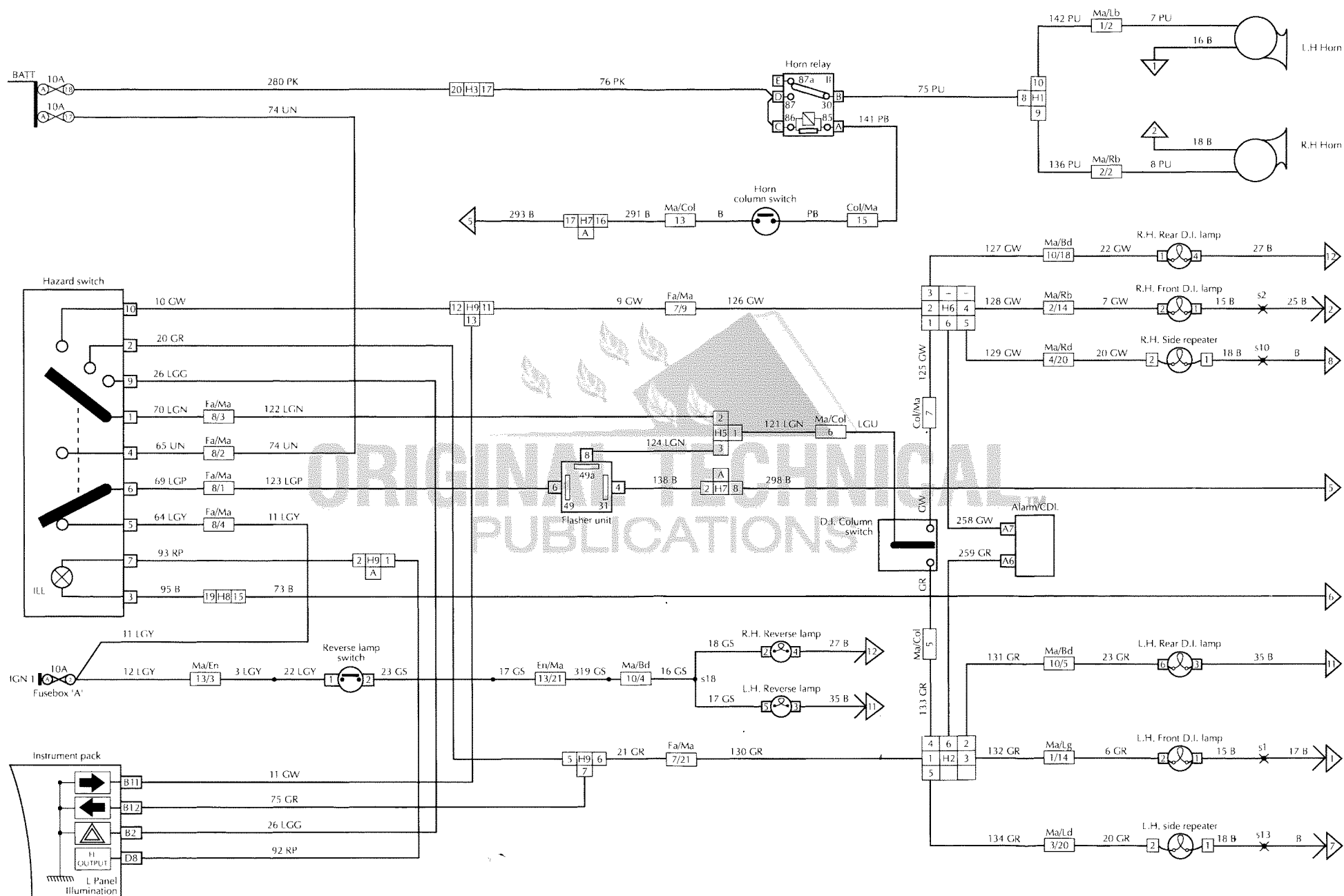


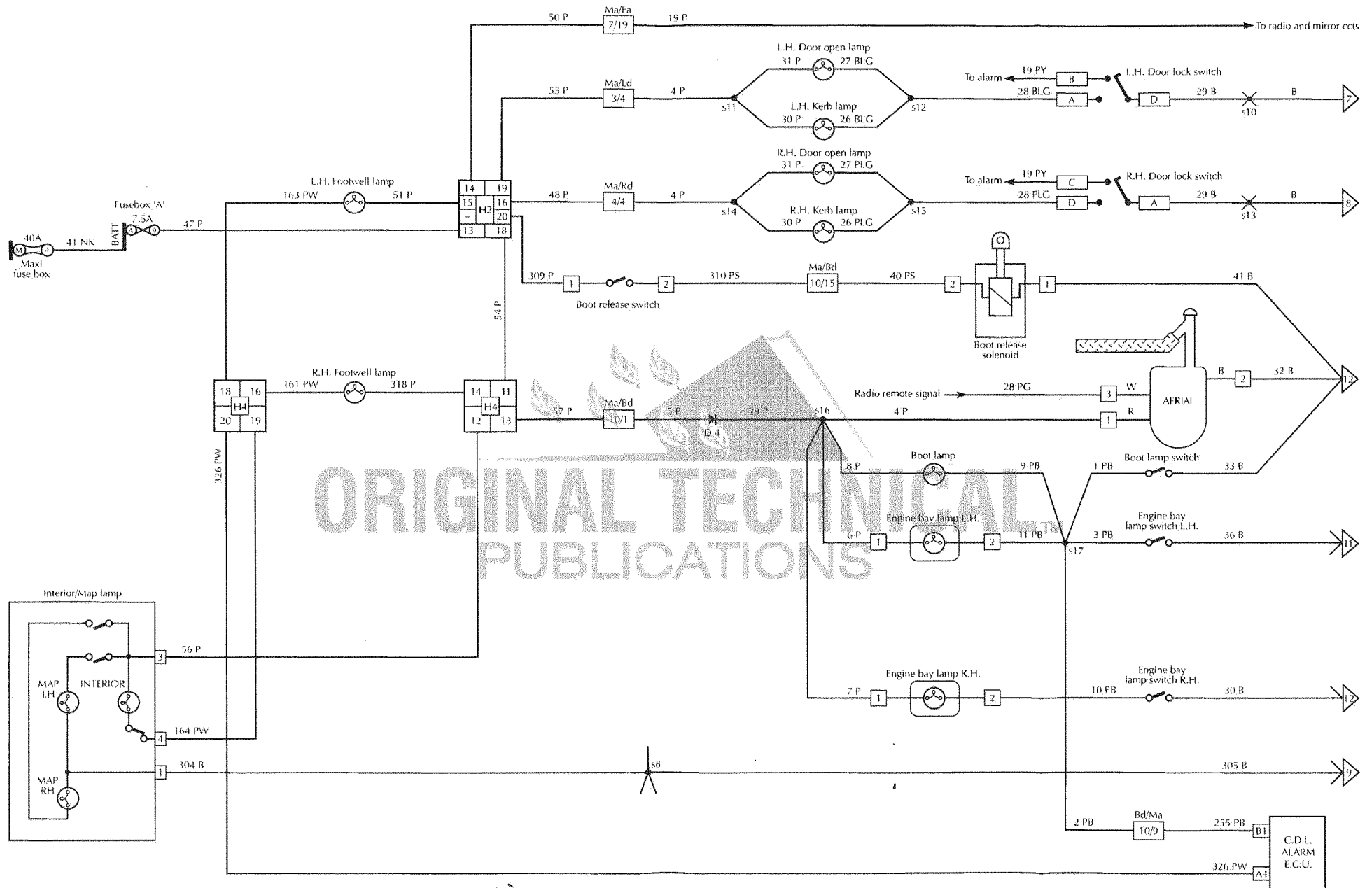
LIGHTING HEADLAMP CIRCUIT (LATER CARS)

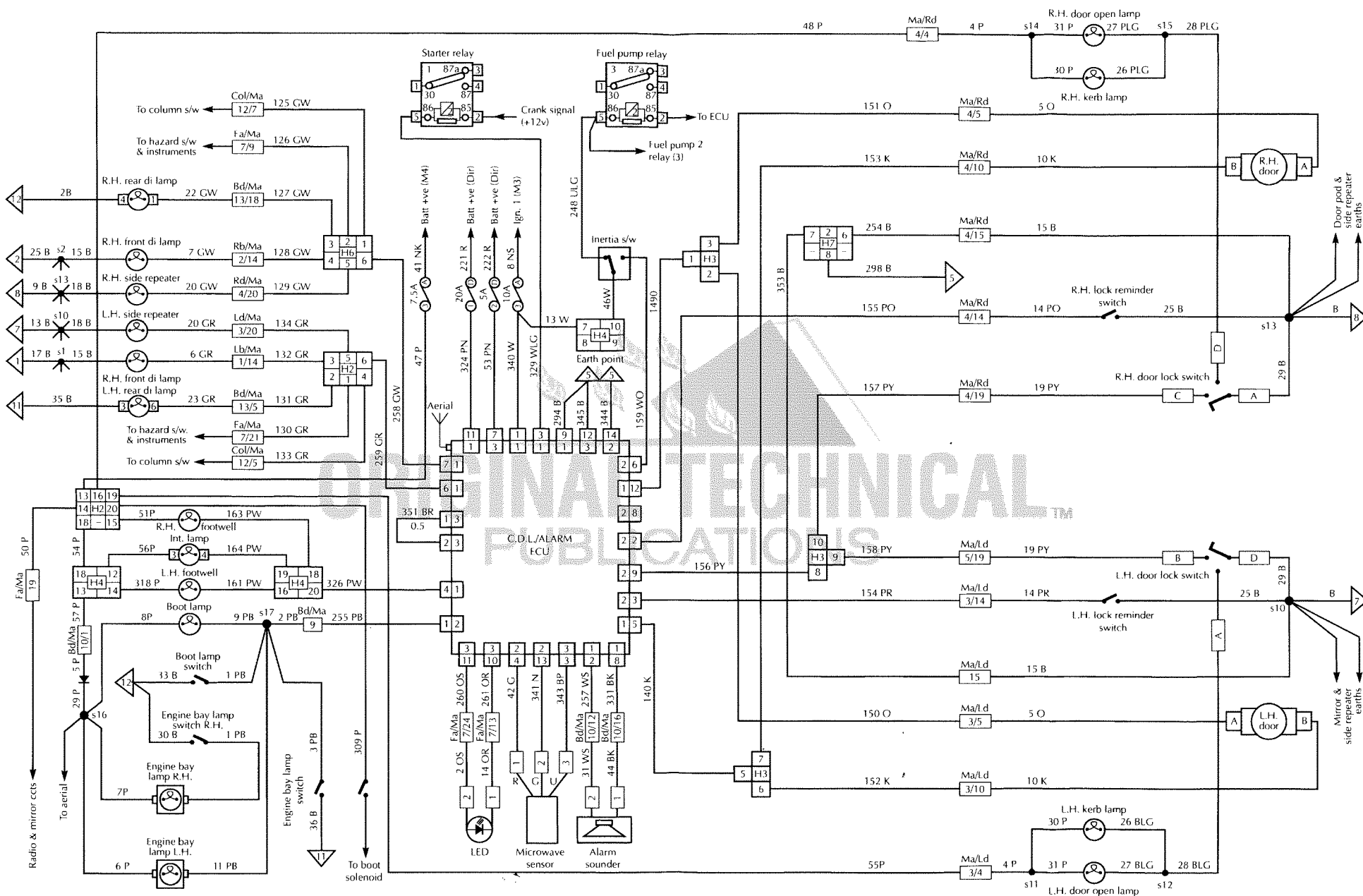


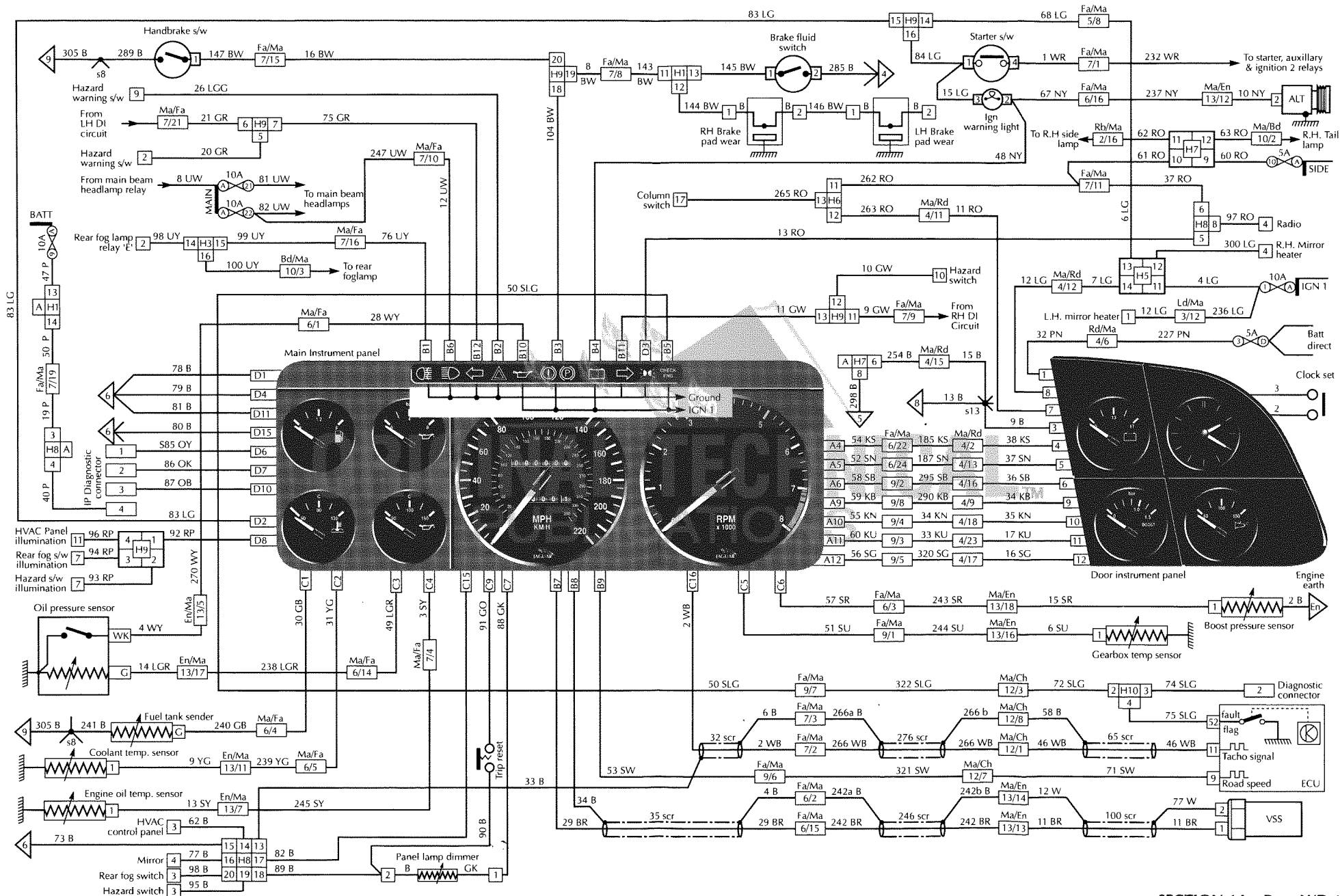


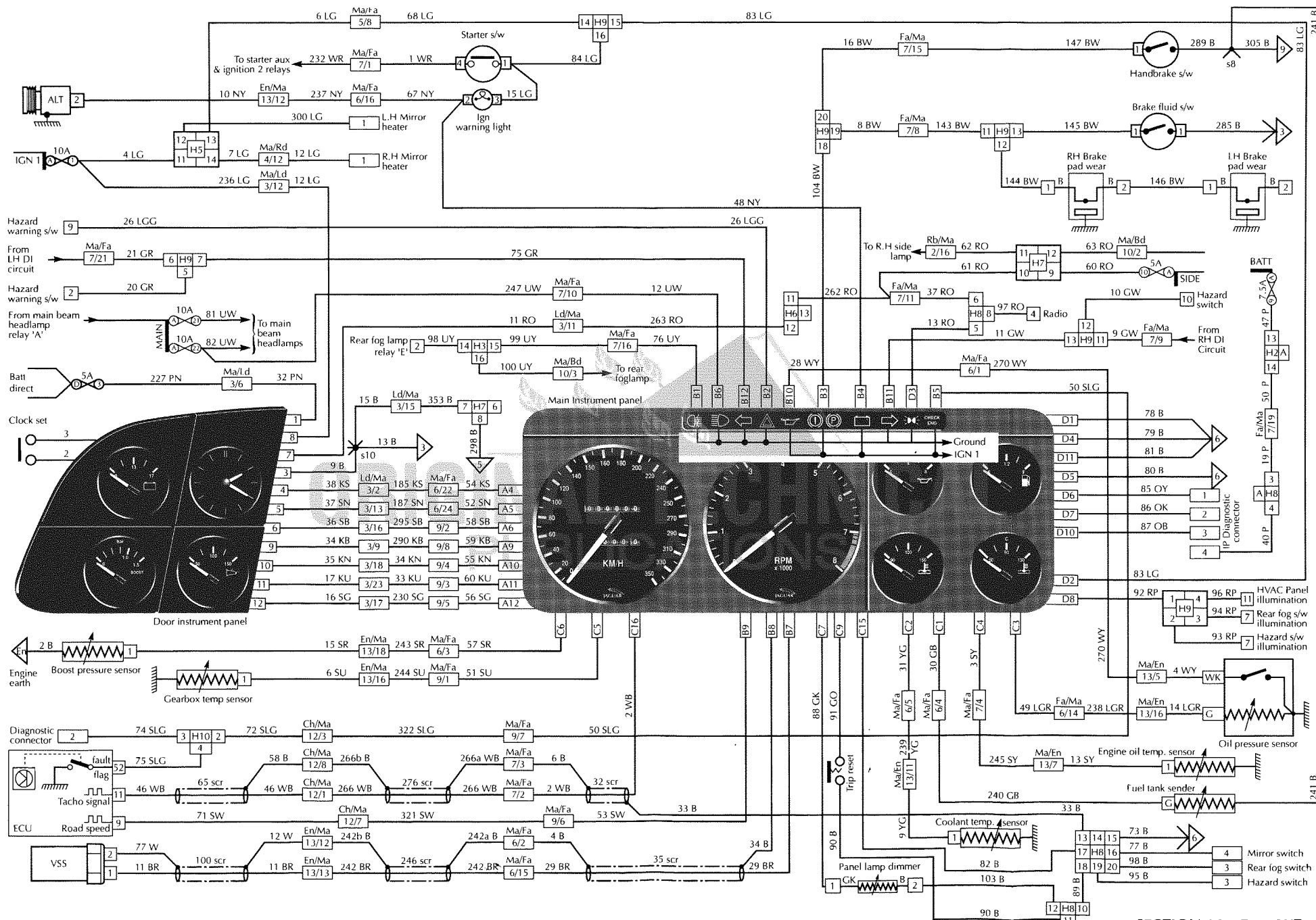
HORNS, DIRECTION INDICATORS/ HAZARD LAMPS & REVERSE LAMPS

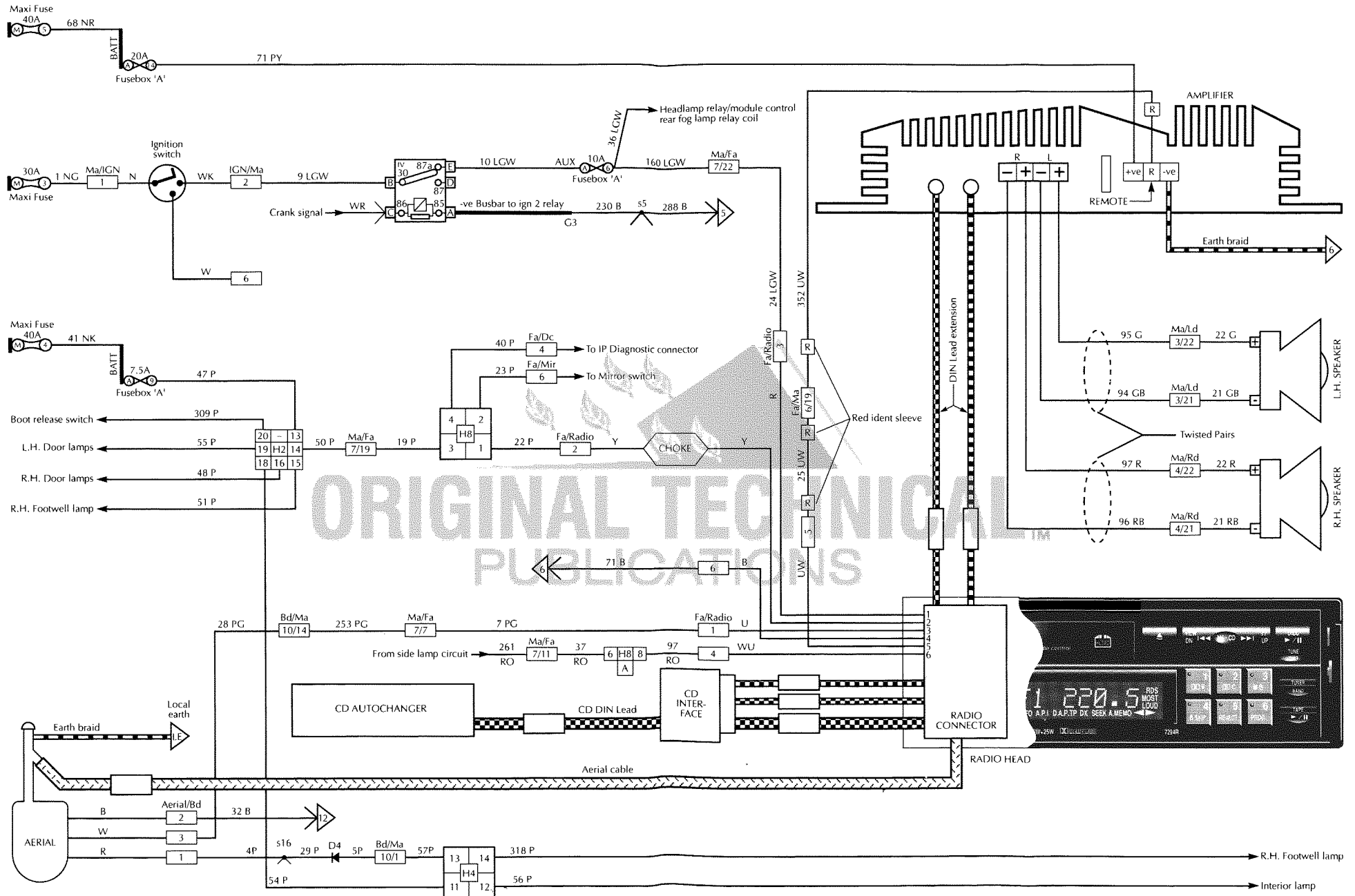


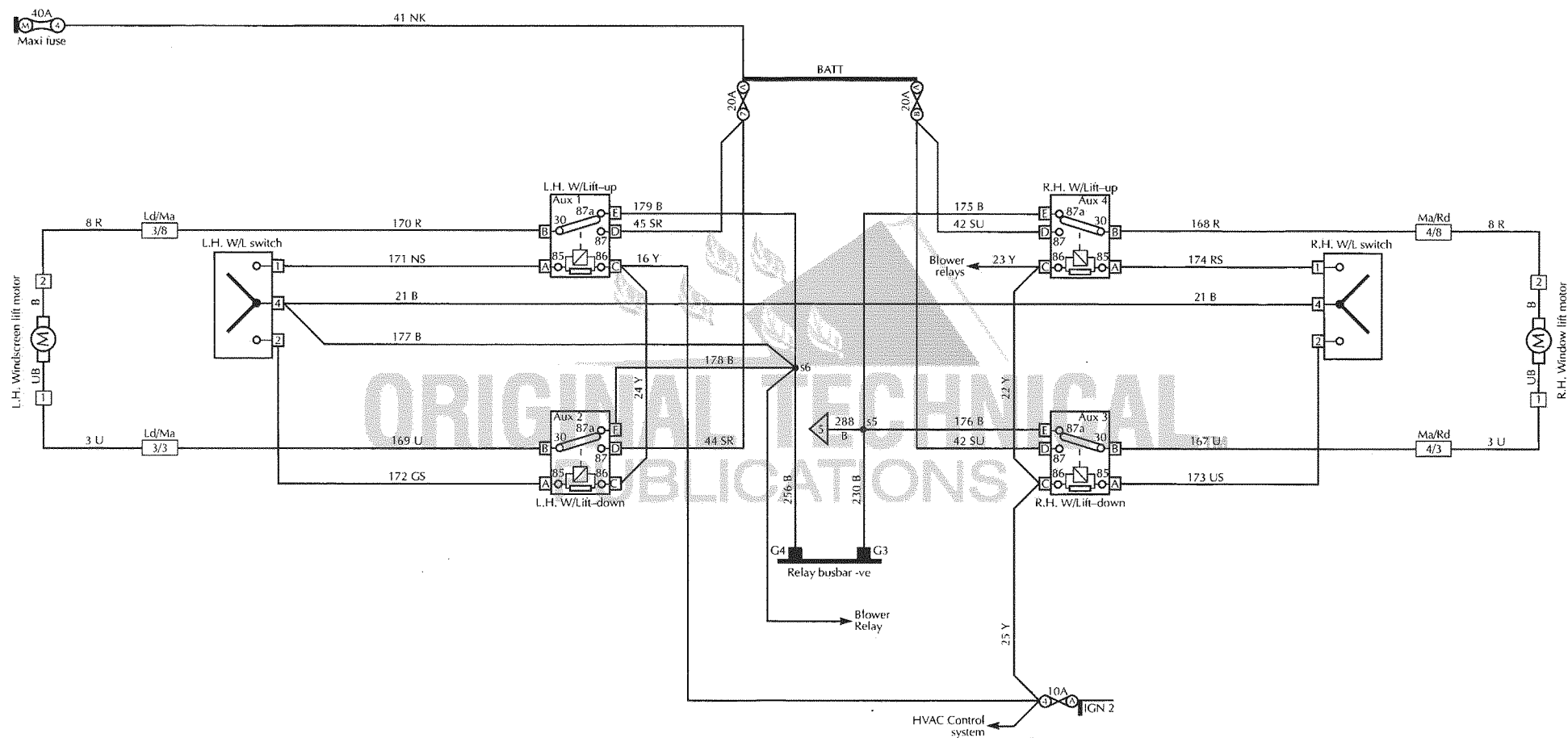




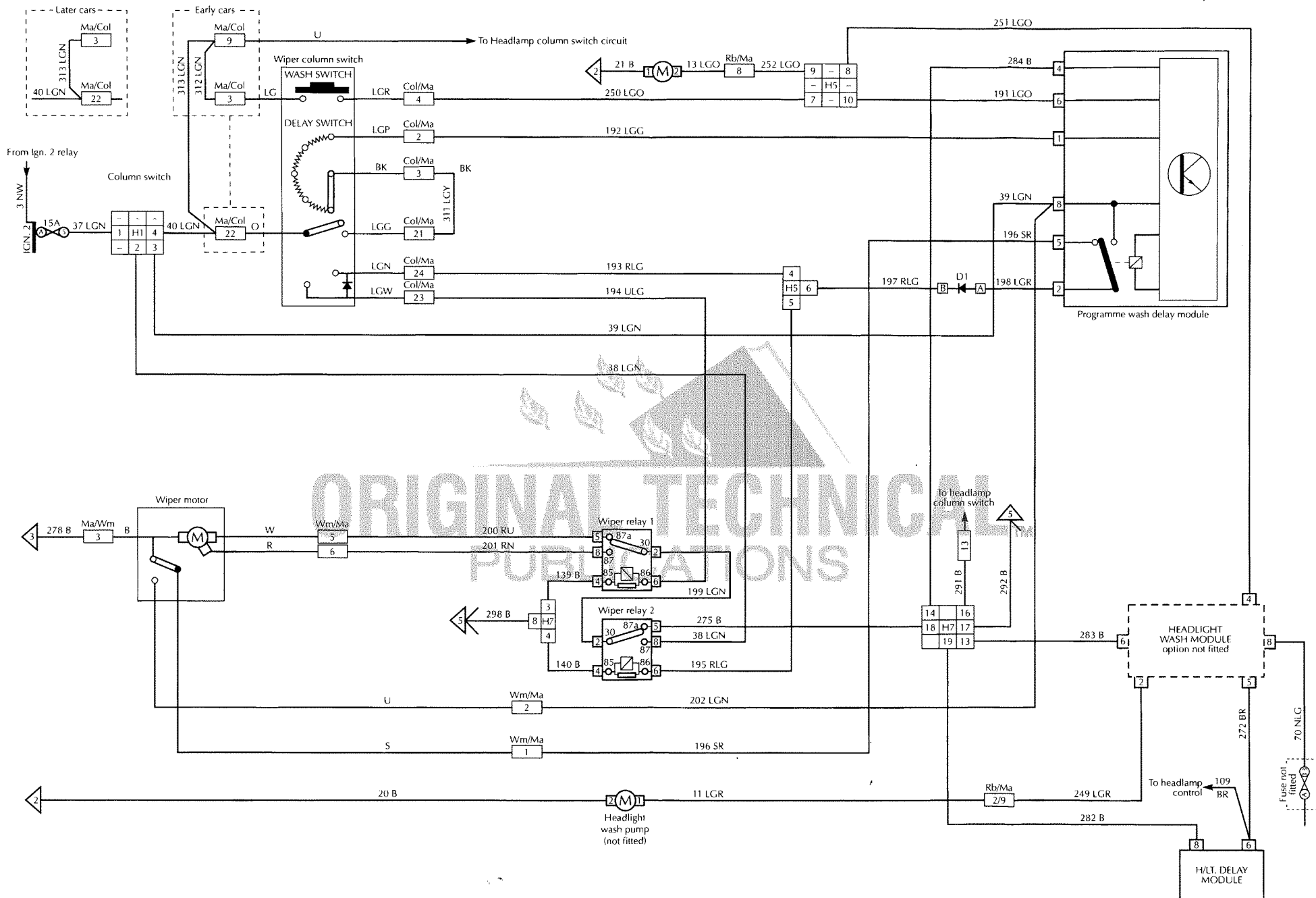


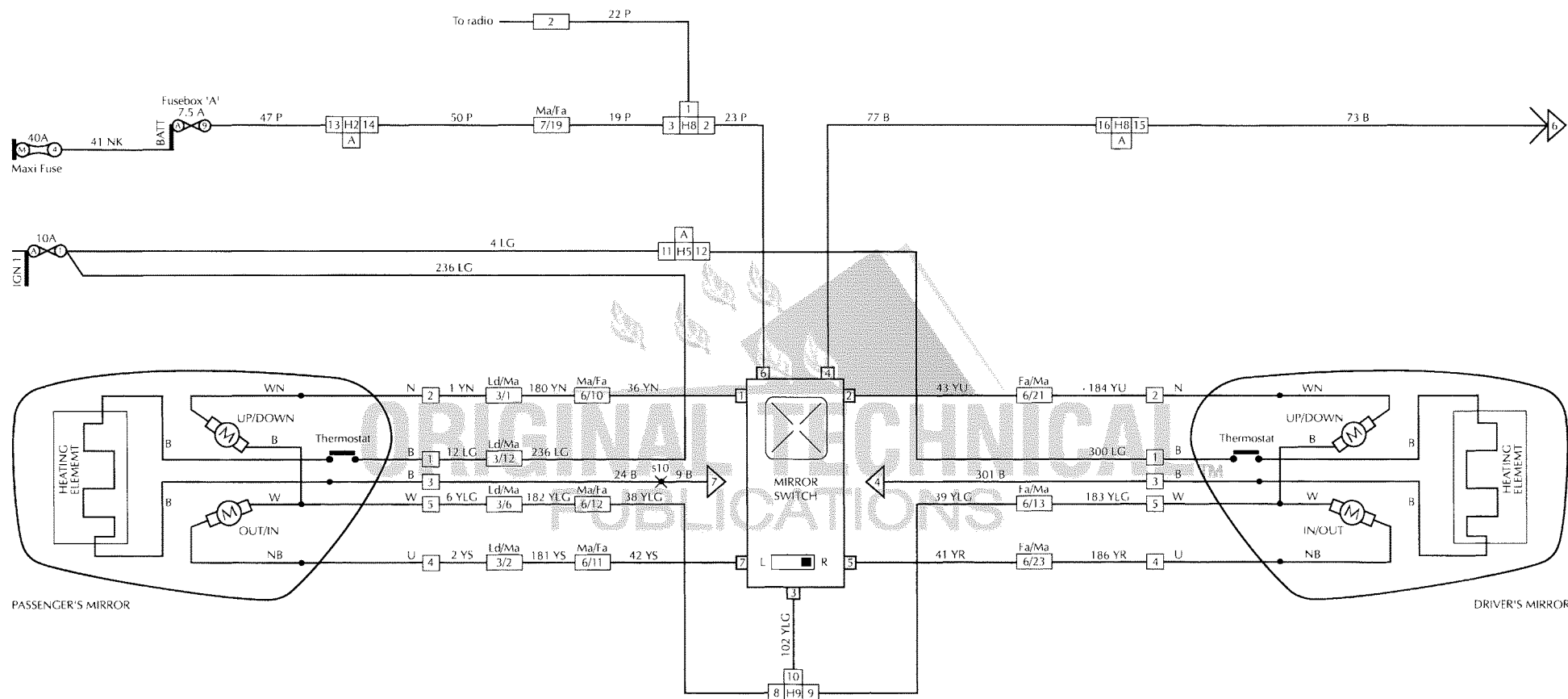


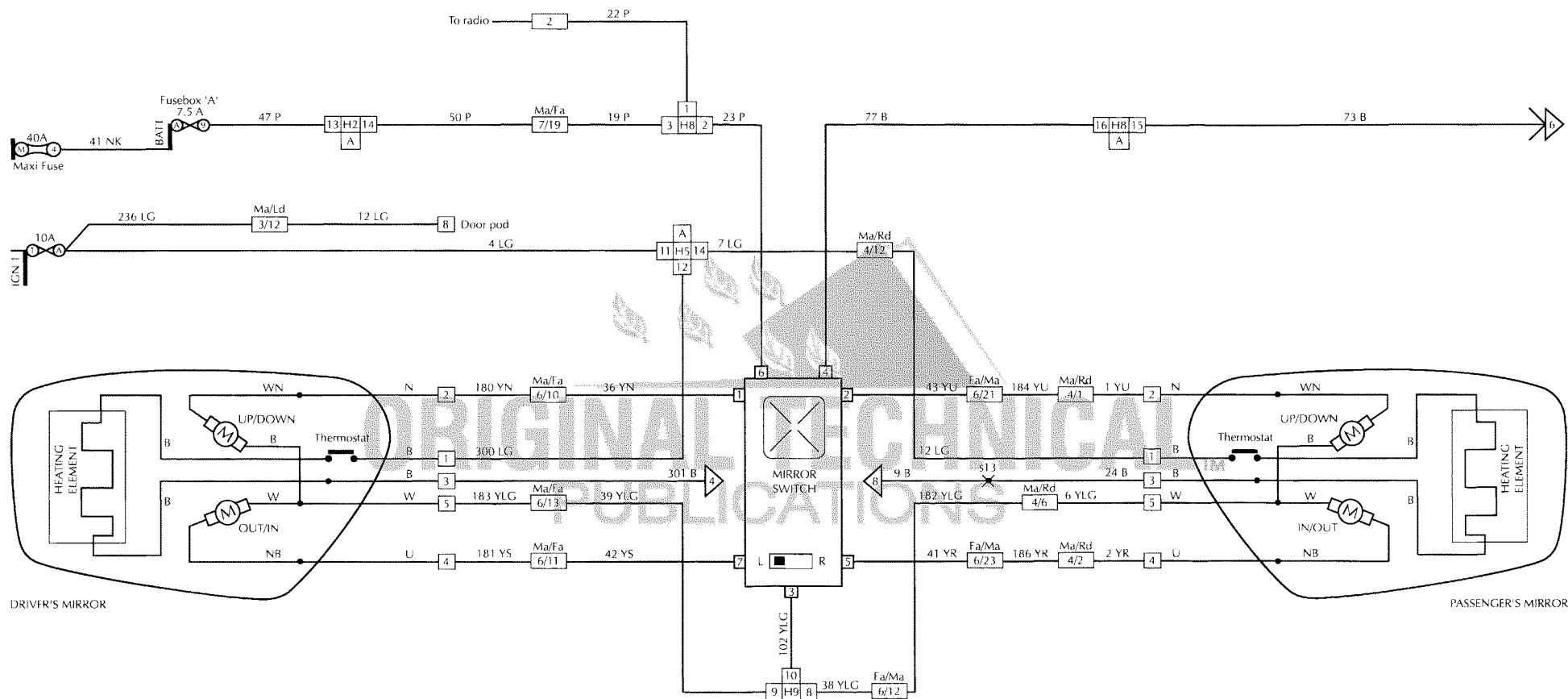


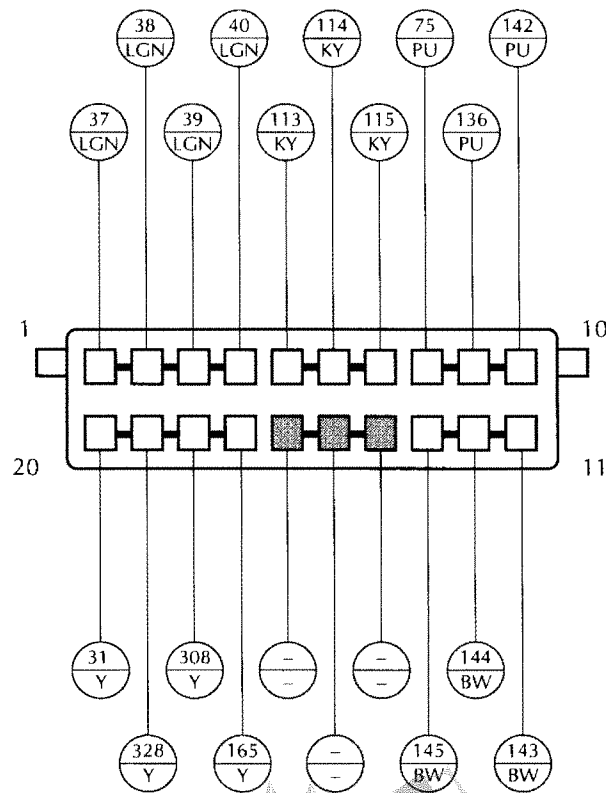


SECTION 14 Page WD 15







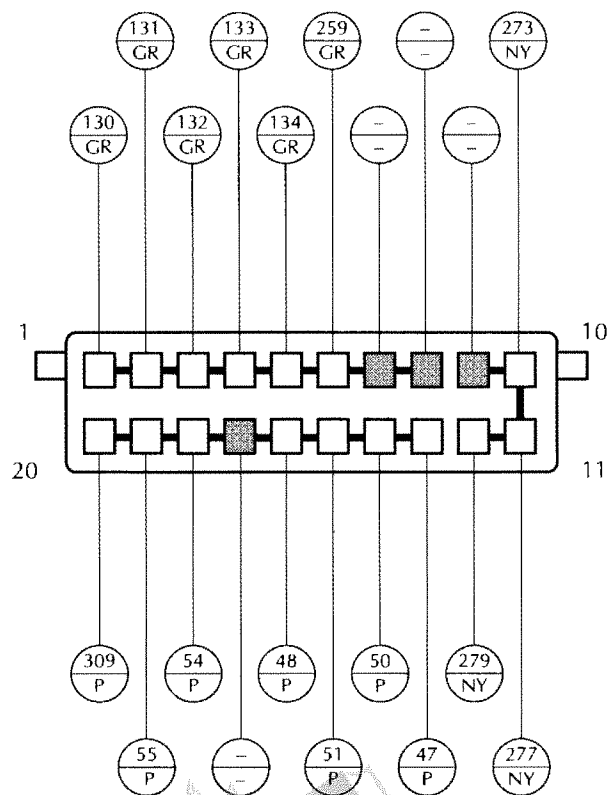


Header No.1(Orange)
(Main Harness)

Location RHD - in header retainer under Fusebox 'A' relay base
Location LHD - Passenger footwell to the side of the harness shield

Cav No.	Cable No.	Cable colour	Circuit Description	Function
1	37	LGN	Ign 2 feed from fuse A5	Ign 2 feed to wiper circuit Fuse A5
2	38	LGN	Feed to wiper relay2 pin 8	
3	39	LGN	Feed to intermittent wipe module pin 8	
4	40	LGN	Feed to column switch	
5	113	KY	To headlamp change over relay C pin 2	Headlamp flash circuit
6	114	KY	Headlamp delay module pin 5	
7	115	KY	To column lighting switch connector	
8	75	PU	From horn relay 11 pin B	Relay controlled feed to horn Fuse A18 (Batt)
9	136	PU	Feed to R.H. front body connector	
10	142	PU	Feed to L.H. front body connector	Brake warning lamp circuit
11	143	BW	To facia connection - Ma7 pin 8	
12	144	BW	To R.H. brake pad sensor	
13	145	BW	To brake fluid switch	
14				
15				
16				
17	165	Y	Ign 2 feed from fuse A4	Ign 2 feed to HVAC control circuit Fuse A4
18	308	Y	Feed to heater water pump	
19	328	Y	To facia connection Ma6 pin 17 (control panel)	
20	31	Y	Feed to HVAC delay module connector A pin 5	

HEADER 2

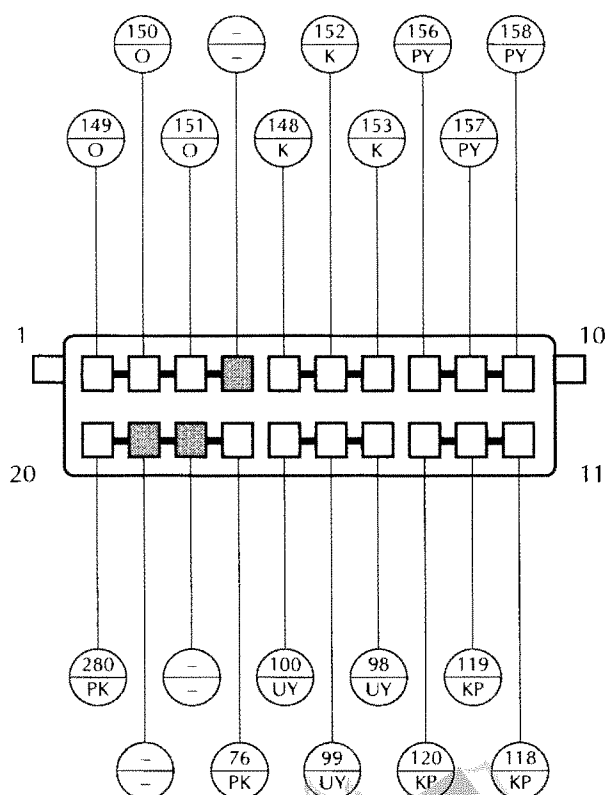


Header No. 2 (Grey)
(Main Harness)

Location RHD - in header retainer under Fusebox 'A' relay base

Location LHD - Passenger footwell to the side of the harness shield

Cav No.	Cable No.	Cable No.	Circuit Description	Function
1	130	GR	To facia harness connector Ma7 pin 21 - hazard switch & warning lamp	LH direction indicator circuit
2	131	GR	To rear body harness connector Ma 10 pin 5 - rear LH direction indicator	
3	132	GR	To LH front body connector Ma 1 pin 14 - L.H direction indicator	
4	133	GR	Feed from column switch connector pin 5 - direction indicator switch	
5	134	GR	To L.H door harness connector Ma3 pin 20 - side repeater lamp	
6	259	GR	From CDL/Alarm connector 1 pin 6	
7				
8				
9				
10	273	NY	Not used	Feed courtesy lamp circuits radio and mirror circuits Batt fuse A9
11	277	NY	Not used	
12	279	NY	Not used	
13	47	P	Feed from fuse A9	
14	50	P	To facia harness connector Ma7 pin 19 - radio and mirror feed	
15	51	P	To L.H footwell lamp	
16	48	P	To R.H door connector Ma4 pin 4 - door kerb/open lamp	
17				
18	54	P	To header 4 pin 11 - loop circuit	
19	55	P	To L.H door harness connector Ma3 pin 4 - door kerb/open lamp	
20	309	P	To boot release switch	



Header No. 3 (Orange)

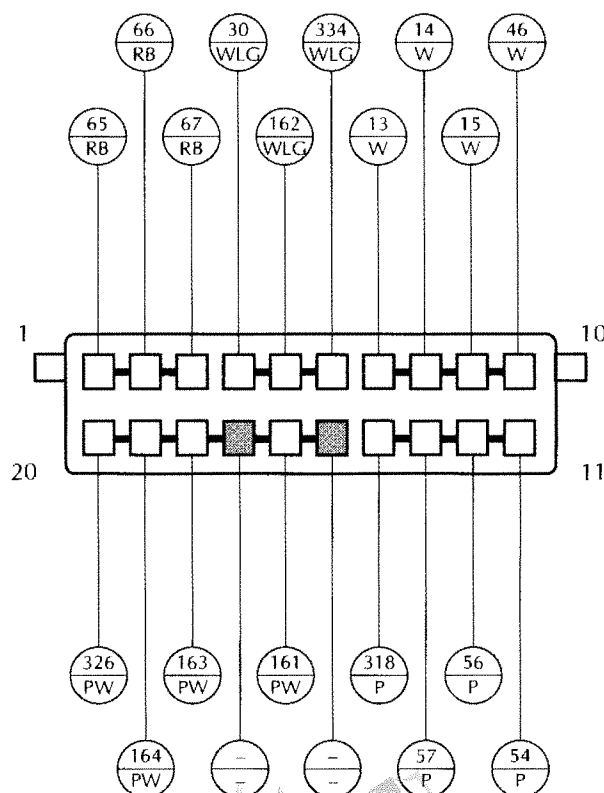
(Main Harness)

Location RHD - in header retainer under Fusebox 'A' relay base

Location LHD - Passenger footwell to the side of the harness shield

Cav No.	Cable No.	Cable colour	Circuit Description	Function
1	149	O	From CDL/Alarm unit	Door lock circuit lock ground
2	150	O	To L.H door lock actuator	
3	151	O	To R.H door lock actuator	
4				
5	148	K	From CDL/Alarm unit	Door lock circuit lock 12V batt
6	152	K	To L.H door lock actuator	
7	153	K	To R.H door lock actuator	
8	156	PY	From CDL/Alarm unit	Courtesy lamp circuit switched to ground when door closed
9	157	PY	To R.H door lock switch	
10	158	PY	To L.H door lock switch	
11	118	KP	To headlamp dip beam relay B pin 3	Headlamp cover module control circuit
12	119	KP	To headlamp delay module pin 9	
13	120	KP	To Headlamp door module pin A	
14	98	UY	From rear fog lamp relay E pin 2	Rear fog lamp switched feed Side light controlled fuse A12
15	99	UY	To facia connector Ma7 pin 16 - rear fog warning lamps	
16	100	UY	To body connector rear fog lamps	
17	76	PK	To horn relay 11 pin D	Horn relay Batt feed fuse A18
18				
19				
20	280	PK	From fusebox 'A' 18	

HEADER 4



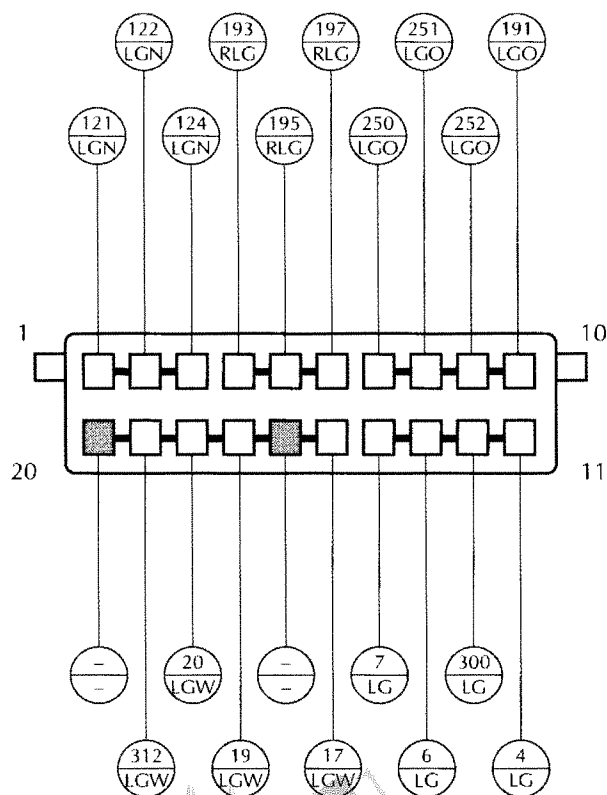
Header No. 4 (Black)

(Main Harness)

Location RHD - in header retainer under Fusebox 'A' relay base

Location LHD - Passenger footwell to the side of the harness shield

Cav No.	Cable No.	Cable colour	Circuit Description	Function
1	65	RB	From fuse A 11	LH side lamp circuit Fuse A11
2	66	RB	To L.H front front body harness connector Ma1 pin 16 - side lamp	
3	67	RB	To L.H body connector Ma10 pin 13 - rear tail lamp	A/C compressor request control circuit
4	30	WLG	To RH front body harness connector Ma2 pin 3 - HVAC binary switch	
5	162	WLG	To HVAC realy panel pin A3	
6	334	WLG	From facia harness connector Ma5 pin 7 - HVAC control panel A/C request	
7	13	W	From fuse A3	Ignition 1 control circuit fuse A3
8	14	W	To engine harness connector Ma13 pin 20 - ignition amplifier, coil and alternator	
9	15	W	To ECU relay pin 5	
10	46	W	To inertia switch pin 1	Battery feed courtesy lamp circuits and aerial fuse A5
11	54	P	To header 2 pin 18 Loop circuit	
12	56	P	To inertior lamp & map reading lamps	
13	57	P	To rear body harness connector Ma10 pin 1 - aerial, engine bay lamps & boot lamp	Courtesy lamp switched circuit switched to earth when door open
14	318	P	To R.H footwell lamp	
15				
16	161	PW	To R.H footwell lamp	Courtesy lamp switched circuit switched to earth when door open
17				
18	163	PW	To L.H footwell lamp	
19	164	PW	To interior lamp	
20	326	PW	To C.D.L/Alarm unit connector 1 pin 4	

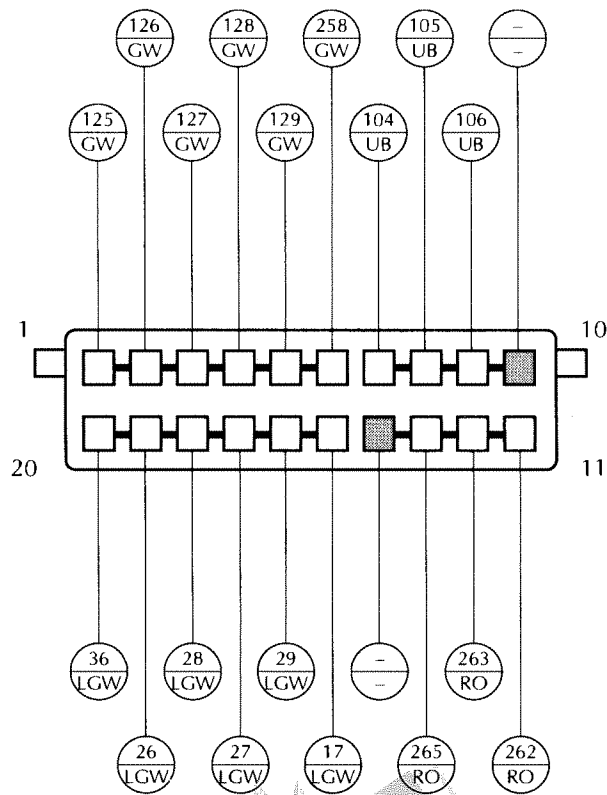


Header No. 5 (Black)
(Main Harness)

Location RHD - Passenger footwell to the side of the harness shield
Location LHD - In header retainer under Fusebox 'A' relay base

Cav No.	Cable No.	Cable colour	Circuit Description	Function
1	121	LGN	Feed to column switch	Flasher unit switched feed for Di/hazard circuit
2	122	LGN	Feed to facia connector Ma8 pin 3 - hazard switch	
3	124	LGN	Feed from flasher unit	
4	193	RLG	Feed from column switch	Wiper circuit slow speed
5	195	RLG	To wiper relay 2 pin 6	
6	197	RLG	To diode 1 pin B	
7	250	LGO	Switched feed from column switch connector pin 4	Wash/wipe control circuit connector
8	251	LGO	To Headlamp wash relay (not used)	
9	252	LGO	To R.H front body connector Ma2 pin 9 - washer pump	
10	191	LGO	To intermittent wash/wipe module pin 5	
11	4	LG	Ignition 1 feed fuse A1	Ignition 1 feed fuse A1
12	300	LG	To mirror connector-heater circuit	
13	6	LG	To facia connector Ma5 pin8 - instruments/start button	
14	7	LG	To R.H door connector Ma4 pin 12 - instruments	
15	17	LGW	Loop to header 6 pin 15	Auxiliary feed fuse A6 headlamp control and rear fog circuits Auxiliary feed fuse A6
16				
17	19	LGW	To headlamp changeover module pin 5	
18	20	LGW	To headlamp delay module pin 2	
19	312	LGW	To column switch connector pin 9 - headlamp flash	
20			(Later headlamp circuit only)	

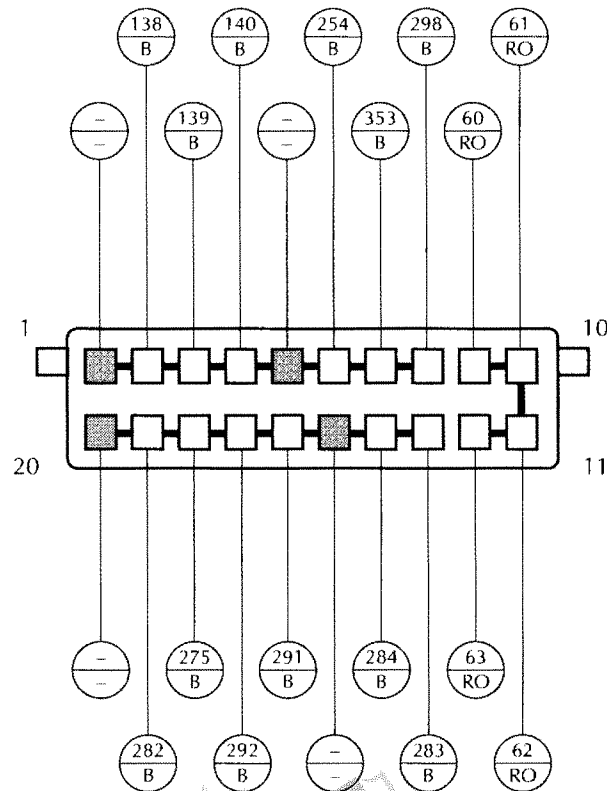
HEADER 6



Header No. 6 (Blue)
(Main Harness)

Location RHD - Passenger footwell to the side of the harness shield
Location LHD - In header retainer under Fusebox 'A' relay base

Cav No.	Cable No.	Cable colour	Circuit Description	Function
1	125	GW	Feed from column switch connector pin 7 - direction indicator switch	RH direction indicator circuit
2	126	GW	To facia harness connector Ma7 pin 9 - hazard switch and warning lamp	
3	127	GW	To rear body connector Ma10 pin 18 - rear R.H indicator	
4	128	GW	To R.H front body connector Ma2 pin 14 - R.H indicator	
5	129	GW	To R.H door harness connector Ma4 pin 20 - side repeater	
6	258	GW	From CDL/Alarm connector 1 pin 7	Switched ground headlamp circuit
7	104	UB	To facia harness connector Ma7 pin 6 - rear fog lamp switch	
8	105	UB	From column switch connector pin 10 - Headlamp switch	
9	106	UB	To headlamp changeover module pin 1	
0				
11	262	RO	From facia harness connector Ma7 pin 11	RH side light switched feed fuse A10
12	263	RO	R.H door harness connector Ma4 pin 11 - instrument door pod illumination	
13	265	RO	Column switch connector pin 17 - switch illumination	
14				
15	17	LGW	Feed to header 5 pin 15	Auxiliary feed fuse A6
16	29	LGW	To headlamp change over relay C pin 3	
17	27	LGW	To headlamp control relay D pin 3	
18	28	LGW	To headlamp control relay D pin 1	
19	26	LGW	To rear fog lamp relay E pin 3	
20	36	LGW	Feed in from fusebox A6	



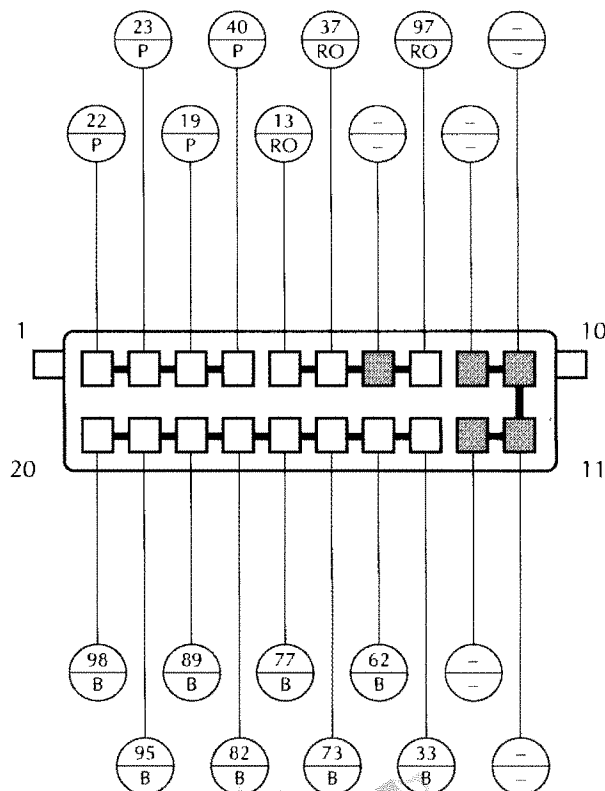
Header No. 7 (Grey)

(Main Harness)

Location - Under relay/module bases 1 and 2

Cav No.	Cable No.	Cable colour	Circuit Description	Function
1				
2	138	B	Earth return from flasher unit	Earth returns Earth point 5
3	139	B	Earth return from wiper relay 1 pin 4	
4	140	B	Earth return from wiper relay 2 pin 4	
5				
6	254	B	Earth return from R.H door connector Ma4 pin 15	RH side lamp switched feed fuse A10
7	353	B	Earth return from L.H door connector Ma3 pin 15	
8	298	B	To earth point 5	
9	60	RO	Feed from fuse A10	RH side lamp switched feed fuse A10
10	61	RO	To facia connector Ma3 pin 11 - instrument illumination	
11	62	RO	To R.H front body connector Ma2 pin 16 - R.H side lamp	
12	63	RO	To body connector Ma10 pin 2 - R.H side rear lamp & No. plate lamps	Earth returns circuit Earth point 5
13	283	B	Earth return from Headlamp wash relay	
14	284	B	Earth return from programme wash/wipe delay module pin 4	
15				
16	291	B	Earth return from column switch connector 13 - Headlamp cct	Earth returns circuit Earth point 5
17	292	B	Earth point 5	
18	275	B	Earth return from wiper relay 2 pin 5	
19	282	B	Earth return from headlight delay module pin 8	
20				

HEADER 8

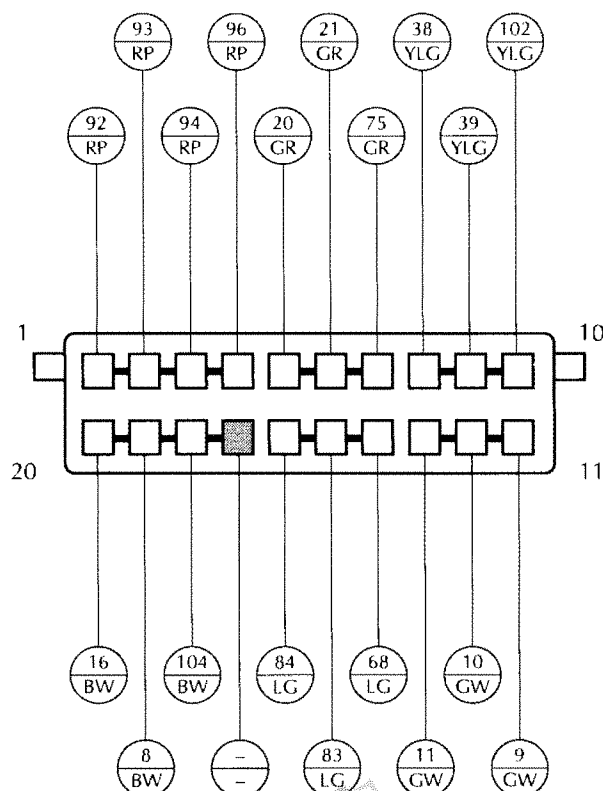


Header No 8 (Green)

(Facia Harness)

Location - Behind main instrument panel

Cav No.	Cable No.	Cable colour	Circuit Description	Function
1	22	P	To radio connector pin 2 - radio	Battery feed fuse A9
2	23	P	To mirror switch connector pin 6	
3	19	P	Feed from main facia harness connector Fa7 pin 19	
4	40	P	To instrument panel diagnostics connector pin 4	RH side lamp feed fuse A10
5	13	RO	To instrument panel connector D pin 3 - panel illumination	
6	37	RO	From main facia harness connector Fa7 pin 11 - side lamp feed	
7	97	RO	To radio connector pin 4 - panel illumination	
8				
9				
10				
11				
12				
13	33	B	Earth return from tachometer screen lead	Earth returns Earth point 6
14	62	B	Earth return from HVAC control panel connector pin A3	
15	73	B	To earth point 6	
16	77	B	Earth return from mirror switch connector pin 4	
17	82	B	Earth return from instrument connector C pin 15	
18	89	B	Earth return from dimmer pot switch pin 2	
19	95	B	Earth return from hazard switch connector pin 3 - illumination	
20	98	B	Earth return from rear fog switch connector pin 3 - illumination	



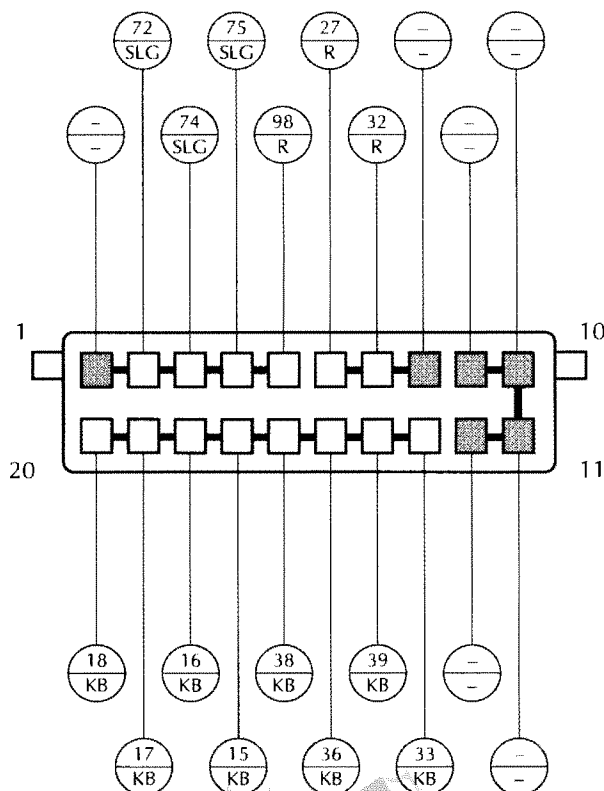
Header No. 9 (Orange)

(Facia Harness)

Location - Behind main instrument panel

Cav No.	Cable No.	Cable colour	Circuit Description	Function
1	92	RP	Dimmed illumination from instruments	Instrument panel dimmer control circuit
2	93	RP	To hazard switch connector pin 7 - illumination	
3	94	RP	To rear fog switch connector pin 7 - illumination	
4	96	RP	To HVAC control panel pin 11 - illumination	
5	20	GR	From hazard switch connector pin 2	LH DI circuit
6	21	GR	To main facia harness connector Fa7 pin 21	
7	75	GR	To instrument panel connector B pin 12	
8	38	YLG	To main facia harness connector Fa6 pin 12	
9	39	YLG	To main facia harness connector Fa6 pin 13	Mirror motor control circuit (common)
10	102	YLG	From mirror switch connector pin 3	
11	9	GW	To main facia harness connector Fa7 pin 9	RH DI circuit
12	10	GW	From hazard switch connector pin 10	
13	11	GW	To instrument connector B pin 11	
14	68	LG	To main facia harness connector Fa5 pin 8	Ignition 1 circuit fuse A1
15	83	LG	To instruments connector D pin 2	
16	84	LG	To starter button +ign. warning light	
17				Brake warning lamp circuits
18	104	BW	To instrument connector B pin 3	
19	8	BW	From main facia harness connector Fa7 pin 8	
20	16	BW	From main facia harness connector Fa7 pin 15	

HEADER 10

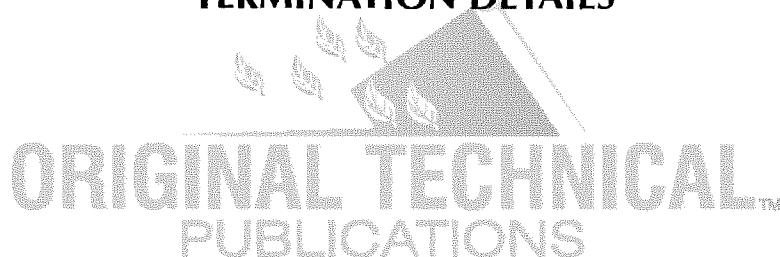


Header No. 10 (Green)
(Chassis Harness)

Location - Behind RH ECU sill panel trim panel

Cav No.	Cable No.	Cable colour	Circuit Description	Function
1				
2	72	SLG	To main chassis harness connector Ch12 pin 3 - Engine check lamp	Fault flag circuit Grounded via ECU
3	74	SLG	To ECU diagnostic plug	
4	75	SLG	From E.C.U. connector pin 52 - Engine check lamp	12V ECU feed to speed sync circuit
5	98	R	From ECU connector pin 17 - ECU +12V supply	
6	27	R	To main chassis harness connector Ch15 pin 30 - ECU +12V supply to synch sensor	
7	32	R	To main chassis harness connector Ch15 pin 36 - ECU +12V supply to speed 1 sensor	
8				
9				
10				
11				
12				
13	33	KB	To main chassis harness connector Ch11 pin 2 - purge valve feed	ECU/Ign 1 controlled feed fuse C4
14	39	KB	To main chassis harness connector Ch11 pin 1 - main ECU feed	
15	36	KB	To ECU diagnostic plug pin 1	
16	38	KB	To ECU connector pin 57 - main feed to ECU	
17	15	KB	To chassis engine harness connector Ch15 pin 16 - feed to cylinders 1,2,3 secondary injectors	
18	16	KB	To chassis engine harness connector CH15 pin 17 - feed to cylinders 1,2,3 primary injectors, ISCV, boost control bank B	
19	17	KB	To chassis engine harness connector Ch15 pin 18 - feed to cylinders 4,5,6 secondary injectors	
20	18	KB	To chassis engine harness connector CH15 pin 19 - feed to cylinders 4,5,6 primary injectors and boost control bank A	

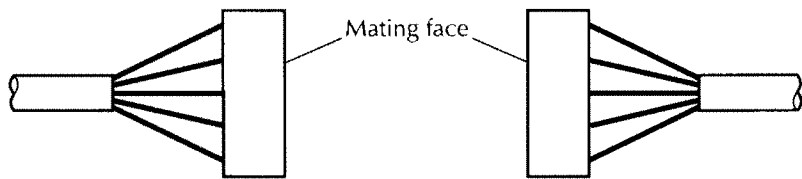
**MAIN HARNESS CONNECTORS
AND
TERMINATION DETAILS**



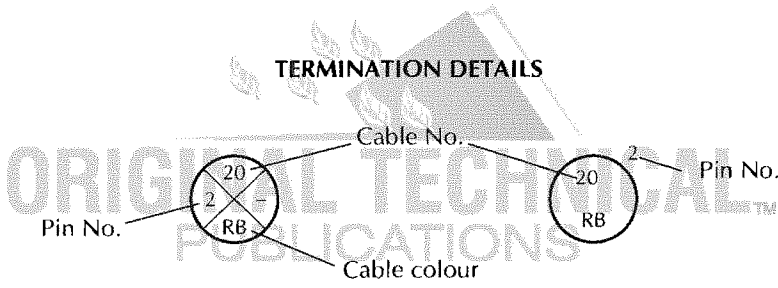
HARNESS CONNECTOR KEY

TYPICAL CONNECTOR

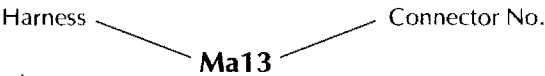
All connectors viewed on mating face



TERMINATION DETAILS

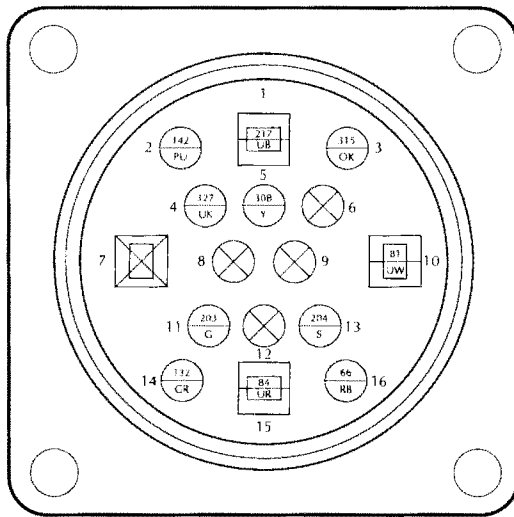


INTERFACE CONNECTORS



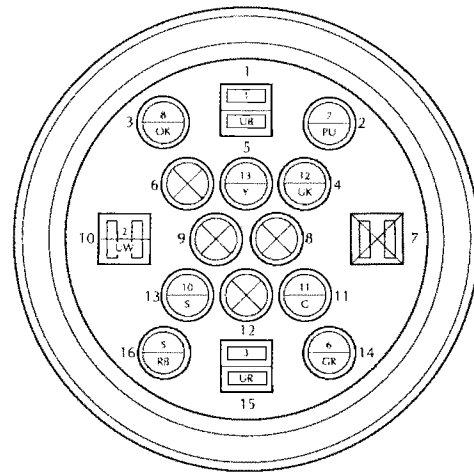
LH FRONT BODY HARNESS CONNECTOR

Main Harness



Ma1

LH Front Body Harness



Lb1

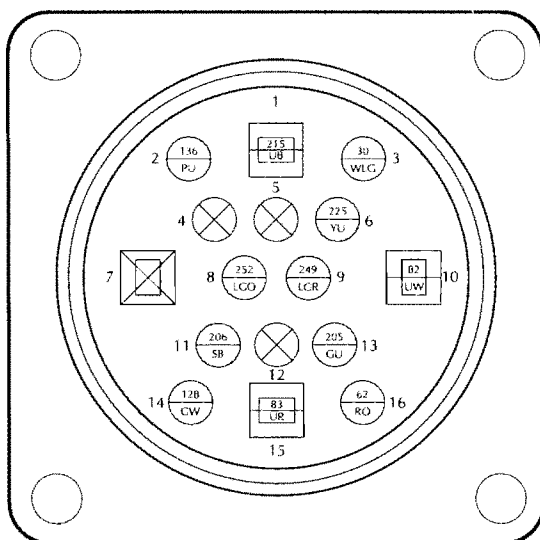
Connectors viewed on mating face

Location - LH Front suspension plate

Cav No.	Cable No.		Cable colour	Circuit Description	Function
	Ma1	Lb1			
1	217	1	UB	Rad fan L.H	Rad fan relay controlled batt feed, fuse M7
2	142	7	PU	Horn L.H.	Horn relay controlled batt feed, fuse A18
3	315	8	OK	Heater water pump	Switched ign 2 feed fuse A4 via HVAC control panel
4	327	12	UK	Water temperature control valve - return	Switched earth return via HVAC control panel (PWM signal)
5	308	13	Y	Water temp control valve - feed	Ign 2 feed fuse A4
6					
7					
8					
9					
10	81	2	UW	Headlamp main beam LH	Main beam relay controlled batt feed fuse A21
11	203	11	G	Headlamp cover motor LH	Module controlled batt feed fuse B1 (+ve for Down) or (-ve for UP)
12					
13	204	10	S	Headlamp cover motor LH	Module controlled batt feed fuse B1 (+ve for Up) or (-ve for Down)
14	132	6	GR	Direction indicator lamp LH	Output from DI/hazard unit
15	84	3	UR	Headlamp dip beam LH	Dip beam relay controlled batt feed fuse A19
16	66	5	RB	Side lamp LH	Side lamp relay controlled batt feed fuse A11

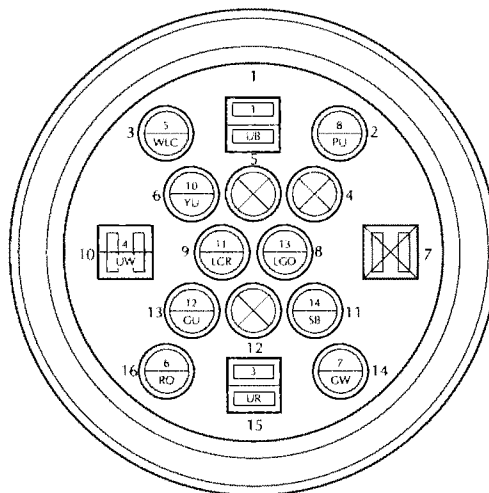
RH FRONT BODY HARNESS CONNECTOR

Main Harness



Ma2

RH Front Body Harness



Rb 2

Connectors viewed on mating face

Location - RH Front suspension plate

Cav No.	Cable No.	Cable colour	Circuit Description	Function
	Ma2	Rb2		
1	215	1	UB	Rad fan R.H
2	136	8	PU	Horn R.H
3	30	5	WLG	Binary switch - feed (A/C request)
4				
5				
6	225	10	YU	Binary switch - return feed (A/C request)
7				
8	252	13	LGO	Windscreen wash pump
9	249	11	LGR	Headlamp wash (not fitted)
10	82	4	UW	Headlamp main beam RH
11	206	14	SB	Headlamp cover motor RH
12				
13	205	12	GU	Headlamp cover motor LH
14	128	7	GW	Direction indicator lamp RH
15	83	3	UR	Headlamp dip beam RH
16	62	6	RO	Side lamp RH

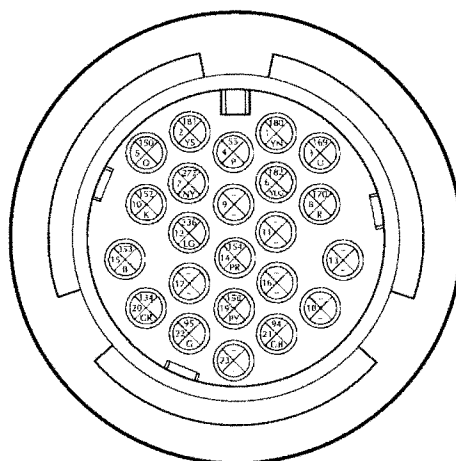
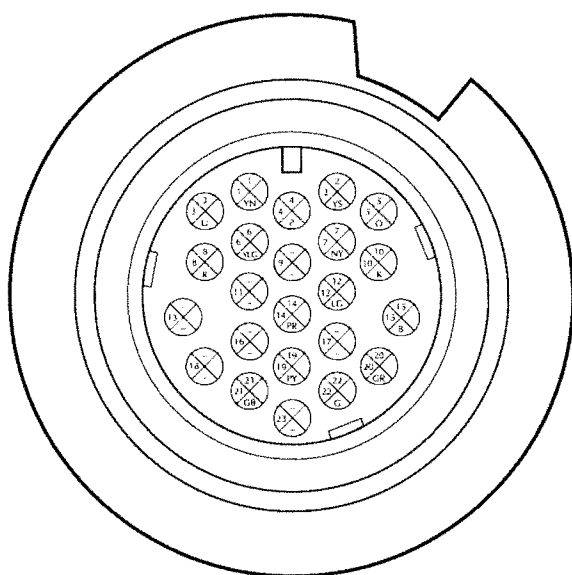
RH PASSENGER DOOR HARNESS CONNECTOR

Door Harness

Main Harness

Door harness

Main harness



Ld3 R.H.D

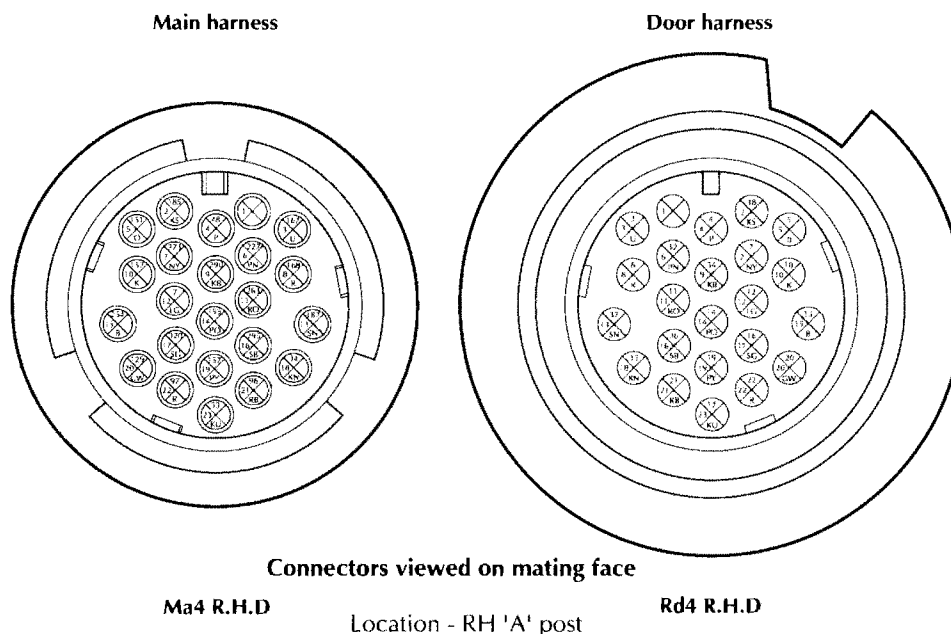
Ma3 R.H.D

Connectors viewed on mating face

Location - LH 'A' post

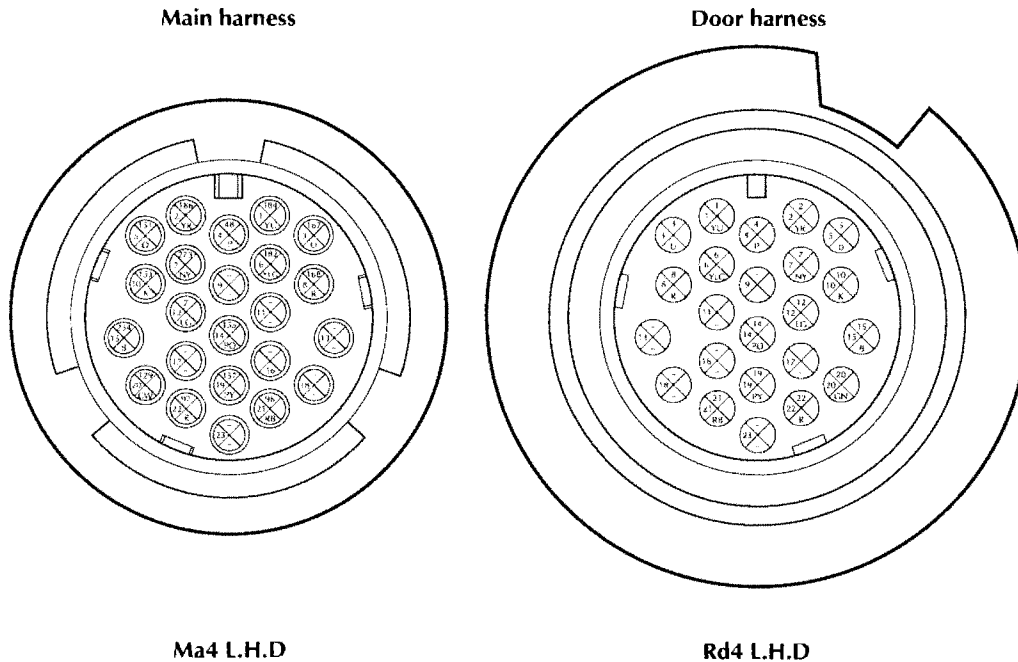
Cav No.	Cable No.	Cable colour	Circuit Description	Function
1	1	180 YN	Door mirror	When negative - mirror glass turns down
2	2	181 YS	Door mirror	When negative - mirror glass turns out
3	3	169 U	Window lift motor LH	Window relay controlled batt feed fuse A7 for Up (-ve when Down)
4	4	55 P	Door open/kerb lamps feed	Batt feed fuse A9
5	5	150 O	Door lock actuator	CDL/Alarm module controlled batt (dir) feed fuse D1 (+ve for Unlock)
6	6	182 YLG	Door mirror - common	When 12v positive - mirror glass turns out/down
7	7	277 NY	Not used	
8	8	170 R	Window lift motor LH	Window relay controlled batt feed fuse A7 for Down (-ve for Up)
9				
10	10	152 K	Door lock actuator	CDL/Alarm module controlled batt (dir) feed fuse D1 (+ve for unlock)
11				
12	12	236 LG	Heater element - external rear view mirror	Ign 1 feed fuse A1
13				
14	14	154 PR	Lock reminder switch	Switched to earth when doors unlocked
15	15	353 B	Earth return	Earth point 5 via header No. 7
16				
17				
18				
19	19	158 PY	Door lock switch - CDL/Alarm	Switched to earth when door open
20	20	134 GR	Side repeater lamp LH	Output from DI/hazard unit
21	21	94 GB	Door speaker, -ve	Negative output from amp
22	22	95 G	Door speaker, +ve	Positive output from amp
23				

RHD DRIVER'S DOOR HARNESS CONNECTOR



Cav No.	Cable No.		Cable colour	Circuit Description	Function
	Ma4	Rd4			
1					
2	185	38	KS	Door pod instruments - Serial data(data signal)	Communication line from main instrument panel
3	167	3	U	Window lift motor RH - feed	Window lift relay controlled batt feed fuse A8 for UP (-ve when Down)
4	48	4	P	Door open & kerb lamps - feed	Batt feed fuse A9
5	151	5	O	Door lock actuator - feed	CDL/Alarm module controlled batt (dir) feed fuse D1 (+ve for Unlock)(-ve for Lock)
6	227	32	PN	Door pod instruments - analogue clock	Batt (dir) feed fuse D3
7	273	7	NY	Not used	
8	168	8	R	Window lift motor RH - feed	Window lift relay controlled batt feed fuse A8 for DOWN (-ve when Up)
9	290	34	KB	Door pod instruments - Over charge warning lamp	Batt voltage high LED 2
10	153	10	K	Door lock actuator - feed	CDL/Alarm module controlled batt (dir) feed fuse D1 (+ve for Lock)(-ve for Unlock)
11	263	11	RO	Door pod instruments - panel illumination	Side lamp relay controlled batt feed fuse A10
12	7	12	LG	Door pod instruments - feed	Ign 1 feed fuse A1
13	187	37	SN	Door pod instruments - serial data (clock)	Communication line from main instrument panel
14	155	14	PO	Lock reminder switch	Switched to earth when doors unlocked
15	254	15	B	Earth return	Earth point 5 via header No. 7
16	295	36	SB	Door pod instruments - serial data (enable)	Communication line from main instrument panel
17	320	16	SG	Door pod instruments - high boost pressure warning lamp	Not used
18	34	35	KN	Door pod instruments - low charge warning lamp	Batt voltage low LED 1
19	157	19	PY	Door lock switch - CDL/Alarm	Switched to earth when door open
20	129	20	GW	Side repeater lamp RH	Output from DI/hazard unit
21	96	21	RB	Door speaker -ve	Negative output from amp
22	97	22	R	Door speaker +ve	Positive output from amp
23	33	17	KU	Door pod instruments - gearbox oil high temperature warning lamp	Temp high LED 4

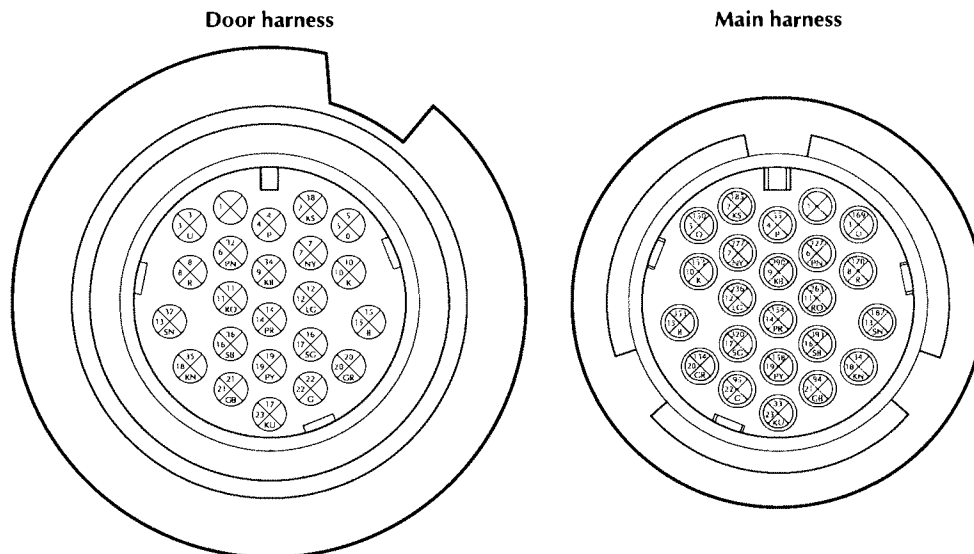
LHD PASSENGER DOOR HARNESS CONNECTOR



Connectors viewed on mating face
Location - RH 'A' post

Cav No.	Cable No.		Cable colour	Circuit Description	Function
	Ma4	Rd4			
1	184	1	YU	Door mirror	When negative - mirror glass turns down
2	186	2	YR	Door mirror	When negative - mirror glass turns in
3	167	3	U	Window lift motor RH	Window relay controlled batt feed fuse A7 for Up (-ve for Down)
4	48	4	P	Door open/kerb lamps feed	Batt feed fuse A9
5	151	5	O	Door lock actuator	CDL/Alarm module controlled batt (dir) feed fuse D1 (+ve for Unlock)(-ve for Lock)
6	182	6	YLG	Door mirror - common	When positive - mirror glass turns inwards/down
7	273	7	NY	Not used	
8	168	8	R	Window lift motor RH	Window relay controlled batt feed fuse A7 for Down (-ve for Up)
9					
10	153	10	K	Door lock actuator	CDL/Alarm module controlled batt (dir) feed fuse D1 (+ve for Lock)(-ve for Unlock)
11					
12	7	12	LG	Heater element - external rear view mirror	Ign 1 feed fuse A1
13					
14	155	14	PR	Lock reminder switch	Switched to earth when doors unlocked
15	254	15	B	Earth return	Earth point 5 via header No. 7
16					
17					
18					
19	157	19	PY	Door lock switch - CDL/Alarm	Switched to earth when door open
20	129	20	GW	Side repeater lamp RH	Output from DI/hazard unit
21	96	21	RB	Door speaker, -ve	Negative output from amp
22	97	22	R	Door speaker, +ve	Positive output from amp
23					

LHD DRIVER'S DOOR HARNESS CONNECTOR



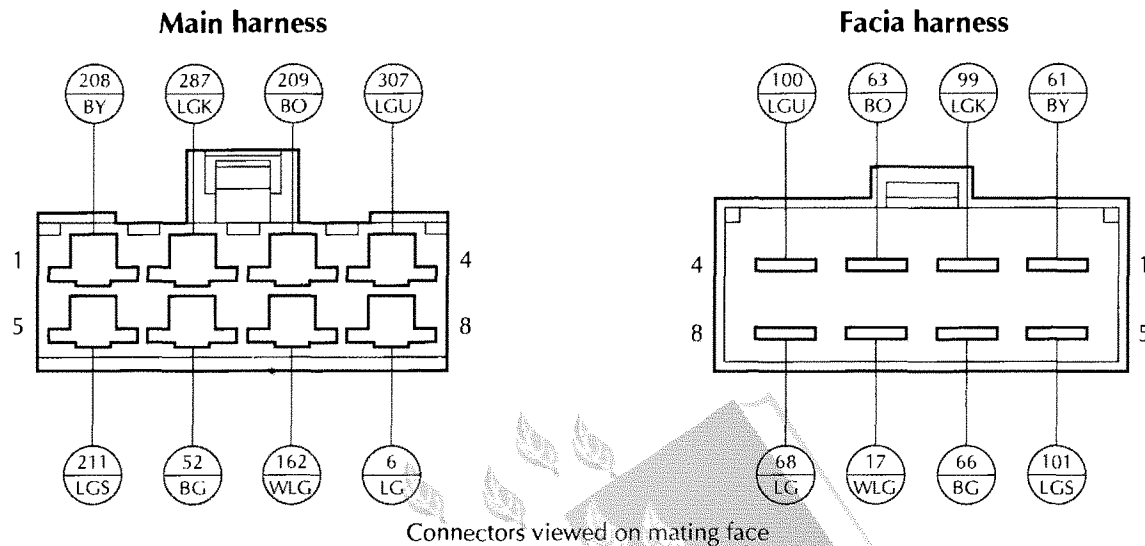
Connectors viewed on mating face

Ld3 L.H.D

Ma3 L.H.D

Location - LH 'A' post

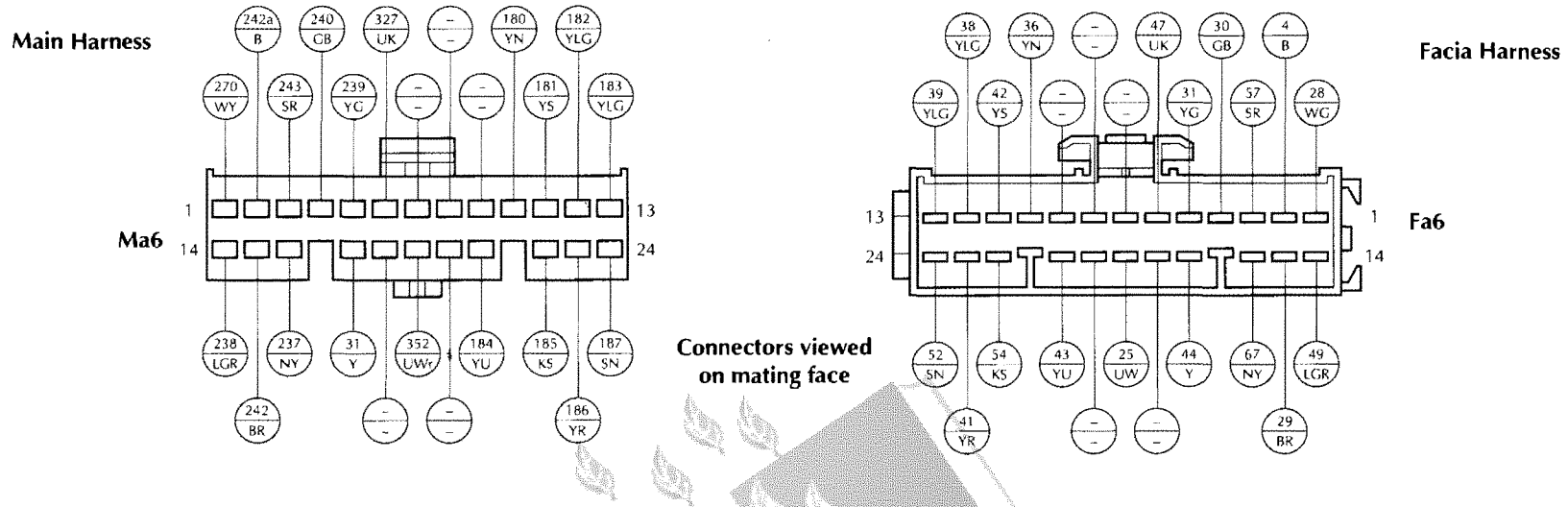
Cav No.	Cable No.	Cable colour	Circuit Description	Function
	Ld3	Ma3		
1				
2	38	185	KS	Door pod instruments - Serial data(data line)
3	3	169	U	Communication line from main instrument panel
4	4	55	P	Window lift motor LH
5	5	150	O	Window lift relay controlled batt feed fuse A8 for Up (-ve when Down)
6	32	227	PN	Batt feed fuse A9
7	7	277	NY	CDL/Alarm module controlled batt (dir) feed fuse D1 (+ve for Unlock)(-ve for Lock)
8	8	170	R	Batt (dir) feed fuse D3
9	34	290	KB	Window lift relay controlled batt feed fuse A8 for Down (-ve when Up)
10	10	152	K	Batt voltage high LED 2
11	11	263	RO	CDL/Alarm module controlled batt (dir) feed fuse D1 (+ve for Lock)(-ve for Unlock)
12	12	236	LG	Side lamp relay controlled batt feed fuse A10
13	37	187	SN	Ign 1 feed fuse A1
14	14	154	PR	Communication line from main instrument panel
15	15	353	B	Switched to earth when doors unlocked
16	36	295	SB	Earth point 5 via header No. 7
17	16	320	SG	Communication line from main instrument panel
18	35	34	KN	Not used
19	19	158	PY	Batt voltage low LED 1
20	20	134	GR	Switched to earth when door open
21	21	94	GB	Output from DI/hazard unit
22	22	95	G	Negative output from amp
23	17	33	KU	Positive output from amp
				Temp high LED 4

**Ma5****Fa5****Facia Harness Connectors**

Location - Behind Driver's underscuttle trim panel

Cav No.	Cable No.		Cable colour	Circuit Description	Function
	Ma5	Fa5			
1	208	61	BY	Blower fan switch - speed 1	Switched earth return
2	287	99	LGK	HVAC, mode switch (floor actuator)	Switched Ign 2 feed, fuse A4 via HVAC control panel
3	209	63	BO	Blower fanswitch - speed 2	Switched earth return
4	307	100	LGU	HVAC mode switch (screen/vent actuator)	Switched Ign 2 feed, fuse A4 via HVAC control panel
5	211	101	LGS	HVAC, mode switch (recirc actuator)	Switched Ign 2 feed, fuse A4 via HVAC control panel
6	52	66	BG	Blower, fans witch - speed 3	Switched earth return
7	162	17	WLG	HVAC, mode switch (A/C request/blower slow)	Switched Ign 2 feed, fuse A4 via HVAC control panel
8	6	68	LG	Start button and instruments	Ign 1 feed, fuse A1

Facia Harness Connectors
Location - Behind driver's underscuttle trim panel



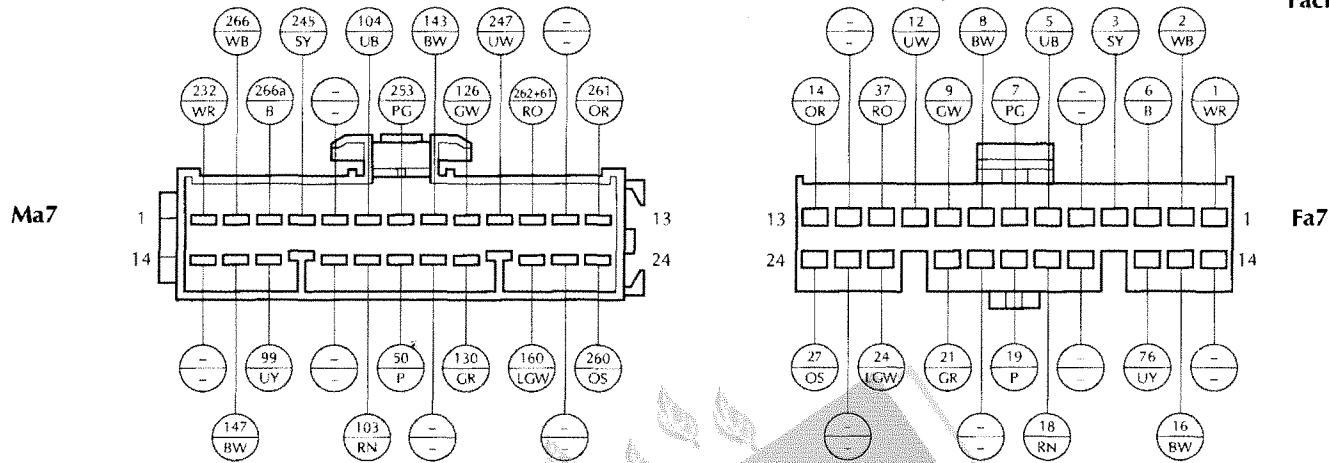
Cav No.	Cable No.		Cable colour	Circuit Description	Function
	Ma6	Fa6			
1	270	28	WY	Oil pressure warning light switch	Signal lead - switched to ground when pressure less than 0.5 bar
2	242a	4	B	Vehicle speed sensor screen -ve	Earth screen lead
3	243	57	SR	Boost gauge transducer	Signal lead (analogue resistive)
4	240	30	GB	Fuel tank sensor unit	Signal lead (analogue resistive)
5	239	31	YG	Water temperature gauge transducer	Signal lead (analogue resistive)
6	327	47	UK	HVAC water temperature control valve - return	Switched earth return via HVAC control panel (P.W.M. signal)
7					
8					
9					
10	180	36	YN	Door mirror LH	When negative - mirror glass turns down
11	181	42	YS	Door mirror LH	When negative - mirror glass turns out
12	182	38	YLG	Door mirror, common, passenger	When 12v positive - mirror glass turns out(RHD) - in(LHD)/down
13	183	39	YLG	Door mirror, common, driver	When 12v positive - mirror glass turns in(RHD) - out(LHD)/down
14	238	49	LGR	Oil pressure gauge transducer	Signal lead (analogue resistive)
15	242	29	BR	Vehicle speed sensor	Road speed signal (analogue - sign wave)
16	237	67	NY	Ignition warning lamp	Feed to alternator field/charging circuit
17	31	44	Y	HVAC control panel feed	Ign 2 fuse A4
18					
19	352	25	UWr	Audio amplifier	Remote signal + 12Volts from tuner head unit when on
20					
21	184	43	YU	Door mirror RH	When negative - mirror glass turns down
22	185	54	KS	Door pod instruments - serial data (data signal)	Communication line from main instrument panel
23	186	41	YR	Door mirror RH	When negative - mirror glass turns in
24	187	52	SN	Door pod instruments - serial data (clock)	Communication line from main instrument panel

Main Harness

Facia Harness Connectors

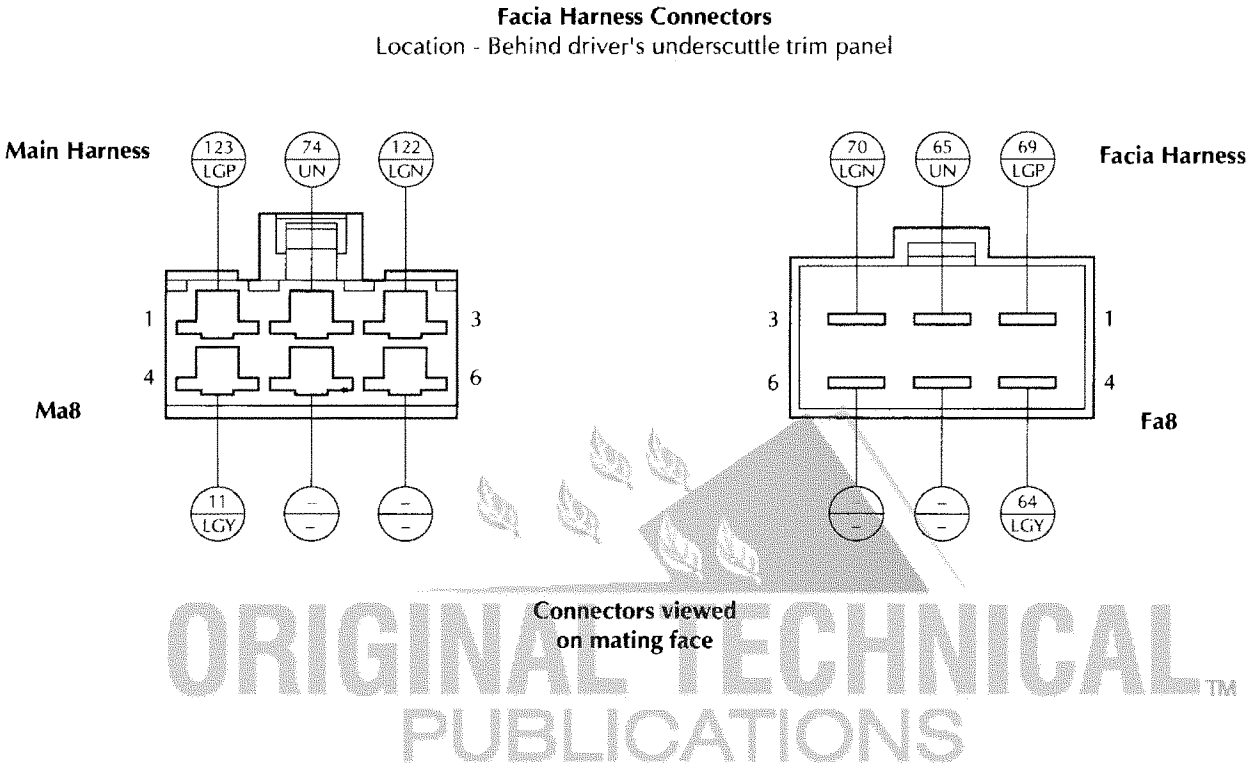
Location - Behind driver's underscuttle trim panel

Facia Harness



Connectors viewed on mating face

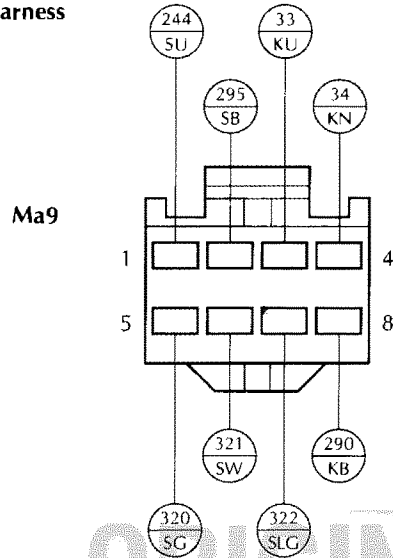
Cav No.	Cable No.	Cable colour	Circuit Description	Function
1	Ma7 232	Fa7 1 WR	Start switch relay	Crank signal - switched Ign 1 feed fuse A1
2	266	2 WB	Tacho input signal	Output signal from ECU (square wave digital 0-12 volts)
3	266a	6 B	Tacho screened lead	Earth return
4	245	3 SY	Engine oil temperature gauge transducer	Signal lead (analogue resistive)
5				
6	104	5 UB	Rear foglamp switch	Switched earth via column headlamp switch
7	253	7 PG	Aerial trigger	Control signal (+ 12 volts - aerial up)
8	143	8 BW	Brake warning lamp	Switched earth via brake fluid level/brake pad wear sensor
9	126	9 GW	Direction indicator lamp R.H	Output from DI/hazard unit
10	247	12 UW	Main beam warning lamp	Main beam relay controlled feed fuse A22
11	62+6	57 RO	Instrument illumination feed	Side lamp relay controlled base feed fuse A10
12				
13	261	14 OR	Alarm LED indicator	Switched negative via CDL/Alarm unit when armed
14				
15	147	16 BW	Brake warning lamp	Switched earth via handbrake switch
16	99	76 UY	Rear fog lamp relay output to rear fog lamp	Rear fog relay controlled feed fuse A12
17				
18	103	18 RN	Rear fog lamp relay coil	Switched earth via rear fog switch
19	50	19 P	Audio unit door mirror switch	Batt feed fuse A9
20				
21	130	21 GR	Direction indicator lamp L.H	Output from DI/hazard unit
22	160	24 LGW	Audio unit	Aux feed fuse A6
23				
24	260	27 OS	Alarm LED indicator	Switched positive via CDL/Alarm unit when armed



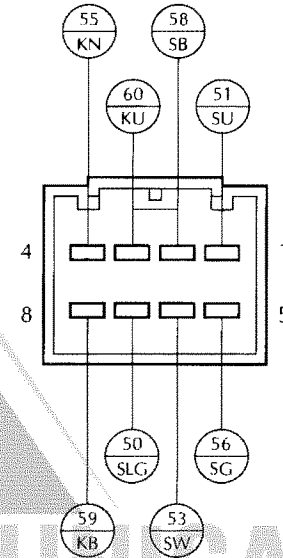
Cav No.	Cable No.		Cable colour	Circuit Description	Function
	Ma8	Fa8			
1	123	69	LGP	Flasher unit feed switched via hazard switch	Switched Ign 1 feed fuse A10 or batt feed fuse A17
2	74	65	UN	Hazard switch feed	Batt feed fuse A17
3	122	70	LGN	Flasher unit switched output	Flasher unit output to DI/hazard circuit
4	11	64	LGY	Hazard switch feed	Ign 1 feed fuse A10
5					
6					

Facia Harness Connectors Location - Behind driver's underscuttle trim panel

Main Harness

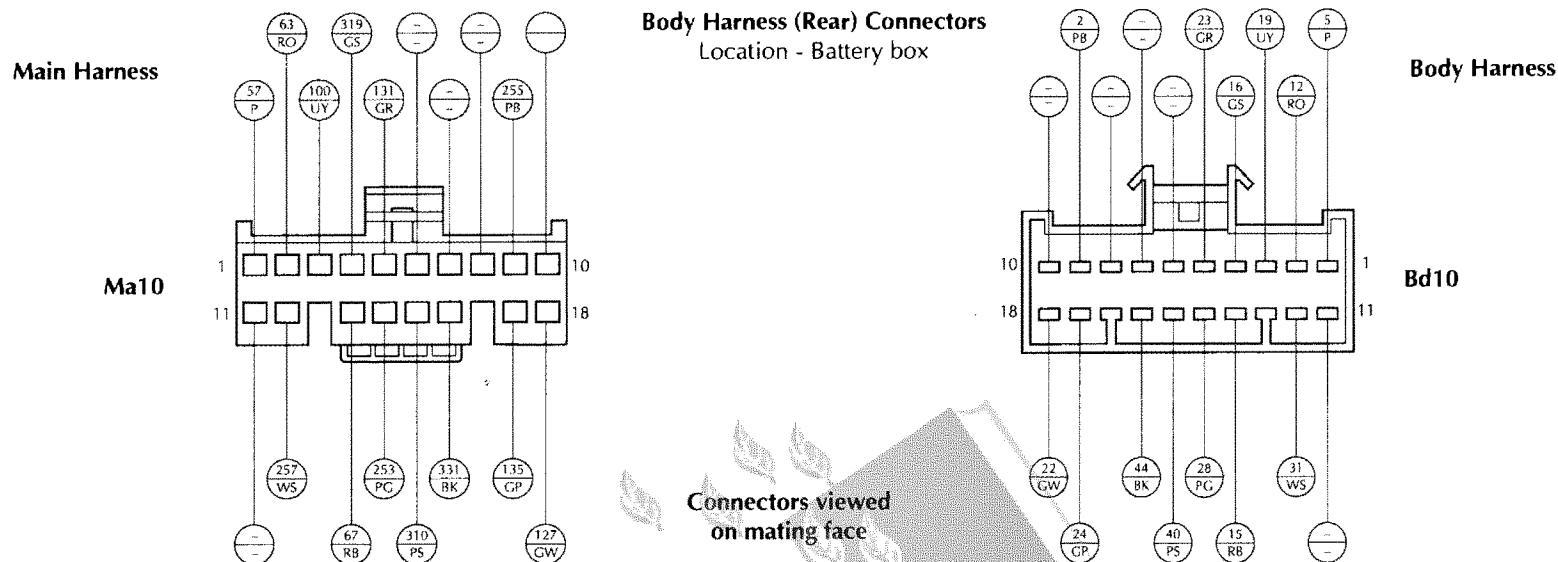


Facia Harness



Connectors viewed
on mating face

Cav No.	Cable No.		Cable colour	Circuit Description	Function
	Ma9	Fa9			
1	242	51	SU	Gearbox oil temperature gauge transducer	Signal lead (analogue resistive)
2	295	58	SB	Door pod instruments - serial date (enable)	Communication line from main instrument panel
3	33	60	KU	Door pod instruments - gearbox oil high temperature warning lamp	Temp high LED 4
4	34	55	KN	Door pod instruments - low charge warning lamp	Batt voltage low LED 1
5	320	56	SG	Door pod instruments - high boost pressure warning lamp	Not used
6	321	53	SW	Road speed signal	Digital output from instrument panel to ECU (VSS signal divide by 2)
7	322	50	SLG	Engine check light	Output from ECU (+ 12 volts when enabled)
8	290	59	KB	Door pod instruments - over charge warning lamp	Batt voltage high LED 2

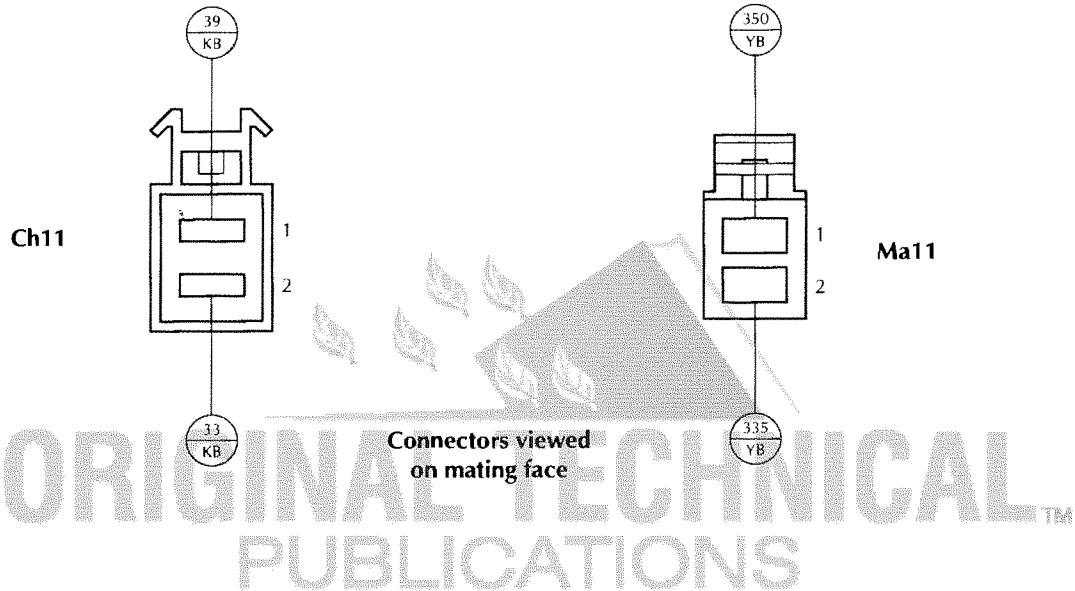


Cav No.	Cable No.		Cable colour	Circuit Description	Function
	Ma10	Bd10			
1	57	5	P	Engine bay lamps, boot lamp, aerial batt feed	Batt feed fuse A9
2	63	12	RO	R.H rear side lamp No. plate lamps	Side lamp relay controlled batt feed fuse A10
3	100	19	UY	Rear fog lamps	Side/fog lamp relay controlled batt feed fuse A12
4	319	16	GS	Rverse lamps	Ign 1 feed fuse A2 switched via reverse switch
5	131	23	GR	Direction indicator lamp LH rear	Output from DI/hazard unit
6					
7					
8					
9	255	2	PB	Tailgate and boot lamp switches	Alarm trigger
10					
11					
12	257	31	WS	Alarm sounder	Analogue output from CDL/Alarm unit
13	67	15	RB	Side lamp	Side lamp relay controlled batt feed fuse A11
14	253	28	PG	Aerial	Control signal (+ 12 volts - aerial up)
15	310	40	PS	Boot release solenoid	Batt feed fuse A9
16	331	44	BK	Alarm sounder	Analogue output from CDL/Alarm unit
17	135	24	GP	Brake lamps	Brake switch controlled batt feed fuse B3
18	127	22	GW	Direction indicator lamp RH rear	Output from DI/hazard unit

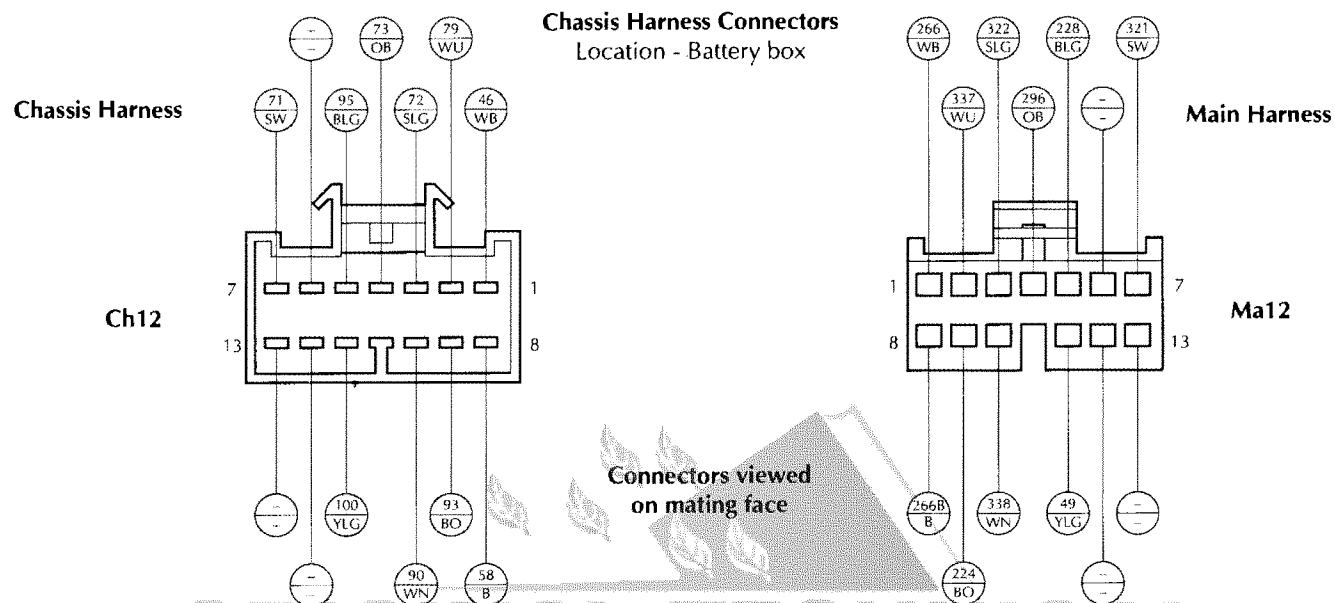
Chassis Harness Connectors
Location - Battery box

Chassis Harness

Main Harness



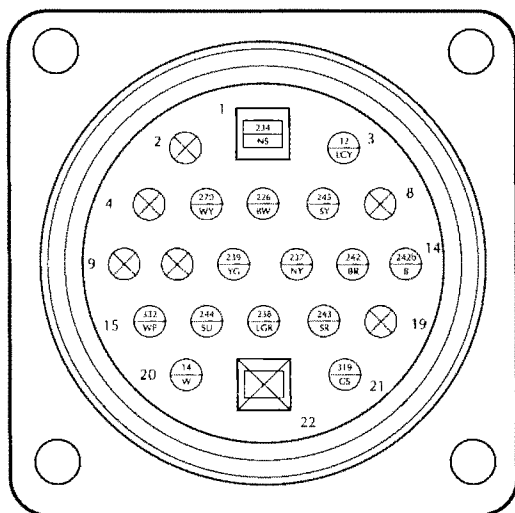
Cav No.	Cable No.	Cable colour	Circuit Description	Function
	Ch11	Ma11		
1	39	350	KB/YB	Feed to header No. 10
2	33	335	KB/YB	Return feed to purge valves
				ECU controlled relay Ign 1 feed fuse C4
				ECU controlled relay Ign 1 feed fuse C4



Cav No.	Cable No.	Cable colour	Circuit Description	Function
	Ch12	Ma12		
1	46	266	WB	Tacho output signal
2	79	337	WU	Output from ECU (square wave digital 0-12 volts)
3	72	322	SLG	Purge valve
4	73	296	OB	Switched earth return via ECU
5	93	228	BLG	Engine check light
6				Output from ECU (+12 volts when enabled)
7	71	321	SW	Fuel pump relay coil
8	58	266b	B	Switched earth return via ECU
9	93	224	BO	HVAC compressor relay coil
10	90	338	WN	Switched earth return via ECU
11	100	49	YLG	Digital output from instrument panel to ECU (VSS signal divide by 2)
12				Earth return
13				Switched earth return via ECU
				Switched earth return via ECU
				Output from HVAC control panel Ign 2 feed fuse A4 (+12 volts when enabled)

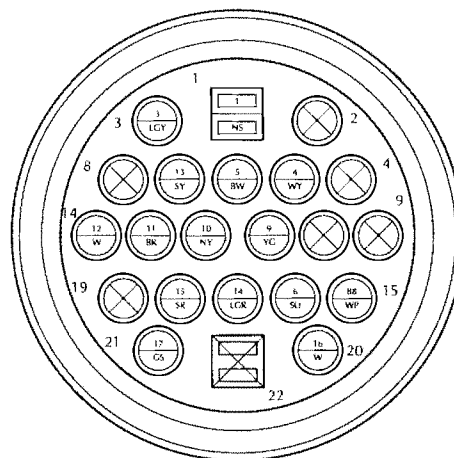
ENGINE HARNESS CONNECTOR 13

Main Harness



Ma13

Engine Harness



En13

Connectors viewed on mating face

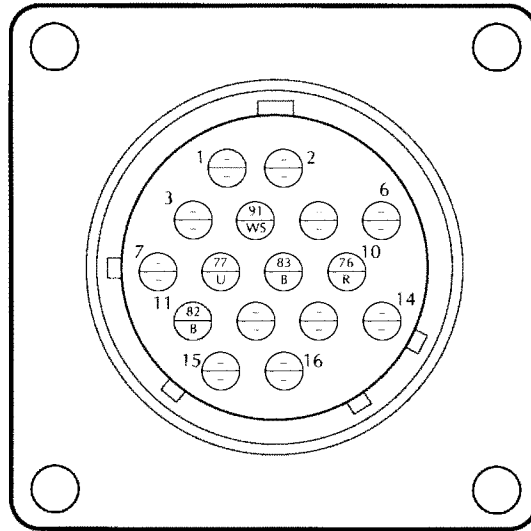
Engine Harness Connector

Location - Battery box/Rear bulkhead

Cav No.	Cable No.	Cable colour	Description	Function
1	Ma13 234	En13 1 NS	Starter solenoid	Starter relay controlled batt feed fuse M8
2				
3	12	3 LGY	Reverse switch feed	Ign 1 feed fuse A2
4				
5	270	4 WY	Oil pressure warning light switch	Signal lead - switched to ground when pressure less than 0.5 bar
6	226	5 BW	A/C compressor switched feed	Compressor relay controlled batt fed fuse A15
7	245	13 SY	Engine oil temperature guage transducer	Signal lead (analogue resistive)
8				
9				
10				
11	239	9 YG	Water temperature guage transducer	Signal lead (analogue resistive)
12	237	10 NY	Ign warning light	Feed to alternator filed/charging circuit
13	242	11 BR	Vehicle speed sensor	Road speed signal (analogue sign wave)
14	242b	12 B/W	Vehicle speed sensor screen -ve	Earth screen lead
15	332	88 WP	Oxygen sensor heater element	Fuel pump relay controlled batt feed fuse C1
16	244	6 SU	Gearbox oil temperature guage transducer	Signal lead (analogue resistive)
17	238	14 LGR	Oil pressure guage transducer	Signal lead (analogue resistive)
18	243	15 SR	Boost guage transducer	Signal lead (analogue resistive)
19				
20	14	16 W	Ign coil & alternator feed	Ign 1 feed fuse A3
21	319	17 GS	Reverse lamps switched feed	Ign 1 feed fuse A2 switched via reverse switch
22				

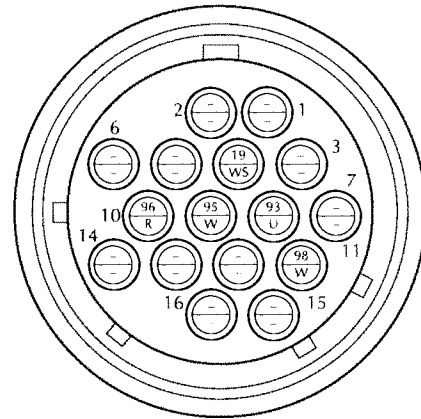
ENGINE HARNESS CONNECTOR 14

Chassis Harness



Ch14

Engine Harness



En14

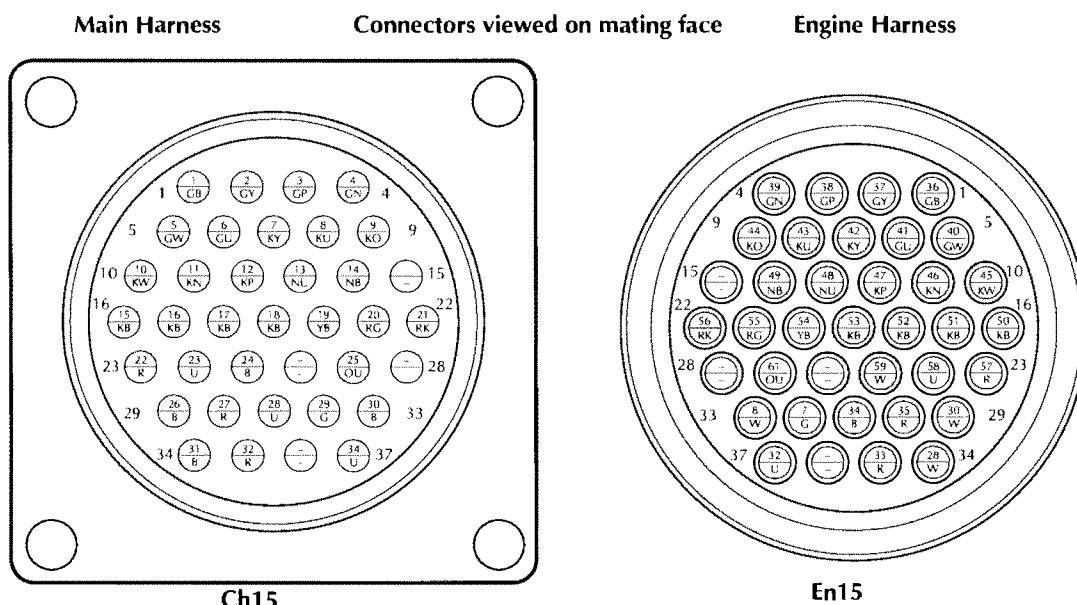
Connectors viewed on mating face

Engine Harness Connector

Location - Battery box/Rear bulkhead

Cav No.	Cable No.		Cable colour	Description	Function
	Ch14	En14			
1					
2					
3					
4	91	19	WS	Idle speed control valve(ISCV)	P.W.M. signal - switched to earth via ECU
5					
6					
7					
8	77	93	U	Oxygen sensor signal 'A' bank	Analogue signal - continuously variable
9	83	95	B/W	Oxygen sensor screen 'A' bank	Screened earth return
10	76	96	R	Oxygen sensor signal 'B' bank	Analogue signal - continuously variable
11	82	98	B/W	Oxygen sensor screen 'B' bank	Screened earth return
12					
13					
14					
15					
16					

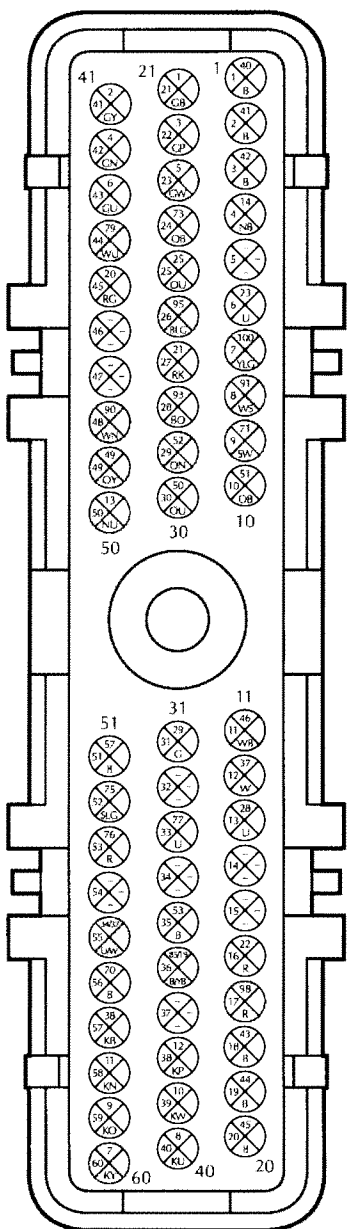
ENGINE HARNESS CONNECTOR 15



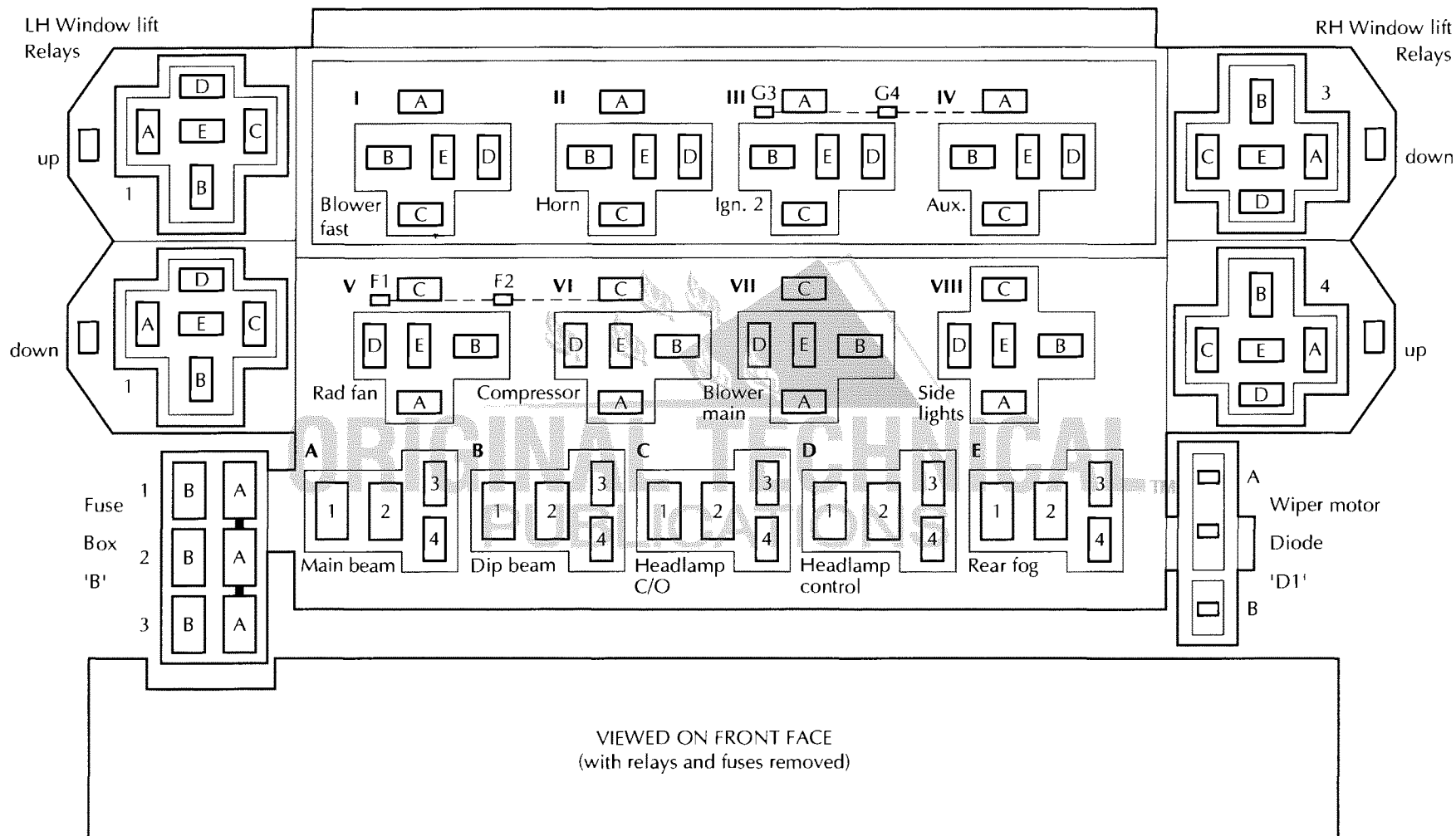
Engine Harness Connector
Location - Battery box/Rear bulkhead

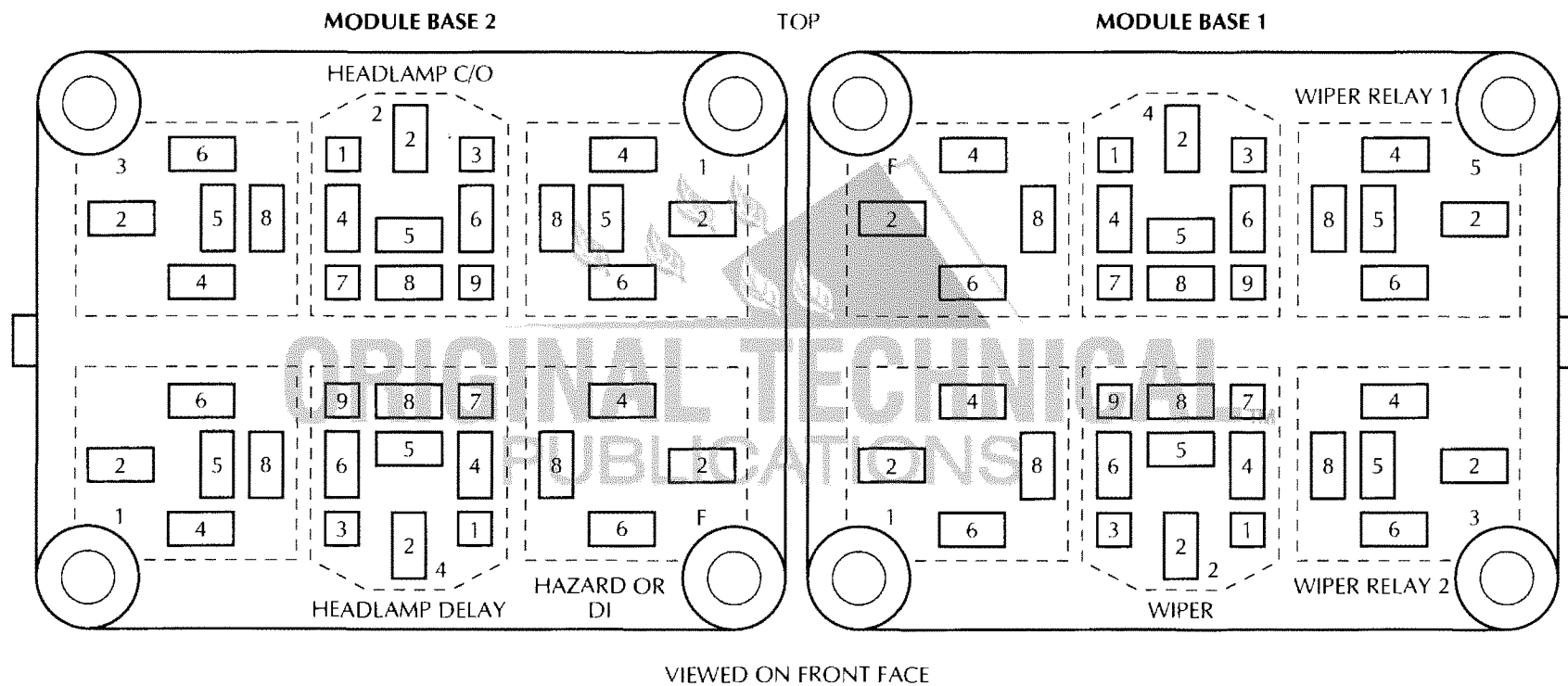
Cav No.	Cable No.		Cable colour	Description	Function
	Ch15	En15			
1	1	36	GB	Primary injector cylinder 1	Pulse width modulated signal - switched via ECU
2	2	37	GY	Primary injector cylinder 6	Pulse width modulated signal - switched via ECU
3	3	38	GP	Primary injector cylinder 3	Pulse width modulated signal - switched via ECU
4	4	39	GN	Primary injector cylinder 5	Pulse width modulated signal - switched via ECU
5	5	40	GW	Primary injector cylinder 2	Pulse width modulated signal - switched via ECU
6	6	41	GU	Primary injector cylinder 4	Pulse width modulated signal - switched via ECU
7	7	42	KY	Secondary injector cylinder 1	Pulse width modulated signal - switched via ECU
8	8	43	KU	Secondary injector cylinder 6	Pulse width modulated signal - switched via ECU
9	9	44	KO	Secondary injector cylinder 3	Pulse width modulated signal - switched via ECU
10	10	45	KW	Secondary injector cylinder 5	Pulse width modulated signal - switched via ECU
11	11	46	KN	Secondary injector cylinder 2	Pulse width modulated signal - switched via ECU
12	12	47	KP	Secondary injector cylinder 4	Pulse width modulated signal - switched via ECU
13	13	48	NU	'A' bank boost solenoid	Pulse width modulated signal - switched via ECU (Nominal 12Hz signal)
14	14	49	NB	'B' bank boost solenoid	Pulse width modulated signal - switched via ECU (Nominal 12Hz signal)
15					
16	15	50	KB	Secondary injectors - cylinders 1, 2 & 3	ECU relay controlled batt feed fuse M1 via header 10
17	16	51	KB	Primary injectors - cylinders 1, 2, 3, B bank boost solenoid and ISCV	ECU relay controlled batt feed fuse M1 via header 10
18	17	52	KB	Primary injectors - cylinders 4,5,6 and A bank boost solenoid	ECU relay controlled batt feed fuse M1 via header 10
19	18	53	KB	Secondary injectors - cylinders 4,5 & 6	ECU relay controlled batt feed fuse M1 via header 10
20	19	54	YB	Thermistor common (air A, air B and water)	Earth return via ECU
21	20	55	RG	'B' bank air temperature thermistor - feed	ECU feed + 5 volts
22	21	56	RK	'A' bank air temperature thermistor - feed	ECU feed + 5 volts
23	22	57	R	Throttle pot - feed	ECU feed + 5 volts
24	23	58	U	Throttle pot - signal	0 to +5 volts return signal
25	24	59	W/B	Throttle pot - earth	Earth return via ECU
26					
27	25	61	OU	Water temperature thermistor - feed	ECU feed + 5 volts
28					
29	26	30	W/B	Synch sensor - screened earth return	Earth return via ECU
30	27	35	R	Synch sensor - feed	ECU feed + 12 volts
31	28	34	B/U	Synch sensor - signal	Digital signal 0 to 12 volts (Hall effect)
32	29	7	G	Ignition amp signal	Digital signal 0 to 12 volts
33	30	8	W/B	Ignition amp screened earth return	Local amp earth
34	31	28	W/B	Speed sensor 1 - screened earth return	Earth return via ECU
35					
36	32	33	R	Speed sensor 1 - feed	ECU feed + 12 volts
37	34	32	U	Speed sensor 1 - signal	Digital signal 0 to 12 volts (Hall effect)

ECU Harness Connector Location - Right hand sill panel



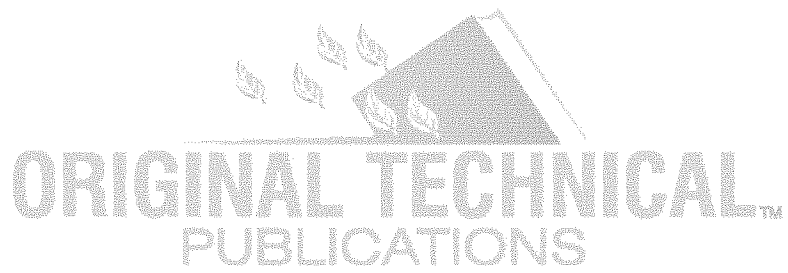
Cav No.	Cable No.	Cable colour	Description	Function
1	40	B	ECU ground	Earthed via earth point '10'
2	41	B	ECU ground	Earthed via earth point '10'
3	42	B	ECU ground	Earthed via earth point '10'
4	14	NB	B' bank boost solenoid	Pulse width modulated signal switched via ECU (nominal 12Hz signal)
5	-	-	-	0 to +5 volts return signal
6	23	U	Throttle pot - Signal	Output from HVAC control panel Ign 2 fuse 'A4' (+12 volts when enabled)
7	100	YLG	A/C request	Pulse width modulated signal switched via ECU
8	91	WS	Idle speed control valve (ISCV)	Digital output from instrument panel (VSS signal divide by 2)
9	71	SW	Road speed signal	Data transmission line
10	51	OB	ECU diagnostic connector	-
11	46	WB	Tacho output signal	Speed signal input from speed 1
12	37	W	Speed 2 - signal input	Digital signal 0 to 12 volts (half effect)
13	28	U	Sync sensor - signal	-
14	-	-	-	-
15	-	-	-	-
16	22	R	Throttle pot - feed	ECU feed +5 volts
17	98	R	Speed/sync sensor feed	ECU feed +12 volts
18	43	B	ECU ground	Earthed via earth point 10
19	44	B	ECU ground	Earthed via earth point 10
20	45	B	ECU ground	Earthed via earth point 10
21	1	GB	Primary injector cylinder 1	Pulse width modulated signal - switched via ECU
22	3	GP	Primary injector cylinder 3	Pulse width modulated signal - switched via ECU
23	5	GW	Primary injector cylinder 2	Pulse width modulated signal - switched via ECU
24	73	OB	Fuel pump relay coil	Switched earth return via ECU
25	25	OU	Water temperature thermister - feed	ECU feed +5 volts
26	95	BLG	HVAC compressor relay coil	Switched earth return via ECU
27	21	RK	A' bank air temperature thermister - feed	ECU feed +5 volts
28	93	BO	Red fan relay coil	Switched earth return via ECU
29	32	ON	ECU diagnostic plug	SW1 (data switch) not used
30	50	OU	ECU diagnostic plug	Write enable
31	29	G	Ignition amp signal	Digital signal 0 to 12 volts
32	-	-	-	-
33	77	U	Oxygen sensor signal 'A' bank	Analogue signal - continuously variable
34	-	-	-	-
35	53	B	Throttle pot - earth	Earth return via ECU
36	85 + 19	B/YB	Thermistor common (air A, air B and water)	Earth return via ECU
37	-	-	-	-
38	12	KP	Secondary injector cylinder 4	Pulse width modulated signal - switched via ECU
39	10	KW	Secondary injector cylinder 5	Pulse width modulated signal - switched via ECU
40	8	KU	Secondary injector cylinder 6	Pulse width modulated signal - switched via ECU
41	2	GY	Primary injector cylinder 6	Pulse width modulated signal - switched via ECU
42	4	GN	Primary injector cylinder 5	Pulse width modulated signal - switched via ECU
43	6	GU	Primary injector cylinder 4	Pulse width modulated signal - switched via ECU
44	79	WL	Purge valve	Switched earth return via ECU
45	20	RG	B' bank air temperature thermister - feed	ECU feed +5 volts
46	-	-	-	-
47	-	-	-	-
48	90	WN	EVAP boost valve	Switched earth return via ECU
49	49	OY	ECU diagnostic plug	RYA
50	13	NU	A' bank boost solenoid	Pulse width modulated signal switched via ECU (nominal 12Hz signal)
51	57	B	Ignition amp screened earth return	Local amp earth
52	75	SLG	Engine check lamp	Output from ECU (+12 volts when enabled)
53	76	R	Oxygen sensor signal 'B' bank	Analogue signal - continuously variable
54	-	-	-	-
55	34 + 37	U/W	Speed sensor 1 - signal	Digital signal 0 to 12 volt (half effect)
56	70	B	ECU ground	Earthed via earth point 10
57	38	KB	Main ECU feed	ECU relay controlled battery feed fuse 'C4'
58	11	KN	Secondary injector cylinder 2	Pulse width modulated signal switched via ECU
59	9	KO	Secondary injector cylinder 3	Pulse width modulated signal switched via ECU
60	7	KY	Second injector cylinder 1	Pulse width modulated signal switched via ECU





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INTRODUCTION

This section details the procedures for replacement of lighting system components. Replacement of components or disassembly instructions not covered in this section require the component be returned to JaguarSport.

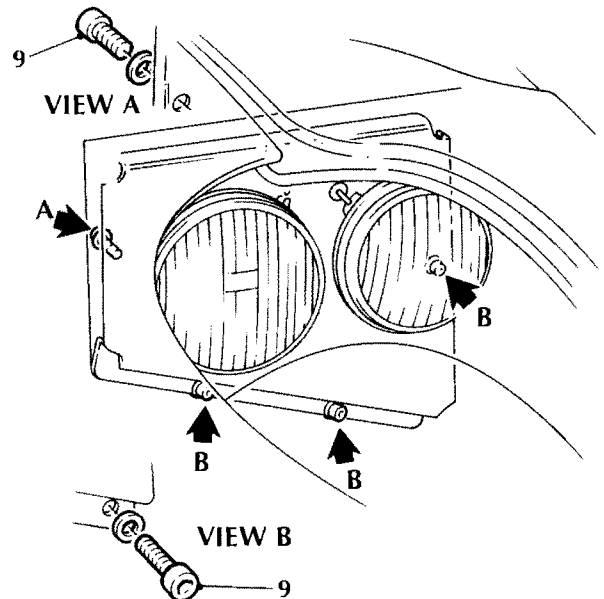
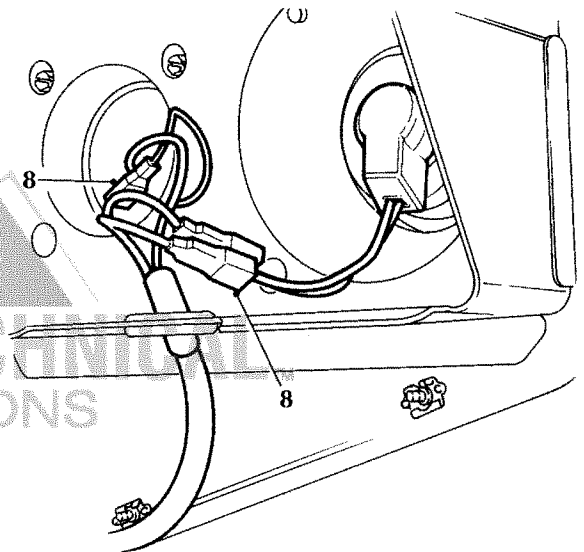
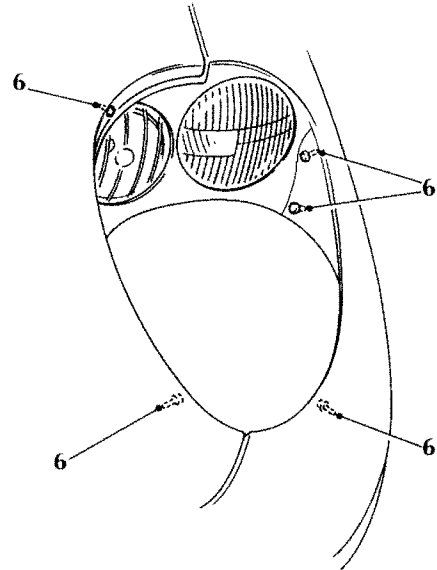
1 HEADLAMP ASSEMBLY

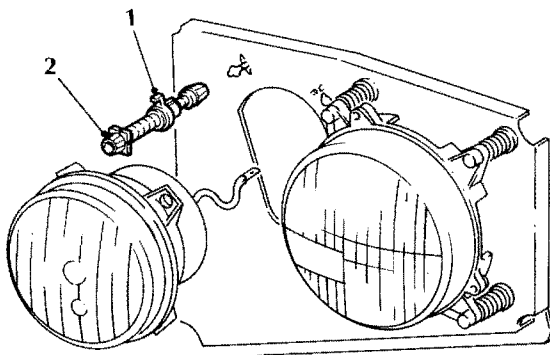
A Removal Procedure

NOTES:

- 1 Ensure absolute cleanliness of work area and tools.
- 2 Observe correct use of tools.

- (1) Position the vehicle on lifting apparatus. (Refer to Lifting and Jacking, Section 1.1).
- (2) Fit JaguarSport protective covers.
- (3) Open the bonnet.
- (4) Disconnect the battery.
- (5) Lower the headlamp covers manually.
- (6) Remove the five screws securing the headlamp finishers.
- (7) Remove the relevant front wheel and wheel arch liner. (Refer to Section 4).
- (8) Working through the wheel arch, disconnect the electrical cables to the headlamps.
- (9) Remove the four screws and washers attaching the headlamp assembly to the mounting bracket and remove the assembly from the vehicle.



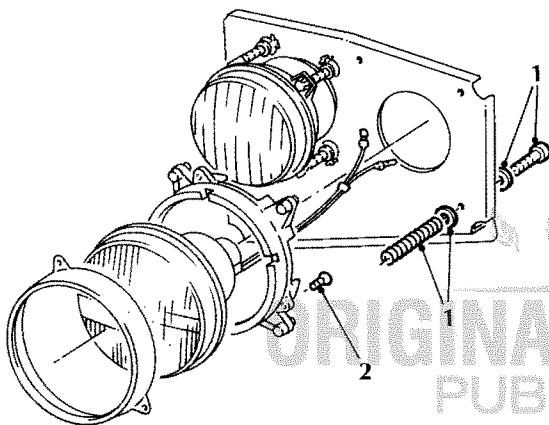


B Disassembly Procedure - Main Beam

NOTE:

Prior to removal of the unit, measure and record the distance of each adjusting screw.

- (1) Turn the mounting screws sleeves through 90° and remove the main beam lamp from the mounting bracket.
- (2) Prise the ball end clips from the main beam lamp brackets.

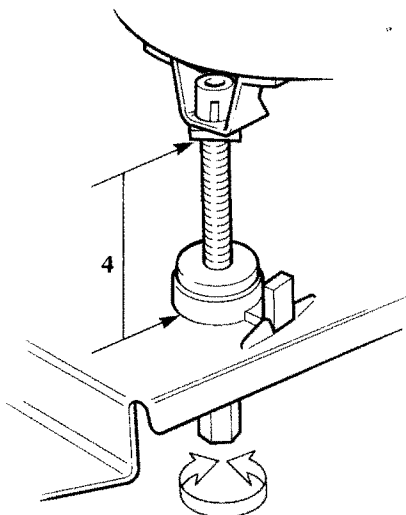


C Disassembly Procedure - Dipped Beam

NOTE:

Prior to removal of the unit, measure and record the distance of each adjusting screw.

- (1) Remove the three screws, washers and springs attaching the dipped beam headlamp to the mounting bracket.
- (2) Remove the three screws attaching the headlamp surround and remove the surround.



D Assembly Procedure - Main Beam

- (1) Wind the sleeves approximately 10mm (0.39in) onto the mounting screws.
- (2) Fit the screws to the mounting bracket and turn the sleeves through 90° to lock.

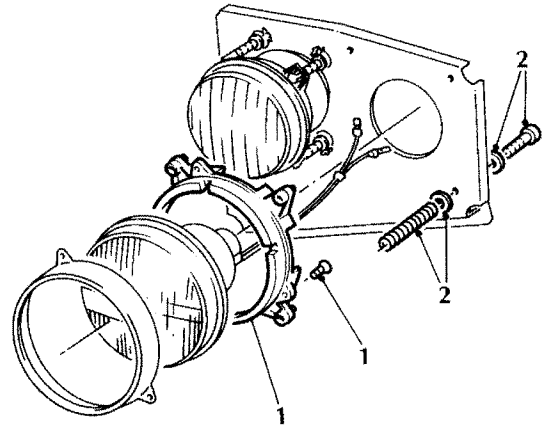
NOTE

Ensure the word "TOP" on the lamp is at the top of the panel.

- (3) Locate the main beam headlamp onto the mounting screws and press the ball end clips into the lamp brackets.
- (4) Adjust the headlamp screw to lengths measured before removal.

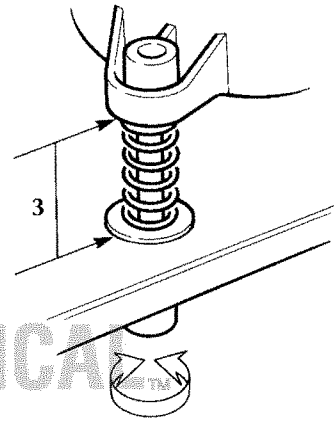
E Assembly Procedure - Dipped Beam

- (1) Fit the headlamp surround aligning the lugs on the surround and lamp (as illustrated) and secure using three screws.
- (2) Assemble the screws, washers and springs and fit the headlamp to the mounting bracket.
- (3) Adjust the headlamp screw to lengths measured before removal.



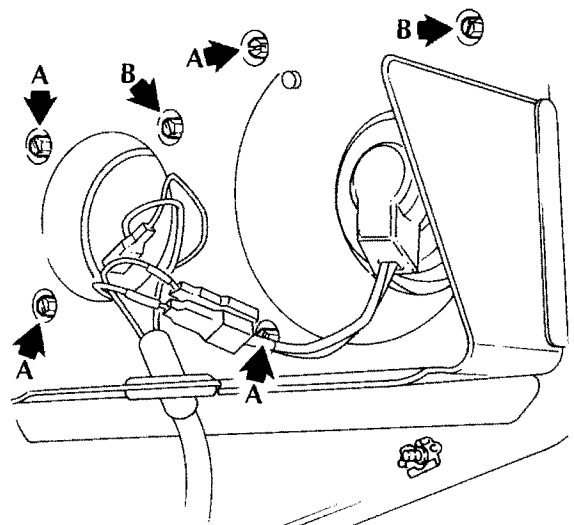
F Installation Procedure

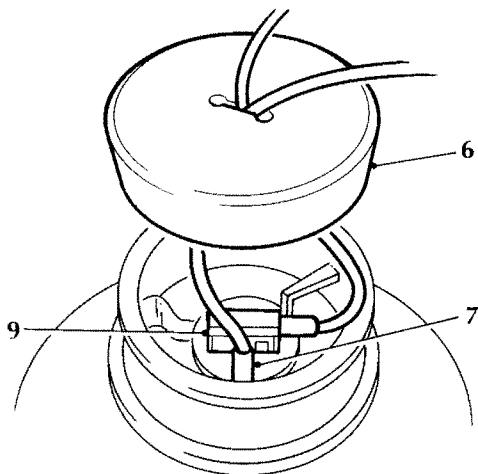
- (1) Locate the headlamp assembly in the vehicle and secure with four washers and screws.
- (2) Connect the electrical cables to the headlamps.
- (3) Fit the wheel arch liner and wheel.
- (4) Ensure the headlamp covers are lowered and fit the headlamp finishers.
- (5) Raise the headlamp covers manually to ensure clearance between the headlamp covers and lamps.
- (6) Connect the battery and ensure the headlamps function correctly.
- (7) Close the bonnet.
- (8) Remove the protective covers and remove the vehicle from the lifting apparatus.



G Headlamp Alignment/Adjustment Procedure

- (1) Ensure the ride height is correct and the vehicle is level.
- (2) Set up the alignment equipment.
- (3) Switch on the headlamps and check the alignment.
- (4) If adjustment is required, disconnect the headlamp motor electrical connections.
- (5) To adjust the vertical alignment turn screw A.
- (6) To adjust laterally, turn screw B.
- (7) Switch off the headlamps on completion.
- (8) Raise the headlamp covers manually to ensure clearance between the headlamp covers and lamps.
- (9) Reconnect the headlamp motor connectors and test for correct operation.





H Bulb Replacement Procedure - Main Beam

- (1) Open the bonnet.
- (2) Remove the headlamp access panel in the radiator duct side.
- (3) Remove the tyre weld aerosol if left hand bulb is to be replaced.
- (4) Switch the battery master switch to the off position.
- (5) Working through the access duct, disconnect the electrical cable to the bulb.
- (6) Pull off the headlamp grommet.
- (7) Remove the earth lead from the headlamp .
- (8) Release the spring retaining clip by pressing the legs of the clip together and swinging outwards.
- (9) Remove the bulb from the headlamp.
- (10) Fit the new bulb electrical cable through the grommet.

CAUTION

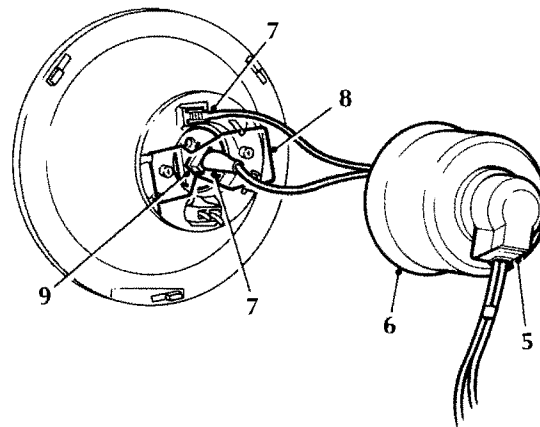
Do not touch the bulb . The bulb may be cleaned with a soft cloth dampened in white spirit.

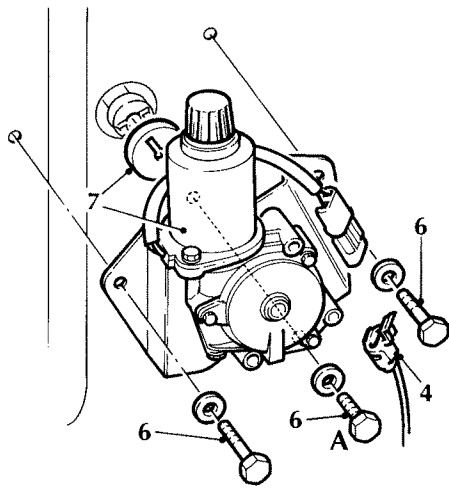
- (11) Fit the bulb to the headlamp ensuring the bulb holder cut-out aligns with the headlamp lug.
- (12) Secure the spring retaining clip.
- (13) Fit the earth lead to the headlamp.
- (14) Fit the grommet to the headlamp.
- (15) Connect the bulb electrical cable to the harness.
- (16) Switch the battery master switch to the on position on check the operation of the headlamp.
- (17) Refit the tyre weld aerosol and refit the headlamp access panel.
- (18) Close the bonnet.

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1 Bulb Replacement Procedure - Dipped Beam

- (1) Open the bonnet.
- (2) Remove the headlamp access panel in the radiator duct side.
- (3) Remove the tyre weld aerosol if left hand bulb is to be replaced.
- (4) Turn the battery master switch to the off position.
- (5) Working through the access duct, remove the cable boot from the grommet and slide back along the cables.
- (6) Pull the grommet from the headlamp.
- (7) Disconnect the earth and bulb electrical leads.
- (8) Release the spring clip and swing outwards.
- (9) Remove the bulb from the headlamp.
- (10) Fit the new bulb to the headlamp ensuring the bulb cut-outs align the headlamp lugs.
- (11) Secure the spring clip.
- (12) Fit the earth and bulb electrical leads.
- (13) Fit the headlamp grommet and cable boot.
- (14) Switch the battery master switch to the on position and ensure the headlamp functions correctly.
- (15) Refit the tyre weld aerosol and refit the headlamp access panel.
- (16) Close the bonnet.





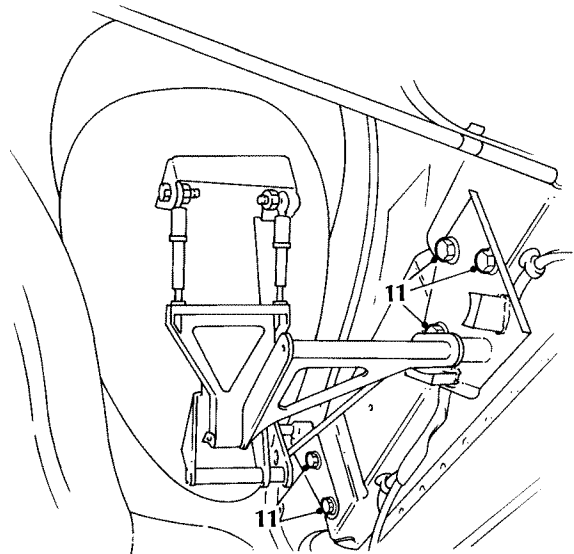
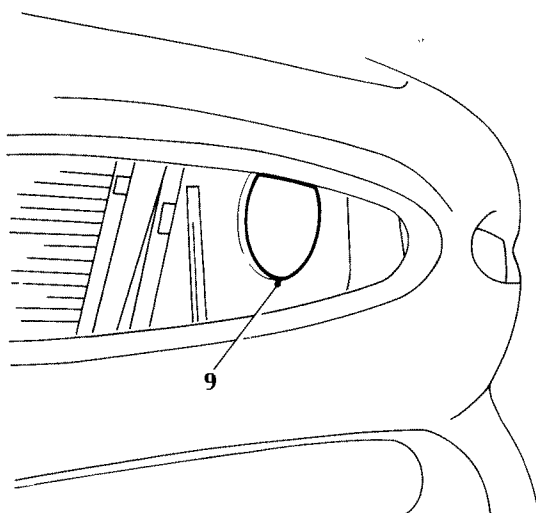
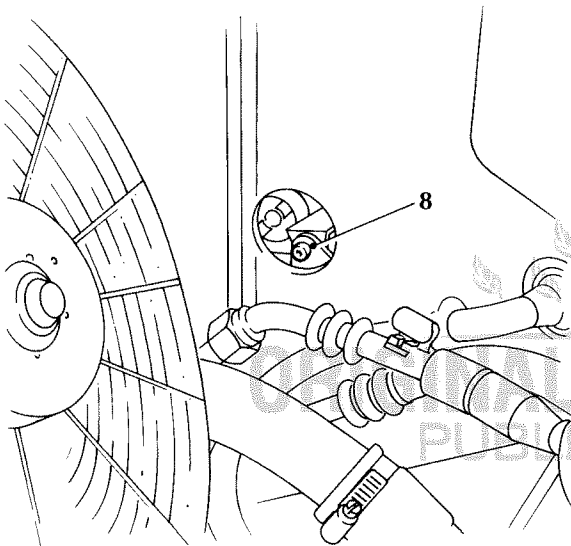
2 HEADLAMP COVER MECHANISM

A Removal Procedure

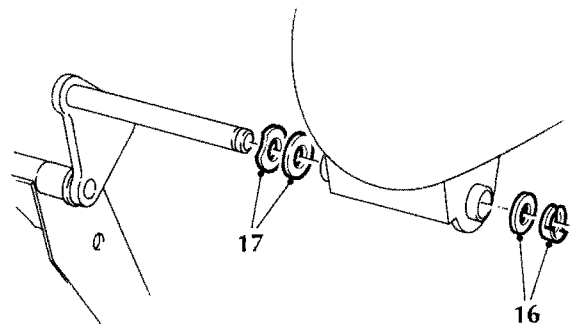
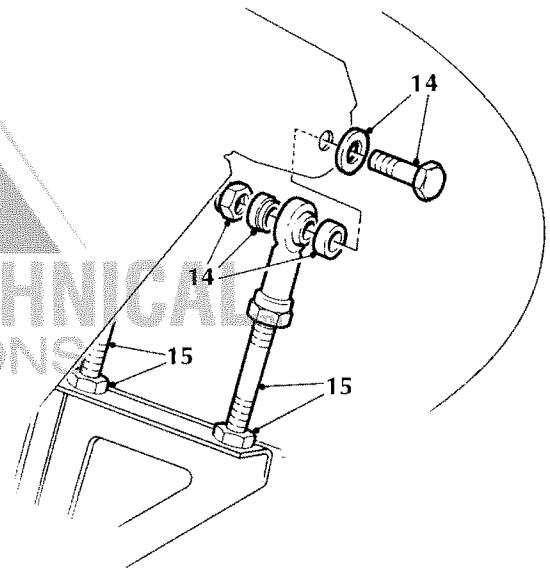
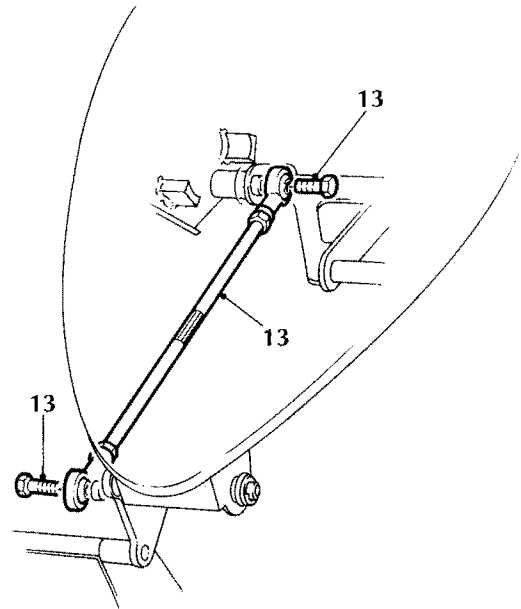
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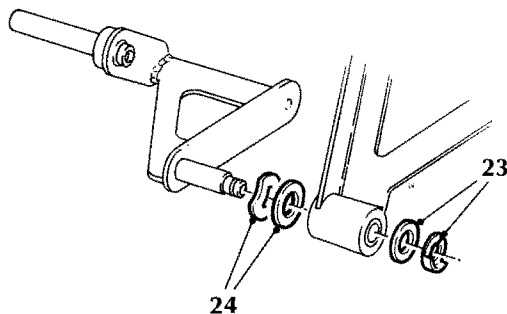
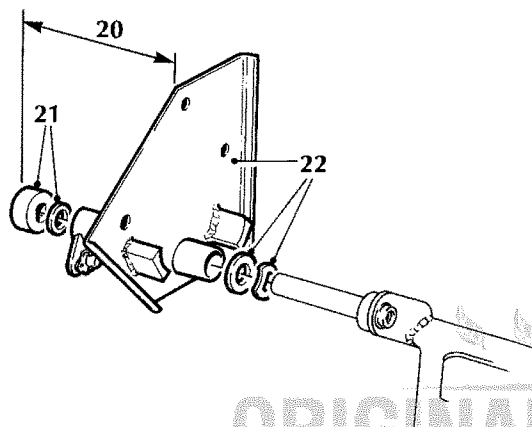
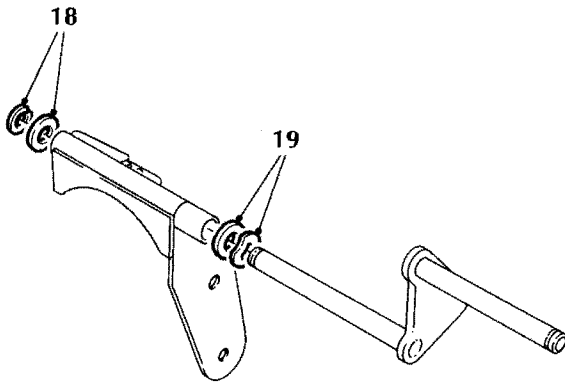
- 1 Ensure absolute cleanliness of work area and tools.
- 2 Observe correct use of tools.

- (1) Open the bonnet.
- (2) Fit JaguarSport protective body covers.
- (3) Disconnect the battery.
- (4) Disconnect the electrical connections to the headlamp cover motor.
- (5) Lower the headlamp covers manually and remove the headlamp finishers.
- (6) Remove the three screws attaching the motor bracket to the radiator duct assembly. Note the position of the shorter screw (A).
- (7) Remove the motor and bracket and collect the torque disc.
- (8) Remove the M5 screw attaching the mechanism to the radiator duct inside the bonnet compartment.
- (9) Remove the screw from inside the HVAC intake duct.
- (10) Remove the front splitter. (Refer to Section 2).
- (11) Remove the 6 screws attaching the mechanism to the structure.



- (12) Remove the mechanism and cover from the vehicle.
- (13) Remove the screws attaching the connecting rod to the operating shaft and remove the rod.
- (14) Remove the screws, nuts and washers attaching the connecting rods to the headlamp cover and collect the spacers.
- (15) Loosen the connecting rod locknut and remove the connecting rod from the rear upper bearing support.
- (16) Remove the spring clip and thrust washer from the front crankshaft and remove the headlamp cover.
- (17) Remove the thrust washer and wavey washer from the crankshaft.





- (18) Remove the spring clip and thrust washer from the front crankshaft and remove the front bearing support.
- (19) Remove the thrust washer and wavey washer from the crankshaft.
- (20) If the motor hub has not been removed from the motor shaft, measure the distance from the end of the bearing support plate to the end of the hub.
- (21) Loosen the grubscrew and remove the hub and thrust washer from the operating shaft.
- (22) Remove the rear bearing support and collect the thrust washer and wavey washer from the operating shaft.
- (23) Remove the spring clip and thrust washer from the operating shaft and remove the rear upper bearing support.
- (24) Remove the thrust washer and wavey washer from the operating shaft.

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B Installation Procedure

- (1) Fit a wavey washer and thrust washer to the operating shaft.
- (2) Fit the operating shaft to the rear upper bearing support and secure with thrust washer and spring clip.
- (3) Fit a wavey washer and thrust washer to the operating shaft and fit the rear bearing support.
- (4) Fit the thrust washer to the operating shaft. Apply bearing lock to the end of the shaft and fit the hub. Lightly pre-load the wavey washer and secure the hub and grubscrew. The hub must be reset at the distance measured in the removal procedure operation 20.

- (5) Fit the wavey washer and thrust washer to the crankshaft.
- (6) Fit the front bearing support to the crankshaft and secure with thrust washer and spring clip.
- (7) Fit a wavey washer and thrust washer to the crankshaft.
- (8) Fit the headlamp cover to the crankshaft and secure with thrust washer and spring clip.
- (9) Fit the connecting rods to the rear upper bearing support.

NOTE

Do not tighten the locknut at this stage.

- (10) Fit the connecting rods to the headlamp cover and secure with nuts, washers and screws.
- (11) Fit the connecting rod to the operating shaft and crankshaft and secure with screws.
- (12) Fit the mechanism to the vehicle loosely assemble with 6 screws.
- (13) Fit and secure the screw inside the HVAC intake duct.
- (14) Secure the remaining screws.
- (15) Fit the torque disc to the operating shaft and fit the motor and bracket, secure the bracket with three screws. Ensure the shorter screw is fitted in the correct position.
- (16) Adjust the mechanism as follows;
 - a) Wind the headlamp covers manually to the fully closed position.
 - b) Loosen the locknuts on vertical and longitudinal connecting rods.

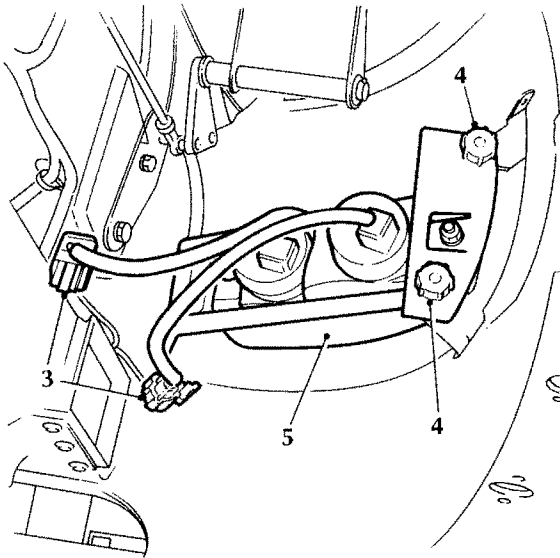
NOTE

The vertical connecting rod upper locknut has a LH thread.

The longitudinal connecting rod has both RH and LH threads

- c) Adjust the connecting rod length to align the cover flush with the vehicle body.
- d) Lock the connecting rod locknuts.

- (17) Manually lower the headlamp covers to ensure clearance between the lamps and cover.
- (18) Refit the headlamp finishers and secure the 5 screws.
- (19) Manually raise the headlamp covers to ensure clearance between the cover and finishers.
- (17) Connect the headlamp motor electrical connectors.
- (18) Connect the battery.
- (19) Switch on the ignition and check the operation of the headlamp covers.
- (20) Re-adjust if necessary.
- (21) Close the bonnet.
- (22) Fit the front splitter. (Refer to Section 2).
- (23) Remove the protective body covers.

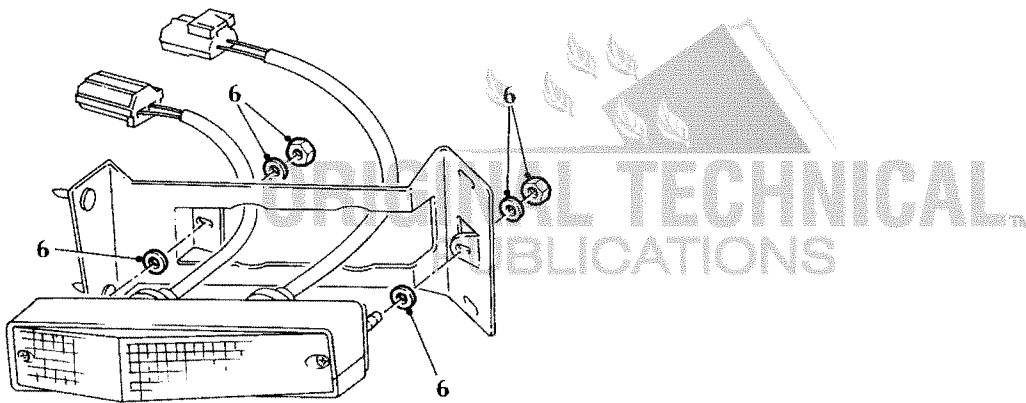


3 FRONT SIDE AND INDICATOR LAMPS

A Removal Procedure

NOTES:

- 1 Ensure absolute cleanliness of work area and tools.
 - 2 Observe correct use of tools.
- (1) Remove the relevant wheel arch liner (Refer to Section 2).
 - (2) Switch the battery master switch to the off position.
 - (3) Working through the wheel arch, disconnect the lamp electrical harness plugs.
 - (4) Remove the two knurled nuts and washers attaching the lamp mounting bracket.
 - (5) Remove the lamp assembly from the vehicle.
 - (6) Remove the two nuts and washers attaching the lamp to the mounting bracket.

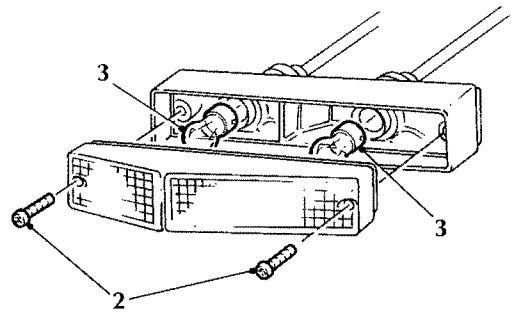


B Installation Procedure

- (1) Locate the lamp in the mounting bracket and secure with two washers and nuts.
- (2) Fit the mounting bracket to the vehicle and secure with two washers and knurled nuts.
- (3) Connect the lamp electrical harness plugs.
- (4) Switch the battery master switch to the on position and carryout a functional check of the lamps.
- (5) Fit the wheel arch liner. (Refer to Section 2).

C Bulb Replacement Procedure

- (1) Remove the lamp assembly as described in Sub-section 3.A paragraphs 1 to 5 of this section.
- (2) Remove the two screws securing the lens and remove the lens.
- (3) Remove the defective bulb and replace with one of the correct specification.
- (4) Fit the lens cover and secure the two screws.
- (5) Fit the lamp assembly as described in Sub-section 2.B paragraphs 3 to 5 of this section.



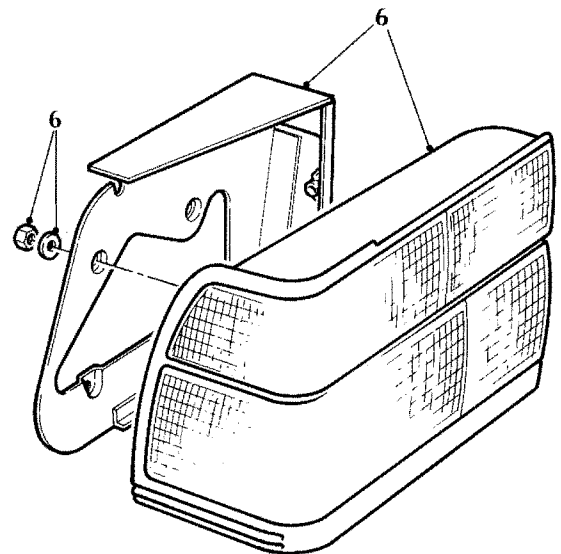
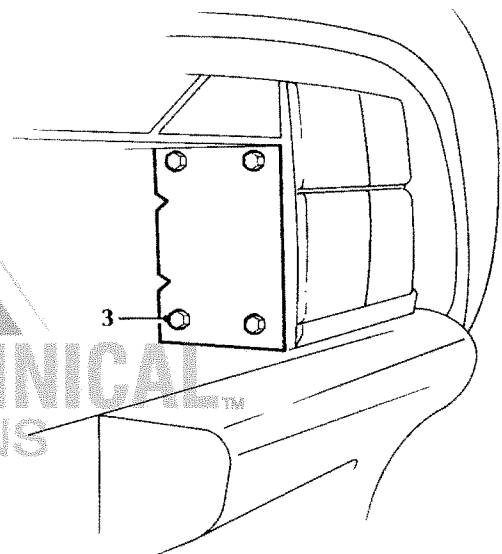
4 REAR LAMPS

A Removal Procedure

NOTES:

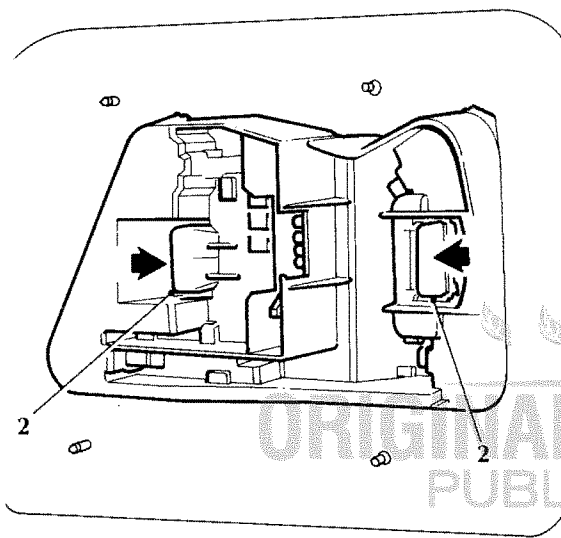
- 1 Ensure absolute cleanliness of work area and tools.
- 2 Observe correct use of tools.

- (1) Remove the rear grille. (Refer to Section 2).
- (2) Turn the battery master switch to the off position.
- (3) Support the assembly and remove the four screws attaching the rear lamp assembly to the vehicle.
- (4) Ease the lamp assembly rearwards and disconnect the electrical harness plug.
- (5) Remove the assembly from the vehicle.
- (6) Remove the four washers and nuts securing the lamp assembly to the mounting bracket and separate the assembly and bracket.



B Installation Procedure

- (1) Locate the lamp assembly on the mounting bracket and secure with four nuts and washers.
- (2) Locate the assembly in the vehicle and connect the electrical harness plug.
- (3) Secure the assembly to the vehicle using four washers and screws.
- (4) Switch the battery master switch to the on position and carryout a functional check of the rear lamps.
- (5) Fit the rear grille. (Refer to Section 2).



C Bulb Replacement Procedure

- (1) Remove the lamp assembly as described in Sub-section 4.A paragraphs 1 to 5 of this section.
- (2) On the bulb holder assembly, press the two retaining clips inwards and remove the bulb holder assembly.
- (3) Remove the defective bulb(s) and replace with one of the correct specification.
- (4) Fit the bulb holder to the lamp assembly. Ensure the retaining clips are fully secure.
- (5) Fit the lamp assembly to the vehicle as described in Sub-section 4.B of this section.

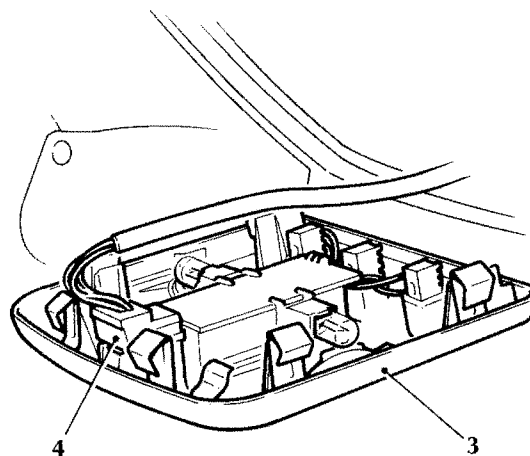
5 INTERIOR/MAP LAMP CONSOLE

A Removal Procedure

NOTES:

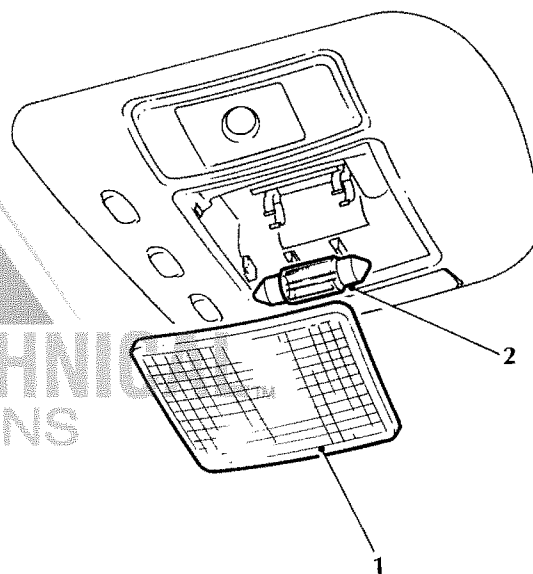
- 1 Ensure absolute cleanliness of work area and tools.
- 2 Observe correct use of tools.

- (1) Fit JaguarSport interior protective covers.
- (2) Move the battery master switch to the off position.
- (3) Carefully lever out the lamp console.
- (4) Disconnect the electrical leads to the lamp unit.
- (5) Remove the unit from the vehicle.



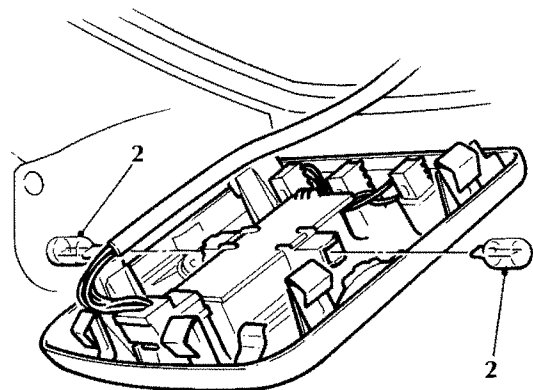
B Bulb Replacement Procedure - Main Interior

- (1) Carefully lever out the lamp lens from both ends.
- (2) Prise the bulb from the bulb holder.
- (3) Fit a new bulb and refit the cover.



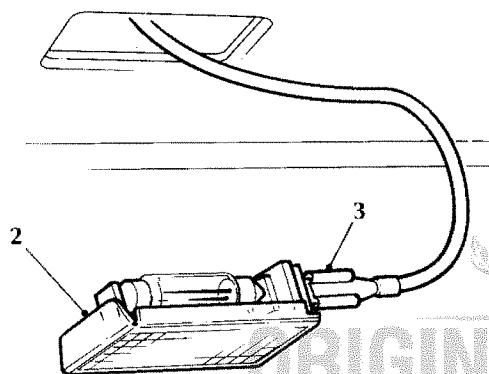
C Bulb Replacement Procedure - Map Lights

- (1) Carefully lever out the lamp console.
- (2) Pull the defective bulb(s) from the holder(s).
- (3) Fit a new bulb.
- (4) Refit lamp console.



D Installation Procedure

- (1) Fit the electrical leads to the lamp console.
- (2) Fit the console to the roof trim and ensure it is properly engaged.
- (3) Move the battery master switch to the on position.
- (4) Carryout a functional check of the lamp console. Rectify faults if necessary.
- (5) Remove the interior protective covers.

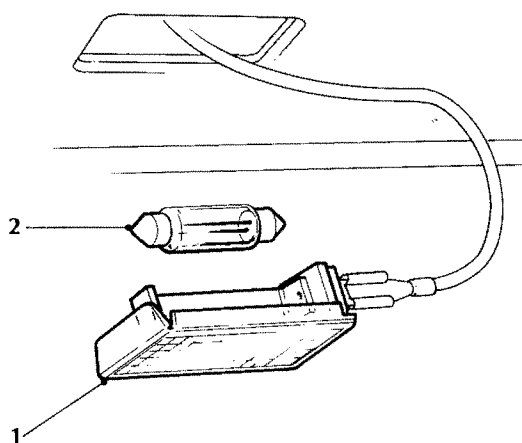


6 FOOTWELL LAMPS

A Removal Procedure

NOTES:

- 1 Ensure absolute cleanliness of work area and tools.
 - 2 Observe correct use of tools.
- (1) Move the battery master switch to the off position.
 - (2) Carefully lever out one of the short ends of the lamp holder.
 - (3) Disconnect the electrical connectors and remove the lamp assembly.



B Bulb Replacement Procedure

- (1) Lever out the bulb holder.
- (2) Replace the defective bulb and refit the bulb holder.

C Installation Procedure

- (1) Connect the electrical connectors to the bulb holder.
- (2) Fit the bulb holder into the underscuttle.
- (3) Move the battery master switch to the on position.
- (4) Carryout a functional check of the footwell lamps and rectify any faults as necessary.

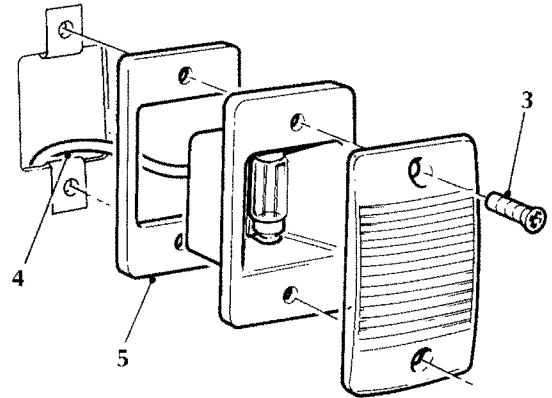
7 DOOR WARNING LAMPS

A Removal Procedure

NOTES:

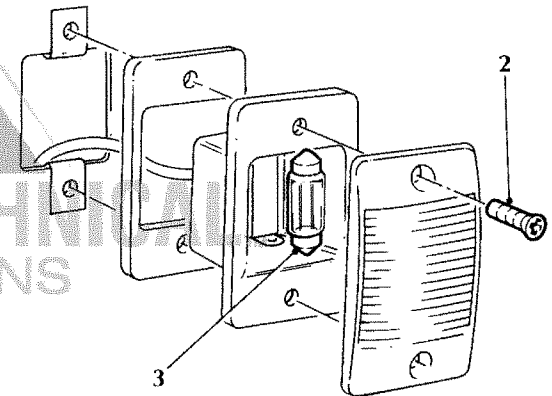
- 1 Ensure absolute cleanliness of work area and tools.
- 2 Observe correct use of tools.

- (1) Open the relevant door.
- (2) Move the battery master switch to the off position.
- (3) Remove the two screws attaching the lamp assembly to the door.
- (4) Withdraw the lamp assembly and disconnect the lamp electrical connections.
- (5) Remove the gasket.



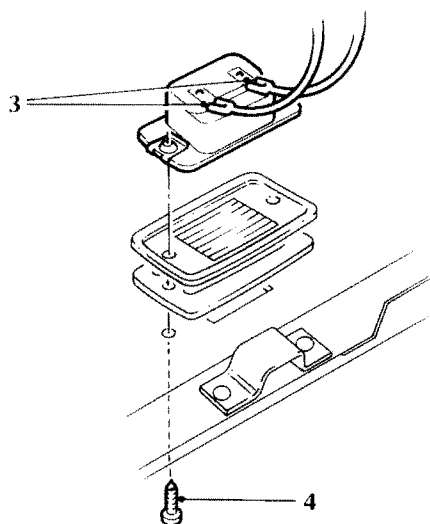
B Bulb Replacement

- (1) Move the battery master switch to the off position.
- (2) Remove the two screws securing the lens and remove the lens.
- (3) Replace the bulb with one of the correct type and refit the lens.
- (4) Move the battery master switch to the on position and ensure the lamp functions correctly.



C Installation Procedure

- (1) Fit a new gasket to the lamp assembly.
- (2) Connect the lamp electrical connectors.
- (3) Fit the lamp assembly to the door and secure with the two screws.
- (4) Move the battery master switch to the on position and ensure the lamp functions correctly.



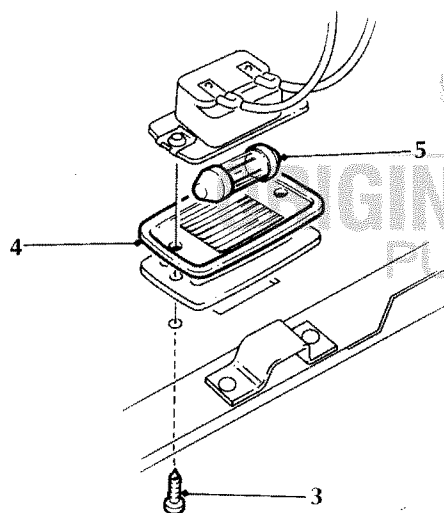
8 KERB LAMP

A Removal Procedure

NOTES:

- 1 Ensure absolute cleanliness of work area and tools.
- 2 Observe correct use of tools.

- (1) Move the battery master switch to the off position.
- (2) Remove the relevant door trim. (Refer to Section 2).
- (3) Disconnect the electrical leads to the lamp assembly.
- (4) Remove the two screws securing the lamp assembly to the door and lift out the assembly.



B Bulb Replacement Procedure

- (1) Move the battery master switch to the off position.
- (2) Remove the door trim (Refer to Section 2).
- (3) Remove the two screws securing the lamp to the door.
- (4) Lift off the lens.
- (5) Replace the bulb with one of the correct type and refit the lens.
- (6) Fit the lamp to the door and secure with the two screws.
- (7) Move the battery master switch to the on position and ensure the lamp functions correctly.
- (8) Fit the door trim (Refer to Section 2).

C Installation Procedure

- (1) Fit a new gasket to the lamp assembly.
- (2) Fit the lamp to the door and secure with the two screws.
- (3) Connect the electrical leads to the lamp.
- (4) Move the battery master switch to the on position and ensure the lamp functions correctly.
- (5) Fit the door trim. (Refer to Section 2).

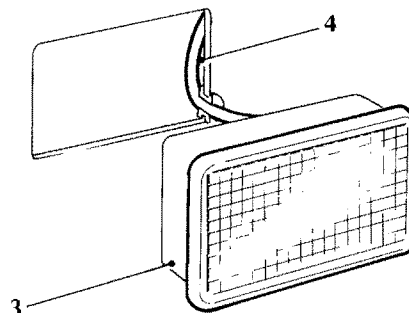
9 ENGINE COMPARTMENT LAMPS

A Removal Procedure

NOTES:

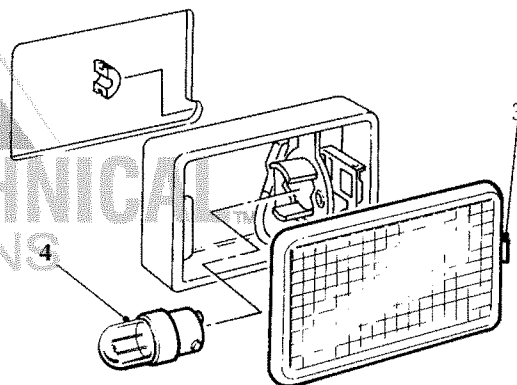
- 1 Ensure absolute cleanliness of work area and tools.
- 2 Observe correct use of tools.

- (1) Open the engine compartment hatch and fit JaguarSport protective covers.
- (2) Move the battery master switch to the off position.
- (3) Prise the lamp assembly out of the body aperture.
- (4) Disconnect the electrical connections to the lamp.
- (5) Remove the lamp from the vehicle.



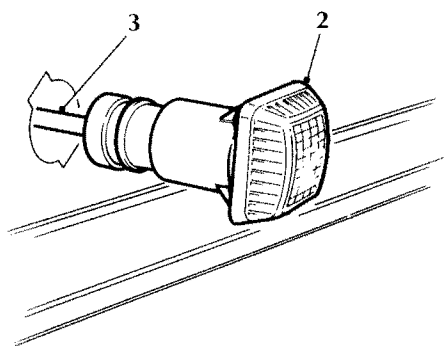
B Bulb Replacement Procedure

- (1) Open the engine compartment and fit JaguarSport protective covers.
- (2) Move the battery master switch to the off position.
- (3) Push the spring clip on the side of the lens and remove the lens.
- (4) Twist and remove the bulb.
- (5) Replace the bulb with one of the correct type and refit the lens. Ensure the lens is securely engaged.



C Installation Procedure

- (1) Connect the electrical connections to the lamp.
- (2) Fit the lamp to the vehicle and ensure the electrical leads are located in the half grommet.
- (3) Move the battery master switch to the on position and ensure the lamp functions correctly.
- (4) Remove the protective covers and close the engine compartment hatch.



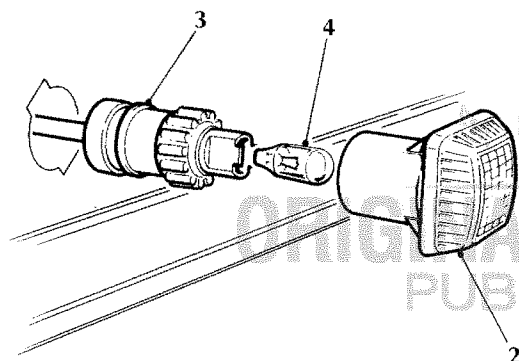
10 SIDE REPEATER LAMPS

A Removal Procedure

NOTES:

- 1 Ensure absolute cleanliness of work area and tools.
- 2 Observe correct use of tools.

- (1) Move the battery master switch to the off position.
- (2) Rotate the lens one quarter of a turn clockwise by hand and pull the lamp assembly out of the door.
- (3) Disconnect the electrical connections and remove the lamp assembly.



B Bulb Replacement Procedure

- (1) Move the battery master switch to the off position.
- (2) Rotate the lens one quarter of a turn clockwise and remove the lamp assembly from the door.
- (3) Hold the base of the assembly and rotate the lens one quarter of a turn counter-clockwise to separate it from the base.
- (4) Remove the bulb and replace it with one of the correct type.
- (5) Refit the lens on the base and rotate it clockwise until it locks.
- (6) Insert the assembly into the door and turn counter-clockwise to secure.
- (7) Move the battery master switch to the on position.
- (8) Switch on the ignition and ensure the side repeaters function correctly.

C Installation Procedure

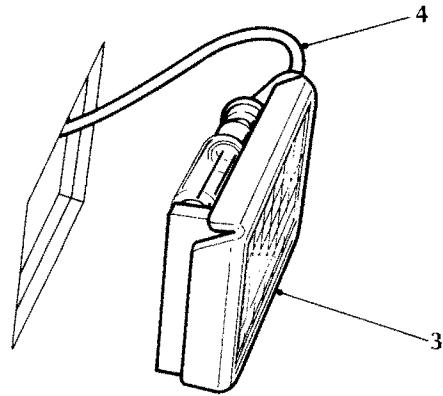
- (1) Fit the electrical connections to the lamp assembly.
- (2) Fit the lamp assembly to the door and turn counter-clockwise to secure.
- (3) Move the battery master switch to the on position.
- (4) Switch on the ignition and ensure the side repeaters function correctly.

11 LUGGAGE COMPARTMENT LAMP

A Removal Procedure

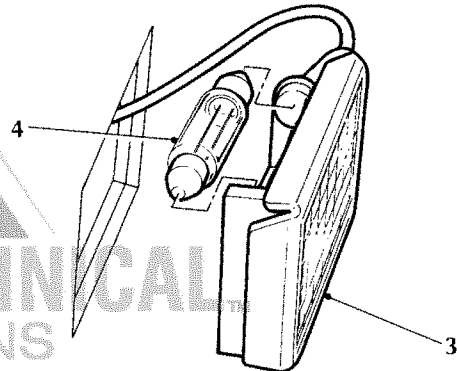
NOTES:

- 1 Ensure absolute cleanliness of work area and tools.
 - 2 Observe correct use of tools.
-
- (1) Move the battery master switch to the off position.
 - (2) Open the luggage compartment.
 - (3) Carefully lever one of the ends of the lamp holder from the side panel.
 - (4) Disconnect the electrical connections from the lamp and remove the assembly from the vehicle.



B Bulb Replacement Procedure

- (1) Move the battery master switch to the off position.
- (2) Open the luggage compartment.
- (3) Carefully lever one of the ends of the lamp holder from the side panel.
- (4) Remove the bulb and replace it with one of the correct type.
- (5) Refit the lamp holder and ensure it is fully engaged in the side panel.
- (6) Move the battery master switch to the on position.
- (7) Switch on the ignition and ensure the luggage compartment lamp functions correctly.
- (8) Close the luggage compartment.



C Installation Procedure

- (1) Fit the electrical connections to the lamp assembly.
- (2) Fit the lamp assembly and ensure it is fully engaged in the side panel.
- (3) Move the battery master switch to the on position.
- (4) Switch on the ignition and ensure the luggage compartment lamp functions correctly.
- (5) Close the luggage compartment.